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BASIC INDUSTRIAL BIOTECHNOLOGY



S.M. Reddy • S.Ram Reddy
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Basic Industrial Biotechnology

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Preface

Developments in the area of life sciences such as recombinant DNA technology and human genome project have changed our basic concepts which has profound influence on the quality of life. It is likely that such changes will be accelerated in the future and such advances probably rely on basic knowledge along with its transformation into products that can be produced eco-friendly as well as in cheapest possible manner. The fundamentals of biotechnology remains strong, the production of goods and services that are needed without risk involvement that too at economic price.

Biotechnology is not mere recombinant of DNA and cloning, but production of more prosaic materials like organic acids, amino acids, beverages, fermented foods, antibiotics, biosurfactants, polysaccharides and the like. The main aim of this discipline is to provide clear technology for 21st century which will sustain the growth and development all over the world to improve the quality of life. It is likely to influence the health care, foods supply and environment. In short no aspect of our life will remain unaffected.

The main aim of this book is, to provide an overview of many of the fundamental aspects that underpin all biotechnology and provide examples of how these principles are put into operation starting from substrate through final product. Since the biotechnology is huge multidisciplinary activity, we restricted ourselves to provide an mainstream account of the current state of industrial bioprocesses which may provide the reader with insight, inspiration and instruction in skills and art of the subject. We also hope that it will provide some understanding of promise, limits of biotechnology to policy makers, regulators and corporate decisions.

Precise care has been taken while writing the book in simple and lucid language, keeping in view the standard of graduate and postgraduate students of Indian universities in the discipline of basic biotechnology, engineering, pharmaceutics and chemical technology. An attempt has also been made to cover the syllabi of as many universities as possible including JNTU, Hyderabad.

For the last fifteen years we have offered a course on applied microbiology and industrial microbiology to the students of Kakatiya University, Warangal and Jawaharlal Nehru Technological University, Hyderabad, which provided an inspiration to develop much of the

material herein. We thank all these students profusely. The task of this magnitude needs help of many, either directly or indirectly are worthy of our thanks. We also thank those who graciously permitted us to use published materials. We thank the Chairman Sri M Laxman Reddy and management of St. Martin's Engineering College, Secunderabad and the authorities of Kakatiya University, for kind encouragement and facilities.

S. M. REDDY
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1

History of Industrial Microbiology

Industrial microbiology came into existence, primarily, based on a naturally occurring microbiological process called **fermentation**. There are many evidences which clearly shows that ancient man knew fermentation process and practiced it more as an art rather than as a science. Early fermentation process practiced by man included the leavening of bread, retting of flax, preparation of vinegar from wine, production of various alcoholic beverages like beer, wine, mead and the production of various fermented foods and milk. Due to invention of microscope, discovery of microorganisms and understanding of their metabolic processes, lead to clear understanding of the fermentation, which paved the way for the development of Industrial Microbiology.

The history of industrial microbiology can be divided into five phases, which are précised in table 1.1 Phase I up to 1900 Alcohol fermentation period, Phase II 1900-1940 Antibiotic period, Phase III 1940-1964 Single cell protein period, Phase IV 1964-1979 Metabolite production period, and Phase V 1979 onward Biotechnology period.

Table 1.1: The phases in the history of Industrial Microbiology

Phase	Main products	Fermenters	Process control	Culture method	Quality control	Pilot plant facilities	Strain selection
I Period before 1900	Alcohol	Wooden upto 1500 barrels capacity	Use of thermo-meters, hydrometer and heat exchangers	Batch	Prac-tically nil	Nil	Pure yeast culture used at some of the breweries
	Vinegar	Barrels-shallow trays-trickle filters	---	Batch	Practi-cally nil	Nil	Process inoculated with good vinegar

contd...

Phase	Main products	Fermenters	Process control	Culture method	Quality control	Pilot plant facilities	Strain selection
	Bakers yeast, glycerol, citric acid, lactic acid and acetone/ butanol	Steel vessels upto 200 m^3 for acetone / butanol. Air sprayers used for bakers yeast.	pH electrodes with off-line control. Temperature control	Batch and fed-batch systems	Practically nil	Nil	Pure cultures used
II Period between 1900-1940	Penicillin, streptomycin other antibiotics	Mechanical stirring used in small vessels, mechanically aerated vessels	Sterilizable pH and oxygen electrodes	Batch and fed-batch common	Very important	Becomes common	Mutations and selection programme essential
III Period between 1940-1964	Gibberellins, amino acids, nucleotides, enzymes, transformations	Vessels operated aseptically, true fermentations	Use of control loops which were later computerised	Continuous culture introduced for brewing and some primary metabolites	Very important	Becomes common	Mutation and selection programme essential
IV Period between 1964-1979	Single cell protein using hydrocarbons and other feed stocks	Pressure cycle and pressure jet vessels developed to overcome gas and heat exchange problems	Use of computer linked control loops	Continuous culture with medium recycle	Very important	Very important	Genetic engineering of producer strain attempted
V 1979-onward	Production of heterogenous proteins by microbial and animal cells; Monoclonal antibodies produced by animal cells	Fermenters developed in phase 3 and 4. Animal cell reactors developed	Control and sensors developed in phases 3 and 4	Batch, fed-batch or continuous fermentation developed for animal cell processes	Very important	Very important	Introduction of foreign genes into microbial and animal cells. In vitro recombinant DNA techniques used in the improvement of phase 3 products

1.1 ALCOHOL FERMENTATION PERIOD (BEFORE 1900)

The period before 1900 is marked by the production of primarily alcohol, vinegar and beer, although without the knowledge of biochemical processes involved in it. Though beer, which

represents the phase-I in fermentation process, was produced by ancient Egyptians, large scale brewing in large wooden vats of 1500-barrel capacity was started in the early 1700. An attempt was also made for process control by the use of thermometers and heat exchangers in these early breweries.

In the middle of 18th century, the chemist **Liebig** considered fermentation purely as a chemical process. He believed fermentation as a disintegration process in which molecules present in the starter substance like starch or sugar underwent certain changes resulting in the production of alcohol. Other eminent chemists of this period like **Berzelius** (1779–1848) and **Bertholet** (1827–1907) have also supported this view. **Cagniard Latour, Schwan** and **Kutzilog** while working independently concluded that alcoholic fermentation occurs due to action of yeast which is an unicellular fungus. But, it was **Louis Pasteur** who eventually convinced the scientific world that the fermentation is a biological process. By conducting series of experiments, **Louis Pasteur** conveniently proved that yeast is required for conversion of sugars into alcohol. In 1857, he discovered the association of different organisms other than yeasts in the conversion of sugars into lactic acid. These observations led **Pasteur** to conclude that different kinds of organisms are required for different fermentations.

While working on butyric acid fermentation in 1861, **Pasteur** made another important discovery that the fermentation process can proceed in the absence of oxygen. The rod shaped organisms responsible for butyric acid fermentation, remains active in the absence of oxygen. This organism was later on identified as butyric acid bacterium. This observation subsequently lead to the emergence of a new concept of anaerobic microorganisms and a classification of three organisms broadly into two categories, viz., **aerobic** and **anaerobic** microorganisms.

During this period, wine Industry in France was incurring heavy losses due to soaring of wine. Pasteur was requested by the Government of France to study this problem. After careful study, he reported that the soaring of wine was due to the growth of other unwanted microorganisms, other than yeast, which invaded the wine and changed its chemical and physical properties leading to soaring. He showed that these unwanted organisms could be eliminated from the wine by partially sterilizing the juice from which wine is produced, below the boiling point. This process is now called as **Pasteurization**. Pasteurization kills all the bacteria but does not alter the desirable qualities of juice. This proposition of ***Pasteur*** saved the wine industry of France from heavy losses. Later on ***Pasteur*** has also studied the fermentation of acetic acid and beer. He disproved the concept of chemical basis of fermentation.

During the late 19th century ***Hansen***, working at Carlsberg Brewery, developed methods for production of pure cultures of yeast and techniques for production of starter cultures. Thus, by the end of nineteenth century, the concept of involvement of microorganisms in fermentation process and its control were well established in brewing industry.

1.2 ANTIBIOTIC PERIOD (1900–1940)

Important advances made in the progress of industrial microbiology were the development of techniques for the mass production of bakers yeast and solvent fermentations. However, the growth of yeast cells in alcoholic fermentation was controlled by the addition of Wort periodically in small amounts. This technique is now called as ***fed batch culture*** and is widely used in the fermentation industry specially to avoid conditions of oxygen limitation. The aeration of early yeast cultures was also improved by the introduction of air through sparging tubes.

The other advancement during this period was the development of acetonebutanol fermentation by ***Weisman***, which was considered to be truly aseptic and anaerobic fermentation. The techniques developed for the production of these organic solvents were major advances in fermentation technology, which led to the successful introduction of aseptic aerobic processes, which facilitated in the production of glycerol, citric acid and lactic acid.

Another remarkable milestone in the industrial microbiology was the large-scale production of an antibiotic called penicillin, which was in great demand to save lives of thousands of wounded soldiers of Second World War. The production of penicillin is an aerobic process which is carried out by submerged culture technique under aseptic conditions. The inherent problems of contamination, requirement of large amount of liquid medium, sparging the culture with large volume of sterile air, mixing of highly viscous broth were solved. The technology established for penicillin fermentation paved the way for the development of a wide range of new processes such as production of other antibiotics, vitamins, amino acids, gibberellins, enzymes and steroid transformations.

At about the same time ***Dubos*** at Rockfeller Institute, discovered a series of microbial products which showed antimicrobial properties and hence useful in treating certain human diseases. ***Waksman***, a soil microbiologist, and his associates have discovered many antibiotics produced by species of *Streptomyces*, soil inhabiting, which is now widely used (**table 1.2**).

Table 1.2: List of antibiotics and the year of their discovery

Name of the antibiotic	Name of the discoverer	Year of discovery	Producing organism
Penicillin	Alexander Fleming	1929	Penicillium Chrysogenum
Tyrothricin	–	1939	Bacillus
Griseofulvin	–	1939	Penicillium griseofulvum
Streptomycin	S.A. Waksman *et al.*	1943	Bacillus licheniformis
Bacitracin	Johnson *et al*.	1945	Streptomyces griseus
Chloramphenicol	Ehrlich	1947	St. Venezuelae
Polymyxin	–	1947	Bacillus polymyxa
Chlortetracycline	Duggar	1948	St. aureofacieus
Cephalosporin, C, N, P	Brolzu	1948	Cephalosporium acremonium
Neomycin	Waksman *et al.*	1949	St. fradiae
Oxytetracycline	Finley *et al.*	1950	St. rimosus
Nystatin	–	1950	St. noursei
Erythromycin	Clark	1952	St. erythreus
Novobiocin	–	1955	St. niveus
Kanamycin	–	1957	St. kanamyceticus
Fusidic Acid	–	1960	Furidium calcineurin
Ampicillin	–	1961	Semi synthetic
Cephalothin	–	1962	Semi synthetic
Lincomycin	–	1962	St. lincolensis
Gentamycin	–	1963	Micromonospora purpurea
Carbenicillin	–	1964	Semi synthetic
Cephalexin	–	1967	Semi synthetic
Clindamycin	–	1968	Semi synthetic

1.3 SINGLE CELL PROTEIN PERIOD (1940–1964)

This period is marked by the production of proteinaceous food from the microbial biomass. As the cost of the resultant product was very low there was a need for large-scale production of microbial biomass. This led to the development of largest mechanically stirred fermenters ranging from 80,000 to 1,50,000 liters or even more in diameter, which were to be operated continuously for several days, if they were to be economical. Thus, a new fermentation process called continuous culture fermentation came into existence. The most long-lived continuous culture fermentation was the ICI Pruteen animal feed process employing the culture of *Methylophillus methylotrophus.*

1.4 METABOLITE PRODUCTION PERIOD (1964–1979)

During this period, new microbial processes for the production of amino acids and 5^1- nuclosides as flavour augmenters were developed in Japan. Numerous processes for enzyme production, which were required for industrial, analytical and medical purposes, were perfected. Techniques of immobilization of enzymes and cells were also developed. Commercial production of microbial biopolymers such as Xanthan and dextran, which are used as food additives, had been also started during this period. Other processes that were developed during this period includes the use of microorganisms for tertiary oil recovery.

1.5 BIOTECHNOLOGICAL PERIOD (1980 ONWARDS)

Rapid strides in industrial microbiology have taken place since 1980, primarily because of development of new technique like genetic engineering and hybridoma technique. By genetic engineering it was made possible to *in vitro* genetic manipulations which enabled the expression of human and mammalian genes in microorganisms so thereby facilitating large scale production of human proteins which could be used therapeutically. The first such product is the human insulin used for treating the ever growing disease, diabetes. This was followed by the production of human growth hormone, erythropoietin and myeloid colony stimulating factor (CSFs), which control the production of blood cells by stimulating the proliferation, Erythro-poietin used in the treatment of renal failures, anemia and platelet deficiency associated with cancer, gametocyte colony stimulating factor (GCSF) used in cancer treatment and several growth factors used in wound healing processes. The hybridoma technique, which is employed for the production of monoclonal antibodies which aid in medical diagnosis and therapeutics, is also developed during this period.

Perfection of production of microbial secondary metabolites related fermentation processes and their large-scale production is the other major development of this period. Some of such secondary metabolites released into the market includes:

1. Cyclosporine, an immunoregulant used to control rejection of transplanted organs.
2. Imipenem, a modified carbapenem used as a broad-spectrum antibiotic.
3. Lovastatin, a drug used for reducing blood cholesterol levels.
4. Ivermectin, an antiparasitic drug used to prevent African River Blindness disease.

This brief account of history of development of industrial microbiology justifies the statement of Foster (1949), "Never underestimate the power of microbes".

REVIEW QUESTIONS

I. Essay Type Questions

1. Trace the history of use of microorganisms in industry.
2. Discuss the role of microorganisms in food industry.
3. Discuss milestones in the development of industrial microbiology.

II. Write short notes on:

(a) Antibiotic era
(b) Alcoholic beverage period
(c) Microbial metabolites era
(d) Biotechnology era
(e) Single cell protein concept
(f) Monoclonal antibody era
(g) Pasteurization
(h) cyclosporin
(i) lovastatin

FURTHER READING

1. Bader, F.G. (1992). Evolution in fermentation facility design from antibiotics to recombinant proteins in Harnessing Biotechnology for the 21st century (eds. Ladisch, M.R. and Bose, A.) American Chemical Society, Washington DC pp. 228–231.
2. Bushell, M.E. (1998). Application of the principles of industrial microbiology to biotechnology (ed. Wiseman, A.) Chapman and Hall, New York pp. 5–43.
3. Rehm, H.J. and Reed, G. (1993), Biotechnology (2nd edition) Vol. 1–12, VCH, Weinheim.

2

Fermentation Process

Fermentation term for the first time was coined by **Louis Pasteur** for a phenomenon of bubbling of sugar solution. Later on, it has been applied for the phenomenon of production of different chemicals involving microorganisms. Presently, the term is used solely to any phenomenon involving microorganisms. Many products are made by large-scale fermentation including amino acids, enzymes, organic acids, vitamins, antibiotics, solvents and fuels. The typical fermentation process is depicted in Fig. 2.1.

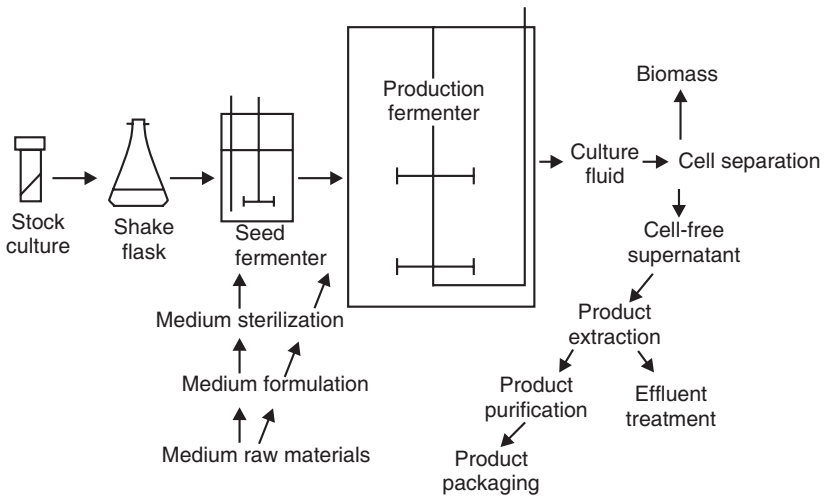


Fig. 2.1: A schematic representation of a typical fermentation process

The advantages in producing materials by fermentation are as follows:

1. Complex molecules such as antibiotics, enzymes and vitamins are impossible to produce chemically.
2. Optically active compounds such as amino acids and organic acids are difficult to prepare chemically.

3. Though some of the products that can be economically derived by chemical processes, but for food purpose they are better produced by fermentation such as beverages, ethanol and vinegar (acetic acid).
4. Fermentation usually uses renewable feed stocks instead of petrochemicals.
5. Reaction conditions are mild, in aqueous media and most reaction steps occur in one vessel.
6. Byproducts of fermentation are usually chemicals. The cell mass and other major by products are highly nutritious and can be used in animal feeds.

However, it is beset with some drawbacks, which are as follows:

1. The products are made in complex solutions in low concentrations as compared to chemically derived compounds.
2. It is difficult and expensive to purify the product.
3. Microbial processes are much slower than chemical processes, increasing the fixed cost of the process.
4. Microbial processes, are subjected to contamination by competiting microorganisms, requires the sterilization of the raw materials and the containment of the process to avoid contamination.
5. Most microorganisms do not tolerate wide variation in temperature, pH and are also sensitive to upsets in the oxygen and nutrient levels. Such upsets not only slow the process, but fatal to microorganism. Thus careful control of pH, nutrients, air and agitation require close monitoring and control.
6. Although nontoxic, waste products have high BOD and requires extensive sewage treatment.

Though microorganism belonging to bacteria, fungi and yeasts are extensively used in these fermentation, few fermentations are also based on algae, plants and animal cells. Several cellular activities contribute to fermentation products such as:

1. **Primary metabolites:** Ethanol, lactic acid and acetic acid.
2. **Energy storage compounds:** Glycerol, polymers and polysaccharides.
3. **Proteins:** SCP, enzymes of both extra and intracellular nature and foreign protein.
4. **Intermediate metabolites:** Amino acids, citric acid, vitamins and malic acid.
5. **Secondary metabolites:** Antibiotics.
6. **Whole cell products:** SCP, bakers yeast, *brewers* yeast, bioinsecticides.

Some of the products such as ethanol, lactic acid and cell mass products are generally growth associated, while secondary metabolites, energy storage compounds, and polymers are non-growth associated. Other products, such as protein depends on the cellular or metabolic function. Unlike primary metabolites which are essential for growth and reproduction, secondary metabolites are not essential for the growth and development of reproducing organism and are produced only in luxuriant conditions (**Bu Lock, 1961**). The secondary metabolites are basically are:

1. Secondary metabolites are produced only by few organisms.
2. Secondary metabolites are needed depending on environmental conditions.
3. Secondary metabolites are produced as a group of closely related structures.
4. Some organisms forms a variety of different classes of substances such as secondary metabolites.
5. The regulation of biosynthesis of secondary metabolites differs significantly from that of primary metabolites.
6. Secondary metabolites are mostly produced in iodophase (Fig. 2.3)

   Origin and production of different secondary metabolites are depicted in Fig. 2.2 and 2.2 a.

Fermentative products are in use by man since ancient times. Fermentation of grains or fruit produce, bread, beer and wine that retained much of the nutrition of raw materials, while keeping the product from spoiling. The natural yeasts that caused fermentation added some vitamins and other nutrients to the bread or beverage. Lactic acid producing bacteria ferment milk to *yogurt* and *cheese* and extend the life of milk products. Other food products such as pickles, vegetables and the fermentation of tea leaves and coffee beans were preserved or enhanced in flavor by fermentation.

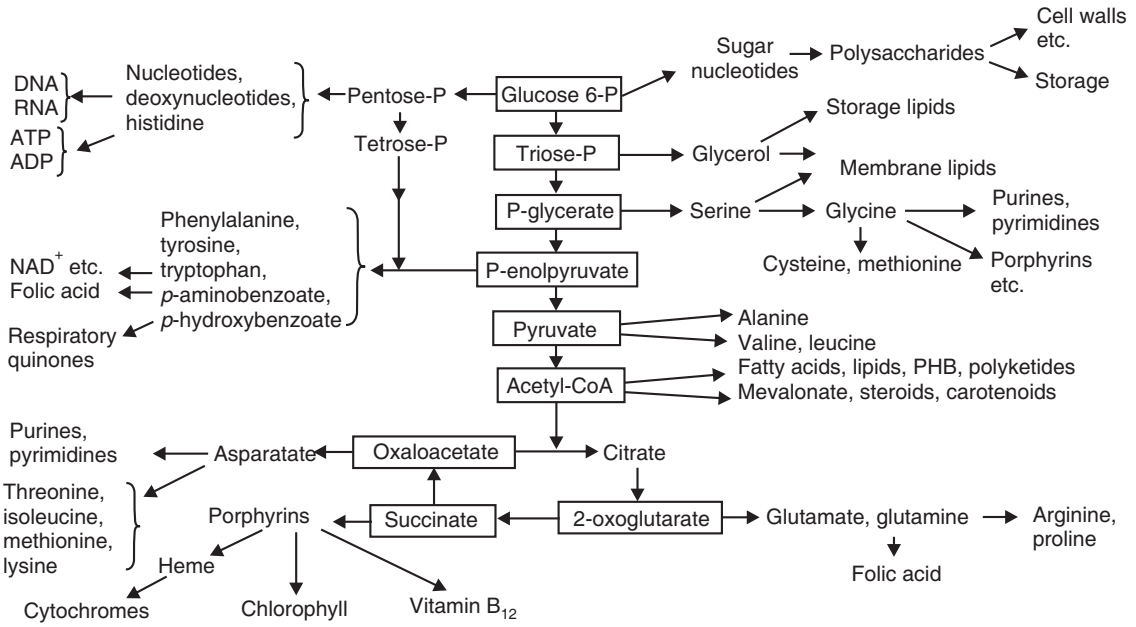


Fig. 2.2: Primary metabolites giving rise to variety of cell substances

Fermentation was an art until the second half of the 19th century. A batch was begun with either a starter, a small portion of previous culture, or with culture residing in the products or vessel. **Pasteur** (1775) made it clear that fermentation needs, heat treatment to improve storage quality and thus formed the basis for sterilization of medium. **Emil Christian Hansen** (1883) used for the first time pure culture of yeast for production of yeast in Denmark. During 1920–30

the emphasis in fermentation shifted to organic acids primarily lactic acid and citric acid. The discovery of penicillin in 1929 and commercialized in 1942, gave a boost to fermentation industry and led to the development of big fermenters and submerged cultivation. Success of penicillin inspired pharmaceutical companies to launch massive efforts to discover and develop many other antibiotics. In 1960s amino acid fermentations were developed in Japan. Commercial production of enzymes for use in industrial process began on a large scale in 1970. The discovery of the tools of genetic engineering expanded the possibilities for products made by fermentation *in situ,* and the first genetically engineered fermentation product was developed and commercialized in 1977. The historical events developed in the progress of fermentations are précised in table 2.1.

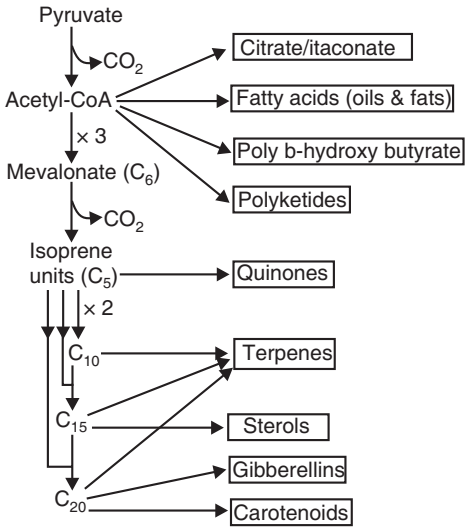


Fig. 2.2(a): Production of secondary metabolites

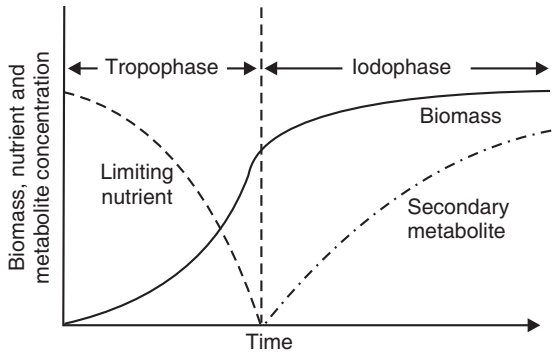


Fig. 2.3: The growth phases of biomass production and secondary metabolite production

Table 2.1: Historical events in the progress of fermentation

Year of commer-cialization	Fine Chemicals	Enzymes	Therapeutic substances	Microbial transformations	Other
1880-1900	Lactic acid	-	-	-	-
1900-1910	Ethanol Glycerol	Amylases	-	-	-
1910-1920	Acetone n-butanol	Invertase	-	-	-
1920-1930	Citric acid	-	-	-	-
1930-1940	Gluconic acid	Proteinases	Riboflavin	Sorbose Acyloions	-
1940-1950	Itaconic acid 2-keto-D-gluconic acid	Cellulases Pectinases	Vitamin B_{12} Penicillin G Other antibiotics	-	-
1950-1960	Kojic acid L-glutamic acid lysine	Oxidase Glucose Catalase	Ampholericin B Erythromycin Semi-synthetic penicillins	Steroid oxidation	Gibberellins Dextran SCP
1960-1970	Valine	Glucose Isomerase Glucamylase Lipases Lactase	Amphomycin Cephalosporins Fusidic acid Linomycin Some more antibiotics	Acetone dihydroxy	Xanthan 5'nucleosides bioinsecticide
1970-1980	Comenic acid	Microbial enzyme Rennet Replacement Melibiase Dextranase	Salinomycin Validanycin A	Sterol cleavage Xylitol, malic acid	Ribose zearalenol

Fermentation may be aerobic if it is operated in the presence of oxygen, while it may be anaerobic if carried out in the absence of oxygen. Anaerobic fermentations can be carried out either by use of fresh medium, covered with an inert gas such as nitrogen or argon or accumulation of CO_2 or foam (Fig. 2.4).

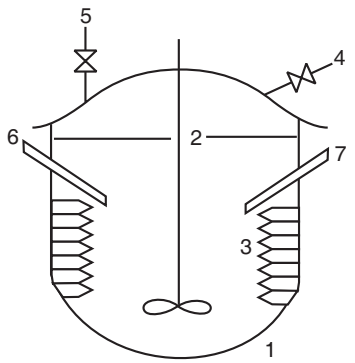


Fig. 2.4: Anaerobic fermenter

The fermentation is called *batch fermentation* when it is operated for a definite period. On the other hand, fermentation which is operated for a indefinite period it is called **continuous fermentation**. Some of the organisms are sensitive to substrate concentration and they are inhibited when the substratum is in high concentration. Under such conditions, fermentation can be carried by addition of substrates in installments and the process is called **Fed batch fermentation.**

Fermentations can be carried out under non-aseptic conditions where the risk of contamination is not a major concern. However, fermenters must be designed for prolonged aseptic operation. The design rules for an aseptic bioreactor demand that there is no direct contact between the sterile and non-sterile sections to eliminate microbial contamination.

Similarly, fermentation based on number of organisms involved can be classified into simple fermentation when only one organism is involved to produce a product from substratum. On the other hand, in some fermentations two organisms are involved in order to get a fermentation product from a substratum. The product of first phase of fermentation serves as substratum for second phase in order to yield desired product. In this type of fermentation, two organisms may grow simultaneously and product is formed instantly. Commercial growth of lichens involving algae and fungi is a good example for simultaneous fermentation. Production of glutamic acid from glucose firstly gets oxidized to ketoglutaric acid, which inturn get aminated to produce glutamic acid and production of lactic acid from glucose by yeast and *Lactobacillus lactis,* production of β-carotene jointly by (+) and (–) strains of either *choaenophoracucurbitarum* or *Blakesleea trispora are three very good examples*. On the other hand, the two organisms involved in a fermentation are separated widely in time and space, such fermentation is called successive fermentation. For example, production of acetic acid from glucose. First glucose is acted by yeast to produce ethyl alcohol, which is oxidized to acetic acid by *Acetobacter aceti*. Similarly production of lysine from glycerol. Glycerol is fermented to Diaminopimelic acid (DAPA) by an auxotrophic mutant of *E. coli* which gets aminated to form L-Lysine by *Aerobacter aerogenes*. When more than two organisms are involved in a fermentation it is called as mixed fermentation or multiple fermentation. In this fermentation, the substratum is heterogeneous and organisms with different potentialities of producing enzymes are involved in the fermentation. For instance, degradation of municipal wastes and decomposition of dead plants and animals can be taken as mixed or multiple fermentation. Similarly, remediation of waste water comes under this fermentation.

2.1 DESIGN AND BASIC FUNCTIONS OF A FERMENTER

A process in which a chemical product of human utility is produced involving microorganisms is called as fermentation. The vessel in which it is carried out is known as fermenter. An ideal fermenter should provide congenial environmental conditions, which promote the optimum growth of an organism and produce maximum product. Fermenter plays a critical role in the product yield. Hence design of a fermenter is important in the process of fermentation.

The fermenter may be simple for fermentation products, which does not require aseptic, conditions, while fermenter requiring an aseptic conditions have to be designed to prevent interference by the contaminating organisms.

Fermenter supports best possible growth, biosynthetic conditions and ease of manipulation for all operations associated with the use of it. For the better production of a desired product an ideal fermenter should have following provisions:

1. Fermenter body
2. Agitator
3. Coil
4. Gas outlet
5. Inoculation port
6. Thermometer
7. pH electrode

(i) Characteristic features of a good fermenter:

1. It must be strong enough to withstand pressure of large volume of aqueous medium.
2. It must not corrode and contribute toxic ions to the growing microorganism.
3. Make provision for control or prevention of growth of contaminating microorganisms.
4. Provision for rapid incorporation of sterile air into the medium.
5. Carbon dioxide released during fermentation must be flushed out.
6. Stirrer must be available to mix the medium and microorganisms to facilitate the availability of nutrients and oxygen.
7. Intermittent addition of antifoaming agent.
8. Provision for controlling temperature.
9. Aseptic withdrawal of culture sample during fermentation and introduction of inoculum at the initiation of fermentation.
10. Determination of the pH and its adjust mentill, if required.
11. Some means of sterilization of medium and addition of antifoaming agent.
12. Air filter.
13. Drain in the bottom.
14. Access to the inside of the fermenter to clean it.

(ii) Fermentation vessel:

Fermenters used in microbial fermentation represent a wide range of devices from a simple type of tube aeration system fermenter operating on the air lift system or the deep jet principle or devices with rotary stirrers in which the air is sucked in or distributed under pressure into the stirrer space.

There are wide variety of designs of fermenter (bioreactor) available. Selection of a fermenter design for a particular process depends on a variety of factors such as mass transfer considerations, mixing, sheer sensitivity, broth viscosity, oxygen demand, reliability of operations, sterilization considerations, the cost of construction and operation.

Stirred tank reactors use sparged air and submerged impellers to aerate and mix the broth. They are versatile and are specially adapted to highly aerobic cultures and highly viscous fermentations. Even in this, there are many variations in design such as the style, number and placement of impellers, the height to diameter ratio, the number and placement of coils or baffles that affect the mixing characteristics of the vessels. The main drawbacks are high-energy input and the use of rotating scale on the agitator shaft which may cause contamination risk.

Airlift fermenters (Fig. 2.5) mix broth with air from the sparger. Some designs have an internal shaft tube to direct the flow of fluid. Most airlift designs have a much greater height to diameter ratio than stirred tank vessels to improve oxygen transfer. The mixing is not as good as in a stirred tank but the energy input and shear forces are much lower, thus, useful for shear sensitive cultures or in processes where the energy cost of agitation is a significant factor. Ability to clean the vessel and maintenance cost are important factors for the selection of a bioreactor. Some reactor designs have excellent characteristics but in the pilot plant are not good choice for larger scale operation due to mechanical complexity that causes sterility and maintenance

problem on scale up. Most large-scale airlift fermenters are used for plant effluent treatment production or for baker's yeast or for fungal fermentations where the size of the mycelial pellets is controlled by shear forces.

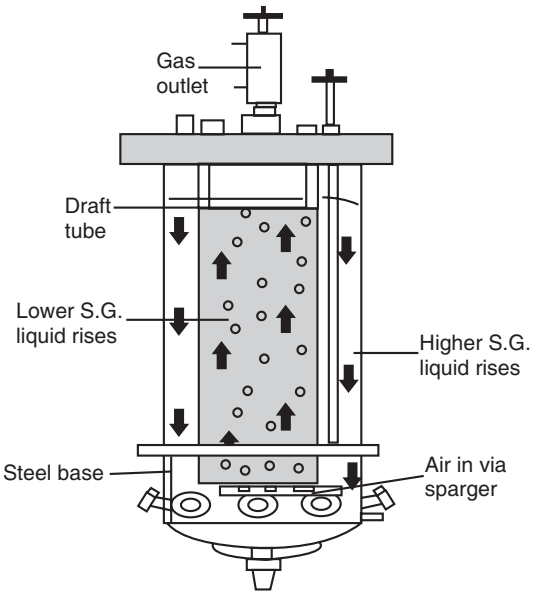


Fig. 2.5: A diagram illustrating the principle of an airlift fermenter

Fermenters with mechanical stirrers are used to mix the reactant mixture and they are called **stirred tank fermenters** (Fig. 2.6a).

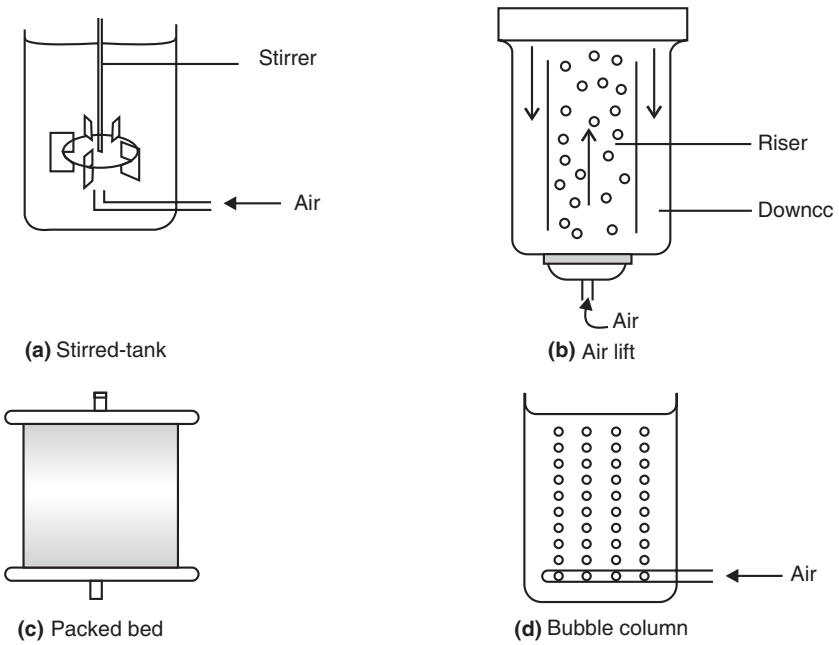


Fig. 2.6: Stirred tank fermenters

Fermenter with a draft tube is a hollow perforated tube that improves circulation and oxygen transfer. The air is introduced from the bottom of the fermenter that lifts the draft tube and it is known as **Airlift Fermenter (Fig. 2.6b)**. The fermenters can also fluidize its bed where the microbial cells are immobilized on small particles. These particles move along with the fluid and as a result, nutrient easily stick to it that enable high rate of oxygen and nutrient transfer to cells. On the other hand, **flocculated or packed bed reactor (fermenter)** contains larger particles which immobilize cells and cannot move along with the liquids (Fig. 2.6c). The reactor can be operated in either upflow or downflow mode, that is the liquid containing the substrate can be introduced either at the top or the bottom of the reactor. These type of fermenters are employed in sewage treatment where cells are immobilized by flocculation. They may also be used for bioconversion of small molecule. **Bubble column fermenter** (reactor) is another type of reactor in which the agitation and aeration are provided by a bottom sparger. To ensure even agitation, the sparger nozzles must be distributed uniformly over the cross-section of the bottom. Either a ring with regularly spaced holes, a small number of parallel pipes or star like arrangement of pipes is used (Fig. 2.6d).

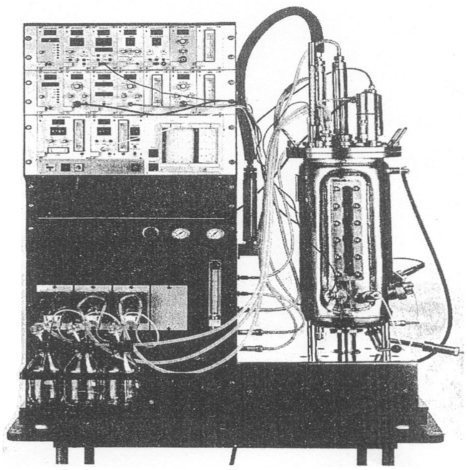


Fig. 2.7: Laboratory fermenter

According to the size they may be classified as laboratory fermenter (Fig. 2.7), 500 ml to 50 liters in volume, pilot plant with 50 to 500 liters in volume and production fermenter with 500 liters and above one lakh liters (Fig. 2.8).

The fermenter shape may vary from cylindrical to spherical, to tubular usually with a D-shaped bottom. It is closed at the top and bottom.

The material with which a fermenter is constructed vary according to the type of fermentation process. For example, fermentation of alcohol and lactic acid is carried out in a wooden fermenter, where sterilization is not necessary or where there is no chance of corrosion of inner lining of the fermenter. However, present day fermenters are constructed with inner surface lined with stainless steel or copper or iron or glass, which are chemically inert. Normally fermenters upto 1000 litres capacity have an external jacket and larger vessels have internal coils. Both, provide a mechanism for vessel sterilization and temperature control during the fermentation.

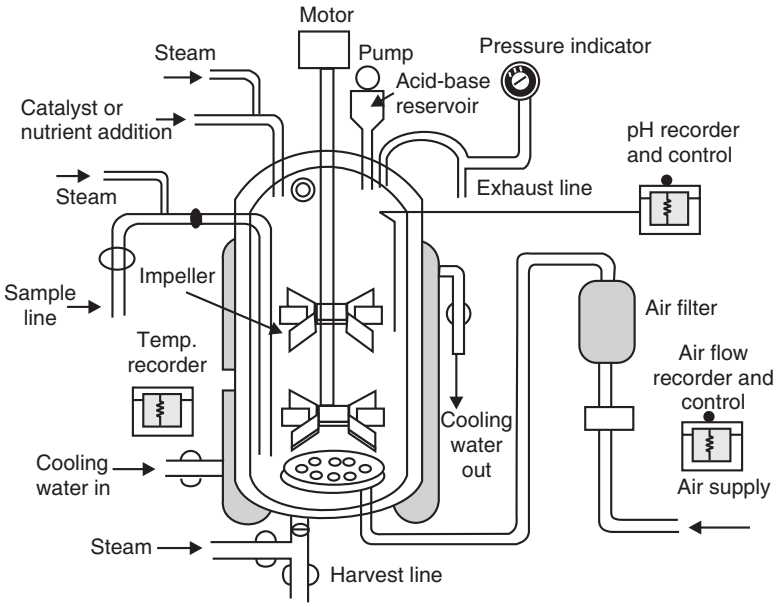


Fig. 2.8: Typical fermenter (stirred tanker fermenter)

2.1.1 **Provision for control of microbial growth:** Since most industrial fermentations utilize pure cultures, fermenters should be designed in such a way that promotes luxuriant growth of microbe but prevent the growth of contaminating microorganisms.

2.1.2 **Provision for incorporation of sterile air** (oxygen)**:** Most aerobic fermentation processes requires oxygen supply, which is called as aeration. Aeration is done by passing sterile air under pressure into the fermenter. The required air is sterilized by passing it through a sterile filter consisting of glass wool or some other finely powdered material that help in trapping microorganisms present in the air. The sterile air bubbled into the liquid medium through a sparger in order to make oxygen distributes uniformly in the medium.

2.1.3 **A device for removal of CO_2:** During fermentation process carbon dioxide and hydrogen gases are liberated which collect in the head space of the fermenter. The fermenter should be provided with a device to release these gases outside aseptically.

2.1.4 **An impeller:** An impeller, a rotating device, is generally provided to most of the fermenters, which accomplishes vigorous stirring and agitation of the medium. The rotation is carried out either by indirect or direct methods. In indirect method, the impeller is mounted on a shaft, which is driven by an electric motor fitted at the top of the fermenter. In direct method, impeller action is varied by using different impeller blades and is driven by a magnetic coupling fitted to a motor which is mounted beneath the fermenter. The impeller blades are arranged at different heights to achieve vigorous stirring and agitation of the medium.

2.1.5 **A device for addition of antifoam agent:** Aeration and agitation of the liquid medium causes the production of foam. Media with high levels of proteins or peptides cause more foam than with pure sugars and inorganic salts. Proteolytic bacteria that degrade proteins into peptides and amino acids also produce more foam. Appearance of foam leads to problems like contamination of the medium and impediment of aeration.

The formation of foam is undesirable, can be prevented by the addition of antifoam agents. An antifoam agent lowers surface tension of the foam and thereby it collapses, which leads to the disappearance of foam. Antifoam agents may be added to the medium either manually or mechanically. Manual addition requires the presence of some device in the fermenter to add antifoam agent aseptically, whenever needed. In mechanical addition, which is done automatically, an electrical sensing mechanism is provided at the top of the fermenter. It consists of two electrodes projecting into the headspace of the fermenter. They are connected to a pump of antifoam reservoir. When foam builds up in the headspace and touches the electrodes, current flows between the electrodes and activates the pump for addition of antifoam. When foam collapses the electrodes get disconnected and addition of antifoam ceases.

2.1.6 A device for temperature control: Microorganisms widely differ in their temperature dependence for growth. However, they grow well at optimum fermentation temperature, which may be below or above ambient temperature. During fermentation lot of temperature is generated due to metabolic activities of microorganism, which leads to a rise in the temperature in the fermenter. For maintaining optimum temperature in the fermenter, one of the following devices is provided (Fig. 2.9).

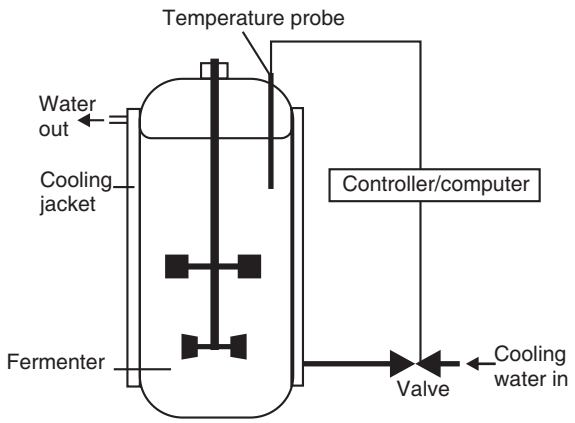


Fig. 2.9: A scheme for controlling fermenter temperature

1. Sparging cold water on the fermenter,
2. By circulating cold water through the jacketed walls of fermenter, or
3. Through coils arranged along the inside walls of the fermenter.

2.1.7 A device for addition or withdrawal of inoculum: The fermenter should also be provided with a device to introduce inoculum at the beginning of the fermentation, and its withdrawal aseptically during fermentation.

2.1.8 A provision for pH adjustment: pH, which plays an important role in the growth and metabolism of microbes, influences fermentation process. A mechanism for determining pH of the fermentation broth intermittently and adjusting the values is often required. This is usually accomplished by withdrawing a sample from the fermenter for pH measurement, followed by addition of alkali or acid to the fermentation medium to adjust pH.

2.1.9 **Seed tanks:** Inoculum of 1-10% is required to inoculate production tanks to reduce incubation period. They are also called as inoculum tanks. They are generally small sized fermenters in which inoculum is produced under controlled conditions.

2.1.10 **Medium preparation vessel:** Fermentation requires additional vessels for the preparation of medium. Required nutrients for the medium is transferred to the fermenter from these vessels.

2.1.11 **Sterilization of the medium:** Most of the fermentations require pure culture and needs sterilized medium. For this purpose, the medium is passed through retention tubes and heat exchangers before passing into the large, empty and sterilized fermentation tank. The retention tubes contain steam waters jet that inject high pressure steam into the medium to sterilize it as it passes through the pipes and the rate of passage is adjusted in such a way that there will be complete sterilization. The heat exchanger, which consists of a pipe containing the medium within a second pipe containing cool water moving in the opposite direction, cool the medium before it is passed into the fermenter. After entry into the fermenter the medium is diluted with sterile water.

2.1.12 **Device for withdrawal of used medium:** There must be a device at the bottom of the fermenter or some mechanism may be provided for removing the completed fermentation broth from the tank. The fermenter should be accessable for cleaning after fermentation is completed.

Fermentation system must be efficiently controlled in order to optimize productivity, product yield and ensure reproducibility. The key physical and chemical parameters involved largely depend on the bioreactor, its mode of operation and the microorganism being used. They are primarily aeration, mixing, temperature, pH and foam control. The control and maintenance at optimum levels inside the reactor is mediated by sensors (electrodes) along with compatible control systems and data logging.

2.2 FERMENTATION PROCESSES

Fermentation process can be conveniently divided into six stages regardless of the type of process. They are:

1. The formulation media used for the growth of the microorganism to be employed as inoculum and also in the production of fermentation products.
2. The sterilization of the medium, fermenter and other associated equipment.
3. The preparation of adequate quantities of pure culture that is to be inoculated into the fermenter.
4. The creation of optimum conditions in the fermenter for optimum growth of the organism and for optimum output of the desired product.
5. The extraction of the product and its purification.
6. The disposal of effluents generated during fermentation.

The inter relationships among these six phases are diagrammatically illustrated in Fig. 2.10. This process varies with the type of organism used and product to be produced. The entire process can be discussed under two headings.

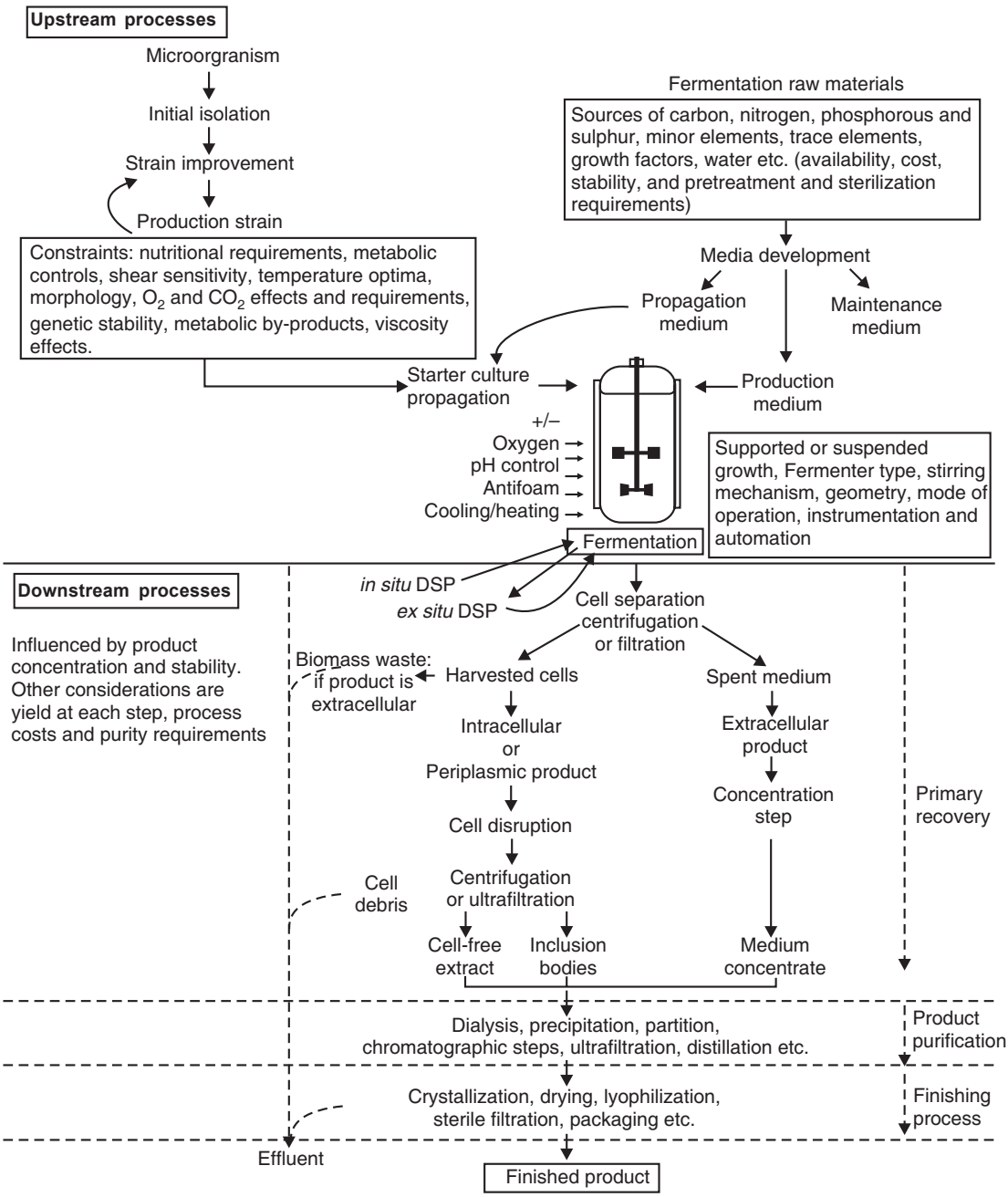


Fig. 2.10: The upstream and downstream process of a typical fermentation process

(a) Upstream process: It includes selection of organism and medium, medium sterilization, inoculation and ends with monitoring of fermentation process and product formation. This involves selection of microorganism. The selection of

microorganisms for fermentation should be critically done. At first it should have potential to produce particular substance in an economic amounts. It should be non-pathogenic and non-hazardous. Further it should be amenable to growth in a fermenter and produce the product in good amounts.

(b) Downstream process: It includes the product separation and purification and effluent treatment.

2.3 TYPES OF FERMENTATIONS

The vessel in which fermentation is carried out is called fermenter. The yield of the product is atleast partly dependent on the type of fermenter. Generally, fermenters are designed to provide best possible growth and biosynthetic conditions and ease of manipulations for all operations associated with the use of fermenters. A good fermenter is that which fulfills the following characteristics.

It should provide control and observation of many facets of microbial growth and biosynthesis. Fermentation processes can be classified into the following three categories. They are:-

1. Batch fermentation
2. Continuous culture fermentation
3. Fed batch fermentation

The choice of the operation depends largely upon the organism and the type of product being produced.

2.3.1 Batch fermentation: A batch fermentation is a closed culture system, because initial and limited amount of sterilized nutrient medium is introduced into the fermenter. The medium is inoculated with a suitable microorganism and incubated for a definite period for fermentation to proceed under optimal physiological conditions. Oxygen in the form of air, an antifoam agent and acid or base, to control the pH, are being added during the course of fermentation process (Fig. 2.11).

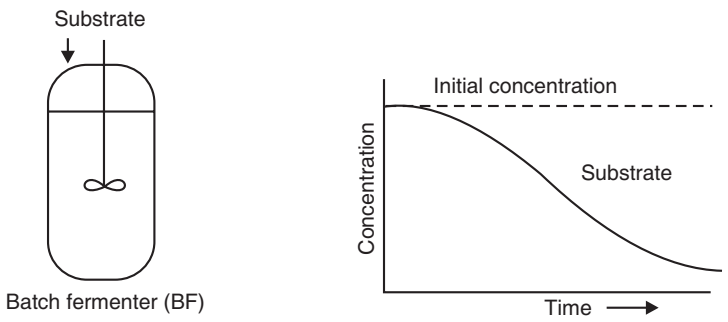


Fig. 2.11: A typical batch fermenter

During the course of incubation, the cells of the microorganism undergo multiplication and pass through different phases of growth and metabolism due to which there will be change in

the composition of culture medium, the biomass and metabolites. The fermentation is run for a definite period or until the nutrients are exhausted. The culture broth is harvested and the product is separated.

Batch fermentation may be used to produce biomass, primary metabolites and secondary metabolites under cultural conditions supporting the fastest growth rate and maximum growth would be used for biomass production. The exponential phase of growth should be prolonged to get optimum yield of primary metabolite, while it should be reduced to get optimum yield of secondary metabolites.

The used medium along with cells of microorganism and the product is drawn out from the fermenter. When the desired product is formed in optimum quantities, the product is separated from the microorganism and purified later on. It has both advantages and disadvantages which are detailed below.

(i) Merits:

(a) The possibility of contamination and mutation is very less.

(b) Simplicity of operation and reduced risk of contamination.

(ii) Demerits:

(a) For every fermentation process, the fermenter and other equipment are to be cleaned and sterilized.

(b) Only fraction of each batch fermentation cycle is productive.

(c) It is useful in fermentation with high yield per unit substratum and cultures that can tolerate initial high substrate concentration.

(d) It can be run in repeated mode with small portion of the previous batch left in the fermenter for inoculum.

(e) Use of fermenter is increased by eliminating turn round time or down time.

(f) Running costs are greater for preparing and maintaining stock cultures.

(g) Increased, frequency of sterilization may also cause greater stress on instrumentation and probes.

(h) Fresh sterilized medium and pure culture are to be made for every fermentation process.

(i) Yield of the desired product may also vary.

(j) There will be a non-productive period of shutdown between one batch productive fermentation to the other.

(k) More personal are required.

2.3.2 Continuous fermentation: It is a closed system fermentation, run for indefinite period. In this method, fresh nutrient medium is added continuously or intermittently to the fermenter and equivalent amount of used medium with microorganisms is withdrawn continuously or intermittently for the recovery of cells or fermentation products (Fig. 2.12). As a result, volume of the medium and concentration of nutrients at optimum level are being maintained. This has been operated in an automatic manner.

The continuous fermenter has its maximum use that take long time to reach high productivity, reduces down time and lowers the operating costs.

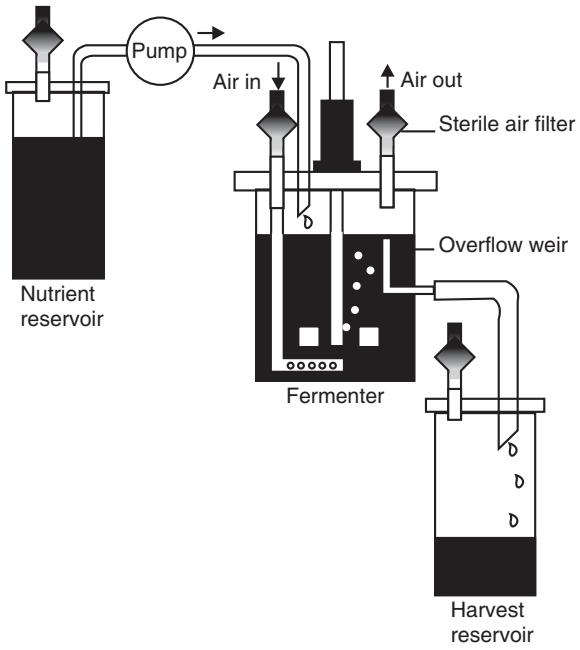


Fig. 2.12: Continuous fermentater

In continuous mode, starting medium and inoculum are added to the fermenter. After the culture is grown the fermenter is fed with nutrients and broth is withdrawn at the same rate maintaining a constant volume of broth in the fermenter. In continuous mode with cell cycle, the cell mass is returned to the fermenter using micro filtrations with bacteria or screens with fungal mycelium.

A continuous fermentation is generally carried out in the following ways:

(a) Single stage fermentation

(b) Recycle fermentation

(c) Multiple stage fermentation

(a) Single stage fermentation: In this process, a single fermenter is inoculated and the nutrient medium and culture are kept in continuous operation by balancing the input and output of nutrient medium and harvested culture, respectively.

(b) Recycle fermentation: In this method, a portion of the medium is withdrawn and added to the culture vessel. Thus, the culture is recycled to the fermentation vessel. This method is generally adopted in the hydrocarbon fermentation process. The recycling of cells provides a higher population of cells in the fermenter which results in greater productivity of the desired product.

(c) Multiple stage fermentation: In this process, two or more fermenters are employed simultaneously and the fermentation is operated in a sequence. Different phases of fermentation process like growth phase and synthetic phase are carried out in different fermenters. Generally, growth phase is allowed in the first fermenter, synthetic phase in the second and subsequent fermenters. This process is adapted

particularly to those fermentations in which growth and synthetic activities of the microorganisms are not simultaneous. Synthesis is not growth related but occurs when cell multiplication rate has slowed down.

The process of continuous fermentation is monitored either by microbial growth activity or by product formation and these methods are called:

(i) Turbidostat method, and (ii) Chemostat method.

(i) Turbidostat Method: In this method the total cell content is kept constant by measuring the culture turbidity at a regular interval of fermentation process. By turbidity measurment it is possible to the fermenter to regulate both the nutrient feed rate and the culture withdrawal rate. Fermentation, in which this method is employed, must be carried out at a low maximum cell population which leads to the usage of less amount of substrate and wastage of greater amount of substrate as unused and residual medium, which is removed from the fermenter along with the harvested culture (Fig. 2.13).

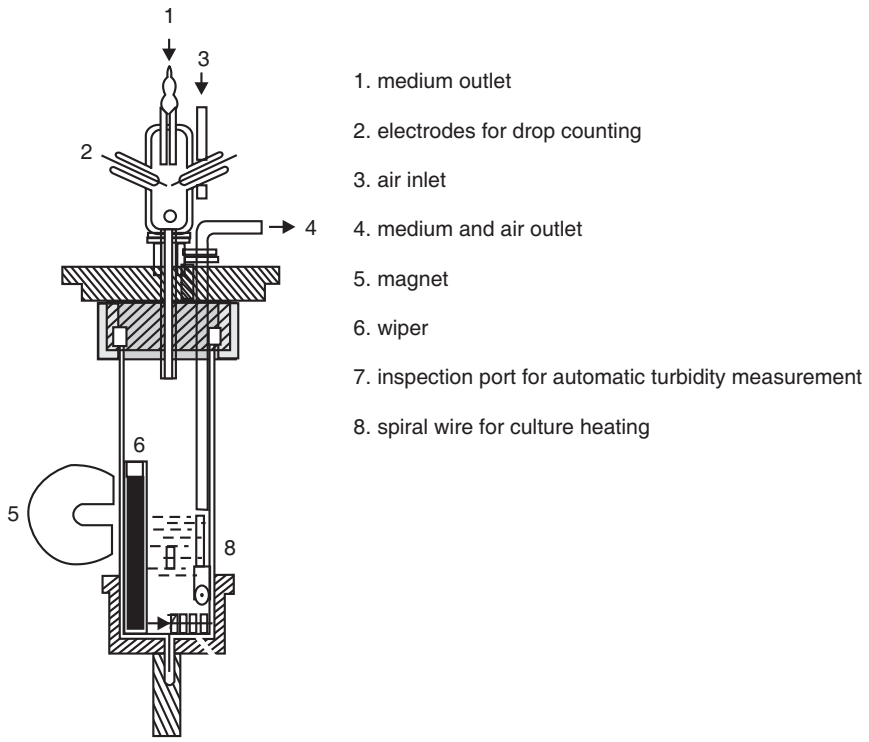


Fig. 2.13: Turbidostat

(ii) Chemostat Method: In this method nutrient feed rate and harvest culture withdrawal rate are maintained at constant value. This is achieved by controlling the growth rate of the microorganism by adjusting the concentration of any one of the chemicals of the medium, like carbon source, nitrogen source, salts, O_2 etc. which acts as a growth limiting factor. Apart from the above chemicals, sometimes the concentration of the toxic product generated in the fermentation process, the pH values and even temperature also act as growth limiting factors. This method is

employed more often than turbidostat method because of fewer mechanical problems and presence of less amount of unused medium in the harvested culture (Fig. 2.14).

However, continuous fermentations have certain advantages and limitations which are as follows:

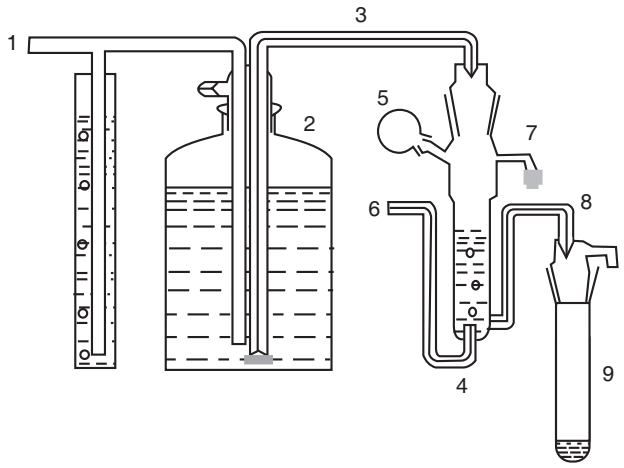


Fig. 2.14: Chemostat

1. Air inlet
2. Mariotte's bottle
3. Capillary for medium inlet
4. Culture vessel
5. Inoculation port
6. Air inlet
7. Air outlet
8. Overflow capillary
9. Sampling tube

(i) Merits:

1. The fermenter is continuously used with little or no shutdown time.
2. Only little quantity of initial inoculum is needed and there is no need of additional inoculum.
3. It facilitates maximum and continuous production of the desired product.
4. There is optimum utilization of even slow utilizable substances like hydrocarbons.

(ii) Demerits:

1. Possibility of contamination and mutation because of prolonged incubation and continuous fermentation, are more.
2. Possibility of wastage of nutrient medium because of continuous withdrawal for product isolation.
3. The process becomes more complex and difficult to accomplish when the desired products are antibiotics rather than a microbial cells.
4. Lack of knowledge of dynamic aspects of growth and synthesis of product by microorganism used in fermentation.

(iii) Applications:

Continuous culture fermentation has been used for the production of single cell protein, antibiotics, organic solvents, starter cultures etc. (table 2.2).

Table 2.2: Chemical products produced in continuous fermentation

Growth associated	Non-growth associated
Acetic acid	Acetone
Butanediol	Butanol
Ethanol	Glycogen
Gluconic acid	Subtilin
Hydrogen sulfide	Chloramphenicol
Lactic acid	Penicillin
	Streptomycin
	Vitamin B_{12}

Pilot plants or production plants have been installed for production of beer, fodder yeast, vinegar, baker's yeast. A wide variety of microorganisms are used for this type of fermentation (table 2.3).

Table 2.3: Microorganisms used in continuous fermentation

Group of organism	Genera
Actinomycetes	*Streptomyces,*
Algae	*Chlorella, Euglena and Scenedesmus.*
Bacteria	*Aerobacter, Azotobacter, Bacillus, Brucella, Clostridium and Salmonella.*
Fungi	*Ophiostoma and Penicillium.*
Protozoa	*Tetrahymena*
Yeast	*Saccharomyces, Torula*

2.3.3 Fed batch fermentation: It is a modification to the batch fermentation. In this process substrate is added periodically in instalments as the fermentation progresses, due to which the substratum is always at an optimal concentration. This is essential as some secondary metabolites are subjected to catabolite repression by high concentration of either glucose, or other carbohydrate or nitrogen compounds present in the medium. For this reason, the critical elements of the nutrient medium are added in low amount in the beginning of the fermentation and these substrates continue to be added in small doses during the production phase. This method is generally employed for the production of substances such as penicillin. Yoshida (1973) introduced this term for the first time for feeding the substrates to the medium as the nutrients are exhausted, so as to maintain the nutrients at an optimum level. The fed-batch fermentation may be of three types:

(i) Variable volume fed batch culture. The same medium is added resulting in an increase in volume.

(ii) Fixed volume fed batch culture. A very concentrated solution of the limiting substrate is added at a very little amount resulting in an insignificant increase in the volume of medium.

(iii) Cyclic fed batch culture.

As it is not possible to measure the substrate concentration by following direct methods during fermentation, which is necessary for controlling the feeding process, generally indirect methods are employed. For example, in the production of organic acids, the pH value may be used to determine the rate of glucose utilization.

(i) Advantages:

1. Production of high cell densities due to extension of working time (particularly growth associated products).
2. Controlled conditions in the provision of substrates during fermentation, particularly regarding the concentration of specific substrates for e.g. the carbon source.
3. Control over the production of, by products or catabolite repression, effects due to limited provision of substrates solely required for product formation.
4. The mode of operation can overcome and control deviations in the organism's growth pattern as found in batch fermentation.
5. Allows the replacement of water loss, by evaporation.
6. Alternative mode of operation for fermentations dealing with toxic substances or low solubility compounds.
7. Increase of antibiotic marked plasmid stability by producing the correspondent antibiotic during the time span of the fermentation.
8. No additional special piece of equipment is required as compared with the batch fermentation.
9. It is an effective method for the production of certain chemicals, which are produced at optimum level when the medium is exhausted like penicillin.

(ii) Disadvantages:

1. It is not possible to measure the concentration of feeding substrate by following direct methods like chromatography.
2. It requires precious analysis of the microorganism. Its requirements and the understanding of its physiology with productivity is essential.
3. It requires a substantial amount of operator skill for the set up of fermentation and development of the process.
4. In a cyclic fed batch culture, care should be taken in the design of the process to ensure that toxins do not accumulate to inhibitory levels and that nutrients other than those incorporated into the fed medium become limited also, if many cycles are run. The accumulation of non-producing or low producing variants may result.
5. The quantities of components to control must be above the detection limits of the available measuring equipment.

Fed-batch with recycle of cells can also be used for specific purpose such as ethanol fermentation and waste water treatment.

At present following products are being produced under fed batch culture.

1. Production of baker's yeast.
2. Penicillin production.
3. Production of *Thiostrepton* by *Streptomyces laurentii*
4. Production of industrial enzymes, histidine, glutathione (*Brevibacterium flavum*), Lysine (*Corynebacterium glutamicum*)

(iii) Applications:

1. It facilitates in avoidance of repressive effect.
2. It has control over organisms growth rate and O_2 requirement.
3. In maintaining concentration of both the biomass and non-limiting nutrient substrates constant.
4. Production phase may be extended under controlled conditions and overcome problems associated with the use of repressive rapidly metabolized substrates.
5. Shift in growth rate may provide an opportunity to optimum product synthesis.
6. It facilitates to overcome viscosity problems or its toxicity at higher concentration.

2.3.4 Anaerobic fermentation: A fermentation process carried out in the absence of oxygen is called as anaerobic fermentation. There are two types of anaerobic microorganisms viz, obligate anaerobic microorganisms and facultative anaerobic microorganisms. The former like *Clostridium* sp. cannot withstand oxygen or remain active only in the absence of oxygen. They remain active in the absence of oxygen and produce optimum amount of the desired product. The facultative anaerobes like lactic acid bacteria are able to withstand small amount of oxygen. However, certain organisms like yeast require an initial aeration to build up high cell yield before anaerobic conditions are created.

Anaerobic conditions in the fermenter are created either by withdrawing the oxygen present in the head space by an exhaust pump and pumping some inert gases like nitrogen, argon etc. or by flushing it out, by the emergence of certain gases like carbon dioxide or hydrogen (Fig. 2.4).

Stationary medium and viscous medium also creates anaerobic conditions. Sometimes in order to create anaerobic condition, medium is inoculated at the bottom of the fermenter soon after sterilization.

(i) Merits:

1. Production of economically valuables byproducts like carbon dioxide and hydrogen gas during anaerobic fermentation, which may fetch some profits to the manufacturers.

(ii) Demerits:

1. Manufacturers may have to spend more money in providing extra provisions to the fermenter like exhaust pump in order to enforce anaerobic conditions.
2. It requires special media like viscous media whose preparation requires certain costly chemicals.

2.3.5 Aerobic fermentation: A fermentation process carried out in the presence of oxygen is called as aerobic fermentation. In most of the commercial processes and majority of the products of human utility are produced by this type of fermentation.

Fermentation can be surface culture or static and submerged.

2.3.6 Surface fermentations: are those where the substratum may be solid or liquid. The organism grows on the substratum and draws the nutrients from the substratum. These types of fermentations are desirable where the products are based on sporulation. But it has several disadvantages such as it exposes the organism to unequal conditions, both oxygen and nutrients.

Table 2.4: History and development of solid-state fermentation

Period	Development
2000 BC	Bread making by Egyptians
(Recorded history 1000BC)	Cheese making by *penicillium roquefortii*
2500 BC	Fish fermentation/preservation with sugar, starch, salts, etc.
2500 BC	Koji process
7th Century	Koji process from China to Japan
18th Century	Vinegar from pomace
18th Century	Use of gallic acid tanning, printing, etc.
1860-1900	Sewage treatment
1900-1920	Production of fungal enzymes, kojic acid
1940-1950	Fantastic development in fermentation industry, penicillin production
1950-1960	Steroid transformation
1980-1990	Production of various primary and secondary metabolites, development of column type of fermenter, work on kinetics and modeling aspects of SSF
1990-present	Developments on fundamental aspects of SSF, bioprocesses/products developments: *A. Bioprocess:* Bioremediation and biodegradation of hazardous compounds, biological detoxification of agroindustrial residues, biotransformations, biopulping etc. *B. Products:* Bioactive compounds: Aflatoxins, Ocharatoxin A, bacterial endotoxins, gibberellic acid, zearalenone, ergot alkaloids, penicillin, cephalo-sporin, cephamycin C, tetracycline, chlorotetracycline, oxytetracycline, actinohodin, methylenomycin, surfactin, monorden, cyclosporin A, ustiloxins, antifungal volatiles. Calvulanic acid, mycophenolic acid.
	C. Enzymes: Cellulase, β-glucosidase, CMCase, laccase, xylanase, polygalacturonase, ligninase, xylanases, β-xylosidase, α-arabinofurano-sidase, L-peroxidase, Mn-peroxidase, arylalachol oxidase, catalase, phenol oxidase, proteases (acidic, neutral and alkaline), lipases, α-galactosidase, β-galactosidase, α-amylase, β-amylase, glucoamylase, glutaminase, inulinase, hytase, tannase, feruloyl para-coumaroyl esterase. *D. Organic acids:* Citric acid, fumaric acid, lactic acid, oxalic acid, gallic acid. *E. Other products:* L-glutamic acid, pigments, carotenoid, xanthan gum, succino- glycan, ethanol, aroma compounds, vitamins : B-12, B-6, riboflavin, thiamine, nicotinic acid, nicotinamide, gammalinolenic acid, biosurfactants, biopes- ticides/bioherbicides.

2.3.7 Submerged fermentations: are those in which the nutrient substratum is liquid and the organism grows inside the substratum. The culture conditions are made uniform with the help of spargers and impeller blades. Most of the industrial fermentations are of this type. The substratum which is in a liquid state and such medium is also called as broth.

2.3.8 Solid substrate/state fermentation: Solid state (substratum) fermentation (SSF) is generally defined as the growth of the microorganism on moist solid materials in the absence or near the absence of free water. In recent years SSF has shown much promise in the development of several bioprocesses and products, SSF has been ambiguously used as solid-state fermentation or solid-substrate fermentation. However, it is proper to distinguish between two processes. Solid substrate fermentation should be used to define only those processes in which the substrate itself acts as carbon source occurring in absence or near absence of free water. On the other hand, the solid state fermentation is that fermentation which employs a natural substrate as above or an inert substrate used as solid support. Solid substrate fermentation are normally many step process involving. SSF has a long history and some of the main events are precised in table 2.4. Comparison of solid state and submerged fermentation is given in table 2.5.

Table 2.5: Comparison of characteristics of SSF and submerged fermentation

Characteristics	Solid State	Submerged
Microorganism, substrate	Static	Agitated
Water usage	Limited	Unlimited
Oxygen supply	Diffusion	Aeration
Volume of fermentation mash	Smaller	Larger
Liquid waste produced	Negligible	Significant volume
Physical energy requirement	Low	High
Human energy requirement	High	Low
Capital investment	Low	High

Based on the need for aeration and agitation, SSF can be divided into two groups.

(a) Fermentation without agitation.

(b) Fermentation with occasional or continuous agitation. Second group can be further divided into:

(i) Fermentation with occasional agitation, without forced aeration.

(ii) Fermentation with slow continuous agitation with forced agitation.

(iii) Pretreatment of a substratum that often requires either mechanical, chemical or biological processing.

(iv) Hydrolysis of polymeric substrates such as polysaccharides and proteins.

(v) Utilization of hydrolysis products.

(vi) Separation and purification of end products.

(vii) Fermentation with occasional agitation and forced aeration.

(viii) Fermentation with slow continuous agitation and forced aeration.

Several types of fermenters have been used for solid state fermentation. Laboratory studies have generally been carried out in flasks, beakers, Roux bottles, petri dishes, glass jars and columns. Inoculum is added after substrate autoclaving and incubated without any agitation and aeration. For large-scale SSF bioprocess, three types of fermenters are in operation.

(a) Drum fermenter: It basically consists of drum type vessel usually equipped with a rotating device and arrangements for air circulation (Fig. 2.15a). The air inlet pipe may run parallel to the bottom or center or it may branch at several points over the whole length of the drum to facilitate air distribution which is normally attained by forced aeration, thus achieving the mixing of the fermenting substratum. Growth of the microorganism in this type of fermenter is considered to be better and more uniform than the tray fermenter.

(b) Tray fermenter: Tray fermenters are the simplest and can be constructed using wood, metals or plastic material. The bottom of tray is perforated in such a way that it holds substrate and allows aeration (Fig. 2.15b). Kofi fermentation has traditionally been carried out in tray fermenter. Tray fermenter, however, require a large operational area and labour intensive. Their design does not lead readily to mechanical handling. The substrate requires separate sterilization.

(c) Column fermenter: Column fermenter consists of a glass or plastic column with lids at both ends. It may be fitted with a jacket for the circulation of water to control the temperature of fermenting substrate. Alternatively, the whole column may be placed in temperature controlled water bath. Usually air is circulated from bottom to top (Fig. 2.15c). The column may be vertical or horizontal as per convenience. Bed reactor is simple in design in which humidified air is pumped into substratum and the used waste gases goes out through the outlet provided continuous agitation with forced air to prevent adhesion and aggregation of substrate particles. These systems are very useful for biomass production for animal feed.

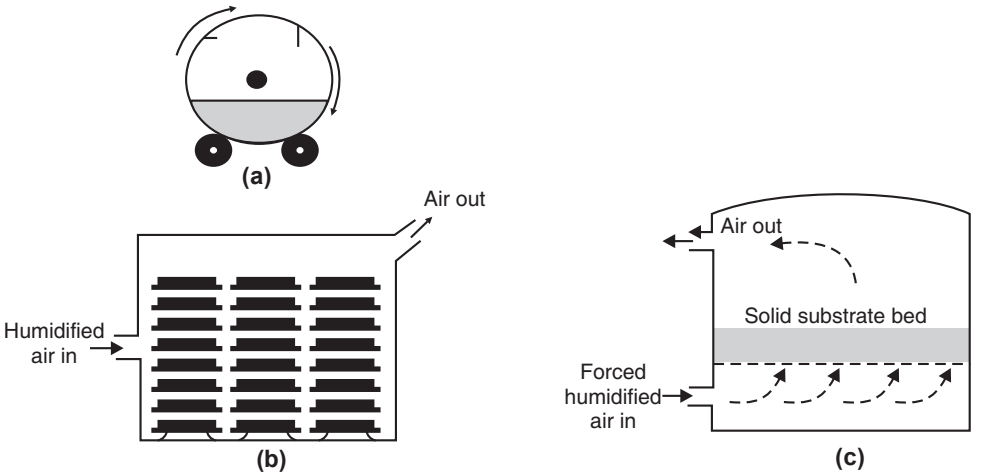


Fig. 2.15 (a), (b), (c): Three types of solid-substrate fermenters,
(a) Rotating-drum fermenter, (b)Tray fermenter, and (c) Column fermenter

Microorganisms associated with solid substrate fermentation are those that tolerate relatively low water activity down to 0.7. They may be employed in the form of monocultures as in mushroom production e.g. *Agaricus bisporus.* Dual cultures e.g. straw conversion using *Chaetomium cellulolyticum* and *Candida tropicalis.* Mixed cultures as used in compositing and the preparation of silage where the microorganisms may be indigenous or added as mixed starter cultures.

For some fermentation, SSF is desirable because of following reasons:

1. In several productions, the product formation has been found superior in solid culture process.
2. The most commonly used microorganisms in the production of secondary metabolites are fungi and actinomycetes and the mycelial morphology of such organisms is ideal for their invasive growth on solid and insoluble substrates.
3. The fungal morphology is responsible for considerable difficulties in large scale submerged processes. These include highly viscous non-Newtonian broths and foam production. This results in very high power requirements for mixing and oxygen transfer. The presence of chemical antifoam in fermentation broth reduces oxygen transfer efficiency and can lead to problems in the product recovery.
4. In some processes the final product is required in solid form, such as antibiotics in animal feed.
5. The capital cost of overall production process is claimed to be significantly less.
6. The yields of certain secondary metabolites such as aflatoxin B_1 and ochratoxin A obtained from liquid culture were found to be very poor. This led to the use of SSF to get higher yield of mycotoxins (100 g).
7. The fungus possess tremendous turgor pressure at the mycelial tips.
8. Microbial cells attach to solid substrate particles and completely surrounds the particle in mycelial webs.
9. It provides optimum quantity of water (a_w) for growth.
10. Crude substrates can be used as the organisms can tolerate high concentration of metal ions and mineral ions.
11. Overcome catabolite repression and can be provided high substrate concentration.
12. Enzymes become extracellular otherwise intracellular in SMF. E.g.: Galactase, tannase and invertase.
13. Metabolite production phase is long.
14. Co-production of carbohydrates and proteases.
15. Enzymes produced by this will be with better properties and extra desirable components.
16. Fermentation of straw eliminates costly centrifugation and dewaleing.
17. Lower capital and recurring expenditure.
18. Low waste water output/less water need.
19. Reduced energy requirement.
20. Absence of foam formation.
21. Simplicity.
22. High reproducibility.
23. Simpler fermentation media.
24. Lesser fermentation space.
25. Absence of rigorous control of fermentation parameters.

26. Easier aeration.
27. Economical to use even in smaller scales.
28. Easier control of contamination.
29. Applicability of using fermented solids directly.
30. Storage of dried fermented matter.
31. Lower cost of downstream processing

Some of the substances produced by SSF are precised in table 2.6

Table 2.6: Production of different substances in Solid State Fermentation

Compound	Substrate	Microorganism
Aroma compounds	Sugarcane bagasse, cassava bagasse, coffee –husk, soya beans	*Rhizopus oryzae, Ceratocystis fimbriata, Bacillus subtilis*
Biosurfactants	Agro –industrial residues, molasses	*Bacillus subtilis*
Biopesticides, bioherbicides	Sweet sorghum, rice flour, perlite-corn meal agar	*Entomopathogenic* and *mycoparasitic fungi*
Carotenoids	Corn meal	*Penicillium species*
Ethanol	Apple-pomace, sorghum, sugarbeet, sweet sorghum, sweet potato, potato, wheat flour, rice starch.	*Schwanninomyces castellii, Zymomonas mobilis, Candida utilis, Torula utilis, Saccharomyces cerevisiae*
Gamma-linolenic acid	Cereals, soya beans	*Cunninghamella japonica* *Rhizopus* sp.
L-glutamic acid	Sugarcane bagasse	*Brevibacterium* sp
Pigments	Sugarcane bagasse	*Monascus purpureus*
Succinoglycan	Agar-medium, spent multi-grain, ivory nut shavings, grated carrots	*Agrobacterium tumefaciens, Rhizobium hedisary*
Vitamin B_{12}, B_6, riboflavin, thiamine, nicotinic acid, nicotinamide	Soya beans tempeh	*Citrobacter freundii, Klebsiella pneumoniae, Rhizopus oligosporus, R. arrhizus, R. stolonifer*
Xanthan gum	Apple pomace, grape pomace, citrus peels, spent multi-grain.	*Xanthomonas compestris*

2.4 SCREENING

The economics of a fermentation process largely depends upon the type of microorganism used. If fermentation process is to yield a product at a cheaper price the chosen microorganism should give the desired product in a predictable and economically adequate quantity. The microorganism with a desired characters is generally isolated from natural substrates like soil etc. Such an organism is generally called as a producer strain. A producer strain should possess the following characters.

1. It should be able to grow on relatively cheaper substrates.

2. It should grow well in an ambient temperature preferably at 30–40°C. This reduces the cooling costs.
3. It should yield high quantity of the end product.
4. It should possess minimum reaction time with the equipment used in a fermentation process.
5. It should possess stable biochemical characteristics.
6. It should yield only the desired substance without producing undesirable substances.
7. It should possess optimum growth rate so that it can be easily cultivated on a large scale.

Detection and isolation of a microorganism from a natural environment like soil containing large number of microbial population is called as screening. It is very time consuming and expensive process. For example, Eli Lilly & Co. Ltd discovered three species of antibiotic producing organisms in a span of 10 years and after screening 4,00,000 organisms. Although there are many screening techniques, all of them are generally grouped into two broad categories. They are:

1. Primary screening, and
2. Secondary screening.

2.4.1 **Primary screening**: Primary screening may be defined as detection and isolation of the desired microorganism based on its qualitative ability to produce the desired product like antibiotic or amino acid or an enzyme etc. In this process desired microorganism is generally isolated from a natural environment like soil, which contains several different species. Sometimes the desired microorganism has to be isolated from a large population of different species of microorganisms. The following are some of the important primary screening techniques.

(i) The crowded plate technique
(ii) Indicator dye technique
(iii) Enrichment culture technique
(iv) Auxanographic technique
(v) Technique of supplementing volatile and organic substrates.

(i) **The crowded plate technique:** This technique is primarily employed for detecting those microorganisms, which are capable of producing antibiotics. This technique starts with the selection of a natural substratum like soil or other source consisting of microorganisms. Progressive serial dilution of the source is made. Suitable aliquot of the serial dilution is chosen which is able to produce 300 to 400 individual colonies when plated on an agar plate, after incubation. Such a plate is called as crowded plate. The antibiotic producing activity of a colony is indicated by no growth of any other bacterial colony in its vicinity. This region of no growth is indicated by the formation of a clear and colorless area around the antibiotic producing microorganism's colony on the agar plate. This region is called as growth inhibitory zone. Such a colony is isolated from the plate and purified either by making repeated subculturing or by streaking on a plate containing a suitable medium, before stock culture is made. The purified culture is then tested for its antibiotic spectrum.

However, the crowded plate technique has limited applications, as it will not give indication of antibiotic producing organism against a desired organism. Hence, this technique has been improved later on by employing a test organism to know the specific inhibitory activity of the antibiotic. In this modified procedure, suitable serially diluted soil suspension is spread on the sterilized agar plate to allow the growth of isolated and individual microbial colonies (approximately 30 to 300 per plate) after incubation. Then the plates are flooded with a suspension of test organism and the plates are incubated further to allow the growth of the test organism. The formation of inhibitory zone of growth around certain colonies indicates the antibiotic activity against the test organism. A rough estimation of the relative amounts of antibiotic produced by a microbial colony can be estimated by measuring the diameter of the zone of inhibited test organism's growth. Antibiotic producing colonies are later on isolated from the plate and are purified before putting to further testing to confirm the antibiotic activity of a microorganism.

(ii) **Indicator dye technique:** Microorganisms capable of producing acids or amines from natural sources can be detected using this method by incorporating certain pH indicator dyes such as neutral red or bromothymol blue into nutrient agar medium. The change in the color of a particular dye in the vicinity of a colony will indicate the ability of that colony to produce an organic acid or base.

Production of an organic acid can also be detected by an alternative method. In this method calcium carbonate is incorporated into the agar medium. The production of organic acid is indicated by the formation of a clear zone around those colonies which release organic acid into the medium.

The identified colonies are isolated and purified either by repeated subculturing or by streaking methods and a stock culture is made which may be used for further qualitative or quantitative screening tests.

(iii) **Enrichment culture technique:** This technique is generally employed to isolate those microorganisms that are very less in number in a soil sample and possess specific nutrient requirement and are important industrially. They can be isolated if the nutrients required by them is incorporated into the medium or by adjusting the incubation conditions.

(iv) **Auxanotrophic technique:** This technique is employed for the detection and isolation of microorganisms capable of producing certain extracellular substances such as growth stimulating factors like amino acids, vitamins etc. A test organism with a definite growth requirement for the particular metabolite is used in this method.

For this purpose, spread a suitable aliquot on the surface of a sterilized agar plate and allow the growth of isolated colonies, after incubation. A suspension of test organism with growth requirement for the particular metabolite is flooded on the above plate containing isolated colonies, which are subjected to further incubation. The production of the particular metabolite required by the test organism is indicated by its increased growth adjacent to colonies that have produced the required metabolite. Such colonies are isolated, purified and stock cultures are prepared which are used for further screening process.

(v) **Technique of supplementing volatile organic substances:** This technique is employed for the detection and isolation of microorganisms capable of utilizing

carbon source from volatile substrates like hydrocarbons, low molecular weight alcohols and similar carbon sources. Suitable dilution of a microbial source like soil suspension are spread on to the surface of sterile agar medium containing all the nutrients except the one mentioned above. The required volatile substrate is applied on to the lid of the petri plates, which are incubated by placing them in an inverted position. Enough vapors from the volatile substrate spread to the surface of agar within the closed atmosphere to provide the required specific nutrient to the microorganism, which grows and form colonies by absorbing the supplemented nutrient. The colonies are isolated, purified and stock cultures are made which may be utilized for further screening tests.

2.4.2 Secondary screening: Primary screening helps in the detection and isolation of microorganisms from the natural substrates that can be used for industrial fermentations for the production of compounds of human utility, but it cannot give the details of production potential or yield of the organism. Such details can be ascertained by further experimentation. This is known as secondary screening, which can provide broad range of information pertaining to the

1. Ability or potentiality of the organism to produce metabolite that can be used as an industrial organism.
2. The quality of the yield product.
3. The type of fermentation process that is able to perform.
4. Elimination of the organisms, which are not industrially important.

To evaluate the true potential of the isolated microorganisms both qualitative and quantitative analysis are generally conducted. The sensitivity of the test organism towards a newly discovered antibiotic is generally analysed during qualitative analysis, while the quantum yield of newly discovered antibiotic is estimated by the quantitative analysis.

A. EVALUATION OF POTENTIALITIES OF MICROORGANISMS

Microorganisms isolated in the primary screening are critically evaluated in the secondary screening so that industrially important and viable potentialities can be assessed. They include:

1. To determine the product produced by an organism is a new compound or not.
2. A determination should be made about the yield potentialities of various isolated microorganisms that are detected in primary screening for that new compound.
3. It should determine about the various requirements of the microorganism such as pH, aeration, temperature etc.
4. It should detect whether the isolated organism is genetically stable or not.
5. It should reveal whether the isolated organism is able to destroy or alter chemically their own fermentative product by producing adaptive enzymes if they accumulate in higher quantities.
6. It should reveal the suitability of the medium or its constituent chemicals for the growth of a microorganism and its yield potentialities.
7. It should determine the chemical stability of the product.
8. It should reveal the physical properties of the product.
9. It should determine whether the product produced by a microorganism in a fermentative process is toxic or not.

10. Secondary screening should reveal that whether the product produced in fermentation process exists in more than one chemical form. If so, the amount of formation of each chemical formation of these additional products is particularly important since their recovery and sale as byproducts can greatly improve the economic status of the fermentation industry.
11. The new organism should be identified to the species level. This will help in making a comparison of growth pattern, yield potentialities and other requirements of test organism with those already described in the scientific and patent literature, as being able to synthesize products of commercial value.
12. It should select industrially important microorganisms and discard others, which are not useful for fermentation industry.
13. It should determine the economic status of a fermentation process undertaken by employing newly isolated microorganism.

B. METHODS OF SECONDARY SCREENING

As described above secondary screening gives very useful information pertaining to the newly isolated microorganisms that can be employed in fermentation processes of commercial value. These screening tests are conducted by using petri dish containing solid media or by using flasks or small fermenters containing liquid media. Each method has some advantages and disadvantages. Sometimes both the methods are employed simultaneously.

Liquid media method is more sensitive than agar plate method because it provides more useful information about the nutritional, physical and production responses of an organism to actual fermentation production conditions. Ehrlenmayer flasks with baffles containing highly nutritive liquid media are used for this method. Flasks are fully aerated with glass baffles and continuously shaken on a mechanical shaker in order to have optimum product yield.

There are several techniques and procedures that can be employed for secondary screening. However, only a specific example of estimation of antibiotic substance produced by species of *Streptomyces*, is described in the following paragraph. Similar methods could be used for the detection and isolation of microorganisms capable of producing other industrial products.

(i) Giant colony technique: This technique is used for isolation and detection of those antibiotics, which diffuse through solid medium. Species of *Streptomyces,* is capable of producing antibiotics during primary screening. The isolated *Streptomyces* culture is inoculated into the central area of a sterilized petri plates containing nutrient agar medium and are selected. The plates are incubated until sufficient microbial growth takes place. Cultures of test organism, whose antibiotic sensitivity is to be measured are streaked from the edges of plates upto but not touching the growth of *Streptomyces* and are further incubated to allow the growth of the test organisms. Then the distance over which the growth of different test organisms is inhibited by the antibiotic secreted *Streptomyces* is measured in millimeters. The relative inhibition of growth of different test organisms by the antibiotic is called inhibition spectrum. Those organisms whose growth is inhibited to a considerable distance are considered more sensitive to the antibiotic than those organisms, which can grow close to the antibiotic. Such species of *Streptomyces,* which have potentiality of inhibiting microorganisms is preserved for further testing.

(ii) Filtration method: This method is employed for testing those antibiotics which are poorly soluble in water or do not diffuse through the solid medium. The *Streptomyces*

is grown in a broth and its mycelium is separated by filtration to get culture filtrate. Various dilutions of antibiotic filtrates are prepared and added to molten agar plating medium and allowed to solidify. Later on cultures of various test organisms are streaked on parallel lines on the solidified medium and such plates are incubated. The inhibitory effect of antibiotic against the test organisms is measured by their degree of growth in different antibiotic dilutions.

(iii) Liquid medium method: This method is generally employed for further screening to determine the exact amount of antibiotic produced by a microorganism like *Streptomyces.*

Ehrlenmayer conical flasks containing highly nutritive medium are inoculated with *Streptomyces* and incubated at room temperature. They are also aerated by shaking continuously and vigorously during incubation period to allow *Streptomyces* to produce the antibiotic in an optimum quantity. Samples of culture fluids are periodically withdrawn aseptically for undertaking the following routine checks:

1. To check the suitability of different media for maximum antibiotic production.
2. To determine the value of pH at which there will be maximum growth of the microorganism and antibiotic production.
3. To check for contamination.
4. To determine whether the antibiotic produced is new or not.
5. To check the stability of the antibiotic at various pH levels and temperatures.
6. To determine the solubility of the antibiotic in various organic solvents.
7. To check about the toxicity of the antibiotic against the experimental animals.

After carrying out the above mentioned routine tests further studies are also conducted to know the following additional information.

1. Effect of incubation temperature and antifoaming agents on fermentation.
2. Rate of resistance developed among the test organisms.
3. Checking the antibiotic for its bacteriostatic or bactericidal properties. Its ability to precipitate serum proteins to cause hemolysis of blood or to harm phagocytes.
4. Checking for possibility of inclusion of precursor chemical of the antibiotic production in the medium.
5. Suitability of the organism for mutation and other genetic studies.

2.5 PRESERVATION OF INDUSTRIALLY USEFUL ORGANISMS

Isolation, preservation and detection of industrially useful microorganisms is a time consuming and very expensive process. Therefore, it is essential to keep the isolated organisms in a viable condition so that it retains the desirable characters and it can be used whenever required for industrial production. This is done by storing it by creating certain special environmental conditions by which it remains in a viable condition but in an inactive state. This phenomenon is called as preservation of culture. The preservation of culture should be done in such a way that it eliminates the genetic changes, prevents contamination and retains the viability. Though there are several methods of preservation of industrially useful organisms, a description of only some important methods are given below.

2.5.1 Repeated subculturing: This is the most common, simplest and routine method of preservation of microorganisms. Selected microorganisms are initially grown on agar slants. After sufficient growth has taken place, they are transferred to fresh medium before they loose their viability. The appropriate time period for such transfer ranges from a week to few months (generally four to eight months). Though an organism may be kept viable by this method but there is a probability of occurrence of mutations in the organism, which may lead to strain degeneration and subsequent uselessness of the organism for commercial usage. That is why it is less frequently used for preserving microorganism.

(a) Advantages:

1. This method is cheap,
2. Needs no special equipment,
3. Recommended for small collection centers, and
4. Retrieval easy

(b) Disadvantages:

1. Change in physiological and genetical characters, and
2. Time consuming.

2.5.2 Storage under liquid nitrogen: This method is also called as cryogenic storage method, because a cryoprotective agent in the form of 10% glycerol is used. Industrially useful microorganisms are stored under very low temperature ranging from 150°C - 196°C. In this method ranging, low temperatures are created by employing liquid nitrogen. Metabolic activities of microorganisms are reduced considerably at this low temperature. This method is generally employed for the preservation of fungi, bacteriophages, viruses, algae, yeasts, animal and plant cells, and tissue cultures.

This technique involves growing the desired microorganism in sufficient quantity either in the form of cells or spores or fragments of fungal mycelium. The grown up culture is suspended in 10% glycerol. The suspension is then introduced in to small ampoules at the rate of approximately 0.5 ml each. The ampoules are usually made up of borosilicate glass. The ampoules containing culture suspension are frozen and sealed hermitically. Freezing is done either by directly dipping the ampoules into the liquid medium or hanging the ampoules initially over the column of liquid nitrogen for sometime and finally dipping into the liquid. The frozen ampoules are then dipped one above the other on small aluminum containers at the rate of six ampoules per can. The cans are then packed in aluminum boxes, 20 per each box. The perforations allow free flow of liquid nitrogen. There may be loss of viability in few cells during freezing process but there is virtually no loss of viability during storage phase.

(a) Advantages:

1. Viable cultures may be preserved for many years by this method, especially those cultures which do not withstand preservation by freeze drying.
2. Though the equipment is costly, the process is economical.
3. The cultures remain viable under these conditions for 10-30 years without undergoing any change in their characteristics.

(b) Disadvantages:

1. Evaporation of liquid nitrogen and replacement of lost liquid nitrogen regularly and periodically. If this is not done the apparatus will fail due to which there will be loss of valuable cultures, and
2. The method is relatively expensive.

2.5.3 Employment of dried cultures: This technique has been used extensively for the storage of fungi and actinomycetes particularly for sporulating mycelial organisms. Moist soil is generally used as a preservating medium. Moist soil is first sterilized and then inoculated with a desired culture and incubated for several days to allow some growth to occur. The soil with growing organism is dried at room temperature for a period of two weeks. The dried soil is then stored in a dry atmosphere or in a refrigerator.

Silica gel and porcelain beads may be used alternatively for soil. It is possible to preserve a culture for more than 20 years. Pridham and his associates reported that out of 1800 actinomycetes preserved by this method about 50% were viable, after 20 years storage.

2.5.4 Lyophilization: It is one of the best methods for long-term preservation of micro-organisms. It is generally used for the preservation of fungi, viruses, bacteria, enzymes, toxins, sera and other microorganisms. It is a convenient method for the preservation of large number of cultures.

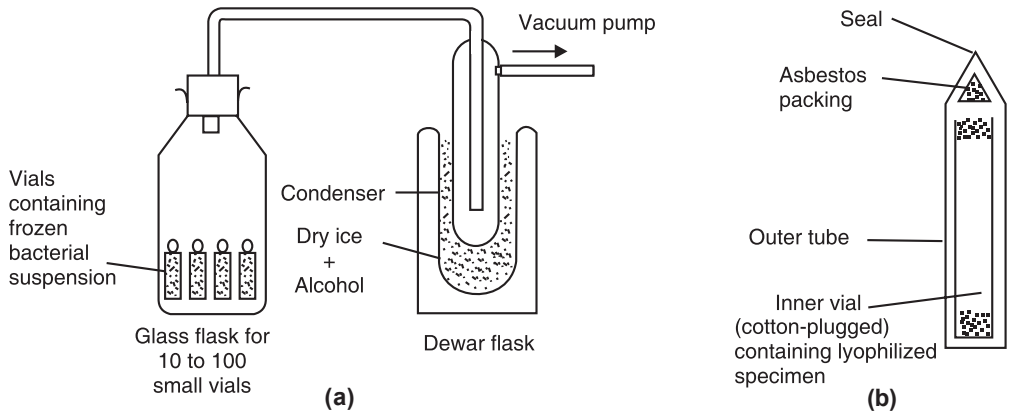


Fig. 2.16: (a) Small cotton-plugged vials containing frozen suspension of the microorganisms are placed in the glass-flask, which is attached to a condenser. The condensor is connected with a high-vacuum pump and this system brings about desiccaton of the cultures.

(b) After desiccation of the cultures as in (a) the vials are removed, placed individually in a large tube covered with asbestos packing and under vacuum.

Lyophilization, which is also called as freeze drying involves freezing of a culture followed by its drying under vacuum which results in the temporary inhibition of metabolic activities of microorganisms. The technique consists of the following stages :

1. The organism is allowed to grow to the maximum stationary phase on a suitable sterilized medium.
2. The cells are suspended in a protective medium like milk, serum or sodium glutamate.
3. A few drops of suspension are transferred to a glass ampoule.

4. The ampoules are then frozen by immersing into a freezing mixture of dry ice and alcohol at 78°C and are subjected to high vacuum until evaporation takes place completely (Fig. 2.16a).
5. The ampoules are then sealed and stored in a refrigerator (Fig. 2.16b).
6. The method of revival vary from laboratory to laboratory. Generally, during revival process the ampoules are decaped under sterile conditions and the dried pellets consisting of cells of the culture are transferred to a suitable liquid medium and are allowed to dissolve in order to make a suspension of cells. Then the cells are streaked on to agar plates. Sometimes there may be a need to undertake repeated subculturing for getting a culture exhibiting all characters. Cells of a lyophilized culture may remain viable for 10 years or more.

(a) Advantages:

1. Culture once dried needs no further attention,
2. It needs very cheap storage equipment like refrigerator, and
3. It is easy to transport freeze-dried ampoules to far off places in large numbers in relatively small boxes.

(b) Disadvantages:

1. This is expensive and needs expertise.

2.5.5. With mineral oil: This is one of the cheap and easy methods of preservation. Many microbes can be successfully preserved for longer time.

In this method tubes with sterile agar slants are inoculated with a given culture. The tubes are incubated till sufficient growth of the given microbe takes place. The grown up culture is covered with a suitable mineral oil to a depth of about 1 cm above the top of the slanted surface using sterile technique. Thus, over laid cultures can be stored at room temperature or preferably at low temperature by about 15°C.

Paraffin oil of specific gravity 0.865 to 0.890 is generally used in this method. The oil is sterilized either in McCartney bottles for 15 minutes at 103.41×10^3 Nm^{-2} pressure or in an autoclave at 103.41×10^3 Nm^{-2} pressure for 2 hours and then dried in an oven at 170°C for 1-2 hours, before it is used. Maintenance of viability of a culture under this treatment largely varies with the species and generally ranges from 10-20 years.

(a) Advantages: This method of maintenance has the unique advantage that you can remove some of the growth under the oil with a transfer needle, inoculate a fresh medium and still original culture can be preserved. It is easy to control mites problem.

(b) Disadvantages:

1. Chances of air-borne contamination during subculturing are more,
2. Chances of mutations are more, and
3. Retarded growth or inability to sporulate on retrieval.

2.5.6. Storage in soil: Spore suspension in sterile water is poured into a culture bottle containing twice autoclaved loam soil (20% moisture). Fungal growth is allowed for few days and then stored with loose caps in a refrigerator. Fungi like *Rhizopus, Alternaria, Aspergillus, Penicillium* and *Fusarium* which have long viability and stability can be maintained by this method.

Advantages:

1. Cheap and convenient,
2. Free from mites infestation, and
3. Though some variations may occur but the cultures are stable and survive upto 10 years.

2.5.7. Silica gel storage: Mc Cortney bottle is filled partly with medium grain non-indicating silica gel and sterilized by dry heat. The bottles are kept in a tray of water to the depth above the level of the gel. The water is frozen by placing the tray in a deep freeze with the temperature 17°C to 102°C. Spore suspensions are prepared in sterilized and cooled 5% skimmed milk and added to silica gel crystals in the tray of frozen water using Pasteur pipette and wetted three quarters to avoid over saturation. The gel bottles are left in the ice bath for about 20 minutes until the ice around them have melted a little. The crystals are agitated to ensure thorough dispersion of the suspension. Bottles are dried with the caps loose for 10-14 days at 25°C until the silica gel crystals separate. Bottles are reversed down from tightly and stored over indicator silica gel in air tight container at 4°C. The indicator gel requires replacement once or twice in a year.

(a) Advantages:

1. This method is simple and mites free.
2. Suitable for oomycetous fungi.
3. Stability in some cultures like *Neurospora* and *Aspergillus* is more.

(b) Disadvantages:

1. Repeated retrieval can result contamination, and
2. Suitable only for fungi.

2.6 FERMENTATION MEDIUM

Designing a suitable medium is an important stage for carrying out successful, laboratory experiments, scale up fermentations and manufacturing processes. On small scale, it is relatively simple to device a medium containing pure compounds but it may be unsuitable for use in large scale process. A medium to be employed in a large-scale process will have to meet the following aspects.

1. It has to produce maximum quantity of biomass for every gram of substrate used.
2. It has to yield maximum amount of product.
3. It has to produce minimum amount of undesirable products.
4. The sources of the medium should be of consistent quality and be available throughout the year.
5. It should cause minimal problems while making sterilization, aeration, agitation, extraction, purification and waste treatment.
6. It should supply the energy, nutrients for growth, building of cell substances and biosynthesis of fermentation products.

An ideal medium should contain carbon source, nitrogen source, inorganic salts, water, vitamins, growth factors, buffers, antifoaming agents and dissolved oxygen. The following

equation is generally considered in formulating a medium employed in an aerobic fermentation process.

$$\text{Carbon} + \text{Nitrogen} + O_2 + \text{others} \rightarrow \text{Biomass} + \text{products}$$
$$\text{Energy source requirements} + CO_2 + H_2O + \text{heat}$$

Quantification of the nutrients indicated in the equation is generally made for designing an economical medium and for minimizing the wastage of the component substrates. Thus, it will be possible to calculate the minimal quantities of nutrients, which will be needed to produce a specific amount of biomass and optimal yield of the product. However, it is not always easy to quantify all the factors very precisely.

2.6.1 Types of media: There are basically two types of fermentation media. They are:

1. Simple media, and 2. Complex media

(i) Simple media: They are also called as inorganic media because they contain inorganic salts, water and a nitrogenous source in the form of inorganic salts of nitrates or ammonia. They are used for autotrophic microorganisms and their carbon requirement is fulfilled by the carbon dioxide of the air or carbonates. Thus, from simple inorganic nutrients autotrophic microorganisms are able to synthesize their required and life sustaining organic compounds, which allow growth and multiplication of their cells to meet their energy requirements.

(ii) Complex media: These are also called as organic media. These are employed for the growth of heterotrophic and saprophytic microorganisms. These organisms require the presence of many types of simple to complex preformed nutrients in the medium. Most important of them is the organic substance which help in the synthesis of cell substances and release of metabolic energy. Complex media are further subdivided into two categories. They are: (a) Synthetic media and (b) Crude media.

(a) Synthetic medium: It contains purified sugar, nitrogen source (ammonium/nitrate compounds/amino acids), inorganic salts and water. The medium constituents are known both qualitatively and quantitatively. Hence, these media are also called as defined media. Synthetic media are very important for certain types of studies.

1. It is easy to know the specific effect of an individual component or several components of the medium on cell growth and product yield.
2. They help in obtaining reproducibility of growth and product yield from one fermentation run to another so that errors due to medium composition is held at a minimum.
3. There will be, usually, no foaming because they do not contain any protein or high molecular weight peptides.
4. The recovery and purification of fermentation products are relatively simple because most of the compounds that might interfere with the recovery are known.

In spite of above advantages synthetic media are less frequently employed because of high cost and low product yield.

(b) Crude media: They are also called as undefined media because neither the type of chemical substances present in the medium nor their exact composition is known. They are usually made from the cheapest agricultural products like Soyabean meal, black strap molasses, corn steep liquor, sulfite waste liquor etc. They are not only

cheap but also provide cheap source of major carbon source, unknown nutrients and growth factors. Crude media, apart from providing nutrients for growth and product formation, also meet certain requirements of fermentation, which include:

1. Buffering capacity.
2. Lack of foam production.
3. Control of oxidation and reduction potential.
4. Inhibition of the growth of contaminating microorganisms.
5. Neutralization of acidic or alkaline products.
6. Contribute to the maintenance of genetic stability of the microorganisms.
7. Promotion of vigorous aeration and agitation.
8. Allowing the recovery of the fermentation product without resort to complex recovery procedures.

Simple or complex carbohydrates, alcohols, organic acids, proteins, peptides, amino acids and event hydrocarbons can be used as carbon source for fermentation. These are usually used in crude form. Crude source of simple carbohydrates include beet and sugarcane molasses, corn molasses or hydrol, whey, sulfite waste liquor, Cannery wastes etc. Crude form of complex carbohydrates include corn, wheat, rye, rice, potatoes, sweet potatoes and others which also contain complex carbohydrates such as agricultural waste products are also used as a carbon source in some of the fermentation processes. Thus, several agricultural waste products can be used as carbon source. But, only few and important substrates used as carbon source are described.

2.6.2 Substrates used as carbon source:

(i) Molasses: Molasses are some of the cheapest sources of carbon. It is a by-product in sugar production. Apart from containing a large amount of sugar, it also contains nitrogenous substances, vitamins and trace elements. The chemical composition of a molasses, however, varies depending upon the raw material from which it is produced. A comparative chemical composition of sugar beet molasses and sugar cane molasses is given in table 2.7. Considerable variation in the quality of molasses occurs depending on the location, climatic conditions and the production process of each sugar factory. These molasses are extensively used in different fermentations (Fig. 2.17).

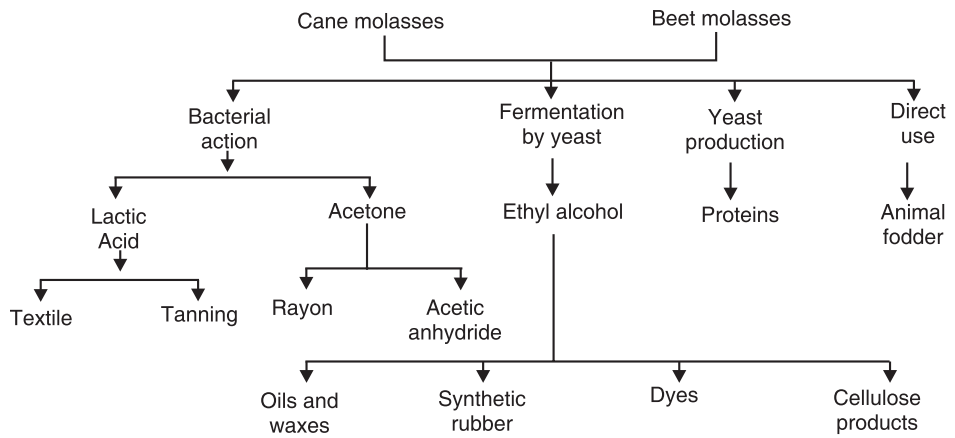


Fig. 2.17: Uses of molasses in different fermentation industries

Table 2.7: Composition of sugar beet and sugar cane molasses

Name of the chemical	Beet molasses/g	Sugarcane molasses/g
Sucrose	48.5	33.4
Raffinose	1.0	–
Invert sugar	1.0	21.2
Miscellaneous organic matter	20.7	19.6
Nitrogen	0.2 – 2.8	0.4 – 1.5
P_2O_5	0.02 – 0.07	0.6 – 2.0
Calcium oxide	0.15 – 0.7	0.1 – 1.1
MgO	0.01 – 0.1	0.03 – 0.1
K_2O	2.2 – 4.5	2.6 – 5.0
SiO_2	0.1 – 0.5	–
Al_2O_3	0.005 – 0.06	–
Fe_2O_3	0.001 – 0.02	–
Ash	4 – 8	7 – 11
Dry weight	78 – 85	77 – 84
Thiamine	130 mg/ 100g	830 mg/ 100g
Riboflavin	4.1 mg/ 100g	250 mg/ 100g
Pyridoxine	540 mg/ 100g	650 mg/ 100g
Niacin amide	5100 mg/ 100g	2100 mg/ 100g
Pontothenic acid	130 mg/ 100g	2140 mg/ 100g
Folic acid	21 mg/ 100g	3.8 mg/ 100g
Biotin	5.3 mg/ 100g	120.0 mg/ 100g

(ii) Cornsteep liquor: It is a byproduct resulting from the steeping of corn during the commercial production of corn starch, gluten and other corn products. Cornsteep liquor is formed when the used or spent steep water are concentrated to approximately 50% solids. Chemically, cornsteep liquor contains lactic acid in half quantity and in the other half there are amino acids, glucose, reducing sugars, salts, vitamins and some precursors which help in the synthesis of penicillin. It was first extensively used for making medium for the production of penicillin.

(iii) Sulfite waste liquor: It is a waste product of paper industry. It contains sugars. The relative amount of sugars present in sulfite waste liquor depends on the type of wood employed in the paper production. It contains approximately 2-3% of sugars. Generally soft woods contain higher amounts of hexose's sugars, while hard wood contain pentose sugars. D-glucose, D-galactose and D-mannose are the predominant hexose sugars, while D-xylulose and L-arabinose are the predominant pentose sugars present in the sulfite waste liquor. It is generally used for making fermentation media for the production of ethanol by *Saccharomyces cerevisiae* and in the growth of *Torula utilis* for feed.

(iv) Malt Extract: It is an aqueous extract of malted barley and consists of about 90-92% of carbohydrates which include monosaccharides (glucose and fructose), disaccharides (maltose and sucrose), trisaccharides (maltotriose) and dextrins. It also contains some nitrogenous substances in the form of proteins, peptides, amino acids, purines, pyrimidines and vitamins. However, the amino acid content of different malt extracts varies according to the type of grains used. But of all the amino acids, proline makes up about 50% of the total amino acids present (table 2.8). It is an excellent substratum for many fungi, yeasts and actinomycetes.

Table 2.8: Chemical composition of malt extract

Chemical	% of dry weight
Maltose	52.2
Hexose (glucose, fructose)	19.1
Sucrose	1.8
Dextrin	15.0
Other carbohydrates	3.8
Nitrogenous materials	4.6
Ash	1.5
Water content	2.0

2.6.3 Substrates used as nitrogen source: Inorganic or organic substrates can be used as source of nitrogen. Inorganic nitrogen may be supplied as ammonia gas, ammonium salts or nitrates. However, ammonium salts like ammonium sulphate or ammonium nitrates are more frequently employed in the fermentation processes. Ammonium salts such as ammonium sulphate will usually produce acidic conditions, as the ammonium ion is used and the free acid will be liberated. This lowers the pH and leads to optimal production of desired product. On the other hand, when nitrates are utilized as nitrogen source, acidic and alkaline conditions are created alternatively.

It has been observed that on organic nitrogen source, the growth of industrial microorganisms is faster than on inorganic nitrogen. Organic nitrogen may be supplied either in the form of pure and synthetic amino acids or protein or urea or in the form of agriculture byproducts like Soya meal, Corn steep liquor, peanut meal, cotton seed meal etc. Description and utility of some of the organic nitrogen sources derived from agricultural products and other sources are given below.

(i) Corn Steep Liquor: It is formed during starch production from corn. The production and composition of cornsteep liquor is given earlier. The concentrate contains about 4% of nitrogen. Many amino acids like alanine, arginine, glutamic acid, isoleucine, threonine, valine, phenylalanine, methionine and cystine are present in considerable amounts.

(ii) Yeast Extract: It forms an excellent source of organic nitrogen for many industrial microorganisms. It is produced from baker's yeast by autolysis at 50° to 55°C or by plasmolysis in the presence of high concentration of sodium chloride. It contains amino acids, peptides, water soluble vitamins and carbohydrates (table 2.9). However, the chemical composition of yeast extract depends upon the substrates used for yeast cultivation. The complex carbohydrates, glycogen and trehalose

present in the yeast cells are hydrolyzed into glucose during yeast extract production.

Table 2.9: Chemical composition of yeast extract

Composition	Yeast extract produced by means of	
	Autolysis	Plasmolysis with NaCl
Dry matter	70.0	80.0
Total nitrogen	8.8	7.4
Protein	55.0	46.0
NaCl	1.0	18.0
AMINO ACIDS (mg/g)		
Alanine	3.4	2.3
Amino butyric acid	0.1	0.1
Arginine	2.1	1.1
Asparagine	3.8	3.1
Cystine	0.3	0.2
Glutamic acid	7.2	5.1
Glycine	1.6	1.6
Histidine	0.9	0.8
Isoleucine	2.0	1.6
Leucine	2.9	2.3
Lysine	3.2	2.9
Methionine	0.5	0.5
Ornithine	0.3	0.9
Phenylalanine	1.6	1.6
Proline	1.6	1.5
Serine	1.9	1.5
Threonine	1.9	1.4
Tyrosine	0.8	0.5
Valine	2.3	1.3
VITAMINS (in ppm)		
Thiamine	20 – 30	10 – 15
Riboflavin	50 – 70	50 – 70
Pyridoxine	25 – 35	20 – 30
Niacin amide	600	100
Pantothenic acid	200	350

(iii) **Soya meal:** It is a residue formed from soyabean after the extraction of soyabean oil. It is chemically complex substance containing 50% proteins, 30% carbohydrates which include acidic polysaccharide, arabinan, arabinoglucan, raffinose, stachyose and sucrose, 1% of residual fats and 1.8% lecithin. It is frequently used in antibiotic fermentations.

(iv) Peptones: Peptones are the products of protein hydrolysis. Though, they are relatively expensive, they are utilized as organic nitrogen source by many industrial microorganisms. Sunflower seeds, cotton seeds, soyameal, peanut seeds, gelatin, keratin and casein are the rich sources of peptones. Chemical composition of peptone largely depends upon its source. For example, peptones from keratin has a large proportion of proline and cystine but lacks lysine. On the other hand, peptone from gelatin largely contains proline and hydroxyproline but does not contain sulfur containing amino acids. Peptones of soya meal and cotton seeds have large content of carbohydrates (table 2.10). The composition also varies according to the type of hydrolysis employed whether acidic or enzymatic especially with regard to its tryptophan content.

Table 2.10: Chemical composition of some peptones

Composition (%)	Casein (enzymatic)	Casein (Acid method)	Soya meal (enzymatic)
Water content	4.8	3.5	4.0
Ash	5.8	39.7	14.0
Total nitrogen	12.8	8.3	9.2
Amino nitrogen	6.7	6.4	1.8
NaCl	3.0	38.0	5.9
Carbohydrate	–	–	29.0
Free amino acids (mg/g)			
Alanine	16.8	14.1	3.3
Arginine	30.2	5.0	1.1
Aspartic acid	8.7	33.2	2.4
Cystine	–	0.7	–
Glutamic acid	38.6	79.6	5.6
Glycine	4.4	9.3	2.3
Histidine	13.3	4.6	1.1
Isoleucine	29.9	10.7	2.8
Leucine	71.9	29.7	9.4
Lysine	61.1	12.8	7.9
Methionine	22.7	11.2	3.3
Phenylalanine	33.0	9.7	3.7
Proline	7.4	34.5	0.8
Serine	28.7	19.8	4.8
Threonine	21.5	13.1	1.9
Tryptophan	8.6	0.05	1.6
Tyrosine	14.0	4.9	0.7
Valine	36.4	16.7	3.8

2.6.4 **Minerals:** All microorganisms require certain mineral elements for growth and metabolism. Mineral elements required for microorganisms are classified into two categories:

1. Macroelements, 2. Trace elements

The important **macroelements,** which are required in large quantities, include magnesium, phosphorus, potassium, sulphur, calcium and chlorine. They are added into the fermentation medium generally in the form of inorganic salts. Cobalt, copper, iron, manganese, molybdenum and zinc are the important trace elements which are generally required in low quantities. The **trace elements** generally present as impurities in other major ingredients of fermentation media, are sufficient to meet the requirement of micro organisms. Though they are present in extremely low concentration, they play an important role in the metabolic activities of microorganism by activating the relevant enzymes. However, when synthetic media are used, they have to be added to the medium in the form of salts. The form in which these minerals are usually supplied and their concentration ranges are given in table 2.11.

Table 2.11: Commonly used salts as source of minerals

Constituent	Quantity (in g / liter)
$KH_3\ PO_4$	1.0 – 4.0
$Mg\ SO_4 \cdot 7H_2O$	0.25 – 3.0
KCl	0.5 – 12.0
$CaCO_3$	5.0 – 17.0
$FeSO_4 \cdot 4H_2O$	0.01 – 0.1
$ZnSO_4 \cdot 8H_2O$	0.1 – 1.0
$MnSO_4 \cdot H_2O$	0.01 – 0.1
$CuSO_4 \cdot 5H_2O$	0.003 – 0.01
$Na_2\ MoO_4 \cdot 2H_2O$	0.01 – 0.1

2.6.5 **Growth factors:** Some microorganisms require preformed compounds called growth factors for the synthesis of full complement of cell components. The growth factors more commonly required are vitamins, but certain specific amino acids, fatty acids and sterols sometimes also act as growth factors. Many of the natural carbon and nitrogen sources used in medium preparation contain all or some of the required growth factors. If only one vitamin is required to be added to the medium it is economical and beneficial to add the pure vitamin instead of using a larger amount of a cheaper multiple vitamin source. Calcium pantothenate is used in vinegar production.

2.6.6 **Buffers:** Chemical substances that control pH values are called as buffers. The control of pH may be extremely important if optimal productivity is to be achieved. An increase or decrease in pH during the fermentation may affect product yield because of specific effects of acidity or alkalinity on the metabolism of the microorganisms.

A compound may be added to the medium to serve specifically as a buffer or may also be used as a nutrient source. Many media are buffered at about pH 7.0 by the incorporation of calcium carbonate. Addition of an inorganic salt into a medium can cause pH change. For example, ammonium sulphate supplies nitrogen as the ammonium ion whereas sulphuric acid, liberated may make the medium more acidic. Phosphates

which are part of many media also play an important role in buffering. The pH may also be controlled externally by addition of sodium hydroxide or sulphuric acid as required.

The balanced use of carbon and nitrogen sources will also form the basis for pH control. Buffering capacity can be provided by the proteins, peptides and amino acids, such as in cornsteep liquor.

2.6.7 Antifoam agents: Aeration and agitation of a medium especially liquid medium causes formation of foam in the fermenter. The most common cause of formation of foam is due to proteins in the medium such as cornsteep liquor, pharmamedia, peanut meal, soyabean meal, yeast extract or meat extract. It may also be due to some factors produced by the microorganism that is employed in the fermentation. If foaming is not controlled it may lead to physical and biological changes in fermenter causing some problems in the fermentation.

(a) The important physical changes include.

1. Reduction into the working volume of the fermenter due to oxygen exhausted gas bubbles circulating in the system.
2. Changes in bubble size.
3. Lower mass and heat transfer.
4. Incorrect monitoring and control.

(b) The important biological changes include

1. Removal of cells from the medium which will lead to autolysis and release of cell proteins resulting in the stability of the foam.
2. Deposition of cells in upper part of the fermenter.
3. Danger of contamination due to loss of sterile conditions in the fermenter because of wetting of air filter exit of the fermenter.
4. Possibility of siphoning, leading to loss of product.

The formation of foam must be controlled if a fermentation process is to be carried out in a proper manner. Foam formation has adverse effect on fermentation and can be avoided by using a defined medium. Use of mechanical foam breaker will also be efficient. If the foam is unavoidable it can be controlled by adding antifoaming agent. This is accomplished by adding antifoam agents to the fermentation medium. Antifoam agents are surface-active agents reducing the surface tension in the foam and destabilizing foam protein films by:

1. Hydrophobic bridge between two surfaces.
2. Displacement of the absorbed protein.
3. Rapid spreading on the surface of the film.

It is quite important to use a good antifoaming agent as it may effect the fermentation process. An ideal antifoam should possess the following properties:

1. Should be heat sterilizable.
2. Should have no effect on oxygen transfer.
3. Should be cheap.
4. Should not cause any problem in the extraction and purification of the product.
5. Should not cause any handling problems.

6. Should be non-toxic to humans and animals.
7. Should not be metabolized by the microorganisms.
8. Should be long acting in preventing new foam formation.
9. Should be active at low concentration.
10. Should be non-toxic to the microorganism.
11. Should disperse readily and have fast action on existing foam.
12. It should be compatible with other components of the medium and the process.

The following compounds which meet most of the above requirements have wide acceptance and most suitable as antifoaming agents in different fermentation processes.

1. Silicones, 2. Sulphonates, 3. Esters, 4. Fatty acids and their derivatives like glycerides, which include cotton seed oil, linseed oil, soyabean oil, olive oil, castor oil, sunflower oil, rapeseed oil and codliver oil, 5. Alcohols-stearyl and octyl decanol, 6. Miscellaneous like Alkaterge, exazaline, polypropylene glycol.

Sometimes a mixture of oil and alcohol, such as lard oil mixed with octyl decanol, is also used as an antifoam agent, especially in penicillin fermentation. They remain inactive if added before sterilization of the medium. That is why they are sterilized separately and then added to the medium. It is necessary that correct amount and form of antifoam must be added each time. Use of excess amount of antifoam agent not only becomes toxic to the fermenting microorganism but also becomes ineffective in controlling foam. Sometimes they also provide nutrients like fatty acids to fermenting microorganism. For example *Streptomyces* in the antibiotic fermentation, antifoam also helps in lowering the pH of the medium.

2.6.8 **Water:** Water is crucial in any fermentation industry and required in quantity ranging from 70% to 90%. It is also needed in carrying out many of the fermentation processes such as heating, cooling, cleaning and soaking. Clean water of consistent composition is, therefore, required in large quantities from usable permanent source. Water also supplies certain trace elements such as copper, molybdenum, zinc, boron and other substances that may be required in minute quantities by microorganisms.

2.6.9 **Oxygen requirements:** Oxygen although not added to an initial medium, is a very important component of the medium and its availability can be extremely important in controlling the growth rate of fermenting microorganisms and subsequent desired metabolite production. The medium may influence the oxygen availability in a number of ways. For example, certain substance of a medium like rapidly metabolized sugars with a high oxygen requirement create oxygen deficient conditions in the medium which leads to non availability of oxygen to the fermenting microorganism due to which its growth and product yield is affected.

2.7 OPTIMIZATION OF MEDIUM COMPONENTS

Determination of different combinations and sequences of process conditions need to be investigated to assess the growth conditions which produce the biomass with the physiological state best constituted for product formation is called as optimization of medium components.

In a classical method, medium optimization is carried out by changing one independent variables like nutrient, antifoam, pH, temperature etc., while fixing all others at a certain level. But this method cannot be used where a large number of variables are employed simultaneously

as it could be extremely time consuming and expensive. Industrially, the aim is to perform the minimum number of experiments to determine optimal conditions.

When more than five independent variables are to be investigated, the Plackett-Burman design may be used to find the most important variable in a system, which are then optimized in further studies. These authors gave a series of designs for up to one hundred experiments using an experimental rational known as balanced incomplete blocks. This technique allows for the evaluation of X-1 variables by X experiments. X must be a multiple of 4, for example 8, 12, 16, 20 etc. Normally one determines how many experimental variables need to be included in an investigation and then selects the Plackett-Burman design, which meets the requirement most closely in multiples of 4. Any factor known to have no effect may be included and designated as a dummy variable, whose incorporation into an experiment makes it possible to estimate the variance of an effect (experimental error).

Plackett-Burman design for seven variables A to G at high and low levels is shown in table 2.12, in which two factors E and G are designated as dummy variables. These can then be used in the design to obtain an estimate of the error. Normally, three dummy variables will provide an adequate estimate of the error. However, more can be used if fewer real variables need to be studied in an investigation. Each horizontal row in the table 2.12 represents a trial and each vertical column represents the H (high) and L (low) values of one variable in all the trials. This design requires the frequency of each level of a variable in a given column to be equal. So that in each test (horizontal row) the number of high and low variables should be equal. Consider variable A; for the trials in which A is high, B is high in two of the trials and low in the other two, as will all the remaining variables. For those trials in which A is low, B will be high two times and low two times. This will also apply to all the other variables. Thus, the effects of changing the other variables cancel out when determining the effect of A. The same logic then applies to each variable. However, no changes are made to the high and low values for E and G columns.

Table: 2.12: Plackett-Burman design for seven variables

Trial	Variables							Yield
	A	B	C	D	E	F	G	
1	H	H	H	L	H	L	H	1.1
2	L	H	H	H	L	H	L	6.3
3	L	L	H	H	H	L	H	1.2
4	H	L	L	H	H	H	L	0.8
5	L	H	L	L	H	H	H	6.0
6	H	L	H	L	L	H	H	0.9
7	H	H	L	H	L	L	H	1.3
8	L	L	L	L	L	L	L	1.4

H denotes a high level: L denotes a low level value.

The effect of the dummy variables are calculated in the same way as the effects of the experimental variables. If there are no interactions and errors in measuring the response, the effect shown by a dummy variable should be zero. If the effect is not equal to zero it is assumed to be a measure of the lack of experimental precision plus any analytical error in measuring the response.

This procedure will identify the important variables and allow them to be ranked prioritywise which soever is investigated in detail to determine the optimum value for use.

2.8 MEDIA STERILIZATION

Media may be sterilized by any one of the three following methods:

1. Sterilization by boiling.
2. Sterilization by passing live steam.
3. Sterilization by subjecting the medium to steam under pressure.

However, it is important to evaluate each medium to determine its particular requirements for sterilization. Sterilization of certain media is achieved by simply boiling them directly. But, care should be taken that there will be no overcooking of the medium. Because overcooked medium neither supports good microbial growth nor yield good quantity of the desired product. Most of the fermentation media are sterilized by passing live steam. This is accomplished either by batch sterilization or continuous sterilization.

2.8.1 Batch sterilization: This process is employed for large-scale pure culture fermentation. The medium constituents, doubled to the strength, are mixed with water in a separate mixing tank, then passed through retention tubes and heat exchangers before passing into the large empty and steam sterilized fermentation tank. The retention tubes contain steam jet heaters that inject high pressure steam into the medium to sterilize it as it passes through the pipes and rate of passage is adjusted to provide complete medium sterilization without overcooking. The heat exchangers consist of a pipe filled with the medium within a second pipe containing cool water moving in the opposite direction. They are employed for cooling the medium before delivery to the fermenter. The medium is subjected to a temperature of 121°C for about 20 minutes. A sterilization profile of typical fermenter is shown in Fig. 2.18. Destruction of cells occurs during heating and cooling (from 12°C to the fermentation operating temperature). A mechanical vacuum breaker on the tank, prevents a vacuum build up and consequent collapse of the tank Fig. 2.18. It has an advantage of saving production time because fermenter is kept empty between two fermentation runs.

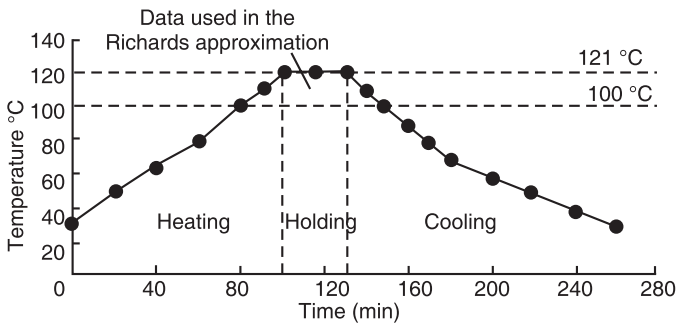


Fig. 2.18: Sterilization profile of typical fermenter

2.8.2 Continuous sterilization: In this method the medium is passed through a heat exchanger, where the temperature is raised to a required level usually greater than 120°C. The medium from heat exchanger is then allowed to pass through a holding coil where it is held at sterilizing temperature for pre-determined time period. The medium is finally cooled by circulating it again through the cooling pipes of heat exchangers. Subjecting the medium to shorter holding time, will allow the use of sterilization temperature

greater than 120°C without causing any injurious effect on the nutritional value of the medium (Fig. 2.19 a and b).

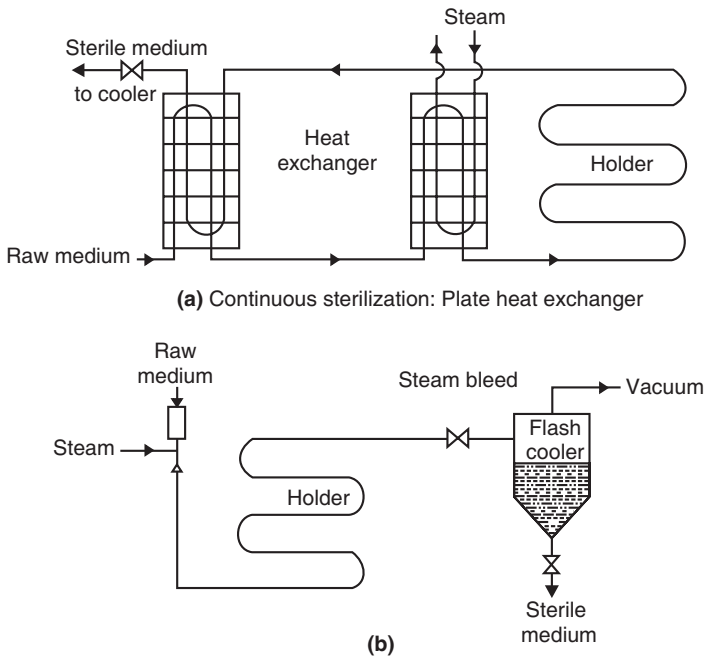


Fig. 2.19: (a) Flowsheet of continuous steam sterilization
(b) Continuous sterilization: Steam injection and flash cooling

The method and duration of sterilization generally varies from medium to medium. Synthetic media require shorter sterilization time. Crude media, however, require longer time of sterilization because viscous nature of the medium and because of presence of heat resisting spores in it. Prolonged heating of a medium containing sugar not only leads to carmalization but also results in reaction between sugars and phosphates that leads to degradation of medium components of low or high pH values. Medium containing vitamins and enzymes cannot be sterilized by heat treatment. But, sterilization of such media can be accomplished by filtration and by employing specialized filters like Seitz filters. Similar methods of filtration sterilization methods can also be adapted to media containing volatile chemicals.

However, media employed in certain fermentation processes, like processes in which antibiotics are produced, and certain yeast fermentations which are conducted at low pH values, need no medium sterilization because the aseptic conditions either provided or generated during the coarse of fermentation will help in controlling the contaminating organisms that might grow in the medium.

2.9 INOCULUM PREPARATION

Any culture that may be used, as inoculum should posses the following qualities.

1. It must be healthy and in an active state to minimize the duration of lag phase in the subsequent growth.

2. It must be available in sufficiently large volumes to provide an inoculum of optimum size.
3. It must be in a suitable morphological form.
4. It must be free of contamination.
5. It must retain its product formation capabilities.

Table 2.13: Composition of inoculum development and production medium

Process	Inoculum development medium (in %)	Production medium (in %)
Griseofulvin	Whey powder lactose 3.5 Whey powder nitrogen 0.05 Corn steep liquor solids to give approx. 0.04 N. 0.38 KH_2PO_4 0.4 KCl 0.05	**Lactose 7** Corn steep liquor solids to give N 0.2 Lime stone 0.8 KH_2PO_4 0.4 KCl 0.1
Lysine	Corn molasses 5% Corn steep liquor 1% $CaCO_3$ 1% Soya bean meal hydrogen-	20% - - 1.8
Vitamin B_{12}		(g/lit)
	Sugar beat molasses 70 Sucrose - Betaine - $NH_4H_2PO_4$ 0.8 $(NH_4)_2SO_4$ 2.0 $MgSO_4$ 0.2 $ZnSO_4$ 0.02 5-6 dimethyl benzimideizole 0.005	105 15 3 - 2.5 0.2 0.08 0.025
Penicillin		(g/lit)
	Glycerol 7.5 Cane molasses 7.5 $MgSO_4 \cdot 7H_2O$ 0.05 Corn steep liquor 2.5 KH_2PO_4 0.06 Peptone 5.00 NaCl 4.00 Fe-tartarate 0.005 $CuSO_4 \cdot 5H_2O$ 0.004 Agar 2.50 Distilled water 1000 ml	Corn steep liquor 3.5 Lactose 3.5 Glucose 1.0 $CaCO_3$ 1.0 KH_2PO_4 0.4 Edible oil 0.25 Penicillin precursor Distilled water 1 liter

The method employed to produce an inoculum is called, as inoculum development. An important factor in obtaining a suitable inoculum is the choice of the culture medium. The chosen medium should not be only in a position to support maximum growth of the inoculum but should also yield optimal quantity of the product and minimize the lag phase of the microbial growth. This can be achieved if the medium for inoculum development and production

medium should be sufficiently similar in chemical composition. Similarity the chemical composition of these two media will minimize any period of adaptation of the culture to the production medium, thus reducing the lag phase and the fermentation time. However, though sufficiently similar in chemical composition, the inoculum medium shall have to be developed in such a way that it becomes less nutritious than production medium and contains lower levels of carbon source so as no precursor, required for a product formation. Composition of some of the inoculum development and production medium are given below (table 2.13).

The quantity of inoculum normally used is between 3 to 10% of the medium volume. A relatively large inoculum volume is used to minimize the length of the lag phase and to generate the maximum biomass in the production fermentor in short time is possible, thus increasing vessel productivity.

2.9.1 Inoculum development: A pure culture with good growth and morphological characters is selected as master culture, is plated on a solid medium and the plates are incubated. Approximately ten colonies of typical morphology are selected and inoculated on to the slants and incubated. They are used as submaster cultures.

A submaster culture is inoculated into a shake flask of approximately 250 or 500 ml size containing 50 to 100 ml liquid medium and is incubated at suitable temperature. After sufficient microbial growth occurs it is used as an inoculum for a larger flasks or a laboratory fermentor containing 3 to 10% by volume medium. Culture purity checks are carried out at each stage to detect contamination as early as possible.

Throughout the procedure there is a possibility of contamination and strain degeneration especially, if the employed fermentation is continuous culture process. The greater the number of stages between the master culture and the production fermentor, the greater the risk of contamination and strain degeneration. To avoid this situation, stringent quality control measures may have to be employed either with the employment of suitable quantity of inoculum to prevent contamination and strain degeneration. It has also been observed that inoculum development becomes economical if in a continuous culture fermentation a small scale fermentor, ten times smaller than the production fermentor, is employed together with usage of appropriate quantity of the medium.

2.9.2 Inoculation: The physiological condition of the inoculum when it is transferred to the next production stage can have major effect on the performance of fermentation. The optimum time at which the inoculum reaches to good physiological condition is an important factor in the transfer of inoculum to the next production stage. The optimum time of transfer must be determined experimentally and then standard procedures are established so that inoculation with ideal culture may be achieved decisively. This determination is done by:

1. Standardization of culture conditions.
2. On line monitoring of fermentation.
3. Status of quality and enzyme profile.

(*i*) Standardization of culture conditions: These procedures include the standardization of cultural conditions and monitoring the state of an inoculum

culture. So that it is transferred at the optimum time, when it is, in the correct physiological state.

The most widely used criterion for the transfer of vegetative inoculum is biomass and such parameters as packed cell volume, dry weight, net weight, turbidity, respiration, residual nutrient concentration and morphological form have been used. **Ettler (1992)** demonstrated that the physiological behaviour of *Streptomyces noursei* could be used as the transfer criteria in nystatin fermentation. The rhedogy of the seed fermentation changed from Newtonian to non-Newtonian behaviour and the optimum inoculum transfer time corresponded with this transformation.

(*ii*) **On line monitering of fermentation:** Criteria which may be monitored on line are the most convenient parameters to use as indicator of inoculum quality. These criteria would include dissolved oxygen, pH and oxygen or carbon dioxide in the effluent gas. Parton and Willis (1990) advocated the use of carbon dioxide production rate (CPR) as a transfer criterion, which requires analysis of the fermenter effluent gas. These workers provide an excellent example of the effect of inoculum transfer time on the production of a streptomycetes secondary metabolite.

The CPR of the inoculum fermentation and the point at which inoculum was transferred is shown in (Fig. 2.20a). The CPR of the subsequent production fermentation is shown in (Fig. 2.20b) from which it may be seen that these three fermentations performed similarly. However, (Fig. 2.20c) illustrates the very different secondary metabolite production of the three fermentations. Thus, although the time of transfer had only a marginal influence on biomass in the production fermentation, the effect on the product formation is critical. It should be emphasized that the amount of biomass transferred was standardized for the three fermentations and the difference in performance were due to the physiological states of the inoculum. Monitoring of fermentation conditions through computer is illustrated in Fig. 2.21.

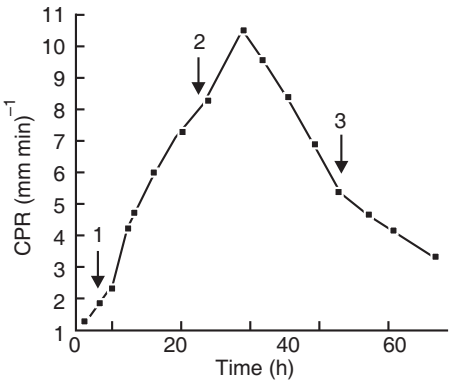


Fig. 2.20 (a)

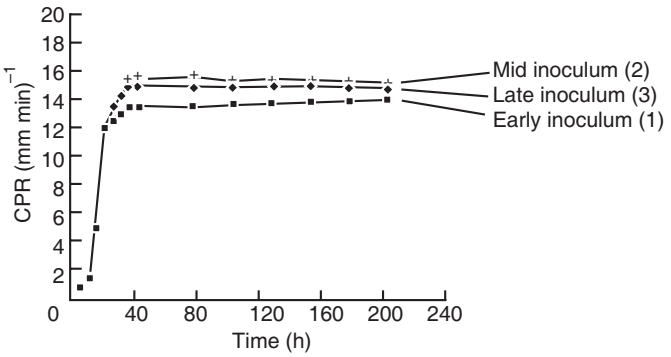


Fig. 2.20 (b)

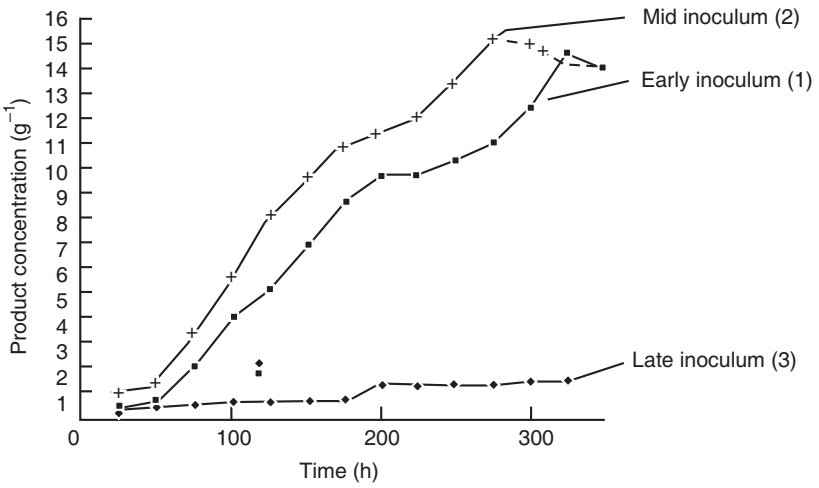


Fig. 2.20 (c)

Fig. 2.20 : The effect of inoculum age on growth and productivity in a streptomycetes fermentation.
(a) The carbon dioxide production rate (CPR) profile of the inoculum culture showing the points (1, 2 & 3) at which inocula were removed.
(b) The effect of inoculum age on the CPR of the production fermentation.
(c) The effect of inoculum age on productivity in the production fermentation.

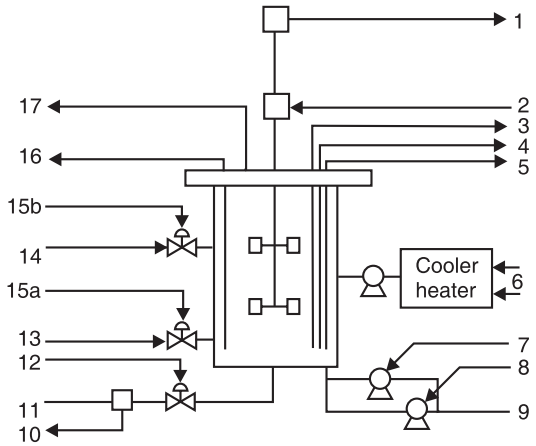


Fig. 2.21: Bioreactor configuration for computer monitoring and control.

1. Agitation measurements, 2. Agitation feedback, 3. Dissolved O_2 measurements, 4. Volume measurements, 5. Temperature measurements, 6. Temperature feedback, 7. and 8. Volume feedback, 9. Effluent, 10. Air-flow measurements, 11. Air, 12. Air-flow feedback, 13. Acid, 14. Base, 15a. and 15b. pH feedback, 16. pH measurements, 17. Off gas O_2+CO_2 measurements

(*iii*) **Status of quality and enzyme profile: Smith** and **Calam (1980)** compared the quality and enzyme profile of differently prepared inoculum of *Penicillium patulum* producing griseofulvin and demonstrated that the low level of glucose-6-phosphate dehydrogenase was indicative of a good quality inoculum. The enzyme profile of good quality inoculum was established early in the growth of the seed culture. Thus, this approach could be used to assess the cultural conditions giving rise to satisfactory inoculum, but would be of less value in determining the time of transfer.

2.10 SCALE UP OF FERMENTATION

The conversion of a laboratory fermentation process to an industrial process is termed as scale up. A process which works well at the laboratory scale may work poorly or may not work at all when attempted first at large scale, because the fermentation conditions required by a laboratory fermentation differs widely with the one required by fermentation at industrial scale (Fig. 2.22). The yield of the fermentation product is made to increase in quantity during scaling up and all the factors required for this scaling up are suitably monitored. The major factors involved in a scale up are:

1. Sufficient quantities of inoculum is to be developed.
2. The process of sterilization must be adjusted according to the scale up, which may result a change in the medium after sterilization.
3. Environmental parameters have to be changed according to the requirements of the organism. These parameters include nutrient availability, pH, temperature, concentration of dissolved oxygen, shear conditions, concentration of dissolved carbon dioxide and foam production.

Agitation and aeration play a determinative role in a scale up process. Agitation is generally carried out by bulk mixing and aeration by the provision of oxygen. The parameters, nutrient availability, pH and temperature are affected by agitation, while concentration of dissolved oxygen, shear conditions, concentration of dissolved carbon dioxide and foam production are affected by aeration. In a laboratory fermentation process, it is extremely difficult to carry out suitable agitation, aeration and monitoring of various fermentation conditions that may influence the fermentation microorganism, for greater yield of the desired product. That is why it is important to device suitable criteria of particular fermentation and scale up is conducted based on these criteria.

The problems of scale up can be overcome to some extent, if the following points are taken into consideration.

1. The identification of the principal environmental conditions affected by aeration and agitation in the fermentation specially oxygen concentration and shear.
2. The identification of process variables like power consumption which affects the identified environmental conditions.
3. The calculation of the value of the process variable, to be used on the large scale which will result in the replication of some environmental conditions on both scales.

The important process variables are concentration of dissolved oxygen and power consumption. The process variables which affect mixing and mass transfer are summarized in table 2.14.

Table 2.14: The effect of process variables on mass transfer or mixing characteristics

Process variable	Mixing or mass transfer characteristics affected
Power consumption per unit volume	Oxygen transfer rate
Volumetric air flow rate	Oxygen transfer rate
Impeller tip speed	Shear rate
Pumping rate	Mixing time
Reynolds number	Heat transfer

If concentration of dissolved oxygen (DO) is considered as the determining environmental condition, then power consumption per unit volume and volumetric air flow rate per unit volume should be maintained constant on scale up. This may lead to a change in other parameters as well which are to be then controlled suitably.

If power consumption per unit volume is kept constant then the impeller tip speed (that is shear) increases and mixing decreases. If mixing is kept constant an enormous increase in power is required and shear increases 5 fold. If impeller tip speed (shear) is kept constant then power consumption and mixing decreases. This analysis indicates that it is economically impossible to maintain the same degree of mixing on scale up and therefore, a decrease in yield may be due to mixing anomalies.

The most important environmental conditions affected by aeration and agitation for the majority of fermentations are oxygen concentration and shear. Thus, the most widely used scale up criteria are the maintenance of a constant power consumption or constant shear conditions. Constant shear may be achieved by scaling up on the basis of constant impeller tip speed. Constant power consumption may be achieved on the basis of constant power consumption per unit volume and constant volumetric air flow rate. The operating variable dictating constant power consumption, in geometrically similar vessels, is the agitator speed. The agitator speed on the large scale is then calculated from the correlation between power consumption and operating variables. **Hubbard *et al.* (1988)** summarized the procedure for scaling up of fermentations and proposed two methods to determine the large-scale conditions.

Method–I

1. Determine the volume time airflow rate Q on the large scale based on maintaining Q/V constant (V = Working volume of the fermenter)
2. Calculate the agitator speed that will give same value of power consumption on the large scale. This is achieved using the correlation between power consumption and impeller rotation number.

Method–II

1. Calculate the agitator speed keeping impeller tip speed constant.
2. Calculate the volumetric air flow rate Q from power correlation.

2.10.1 Vessels used for scale up: Different fermentation vessels are being used for determination of scale up of different fermentation processes ranging from laboratory process to commercial process. But, each vessel has advantages and disadvantages as far as scale up of fermentation is concerned.

For laboratory process development, Ehrlenmeyer flasks are the conventional vessels. However, they provide a poor estimate of the fermentation potential of a microorganism and its

medium because of the poor aeration characteristics, even though they are put to vigorous rotatory shaking.

Ehrlenmeyer flasks with glass baffles mounted in the bottom provide better aeration than normal flasks but fail to provide favourable aeration conditions that are obtainable in a large tank. Small laboratory fermentation tank of 1 to 12 liters capacity are more ideal for fermentation studies, since their aeration and agitation conditions can be varied.

Since overall conditions of these vessels resembles more closely with the larger production tank. Pilot plant tanks with a capacity of 100 to 400 lit. are still better suited for carrying out scale up studies because these tanks allow fermentation studies in relation to production tank conditions. At times production tanks are also employed in scale up processes to gain additional information on a particular fermentation before it is carried out on a commercial scale. In general it is that experience with particular fermentation equipment and previous fermentations is the only real guide in translating scale up information to production tanks. General layout of industrial fermentation plant is illustrated in Fig. 2.22.

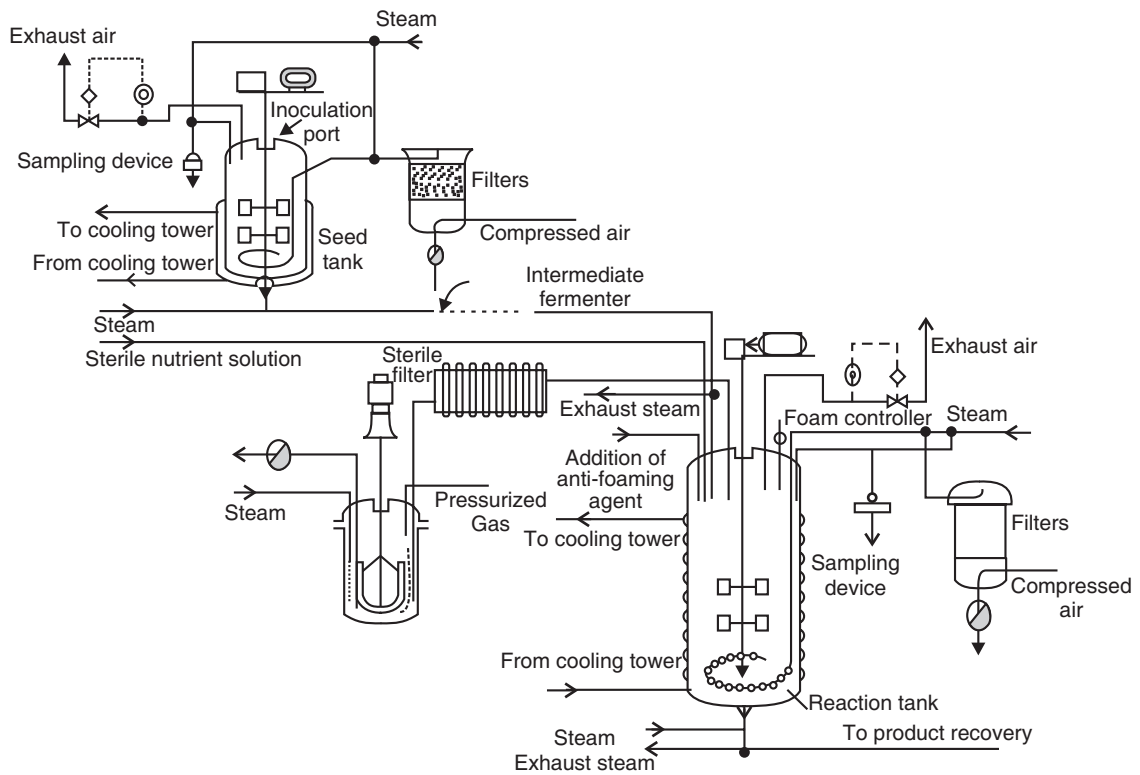


Fig. 2.22: General layout of industrial fermentation plant

2.11 DOWNSTREAM PROCESS

The recovery and purification of fermentation products is one of the most important aspects of industrial fermentation processes. The selection of suitable process of recovery and purification depends upon the nature of the end product, their concentration, the by-products present, the

stability of the product and degree of purification. However, the selected procedure must be quick, simple, reliable, accurate and must measure the desired compound accurately from among the different chemicals that may be present in the growth medium. It is generally considered that the fermentation and product recovery are integral parts of an overall process. Because of interaction between the two, neither of them be developed independently, is necessary as this must not result in problems and unnecessary expenditure.

2.11.1 General pattern of a recovery process: The following flow chart shows the various stages that are generally involved in the recovery process of a biological product. The recovery of an extracellular product of a fermentation broth is taken as an example to describe the various stages involved in a recovery process (Fig. 2.23).

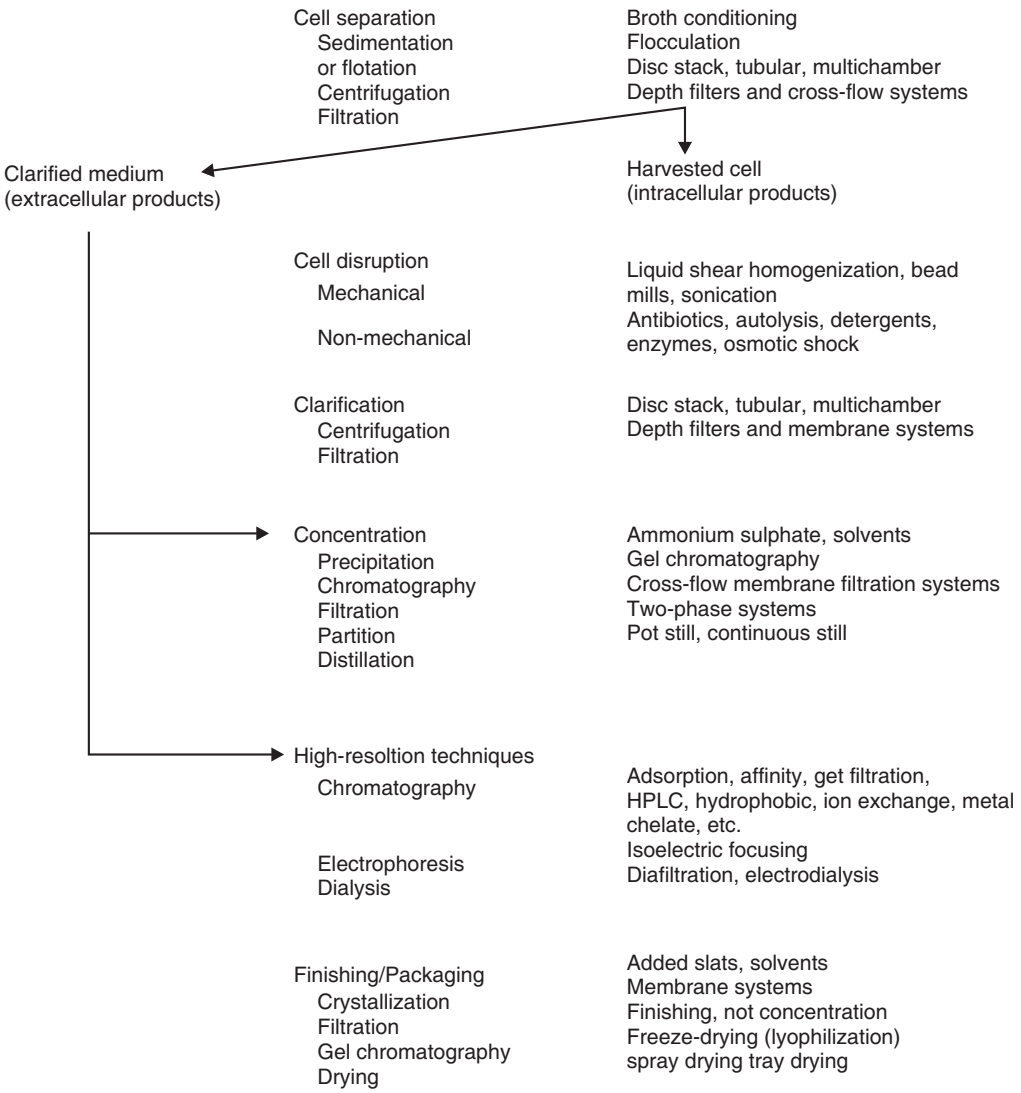


Fig. 2.23: Typical unit processes used in downstream processing.

The first stage for the recovery of an extracellular product is the removal of the large solid particles and microbial cells by centrifugation or filtration process. The broth is then fractionated or extracted into major fractions. Fractionation is carried out by employing either ultrafiltration or reverse osmosis or adsorption/ion exchange/gel filtration or affinity chromatography or liquid – liquid extraction or two-phase aqueous extraction based on chemical and physical properties of substances or precipitation. The product-containing fraction is then purified by fractional precipitation. Further, more precise chromatographic techniques or crystallization process is employed to obtain a product, which is highly concentrated and devoid of any impurities (table 2.15). Various recovery processes can broadly be classified into the following categories:

1. Physico-chemical processes.
2. Chromatographic partition processes.
3. Biological assays.

Table 2.15: Basic operation in downstream process

	Basic operation	Method
Fermentation broth	Removal of biomass and other solids	Centrifugation Filtration Sedimentation Sieving
Raw product	Concentration and Fractionation	Extraction Ion-exchange Adsorption Precipitation Distillation
Final product	Final purification	Chromatography Distillation Decolorization Crystallization Drying

2.11.2 Physico-chemical processes: Though there are many physico-chemical recovery processes employed for the isolation of fermentation products, major of them are described here. However, it is to be remembered that the actual choice of recovery process largely depends upon chemical reaction or chemical analysis involved because the fermentation broth contains many compounds in addition to the desired one.

(*a*) Turbidity analysis: This technique is employed for the determination of cell yield or biomass, which is the primary product of the fermentation process. Centrifugation and determination of packed cell volume is the simplest method by which estimation of cell yields of a microorganism can be made quickly. Fermentation broth with the cells is centrifuged in graduated centrifuge tubes and the volume of sedimented cells are measured in centimeters.

The other method employed for measuring the cell yields is the turbidity analysis of a fermentation broth, containing little insoluble debris other than cells. The cells present in the growth medium are diluted to a turbidity range and the cell suspension is exposed to a range of visible light or a monochromatic light with wave-length, (660 nm) in a colorimeter. Quantitative cell measurements are made as optical density by deflection of light that is caused by the microbial cells present in the diluted suspension. For determining number of cells, trubidimetric measurements of cell numbers are usually standardized against some other procedures, such as plate count. A standard curve is then drawn with the help of values of optical density of plate count which may be derived from a series of dilutions of the cell suspension. Once the standard curve is prepared, it is easy to determine the cell number by converting reading of optical density.

(*b*) Titration and gravimetric analysis: This technique is generally employed for the recovery of organic acids and the choice of recovery method varies according to the nature of the acid. For example the amount of an organic acid like lactic acid produced in a fermentation can easily be determined by adding bromothymol blue or other pH indicating dye to a sample fermentation broth, followed by titration with an alkali of known concentration. If the acid formed in a fermentation broth is able to form an insoluble salt with an alkali it can be isolated by precipitation, which can be washed, dried and weighed. Volatile acids with low molecular weight can be isolated by direct distillation of fermentation broth and distillate is analyzed by titration.

Similarly acids of high molecular weight can be separated from fermentation broth by adsorption to a suitable anion exchange resin and subsequent elution.

(*c*) Spectrophotometric determinations: Different types of spectrophotometers are used during spectrophotometric determinations. The difference in these analytical instruments lies in the kind of light absorbed. Visible light spectrophotometers absorb visible light in the range of 350 to 1000 nm. They measure the amount of light at a specific wavelength which is absorbed as light beam passes through a colored solution. Ultraviolet spectrophotometers absorb light within the wavelength range of 200 to 380 nm, while in a fluorescent spectrophotometer the amount of light emitted by a fluorescent compound is measured.

Fermentation products which are colored can be measured directly by visible light spectrophotometers. A specific wavelength of visible light is allowed to be absorbed by a fermentation product and the absorption of light is measured as optical density. A standard curve is prepared with the help of the optical density of known concentration of a pure concerned chemical. By substituting the values of optical density of the fermentation product in the standard curve, the amount of fermentation product can be determined.

The colorless fermentation products are made coloured by treating with a colouring reagent. For example, amino acids are generally colorless chemicals but they develop purple colour when treated with ninhydrin under suitable conditions. The amount of fermentative production of amino acid is then determined by following the method described above.

Some fermentation products are basically colourless and do not develop colours even after chemical treatment. Determinations of such products are made with the help of ultraviolet spectrophotometers. By allowing them to absorb wavelength of ultraviolet

light and measuring the optical density of the absorbed light.

Still some fermentation products like riboflovin fluoresce when exposed to ultraviolet light. The intensity of fluorescent light is measured by employing fluorescent spectrophotometers or fluorometer. By determining the optical density of the fluorescent light, the amount of fermentation product can be determined. The principles of analysis for ultraviolet and florescent spectrophotometers are similar to those for visible spectrophotometers.

Chromatographic partition determinations: The term chromatography was originally used in earlier times for a technique to separate coloured compounds. Although partition is the predominant cause of separation, other factors are also involved though in a minor fashion.

Employment of partition chromatography on paper or thin layer plates, not only made marked strides in the fermentation research and technology, but also allowed to detect and identify many types of new fermentation products even though they are in extremely smaller amounts. An unknown compound can easily be detected with the help of this technique by comparing the partition value with already known compounds.

The basic principle involved in the partition chromatography is the continuous partition of the solute or the sample between a stationary phase, such as paper or the silica gel thin layer plates, and a mobile phase consisting of a mixture of solvents, as these solvents move across the paper or silica gel layer. Basically there are two types of partition chromatography:

(*i*) Paper chromatography, and (*ii*) Thin layer chromatography.

(*i*) Paper chromatography: If a cellulose paper is used as a stationary phase it is called as paper chromatography. Water soluble compounds are generally separated by employing paper chromatography.

In this technique the fermentation product to be separated is applied as series of spots about ¾ apart along a line, about few centimeters away from one end of a rectangular filter paper. Sometimes sample is also applied as a single spot on the paper. The paper is kept dipped in a mixture of solvent system consisting of a hydrophobic and a hydrophilic solvent. The solvent mixture starts ascending the paper and thereby wetting it by capillary action. But the cellulose, which is the component of a paper, can absorb only the hydrophilic solvent and not the other. Thus there exists a phase separation at the microlevel. The compounds present in the fermentation broth get gradually separated when the solvent front reaches the spot, which contains the fermentation products to be separated. As the solvent front moves further the fermentation products get separated as consolidated spots on the filter paper.

For carrying out the solvent migration on the filter paper, (also called as chromatogram,) it is placed in a closed tank so that its atmosphere becomes saturated with the components of solvent system and helps in the perfect separation of the product. The developed chromatogram is removed from the tank and allowed air dry to remove extra solvent. The separation is measured in terms of a unit called Rf (Resolution factor) which is described as relative to front. Its value is calculated by the following formula.

$$Rf = \frac{\text{Distance travelled by the solute}}{\text{Distance travelled by the solvent}}$$

The Rf value of a given compound in a particular solvent system is constant under given set of physical conditions such as temperature, pH etc. and this can be used to identify the unknown compound.

Based on the method adopted, the processes of paper chromatography are classified into following categories:-

(*a*) **Ascending paper chromatography:** In this method, the solvent system is placed as a shallow layer at the bottom of the tank and the filter paper strip is suspended in the solvent system in such a manner that the spotted sample does not dip in the solvent system, so that solvent ascends the paper (Fig. 2.24 a).

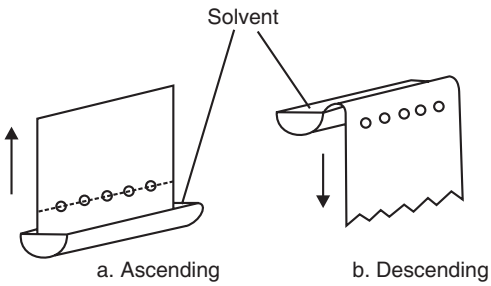


Fig. 2.24: (a) Ascending, and (b) Descending chromatography

(*b*) **Descending paper chromatography:** In this method, a trough containing solvent mixture is placed near the top of the tank and spotted chromatographic paper is placed in this trough so that the solvent migrates down the paper (Fig. 2.24 b).

(*c*) **Horizontal or circular chromatography:** In this method, a circular filter paper is chosen and a hole is made at its center. A wick placed in the central hole elevates the solvent to the chromatograph from a reservoir present beneath the sheet (Fig. 2.24 c) so that solvent migrates all round from the centre. This is also called as disc chromatography.

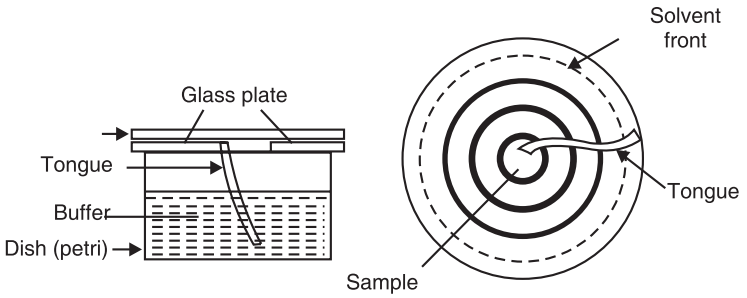


Fig. 2.24: (c) Horizontal or circular chromatography

(*d*) **Two-dimensional chromatography:** It is a modified or improved method of chromatography. It is employed in such situations where there are many compounds present in the sample and the individual spots on the chromatograph tend to coalesce (Fig. 2.25).

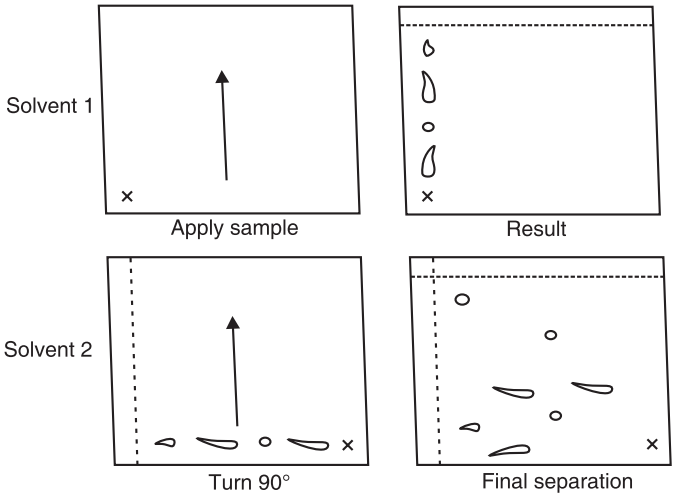


Fig. 2.25: Two dimensional paper chromatography

A single spot of the sample is applied at one corner of the rectangular filter paper and solvent migration is allowed across the paper. This moves out individual compounds along a line from the origin in the direction of solvent migration. Solvents are then evaporated by drying filter paper, it turned 90 degrees and placed in a second different solvent system. Solvent migration with the second solvent system moves individual compound across the filter paper at an angle of 90° from the original direction of travel. This two dimensional chromatography allows the use of two different, even contrasting solvent system on a single chromatographic paper and separates the compounds more clearly. However, it has certain major disadvantages are:

1. Only a single spot can be applied on the filter paper.
2. It is difficult to interpret the Rf values, as the values are associated with two different directions of solvent migration. But, it is still useful in the separation of compounds with small differences in their Rf values.

(*ii*) **Thin layer chromatography:** Thin layer chromatography (TLC) is similar in most respects to paper chromatography except the usage of a thin glass plate in place of filter paper. A thin glass plate is coated with a thin layer of silica gel or aluminum oxide, which acts as a stationary phase. Calcium sulfate or starch is generally incorporated into the stationary phase compound, which helps in binding the latter to the glass plate surface. The layer is allowed to dry at this stage, it is called as chromatogram.

Fermentation broth sample is applied as a series of small spots approximately ¾ inch apart along a line about one inch from one end of the glass plate. The spots are allowed to dry at room temperature or heated with a stream of warm air or an infrared lamp.

After the spots have dried up glass plate is kept in a tank, saturated with solvent mixture, in such a way that spot do not dip in solvent mixture. The solvent mixture is then allowed to migrate through the chromatogram. The individual compounds present in fermentation broth get separated based on their molecular weight as consolidated spots on the chromatogram, when solvent mixture migrates through it. For precise analytical determinations the temperature at which solvent migration occurs must be carefully controlled. When solvent migration is done TLC plates are allowed to air dry.

The compounds thus separated on the chromatogram can be detected with the help of suitable spray reagent or made to absorb ultraviolet light. Amino acids can be detected by spraying the chromatogram with ninhydrin to produce purple coloured spots. Radioactive compounds can be located by exposing the chromatogram to x-rays. The separated compounds on the chromatogram can be identified by calculating the Rf values and subsequently comparing the values with the known compounds. Biologically active compounds such as antibiotics can be detected by placing the chromatogram for short period on the surface of an inoculated agar plate.

(*iii*) **Column chromatography:** In this method, a vertical tube made up of glass or polyacrylate plastic is provided with an inlet and an outlet. The inlet is connected to reservoir containing a solvent to have a continuous flow of the solvent, while the outlet is fitted with rubber tubing to collect the eluting compounds. The column is fitted in the upright position and its bottom is sealed with a glass wool, which supports the stationary phase, which is gradually packed up to ¾ of the column. The stationary phase is made of either gel or adsorbent or resin. The stationary phase is prepared in the form of a thick suspension and is called as slurry. After the slurry settles well in the column, the fermentation broth is applied at its top end to a height of 5-10 cm. The solvent mixture, which acts as a mobile phase, is made to flow through the column from its reservoir (Fig. 2.26).

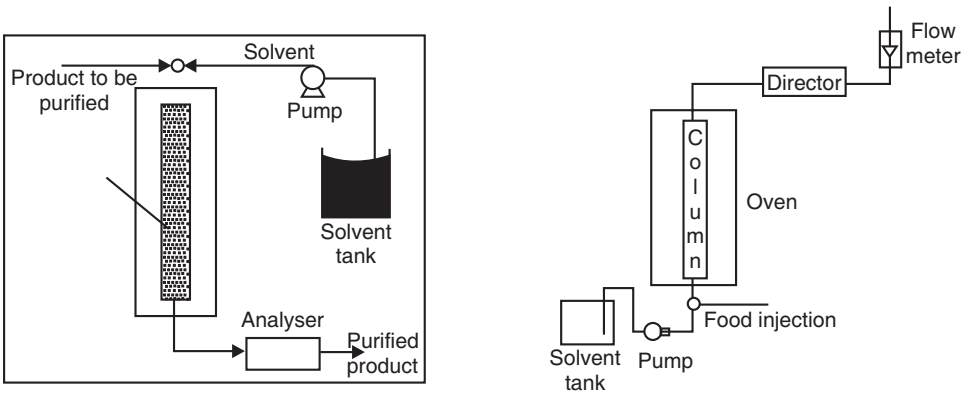


Fig. 2.26: Basic operation of column chromatography

As the solvent mixture moves through the column the products present in the fermentation broth get separated as the products emerge from the column outlet they are analyzed, identified and calculations with regard to their quantity is also made by employing one of the several appropriate analytical methods available like fluorescence detectors, voltameters, refractometric detectors, conductivity detectors etc. This technique is used for the separation of proteins and related compounds.

(*iv*) **Ion exchange chromatography:** Macromolecules with ionic groups can be separated by ion exchange chromatography. The macromolecules are made to adsorb a carrier and are eluted by a solvent of defined strength. The separation mainly depends upon the nature of the macromolecule and the ionic strength of the eluting solvent. This technique is generally used for the purification of antibiotics from fermentation broth, purification of proteins. The ionic material like Dowex, HCR and OCR, Amberlite, IR and IRC are widely used for cation exchange. Anion exchangers used are Dowex, SAR and MSA, Amberlite, IRA and Lewatit M.

(*v*) **Adsorption chromatography:** In adsorption chromatography separation of the contents of biological material is achieved due to hydrophilic and hydrophobic reaction between them. Elution and fractionations are accomplished by means of solutions of higher or lower polarity or strength. Silicates, alumina, activated carbon, cross-linked dextrans and hydroxylapatite are generally employed as adsorption material.

(*vi*) **Affinity chromatography:** It is a specific method for purification of biological material. The desired biological material binds specifically and reversibly to a ligand, which has been fixed to an inert carrier. For example, the nucleotide adenine dinucleotide (NAD) can be purified by allowing it to bind to a carrier containing a dehydrogenase enzyme. The antibiotic bacitracin can be used as a ligand and to isolate aspartine, serine, cystine and metalloproteases from a crude mixture.

(*vii*) **Gel filtration:** In this technique, a gel is used as filtering agent and molecules of different size can be separated by allowing them to pass through the gel with different pore size. As the large molecules cannot penetrate the gel and pass directly through the column, finally elute out along with the mobile front, whereas small molecules penetrate deep into the gel and hence their mobility is inhibited from the solvent front. Standard curves can be developed which relate molecular weight to the elution position. This technique is mainly used at the industrial scale to eliminate salts and separate low molecular weight impurities. It is also used to purify proteins like insulin or interferon, at a smaller scale industrial production. The Biogel types P and A from BioRad and the Sephadex series S, G and LH from Pharmacia are the gels generally employed in gel filtration technique.

(*viii*) **Gas chromatography:** In this technique fermentation products are first volatilized from the fermentation broth and then separated. The product to be separated from the fermentation broth is injected into the gas chromatograph and is converted into its gaseous state by heating. The gas thus formed is pushed by a stream of inert gas through a partition column, and compounds get separated while passing through the column and are detected as they come out of the column by employing proper detecting system. Quantitative assay is obtained by comparison of the areas under peaks on plots of the data with corresponding areas for various quantities of reference compounds.

2.12 BIOLOGICAL ASSAYS

Determining the effect of a fermentation product or an enzyme on the growth and metabolic activities of test organisms is called biological assay or bioassay. Therefore, it is an indirect method of assessing the quality of a fermentation product. Biological assays are, therefore, difficult to perform, less reproducible and usually do not give accurate results.

The microorganisms employed in these assays are called as test organisms. Strains of microorganisms occurring in nature or artificially mutated ones may be employed as test organism. Various categories of microorganisms are used as test organisms. For example, bacteria for assaying amino acids, antibiotics and vitamins; fungi for assaying vitamins and trace elements; yeasts for assaying vitamins and antibiotics.

A microorganism that is employed in a biological assay should possess the following characters:

1. It should genetically be stable and should not exhibit any genetic change in response to the test chemical.
2. It should respond in a graded manner to the test compound and not to other compounds.
3. It should not be a pathogen and should grow on simple medium.
4. It should preferably be aerobic or facultative aerobic but not an anaerobic.
5. It should grow well at a pH that does not affect the stability or toxicity of the fermentation product under assay.

2.12.1 Types of biological assays

Biological assays can be categorized into four general groups. They are Diffusion assays, Turbidimetric assays, Metabolic response assays and Enzymatic assays.

(*i*) **Diffusion assay:** In this assay, the fermentation product to be assayed is allowed to diffuse through medium in a radial fashion from a pad or cup and the adjacent growth of the test organism is either retarded or stimulated. The diameter of this area reflects the concentration of the compound being assayed and the amount of the product is determined by measuring and comparing the area with the various known concentrations of reference compounds. There are two types of diffusion assays, the cylinder method and paper disc method.

(*a*) **Cylinder method:** About 10 ml of molten agar medium is placed in a sterilized petri plate and is allowed to solidify. About 5 ml of the same or a different medium inoculated with a test microorganism is added above the solidified agar medium and is allowed to solidify. This layer is called seeded agar layer. Several small metal, glazed porcelain or glass cylinders are set on the agar surface. The cylinders are filled with appropriate dilutions of the solution to be assayed and are incubated for a specific period of time at constant temperature. The diameter of the zone of stimulated or retarded growth is then measured in millimeters. The concentration of the solution under assay are determined by comparison with a standard curve prepared from the inhibition or stimulation zone (Fig. 2.27).

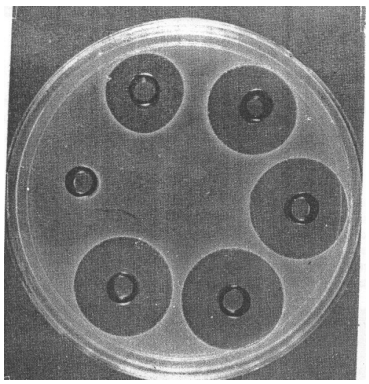
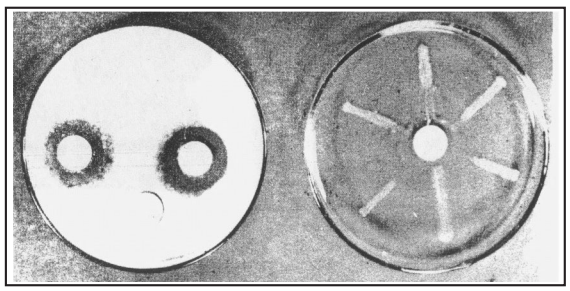


Fig. 2.27: Cylinder cup-plate method of antimicrobial assay

(*b*) **Paper disc method:** Sterile filter paper discs of generally 12.8 mm diameter are used in this method. Each disc is treated with approximately 0.1 ml of reference compound. These treated discs are then placed on the surface of seeded agar medium and incubated as described for the cylinder assay. The calculations of assay results are similar to those for the cylinder assay (Fig. 2.28).



(a) **(b)**

Fig. 2.28: (a) Growth inhibition by antibiotic diffusion through an agar medium
(b) a paper disc method

(*ii*) **Turbidimetric assays:** In this method, the effect of the fermentation product under test in a liquid culture is measured as an increased or decreased turbidity associated with the growth rate or total growth of the microorganism.

A suitable sterilized medium is taken in series of sterilized test tubes and graded amounts of the product to be assayed are added. The tubes are inoculated with an equal amount of vigorously growing young culture of the test organism and then incubated for a predetermined period of time at a constant temperature. The length of incubation period depends on the type of growth measurements, that is, whether growth rate or total growth of test microorganism. For measuring the growth rate, the turbidity of the culture at some point during logarithmic growth is determined, while for measuring the total growth of organism the maximum stationary growth phase is determined.

The relative turbidity produced in the test tubes can be determined either visually or by employing a spectrophotometer. Readings are made as optical densities or absorbance. The optical density is plotted against the concentration of the standard to obtain a standard curve. The concentration of the unknown product is determined by comparing its optical density with the standard curve. (Fig. 2.29 a and b).

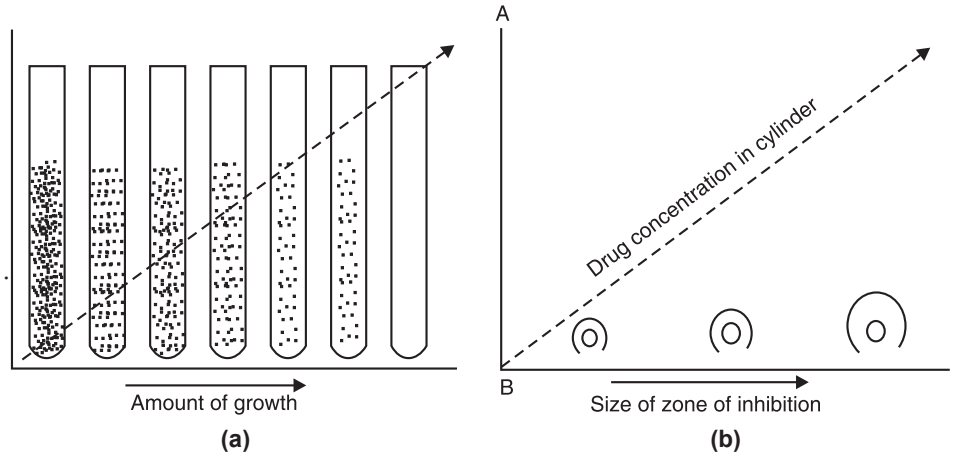


Fig. 2.29: Antimicrobial activity
(a) Turbidimetric assay (Tube dilution technique) (b) Cylinder-plate technique

(*iii*) **Metabolic response assays:** These assays are similar to turbidimetric assays except that, instead of measuring the effect of the fermentation product on the rate of growth or the total growth of the test organism, the effect on some metabolic reaction that test organisms carries out during growth are measured. A few metabolic reactions used for assay of this type are acid production, carbon dioxide evolution, oxygen absorption and enzyme dehydrogenase activity.

(*iv*) **Enzymatic assays:** Enzymatic assays are more specific. They are able to detect quantitatively in extremely minute amounts of fermentable products and differentiate between biologically active and inactive forms of a compound. The required enzymes of these assays are derived from a commercial source or from a microbial culture or from other enzyme source. The enzyme preparation is incubated with a sample of culture broth so as to cause some enzyme mediated changes in the fermentation product such as a partial decomposition with consequent formation of a measurable product.

For example, L-glutamic acid in a fermentation broth can be detected by adding cells of a pure culture of a strain of *Escherichia coli*, which contains the enzyme of glutamic acid decarboxylase. A little amount of toluene is also added to fermentation broth which facilities the release of enzyme from the cells of *E. coli*. The test is carried out at an acidic pH 5.0. One molecule of CO_2 is generally liberated from each molecule of glutamic acid. The CO_2 liberated due to the degradation of L-glutamic acid by the enzyme glutamic acid decarboxylase collects in the atmosphere of the test tube containing the fermentation broth as a gas. Its concentration is calculated with a manometer such as Warburg respirometer.

2.12.2 Separation and purification of the product: Downstream process in fermentation refers to the process of product recovery from the culture. The methods and techniques involved in downstream processing differ from one product to another. However, generally three steps such as separation of the cell from the culture, cell fractionation and

extraction and purification of the product. All these steps are necessary to recover intracellular products, while for extra-cellular products, cell fractionation may not be required as they are excreted into the medium and can be purified directly from culture broth.

(A) Cell separation: This is the first step in the downstream process irrespective of the product is intracellular or extracellular. After separation of cells, the cell free culture broth is taken for extraction of extracellular product, while the separated cells are taken for further extraction of intracellular products. Some of the methods adapted for separation are:

(B) Centrifugation: It employs centrifugal force to separate cells and promote accelerated settling of particles in a solid–liquid mixture. Centrifuge achieves separation by means of accelerated gravitational force by rapid rotation. This is used preferably for separation of bacteria from cultured broth. It employs centrifugal force to separate cells from culture broth, by promoting accelerated settling of particles in a solid-liquid mixture.

Centrifugation may be essential as filtration is slow and difficult. Further, the cells or other suspended matter must be obtained free of filter aids. For continuous separation, a high standard of hygiene is required. Cell aggregation made to flocculate so that they can be easily separated.

1. Cell aggregation can be achieved by neutralization of anionic charges,
2. Reduction of surface hydrophilicity,
3. Use of high molecular weight polymer bridges. Thus, cell mass can easily be separated by employing range of centrifuges.

(*a*) Basket centrifuge: It is used when solid content of the suspension is higher. It consists of a simple drum shaped basket or bowl usually rotating around vertical axis. The solid accumulates are compressed by the effect of centrifugal force. It is widely used simple machine made of tube whose length is several times its diameter rotating between bearings at each end. The bulk of solids will adhere on the walls of the bowl, while liquid exits at the top of centrifuge.

(*b*) Solid bowl scroll centrifuge: Solid bowl scroll centrifuge are not dewatered. The residual liquid drains out. When the rotation of the bowl is stopped, the solids are removed manually by scrapping or shoveling. However, unloading could be done without switching off the machine.

(*c*) Disc-bowl centrifuge: The simplest design of disc bowl centrifuge is a closed bowl containing disc stack with an arrangement to collect residual solids at the outer part of the bowl from where they have to be removed manually after stopping the rotation. The solids are discharged from the bowl through nozzles which are always open.

(*d*) Tubular-bowl centrifuge: Tubular bowl centrifuge is widely used, simple machine made up of a tube of length several times to its diameter rotating between bearings at each end. High centrifugal force act to separate the solids and liquids. The bulk of solids will adhere on the walls of the bowl, while the liquids exit at the top of centrifuge. The tubular bowl centrifuge has dewatering capacity but solids have to be removed manually by dismantling the system and scrapping. Foaming can be a problem unless the system includes special skimming or centripetal pumps (Fig. 2.30).

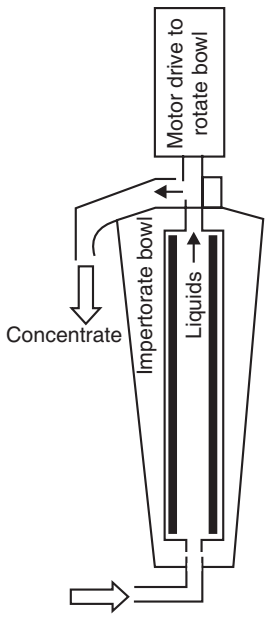


Fig. 2.30: Tubular bowl centrifuge

(*e*) Chamber bowl centrifuge: It has a number of tubular bowls arranged parallelly. The main bowl contains cylindrical inserts that divide the volume of the bowl into a series of annular chambers which operate in sequence. The feed enters the center of the bowl and the suspension passes through each chamber inturn at increasing distances from access (Fig. 2.31). The solids settle on the outer wall of each chamber and clarified liquid overflows from the longest diameter chamber. This centrifuge will help to separate solids based on the size. However, removal of solids requires halting system.

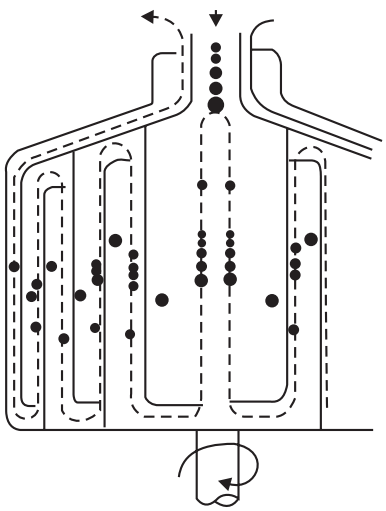


Fig. 2.31: Chamber bowl centrifuge

(*f*) **Imperforate basket centrifuge:** This is used when the solid content of the suspension is higher. Imperforate basket centrifuge consist of a simple drum shaped basket or bowl usually rotating around vertical axis. The solids accumulate and are compressed by the effect of centrifugal force but they are not dewatered and solids are removed by scrapping or shoveling. However, unloading could be done without switching off the machine.

(*g*) **Disc stack separator:** Disc stack separator is simple in design having a closed bowl containing the disc stack with an arrangement to collect residual solids at the outer part of the bowl from where they have to be removed manually after stopping rotation. The solids are discharged from the bowl through nozzles which are always open (Fig. 2.32).

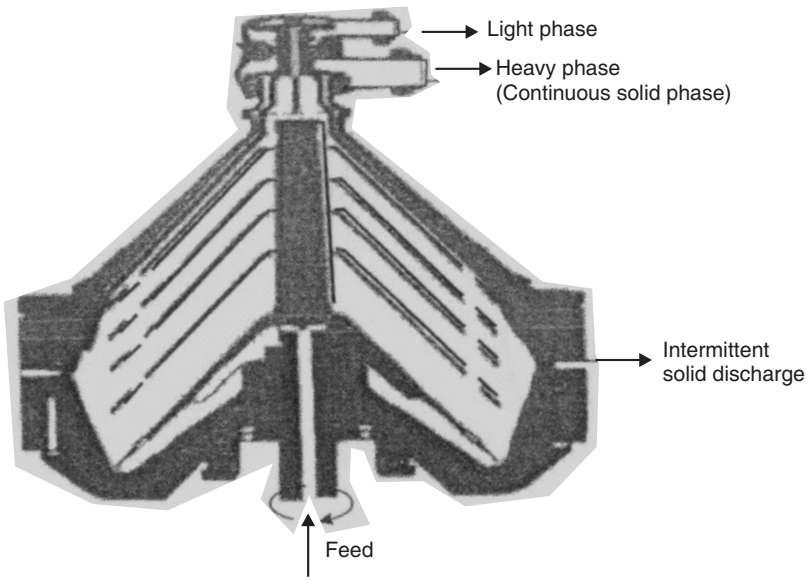


Fig. 2.32: Disc stack separator

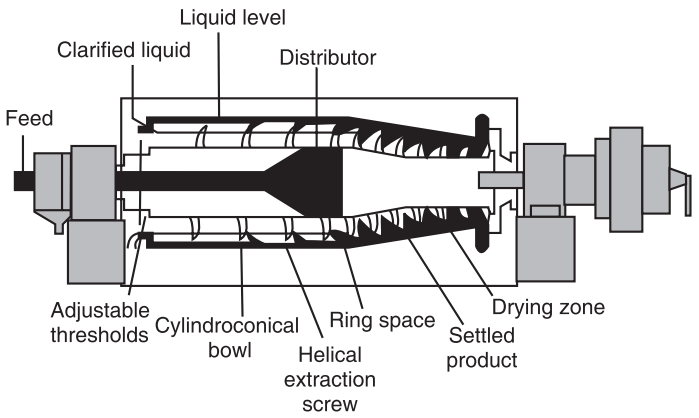


Fig. 2.33: Decanter centrifuge

(*h*) **Decanter centrifuge:** It basically consist of a horizontal cylindrical bowl rotating at a high speed with a helical extractions screw placed co-axially (Fig. 2.33). The screw perfectly fits the internal counter of the bowl only allowing clearance between the bowl and scroll. The differential speed between screw and scroll provides the conveying motion to collect and remove solids that accumulate at the bowl wall.

Filteration: It is one of the mechanical separations based on particle size which is subjected to a force that moves it to pass the retained particles. The particles suspended in the fluid which will pass through the apertures are retained and buildup into what is called a filter cake. Fine aperture used in filtration comprise fabric clothes, meshes and screens of plastics or metals. Sometime, a thin preliminary coat of cake or fine particles is put on the aperture prior to filteration. The preliminary coating known as pre-coat gives sufficiently fine pores on the filter.

The analysis of filteration is largely a question of studying the flow system. When the fluid passes through the filter, the normal flow rate will be less. The flow rate can be increased by giving pressure to the fluid. The filtration rate is greatly influenced by the given fluid pressure and the resistance of the membrane, thus we can use the following equation.

$$\text{Rate of filtration} = \frac{\text{Driving force}}{\text{Resistance}}$$

Resistance usually arises from the filter cloth mesh or bed as well as from the filter cake that accumulates. The basic requirement for equipment in filtration are:

1. Mechanical support for filter medium
2. Flow access to and fro from the filter medium
3. Precision for removing excess filter cake

Sometimes washing the filter cake may be necessary to remove traces of the solution. Pressure can be provided on the upstream side of the filter, or a vacuum can be drawn downstream, or both can be used to drive the wash fluid through. For this filtration some of the instruments used are plate and frame filter press, rotatory filter, cross flow filteration and air filters.

The rate of filteration depends on:

1. The properties of the filterate, particularly its viscosity and density.
2. The nature of solid particle, particularly their size and shape, the size distribution and packing characterstics
3. The solid : liquid ratio
4. The scale of operation
5. The need for batch or continous operation
6. The need for pressure or vacuum suction to ensure an adequate flow rate of the liquid.

In order to accelerate the process some measures such as use of filter aids. The most commonly used filter aids are:

1. A thin layer of kieselguhr is applied to the filter to form a pre-coat prior to broth filteration
2. The appropriate quantities of filter aid is mixed with the harvested broth
3. When vacuum drum filters are to be used which are fitted with advancing knife blades, a thick pre-coat filter is initially built upon the drum.

2.12.3 Cell disruption

Depending on the product different methods of cell disruption are employed. It is being achieved by any of the three methods (Fig. 2.34)

(*a*) Physico-chemical methods

(*b*) Mechanical methods

(*c*) Non-mechanical methods

(*A*) Physico-chemical methods: One of the physico-chemical methods used for breaking cell liquid shear high pressure homogenizer, solid shear–Pressure at –25°C with the help of Hughes press or X-press, agitation with abrasives such as beads, glass, alumina, ceramics, titanium compounds are employed to break cells. Freezing–thawing results in the formation of crystals such as l-glucosidase from *S.cerevisiae*. Ultrasonication with high frequency vibration at the tip leads to cavitation 20 kHz which results in the breakage of cell.

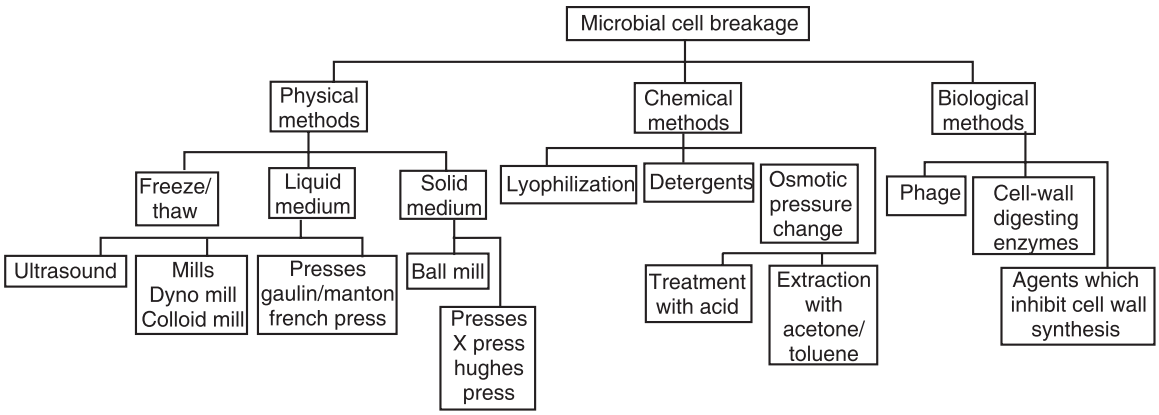


Fig. 2.34: Methods of opening cells to isolate microbial metabolites

(*B*) Mechanical methods:

(*i*) Homogenizers: It uses high pressure up to 500 lbs, pumps slurry through a restricted orifice valve. The cell disruption is followed by an instant expansion through a special exiting nozzle. Cell disruption is accomplished by three different mechanisms: 1. Impingement on the valve, high liquid shear in the orifice and sudden pressure drop upon discharge causes an explosion of the cell. The method is mainly employed for the release of the intracellular molecules. 2. Liquid shear high pressure homogenizer and solid shear pressure at –25°C with the help of titanium compounds are also employed in cell breakage. 3. Freeze thawing results in the formation of crystal releasing β-glucosidase from cells of *Saccharomyces cerevisiae*. Hughes pressure or X-press are also employed for cell disruption. Agitation with abrasives such as beads, glass, aluminium ceramics are also employed for this purpose.

(*ii*) Ball mills: These are also known as centrifugal or planetary mills or devices, used to rapidly grinding of material to colloidal fineness (1 micrometer or below) by developing high grinding energy via centrifugal and/or planetary action. The grinding is carried by pounding and rolling a charge of steel or ceramic balls within the cylinder. Cylinder rotates at a relatively low speed, allowing the balls to cascade through the mill base thus grinding or dispersing the materials.

(*iii*) **Blenders (high speed or warring) :** The french press or even centrifugation in case of weak cell walls also disrupt the cells.

Mechanical disruption of cells has following drawbacks:

1. Because cells break completely, all intracellular materials are released together. Therefore, separation of different cell components will be an additional separation process.
2. Nucleic acids may increase viscosity which makes the solution difficult to clarify.
3. It exposes cell and its products to harsh conditions.

(*i*) **Ultrasonic disruption:** Another widely applied method in which cell membranes are reptured and method is called cell lysis. In this method high frequency sound produced electronically are transported through a metallic tip to an appropriately concentrated cellular suspension. It is not only costly but also used at laboratory scale that too have less resistant cell wall such as bacteria and fungi. Mechanical disruption of cell membrane have some drawbacks:

1. Cell break completely and all intracellular materials are released together. This makes the separation of desired product difficult.
2. The cell debris often consists of cell fragments, which makes the solution difficult to clarify.
3. Mechanical methods expose the cells and extract product to harsh conditions.
4. Some of the proteins get denatured by heat generated unless the device is sufficiently cooled.

Sonicator is an ultrasonic disruption device used commonly**.**

(*C*) **Non-mechanical methods:** Another way to disrupt the cells is permeabilization. This can be achieved by one of the following methods.

(*i*) **Chemical permeabilization:** Many chemical methods are being employed to extract intracellular components from microorganisms by permeabilizing the outer wall barriers. Chemical permeabilization is achieved with organic solvents that can create canals through cell membranes. Eg. toluene, ether, phenylethyl alcohol, dimethyl sulphoxide, benzene, methanol and chloroform. Chemical permeabilization can also be achieved with antibiotics, thiamine, surfactants, chaotropic agents and chelates such as EDTA.

(*ii*) **Chaotropic agents:** Chaotropic agents such as urea and guanidine are capable of brining some hydrophobic compounds into aqueous solutions. This is achieved by disrupting water structure, making it less hydrophilic and weakening the hydrophobic interactions among the solute molecules.

(*iii*) **Detergents:** Disrupts lipoprotein of cell membrane and release intracellular components of *Klebsiella pneumoniae*. Na-dodecyl sulphate, Na-laurylsulphate, quaternary ammonium compounds, TritonX-100 and Guanidine –HCl are widely used for cell disruption.

(*iv*) **Osmotic shock:** Extract of luciferase from *Photobacterium fischeri* is being achieved with the help of osmatic lysis.

(*v*) **Alkali treatment:** Alkali treatment upto pH 11.5 to 12.5 for 20–30 min. will result in the liberation of L-asparaginase.

(*vi*) **Enzyme treatment:** Lysozyme from leucocytes of *Streptomyces spp. Micromonospora spp., Penicillium spp., Trichoderma spp.* and snails are mostly used in cell disruption.

(*vii*) **Mechanical permeabilization:** Exposing cells suddenly to osmotic shock results in the mechanical injury. In this method, cells are exposed to isotonic conditions and subsequently to hypotonic condition resulting in the damage to cell membrane due to increased osmotic pressure. Enzymes released by this method are believed to be periplasmic or atleast located near the surface of the cell.

(*viii*) **Enzymatic permeabilization:** This method is mostly restricted to the release of periplasmic or surface enzyme. EDTA is mostly used for enzymatic permeabilized of β-glucanase, protease and mannase.

(*ix*) **Other techniques:** Basic enzymes or cationic polysaccharides can permeabilize yeast cells. Similarly, mammalian cells can be permeabilized by streptolysin and even viruses. Electrical discharges have also been demonstrated to permeabilize mammalian cells in order to study secretion by exocytosis**.**

(*D*) **Separation of Products:** The separation of products is an important step in bioprocessing. Separation of extracellular products is quite easier than that of intracellular products. In extracellular products, the cells are separated from the culture medium and processed to recover the products in cell free culture medium. In case of intra-cellular products, the cells are fractionated, cell debris are removed, filtration is carried out and the products are recovered from the debris free medium. The general recovery methods are as follows.

(*i*) **Liquid-solid extraction:** In liquid-solid extraction a solvent is added to the solid. Insoluble material can be separated by gravity or vacuum filtration and soluble materials are extracted into the solvent. Solvents of increasing polarity can be used to separate complex mixtures into groups. The solvent can be evaporated to recover the solute in powder or crystalline form.

Table 2.16: Advantages and disadvantages in liquid-liquid extraction:

Unit operation	Advantages	Disadvantages
Mixer settler	Efficient, low head room induces good contacting, can handle any number of stages	Large floor, High setup costs, High operation costs
Columns (without agitation)	Small investment, low operating costs	High head room, Difficult to scale-up from lab
Columns (with agitation)	Good dispersion, low investment, can handle any number of stages	Less efficient than mixer settler, Difficult to separate small density differences, Does not tolerate high flow rates
Centrifugal extraction	Can separate small density differences, short holding time, small liquid inventory	High set up costs, High operating and maintenance costs, can handle many stages.

(*ii*) **Liquid–liquid extraction:** This extraction process is to separate component based on chemical differences rather than difference in physical properties. The basic principle of extraction involves the contacting of a solution with an immiscible solvent. The solvent is also soluble with a specific solute contained in a solution. Two phases are formed after the addition of the solvent due to the differences in densities. A solvent should be so chosen that the solute in the solution has more affinity towards the added solvent. Therefore, mass transfer of the solute from the solution to the solvent occurs. Further, separation of the extracted solute and solvent will be necessary. Some of the advantages and disadvantages in liquid-liquid extraction process are listed in table 2.16**.**

Some of the physical properties such as temperature, pressure, activity, coefficiencey and viscosity are reported to influence the liquid-liquid extraction process.

(*iii*) **Precipitation:** It is widely used for the product recovery from bio-molecules especially proteins. Precipitation is usually induced by the addition of a salt or an organic solvent or by changing the pH of the solution. The most common type of precipitation for proteins is induced by salt such as ammonium and sodium sulphate. Precipitation can also be achieved by addition of organic solvent, change of pH or by addition of non-ionic polymer such as polyethyleneglycol (PEG) or metal ion. Protein binding dyes bind to and precipitate certain classes of proteins. Poly-electrolytes are used in precipitation of a range of components in addition to their use in cell aggregation.

(*a*) **Direct precipitation:** This is an uncommon method. Eg. Recovery of antibiotic cycloserine by adding $AgNO_3$ which result in the formation of insoluble silver salt. This is soon dispersed off due to its loss and dry silver salt may explode. An addition of acetone to recover protease can be used in detergents.

(*b*) **Solvent precipitation:** Fermentative products can be recovered with the help of solvents like methanol, ethanol, propanol and acetone. A good method of precipitating the substances is to add excess of solvent to an aqueous solution**.**

(*c*) **Salt Precipitation:** Ammonium sulphate precipitation is often used on the first purification and concentrating procedure. The protein can be stored in ammonium sulphate which reduces bacterial contamination, denaturation and proteolysis. Ammonium sulphate takes up the water and thus, exposes hydrophobic sites on the protein.

(*iv*) **Dialysis:** It is a process of movement of molecules by diffusion from high concentration to low concentration through semipermeable membrane (Fig. 2.35). Only those molecules that are small enough to pass through the membrane pores moves through the membrane and reach equilibrium with the entire volume of the solution in the system. Once the equilibrium is reached, there is no further net movement of the substance because molecules keep moving through the pores in and only of the dialysis unit. Some of the factors such as dialysis buffer volume, buffer composition changes, time, temperature and particles are reported to influence the dialysis rate**.**

(*v*) **Reverse osmosis:** It is pressure driving membrane separation process that separates dissolved and suspended substances from water. The membrane acts as a selective barrier removing unwanted substances, such as salts.

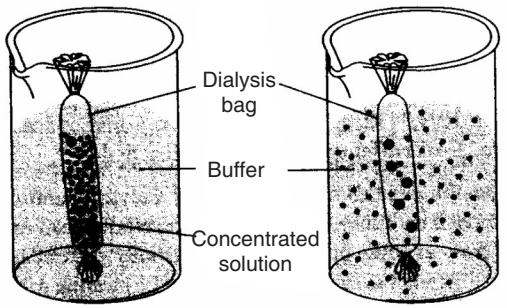


Fig. 2.35: Dialysis

(*vi*) **Adsorption:** The process of binding of molecules or particles to the surface is called adsorption. The binding is usually weak and reversible. Compounds that have color, taste and odor tend to bind strongly. Compounds that contain chromogenic groups are strongly adsorbed on activated carbon. Decolorization can be achieved efficiently by adsorption and with negligible loss of other materials.

(*vii*) **Chromatography:** In a chemical or bioprocessing industry to separate and purify the product is a necessary step and it can be done by a special process such as chromatography. It can separate complex mixture with great precision. It can also separate delicate products. The column chromatography is most extensively used (Fig. 2.26). Gas chromatography, liquid chromatography, reverse phase chromatography, HPLC, ion exchange and affinity chromatography are being used in separation of fermentative products.

(*E*) **Finishing the product:** Finishing is the final step in the downstream processing. After purification the products are dried and crystallized.

(*i*) **Drying and evaporation:** Removal of solvents from purified wet products usually carried out by evaporation and drying. The purpose, principle of evaporation and drying are the same. Based on the nature of the product, drying or evaporation is employed to remove solvents from the product.

The basic factors that affect the rate of evaporation and drying are:

1. Rate at which heat can be transferred to the liquid,
2. Quantity of heat required to evaporate each kg of water,
3. Maximum allowable temperature of the liquid,
4. Pressure at which evaporation takes place,
5. Changes to the food materials during the course of evaporation.

(*a*) **Evaporators:** An evaporator has two principle functions such as exchange of heat and separation of vapors that is formed from liquid. Some of the factors which influence evaporation include:

1. Maximum allowable temperature which may be substantially below 100°C.
2. Promotion of circulation of liquid across heat transfer surfaces to attain reasonably high heat transfer co-efficient and prevent any local over heating.
3. Viscosity of the fluid will often increase substantially as the concentration of the dissolved materials increases.

4. Tendency of foaming makes separation of liquid and vapors difficult.

A typical evaporator is made up of three functional sections:

1. Heat exchangers.
2. Evaporating section where liquid boils and evaporates.
3. Separator in which vapours leaves the liquid and passes to the condenser.

Several types of evaporators are :

(*i*) **Open pans:** liquid is boiled in an open pan.

(*ii*) **Horizontal tube evaporators:** Development of open pan evaporator in which the pan is fixed in a vertical cylinder. The heating tubes are arranged in a horizontal bundle immersed in the liquid at the bottom of the cylinder. Liquid circulation is poor in this type of evaporator.

(*iii*) **Vertical tube evaporators:** These can give good heat transfer. The standard evaporator is an example of this type. Recirculation of liquid is through a large down corner. The liquid rises through vertical tubes of 5–8 cm in diameter, boil in the space just above the upper tube plate and recirculate through down corners (Fig. 2.36). The length to the diameter of the tubes is 15 : 1.

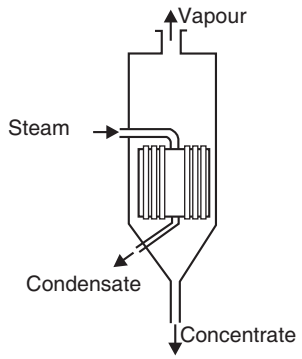


Fig. 2.36: Vertical tube evaporator (Basket type)

(*iv*) **Plate evaporators:** The plate heat exchanges may be adopted for use as an evaporator. The spacing between the plates can be increased and appropriate passages are provided so that much larger volume of vapours can be accomplished. Plate evaporators can provide good heat transfer and they are easy to clean.

(*v*) **Long tube evaporators:** Tall cylinder vertical tubes may be used in evaporators (Fig. 2.37). The tubes which have a length to diameter ratio of 100 : 1 pass vertically upward inside the steam chest. The liquid may either pass down through the tubes called a falling film evaporator or carried up by the evaporating liquor, in which case it is called a climbing film evaporator. Evaporation occurs on the walls of the tubes. Because circulation rates are high and the surface films are thin, good concentrations of heat sensitive liquids can be obtained due to high heat transfer rates and short heating times.

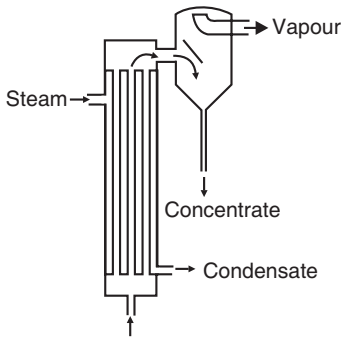


Fig. 2.37: Long-tube evaporator

Generally the liquid is not recirculated and if sufficient evaporation does not occur in one pass, the liquid is fed to another pass. In the climbing film evaporator, the liquid boils inside the tube, slugs of vapour form and this vapour carries up the remaining liquid that continues to boil. Tube diameter may be 2.5 to 5 cm and contact time may be as low as 5–10 seconds. The overall heat transfer coefficients may be up to 5 times as great franc heated surface immersed in a boiling liquid. In the falling film type the diameters are rather greater about 8 cm and these are especially suitable for viscous liquids.

(*vi*) Forced circulation evaporators: Heat transfer coefficient from condensing steam are usually high so the major resistance to heat flow in an evaporator, is in the liquid form. Tubes are generally made of metals with high thermal conductivity though scale formation may occur on the tubes which may reduce tube conductance (Fig. 2.38).

Liquid film coefficient can be increased by improving circulation of the liquid and by increasing its velocity of flow across the heating surfaces. Pumps or impellors can be fitted in the liquid circuit to achieve this. Using pump circulation, the heat exchange surface can be diverted from the boiling and separating section of the evaporator. Forced circulation evaporator is used partially with viscous liquids.

(*b*) Dryers: In an diversified and extensive industry as the bioprocessing industry, a number of different dryers can be used. However, the principles of drying may be applied to any type of dryers.

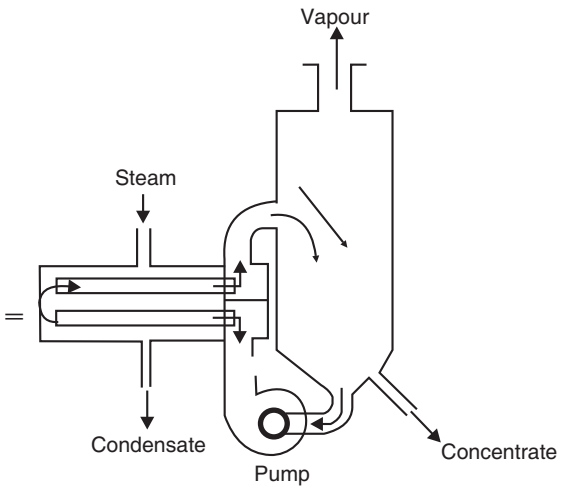


Fig. 2.38: Forced–circulation evaporator

Various types of dryers are:

(*i*) **Tray dryer:** In this type of dryer (Fig. 2.39) the food material spread out quite thinly on trays and heat is supplied by air current sweeping across the trays by conduction from heated trays and shelves or by radiation for heated surfaces.

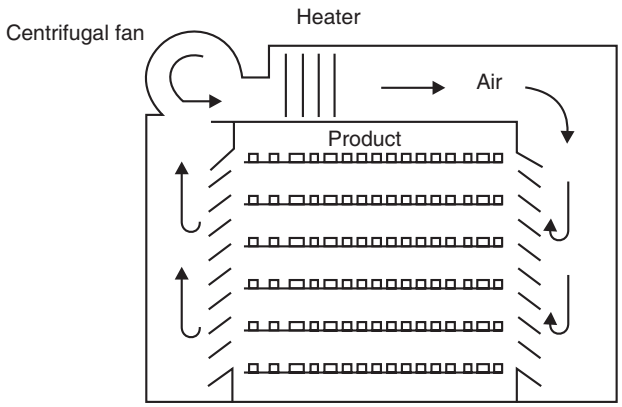


Fig. 2.39: Tray dryer

(*ii*) **Tunnel dryer:** These have diversified from tray dryers. In tunnel dryers, the trays move through a tunnel on trolleys where heat is applied and vapours removed. Usually air current is used in tunnel drying and the materials moves through the dryer either parallel or counter current to the air flow.

(*iii*) **Drum or roller dryer:** In drum dryer, (Fig. 2.40) the food material is spread over the surface of a drum. Drum drying is an example of conduction drying.

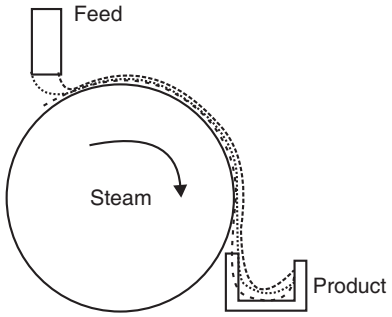


Fig. 2.40: Roller dryer

(*iv*) **Fluidized bed dryer:** In this dryer the product is suspended against gravity in upward flowing air steam. The horizontal flow helps to convey the product through the dryer. Heat is transferred from air to the product mostly by convection (Fig. 2.41).

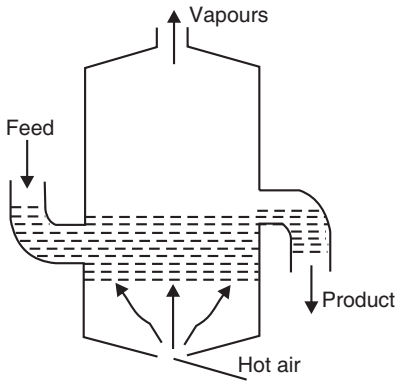


Fig. 2.41: Fluidized bed dryer

(*v*) **Spray dryer:** In this type of dryer liquid or fine solid particles are spread into a current of heated air. The air current and materials may move in parallel or counter current (Fig. 2.42). Drying occurs very rapidly hence this process is very useful for the material that is delicate to heat. The dryer body is so large that particles can settle as they dry without touching the walls. Commercial dryers can be of very large dimensions (10 m diameter and 20 m height).

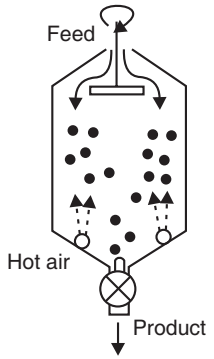


Fig. 2.42: Spray dryer

(*vi*) **Prematic dryer:** In this dryer solid product is conveyed rapidly in an air stream. The velocity and turbulence of stream maintain the particles suspension. Heated air accomplishes drying. In some equipment some form of classifying device is included which separates dried and moist materials which are recirculated for further drying.

(*vii*) **Trough dryer:** The product is placed in trough shaped conveyor belt made out of mesh and air is blown through the bed of product. The movement of the conveyor continuously turn over the material exposing fresh surfaces to hot air.

(*viii*) **Bin dryer:** The product is placed in a bin with a perforated bottom through which warm air is blown upwards which passes through the material and dry it up.

(*ix*) **Belt dryer:** The product is spread as thin layer over a mesh or solid belt and air is passed through or over the material. In some dryer the belt would move and in others the material is transported by scrapper.

(*x*) **Rotary dryer:** In this type of dryer (Fig. 2.43) product is filled in a horizontally inclined cylinder and heated either by air flow through the cylinder or by conduction of heat from the cylinder wall. In some cases cylinder rotates and in others a paddle rotates within the cylinder conveying the product through.

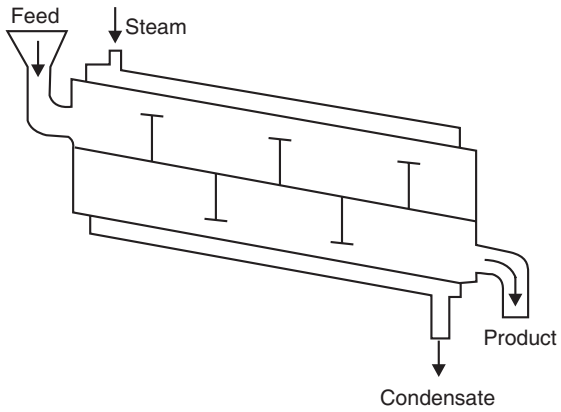


Fig. 2.43: Rotary dryer

(*xi*) **Batch vacuum dryers:** Batch vacuum dryer are similar to tray dryers except these are operated under vacuum and heat transfer is largely by conduction or radiation. The trays are enclosed in a large cabinet in which evacuated water vapour produced is generally condensed so that the vacuum pumps have to deal with only non-condensable gases. Another type consists of an evacuated chamber containing a roller dryer.

(*c*) **Lyophilization:** Freeze-drying technically known as lyophilization which is a process of sublimation in which water molecules in a solid phase specimen are directly converted to free water molecules in vapour phase. The free water molecules are then trapped and removed. Porous dried specimens can easily be rehydrated. The purpose of freeze-drying is to preserve a specimen, the temperature and the drying time used to freeze and sublime differs from one specimen to another. Since lyophilization is the most complex and expensive form of drying its use is restricted to delicate heat sensitive product. Substances that are not usually damaged by freezing can be lyophilized. Many microorganisms and protein survive lyophilization. Hence, lyophilization is favoured method for drying vaccines, pharmaceuticals, blood fractions and diagnostics.

2.13 CONTAINMENT AND ENVIRONMENTAL SAFETY

Safety of employees involved in fermentation industries should be of primary concern. They should not be exposed to potential pathogens and allergens and certain genetically modified microorganisms that are employed in fermentation industries. The classification of microorganisms in terms of danger to humans divides them into four categories which are accepted by World Health Organization (WHO) and they are précised in table 2.17.

Table 2.17: Classification of Biohazardous Agents by Risk Group

Risk Group	Definition	Examples
Risk Group 1 (RG1)	Agents that are not associated with disease in healthy adult humans.	*Bacillus subtilis*, E. coli K-12, Adeno Associated Virus types 1-4
Risk Group 2 (RG2)	Agents that are associated with human disease which are rarely serious and for which preventive or therapeutic interventions are often available.	Forty bacterial genera, including *E. coli* O157:H7; 12 fungal genera; 31 parasites; common viruses in 18 families
Risk Group 3 (RG3)	Agents that are associated with serious or lethal human disease for which preventive therapeutic interventions may be available (high individual risk but low community risk).	Nine types of bacterial agents, including Ricksettsia; 2 fungi; several viruses in 8 families including Hanta, HIV; CJD and other prions.
Risk Group 4 (RG4)	Agents that are likely to cause serious or lethal human disease for which preventive or therapeutic interventions are not usually available (high individual risk and high community risk).	Extremely severe and emerging viruses in 6 families, including Lassa, Ebola, undefined hemorrhagic fever agents.

The cultivation of such microorganisms necessitates the implementation of different levels of barrier/containment systems depending upon the fermentation process and classification of producer organism. Traditional and many established poorer fermentation processes require a minimum level of containment and sterility whereas cultivation of GMMs require high levels of both containment and sterility.

Primary containment barriers protect the personnel and immediate processing facility by preventing the escape of microorganisms from the fermentor or as aerosol generated during downstream processing equipment. This protection is provided by implementing good microbiological techniques and the use of appropriate safety equipment. Some of the problematic areas in large scale fermentations are the stirrer shaft and its seals. Usually two to three seals are now required so that if one breaks another will hold. Also all valves, O rings, taps and pumps must be regularly checked. Secondary barriers involve the use of protective clothing, regular medical supervision and vaccination of laboratory and manufacturing personnel. Further, secondary containment barriers entails specific design criteria for manufacturing plants and laboratories. They include the use of positive and negative air pressure, High Efficiency Particulate Air (HEPA) filters, air locks and changing room for operating personnel along with specific protocols for the sterilization of waste before it leaves the site and its safe disposal.

In recent times attempts are being made to standardize regulations to facilitate international free trade of products.

Table 2.18: Classification of microorganisms on the basis of hazard

	Containment required
Risk group 1 **Low individual and community risk.** A microorganism that is unlikely to cause human disease or animal disease of veterinary importance	Good industrial large scale practices (GILSP)
Risk group 2 **Moderate individual risk, limited community risk.** A pathogen that can cause human disease or animal disease but is unlikely to be a serious hazard to laboratory workers, the community, livestock or the environment. Laboratory exposures may cause serious infection, but effective treatment and preventive measures are available and the risk of spread is limited, e.g. *Salmonella* food poisoning. Vaccines and antibiotics are available.	Level 1 containment
Risk group 3 **High individual risk, low community risk.** A pathogen that usually causes serious human disease, is not contagious. Prophylaxis and treatment may be available	Level 2 containment
Risk group 4 **High individual and community risk.** A pathogen that usually causes serious human or animal disease and may be contagious. No effective prophylaxis or treatment is available, e.g. Ebola virus	Level 3 containment

Table 2.19: Safety precautions required for different levels of containment

Containment category				
Non genetically modified microorganisms	GILSP	1	2	3
Genetically modified microorganisms	B1	B2	B3	B4
Procedures written code of practice biosafety manual restricted access medical surveillance	 + - - -	 + + + +	 + + + +	 + + + +
Primary containment use of closed systems for viable microorganisms treatment of exhaust gases removal of material, products and effluents penetration of the closed system, e.g. agitator shaft, monitoring devices, etc.	 - - - -	 Minimization of release Minimization of release Minimization of release Minimization of release	 Prevention of release Prevention of release Prevention of release Prevention of release	 Prevention of release Prevention of release Prevention of release Prevention of release

contd...

	Containment category			
Secondary containment				
disinfection facility	-	+	+	+
emergency shower facility	-	-	+	+
airlock + compulsory shower	-	-	-	+
decontaminated effluents	-	-	+	+
HEPA filters in air ducts	-	-	+	+
area hermetically sealable	-	-	-	+
controlled negative pressure	-	-	-	+

Before any new industrial process can start operating, most countries now require that an assessment of risk analysis be carried out. This is specially required for processes using GMMS. Part of this risk assessment depends on whether the microorganism contains foreign DNA. If the organisms is non-GMMS, it can be placed in one of four categories. The specific categorization of microorganisms depends upon its pathgenicity, virulence, the infective dose necessary to cause disease, the root of infection, the level of incidence in the community and whether vectors are needed.

Risk group I microorganisms are the least likely to cause a problem and for industrial use simply require good industrial large-scale practices (GILSP). Most industrial microorganisms are in these categories, the use of microorganisms with in risk groups 2, 3 and 4 must be more strictly controlled necessitating containment level 1, 2 and 3 respectively (table 2.18).

Large-scale fermentation of GMMS has specific potential problems, they should escape from the fermentor. For example, they often contain antibiotic resistance selection markers that could pose problems when they will be released into the environment.

GMMS classified under high risk groups must have secondary or integral containment of equipment particularly where aerosols are generated. Inspectors from the regulation agencies must give permission for a process to start and for its continued operation (table 2.19). They inspect the manufacturing plant and laboratories on a regular basis to determine whether the manufacturer is operating in a state of control and in compliance with the laws and regulations.

2.14 FERMENTATION ECONOMICS

The ability to produce a fermentation product in large quantity is only a part of the requirement for a successful fermentation process. If a fermentation process is to yield a product at a competitive price, the chosen microorganism give the desired end product in predicable and economically adequate quantities. Therefore, before establishing a fermentation plant, one has to take into consideration the following facts:

2.14.1 **Market potential**: The product produced should have wide acceptability, if it is a new product it should be superior and withstand market competition. The utility of the product should be increasing rather than reducing. If the competition exists with already existing compound of chemical synthesis, fermentation product should have desirable characteristics.

2.14.2 **Fermentation and product recovery costs:** The economic position of a fermentation product is closely tied to the costs associated with its production and distribution. One of the important compounds of this is that raw material should be as cheap as possible and renewable and can be utilized efficiently. A search for possible alternative materials might be made even when a process is in operation. There should be a saving in labour

whenever possible and automation should be used wherever it is feasible. Further, duration of incubation period is another crucial factor which not only results decrease in raw material requirement as well as labour requirement. This may result increased production in a specified time. When a batch process is operated the growth cycle should be as short as possible to obtain the higher yield of product and allow for maximum utilization of equipment. To achieve this objective, it may be possible to use fed-batch culture. The resistance to contamination is another advantage to keep the cost low. If the risk of contamination is more, one has to take care to prevent contamination which may add to the cost of production as sterilization steps may add to the cost. The ability of fermentation to produce high yields, allow good product, simple recovery and purification is a prime consideration in fermentation economics. If the product is pure this may be more acceptable and purification step can be saved. The effluent discharge should be kept minimum, simple and rapid as much possible. The requirement of heat and power will affect the economics of product formation and they should be used efficiently. The space requirements and capital investment should be minimum. Costs attributable to waste disposal range from a minimal to a major factor in fermentation product costs of all high yielding strain of microorganism plays an important role in viability of a fermentation. Patent position is also a vital factor to be taken into consideration. Possessing a patent will eliminate the market competition and the chances of its success increases considerably. Reappraisal of the process is required, in order to maintain the reduced process cost. Research and development division is required for new and innovative methods, cheaper raw material as well as improvement of strain, which adds to the cost of production. Research and development division is there will be a need of finding out new and innovative methods, cheaper raw material, improvement of strain are needed for which. Further, there should be a saving in labour whenever possible and automation should be used wherever it is feasible. The capital investment in the fermenter and ancillary equipment should be confined to a minimum, provided that the equipment is reliable and may be used in a range of fermentation processes. Space requirements should be kept to a minimum but there should be some allowance for potential expansion in a production capacity.

2.14.3 **Process appraisal:** An appraisal of the economic potential for the fermentation process is required that all of the above considerations be evaluated both under present and future market conditions. This evaluation should be made as early as possible during process development as well as at the time of deciding whether to enter into the market with the fermentation product. The process should be evaluated at later dates during commercial production. For these evaluations it is necessary to consider all present and future costs, profits desired and a selling price for the product that the market will bear. All of these points are used to decide whether the fermentation product can be produced and sold at an acceptable level of profit; or to license; or sell the patent and know how to others who can produce and market the product at an acceptable profit level. All the above aspects must comply with safety guidelines and regulations.

As great deal of money often is at stake in these decisions, a fermentation industry should carefully consider all of the alternative possibilities before it is started.

2.15 COMPUTER APPLICATIONS IN FERMENTATION

Advances in computer technology facilitated their application in fermentation technology. The first description of computer controlled fermentation was given in 1969 by **Craysan** and **Yanashita (1972)**. Three distinct areas of computer function were recognized by **Nyiri**. They are**:**

1. **Logging of process data:** Performed by a data acquisition system comprising of both hardware and software components.
2. **Data analysis:** Data obtained by computer can be analyzed by series of selected mathematical equations. After analysis the results can be printed or stored in a data bank for further use.
3. **Process control:** Computer guided signals via interface to pumps, valves, switches etc.

The first international conference on computer application in fermentation technology was held in 1973 in Dijan, France.

In fermentation process, measurable process output variables fall into three major categories.

1. Environmental variables which include temperature, pH, dissolved oxygen, speed, aeration rate and number of concentrations.
2. Physiological variables which include products of metabolism and state of metabolism.
3. Cell culture composition.

The computer can be under direct control of the computer software and is called Direct Digital Control or DDC. When this is done, separate controller units are not needed. For control of temperature, pH, foam control etc., the sensors are directly interfaced to a computer. The alternative approach is to use independent controllers to manage all control functions of a fermenter and the computer communicates with the controller only to exchange information. This is termed supervisory set-point control or SSC.

In computer control of fermentation process, the most advanced level of control is concerned with process optimization. There is a need for suitable on-line sensors to monitor the process continuously. It is also important to develop a mathematical model that adequately describes the dynamic behavior of a process. This approach with appropriate on-line sensors and suitable programme has been used to optimize baker's yeast production, industrial antibiotic process and lactic acid production at optimum level inside the reactor. A technical and economical appraisal of the venture is important in computer control strategies of industrial fermentative products.

2.16 STRAIN IMPROVEMENT IN INDUSTRIAL MICROORGANISMS

In industrial fermentations, microorganisms are important. By process of primary and secondary screening, a strain, which is more efficient and conditions under which it produces maximum will be determined. In order to make the fermentation economically viable, constant attempts are to be made to increase yield either by increasing the efficiency of organism or simplify the procedure either in the upstream or downstream processes. Thus, selection of organism for fermentation plant is of paramount importance. A microorganism to be used in the fermentation should fulfil the following conditions (Bull *et al*., 1979):

1. Able to metabolise inexpensive substrates.
2. Possess tolerance to high concentration of carbon and nitrogen sources.
3. Resistance to bacteriophages.
4. Able to produce less foam during fermentation.
5. Possess reduced oxygen needs with lower viscosity of culture broth in that oxygenation is less of a problem.
6. Requires shorter fermentation time.
7. Do not produce undesirable pigments.
8. Able to synthesize one compound as the main product.
9. Make possible a simplified process for product recovery.
10. Possess genetic stability, amenable to genetic manipulation.
11. Ready break up of cell, if the product is intracellular.

12. Easy for harvesting from the fermentation.
13. Safety, non-pathogenic and should not produce toxic agents unless, this is a target product.
14. Limited or no need for vitamins and additional growth factors.

In the early phase of development of fermentation selection of efficient microorganism will be attempted. The attempt mainly will be aimed to choose substratum, which is cheaper, renewable and less prone to contamination, shorter incubation period or amenable to simply downstream process. Thus, by repeated attempts, it may be possible to harness the organism for maximum product formation. Once that is achieved further increase in product formation or to make the fermentation sustainable can be achieved by one of the following methods.

1. Selection of mutants
2. Regulation
3. Genetic engineering
4. Improvement of industrial strains by modifying properties other than the product yield.

Table 2.20: Mutagens employed for strain development.

Mutagen	Mutation induced	Impact on DNA	Relative effect
Radiation			
Ionizing radiation			
1. X-rays, gamma rays	Single or double strand breakage of DNA	Deletions, structural changes	High
Short wavelengths			
2. Ultraviolet rays	Pyrimidine dimerization and cross-links in DNA	Transversion, deletion, frameshift, GC → AT transitions	Medium
Chemicals			
Base analogs			
3. 5-Chlorouracil, 5-bromouracil	Faulty base pairing	AT → GC, GC → AT transition	Low
4. 2-Aminopurine	Errors in DNA replication	---	Low
Deaminating agents			
5. Hydroxylamine (NH_2OH)	Deamination of cytosine	GC → AT transition	Low
6. Nitrous acid (HNO_2)	Deamination of A, C, and G	Bi-directional translation, deletion, AT → GC and / or GC → AT transition	Medium
Alkylating agents			
7. NTG (*N*-methyl-*N'*-nitro-*N*-nitrosoguanidine)	Methylation, high pH	GC → AT transition	High
8. EMS (ethyl methanesulfonate)	Alkylation of C and A	GC→AT transition	High
9. Mustrards, di -(2-chloroethyl sulfide)			
Intercalating agents	Alkylation of C and A	GC→ AT transition	High
10. Ethidium bromide, acridine dyes	Intercalation between two base pairs	Frameshift, loss of plasmids, microdeletions	Low
Biological			
11. Phage, plasmid, DNA transposons	Base substitution, breakage	Deletion, duplication, insertion	High

2.16.1 Selection of Mutants: Mutation is an important part of strain improvement programme. The factor which causes a mutation is called as a mutagen. Different categories of radiations like ultraviolet rays, X-rays, γ-rays, and fast neutrons constitute physical mutagens, while nitrogen and mustard gas, methylamine, 2-mercaptopurine, 2-oxetanone, azaserine, streptonigrin, 8-ethoxy caffeine, diethyl sulphate and methyl ethane sulphonate are widely used chemical mutagens (table 2.21). However, they differ widely in their mechanism of action. Therefore, they produce diverse results and must be used under particular conditions. All these mutagens act by producing alterations in deoxyribose nucleic acid (DNA). Generally the first mutagen to be used is UV-light with a lethal level of 90-99%. This treatment may yield a number of isolates with a higher productivity and is repeated 2-3 times. The next treatment is carried out using chemical mutagen like ethylamine followed by UV treatment again.

Experience plays a crucial role in the selection of mutagens, mode of their application and the choice of strain. Mutagen with a more or less defined mode of action are generally preferred over agents with complex effects. Mild mutagens cause only minor changes, whereas potent mutagens induce marked changes which may result in the synthesis of desired product in greater quantities. The change in the productivity of strains may be either positive or negative. Positive changes may range from 5% to 25% or even higher increase in the desired product.

A cell or spore suspension (10^7/ml) in a petri dish at a distance of 10–12 cm is exposed to short wavelength UV light (253.7 nm) for a time ranging 30 seconds to 20 min, while agitating the contents. The death rate is usually 90–99.9% on plating of such suspension, isolated colonies appear which can be picked and transferred to a nutrient agar tube. They will be tested for their improvement in different parameters. Repeated exposure of culture to UV–rays may develop

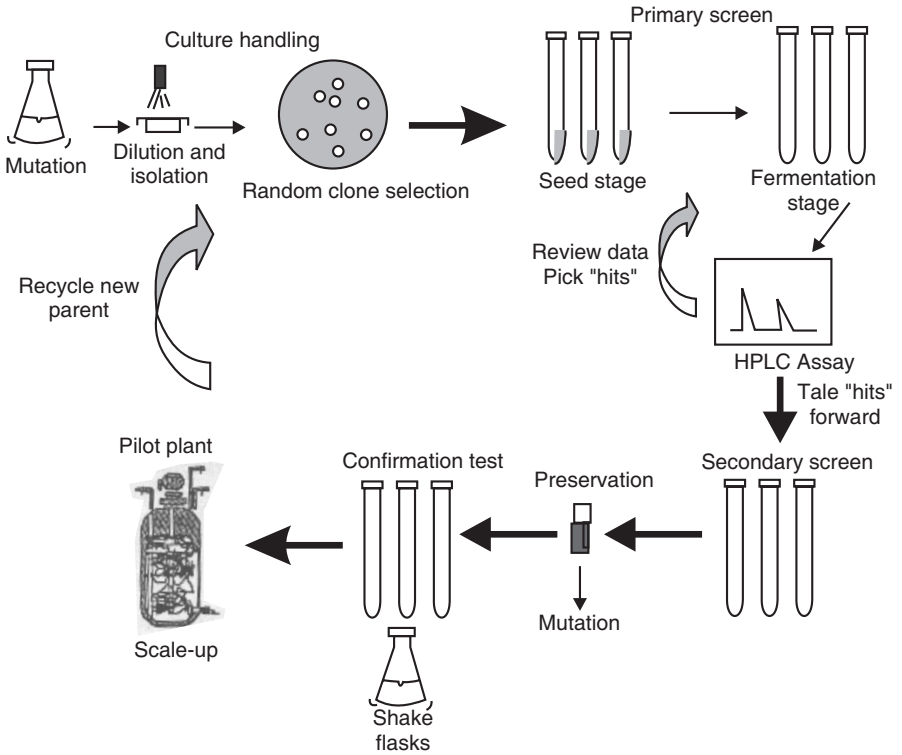


Fig. 2.44: Typical steps in mutation and random strain selection process

resistance and needs increased dose. In such conditions, exposure to day light increases the susceptibility level and the phenomenon is called photoreactivation. X-rays and γ-rays are more powerful mutagens. In still other instances combined treatment of physical with chemical mutagens is most likely to induce more mutations.

In case of chemical mutagenesis, cell or spore suspension is treated with different concentrations of chemical mutagens for a fixed time. Spores thus treated will be plated directly after dilution as required. The colonies thus obtained will be cultured and tested for improvement of the product. Several promising isolates are further evaluated in laboratory, then in pilot plant fermenters and the strain yielding best results is selected for the industrial production (Fig. 2.44). Though isolation of prototroph from natural population is easy, isolation of auxotrophic mutants are difficult, thus they are isolated and characterized only by penicillin method which is illustrated in (Fig. 2.45).

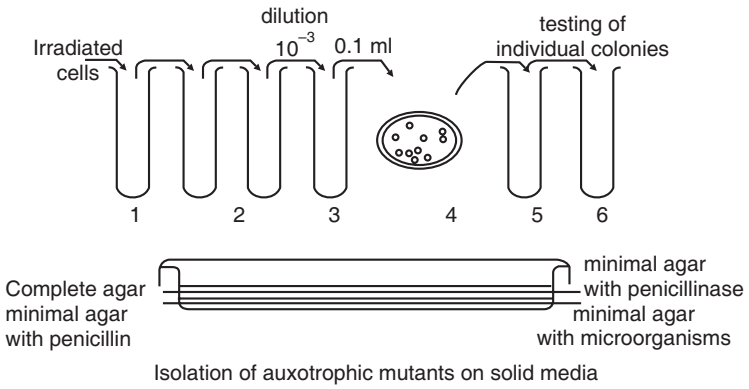


Fig. 2.45: Isolation of Auxotrophic mutants with the help of Penicillin technique

The selection of improved strains by mutagenesis can be illustrated with improvement programme for penicillin production by *Penicillium chrysogenum*. The original fungus *Penicillium notatum* was replaced by *P. chrysogenum* in view of its better performance under submerged fermentation.

Out of parent strain of *P. chrysogenum* NRRL-1951, producing 50-100 1U (international unit) of penicillin, a strain producing 150-200 1U was selected for further improvement. By exposing the strain to X-rays, a strain X-1612, which could produce 400-500 IU was developed. UV-irradiation of this isolate yielded a strain Q176 with 800-1000 1U penicillin yield. However, this strain was not suitable as 25-30% of penicillin was of inferior type (penicillin V) and a pigment chrysogenin was also produced. **Backus** and **Stanffer (1983)** developed a strain BL-3D-10 using UV light which though produced 1000 1U yield of penicillin, it was free of yellow pigment. From this, by selection and mustard gas treatments a best strain 51-20 which had no pigment and gave 2000-3000 1U with 100% benzyl penicillin was produced (table 2.21). The present stage of yield of penicillin took 8 years in strain improvement programme (Fig. 2.46).

Table 2.21: Improvement of Penicillin producing strain of *Penicillium chrysogenum*

Strain	Method	Production14/ml	Remarks
NRRL 1951	Original isolute	50–100	Yellow pigment
NRRL 1951	Selection	150–200	-do-
B 25X-1612	X-rays	300–500	-do-
Q 176	UV light	800–1000	-do-
BL 3-D 10	UV light	--	Pigment free
47–638	Selection	--	-do-
47–1564	Selection	800-1000	Pigment free (95 – 100% G*)
48–701	Selection	--	-do-
49–133	Nitrogen mustard gas	1500-3000	-do-
49–2166	Nitrogen mustard gas	--	-do-
50–935	Nitrogen mustard gas	--	-do-
51–20	Selection	2000-3000	-do-

*Penicillin G (benzyl penicillin)

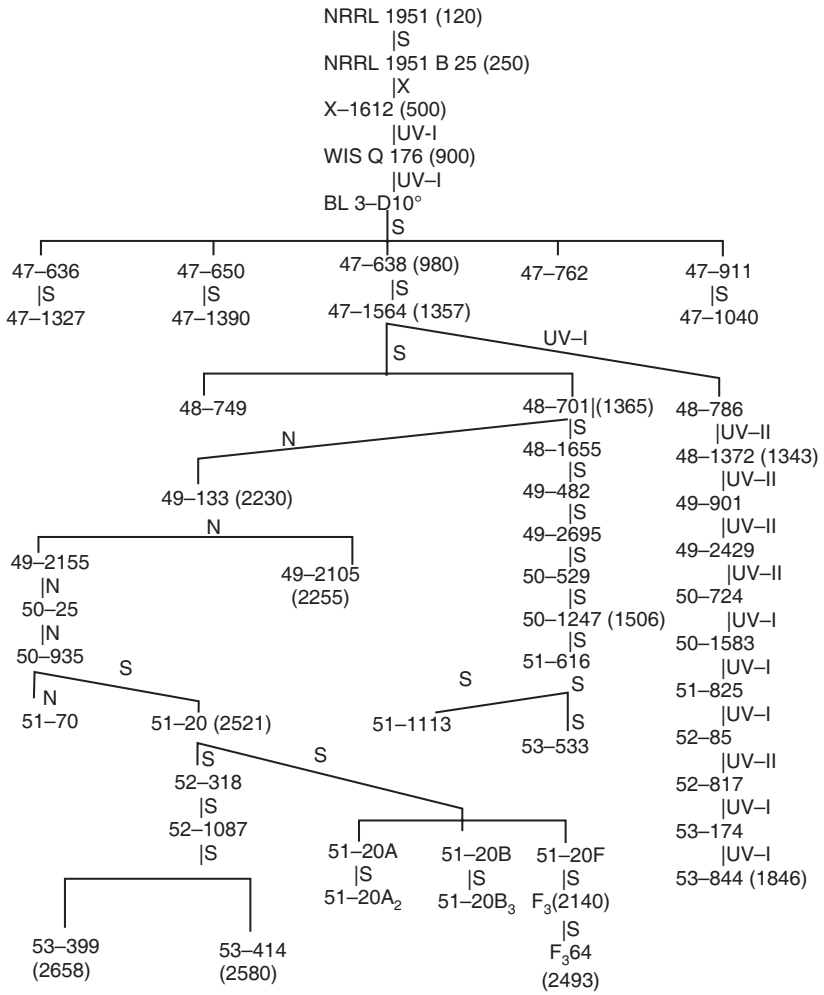


Fig. 2.46: Genealogy of the Wisconsin strain of *Penicillium chrysogenum* S, stages of selection, X, X-ray treatment, UV I, ultraviolet radiation at 275 nm, UV II, ultraviolet radiation at 253 nm, N, treatment with nitrogen mustard. Square brackets show yields in International units/ml. a, pigment-free mutant

In a similar manner intensive efforts have been made to improve cellulolytic strain of *Trichoderma reesei* through mutagenesis. The geneology of this strain is depicted below (Fig. 2.47) :

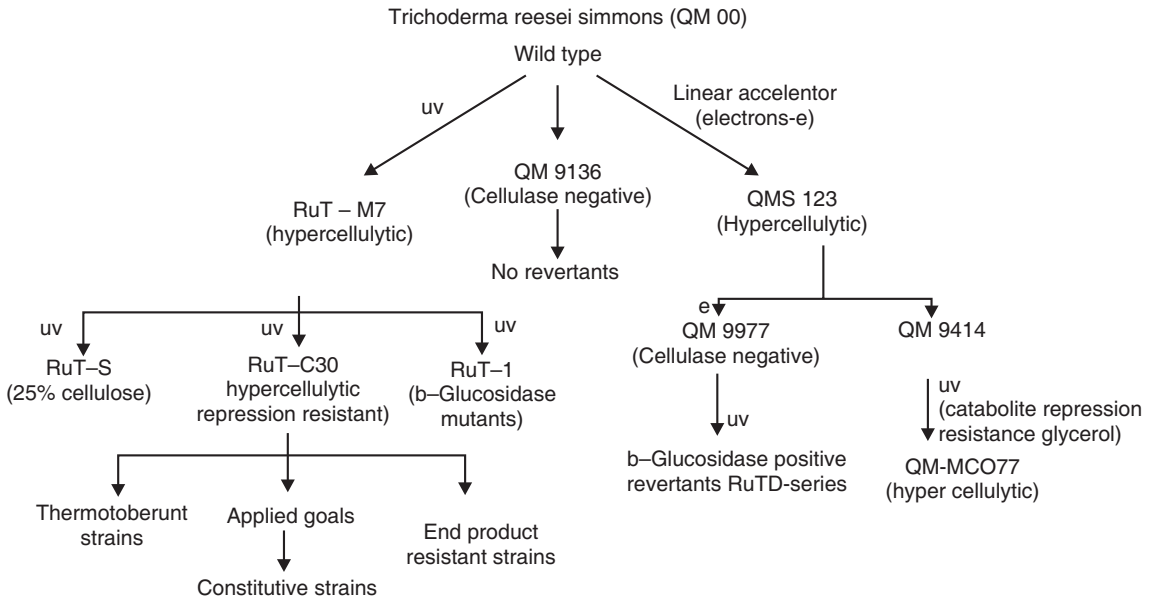


Fig. 2.47: Genealogy of *Trichoderma reesei* cellulytic strains Q.M. uv-ultraviolet irradiation e-high voltage electrons

2.16.2 Regulation: Microbial metabolism, both catabolic and anabolic processes are under the catalytic control of enzymes, and is so efficiently managed that excess metabolites are generally not synthesized. Strain development and optimization of fermentation conditions lead to a relaxation of regulation in the producing strains. A broad understanding of biosynthesis, the enzymes involved in these processes and their regulation is necessary for developing a rational approach to the alteration of the regulation of a fermentation process. Microbial metabolism is controlled by the regulation of both enzyme activity and enzyme synthesis.

(*i*) Regulation of enzyme activity: Enzyme activity can be controlled by different mechanisms. Some of which have been discussed here:

(*ii*) Feedback inhibition: Biosynthetic pathways in many microorganisms, especially in bacteria, take place either in a branched or unbranched manner and are controlled by different enzymes. In both pathways if the end product inhibits the activity of the first enzymes it is called as Feedback inhibition. During feedback inhibition, the end product is attached to a specific site of the enzyme called allosteric site due to which a conformational or structural change occurs in the enzyme. This can be illustrated by the biosynthesis of the amino acid, arginine in a mutant strain of *Corynebacterium glutamicum* (Fig. 2.48).

The mutant strain in which the L–enzyme acting on or methane has been made inactive by allosteric effect or is lost, will excrete the amino acid as long as sufficient amount of arginine is provided for its growth. This process continues as long as enough quantities of ornithine being synthesized to cause feedback inhibition. Optimum synthesis of ornithine occurs in a medium rich in glucose, ammonium salts, arginine and biotin.

Feedback inhibition in a branched biosynthetic pathway, occurs due to the inhibition of the activity of first common enzyme by means of one of the end products, which may affect the synthesis of other end product also. This can be explained by the biosynthesis of lysine by a mutant strain of *Corynebacterium glutamicum* (Fig. 2.48 a).

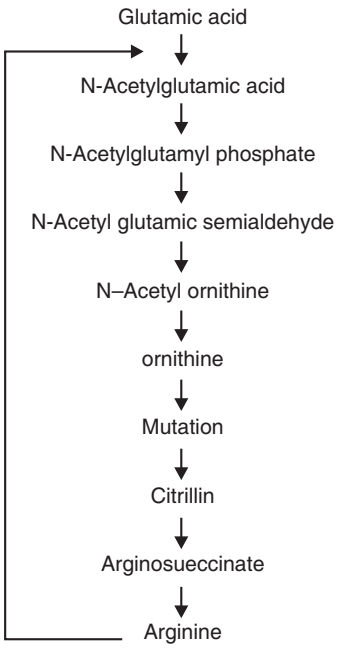


Fig. 2.48: The biosynthetic pathway for arginine in a mutant strain of *Corynebacterium glutamicum*

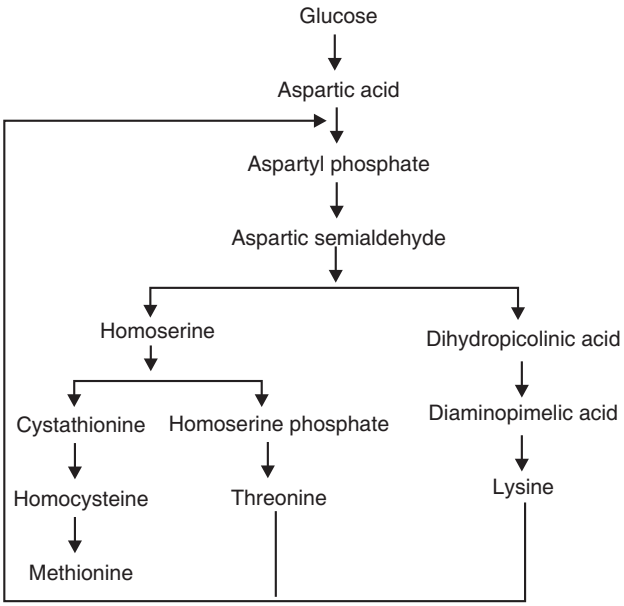


Fig. 2.48 a: Control of lysine biosynthesis in *Corynebacterium glutamicum* in a branched biosynthetic pathway

Biosynthesis of lysine and threonine in the wild strain of *corynebacterium glutamicum* takes place through a biosynthetic pathway in which the initial reactions are common for both the amino acids.

Feedback inhibition of the first enzyme, aspartate kinase occurs due to accumulation of both lysine and threonine in the wild strain. But such feedback inhibition does not occur in a mutant strain requiring homoserine due to the absence of synthesis of threonine. This results in the accumulation of lysine in the medium. Accumulation of methionine in this branched pathway also takes place due to the inhibition of methionine which also affects the aspartate kinase enzyme.

(*iii*) Energy charge: Catabolic pathways in which ATP is the main product are also regulated by feedback inhibition. The energy charge can be calculated by the following formula by measuring the relative concentration of AMP (Adenosine monophosphate), ADP (Adenosine diphosphate) and ATP (Adenosine triphosphate). The value of energy charge will be calculated with the help of following formula which will be between 0 and 1.0

$$EC = \frac{(ATP) + 0.5\,(ADP)}{(AMP) + (ADP) + (ATP)}$$

Generally the activities of enzymes involved in the synthesis of ATP are inhibited if the value of EC is high. For example, the activity of isocitrate dehydrogenase is inhibited when the EC value is > 0.8. Contrary to this, the activities of anabolic enzymes like aspartokinase which consume ATP are stimulated by a high energy charge. Thus, an alteration in the rate of catabolism may cause an increase or decrease in the activity of variety of enzymes, since it affects ATP level which inturn leads to change in energy charge.

(*iv*) Breakdown of enzymes: Enzymes which are no longer needed in metabolism may be broken down through the action of highly specific proteases. One of the best examples is the enzyme tryptophan synthetase in *Saccharomyces cerevisiae* which is broken down specially when cells go into the stationary phase.

(*v*) Modification of enzymes: The activity of some enzymes such as glutamine synthetase in *E.coli* is controlled by conformational changes such as phosphorylation or adenylation.

(*vi*) Regulation of enzymes synthesis: Strain improvement of industrially important microorganisms can also be brought about by the regulation of enzyme synthesis. Regulation of enzyme synthesis can be brought about by any one of the following mechanisms:

(*a*) Induction: Some enzymes are formed in a microbial cell irrespective of the presence of substances in the medium in which they grow, such enzymes are called as *constitutive enzymes.* On the other hand, certain enzymes are formed in response to the presence of substrate in the medium. They are called as *induced enzymes*. The production of one enzyme can inturn induces the synthesis of another enzyme. This process is called as *sequential induction.*

Insolation of mutants requiring an inducer can be achieved by cyclic growth of microorganisms in a medium containing an inducer substrate. In this method the microorganisms are first cultured in an inducer free medium. The constitutive and inducible population grows under these conditions at the same rate on transfer into a medium with an inducer substrate. Constitutive mutants grow at a higher rate

due to the activity of inducible enzymes. Repetition of these growth cycles results in an accumulation of constitutive mutants.

(*b*) **Attenuation:** This is another mechanism for the control of enzyme synthesis, which is brought about by affecting gene expression, which is involved in the biosynthesis of amino acids in bacteria. Attenuation can only be the regulatory step in certain bacteria as in the case of histidine biosynthesis in *Salmonella typhimurium* or it can work in addition to a repressor operator mechanism as with tryptophan in *E.coli.* According to the attenuator model, the transcription rate of an operon is regulated by the secondary structure of the leader sequences of the newly transient mRNA. The structure of the leader sequence determines whether the transcription of the operon is continued by the RNA polymerase or termination occurs. If termination occurs the mRNA transcription ceases and the enzyme or enzymes coded by that mRNA are not made. In the biosynthesis of tryptophan in *E.coli,* both repression and attenuation processes have large effect on the synthesis of related enzymes.

(*c*) **Excess production of primary metabolites:** With more and more understanding of the biochemistry and genetics of microorganism, has led to the isolation of strains which excrete excess primary metabolites like amino acids, vitamins, purines, nucleotides etc. This can be achieved by eliminating feedback inhibition by the following mechanisms.

1. The elimination of end product inhibition or repression is achieved by using auxotrophic mutants that no longer produce the decreased end product due to a block in one of the steps in the pathway. By adding the required end product in low amounts, growth occurs but feedback inhibition is avoided. Excretion of the desired intermediate product thus occurs. Both branched and unbranched pathways can be manipulated in this way (Fig. 2.48).
2. A second method is the selection of mutants that are resistant to antimetabolites. In this case, either the enzyme structure is changed so that the corresponding enzyme lacks the allosteric control site or mutations in the operator or regulator gene resulting in constitutive enzyme production and thus, overproduction.
3. In mutants with a block in an allosterically regulatable enzyme, suppressor mutations can lead to restoration of enzyme activity. However, these enzymes are not allosterically controllable.

(*vii*) **Regulation of secondary metabolites:** When a branched biosynthetic pathway simultaneously leads to the production of primary and secondary metabolites, an auxotrophic mutation in the biosynthesis of the primary metabolite can lead to an increased production of the secondary metabolites. Secondary metabolite production is controlled by five different classes of genes. They are:

1. **Structural genes:** These genes code for enzymes, which help in the synthesis of secondary metabolites.
2. **Regulatory genes:** They control biosynthesis of secondary metabolites by regulating the primary metabolism.
3. **Resistance genes:** They provide resistance to antibiotic producing strains to their own products.

4. **Permeability genes:** They regulate the uptake and secretion of substances.

It has been estimated that 300 genes are involved in chlortetracycline biosynthesis and approximately 2000 genes are directly or indirectly involved in neomycin biosynthesis. Some of the regulatory mechanisms that affect the production of secondary metabolites are described below.

(*viii*) **Induction:** Generally in fermentation processes with readily utilizable carbon and nitrogen sources, synthesis of secondary metabolites takes place after the growth of the microorganism has ceased. The logarthemic growth phase of a microorganism is called as tropophase and the phase of synthesis of secondary metabolites is called as idiophase. In all studies so far made it has been reported that the synthesis of enzymes involved in the biosynthesis of secondary metabolites is repressed during tropophase. There is little information regarding the nature of induction of enzymes involved in the synthesis of secondary metabolites. It has been observed that methionine induces the production of the antibiotics cephalosporin and fosfomycin and factor-A induces production of streptomycin (table 2.22).

Table 2.22: Key enzymes of secondary metabolism which are induced at the end of the tropophase

Enzymes	Secondary metabolite
Amidinatransferase	Streptomycin
A cyltransferase Phenylacetate, activating enzyme	Penicillin Penicillin
Oxidoreductase Transmethylase	Tylosin Tylosin
Synthetase I and II	Gramicidin S
Phenoxazinone synthetase	Actinomycin
Dimethysallyl tryptophoan synthetase	
Chanoclarin-l-cyclase	Ergot alkaloids

(*ix*) **End Product Regulation:** Antibiotics are reported to inhibit their own biosynthesis, for example, penicillin, chloramphenicol, ristomycin, cycloheximide, puromycin, fungicidin, streptomycin etc. Feedback regulation has been explained as a mechanism in few cases to explain such autoinhibition. With chloramphenicol and penicillin, it has been shown that higher concentration of these antibiotics inhibited their further production. Thus if strains could be isolated which are less sensitive to end product inhibition by these antibiotics, they might produce higher yields.

(*x*) **Catabolite Regulation:** When a commonly used substrate added as key enzyme, involved in a catabolic pathway, is repressed, inhibited or inactivated. This phenomenon is called as **catabolite repression**. Both carbon and nitrogen sources bring about catabolite repression.

Biosynthesis of different secondary metabolites like antibiotic gibberellins, ergot alkaloid, is inhibited by rapidly fermentable carbon source specially glucose. The basic mechanism of this carbon catabolite regulation is different, in different organisms and metabolites.

One of the well known carbon catabolite repression seen in many bacteria, yeasts and molds is the inhibition of the synthesis of mRNA. For any enzyme requiring catabolite activator protein (CAP) for its biosynthesis. CAP binds to the promoter site of an operon before RNA polymerase can attach and thereby facilitates the synthesis of enzymes that promote the production of secondary metabolites. CAP can only bind if it is complexed first with cyclic adenosine monophosphate (cyclic AMP). Readily utilizable carbon source such as glucose stimulates an enzyme which causes the breakdown of cyclic AMP, thus causing CAP inactive. In the manufacture of antibiotics and other secondary metabolites by industrial fermentation process the inhibitory effect of glucose is removed by feeding glucose at low concentration. Thus, catabolite repression is minimized.

(*xi*) Autoregulation: It has been shown that cell differentiation and secondary metabolism in certain actinomycetes are selected to a type of self-regulation from low molecular weight substances. For example, the formation of streptomycin, the development of streptomycin resistance and spore formation in *Streptomyces griseus* and *Streptomyces bikiniensis*, are all affected by factor-A, which is produced by *Streptomyces* themselves. The factor-A induces the increased transcription of the gene for the synthesis of the enzyme streptomycin phosphotransferase which imparts streptomycin resistance.

It has been demonstrated that the formation of streptomycin is dependent on the synthesis of factor-A and subsequent shift in the metabolism of carbohydrate source. The activity of the enzyme glucose–6–phosphate dehydrogenase is found to be high in factor–A deficient mutants and the enzyme has not been demonstrated in streptomycin high yielding strains. Moreover, addition of factor–A to mutants leads to a significant decrease in the glucose-6-phosphodehydrogenase enzyme activity. All these experimental evidences clearly indicate that when the pentosephosphate cycle is blocked in the absence of glucose-6-phosphate dehydrogenase, glucose is utilized in the formation of streptomycin.

Similar, autoregulatory mechanisms involving factor (–), A have also been reported in other actinomycetes. For example, formation of the antibiotic Virgimamycin by *Streptomyces virginiae* is regulated by a factor. Butyryl phosphoadenosine has been reported as a regulatory factor in the formation of rifamycin by *Nocardia mediterranean.*

2.16.3 Gene Technology: *In vitro* Recombinant DNA technology by employing restriction endonucleases and ligases, investigators can cut and splice DNA at specific sites. Some endonucleases have the ability to cut precisely and generate what are known as "sticky ends." When different DNA molecules are cut by the same restriction enzymes, they possess similar sticky ends. Through a form of biological "cut and paste" processes, the lower parts of one DNA is made to stick well onto the upper part of another DNA. These DNA molecules are later ligated to make hybrid molecules. The ability to cut and paste the DNA molecule is the basis of "genetic engineering." A useful aspect of this cut and paste process involves the use of plasmid, phage, and other small fragments of DNA (vectors) that are capable of carrying genetic material and inserting it into a host microbe such that foreign DNA is replicated and expressed in the host. A wide array of techniques can now be combined to isolate, sequence, synthesize, modify and join fragments of DNA. It is, therefore, possible to obtain nearly any combination of DNA sequence. The challenges lie in designing sequences that will be functional and useful.

The main events in the genetic engineering are:

1. Isolation and fragmentation of the source DNA,
2. Joining the DNA fragments to a cloning vector with DNA ligase,
3. Introduction and maintenance of introduced DNA in a host organism,
4. Detection and purification of the desired clone.

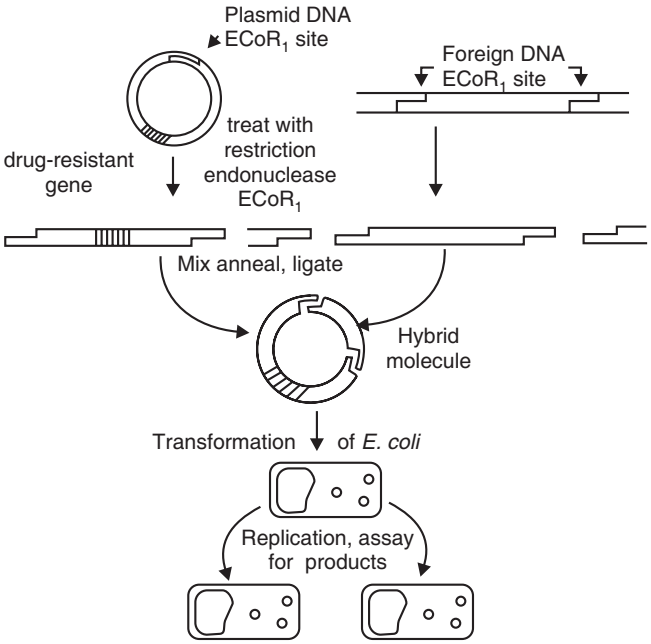


Fig. 2.49: Gene transfer through genetic engineering

The protocol to modify and improve strains involves the following steps:

(a) Isolate the desired gene (DNA fragment) from the donor cells.

(b) Isolate the vector (a plasmid or a phage).

(c) Cleave the vector, align the donor DNA with the vector, and insert gene into the vector.

(d) Introduce the new plasmid into the host cell by transformation or, if a viral vector is used, by infection.

(e) Select the new recombinant strains that express the desired characteristics.

For successful transfer of a plasmid/phage vector, it must contain at least three elements:

1. An origin of replication conferring the ability to replicate in the host cell,
2. A promoter site recognized by the host DNA polymerase, and
3. A functional gene that can serve as a genetic marker. A great deal of literature exists on the theoretical overviews and laboratory manuals on the use of recombinant DNA for strain modification and improvements are available.

   Assemble new combinations of DNA *in vitro*, which are then reinserted into the genome of the microbe, creating new varieties of microbe not attainable through traditional

mutation and rationalized selection approaches (Fig. 2.49). This approach overlaps the other methods to some extent, in that, it involves transformation of microbes with laboratory engineered specific recombinant molecules via plasmid or phage vectors.

2.16.4 Protoplast fusion: Protoplasts are the naked cells formed by removal of cell wall by enzymes chitinase or cellulase of fungi and bacteria, respectively. The protoplasts thus formed are stabilized against lysing by adding on osmotic stabilizing agent like sucrose. Protoplast fusions occur rarely because of presence of reflective charge in the surface of the protoplast. But, it can be achieved in the presence of polyethylene glycol (PEG). Protoplast fusion leads to DNA exchange. After fusion, the cell wall is allowed to regenerate. In the progeny there is significant number of recombinations. In recent years electric charge is being applied to increase percentage of recombinations. With the protoplast fusion both inter and intraspecific hybrid action can be achieved and this technique is better developed. **Hopwood et al.** (1977), **Schaeffer et al. (1976)** and **Foder** and **Alfondi (1976)** have achieved successfully the protoplast fusion in *Streptomyces* **sp.,** *Bacillus subtilis* and B. *megaterium* respectively.

Hamlyn and Ball (1979) proved that a high yielding strain of *Cephalosporium acremonium* could be obtained by protoplast fusion which was not possible by conventional techniques. **Hopwood (1979)** could get high proportion of recombinants of *Streptomyces coelicolor* with the help of protoplast fusion technique. Protoplast fusion has particular advantages over conjugation in that the technique involves participation of entire genome in recombination. **Karaswa** *et al.* (1986) by employing protoplast fusion technique got *Brevibacterium lactofermentum* strain with high lysine producing potential and fast glucose utilization strain which otherwise was difficult.

Pebendy *et al.* (1977) obtained a heterokaryon between *P. chrysogenum* and *P. cyaneofuluum* and demonstrated the formation of diploids which gave rise to recombinants after treatment with *phosphofluorophenylalanine* or benoxyl. **Lein (1986)** obtained a best recombinant of *Penicillium chrysogenum* with the help of protoplast fusion technique (table 2.23). Similarly a high yielding strain of *Streptomyces* **(Hopwood *et al.* 1997)**, *Bacillus* sp. **(Fodor and Alfodi, 1976),** Yeasts **(Sipeczkicv Terenczy, 1979)** were obtained by protoplast fusion technique.

Table 2.23: Improvement of *P. chrysogenum* for production of penicillin V with the help of protoplast fusion

Characteristics	Parent A	Parent B	Best recombinant
Spores per slant ($x10^8$)	2.2	2.5	7.5
Germination frequency (%)	99.0	40.0	49.0
Colour of sporulating colonies	Green	Pale green	Deep green
Seed growth	Good	Poor	Good
Penicillin V yield (mg cm^{-3})	11.7	18.5	18.0
Phenylacetic oxidation	Yes	No	No

(*i*) Advantages of protoplast fusion: Protoplast fusion has the following advantages:

(*a*) The frequency of recombination is significantly increased.

(*b*) The entire genome can be transferred instead of a fragment.

(*c*) The exchange of bacterial genetic materials does not require the presence of fertility factor.

(*d*) Protoplast fusion can be applied to bacteria in which other processes of recombination have been unsuccessful. It is also applicable to *Bacillus* strains, which do not have natural conjugating systems. Gram-negative bacteria such as *E. coli* have also been used where frequency of protoplast regeneration is low.

(*e*) The selection of strains which have tolerance to low O_2. **Mindilin and Zaitseva (1966)** have isolated a lysine producing strain which maintained its productivity under anaerobic conditions, while in the parental strain productivity is very low.

(*f*) **The elimination of undesirable products from a production strain: Dolezilova *et al.* (1965)** isolated a mutant with increased levels of fungicidin but produced no cycloheximide by protoplast technique. The protoplast fusion technique has also been used for the elimination of pencillinase production from pencillin, a producing strain.

2.16.5 Development of strains producing new fermentation products: Feeding of an analogue of the normal precursor of a natural products frequently results in the production of an analogue of natural product. **Hamill *et al.* (1970)** demonstrated that if 5, 6, or 7-substituted tryptophan replaced tryptophan, the normal precursor of an antifungal agent, pyrrolnitrin, in cultures of *Pseudomonas aureofaciens* then a series of substituted pyrrolnitrin were obtained. **Beckman *et al.* (1993)** produced a strain of *P. chrysogenum* capable of accumulating deoxycephalosporin (DAOC) a compound which can be biotransformed into 7-aminodeacetoxy cephaloranic acid (7ADAOC) which is a precursor of atleast three chemically synthesized clinically important cephalosporins. The componential ring of 7ADAOC by the chemical ring expansion of benzyl penicillin which is far more complex than the biotransformation of DAOC.

2.16.6 Strain Optimisation

If desired product is controlled by single or a group of genes then high yielding strains can be produced by:

(*a*) Modification of the gene by use of site directed mutations

(*b*) Manipulation of regulatory mechanism by cloning transcription or translations, by cloning the gene or an expression vector or by inserting the gene into a transposon which has a strong promoter

(*c*) Development and isolation of mutants resistant to inhibition of protein synthesis. Amino acid production has been increased by cloning the entire genome in *E.coli* and later in production strains such as *Corynebacterium, Brevibacterium, Serratia.*

(*d*) **Selection of strain resistant to phage infection:** Bacterial fermentation may be affected very seriously by phage infections, which may result in the lysis of the bacteria. A possible method for reducing the risk of failure due to phage contamination is to select bacterial strains which are resistant to the phages in the fermentation plant. The use of phage resistant mutant does not ensure immunity from phage infection because new host range phages may be introduced into the plant or phage mutants may appear. Plant hygiene is essential to minimize the risk of contamination. It is also possible to utilize chemical agents in the fermentation which selectively inhibit phage replication.

(*e*) **Selection of non-foaming strains:** Foaming during early phase of fermentation may be due to components of the medium, while during later stages of fermentation it is mainly due to the microorganism. Both mutant or raising recombinant may be used to develop non-foaming strains.

(*f*) **Selection of strains which are resistant to components in the medium:** Some media components which are required for product formation may interfere with the growth of organism. Therefore, it is desirable to select a strain which is resistant to medium component. **Polya and Nyiri (1966)** have applied protoplast fusion technique for isolation of a mutant which is resistant to phenylacetic acid, a precursor of penicillin and toxic to the organism at higher concentration.

(*g*) **Selection of morphologically favourable strains:** Morphological form of a filamentous microorganism will affect both the aeration of the system and the ease of filteration of the fermentation broth.

(*h*) **The improvement of industrial strains by modifying properties other than the yield of product:** The design and economics of a commercial process are influenced by properties of the organisms other than its productivity. Although a strain may produce very high level of a metabolite it would be unsuitable for a commercial process if its productivity was extremely unstable, or if the organisms oxygen demand is very high. Therefore, characteristics of the producing strain which affect the process may be critical to its commercial success. Then it may be desirable to modify such characteristics of the producing organism by selecting natural and induced variants and recombinants. Some of the characteristics which may be important in fermentations are strain stability, resistance to phage infection, response to dissolved O_2, tolerance to medium components, production of foaming, production of undesirable products and morphological form of the organism.

(*i*) **The selection of stable strains:** The ability of the producing strain to maintain its high productivity during both culture maintenance and fermentation is a very important quality. Yield decrease during culture storage may be avoided by use of maintenance of good techniques but loss of productivity during the fermentation is far more difficult to control. A decrease in the productivity of a commercial strain is normally due to the occurrence of spontaneous revertant mutants. The yield decay is specially problematic in longer term fermentation such as fed-batch and continuous fermentation. **Woodruff** and **Johnson (1970)** selected a double auxotrophic mutant of *Micrococcus glutamicus* requiring both homoserine and threonine and compared its lysine producing properties with those of a homoserine auxotroph. They claimed that double mutant had two fold advantage that it produced higher levels of lysine compared with single. **Lein (1986)** selected a stable strain of *P. Chrysogenum* by evaluated culture using saint to a slant transfer. If the culture was unstable then the yield of a fermentation from a second slant would be poor resulting in being rejected. This procedure was followed sequentially through the programme with the result that the later strains showed less tendency to degenerate after subculture.

(*j*) **The selection of strains which are tolerant to low O_2 tension: Mindilin and Zoutser (1966)** have isolated a lysine producing strain which maintained its productivity even under anaerobic conditions, while the parental strain productivity decreased by almost half.

(*k*) **The elimination of undesirable products from a production strain: Dolezilova *et al.* (1965)** demonstrated that mutants could be isolated which produced increased levels of fungicidin but produced no cycloheximide by protoplast fusion technique. The protoplast technique has also been used for the elimination of pencillinase production from penicillin G producing strain.

2.16.7 Future research: In industrial fermentations, strain developments plays a crucial role. Discoveries on different strategies in mutations, protoplast fusion, genetic engineering, recombinant DNA technology, modern fermenter design have revolutionized in strain improvement. In future, strain development technology will be supplemented by more knowledge based on scientific methods. With increased understanding of biochemical pathways, elucidation of regulatory mechanisms relating to induction of repression of gene and bioengineering design, it may be possible to apply new strategies which may spare endless possibilities for isolating improved strains. Further, tailoring of genes through *in vitro* DNA recombination techniques in both bacteria and fungi proved to be fruitful in strain improvement. Perhaps these areas will facilitate new strategies and have higher impact on industrial strain improvement.

REVIEW QUESTIONS

I. Essay type questions:

1. Define fermenter. List out characteristic features of an ideal fermenter.
2. Give an account of fermenter types.
3. Briefly describe different processes in an upstream process.
4. Narrate the process of selection of microorganism for industrial fermentation.
5. Trace the history of industrial microbiology.
6. Fermentation economics is based on fermentation medium. Discusss.
7. List out different types of fermentations giving advantages and disadvantages of each.
8. Define continuous fermentation. Give a critical account of organisms employed and products produced.
9. Describe different steps involved in downstream process.
10. Discuss the fermentation economics.
11. List out different carbon sources employed in industrial fermentation.
12. Give an account of nitrogen sources employed in industrial fermentations.
13. Describe different strategies in strain improvement.
14. Give an account of preservation of industrial microorganisms.
15. Define fermentation. Give an account of typical phases of fermentation.
16. Give an account of solid state fermentation.
17. Discuss advantages and disadvantages of continuous fermentation.
18. Describe feed batch fermentation with reference to penicillin production.
19. Define fermenter and describe different parts of fermenter.
20. Give different constituents of typical fermentation medium.
21. Isolation and selection of industrial microorganisms.
22. Discuss the importance of strain improvement of industrial microorganisms.
23. Define patent right. Give the general procedure for obtaining a patent right.
24. Describe the process of getting a patent of microbial culture.
25. Give an account of parameter optimization of fermentation.

26. Explain different types of aeration and their principles.
27. What is the necessity of foam control in fermentation? How it is achieved?
28. Discuss temperature and pH control system.
29. Explain about medium agitation and their principles.

II. Write short notes on:

1. Head space	2. Inoculum	3. Fed-batch fermentation
4. Dual fermentation	5. Impeller blades	6. Sparger
7. Antifoaming agents	8. Turbidostat	9. Chemostat
10. Submerged fermentation	11. Solid state fermentation	
12. Mixed fermentations	13. Containment	14. Down stream
15. Solid state fermentation	16. Dialysis	17. Protoplast fusion

FURTHER READING

1. Baltz, R.H. (1986). **Mutagenesis in *Streptomyces* spp.** In "Manual of Industrial Microbiology and Biotechnology" **(eds.A. Demain and N.A. Solomon),** pp. 184-190. **American Society of Microbiology,** Washington DC.
2. Broadbent, J.F., and Kondo, J.K. (1991). **Genetic construction of nisin producing *Lactococcus cremoris* and analysis of a rapid method for conjugation.** *Appl. Environ. Microbiol.* 57, 517-524.
3. Dorn, P.M. (1995). **Bioprocess Engineering Principles.** Academic Press, London.
4. Elander, R., and Vournakis, J. (1986). **Genetics aspects of overproduction of antibiotics and other secondary metabolites.** In "Overproduction of Microbial Metabolites" (Z. Vanek and Z. Hostalek, eds.), pp. 63-82. Butterworth, London.
5. Lein, J. (1986). **Random thoughts on strain development.** *SIM News* 36, 8-9.
6. Matsushima, P., and Baltz, R. (1986). Protoplast fusion. in "Manual of Industrial Microbiology and Biotechnology" (A. Demain and N. Solomon, eds.), pp. 170-183. **Amercian Society of Microbiology,** Washington, D.C.
7. Nolan, R. (1986). Automation system in strain improvement in "Overproduction of Microbial Metabolites" (Z. Vanek and Z. Hostalek, eds.), pp. 215-230. Butter Worth, London.
8. Parekh, S. (2000) **Strain improvement** In Encyclopedia of microbiology" 4, 428-443.
9. Qyeener, S., and Lively, D. (1986). **Screening and selection for strain improvement.** in "Manual of Industrial Microbiology and Biotechnology" (A. Demain and N. Solomon, eds.), pp. 155-169. **American Society of Microbiology**, Washington, D.C.
10. Reisman, H.B. (1988). **Economic Analysis of Fermentation Processes** CRC Press, BocoRaton, Florida, USA.
11. Schroeder, W., and Johnson, E. (1995). **Carotenoids protect *Phaffia rhodozyma* against singlet oxygen damage.** *Journal of Industrial Microbiology and Biotechnology.* 14, 502-507.
12. Seider, W.D., Seader, J.D. and Lewin, D. R. (1999) **Process Design Principles** : Synthesis, Analysis and Evaluation John Wiley, New York.
13. Van't Riet, K. and Tramper J. (1991) **Banic Bioreactor Design.** Marcel Dekkar, New York.

3

Anaerobic Fermentations

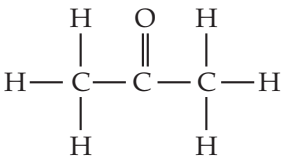
3.1 ACETONE–BUTANOL FERMENTATION

Acetone and butanol are produced through anaerobic fermentation by species of *Clostridium butyricum*. The production of butanol by butyric acid bacteria was first observed by **Louis Pasteur** in the 19th century. Before World War-I processes involving microorganisms were developed for the production of butadiene which is required for the production of synthetic rubber. Later on **Weizmann** reported that *Clostridium acetobutylicum* is capable of producing acetone, butanol and ethanol in an economically feasible quantities. During World War-I, acetone was in great demand to manufacture the explosive trinitrotoluene (TNT). Hence, the acetone-butanol fermentation rapidly expanded. But after war, the demand for acetone decreased and butanol increased, as it was required as a solvent for the rapid drying of nitrocellulose paints in automobile industry. Thus, the commercial process of acetone-butanol survived even after a lack of demand of acetone after World War–I.

But after World War II petroleum based processes replaced biological fermentation processes of acetone-butanol production, which lead to the closure of many industries. However, the fermentative production of acetone-butanol is still being carried out in certain countries where the carbon source material, specially, starchy material are available at cheaper rate. Vitamin B_{12} is produced as a byproduct in this fermentation process. Biosynthesis of acetone-butanol is illustrated in Fig. 3.1.

3.1.1. Chemical structure of acetone and n Butanol.

Structural formula

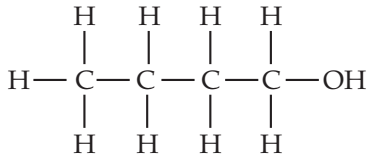


Molecular formula Acetone, C_3H_6O

n-BUTANOL:

Molecular formula - $C_4H_{10}O$

Structural formula



n Butanol, $C_4H_{10}O$

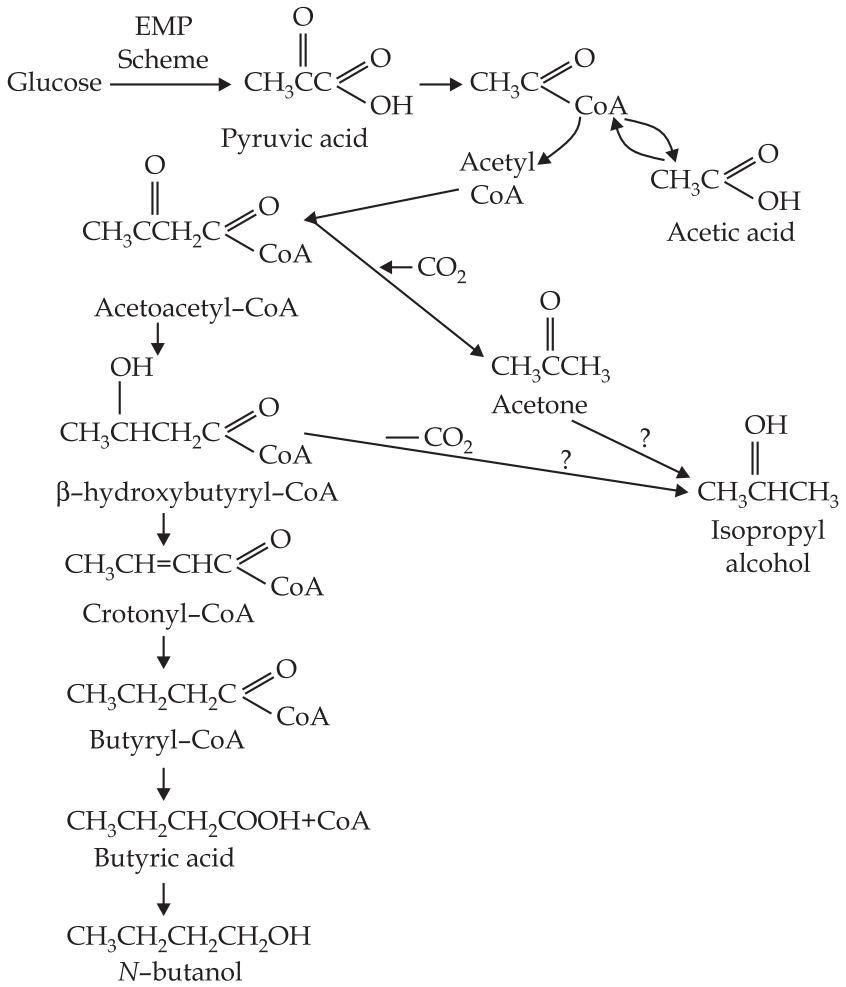


Fig. 3.1: Biosynthesis of acetone-butanol

3.1.2 Fermentation process: Acetone-butanol fermentation process can be described under the following phases:

(*i*) Production of inoculum (*ii*) Preparation of medium

(*iii*) Fermentation process (*iv*) Harvest and recovery

(*i*) Production of inoculum:

Two species of *Clostridium,* which differ slightly in their nutritional requirements and fermentation factors, are generally employed for acetone-butanol fermentation (table 3.1). *Clostridium acetobutylicum* and Cl. *saccharo-acetobutylicum* are the species involved. The fermentation by the former requires corn medium and the later molasses medium for the growth. In general, inoculum growth and fermentative production of the solvents are carried out at 31° to 32°C for *Cl. saccharoacetobutylicum* and at approximately 37°C for *Cl. acetobutylicum.*

Table 3.1: Acetone/butanol/ethanol fermentation of corn cobs, corn stalk and wood by species of *Clostridium*

Culture	Substrate (s)	Time	Yield	Solvent composition (%)		
		(h)	(%)	Butanol	Acetone	Ethanol
Cl. acetobutylicum (S_{25})	Hydrolysate of corn cobs (7%)		26.2	---	---	---
	Hydrolysate of sawdust (7%)		22.2	---	---	---
Cl. acetobutylicum (314)	Hydrolysate of corn cobs (40-60%, 3.7% total), molasses (40-60% sugar)	48-72	40	67.5		
Butyl culture	1 part hydrolysate of corn stalks, 3 parts molasses	50-55	31-37			
	Pentoses (13.5%)					
	Hydrolysates of wood and plants (8%)		25.4	67	33	
			35.5	62	32	6

(*a*) Inoculum of *Cl. saccharoacetobutylicum:* Inoculum of *Cl. Saccharo acetobutylicum* is developed employing molasses, calcium carbonate, ammonium sulphate or phosphate and sometimes cornsteep liquor. Clostridia, being spore formers, are easily maintained as soil stocks in contrast to the vegetative cells, the spores are not very sensitive to oxygen. However, prolonged storage of these spores leads to decrease in the acetone butanol production.

Spores from soil stocks are initially added to deep tubes of semisolid potato-glucose medium for molasses cultures. As the spores are added to the bottom of these tubes they along with soil particles sink to the bottom of the tubes and become submerged. The submerged location of the spores and high reducing power of the medium can protect the vegetative cells from oxygen after germination of spores/vegetative cells. Inoculated tubes are heat shocked and rapidly cooled to incubation temperature to select heat resistant spores. The tubes are then incubated at 31° to 32°C for 20 hours. The growth that occurs in the tubes are used as inoculum for larger batch of molasses medium present in inoculum tanks. Further, increased volumes of inoculum are produced by

successive transfers of approximately 2% to 4% inoculum by volume to larger media with incubation period of 20 to 24 hours at each transfer. At each of these stages of inoculum transfer, anaerobic conditions are produced by:

1. The reducing condition of the medium,
2. Immediate use of freshly sterilized and cooled medium before air becomes incorporated,
3. Evolution of fermentation gasses
4. By filling the head space of the inoculum tank with sterile inert gas with a slight positive pressure.

Various inoculum stages, particularly the last stage, before the inoculum is transferred to production tank, are checked for

1. pH,
2. Density of the inoculum,
3. Rate of gas evolution,
4. Presence of facultative anaerobe tested by aerobic plating on agar medium and
5. Microscopic observation for contamination by hanging drop method.

The inoculum of molasses medium is grown for 24 to 26 hours before addition to production tank.

(*b*) Inoculum of *Cl. acetobutylicum:* Stages of inoculum preparation of this type of *Clostridium* are generally similar to the one described above for Cl. *Saccharo-acetobutylicum* except the following items.

1. Spores of soil stock are added to the deep tubes containing corn medium with about 5% corn meal in water.
2. The tubes are incubated at 37°C for 20 hours.
3. The concentration of corn in the corn mash is adjusted to 6½% during repeated inoculum transfers.
4. In addition, for all the tests described above, titratable acidity test is conducted for corn culture inoculum.
5. The final incubation of inoculum of corn culture is carried out until the titratable acidity test shows acid break up, that is, conversion of acetic acid to acetone and butyric acid to n-butanol.

(*ii*) Preparation of Medium

(*a*) Molasses medium: It provides carbon source for *Clostridium*. Molasses which is formed as a by-product in the sugar industry, is used as a raw material for the preparation of the medium. Sugar content in the form of sucrose is maintained at 6%. For this either black strap or high-test molasses is used. Nitrogen source is added in the form of ammonium sulphate. In addition calcium carbonate, superphosphate and sometimes cornsteep liquor are also added. Calcium carbonate is added to prevent development of gross acidity in the medium. Although, excess addition of this salt lowers the production of the solvents. Ammonium sulphate is added at 18 to 24 hours of fermentation.

(*b*) Corn medium: Corn meal is prepared by passing corn through a magnetic field to remove dust and metallic debris followed by degerming the corn. Corn oil is removed from the sprouted germ. The degermed corn is then ground to a fine powder in roller or

hammer mill. To prepare corn-meal production medium 8% to 10% corn meal is added to water with or without stillage, that is, residue from the preceeding fermentation. It is then heated for 20 minutes at 65°C to gelatinize the starch before sterilization of the medium. The two media described above are then sterilized and employed in fermentation depending upon the type of *Clostridium* species used in the fermentation. Sometimes stillage, that is, residue formed in the previous fermentation is added to the medium approximately at 30% to 40%. It results in the addition of certain nutrients like proteins, carbohydrates and minerals to the freshly prepared medium.

(*iii*) **Fermentation process:** Fermentation is carried out under anaerobic conditions. Production tanks of the capacity of 50,000 to 2.5 lakhs gallons are used in the fermentation. The incubation period is 2 to 2 1/2 days. If the freshly steamed molasses medium is employed, approximately 2 to 4% of inoculum is needed, while for freshly steamed corn medium slightly less inoculum is employed. The inoculum is added first to the production fermenter followed by the addition of medium. This sequence facilitates thorough mixing of inoculum with the medium and maintain anaerobic condition. The yield of different solvents in different media are precised in Table 3.1. However, in an alternative procedure only a part of the medium is added to the inoculum and the inoculum is allowed to initiate growth before the rest of the medium is added. Fermentation generally passes through three phases.

(*a*) **First phase:** In this phase rapid growth of the bacterium and formation of acetic acid and butyric acid in large amounts along with the production of large quantities of carbon dioxide and hydrogen gases. The pH of the medium which was initially 5.0 to 6.5 for corn medium and 5.5 to 6.5 for molasses medium, decreases and then remains constant for the rest of the fermentation process. This phase, lasts for approximately 13 to 17 hours of incubation. The titrable acidity increases to a maximum and adaptive enzymes are produced which convert acids to neutral solvents.

(*b*) **Second phase:** A sharp decrease in the titrable acidity due to conversion of more acids into acetone and butanol. This process is called as acid break, which gets delayed if there is contamination. The rate of gas formation reaches maximum after acid break. However, it gradually slows as the fermentation process proceeds further.

(*c*) **Third phase:** The rate of gas formation decreases substantially along with decreased rate of solvent production. The titrable acidity slowly increases leading to a pH of 4.2 to 4.4 in the corn medium and 5.2 to 6.2 in the molasses medium. Many cells undergo autolysis at this point resulting in the release of riboflavin into the medium.

(*d*) **Yield:** The ratio of yield of acetone, butanol and ethanol differ slightly depending on the fermentation medium. But, generally the yield is 2% by weight of the broth, which is approximately equal to 30% conversion of carbohydrate to solvents. In a corn medium the ratio of butanol, acetone and ethanol are 6 : 3 : 1 respectively, but in molasses medium the ratios are 6 : 5 : 3.

Apart from these solvents, 3 parts of carbon dioxide and 2 parts of hydrogen by volume are also produced as byproducts of the fermentation. They account for approximately half of the sugar medium. Total weight of the gases will be one and half times more than the solvents.

The acetone-butanol fermentation yields several products in addition to the gases described above. They include isopropanol, formic acid, acetic acid, butyric acid,

acetylmethyl carbinol and yellow oil, which is a mixture of higher alcohols and acids, which are industrially very important. Contamination due to bacteriophages and *Lactobacillus* is a common problem which can be prevented by undertaking absolute sterilization.

(*iv*) **Harvest and recovery:** A beer still is used for the recovery of the products from the fermentation broth. The beer still is a tall vertical and continuous still consisting of about 30 perforated plates. The recovery process consists of the following steps:

1. The fermentation broth is allowed to enter the still from top. It descends the still passing through perforated plates.
2. A continuous flow of steam is allowed into the still from its bottom. It moves up the still in a direction opposite to the direction of fermentation broth.
3. Acetone and butanol vaporizes due to the effect of steam.
4. The steam and solvents are then collected and condensed by cooling to get a solution which contains approximately 40% by weight of solvent mixture.
5. The individual solvents present in the solvent mixture are separated by fractional distillation.

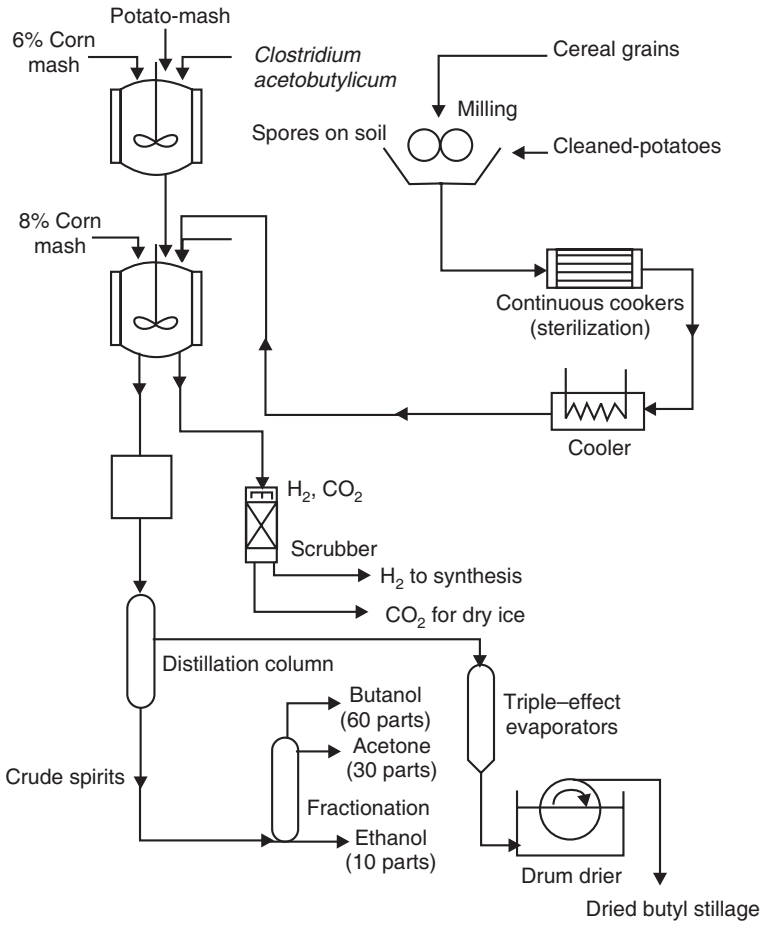


Fig. 3.2: Flow sheet of production of acetone-butanol

6. Acetone and butanol are collected in separate fractions.
7. Ethanol and isopropanol are also collected as a single fraction and sold as a general solvent.
8. The residue contains riboflavin and other B vitamins as well as considerable quantity of bacterial cells. The residue is concentrated and dried and used as vitamin feed supplement. Flow diagram for the production of acetone butanol is shown in the Fig. 3.2.

3.1.3 Uses

1. At present butanol is extensively used in brake fluid antibiotic recovery procedures, urea, formaldehyde resins, amines for gasoline additives and as ester in the protective coating industry.
2. Butanol is also used for the synthesis of butadiene which is used in the preparation of synthetic rubber.
3. Acetone is used as an universal organic solvent and also in the preparation of explosives like trinitrotoluene.

3.2 ETHYL ALCOHOL

Ethyl alcohol has been produced on large scale for centuries. However, much study could not be accomplished because of hazards on human consumption. In 1865 Alcohol Act was passed thereby free sale of alcohol after its denaturing by adding methylated spirit was allowed. Early production was primarily used for human consumption. But today, apart from being used for human consumption, it is also used as universal solvent and as chemical raw material for the production of other industrial products. Along with gasoline it is also used as motor fuel. Because of the above utilities the demand for alcohol increased enormously today, which lead to establishment of many distilleries throughout the world. Ethyl alcohol is produced, besides yeasts, by large number of bacteria and fungi (table 3.2). The biochemical synthesis of alcohol by different microorganisms is depicted in Fig. 3.3. Similarly, besides use of molasses, lignocelluloses can also be used as a sustainable substratum (table 3.3). However, apart from fermentative production it is also produced by chemical processes primarily by catalytic hydration of ethylene.

Chemical convertion of lignocellulose by different fermentations are shown in table 3.5. The whole process of bioconversion of lignocellulose into valuable sugars and other substances are illustrated in Fig. 3.5.

Table 3.2: Microorganisms generally employed in alcohol production

I. Bacteria	Characteristics
Clostridium thermohydrosulfuricum	Extreme thermophile
Clostridium thermocellum	Thermophilic, hydrolyses cellulose
Thermoanaerobacter ethanolicus	Ferments xylose and starch*
Zymomonas mobilis	The wild type ferments only glucose, fructose and sucrose, but with less productivity

contd...

II. Yeasts	Characteristics
Candida pseudotropicalis, C.tropicalis	Ferments xylose
Candida species	Ferment xylose and cellobiose
Kluyveromyces lactis	Ferments lactose in dairy waste
Kluyveromyces marxianus	Hydrolyses inulin (polyfructose)
Pachysolen tannophilus	Ferments xylose
Pichia stipitis	Ferments xylose
Saccharomyces cerevisiae	Most strains ferment only glucose, sucrose, fructose, maltose and maltotriose
S.cerevisiae var. distaticus	Hydrolyses starch
Schwanniomyces alluvius	Hydrolyses starch
III. Filamentous fungi	
Fusarium species	Ferment xylose
Monilia species	Hydrolyse cellulose and xylan
Mucor species	Ferment xylose and arabinose

3.2.1 Fermentative production

On commercial scale ethyl alcohol production process consists of four steps. They are:

(i) Inoculum production, (ii) Production medium,
(iii) Fermentation process, and (iv) Harvest and recovery

(i) **Production of Inoculum:** Both yeast and bacteria are used for the production of ethyl alcohol. Among the bacteria the most widely used organism is *Zymomonas mobilis* and among the yeasts *Saccharomyces cerevisiae, Saccharomyces carlsbergensis,* certain species of *Candida* and *Mucor* are also used, depending upon the raw materials available for ethanol production.

But high yielding and alcohol tolerant strains of *S.cerevisiae* are usually employed. Details of process for production of ethyl alcohol from different substrates are depicted in Fig. 3.3. High yielding strains of yeast are generally used for commercial production of ethanol. These are developed by genetic selection. The strain of yeast that may be used for the industrial production should posses the following characters

1. It should possess uniform and stable biochemical properties.
2. The ability to ferment a broad range of carbohydrate substrates rapidly.
3. It should yield large quantities of ethanol.
4. It should grow fast and should have high osmotolerance.
5. Low levels of by products such as acids and glycerol.
6. It should be tolerant to higher concentration of alcohol.
7. It should possess high temperature tolerance.
8. High cell *viability* for repeated recycling.
9. Appropriate flocculation and sedimentation characteristics to facilitate cell recycle.

However, the choice of microorganism employed in large scale ethanol production depends upon the type of raw material used. For example S. *cerevisiae* is employed when starch or maltose is used as raw material, when whey or sulphite waste liquor is used as raw material *Candida utilis* and *C. albicans* respectively is employed in the fermentation process. Production of ethanol from starch is shown in Fig. 3.3.

(*a*) Inoculum production: A suitable pure strain of yeast is inoculated into a test tube containing approximately 10 ml of sterile medium. It is incubated at 28°C to 30°C till sufficient growth of yeast takes place. The medium employed for the preparation of inoculum and the fermentation process is generally similar.

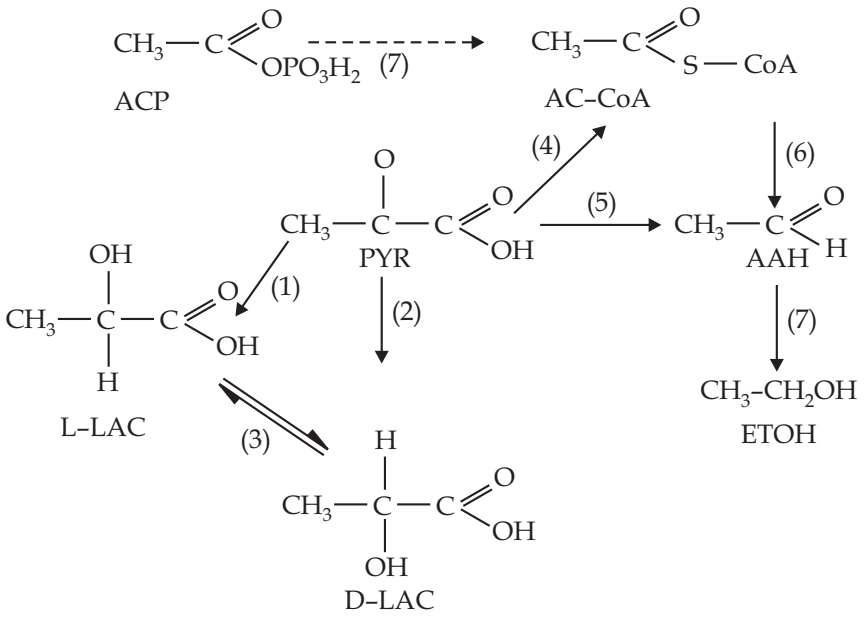


Fig. 3.3: Mechanisms for ethanol (ETOH) synthesis

1. After sufficient growth of the yeast occurs, the culture from the test tube is transferred to a flask containing approximately 200 ml of the medium. The flask is incubated at 28°C to 30°C until predetermined cell mass is formed.
2. The fully-grown culture from the flask is transferred to a glass container containing 4 liters of sterile medium and is incubated at 28° to 30°C till sufficient growth takes place.
3. The culture from the glass container is finally transferred to a small seed tank containing 40 gallons of sterile medium. Then seed tank is to be near the fermentation tank. The culture broth after incubation will be ready for inoculating into the production tank.

   A large amount of yeast culture ranging from 8 to 10% volume of inoculum is required in the industrial production of ethanol. To achieve rapid growth of yeast and large amount of cell mass, high degree of aeration and agitation of the medium is required. The pH and temperature are adjusted at 4.8 and 28° to 30° C respectively during the growth of the yeast.

(*ii*) **Preparation of medium:** Composition of fermentation medium plays an important role in achieving optimum yield of ethanol. It should be prepared in such a way that it contains all the sources of materials that promote optimum growth of yeast and optimum yield of ethanol. Generally, the medium should possess carbon source, nitrogen source and growth factors.

(a) Carbon source: Different varieties of carbohydrates which are produced as waste products in agricultural industries can be used as carbon source. They are grouped into the following categories depending upon their chemical nature.

1. Starchy material - potato starch, cereals like oats, wheat flour and corn starch.
2. Saccharide material - fruit juice, whey, molasses and hydrol.
3. Cellulose material - sulphite waste liquor, lignocellulose.

Molasses is generally used as the main carbon source in the preparation of fermentation medium. However, cane molasses is used in India because of its availability in large quantities from sugar industries. The raw material indicated above, require pretreatment in the form of saccharification. They are put to hydrolysis during saccharification due to which easily fermentable sugars like maltose and glucose are formed.

The optimum sugar concentration should be maintained between 10 to 18% in the fermentation medium. If beet molasses are used, biotin should be added to meet the biotin deficiency. If cane molasses is used as a carbon source, its sucrose concentration should be reduced to 10% by the addition of distilled water which is called a **Wort**. Higher sucrose concentration affects the growth of yeast adversely, while lower sucrose concentration makes the fermentation process uneconomical.

In recent times lignocelluloses as source of carbon proved to be more economical and sustainable. Most of the plants after death are subjected to decomposition releasing fermentable sugars. Large amount of lignocellulose out of waste agricultural products can be converted to sugars by enzymatic hydrolysis. Annual production of these lignocelluloses around the world is given in table 3.3.

Table 3.3: Global production of lingocellulose wastes

Continent / country	Wastes (million tones)			
	Cereal crop	Pulse crop	Oilseed crop	Plantation crop
Africa	**165**	**9**	**11**	**34**
Asia	1135	51	61	174
Australia	35	1	2	12
Europe	**550**	**10**	**8**	**1**
Central America	1500	49	21	84
South America	150	37	10	147
India	**240**	**16**	**14**	**88**
U.S.	440	44	19	15
Canada	60	2	< 1	NA[6]
World	**2946**	**166**	**142**	**548**

Composition of plant lignocellulosic materials on hydrolysis are depicted in table 3.4. Integrated process of conversion of lignocellulose wastes into value added saccharides is presented in Fig. 3.5 and ethanol production from different substrates is depicted in Fig. 3.4.

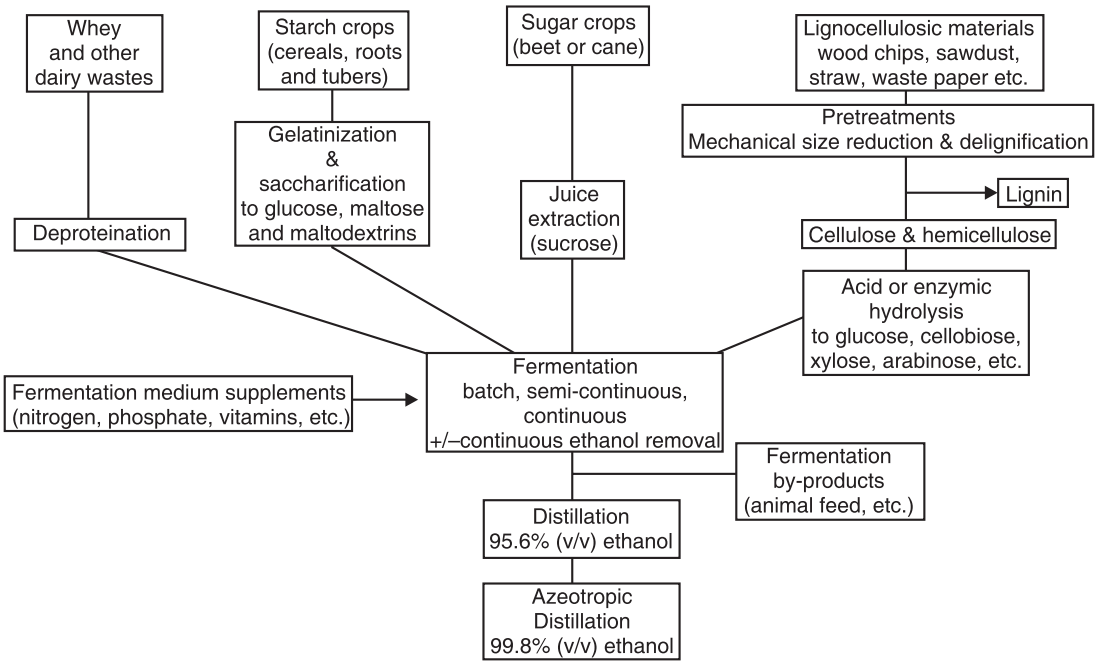


Fig. 3.4: Industrial ethanol production from various substrates

Table 3.4: Chemical Composition (%) of agricultural and wood residues

Substrate	Hexosans	Pentosans	Lignin	Ash
Bagasse	33	30	29	4
Barley straw	40	20	15	11
Birch	40	33	21	4
Corn cobs	42	39	14	2
Corn stalks	35	15	19	5
Ground nut shells	38	36	16	5
Oat straw	41	16	11	12
Pine	41	10	27	8
Rice straw	32	24	13	12
Rice hulls	36	15	19	20
Saw dust	55	14	21	5
Sorghum straw	33	18	15	10
Wheat straw	30	24	18	10

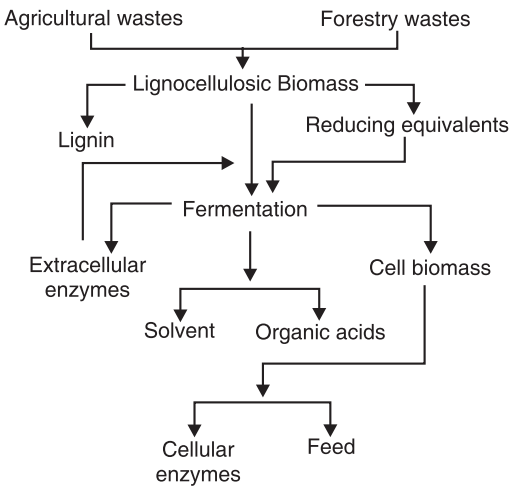


Fig. 3.5: Integrated approach for bioconversion of lignocellulosic wastes into valuable products

(b) **Nitrogen source:** Variety of inorganic and organic nitrogenous compounds may be used as nitrogen source in the medium preparation. Ammonium sulphate is generally used as nitrogen source. Generally 0.15 g of ammonium sulphate is added to 2.5 gallons of molasses. The concentration of nitrogen should be carefully maintained as indicated above. Excess nitrogen promotes rapid growth of yeast cells and decreased production of ethanol.

(c) **Growth factors:** As most varieties of molasses, for example cane molasses, contain suitable concentration of growth promoting substances, hence there is no need for addition of any growth promoting substances separately in the preparation of the medium.

(d) **pH:** The pH of the medium should be adjusted to 4.8 to 5.8 by adding sulphuric acid or lactic acid. As the medium becomes highly anaerobic at this range of pH, this inhibits the growth of contaminating bacteria. There is no need of sterilizing the medium. Further anaerobic fermentation of alcohol discourages growth of many microorganisms. However, pasteurization of the medium can be done.

(*iii*) **Fermentation:** Ethyl alcohol can be produced by any one of the following processes:

1. Batch fermentation,
2. Continuous fermentation with cell, and
3. Continuous fermentation with cell recycling.

A comparison of the operating parameters of the above fermentation processes are indicated in table 3.5.

Table 3.5: Ethanol production by different fermentations

Fermentation process	Sugar concentration (%)	Cell concentration (gl^{-1})	Ethanol concentration (gl^{-1})	Specific productivity (gg^{-1} cells hrs)	Volume specific productivity (gl^{-1} S^{-1})
Batch	16.7	21.3	85.6	0.42	11.8
Continuous	15.6	19.7	79.1	0.72	14.1
Continuous with cell recycling	16.7	100	85.6	0.42	42.5
Continuous immobilised yeast cells	--	--	--	--	> 100

However, batch fermentation is more commonly employed for ethanol production. Production is carried out in a large fermenter with a volume of 600 cm^3. About 30% of inoculum (cell density 3×10^6 ml^{-1}) is generally employed. It is added to the fermenter by pumping or gravity. This addition of inoculum to the fermenter is called as **pitching**. The following environmental factors like incubation time and temperature are to be suitably maintained and controlled in order to achieve optimum yield.

The time required for the maximum yield of ethanol is 30 to 72 hours, which largely depends upon the specific gravity of the fermentation liquid. The fermentation is generally stopped at any hour when the specific gravity of the fermentation liquid becomes constant. At this stage, 6 to 8% of ethyl alcohol will be formed. Fermentation generally starts within few hours of yeasts inoculation. The process becomes rapid after 24 hours.

A temperature range of 25-30°C is favourable for ethanol production. However, as heat is evolved during the fermentation process, the temperature in the fermenter gradually increases and is controlled and maintained at the above indicated range by means of cooling coils or by spraying cold water around the fermenter. Periodical agitation of the fermenter is also required for enforcing uniform cooling of the medium. If higher temperatures are not controlled, contamination of the fermentation broth may occur due to the growth of thermophilic bacteria and also result in the loss of ethanol due to surface evaporation.

When all the fermentation factors are optimum there may be production of 1.9 grams of ethanol one liter of medium per hour. About 90% of carbon source of the fermentation liquid is converted into alcohol.

(*iv*) **Harvest and recovery:** The cell mass is separated before distillation by either centrifugation or sedimentation. It is then distilled in analyzer and rectifier columns to get ethyl alcohol also called rectified spirit and fuel alcohol (higher alcohols). A mixture containing 95.6% ethyl alcohol and 4.4% water is obtained by fractional distillation. After distillation, the spent wash and bottom sludge are drained out as distillary waste. The product is marketed as rectified spirit, denatured spirit or special spirit. *Zymomonas mobilis* produces upto 120 g ethanol per liter per hour. The flow diagram for the production of ethanol is shown in the Fig. 3.6. Some of the byproducts are depicted in Fig. 3.7.

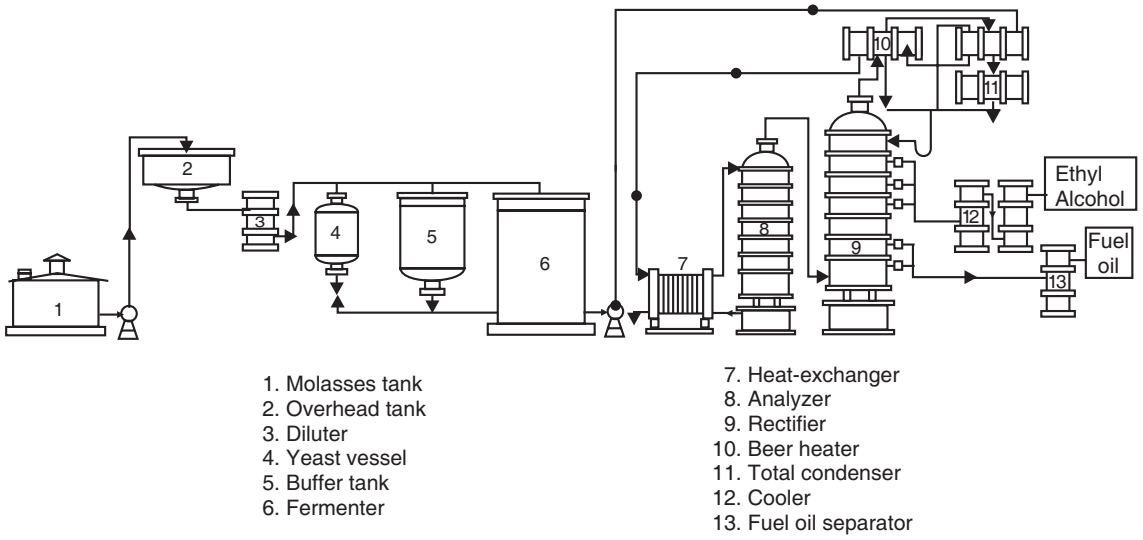


Fig. 3.6: Flow diagram of ethyl alcohol production

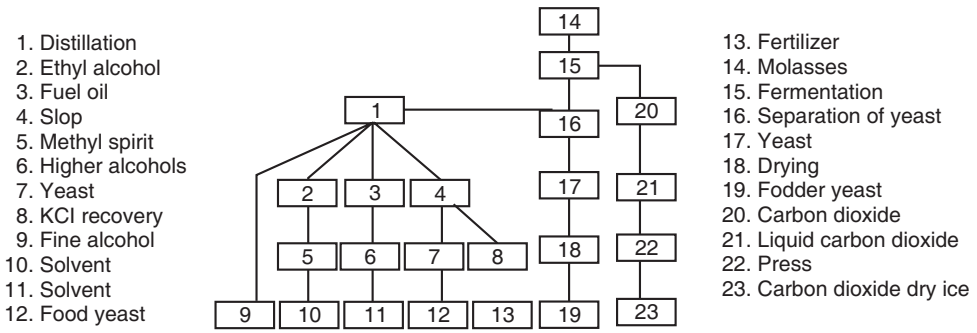


Fig. 3.7: Products in ethyl alcohol fermentation

Large distilleries employ, generally, special rectifier column called as coffey's still which consists of two columns called as the analyzer and the rectifier.

(a) The analyzer: The analyzer is a vertical tower. It consists of columns arranged in a zig-zag manner. The fermentation liquid is allowed to flow down the analyzer and simultaneously steam is allowed to move up the tower from its bottom. Alcohol present in the fermentation liquid vaporizes and its vapours collect at the bottom of the tower and are made to flow into the rectifier.

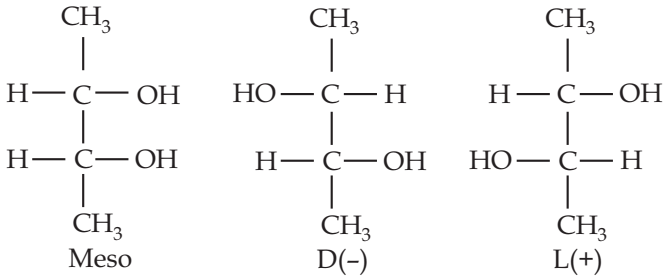
(b) The rectifier: Just like analyzer, the rectifier is also a vertical tower. It consists of specially designed fractionating columns with a number of chambers. The less volatile constituents (the slop and fuel oil) gradually condense and are drawn off from a higher point of the still. The temperature at the higher point of still is roughly equal to the boiling point. The head products, which are very less, contain aldehydes, formic esters etc. owing to their greater velocity, pass out through the top of the column along with a small quantity of uncondensed alcohol. The higher alcohols are removed from the receiver for every two or three days.

3.2.2 Applications:

1. It is extensively used in synthesis of variety of organic compounds and is a universal solvent.
2. As energy source in motor fuel cells.
3. Different byproducts are useful in a variety of ways.

3.3 2, 3–BUTANEDIOL

2, 3-Butanediol (also called diol or 2, 3-Butylene glycol) is a raw material for production of 1, 3 butadiene which is used in synthesis of rubber. It is also used as a permanent type antifreeze. Methyl ethyl ketone, Methyl vinyl carbinol and Methyl vinyl ketone, last two of which are used in plants are produced from 2, 3-Butanediol.



Most bacteria produce meso isomer, while *Bacillus polymyxa* produces more than 98% of the D (–) form.

3.3.1 Fermentation

2,3 butanediol is fermented by using one of the following raw materials

(a) Starchy substances (corn, barley, wheat, sweet potatoes and potatoes)

(b) Sugars and sugar residues (beet and sugarcane molasses)

(c) Wood hydrolysates and paper mill residues (waste sulfite liquor)

Bacterial species which produce economic amount of 2,3-Butanediol are listed in table 3.6.

Table 3.6: Bacteria producing 2,3-Butanediol under anaerobic condition

Bacteria	Diol	Diol formic acid	Diol glycerol	Ethyl alcohol	Lactic acid
Aerobacter aerogenes	52	2	-	23	2
Bacillus polymyxa	68	-	-	33	2
Pseudomonas hydrophilous	56	-	-	26	12
Serratia marcescens	1	56	1	27	6
Bacillus subtlis	-	-	84	3	8

2,3-butanediol fermentations are sensitive to changes in fermentation factors such as air supply, pH and sugar concentration. However, in aerobic fermentation of glucose by above bacteria will give only 2,3-butanediol, 2 mol. of CO_2 and a molecule of water. Similarly under

aerobic conditions *S. marcescens* produces only little formic acid and *B. subtlis* produces only little amount of glycerol. Generally, the yields of 2,3-butanediol or its oxidation product acetoin are increased under aerobic condition. Control of pH is essential to obtain highest yield of 2,3-butanediol. *Bacillus polymyxa* produces 130 moles of 2,3-butanediol and 159 moles of ethyl alcohol per 100 moles of sucrose used during 30 hr of fermentation period at 33°C, pH 6.2 and 8% sugar concentration. *B. polymyxa* has advantage over other organisms as it can produce amylase needed for degradation of starch grain mash.

3.3.2 Recovery

In view of 2,3-butanediol having high boiling point, recovery from fermented broth is difficult. Some of the procedures are spray drying, drum drying, chemical conversion, dialysis, solvent extraction and steam stripping. In most of these procedures of ethyl alcohol is eliminated. Thereafter, liquor is subjected to filtration, followed by concentration by evaporation. Subsequently one of the above mentioned procedures are used to recover 2,3-butanediol. In some cases vapour-phase steam extraction proved to be profitable.

REVIEW QUESTIONS

I. Essay type questions:

1. Describe the process of acetone-butanol fermentation.
2. What are the possible contaminants of acetone-butanol fermentation?
3. Describe the downstream process in acetone-butanol production.
4. Give an account of substrates used in butanol acetone production and the strains of microorganism employed.
5. Discuss the fermentation economics of acetone-butanol production.
6. Describe fermentative production of acetone-butanol in a corn medium.
7. Give an account of production of acetone-butanol in a molasses medium.
8. Describe the process of production of 2,3-Butanediol.
9. Describe the process of ethanol fermentations.

II. Write short notes on:

1. Clostridium
2. Maintenance of *Clostridium* culture.
3. Activation of *Clostridium* culture
4. Method of inoculation
5. Acid break
6. Butadiene
7. Trinitrotoluene
8. Byproducts of acetone-butanol fermentation
9. Pitching
10. Economic importance of ethyl alcohol.

FURTHER READING

1. Lynd, L.R., Cushman, J.H. Nichols R.J. and Wyman C.E. (1991). **Fuel alcohol from biomass.** Science 251: 1318-1323.
2. Klass, D.L. (1990) *The US biofuel industry*. CHEMTECH December, 720-731.
3. Wiegel, J. (1992). *The obligately anaerobic thermophilic bacterial in Kristjansson,* (ed.) J.K. Thermophilic bacteria, pp. 105–184 CRC Press.
4. Hogsett, D.A., Ahn, A.J. Bernandez T.D, South C. R. and Lynd L. R. (1992). *Direct microbial conversion prospects, progress and obstacles.* Applied Biochemistry and Biotechnology 34-35: 527-541.

4

Organic Acids

Different kinds of organic acids are produced during the biochemical activities of microorganisms. Some of them are derived directly from glucose (e.g. gluconic acid), some are derived indirectly from the Kreb's cycle (e.g. itaconic acid) and still others are formed as end products from pyruvate (lactic acid) or ethanol (acetic acid). Though many organic acids are known as microbial products (table 4.1) only a few of them are commercially produced by fermentation process.

Table 4.1: Microbial production of organic acids

Organic Acid	Microorganism
Acetic acid	*Acetobacter aceti*
Citric acid	*Aspergillus niger*
Gluconic acid	*Aspergillus niger*
Itaconic acid	*Lactobacillus delbrucki*
Kojic acid	*Aspergillus oryzae*
Pyruvic acid	*Pseudomonas aeruginosa*
Propionic acid	*Propionobacterium shermani*
Succinic acid	*Bacterium succcinicum*
Tartaric acid	*Gluconobacter suboxydans*
Malic acid	*Lactobacillus brevis*
Fumaric acid	*Rhizopus delemar*
α-Keto glutaric acid	*Candida hydrocarbofimarica*

4.1 CITRIC ACID

Citric acid is formed as an intermediate in the Kreb's cycle, but it is accumulated in greater quantities in the fungus *Aspergillus niger,* may be due to metabolic abnormality. This aspect of the

fungus is being exploited for the fermentative production of citric acid. It is also found as a natural product in many fruits specially citrus fruits. Prior to the development of fermentation technology, citric acid was extracted from the juice of these fruits. Mexico contributes 1% of world production of citric acid. Citric acid can also be produced from glycerol (*Grimause and Adams, 1985*), but it is very expensive. Though *Aspergillus wentii, A. clavatus, Penicillium divaricatum, P. citrinum, P. luteum, Mucor pyriformis, Citromyces pleffencinus, Candida guilleirmondii, Saccharomyces lipolytica, Trichoderma viride, Arthrobacter paraffinicius* and *Corynebacterium spp.* can produce citric acid, *A. niger* is employed extensively. Citric acid production by surface fermentation was started in 1923, while deep fermentation in 1930. Citric acid produced in 1929 was 5000 tons, which has increased to 4.0 lakhs tons by 1992. Sixty percent of citric acid produced is used in food and beverage industry as a flavouring agent and preservative, while 10% in pharmaceutical industry in the form of iron citrate, about 25% of citric acid is used in chemical industry.

4.1.1 Chemical structure: Structurally, citric acid is a hydroxypropane-1, 2, 3 tricarboxylic acid and is shown in Fig. 4.1.

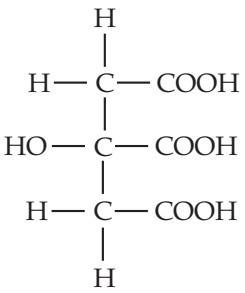


Fig. 4.1: Structure of citric acid

4.1.2 Fermentation process: Various fermentation processes used for the manufacture of citric acid are shown in Fig. 4.2. The ratio of sugar consumed to amount of citric acid produced is normally observed in the ratio of 1:1. *Aspergillus niger* is employed in most of the processes for the following reasons:

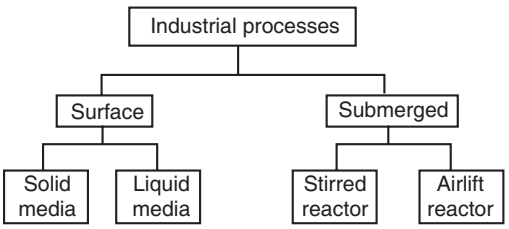


Fig. 4.2: Various fermentation processes for citric acid production

1. Can easily be cultivated.
2. Possesses uniform biochemical properties.
3. Produces only small amount of oxalic acid under controlled conditions.
4. Yield large amount of citric acid

Surface culture process employing solid medium and submerged culture process employing stirred reactor are briefly described here.

4.1.3 Surface culture process: Though it is an old process, it is still employed. It is a kind of stationary fermentation process. This process consists of four phases: (i) inoculum production, (ii) preparation of medium, (iii) fermentation process and (iv) harvest and recovery.

(i) Inoculum production: Spore suspension is used as inoculum for the production of citric acid. Suitable and high yielding strain of *A. niger* is selected from a stock culture. The stock culture is inoculated on to the surface of a sporulating medium present in glass bottles. The bottles are incubated for 10-14 days at 25°C. The composition of trace elements like salts of manganese, zinc or iron in sporulating medium should be suitably maintained, otherwise they will affect the yield of citric acid in the actual fermentation. Suspension of spores is obtained by suspending the grown up spores in a suitable diluents such as water containing a wetting agent, sodium lauryl sulphate. Besides the total number, the viability of spore crop is critical.

(ii) Preparation of medium: The medium, used in the production of citric acid, should have carbohydrate source and inorganic salts. A variety of materials can be used as carbon source. But, generally sucrose and beet molasses are used as carbon source.

Sucrose is the best source of carbon among different organic substances tested. A medium with less than 15% sucrose is reported to give high yield of citric acid. Reduced yield of citric acid is observed when a part of sucrose was substituted by fructose or glucose. Commercially, beet molasses is also extensively used as carbon source in the production of citric acid employing *A.niger*. Apart from sugars, beet molasses also contains excessive amounts of inorganic salts. To eliminate these excessive inorganic salts, it is treated with ferrocyanide or ferricyanide before it is employed in the medium preparation. Alternatively, the inorganic salts are removed by passing the beet molasses through a cation exchange resin.

The elements like nitrogen, potassium, phosphorus and magnesium are also needed in the medium, apart from carbon source. They are added in the form of ammonium nitrate, potassium dihydrogen phosphate or potassium monohydrogen phosphate and magnesium sulphate into the medium in minimum quantity as given in table 4.2. The presence of these elements at higher concentration lowers the yield of citric acid and increases the yield of oxalic acid.

pH of the medium should be adjusted to 3.4–3.5 by using hydrochloric acid. Low pH is reported to be most favourable because it facilitates less contamination, formation of more citric acid, suppression of formation of oxalic acid and easy sterilization of the medium.

Currie (1917) reported a fermentation medium table 4.2 with the following composition for the production of citric acid as the most favourable. Salts and sugars are dissolved in one liter of distilled water. The medium is to be sterilized at $55–10^3$ to $69–10^3$ Nm^{-2} steam pressure per square inch for 30 min.

Table 4.2: Composition of citric acid production medium

Chemical	G/l
Sucrose	125 – 150
NH_4NO_3	2.0 – 2.5
KH_2PO_4	075 – 1.0
$MgSO_4 \cdot 7H_2O$	0.20 – 0.25
pH is to be adjusted to 3.4 – 3.5 by adding 4 – 5ml of HCl	

(iii) Fermentation process: The production medium is placed in shallow pans in such a way that a thin layer of medium with a depth of 1 to 2.5 cm is formed. The spores of inoculum are added to the medium to keep them floating on its surface. This is achieved by suitable modulating devices. Incubation is done in the incubation chambers at 30–40°C. Figure 4.3 shows the layout of the typical fermentation.

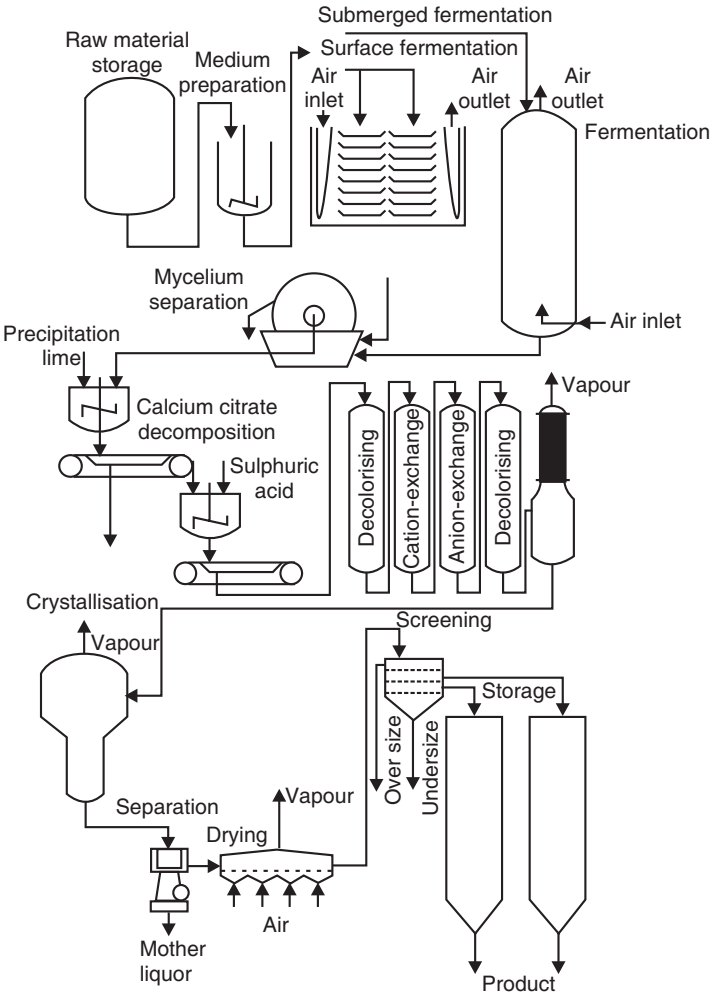


Fig. 4.3: Flow-sheet of citric acid produced by surface or submerged process

The temperature is kept constant at 30°C during the fermentation. Air current ventilation is also important for gas exchange because the rate of citric acid production falls if CO_2 in the atmosphere increases to 10%. Within 24 hrs after inoculation the germinating spore form a thin layer of mycelium on the surface of the nutrient solution. As a result of the uptake of ammonium ions, the pH in the culture liquid fall to 1.5 to 2.0. After 30 hrs of fermentation, if the iron concentration is more oxalic acid and yellowish pigment is formed which inturn hinders the recovery process. The fully developed mycelium floats as thick convoluted white layer on the liquid medium. The fermentation is stopped after 8–14 days.

The rate of bioconversion of sugar to citric acid depends on the ratio of surface area to the volume of the medium. There will be higher yield of citric acid if the ratio is lower. In this shallow pan method, the ratio of surface area to the volume of the medium is lower due to which large surface area of the mycelial mat is exposed to shallow layer of the medium. Under these conditions more and more of sugar is converted into citric acid. That is why this process is considered to be superior to the submerged culture process. Yield per hour from this process amounts to 1.2 – 1.5 kg citric acid monohydrate per square metre of fermentation surface.

(iv) Harvest and recovery: The mycelium is separated from the fermentation broth. Any intracellular citric acid present in the mycelium is obtained by pressing the mycelium. The filtered broth is treated with calcium hydroxide. It is filtered and washed. It is then treated with equal volume of sulphuric acid to liberate citric acid. Calcium sulphate is formed as a precipitate in this process. The precipitate is separated by filtration. An impure solution of citric acid is obtained which is decolorized by treating with activated carbon and also demineralized. Finally pure citric acid crystals are produced by evaporation. It is also recovered alternatively by counter current extraction method.

4.1.4 Submerged culture process: In this method *A. niger* is made to grow uniformly dispersed throughout the liquid production medium. Fermentation is generally carried out in large fermenters having 4klt a capacity of thousands of gallons and are provided with mechanical agitator and sparges. Eighty percent of the worlds supply of citric acid is produced by this process. Cost of production decreases by 25% by this method. It involves low labour cost, longer incubation period, more energy consumption and sophisticated techniques. Three factors are important for production in submerged culture process. They are quality of the metal used for the construction of fermenter, mycelium structure and oxygen supply. *Candida lipolytica,* an alkane utilizing fungus can also be employed in citric acid production under continuous fermentation. It yields, 45% higher than normal citric acid production.

(i) Inoculum production: Mycelial mats called pellets are used as inoculum for fermentation in this process. Suitable and high yielding strains of *A. niger* are selected from a stock culture. The spores are induced to germinate in a seed fermenter. A nutrient solution containing 15% sugar from molasses is used in this seed fermenter. To induce the formation of mycelial pellets, cyanide ions are added to the medium. Pellet formation largely depends upon the concentration of cyanide ions in the medium. Lower yield of citric acid occurs if the cyanide ions are in less concentration. This is because lower concentration of cyanide ions induce formation of normal mycelium instead of pellets. The spores germinate at 32°C and form pellets of 0.2—0.5 mm diameter within 24 hrs. During this period the pH falls to 4.3. These pellets are then used as inoculum for production fermenters.

(ii) Preparation of medium: The medium, employed for surface culture process is also employed in this process.

(iii) Fermentation process: Mostly fermentors used for citric acid production are constructed in the range of 10—220 klt. They must be made of stainless steel to prevent leaching of heavy metals. Normal steel, if it is used in the construction of fermenters, at low pH level of 1-2 may inhibit the formation of citric acid. Small fermenters with a capacity upto 1000 lt should have plastic lining even though they are made of stainless steel because of large surface/volume ratio. However, such plastic lining is not necessary for large stainless steel fermenters.

The mycelial pellets developed in the seed tank are transferred aseptically to the fermenters and incubated at a constant temperature, 30°C. The structure of the mycelium that forms in the fermenter is vital to a successful production process. Little citric acid is produced if the mycelium is loose and filamentous with limited branches and no chlamydospores. Optimal citric acid is formed if the mycelium is in the form of pellets. The ratio of iron to copper in the medium determines the nature of mycelium. In some cases, production fermenters are inoculated directly with spores.

Although *A. niger* requires relatively little oxygen, it is sensitive to oxygen deficiency. There must be minimum oxygen concentration of 20 to 25% of the saturation value throughout the fermentation process. Short interruptions in the oxygen supply ceases the production irreversibly. The aeration rate should be 0.2–1.0 volume per min. during the acid production phase. Due to low viscosity, stirring is not necessary. Thus, although some plants use stirred fermenters, airlift reactors can also be used.

Foaming is a problem in the submerged culture process. However, it can be controlled by adding antifoam agents such as lard oil at frequent intervals. A foam chamber, $1/3^{rd}$ the size of the fermenter volume, is needed in both airlift and stirred bioreacters. Mechanical antifoam devices can also be used. Progress of the fermentation process is monitored regularly by calculating the content of sugar and citric acid in the fermentation.

4.1.5 Uses: Industrially, citric acid is used in the following ways:

1. It is used in the production of carbonated beverages.
2. As a chelating and sequestering agent in the tanning and textile industry.
3. Citrate esters are used as plasticizer.
4. It is abundantly used in food industry as an acidulent in the preparation of food items like jams, preserved fruits and fruit juices etc.
5. It is used in frozen foods to prevent its change in colour and flavour.
6. Metal painting industry
7. In pharmaceutical industry
8. In the manufacture of astringent, hair rinsers and hair setting fluids.
9. In beverage industry as a preservative to prevent oxidation of alcohol, emulsifier of dairy products like cheese and ice creams.
10. It is used as preservative and to prevent change in colour, flavour and in the oxidation of alcohol.

4.2 LACTIC ACID

Lactic acid was the first organic acid produced by fermentation process carried out in 1880. Today, it is manufactured both by chemical as well as fermentation methods. Chemically it is produced from lactonitrile, degradation of sugars, from carbon monoxide and acetaldehyde.

4.2.1 Chemical structure: The structure of lactic acid is shown in Fig. 4.4. It consists of an acid functional group (–COOH) and a hydroxyl group (–OH). Though it contains hydroxyl functional group, it exhibits more acidic properties. It contains an asymmetric carbon atom due to which lactic acid exists in two forms as L (+) lactic acid and D-lactic acid. L-lactic acid is useful as it can be metabolised by animal or human body, while the D-configuration is not metabolized and is largely excreted. However, in industrial fermentations racemic mixture of both D and L is obtained.

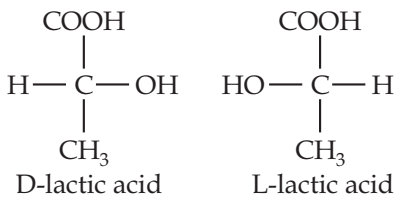


Fig. 4.4: Structure of lactic acid asymmetric carbon

4.2.2 Biosynthesis: The biosynthesis of lactic acid from glucose proceeds via glyceraldehyde-3-phosphate, 1, 3 diphosphoglycerate and pyruvate. The reducing power is produced during the oxidation of glyceraldehyde. Phosphate is transferred with NAD–dependent lactate dehydrogenase to pyruvate which reduces stereospecifically to L(+) or D(–) lactic acid (Fig. 4.5).

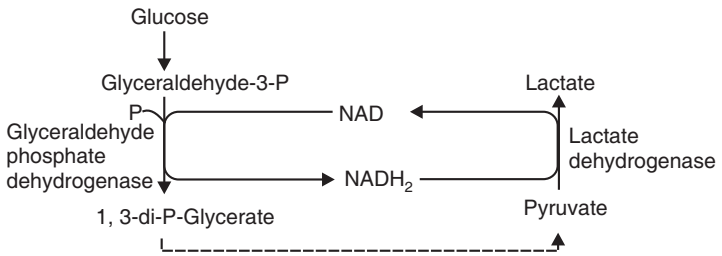


Fig. 4.5: Lactic acid production from glucose

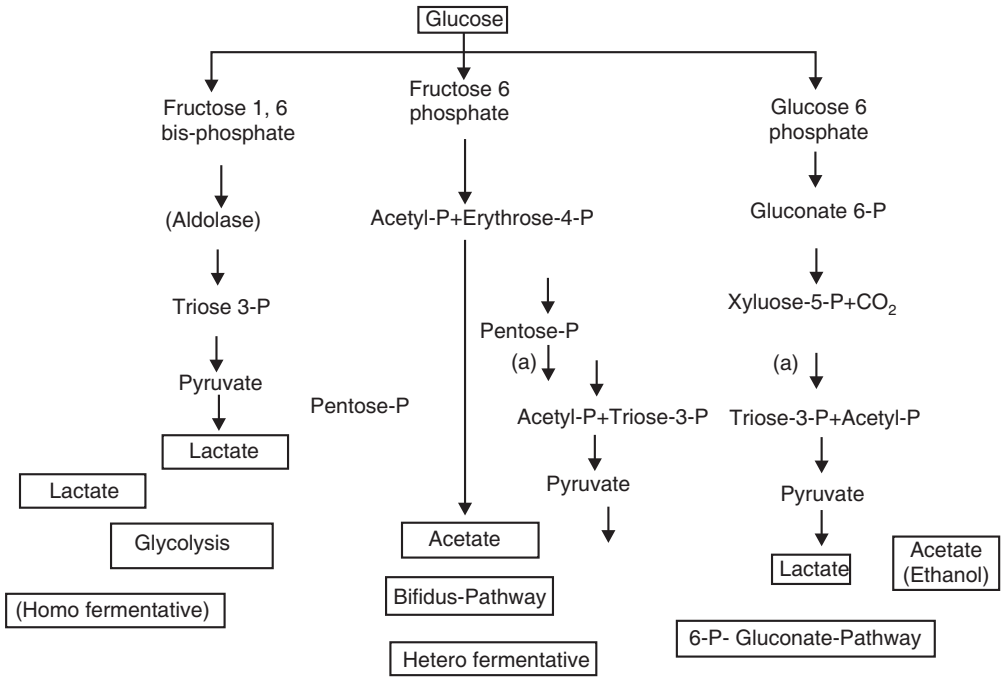
4.2.3 Fermentation process: Both homofermentators like *Lactobacillus bulgaricus, L. plantarum, L. delbrueckii, Streptococcus lactis and Pediococcus sp.* and heterofermentors like *leuconostoc mesenteroides* produces lactic acid. Characteristics of bacteria and fungi producing lactic acid both by batch and continuous fermentation, alongwith organisms employed, type of bioreactor, raw material used, lactic acid productivity are precised in table 4.3 and 4.4, respectively. Different bacteria utilize different metabolic pathway for production of lactic acid (Fig. 4.6).

Table 4.3: Lactic acid production by batch fermentation

Microorganisms	Raw material (gl^{-1})	Lactic acid productivity (gl^{-1} hr^{-1}) (P) Concentration (c) (gl^{-1})	Yield g^{-1} Substrate utilized
Bacteria:			
L. amylophilus	Corn, 45	P : 0.55, C : 31.2	0.68
	glucose, 20	P : 1.56, C : 21	0.93
L. amylovorus	Starch (enzyme thinned)	P : 7.36 C : 96	0.94
L. casei subsp. *thamonus*	Glucose	C : 120–135	0.80–0.90
L. delbrueckii subsp. *bulgaricus*	Cheese, whey and permeate, 50-150	P : 4.4–5 C : 60–155	0.91–0.99
L. helveticus	Cheese, whey and permeate, 39.2.54	P : 3.7–3.8 C: 35–40	0.93
Molds:			
R. arrhizus	Glucose, 100	C : 79	>0.50
R. oryzae	Glucose, 130	C : 93	0.72
	Corn, 150	C : 53.2	0.44

Table 4.4: Lactic acid production by continuous fermentations

Microorganisms	Type of bioreactor, raw material	Lactic acid productivity (P) (gl^{-1} hr^{-1}) Concentration (c), (gl^{-1})	Yield gg^{-1} Substrate utilized
L. amylovorus	Membrane cell recycle, starch	P : 8.4 C : 42	0.88 – 0.92
L. casei sub sp *thamnosus*	CSTR, glucose	P : 8.93 C : 22–25	0.74
	Membrane cell recycle, glucose	P : 10–160 C : 34–59	0.95–0.99
	Immobilized cell, glucose	P : 3–20 C : 5–51	0.76–0.97
L. delbrueckii subsp. *bulgaricus*	CSTR, cheese whey	P : 1.77 C : 55	0.98
	Membrane cell recycle, cheese whey permeate	P : 84–85 C : 43–117	0.99
	Dialysis, cheese whey	P : 11.7 C : 80	0.97
L. helveticus	CSTR, cheese whey permeate	P : 3.7–5 C : 35–84.6	0.83–0.93
	Membrane recycle, cheese whey permeate	P : 15.8 C : 15	0.70
	Immobilized cell, cheese-whey permeate	P : 2.6–13.5	0.50–0.95



(a) Phosphoketolase-pathway

Fig. 4.6: Biosynthetic pathways of lactic acid

(i) Inoculum production: Microorganisms used in the lactic acid production are the species of *Lactobacillus*. Two types of lactic acid bacteria are recognized Heterofermentative and Homofermentative organisms. Heterofermentative organisms produce large number of byproducts and, therefore, not employed in commercial process. The homofermentative organisms produce very little number of byproducts and more lactic acid, therefore, these are employed in lactic acid fermentation (table 4.5).

Table 4.5: Characteristics of selected bacteria and molds of interest in lactic acid production

Microorganism	Lactic acid isomer	Fermentation pattern	Raw material
Bacteria: *Lactobacillus amylophilus*	L (–)	Homofermentative	Starch
L. amylovorus	DL	Homofermentative	Starch
L. casei subsp. rhamnosus (L. delbueckii NRRL B-445)	L (+)	Facultative hetero-fermentative	Glucose, sucrose (molasses)
L.delbueckii subsp. bulgaricus	D (–)	Homofermentative	Cheese whey and permeate (Lactose)
L. helviticus	DL	Homofermentative	Cheese whey and permeate (Lactose)
Molds: *Rhizopus arrhizus* *R. oryzae*	L (+) L (+)	Homofermentative Homofermentative	Glucose, starch Glucose, starch

The selection of species of *Lactobacillus* largely depends upon the nature of the carbon source being used in the fermentation. *Lactobacillus delbrueckii* and *Lactobacillus leichmannii* are employed when glucose is used as substrate, *Lactobacillus bulgaricus* when whey is used as substrate and *Lactobacillus pentosus* when sulfite waste liquor is used as substrate. These organisms are facultative rather than obligate anaerobes and therefore, bioreactors need not be run with complete oxygen exclusion. Suspension of pure bacterial culture is prepared from the suitable high yielding strain of stock culture. The selected *Lactobacillus* culture is transferred to large vessel and the temperature is maintained at 45 to 50°C. Each stage of culture building requires 16-18 hours depending upon the volume of fermentation broth. About 5% inoculum volume is usually used in the fermentation process.

(ii) Preparation of medium: To satisfy the nutritional requirements of the microorganisms the medium should be prepared in such a way as to contain carbon source, nitrogen source, mineral source and source for growth factors. Glucose, maltose, lactose etc. or crude substrates such as corn starch, potato and rice, whey, molasses, sulfite waste liquor etc. are used as carbon source. Pretreatment of starchy material by amylase or by sulphuric acid is necessary to bring about hydrolysis and to change complex starch into simple sugars like glucose and maltose. If sulfite waste liquor is used in the medium, it is essential to steam strip the sulfite to lyse to sulphur dioxide and also to treat it with alkali, to remove lysis during the pretreatment operation.

The sugar content in the medium is maintained at 5-12%. Though various ammonium salts can be used, ammonium hydrogen phosphate is usually employed as nitrogen source in the fermentation of lactic acid. About 0.25% of ammonium salts are added to the medium. Based on the requirement of the selected microorganism, minerals are added to the medium. Calcium carbonate is also added as a buffering agent to control pH.

Lactobacillus requires vitamins, specially vitamin B complexes for proper growth. These are added to the medium in the form of crude vegetable sources like the rootlets of germinating seeds of barley.

(iii) Fermentation process: Lactic acid production is carried in 25-125 klt fermenters. Lactic acid is quite corrosive, therefore, wooden fermenters are used commercially rather than metal fermenters. The pH of the fermentation broth is maintained at 5.5-6.5 by using calcium carbonate. Maintenance of pH is essential because it helps not only in the increase of fermentation rate but also in the increase of the yield. Maintenance of low pH values also helps in the elimination of bacterial contamination and also in the sterilization at low temperatures. As more and more lactic acid is formed it may become toxic to the organism. Hence, neutralization of accumulated acid is done by the continuous addition of calcium hydroxide. Generally, the fermentation duration is 5-10 days. It can also be completed in 72 hours if 12 to 13% of glucose is used in the medium and pH is maintained at 6.3-6.5 by continuous neutralization of the acid formed. However, complete non supply of air is also not advisable because the bacteria employed are facultative aerobes. The fermentation broth is gently agitated to keep the uniform distribution of nutrients and air. Though temperature is generally maintained at 45-50°C, maintenance of particular range of temperature depends to some extent on the type of organism employed. For example *Lactobacillus delbrueckii* and *L.bulgaricus* requires 45-50°C, while *L.casei* and *L.pentosus* requires 30°C. Theoretically two molecules of lactic acid are produced from one molecule of glucose. Over 90% of this theoretical yield is actually attained in practice. The commercially produced lactic acid belongs to L(+) type.

(iv) Harvest and recovery: Technical or crude grade which is coloured and available with 22, 44, 50, 60 and 80% concentrations, while edible grade is straw coloured and is marketed 50 to 80% strength of colourless high purity lactic acid, the plastic grade marketed at 50 to 80% strengths and U.S.P. lactic acid is marketed at 85% strength. Other commercial preparations of lactic acid are calcium lactate, sodium lactate and copper lactate. The recovery method varies depending upon the grade of lactic acid required. However, a general method of recovery process, described here, has the following steps:

1. The fermentation broth is filtered by using porcelain filters to separate the bacterium.
2. The filtrate is acidified with sulphuric acid to regenerate lactic acid, to precipitate calcium as calcium sulphate and is washed.
3. The washed filtrate is treated with activated carbon to remove organic impurities.

4. Repeated refining and evaporation steps are undertaken to get higher percentage of lactic acid to the extent of 50-60%.
5. It is then treated with ferrocyanide to remove heavy metals, if any, like copper.
6. It is finally purified by passing through ion exchange resins.

Lactic acid is also extracted and purified by some other methods such as: 1. Zinc salt of lactic acid is made which has low solubility and can be recovered, 2. Lactic acid is directly extracted with isopropyl ether by counter current method, 3. Methylester of lactic acid is prepared which is easily hydrolysed with boiling in water. Methyl alcohol is recovered, 4. High vacuum distillation of lactic acid.

4.2.4 Uses: It has many uses. Some of them are described below:

1. It is a weak acid with good solvent properties. It is used as a preservative in food industry.
2. It readily polymerizes and, therefore, used in polymer industry.
3. It is also used to treat the fibers in the textile industries and laundry.
4. Calcium lactate is employed as baking powder and as a source of calcium in pharmaceutical industries.
5. Pure lactic acid is used in plastic industry.
6. Ethyl lactate is used in preparation of anti-inflammatory drugs.
7. It is used in manufacture of self-dissolving suture threads in surgery.
8. As a blood coagulant and dietary calcium source.
9. In the manufacture of cellophane resins, biodegradable plastics and some herbicides and pesticides.
10. It is used in food industry as acidulent as it has strong odour or flavour. Ethyl and butyl lactate are used as flavouring ingredients.

4.3 ACETIC ACID

Acetic acid (CH_3COOH) is also called as vinegar. Vinegar fermentation is one of the oldest fermentations known to man. It is formed naturally due to spoilage of wine. Therefore, literally vinegar means "sour wine." Technically vinegar is fermentated food product consisting of about 4 g of acetic acid per 100 ml. Vinegar was produced only for local consumption until the middle ages. The Romans and Greeks, who used diluted vinegar as refreshing drink, produced vinegar by leavening wine open to air. Industrially for the first time it was produced in open vats, which however, was a very slow process. But the process was increased many fold in the 19th century by surface fermentation technique in which trickling generators were used. This process is used still today. From 1949, submerged process was employed in the fermentative production of vinegar. Presently both these processes, that is surface fermentation and submerged fermentation are employed worldwide. The surface fermentation is used still today because of the better flavour of the product.

The production of vinegar actually involves two fermentation processes: the first utilizing yeast to produce alcohol from sugar and the second utilizing acetic acid bacteria to oxidize ethyl alcohol acetic acid through acetaldehyde (Fig. 4.7). The fermentative production of ethanol is explained in the chapter on beverages.

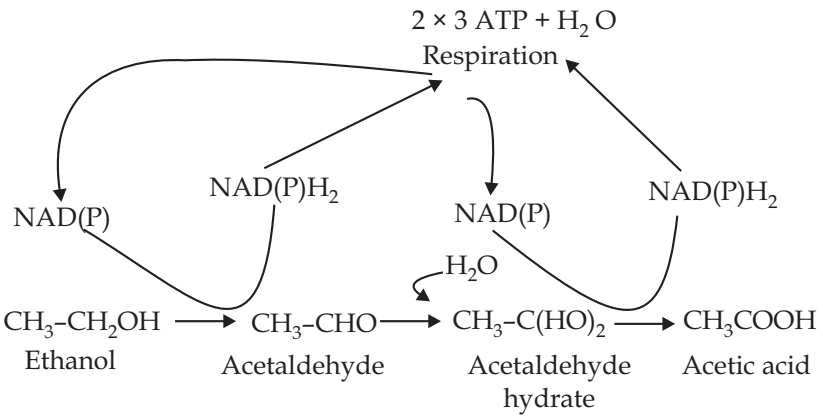


Fig. 4.7: Mechanism of oxidation of ethanol to acetic acid

The microbial oxidation of ethanol to acetic acid is an aerobic fermentation that has high oxygen requirement. *Acetobacter* bacteria are employed for industrial production of vinegar. *Acetobacter* bacteria can be divided into two groups, *Gluconobacter* and *Acetobacter*. *Gluconobacter* oxidizes ethanol to acetic acid, while *Acetobacter* oxidizes ethanol first to acetic acid and then to CO_2 and H_2O. Species of the *Acetobacter* used commercially are *Acetobacter aceti* and *A. pasteurianum*. Similarly, *Gluconobacter oxydans* and its subspecies are employed in the commercial production of vinegar. Mixed cultures, sometimes appear during production even though pure culture is used especially in surface process.

Two oxidation steps occur during the conversion of ethanol to acetic acid. In the first step ethanol is oxidized to acetaldehyde in the presence of NAD or NADP and in the second step acetaldehyde is changed to acetic acid and render the catalytic action of the enzyme alcohol dehydrogenase (Fig. 4.7). In this oxidation one liter of 12% acetic acid is produced from one liter of 12% alcohol that is one mole of acetic acid is formed from one mole of alcohol.

4.3.1 Production process (Fig. 4.8): Commercially acetic acid is produced by two methods, surface fermentation process and submerged fermentation process. If materials with low alcohol content are used such as wine, whey, malt or cider there is no need of addition of any component to constitute a complete nutrient solution. However, if potato or grain spirits or technical alcohol is used, nutrients must be added to obtain optimal growth and acetic acid production. Nutrient concentration that is used in submerged fermentation is generally five times greater than surface fermentation.

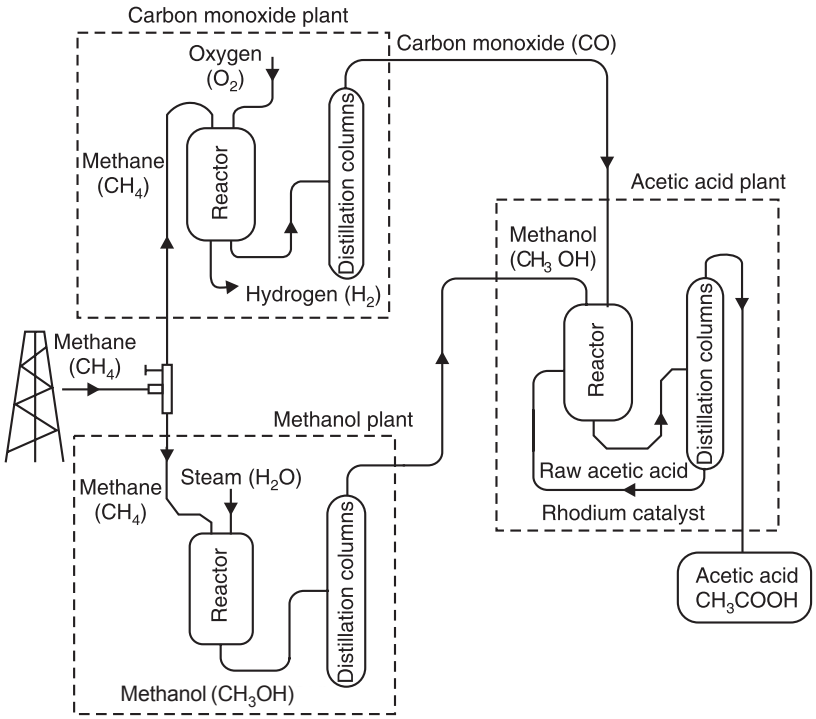


Fig. 4.8: Production process of acetic acid

(a) Surface fermentation process: Trickling generator is generally used in this process (Fig. 4.8a). It is made up of wood and has a total volume upto 60 m^3 and its inner surface is lined with birch wood shavings. The starting material, that is, ethanol is passed into the generator from top which trickles through the birch wood shavings containing bacteria into bottom basin where the partially converted solution is cooled and pumped back again to the top of the generator and passed again through it. Thus, the process is repeated again and again until 88-90% of alcohol is changed to acetic acid. The starting material should contain both acetic acid and ethanol for optimal growth of *Acetobacter*. However, ethanol supply is critical because if it is less than 0.2% (v/v) in solution the death rate of bacteria increases, but its concentration should not increase above 5%. Presently higher yielding strains are employed in vinegar fermentation which are able to yield 13-14% of acetic acid.

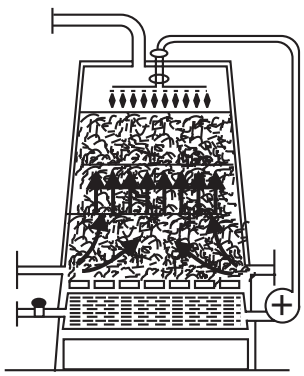


Fig. 4.8(a): Trickling generator for acetic acid production

(b) Submerged fermentation process: The production rate of ethanol with submerged fermentation process is ten times higher per cubic meter than the surface fermentation process. Other advantages include lower capital investment for production, 20% plant area required for installation, conversion to other mashes in a short time and low manual cost because of fully automatic control. Material with low alcohol concentration such as fruits, wines and special mashes were first used in the initial stages of submerged fermentation process, which generally do not require aeration. But presently high yielding materials are employed which are capable of yielding 13% acetic acid. However, the process with such high yielding material requires high aeration upto 50 m^3 oxygen. Fermenters constructed with stainless steel are employed and they are stirred from bottom. Aeration is provided with a suction rotor, with the incoming air coming down through a pipe from the top of the vessel (Fig. 4.9). Heat exchanger is provided to control the temperature along with foam eliminators. Domestic vinegar is produced through semi continuous fully automatic process under continuous stirring and aeration. The starting material 100 ml of 5% ethanol are used in the process to get 7-10 g acetic acid/100 ml. The fermentation process is carried upto 35 hours at 40°C temperature. The ethanol concentration is continuously measured and when its concentration goes down to 0.05%-0.3%. 50%-60% of the solution is removed and replaced with a new mash with 0.2 g acetic acid/100 ml and 10-15% ethanol. The yield of acetic acid is about 98% in fully continuous process.

(c) Recovery: The vinegar produced in a submerged fermentation process is turbid due to presence of bacteria. It is clarified by filtration. Plate filters and filter aids are generally used. After filtration $K_4[Fe(CN)_6]$ is used to decolorize the final product, if required.

The submerged fermentations also utilize two different fermenter designs known as Acetator and Cavitator (Fig. 4.10). Both these fermenters provide the high aeration levels required for acetic acid production. These intact pump recycle their own air so that a compressor is not required. This aeration approach is advantageous for acetic acid fermentation. Since it lessens the loss of ethanol and aromatic substances present in the raw materials. Automatic temperature control is applied to dissipate the heat resulting from the ethanol oxidation. These two fermenters differ in their mechanical operations and in the fermentation programes. The acetator usually operates in semi batch manner, while cavitator in nutrient solution and air is sucked down to a hollow tube extending from the liquid surface so that agitation and cavitation fermentation extends to the bottom of tube resulting in mixing of air with nutrient solution.

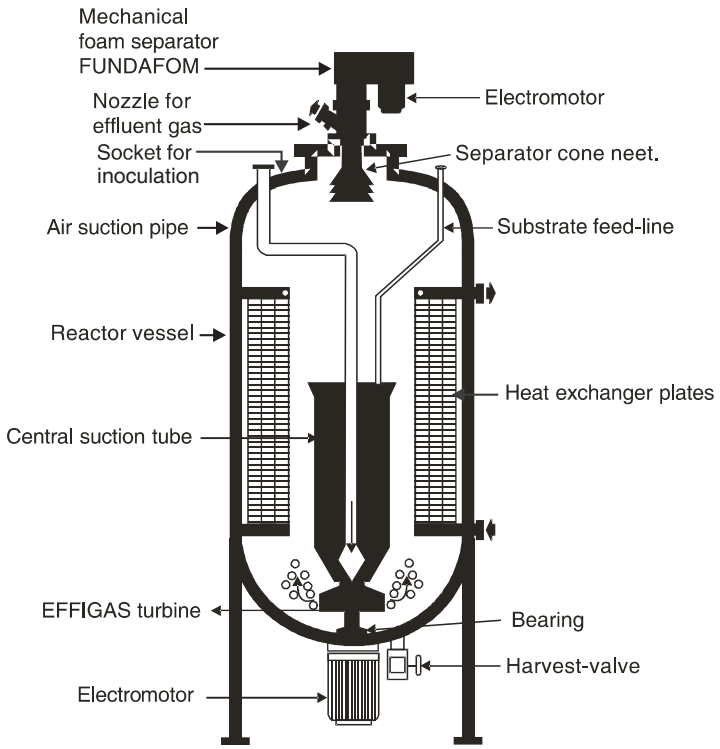


Fig. 4.9: Submerged fermenter for acetic acid production

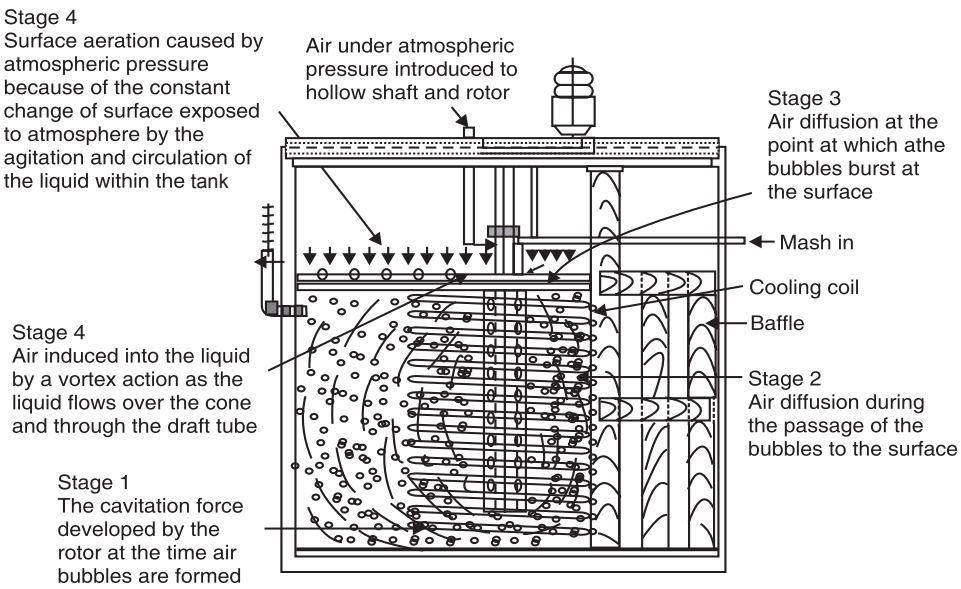


Fig. 4.10: Cavitator

4.4 GLUCONIC ACID

In industrial microbiology gluconic acid has a long history. **Alsberg** in 1911 explained the production of gluconic acid in which *Pseudomonas* was used as fermenting microorganism. In 1928, its production using the fungus *Penicillium luteum* by surface fermentation process was employed for the first time.

Gluconic acid is presently produced commercially either by employing the fungus *Aspergillus niger* or the bacterium, *Acetobacter suboxydans* through submerged fermentation process, in which gluconic acid, sodium and calcium gluconate and glucose oxidase are produced. Apart from the above two microorganisms, other organisms are also reported to produce gluconic acid, however, they are not used for commercial production. The organisms include the fungi *Endomycosis, Pullularia, Penicillium, Gonatobotrys* and *Scopulariosis* and the bacteria *Pseudomonas* and *Vibrio.*

4.4.1 Biosynthesis: Gluconic acid is produced from glucose. This, reaction is catalysed by the enzyme glucose oxidase (Fig. 4.11). The gluconolactone formed at the end of the first step undergoes hydrolysis either spontaneously or enzymatically to produce gluconic acid. During the first reaction, the hydrogen from $FADH_2$ is transferred to oxygen leading to the formation of H_2O_2 which is immediately split into water and oxygen by the enzyme catalase, to prevent its antimicrobial activity.

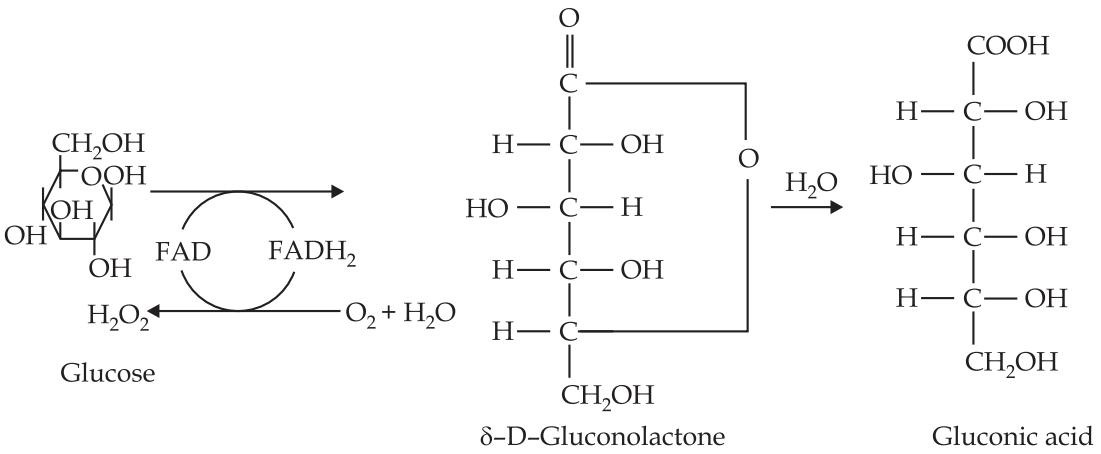


Fig. 4.11: Formation of gluconic acid from glucose by *A. niger*

4.4.2 **Fermentation:** Gluconic acid is produced industrially by employing the fungus or the bacterium. In the former process *Aspergillus niger* and in the later process *Acetobacter suboxidans* (Fig. 4.12) are used. (Fig. 4.13) *A. niger* employs glucose oxidase involving agent like FAD and lactonase in the presence of O_2 resulting in the formation of gluconic acid. On the other hand, *Gluconobacter* employs glucose dehydrogenase with coenzyme pyrroloquinoline quinone (PQQ) and lactonase which help in dissipating hydrogen peroxide.

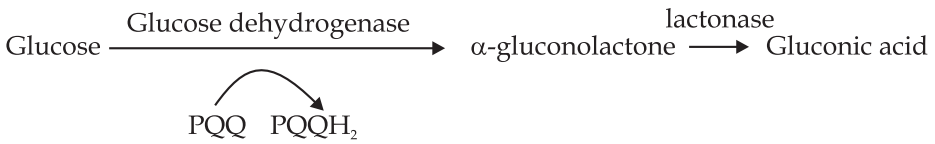


Fig. 4.12: Formation of gluconic acid from glucose by *Acetobacter suboxydans*

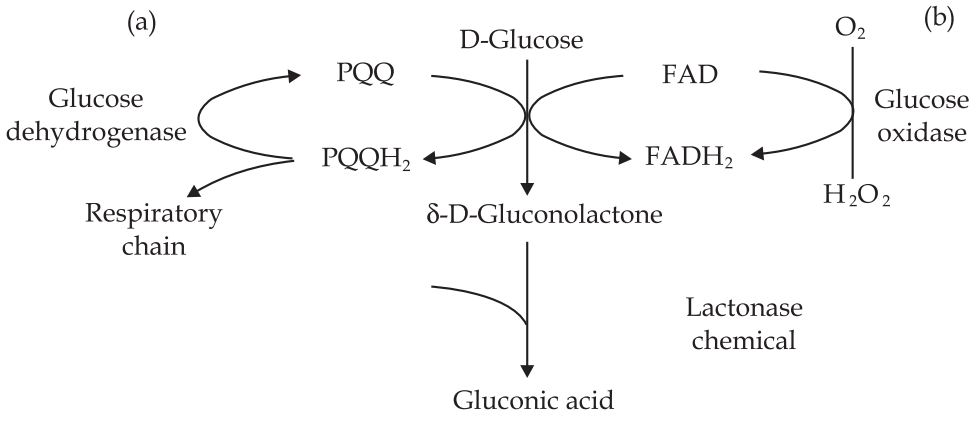


Fig. 4.13: Enzymic reactions leading to gluconic acid formation in (a) *G.suboxydans* (b) *A. niger*

(a) Fungal fermentation: Submerged fermentation process is employed for fungal fermentation. *Aspergillus niger* is used as a microorganism. Either sporulated culture or spores germinated in seed tank is used as inoculum. Each method has its own advantages. For example, use of spore inoculum directly avoids the cost of installation and operation of the seed tank, while use of germinated spores reduces operating cycles for the main fermenter.

Table 4.6: Composition of gluconic acid production medium

Substratum /chemical	Amount (in percentage)
Glucose	28-30%
Corn steep liquor	3.70
$MgSO_4 \cdot 7\,H_2O$	0.17
KH_2PO_4	0.20
Urea	0.10
$(NH_4)_2HPO_4$	0.40
H_2SO_4 to adjust pH at 4.5	
Tap water	1000 ml

Glucose is used as a solution or crystalline glucose or in the form of a syrup prepared from starch or crude starchy material which are treated with amylase and amyloglucosidase (table 4.6). It is necessary to maintain maximum concentration of dissolved oxygen in the form of solution by vigorous agitation with the help of turbido mixture or cavitator. In the manufacture of calcium or sodium gluconate optimum temperature is maintained at 28–30°C and pH 6.5 along with vigorous agitation and aeration.

(i) Recovery and harvest: After fermentation, the fungal mycelium is separated by filteration, the separated mycelium is used for the recovery of glucose. On the other hand, the filtrate is used for the recovery of calcium or sodium gluconate.

(ii) Production of calcium gluconate:

The following steps are employed for the recovery of calcium gluconate:

1. The filtrate is heated with excess of $Ca(OH)_2$.
2. The resulting product is decolorized with activated carbon and filtered.
3. The compound is crystallized by cooling at a temperature below 20°C and seeding with calcium gluconate crystals.
4. The mother liquor left over (about 10–15%) is used for the production of second crop of calcium gluconate through heating followed by treatment with carbon filteration and chilling.

(iii) Production of sodium gluconate: Recovery of sodium gluconate is relatively simple and consists of the following steps.

1. The filterate is concentrated to about 42–45 percent.
2. Then sodium hydroxide is added which not only crystallizes sodium gluconate but also maintains pH at 7.5.
3. The sodium gluconate thus formed is drum dried.

(iv) Production of pure gluconic acid: It is also possible to obtain gluconic acid in pure form by the following procedure, where calcium gluconate is employed:

1. Calcium is precipitated by the addition of sulfuric acid. Calcium sulfate thus formed is separated by filteration.
2. The filtrate is decolorized with activated charcoal.
3. The acid solution is concentrated to 50 percent acid strength. The product thus formed is a mixture of gluconic acid and lactones.
4. The concentrate is then treated with a temperature ranging from 0°C-30°C. The crystals that separate at this temperature consist of pure gluconic acid.
5. When the concentrate is treated with a temperature ranging from 30°-70°C crystals of lactones separate.
6. Y-lactone crystals get separated when the concentrate is treated with 70°C and above temperature.

(b) Bacterial fermentation: Submerged fermentation process is used. The bacterium employed is *Acetobacter suboxydans* or *Gluconobacter suboxydans*. Pure culture of the bacterium is raised by repeated subculturing process which is used in the fermentative production of gluconoic acid and is produced either in the form of calcium gluconate or sodium gluconate.

Glucose is used as a substratum in the fermentation. If calcium gluconate is to be produced 13-15% glucose can be added as a substrate because of low solubility of calcium gluconate (4 g liter^{-1} at 30°C). Higher calcium gluconate levels are formed if glucose is used above 13% levels which would spontaneously crystallize as calcium gluconate, and makes purification difficult. If sodium gluconate is to be produced, a glucose concentration of 28-30% can be used.

The fermentation is carried out at a temperature of 28-30°C, pH 4.5–6.5 with a high aeration rate of 1-1.5 vvm. The gluconic acid yield can be substantially increased (90–95%) by increasing the solubility of oxygen which is achieved by raising the pressure in the system.

4.4.3 Uses:

1. Gluconic acid is used in the manufacture of metal, leather and food.
2. Sodium gluconate is used as a sequestering agent in many detergents.
3. Calcium gluconate is used in medicine.
4. Gluconolactone is used as baking powder and as an additive.

REVIEW QUESTIONS

I. Essay type questions:

1. Describe the fermentative production of citric acid. Add a note on its applications.
2. Describe the fermentative production of lactic acid.
3. Describe the mechanisms of lactic acid production by lactic acid bacteria.
4. Describe the downstream process for lactic acid purification. Add a note on grades of lactic acid.
5. Narrate the different events in acetic acid fermentation.
6. Define dual fermentation and describe the process with reference to acetic acid production.

7. Describe the following fermenters (a) Cavitator (b) Acetator.
8. Describe the process of gluconic acid production.

II. Write short notes on:

1. Heterofermenters
2. Uses of lactic acid
3. Grades of lactic acid
4. Uses of citric acid
5. Calcium gluconate
6. Sodium gluconate
7. Exergonic fermentation
8. Dual fermentation
9. Auxotroph
10. Indirect fermentation of L-Lysine
11. Use of gluconic acid
12. Role of biotin in L-glutamic acid fermentation
13. Aspartine
14. Gene regulation in phenylalanine production

FURTHER READING

1. Kascak, K., Kominek, J. and Roehr, M. (1996). **Lactic acid**. *In Biotechnology*, Products of Primary Metabolism, 2nd edition. Vol. 6: (H.J. Rehm and G. Reed, eds.; M. Roehr, volume editor), pp. 294-306. Verlag Chemie, Weinheim.
2. Roehr, M., Kubicek, C.P. and Kominek, J. (1992). **Industrial acids and other small molecules**. In *Aspergillus: Biology and Industrial Applications* (eds. J.W. Bennett, and M.A. Klich,), pp. 91–131. Buttetworth-Heinemann, Reading, MA.
3. Roehr, M., Kubicek, C.P. and Kominek, J. (1996). **Citric acid**. In Biotechnology, Products of Primary Metabolism 2nd edition, Vol. 6: [eds. H.J. Rehm and G. Reed, ; M. Roehr, volume editor), pp. 308–345. Verlag Chemie, Weinheim.

   Roehr, M., Kubicek, C.P. and Kominek. J. (1996). **Further organic acids.**
4. In Biotechnology, Products of Primary Metabolism 2nd edition, Vol. 6:

   (eds. H.J. Rehm and G. Reed; M. Roehr, volume editor), pp. 364–379. Verlag Chemie, Weinheim.
5. Roehr, M., Kubicek, C.P. and Koninek. J. (1996). **Gluconic acid** (In Biotechnology, Products of Primary Metabolism eds.; H.J. Rehm and G. Reed, 2nd edition, Vol. 6: M. Roehr, volume editor), pp. 347–362, Verlag.
6. Stiles, M.E. and Holzapfel, W.H. (1997). **Lactic acid bacteria in food and their current taxonomy.** International Journal of Food Microbiol. 36, 1-29.

5

Amino Acids

About 20 amino acids are synthesized in the cell of the most microorganisms. They are utilized in the synthesis of proteins and other essential substances required by the cell. Fermentative production of amino acids has started by the discovery of glutamic acid producing bacterium, *Corynebacterium glutamicum (Micrococcus glutamicum),* by **Kinoshita et al. (1957)**. Since then much research work has been carried out on fermentative production of amino acids. Large number of microorganisms were isolated from nature which are capable of producing amino acids by fermentation in commercially feasible quantities, especially from the auxotrophic bacteria (table 5.1).

Table 5.1: Production of amino acids by fermentation

Amino acid	Strain used	Genetic characteristics	Yield (gl^{-1})	Carbon source
D-L-Alanine	*Microbacterium ammoniaphilum*	ArgHx	60	Glucose
L-Arginine	*Serratia marcescens* AT 428 (aru argR2 arg A2)	Transduction Canavanine	50	Glucose
L-Glutamic acid	*Corynebacterium glutamicum* *Brevibacterium flavum* *Arthrobacter paraffineus*	Wild type	100 98 82	Glucose Acetate n-Alkanes
L-Glutamine	*Corynebacterium glutamicum*	Wild type of glutamic acid producer	58	Glucose, with high biotin and NH_4Cl content
L-Lysine	*B. lactofermentum* Aj 11204 *B. flavum*	AEC^r Ala^- $CCL^rML^rFP^s$ Hom^{leaky} $Thre^-$	70 75	Glucose Acetate

contd...

Amino acid	Strain used	Genetic characteristics	Yield (gl^{-1})	Carbon source
L-Proline	*C. acetocidophylum*	Uncharacterized mutant	108	Glucose + Glutamic acid
L-Threonine	*E.coli* VL 344 (pYN7)	$_R$DNA	55	Sucrose
L-Tryptophan	*E.coli JP 4114*	$_R$DNA	23	Glucose
L-Valine	*Brevibacterium lactofermentum* No. 487	TA^r	31	Glucose

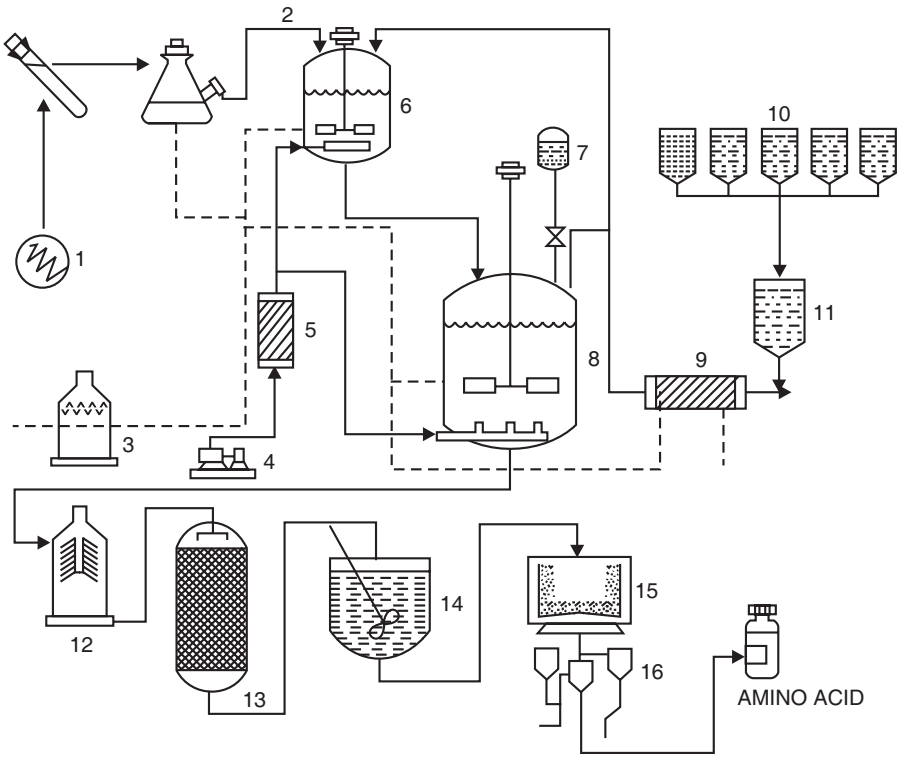


Fig. 5.1: Layout of fermentation plant for production of amino acids

The details of production of different amino acids by fermentation involving different microorganisms are precised in table 5.2. The general layout of fermentation plant for production of amino acids is illustrated in Fig. 5.1. Of the various amino acids, L-glutamic acid and L-lysine are in great demand for commercial production. Auxotrophs are used in the fermentation production of these two amino acids. The major portion of the amino acid accumulation occurs

after log phase of growth of the microorganism. The concentration of specific growth factors is very crucial for the maximum yield of the product because concentration above or below the optimum substantially, reduces the yield.

Microorganisms, generally, do not produce amino acids in surplus or more than required quantities. They accomplish this by regulating cellular metabolism. However, overproduction of the amino acids by microorganisms can be achieved by controlling the complex regulatory systems. This is done by raising auxotrophs by mutation. The enzymes responsible for the

Table 5.2: Production of amino acids by cell free enzymes

Amino acid	Enzyme	Enzyme producing microorganism	Reaction	Yield (g/l)	Efficiency (%)
L-Alanine	Aspartate decarboxylase (RC)	*Pseudomonas dacunha*	L-Aspartate→L-Alanine + CO_2	600	100
L-Aspartic acid	Aspartase (IC)	*Escherichia coli*	Fumarate + NH_4^+ →L-Aspartic acid	560	99
L-Cysteine	Cysteine desulfhydrase (RC)	*Bacillus sphaericus*	β-Choloro- L-Alanine+Na_2S → L-Cysteine + NaCl + NaOH	70	80-85
L-DOPA (L-3,4-Di-hydro-xyphenyl-alanine)	Tyrosinase (RC)	*Erwinia herbicola*	Pyrocatechol + Pyruvate + NH_4^+ → L-DOPA + H_2O	59	95
L-Leucine	L-Leucine dehydrogenase (EMR)	*Bacillus sphaericus*	α-Ketoisocapronic acid + NH_4^+ + NADH → L-Leucine + H_2O + NAD^+	42	99.7
L-Lysine	D-Amino-caprolactam racemase, L-Amino-caprolactam hydrolase (RC)	*Achromobacter obae, Cryptococcus laurentii*	DL-aminocaprolactam + H_2O → L-Lysine	100	100
L-Phenyl-alanine	L-Phenylalanine dehydrogenase Aspartate phenylalanine transaminase (EMR)	*Brevibacterium* *E.coli*	Phenylpyruvate + NH_4^+ L-Aspartate + Phenylpyruvate → Phenylalanine + Oxalocetate	37 30	98
L-Serine	Serine hydroxymethyl transferase (CCE)	*Klebsiella aerogenes*	Glycine + HCHO → L-serine	450	88
L-Tryptophan	Serine hydroxymethyl transferase	*Klebsiella aerogenes*	Glycine + HCHO → L-Serine	200	95

regulatory effect or repressor of an amino acid are made inactive due to mutation which leads to the accumulation of the particular amino acid in commercially feasible quantities which is being exploited commercially for fermentation (table 5.2).

Apart from fermentative processes, some amino acids can be synthesized quite economically by chemical processes. However, these chemical processes generally yield, D-isomers of amino acids, which are biologically inactive and cannot be used as food flavouring substances or food supplements. Only L-isomers are useful as food supplements or flavouring substances and, therefore, most of the L-isomer of amino acids are produced by fermentation process. Some of the amino acids produced by microbial fermentation and their annual production are precised in table 5.3.

Table 5.3: World production of amino acids by different processes and their applications

Amino acid	Annual production (metric tons)	Production methods	Application
L-Arginine	1000	1, 2	Infusions Therapy (liver diseases) Cosmetics
L-Aspartic acid	4000	1, 4	Therapy Flavor enhancer Aspartame® production
L-Glutamic acid	370,000	2	Flavor enhancer
L-Glutamine	500	2	Therapy (ulcer)
L-Histidine	200	4,1	Therapy (ulcer)
L-Isoleucine	150	3	Infusions
L-Leucine	150	1, 2	Infusions
L-Lysine	70,000	2, 4	Feed additive, infusion solution
L-Methionine	150	5, 4	Therapy
L-Ornithine	50	2, 4	Liver therapy
L-Phenylalanine	3000	2, 4	Infusions Therapy Aspartame® production
L-Proline	100	4	Infusions
L-Serine	50	2, 4	Cosmetics
L-Threonine	160	2	Feed additive
L-Tryptophan	200	2, 4	Infusions
L-Tyrosine	100	4	Infusions Precursor for L-DOPA synthesis
L-Valine	150	2, 4	Infusions

1. Extraction of protein hydrolysates;
2. Direct fermentation;
3. Microbial transformation of precursors; and
4. Use of enzymes or immobilized cells.

(i) Economic importance: Aspartame (L-aspartyl-L-phenylalanine methyl ester) which is made up of phenylalanine and L-aspartic acid is used as low-calorie sweetener in soft drinks. Many amino acids are used in medicine particularly as ingredients of infusion solutions in post-operative treatments. In chemical industry, amino acids are used as starting materials for the manufacture of polymers such as polyalanine fibers and lysine-isocyanate resins. Polymethylglutamate is used as a surface layer in the manufacture of synthetic leather. N-acetyl derivatives of some amino acids are used in the manufacture of cosmetics and as surface active substances. Urocanic acid is used as a sun-tanning agent and is produced by the transformation of histidine. Glycine is used in the manufacture of an herbicide, glycophosphate.

5.1 L-LYSINE

L-lysine, 2,6-diaminohexanoic acid, is synthesized by microorganisms either via diaminopimelic acid pathway or the aminoadipic acid pathway. However, in any single organism only one of the two alternative pathways is employed. Bacteria, actinomycetes, cyanobacteria (Blue–green algae), some phycomycetes and protozoa use the DAP (Diaminopimelic acid) pathway (Fig. 5.2), while some phycomycetes, all ascomycetes and basidiomycetes and eukaryotic algae uses the aminoadipic acid pathway.

L-lysine occurs in plant proteins only but too low in concentration. Addition of L-lysine can, therefore, increases the quality of food. The market for L-lysine is increasing day by day. Though, L-lysine is produced today only by microbial processes, a variety of approaches for its production have been developed.

5.1.1 Uses: L-lysine is useful in many fields.

1. L-lysine is an essential amino acid required for the human nutrition.
2. It is used as supplementary for cereal proteins.
3. Protein quality of certain foods like wheat (based foods) is improved by addition of L-lysine which results in the improved growth and tissue synthesis.
4. It is used as a nutraceutical.

5.1.2 Production of lysine: Total world production of L-lysine is around 35,000 metric tons per year. Industrially it is produced by two different fermentation methods. They are:

(a) Indirect fermentation

(b) Direct fermentation.

(a) Indirect fermentation: It is also called as dual fermentation as two different microorganisms are employed in this fermentation process. Auxotrophic mutant of *Escherichia coli* is used in the first half of the fermentation and wild type or prototrophic *E. coli* or *Aerobacter aerogenes* is employed in the second half of the fermentation. Diaminopimelic acid produced in the first half of fermentation by auxotroph of *E. coli*, is converted into L-lysine by *A. aerogenes* in the second half of the fermentation (Fig. 5.2). *A. aerogenes* should also be deficient of lysine decarboxylase so that further decarboxylation of lysine to cadaverine is prevented and accumulation of lysine is facilitated.

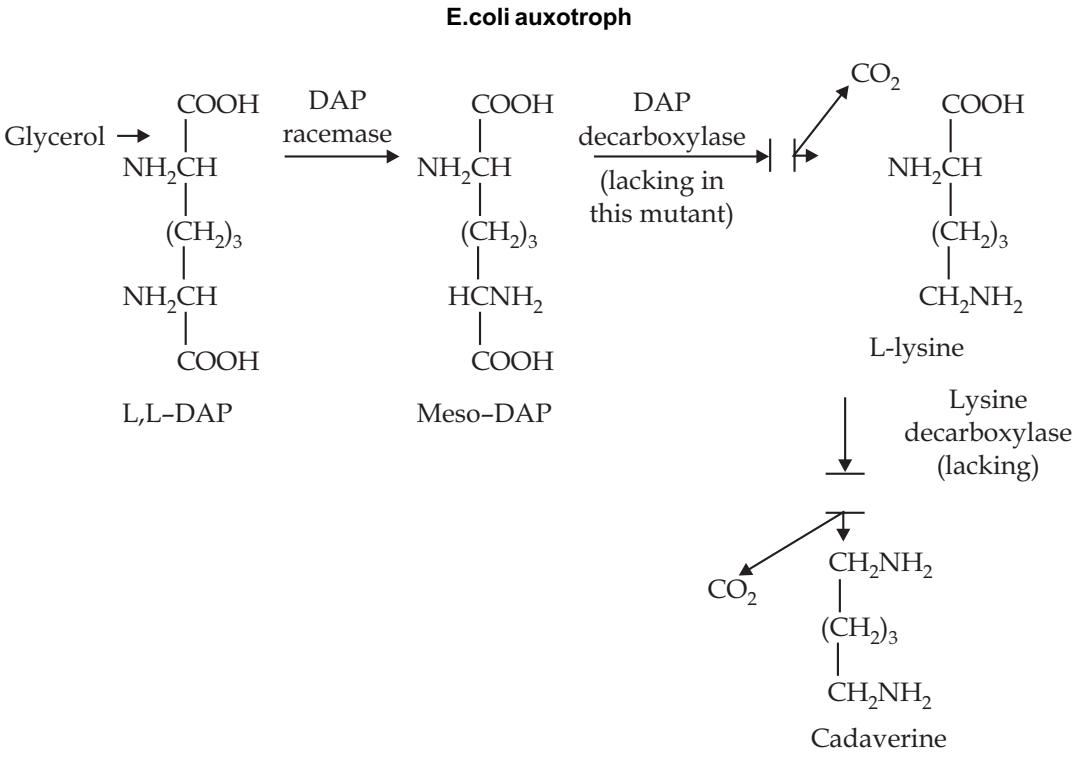


Fig. 5.2: Position of metabolic block in the L-lysine metabolic pathway of an *E.coli* auxotroph which accumulates diaminopimelic acid (DAPA) during growth on glycerol

Fermentation process of L-lysine can be described under four headings. They are

(i) Inoculum production, (ii) Preparation of medium,

(iii) Fermentation process, (iv) Harvest and recovery.

(i) Inoculum production: Pure inoculum of both *E. coli* and *A. aerogenes* is produced from the suitable and high yielding stock culture. These microorganisms should lack the ability to produce diaminopimelic acid (DAPA) decarboxylase and lysine decarboxylase enzymes respectively, so that the DAPA and L-lysine produced will not be further metabolized by respective organisms. Both the organisms should also possess strong, L-diaminopimelic acid racemase activity to convert all residual L, L-diaminopimelic acid to meso-diaminopimelic acid. The composition of the medium which is employed for inoculum production is similar to fermentation medium. The cells of the organisms are separated from growth medium by centrifugation or sedimentation.

(ii) Preparation of medium: Both the inoculum and fermentation media contain glycerol, cornsteep liquor as carbon sources and ammonium hydrogen phosphate, as nitrogen source. In addition, calcium carbonate is also used in the production medium. The levels of all of the nutrients are kept lower in the inoculum medium. Apart from supplying carbon source, the corn steep liquor also provides L-lysine required for the initial growth of auxotroph of *E. coli*. The pH of the medium is maintained at neutral, to slightly alkaline level (pH 8.0).

(iii) Fermentation process: Sufficient quantities of sterilized medium is fed into the fermenter. Pure and required quantities of inoculum of *E. coli* is added (4.0%) to the fermenter. The fermentation is carried out for 3 days at 28 to 30°C temperature. The level of L-lysine quantity provided to *E. coli* is very important, because providing low quantities, less than optimum, results in the back mutation and more quantities results in the feedback control of lysine biosynthesis both of which badly affect the yield.

Through a sequence of enzymatic steps during first stage of fermentation glycerol is converted into L, L-diaminopimelic acid, which is partially converted into D, L- isomer and mesodiaminopimelic acid by the action of diaminopimelic acid recemase enzyme. Thus, the above-mentioned metabolites accumulate in the fermentation broth because auxotrophic *E. coli* lacks diaminopimelic acid decarboxylase enzyme. Hence, it cannot be converted into L-lysine. The broth contains approximately 40% L, L-isomer and 60% mesoisomer of diaminopimelic acid.

In the second half of the fermentation, 1-2 days old culture of *A. aerogenes* is added to the fermentation broth formed at the end of first fermentation process. The microorganism is allowed to grow for one day at 24°C. After sufficient growth occurs toluene is added to the fermentation broth which causes lysis of cells of *A. aerogenes,* due to which the enzyme diaminopimelic acid decarboxylase is liberated into the fermentation broth. By this time most of the L, L-diaminopimelic acid is converted into meso-diaminopimelic acid by the action of diaminopimelic acid racemase enzyme. The meso-diaminopimelic acid is completely converted into L-lysine by the action of diaminopimelic acid decarboxylase. Some of the important features of lysine fermentation are depicted in Fig. 5.3.

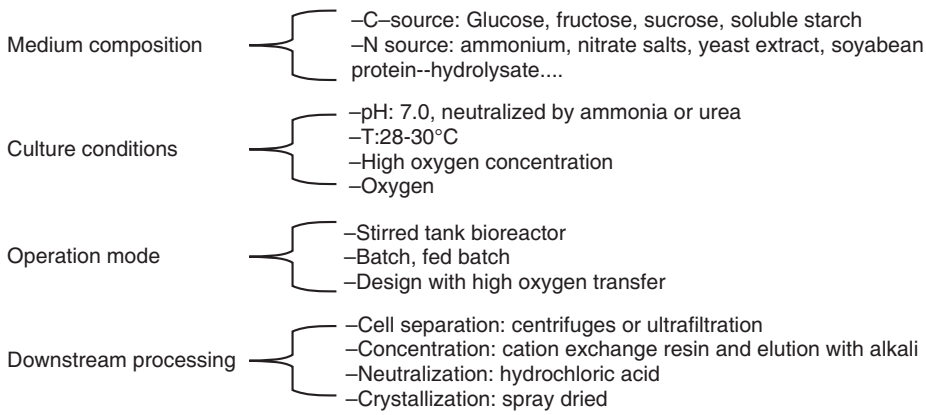


Fig. 5.3: Some important features of L-lysine fermentation

(iv) Harvest and recovery: After sufficient quantities of L-lysine is formed, lysed bacterial cells are removed from the fermentation broth by centrifugation. The L-lysine is obtained in pure form after acidification by any one of the following separating processes.

1. Precipitation at the isoelectric point
2. Ion exchange chromatography

3. Electrophoresis
4. Extraction with organic solvents

(b) Direct fermentation: L-lysine can also be fermentatively produced from any of the substrates directly and the process is called as direct fermentation. Direct fermentation processes are presently employed throughout the world for the production of L-lysine. Direct production of l-lysine from carbohydrate was developed first with a homoserine or threonine plus methionine auxotroph of *Corynebacterium glutamicum*. Production of lysine by this bacterium is regulated by the mechanism as depicted in Fig. 5.4.

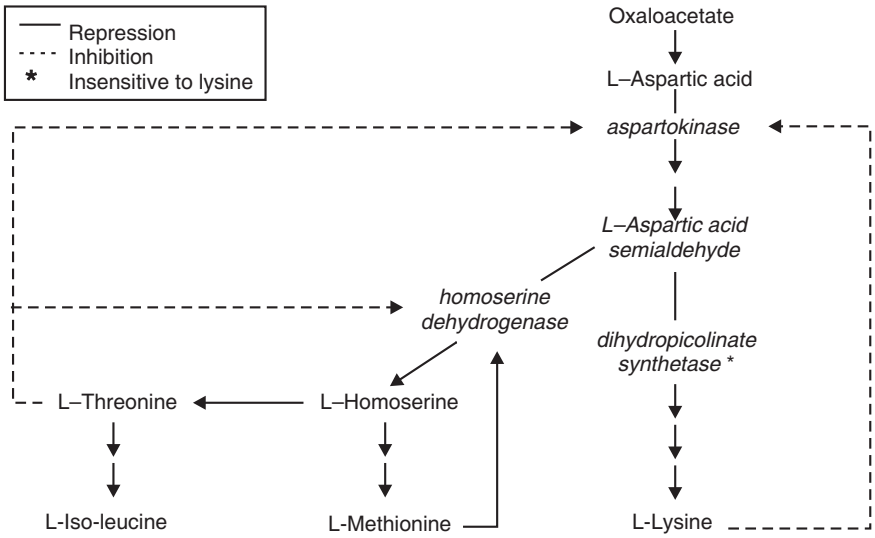


Fig. 5.4: Control of L-lysine production in *Corynebacterium glutamicum*

The prototroph of this bacterium produces L-glutamic acid in large quantities. The same type of process was reported with a homoserine auxotroph of *Brevibacterium flavum.* The homoserine auxotroph was later recognized as threonine sensitive mutant because growth was inhibited by the excess of threonine and the inhibition was released by the addition of methionine. This phenomenon is due to feedback inhibition of residual homoserine dehydrogenase by threonine. Homoserine auxotroph of other bacteria were also found to produce L-lysine but the yields were lower than that from homoserine auxotroph of Coryneform bacteria. Threonine and leucine auxotrophs produce fairly large amounts of L-lysine but they are inferior to the homoserine auxotroph. Other auxotrophs of *Corynebacterium glutamicum* and other bacteria were also inferior to the homoserine auxotroph of *C. glutamicum*. Therefore, this bacterium is extensively used for the L-lysine production on commercial basis by fermentation process.

Double auxotrophs, which require atleast one of the amino acids, threonine or isoleucine or methionine in addition the homoserine, for growth have been found highly stabilized, showing little tendency to revert the homoserine independence. It is possible not only to prevent reversion of the culture to a wild type, but also to produce lysine in higher yields since many of the microorganisms are double mutants in the homoserine pathway.

Fermentation production of L-lysine employing *C. glutamicum* is described here. This process consists of four stages. They are:

(i) Preparation of inoculum, (ii) Preparation of medium,
(iii) Fermentation process, (iv) Harvest and recovery,

(i) Preparation of inoculum: Suitable and high yielding mutant strain of *C. glutamicum* usually (strain 901) is used from the stock culture for the production of inoculum. Seed cultures are raised twice, in which two different media are used. The medium for first seed culture contains.

Glucose	2%
Peptone	1%
Meat extract	0.5%
NaCl	0.25%

The medium is prepared in tap water. The medium for second seed culture contains

Cane molasses	5%
$(NH_4)_2SO_4$	2%
Corn steep liquor	5%
$CaCO_3$	1%

It is also prepared in tap water. This prepared inoculum is employed for fermentation.

(i) Preparation of medium: The medium with the following composition is used as fermentation medium. Reducing sugar (expressed as inverted cane molasses), 20%, Soyabean meal hydrolysate (as weight of meal before hydrolysis with $6NH_2SO_4$ 1.8% and neutralization with ammonia water) are dissolved in tap water and sterilized.

(ii) Fermentation process: The fermentation is carried out at 28°C and is allowed upto 60 hours. The amount of growth factors, homoserine or threonine and methionine should be appropriate for the production of L-lysine and suboptimal quantity to support the optimal growth. The biotin concentration in the medium should be greater than 30 mg per liter. If biotin is supplied in limited quantities there will be accumulation of L-glutamic acid instead of L-lysine. Cane molasses generally supplies enough biotin. There will be 30-40% yield of L-lysine as monohydrochloride in relation to the initial sugar concentration. Foam production in the aerated culture can be controlled by adding suitable antifoam agent.

(iii) Harvest and recovery: The same process of recovery of L-lysine, that is employed in indirect fermentation process is also used in this process.

Mutant strains of *Bacillus licheniformis* are also employed for the production of L-lysine. The mutant strains were obtained by the introduction of both analogue-resistance and auxotrophy. The medium containing 10% cane molasses is used. A temperature of 40°C is suitable for L-lysine production. The sporulation activity which reduces yield, can be suppressed by the addition of certain antibiotics like tetracycline and chloramphenicol. These mutants yield approximately 30 mg of L-lysine per ml of carbon source used.

L-lysine is also produced by enzyme process. Racemase mixture of D and L-amino-caprolactum can be transformed by the L-α-aminocaprolactum hydrolase to lysine. Racemase enzyme converts D-α-aminocaprolactum to L-α-aminocaprolactum (Fig. 5.5). The L-α amino-caprolactic hydrolase and racemase enzymes are obtained from the bacteria, *Achromobacter obae* and yeast, *Cryptococcus lauranti.*

Starting material **:** A racemic mixture of D-and-L-α-aminocaprolactum

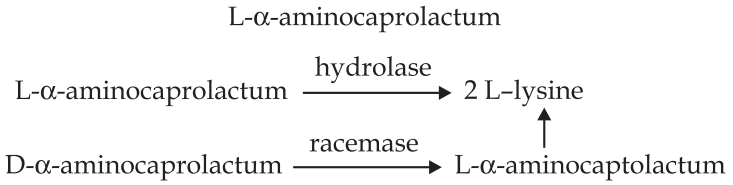


Fig. 5.5: Formation of L-lysine through biotransformation by *Achromobacter* obae or *Cryptococcus lauranti*

5.2 L-GLUTAMIC ACID

L-glutamic acid is a dicarboxylic amino acid and contains two carboxyl groups, along with an amino-group which is attached to the α-carbon atom. Fermentative production of L-glutamic acid was started after the isolation of *Corynebacterium glutamicum* in 1957. Before that, it was used to be produced by chemical synthesis and the product used to contain a mixture of D and L-glutamic acid. Subsequently L-glutamic acid production was found to occur in a wide variety of bacteria like *Corynebacterium*, *Brevibacterium*, *Microbacterium* and *Arthrobacter*. Streptomycetes, yeasts and fungi are also reported to be capable of producing L-glutamic acid upto 30 g per liter. *Corynebacterium glutamicum* which produces good amount of L-glutamic acid possesses the following characteristics:

1. The bacterium is gram (+) positive, non-sporulating and non-motile.
2. Requires biotin for growth.
3. Shows little activity of α-ketoglutaric acid dehydrogenase.
4. Shows increased activity of glutamate dehydrogenase.

Mutants of *C. glutamicum* secrete L-glutamic acid in large quantities even in the presence of high concentration of biotin. For example, a lysozyme sensitive mutant of *C. glutamicum* is able to convert 40% of the added carbon source to L-glutamic acid even in the presence of 100 mg per liter of biotin.

5.2.1 Biosynthesis: The glucose is broken down into C_3 and C_2 fragments by glutamic acid producing microorganisms through the Embden Meyerhof-Parnas (EMP) pathway and the pentose-phosphate pathway and the fragments are channeled into the tricarboxylic acid (TCA) cycle (Fig. 5.6). The reactions of EMP pathway are more common under conditions of glutamic acid production. The key precursor of glutamic acid is α-ketoglutarate, which is formed in the TCA cycle via citrate, isocitrate and α-ketoglutaric acid, which is then converted into L-glutamic acid through reductive amination with free NH_4^+ ions. The last step is catalysed by the NADP dependent glutamate dehydrogenase.

The $NADPH_2$ required at this stage of the reaction is furnished through the proceeding oxidative decarboxylation of isocitrate dehydrogenase. The $NADPH_2$ is then regenerated by the reductive amination of α-ketoglutarate.

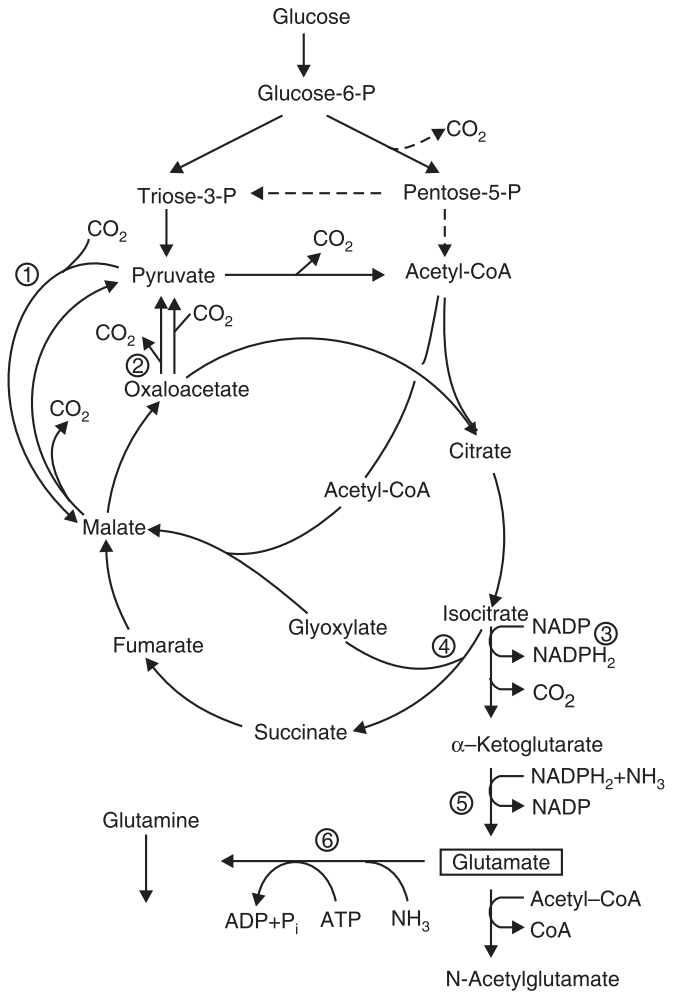


Fig. 5.6: Biosynthesis of L-glutamic acid using glucose as the carbon source. Glyoxylic acid cycle, broken lines, 1. Malic enzyme, 2. Oxaloacetate carboxylase, 3. Isocitrate dehydrogenase, 4. Isocitratelyase 5. Glutamate dehydrogenase

(i) Effect of permeability on glutamic acid production: Production and excretion of glutamic acid is dependent on cell permeability. Increased permeability in glutamic acid producing bacteria can be accomplished by one of the following ways.

(a) Through biotin deficiency.
(b) Through the addition of penicillin.
(c) Through the addition of saturated fatty acids or fatty acid derivatives.
(d) Through the oleic acid deficiency in oleic acid auxotrophs.
(e) Through the glycerol deficiency in glycerol auxotrophs.

(ii) Conditions of production: The following factors affect the glutamic acid fermentation.

(a) Carbon source,

(b) Nitrogen source,

(c) Growth factors,

(d) Oxygen supply, and

(e) pH of the medium

(a) Carbon source: A wide variety of carbohydrates are used as carbon source in the fermentation process. Glucose and sucrose are frequently used. However, starch hydrolysates, fructose, maltose, ribose and xylose are also used less frequently. Moreover sucrose, sugarcane molasses and sugar beet molasses can also be used. Both the molasses contain high biotin content (0.4–1.2 mg kg^{-1} in cane molasses and 0.02–0.08 mg kg^{-1} in beet molasses). Penicillin or fatty acid derivatives (e.g. Tween-66) must be added to the fermentation medium.

When these molasses are used in the medium preparation because they increase the cell permeability of L-glutamic acid. For industrial production, generally cane molasses or starch hydrolysate are used.

(b) Nitrogen source: Ammonium sulphate, ammonium chloride, ammonium phosphate, aqueous ammonia, ammonia gas and urea have been used as nitrogen source. Although large amount of ammonium ions are necessary, a high concentration of it inhibits the growth of the microorganism as well as the yield of L-glutamic acid. Therefore, suitable amount of ammonia is added, as the fermentation progresses. These salts also help in the pH control.

(c) Growth factors: The important growth factor is biotin. Its optimal concentration depends upon the carbon source used. In media with 10% glucose, its requirement is 5 mg $liter^{-1}$. In media with lower glucose concentration, it is considerably lower. Some strains require L-cystine as an additional growth factor.

(d) Oxygen supply: The oxygen concentration should neither be too low nor too high. Optimal L-glutamic acid yields are obtained at kb value of 3.5×10^{-6} mole 02 atm^{-1} min^{-1} ml^{-1}. Excretion of lactate and succinate occurs under oxygen deficiency, whereas excess oxygen under ammonium ions deficiency causes growth inhibition and production of α-ketoglutarate. In both the cases, glutamic acid yields are low.

(e) pH: Optimum pH for growth and glutamic acid production is 7.0–8.0 and it is controlled by the addition of ammonium salts.

(iii) Fermentation production: Important features of L-glutamic acid production are precised in Fig. 5.7. L-glutamic acid can be produced commercially by the following ways.

1. By a two stage fermentation process where α-ketoglutaric acid is produced by one microorganism, is then converted into L-glutamic acid by another microorganism.
2. By one stage fermentation process employing only one microorganism.

The fermentation process of L-glutamic acid is described below:

(i) Inoculum production, (ii) Preparation of medium,

(iii) Fermentation process, (iv) Harvest and recovery

(i) Inoculum production: A medium with the following composition is prepared for inoculum production.

Chemical	Mg l^{-1}
Glucose	40
K_2HPO_4	1.0
$MgSO_4 \cdot 7H_2O$	0.5
Yeast extract	1.0
Urea	8.0
Tap water	1000

A suitable strain of *C. glutamicum* from a stock culture is selected and is inoculated into the above sterilized medium. The culture is incubated upto 16 hours at 35°C. After sufficient growth occurs, approximately 6% by volume of inoculum is added to the production fermenter.

(ii) Preparation of medium: A production medium is prepared with the following composition.

Chemical	Concentration (in %)
Glucose	10
K_2HPO_4	0.05
KH_2PO_4	0.05
$MnSO_4 \cdot 7H_2O$	0.025
$FeSO_4 \cdot 7H_2O$	0.001
$MnSO_4 \cdot 4H_2O$	0.001
Urea	0.5
Biotin	2.5 mg $liter^{-1}$

The medium with the above composition is sterilized and employed for the production of L-glutamic acid.

(iii) Fermentation process: The fermentation is carried out, approximately, for 40-48 hours at 30°C temperature. The pH is adjusted to 7.0–8.0. The urea is added intermitantly during the fermentation. Approximately 50% of the supplied carbohydrate is converted into L-glutamic acid. The general flow diagram for the industrial production of L-glutamic acid is illustrated in Fig. 5.8.

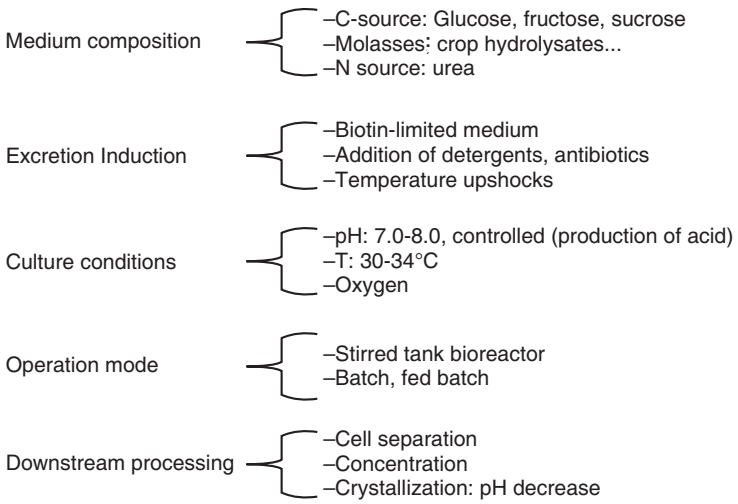


Fig. 5.7: Important features of Glutamic acid production by fermentation

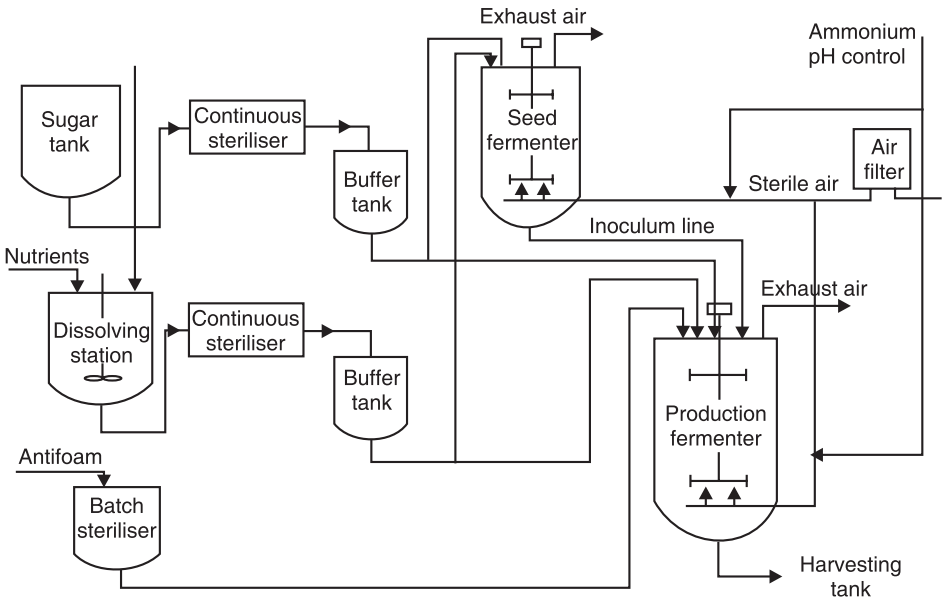


Fig. 5.8: Flow diagram of L-Glutamic acid production

(iv) Harvest and recovery: The same process of recovery that is employed for L-lysine is also employed for the harvest and recovery of L-glutamic acid.

Glutamic acid can also be produced through biotransformation of racemic mixture of D–and L-hydantoin-5-propionic acid with the help of hydantoinase. Simultaneously D-hydantoin is converted to L-hydantoin-5-propionic acid in the presence of hydantoin racemase (Fig. 5.9).

Starting material : A racemic mixture of D-and-L-hydantoin 5-propionic acid

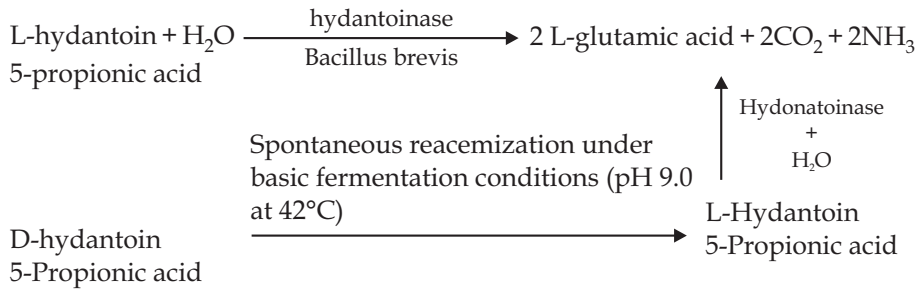


Fig. 5.9: Formation of L-glutamic acid through biotransformation of L-hydantoin 5-propionic acid by *Bacillus brevis*

5.3 L-ASPARTIC ACID

L-aspartic acid is widely used as a food additive and in pharmaceuticals. Since the time of its use in Aspartame as an artificial sweetener production, its demand increased considerably. Although L-aspartic acid was originally produced exclusively using aspartase due to high

productivity and cost effectiveness of the process. Infact this method proved to be highest enzyme used in biotechnology. This method allows the production of 2,20,000 kg of L-aspartic acid per kg of enzyme. The reaction is interconvertible (Fig. 5.10).

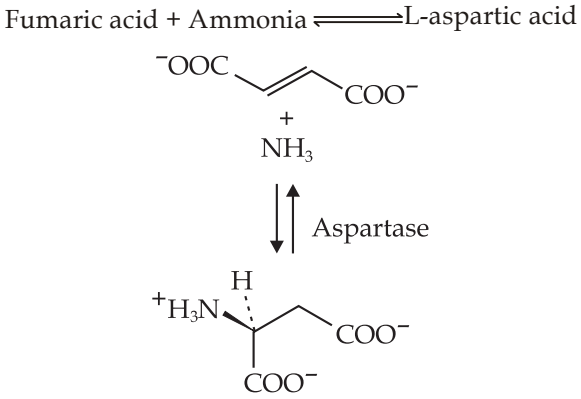


Fig. 5.10: Fumarate and ammonium serves as substrate for the aspartase

Infact reaction favours the ammonification. The enzyme of *E.coli* is a tetramer with a molecular weight of 1,96,000 daltons and has absolute requirement of divalent cation. However, the enzyme was unstable in the beginning which could be overcome by immobilization in polyacrylamide. Subsequently carrageenan, which has half life for about two years proved to be better material for immobilization of aspartase. The initial disadvantage of this enzyme was that it converts part of fumaric acid to L-malic acid. Heat treatment of cells eliminated fumarase activity almost completely using such conditioned and starting with 1m ammonium fumarate. Using such conditioned cells it was possible to produce 987 mM L-aspartic acid 10.7 mM non-reacted fumarate and only trace quantities of L-malic acid of 1.9 mM.

For production of L-aspartic acid, the immobilized cells of *E.coli* are packed into a column designed as a multi stage system. The stages introduced consisting of horizontal tubes serve two purposes. On one hand, they allow effective cooling to prevent decay of the catalytic activity since the asparatase reaction is exergeonic. On the other hand, the flow proportionately of the column are increased. Any compacting of bed over time is prevented and the preferred plug flow characteristics are obtained. The continuous process enables full automation and control to achieve the optimum output with the highest product quality. Yet another advantage of such controlled continuous process reduces waste production. It is estimated that about 3.4 tonnes of aspartic acid per day in a column of 1000 d^{-1} which is 100 tonnes per month. The final product is eventually purified by crystallization.

5.4 L-PHENYLALANINE

L-Phenylalanine can be produced with *E. coli* or *C. glutamicum*. The pathway of L-phyenylalanine synthesis is shared in part with that of L-lyrosine and L-tryptophan. These three aromatic amino acids have in common. The condensation of erythrose - 4 - phosphate and phosphoenol pyruvate to deoxyarabinoheptulosonate phosphate (DAHP) with further conversion in six steps to choristmate. L-Phenylalanine is then finally made in three further steps (Fig. 5.11).

There are three DAHP synthase enzymes in *E.coli* encoded by aroF, aroG and aroH. These enzymes play key role in flux control and regulation of catalytic activity in each case by one of

the aromatic amino acids. About 80% of the total DAHP synthesis activity is contributed by aroG-encoded enzyme. The increased flux towards L-phenylalanine can be obtained by over expression of either aroF or aroG encoding feed-back resistant enzymes. Further, more phe A over expression is essential which encodes the bifunctional corismate mutase pre phenate dehydratase. A second chorismate activity is present as a bifunctional chorismate mutase – prephenate dehydrogenase. The pre A–encoded enzyme acitivities are inhibited by L-phenylalanine and pre A expression is dependent on the level of t-RNA. A pre phenylalanine producer obtain as per rule are tyrosine auxotrophic mutants.

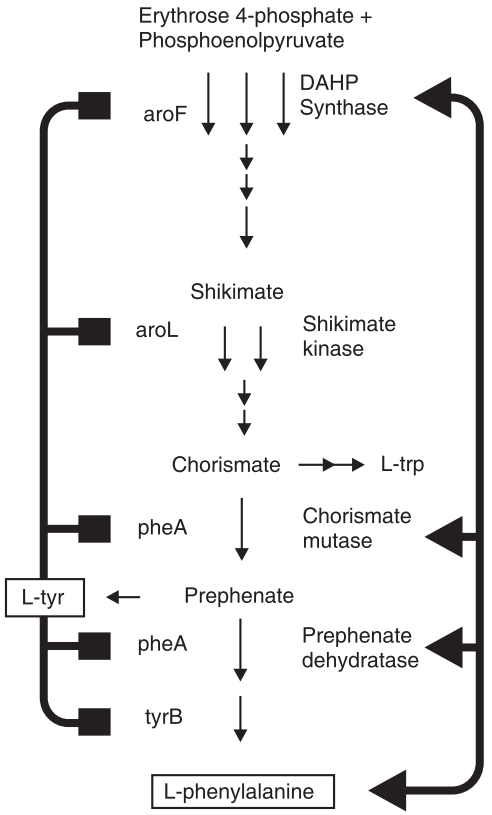


Fig. 5.11: Simplified pathway of L-phenylalanine synthesis and the relevant regulation by L-phenylalanine and L-tyrosine (L-tyr) with feedback control of enzyme activity (arrow head ends) and gene repression (square ends)

Fermentation: As with other amino acids effective phenylalanine production is the joint result of engineering the cellular metabolism and control of production process. Control is necessary for two reasons. First, the carbon flux has to be optimally distributed between the four major products of glucose conversion which are phenylalanine, biomass, acetic acid CO_2. The second reason is that cellular physiology is not constant during the course of *E.coli* and tends to produce acetic acid which has strong negative effect on the process efficiency, which can be prevented by sugar feeding, regulating O_2 concentration, sugar consumption and biomass concentration. Glucose should be added when it is totally exhausted at stage-2 of fermentation. Thus, feeding rate is a compromise where the process run at highest possible feeding rate. When

the tyrosine initially present has consumed, the cells proceeds to stage 3. At this stage, the metabolic capacity of the cells decreases which brings about a consequent decrease of glucose feeding rate. At the end of stage 3, acetic acid excretion begins and the cells enter stage 4 where no further phenylalanine accumulation occurs and the process eventually get terminated (Fig. 5.12).

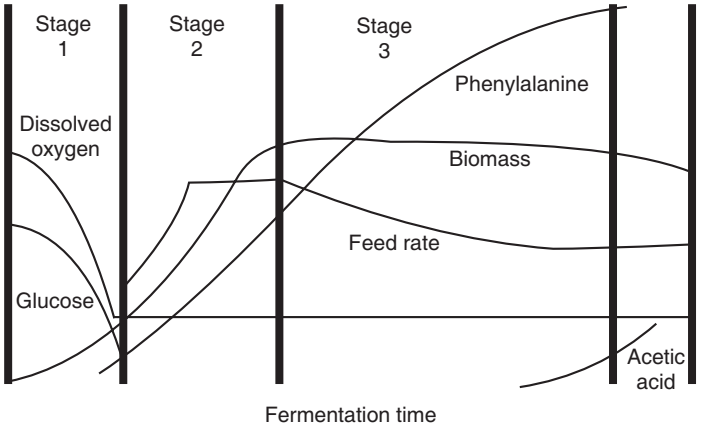


Fig. 5.12: The four stages of L-phenylalanine production

Thus, it reveals that sophisticated feeding strategy with adaptive control stimulates a very high phenylalanine concentration and can be achieved with a high yield of phenylalanine per liter alongwith a yield of 27.5% carbon.

REVIEW QUESTIONS

I. Essay type questions:

1. Discuss metabolic control in amino acid production.
2. Describe fermentative production of L-lysine.
3. Describe production of L-glutamic acid.
4. Give an account of L-aspartic acid production. Add a note on aspartame.
5. Discuss the regulation of production of L-phenylalanine.
6. Describe direct method of production of L-lysine.
7. Describe process of production of L-lysine and L-glutamic acid through biotransformation.

II. Write short notes on:

(a) DAPA decarboxylase,
(b) Lysine decarboxylase,
(c) Biotin role in L-glutamic acid production,
(d) Aspartame,
(e) Chorismic acid,
(f) Aspartase,
(g) α-L-aminocaprolactum,
(h) D-hydantoin

FURTHER READING

1. Eggeling, L. and Sahm, H. (1999). **L-Glutamate and L-lysine: traditional products with impetuous developments.** Applied Microbiology and Biotechnology 52, 146–153.
2. Hodgson, J. (1994). **Bulk amino acid fermentation: Technology and commodity trading.** Biotechnology 12, 152–155.
3. Jetten, M.S.M., Follettie, M.T. and Sinskey, A.J. (1994). **Metabolic engineering of Corynebacterium glutamicum.** New York Academy of Sciences 721, 12–29.
4. Katsumata, R. and Ikeda, M. (1993). **Hyperproduction of tryptophan in Corynebacterium glutamicum by pathway engineering.** Biotechnology 11, 801–806.
5. Kiss, R.D. and Stephanopoulos, G. (1991). **Metabolic activity control of the L-lysine fermentation by restrained growth fed-batch strategies.** Biotechnology Progress 7 (6), 501–509.
6. Konstantinov, K.B., Nishio, N., Seki, T. and Yoshida, T. (1990). **Physiologically motivated strategies for control of the fed-batch cultivation of recombinant *Escherichia coli* for phenylalanine production.** 71, 350–355. J. Ferment, Bioeng, Journal of Fermentation and Bioengineering.
7. Kramer, R. (1994). **Secretion of amino acids by bacteria: Physiology and mechanism.** FEMS 13, Microbiology Reviews. 75–79.
8. Leuchtenberger, W. (1996). **Amino acids, technical production and use.** In: Products of Primary Metabolism (eds. Rehm, H.J. and Reed G.). Biotechnology 6, 455–502.
9. Li, K., Mikola, M.R., Draths, K.M., Worden, R.M. and Frost, J.W. (1999). **Fed-batch fermenter synthesis of 3-dehydroshikimic acid using *Escherichia coli*.** Biotechnology and Bioengineering. 64, 61–73.
10. Peters-Wendisch, P., Kreutzer, C., Kalinowski, J., Patem M., Sahm, H. and Eikmanns, B.J. (1998). **Pyruvate carboxylase from** *Corynebacterium glutamicum:* **Characterization, expression and inactivation of the pyc gene**. Microbiology, 44, 915-927.
11. Schilling, B.M., Pfefferle, W., Bachmann, B., Leuchtenberger, W. and Deckwer, W.D. (1999). **A special reactor design for investigations of mixing time effects in a scaled-down industrial L-lysine fed-batch fermentation process.** Biotechnology and Bioengineering 64, 599–606.
12. Vrljic, M. Sahm, H. and Eggeling, L. (1996). **A New type of transporter with a new type of cellular function: L-lysine export in** *Corynebacterium glutamicum.* Molecular Microbiology 22, 815–826.

6

Antibiotics

Antibiotics are the secondary metabolites of one organism which inhibits the growth of other organisms at very low concentrations. They can be obtained either from natural sources, viz, microbes or can be synthesized chemically. They are used widely in different fields like medicine, veterinary, agriculture etc.

The first antibiotic, the penicillin, was discovered by **Alexander Fleming** in 1929, when he was working at St. Mary's Hospital, London. He observed the inhibition of growth of *Staphylococcus aureus* in one of the petri plates by a contaminating microorganism which was later identified as *Penicillium notatum*. The inhibition of growth of the microbe happened due to secretion of a chemical by the mold. The chemical was named as the penicillin for which **Alexander Fleming** and **Chain**, a biochemist, shared the noble prize in 1941. During World War II the demand for penicillin to treat wound infections led to the development of a production process for penicillin and the beginning of the era of antibiotic research, which is known as the golden era of industrial microbiology.

Since the discovery of penicillin, thousands of different antibiotics produced by different groups of microbes like fungi, actinomycetes and bacteria have been isolated and identified (table 6.1).

Table 6.1: Number of antibiotics produced by different groups of microorganisms

Taxonomic group	Number of antibiotics
Fungi	1,600
Bacteria (other than actinomycetes)	950
Actinomycetes	4,600

Out of them only 123 have been currently produced by fermentation. In addition, more than 50 antibiotics are produced as semi-synthetic compounds, three antibiotics chloramphenicol, phosphonomycin and pyrrolnitrin are produced completely synthetically.

Classification of antibiotics: Although antibiotics are classified according to their antimicrobial spectrum, mechanism of action and producer organism and manner of biosynthesis, they differ in their molecular weight, chemical structure, elemental composition and physicochemical characteristics. These properties are also taken into consideration for the classification of antibiotics. Based on the similarity in their chemical structure **Berdy et al**., 1985 classified antibiotics into:

1. **Aminoglycoside antibiotics:** They contain amino sugars linked together by glycoside linkage. Important antibiotics belonging to the group include streptomycin, neomycin, kanamycin, paromomycin, gentamycin, tobramycin and amikacin.
2. **Antifungal antibiotics:** They include two different categories of antibiotics:
   (i) Polyenes with large ring containing a conjugated double bond system. Nystatin and amphotericin B are the important antibiotics belonging to this category.
   (ii) Other antifungal antibiotics, which include 5-fluorocytosine, clotrimazole and griseofulvin.
3. **Chloramphenicol:** It forms a separate group by itself and contains nitrobenzene derivative of dichloroacetic acid.
4. **Macrolide antibiotics:** They contain macrocyclic lactone ring to which sugars are bonded. The important antibiotics of this group are spiromycin, oleandomycin and erythromycin.
5. **β-lactum antibiotics:** They have β–lactum ring in their chemical structure. The natural penicillin's, the semisynthetic penicillins and cephalosporins belong to this group.
6. **Peptide antibiotics:** They are made up of peptide-linked amino acids of both dextro and laevo forms. These include bacitracins, niacin, gramicidin and polymyxin.
7. **Tetracycline (quinines) antibiotics:** They are the derivatives of the polycyclic naphthacene carboxamide. The important antibiotics include tetracycline, chlorotetracycline, oxytetracycline, demeocycline and minocycline.
8. **Unclassified antibiotics:** They differ widely in their chemical structure and, therefore, not grouped in the above-described major groups. The important antibiotics include cycloserine, fusidic acid (steroid), novobiocin, prosinomycin, spectinomycin, vancomycin.

Antibiotics are also classified based on the target organism.

1. Antibacterial - if they are capable of inhibiting bacteria.
2. Antifungal - if they are active against fungi.
3. Antiprotozoan - If they are active against protozoa.

Mechanism of action : Though very large number of antibiotics have been discovered, less than 1% have been of practical value in medicine. However, the useful antibiotics have made dramatic impact on the treatment of infectious diseases. Further, many antibiotics are made more effective by chemical modifications in the laboratory, these are called as semisynthetic antibiotics. Some antibiotics can stop only growth and are incapable of killing the organism. Such antibiotics are called static (bacteriostatic or fungistatic), while the antibiotics which kill organism are called as bactericidal or fungicidal.

The sensitivity of microorganisms to antibiotics and other chemotherapeutic agents varies. Gram (+) positive bacteria are usually more sensitive to antibiotics than gram (–) negative bacteria, although some antibiotics act only on gram (–) negative bacteria. An antibiotic that acts on both gram (+) positive and gram (–) negative bacteria is called as a broad spectrum antibiotic, which generally finds wider medical use than a narrow spectrum antibiotic, which acts only on a single group of organisms, either gram (+) ve or gram (–) ve. An antibiotic with limited spectrum of activity may, however, be quite valuable for the control of specific microorganism that fail to respond to other antibiotics. For instance vancomycin, a glycopeptide, that is a bactericidal agent that acts against gram (+) positive bacteria such as *Staphylococcus*, *Bacillus* and *Clostridium.*

Antibiotics and other chemotherapeutic agents are classified based on their mode of action (Fig 6.1) into five types (table 6.2).

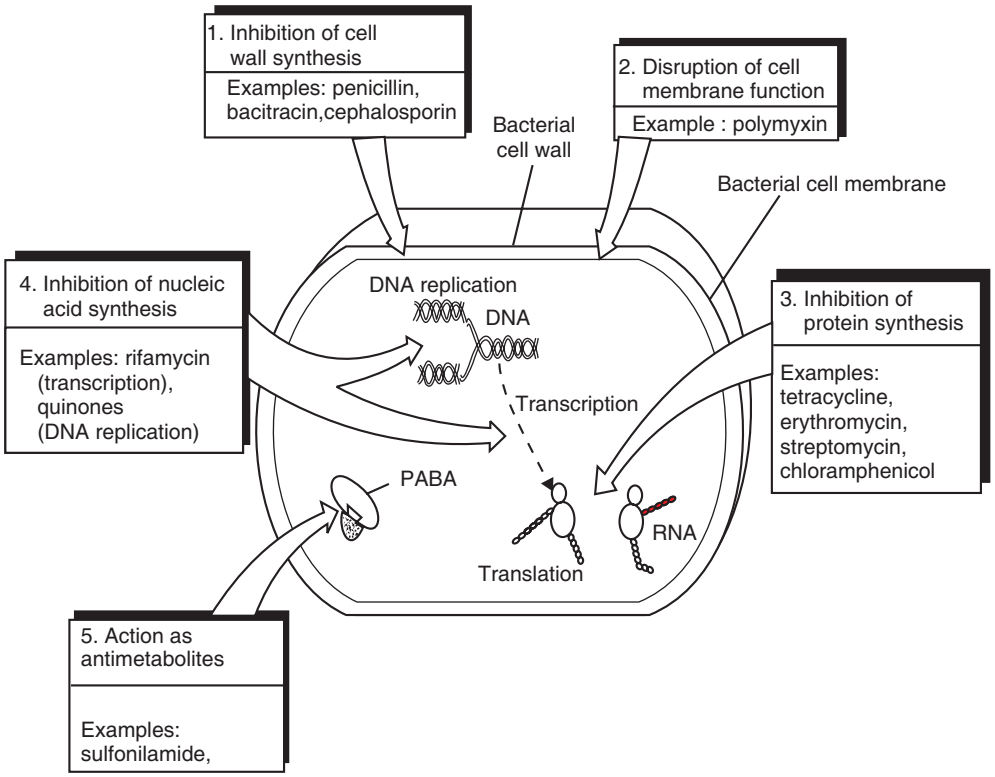


Fig. 6.1: Spectrum and mode of action of some important antibiotics

Table 6.2: Classification of antibiotics based on mechanism of action

Process inhibited	Antibiotic
Cell wall inhibition	Cycloserine Vancomycin Bacitracin Penicillins (β-lactams) Cephalosporins Monobactam Carbapeneme
Cytoplasmic membrane	Polymyxins
Protein synthesis	Tetracycline Spectinomycin Streptomycin Gentamycin Tobramycin Kanamycin Amikacin Nitrofuran Mupirocin Puromycin
Nucleic acid synthesis	Nalidixic acid Norfloxacin Novobiocin Rifampicin
Folic acid metabolism	Trimethoprim Sulfonamides

Out of these antibiotics, fermentative production of penicillin, cephalosporins, tetracyclines, erythromycin, streptomycin, bacitracins, chloramphenicol, fusidic acid, interferon and nisin are described.

6.1 PENICILLIN

Chemically the natural penicillin is 6-amino penicillanic acid (6 - APA), which consists of thiazolidine ring with a condensed β-lactum ring. The various penicillins differ primarily in the nature of R-side chain which are attached by an amido linkage to the chemical nucleus of the molecule. Fleming's original *Penicillium notatum* strain, when grown on his medium produced penicillin-F, which is known as 2-pentinyl penicillin. Subsequently *P. chrysogenum* proved to be better fungus and more suitable for submerged fermentation. The basic structure of penicillin and different types of natural penicillin's differing in the composition of side chain are shown in Fig. 6.2.

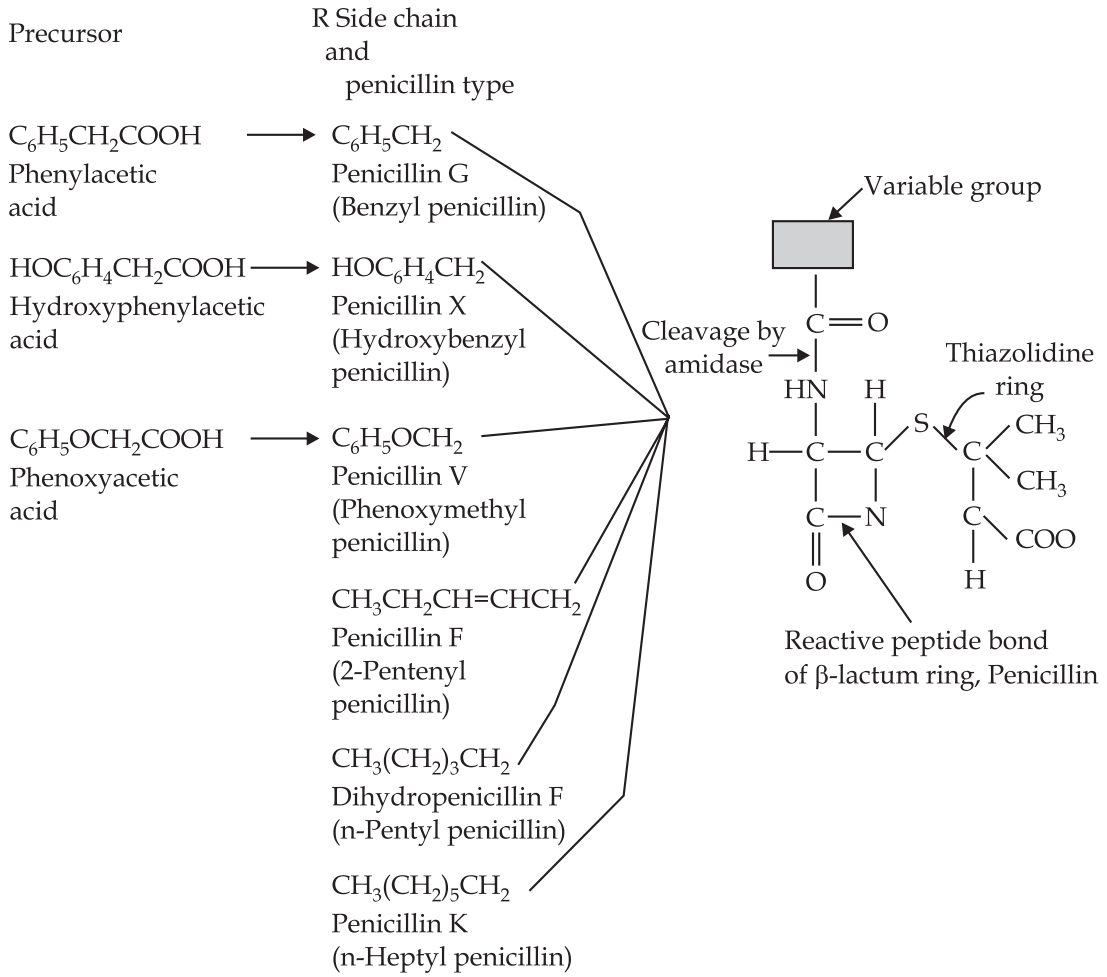


Fig 6.2: Different Penicillins with their precursor and R side chain

If penicillin fermentation is carried out without the addition of side chain precursor, the natural penicillins are formed from which only benzyl penicillin can be isolated. However, the desired penicillin can be obtained by adding suitable side chain precursor into the medium. Such penicillins are called as semi-synthetic penicillins. Penicillin-G and Penicillin-V are generally produced commercially. When compared to natural penicillins, semisynthetic penicillins have improved characters viz, acid stability, resistance to plasmid or chromosomally coded β-lactamases, expanded antimicrobial effectiveness and are therefore, extensively used in therapy.

6.1.1 Biosynthesis of penicillin: The β-lactum thiazolidine ring of penicillin is formed by the condensation of L-cystine and L-valine. The biosynthesis occurs in a non-ribosomal process by means of dipeptide composed of (α-α-AAA) and α-cystine or a breakdown product of cystothiamine. Subsequently L-valine is connected via epimerization reaction resulting in the formation of tripeptide. The first product of cyclization of the tripeptide which can be isolated is isopenicillin N but the biochemical reactions leading to this intermediate is not understood. Benzyl penicillin is produced in exchange of a-a-AAA with activated phenylacetic acid (Fig. 6.3).

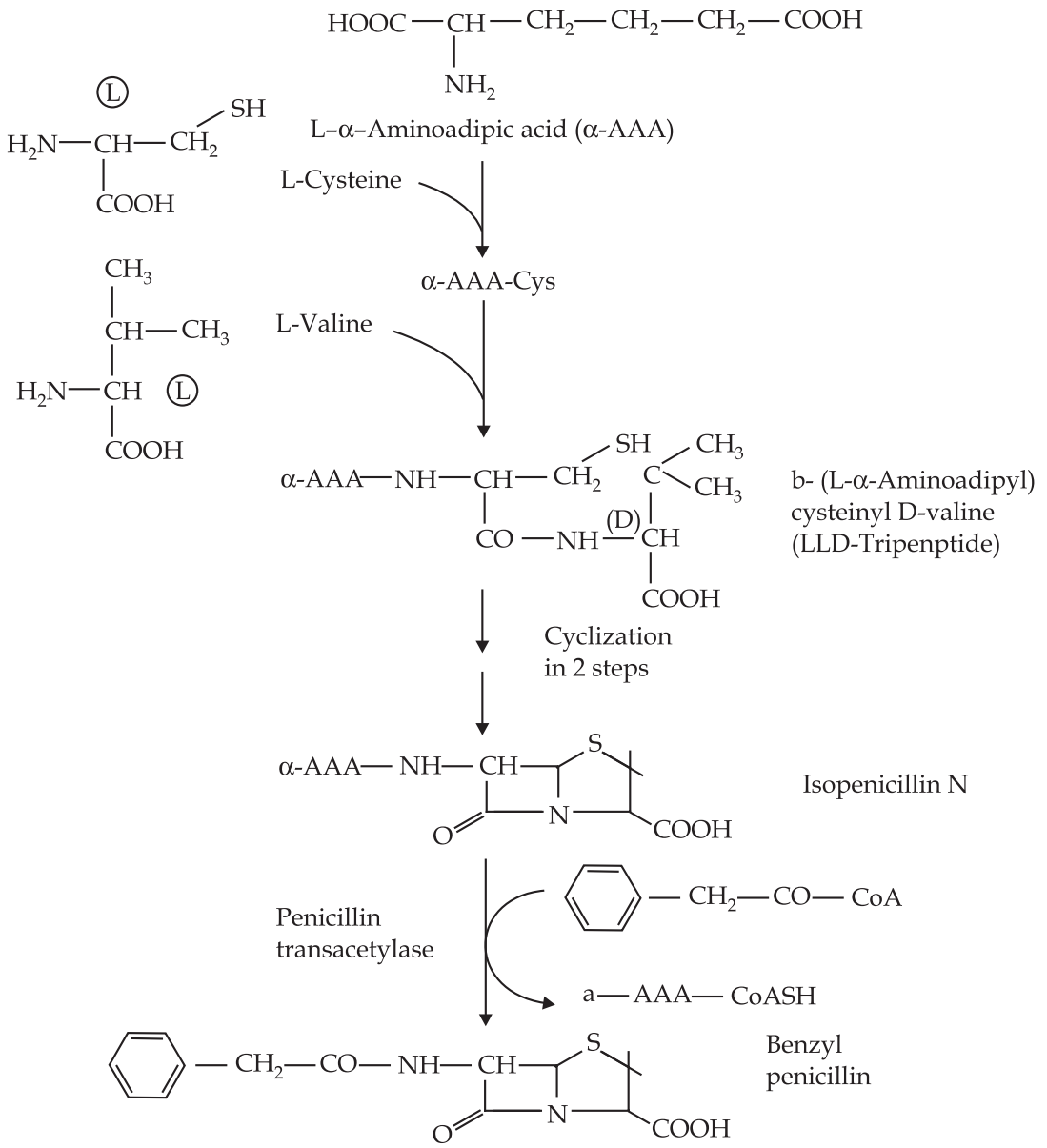


Fig. 6.3: Biosynthesis of Penicillin

About 38% of the penicillins produced commercially are used as human medicine, 12% in veterinary medicine and 43% as starting materials for the production of semi-synthetic penicillins.

6.1.2 Fermentation: Penicillin fermentation is an aerobic process with a volumetric oxygen absorption rate of 0.4 - 0.8mm min^{-1}. The required aeration rate varies according to the strain, the type of fermenter used and on the impellor system. However, the aeration rate varies between 0.5 and 1.0 vvm. It is produced by fed batch submerged fermentation in a stirred tank fermenter. This process can be described under following headings.

1. Strain development, 2. Inoculum production,
3. Inoculation, 4. Penicillin production
5. Extraction and purification

6.1.3 Strain development: The variety of molds which yield greater amount of penicillin is called as high yielding strain. They are generally developed from the wild *P. chrysogenum* by a process called sequential genetic selection. This process consists of stepwise development of improved mutant by treating the wild strain of *P. chrysogenum* with a series of mutagenic agents or exposing to ultraviolet radiation either individually or in combination, such as X-rays and chemical mutagens, is called as strain improvement. Strain development is a laborious and time-consuming process. The selected mutant possesses greater capacity for antibiotic production than the wild type. Details of some high yielding strains which are developed from wild *P. chrysogenum* (N R R L 1951) are shown in Fig. 2.46 (Page 94) and table 6.3.

The expanded role for penicillins came from the discovery that different biosynthetic penicillins can be formed by the addition of side chain precursors to the fermentation medium and that natural penicillins can be modified chemically to produce penicillins with improved characteristics. Most penicillins are now semisynthetic produced by chemical modification of natural penicillin obtained by fermentation using strains of *P chrysogenum.* Modification is achieved by removing their natural acyl group, leaving 6 APA to which other acyl groups can be added to confer new properties. This is achieved by passage through a column of immobilized penicillin acylase usually obtained from *E.coli* at neutral pH. Penicillin G for example converted

Table 6.3: Significant stages in strain improvement programme in *P. chrysogenum*

Strain	Method	Penicillin (mg ml^{-1})	Characteristics
P. notatum (Fleming)	**P. chrysogenum**	3	
NRRL 1951	Original isolate	60	Yellow pigment
NRRL 1951.B25	Selection	125	Yellow pigment
X-1612	X rays	300	Yellow pigment
WIS Q 176	UV-light	800–1000	Yellow pigment
B 13-D10	UV-light	-	Pigment free
WIS 47-638	Selection	-	Pigment free
WIS 47-1564	Selection	800–1000	Pigment free, 95-100% G*
WIS 48-701	Selection	-	Pigment free, 95-100% G*
WIS 49-133	Nitrogen mustard gas	1500–3000	Pigment free, 95-100% G*
WIS 49-2166	Nitrogen mustard gas	1500–3000	Pigment free, 95-100% G*
WIS 50-535	Nitrogen mustard gas	1500–00	Pigment free, 95-100% G*
WIS 51-20	Selection	2000–3000	Pigment free, 95-100% G*
EIS.1	NM & Selection	7000	Pigment free, Penicillin G

to 6-APA and phenylacetic acid. The 6-APA is then *ethically* acylated with an appropriate side chain to produce a semi-synthetic penicillin. Some of such semisynthetic penicillins along with strucuture and biological activity are presented in table 6.4 and Fig. 6.4. Hetacillin, bacampicillin, epicillin, pivampicillin, and talampicillin are converted to ampicillin in the body. These penicillins exhibit various improvements including resistance to stomach acids to allow oral administration, to pencillinase and an extended range of activity against gram(+)positive bacteria.

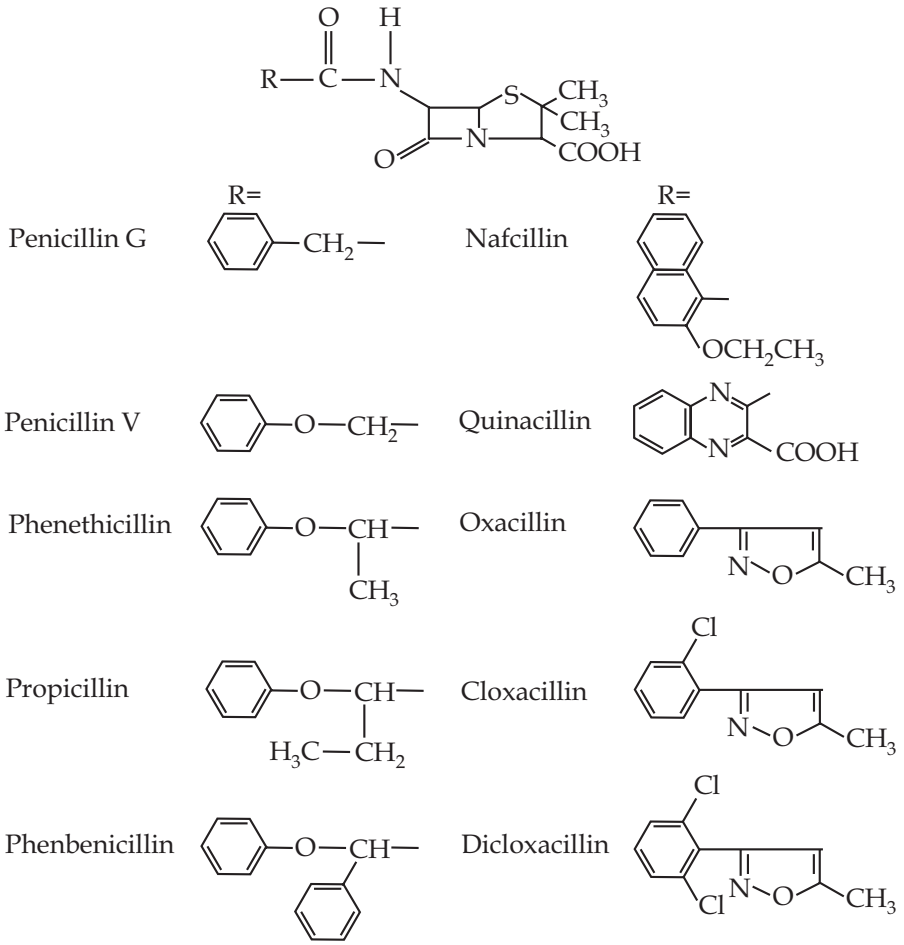


Fig. 6.4: Structure of some clinically useful penicillins

Table 6.4: Characteristics of clinically used semi-synthetic Penicillins

		Specific activity against			
Name	**Method of preparation**	**Gram (+) cocci**	**Gram (–) rods**	**Acid stability**	**Resistance to staphylococcal penicillinase**
Amoxicillin	Semi synthetic	+	+	+	Low
Ampicillin	Semi synthetic	+	+	+	Low
Azidocillin	Semi synthetic	+	+	+	Low
Azolocillin	Semi synthetic	+	Some	+	Low
Bacampicillin	Semi synthetic	+	+	+	Low
Carbenicillin	Semi synthetic	+	Some	-	Low
Cloxacillin	Semi synthetic	+	Nil	+	High
Cyclacillin	Semi synthetic	+	+	-	High
Dicloxacillin	Semi synthetic	+	Nil	+	High
Epicillin	Semi synthetic	+	+	++	Low
Flucloxacillin	Semi synthetic	+	Nil	+	High
Hetacillin	Semi synthetic	+	+	+	Low
Mecillinam	Semi synthetic	+	+	+	High
Metamacillin	Semi synthetic	+	+	+	Low
Methicillin	Semi synthetic	+	-	+	High
Mezlocillin	Semi synthetic	+	+	+	Low
Nafcillin	Semi synthetic	+	Nil	+	High
Oxacillin	Semi synthetic	+	Nil	+	High
Penicillin G	Natural	+	Nil	–	Nil
Penicillin V	Natural	+	Nil	+	Nil
Phenethicillin	Semi synthetic	+	Nil	+	Nil
Pirbenicillin	Semi synthetic	+	+	+	Medium
Pivampicillin	Semi synthetic	+	+	+	Low
Pivmecillinam	Semi synthetic	+	+	+	High
Propicillin	Semi synthetic	+	Nil	+	Low
Sulbenicillin	Semi synthetic	+	Some	+	Low
Talampicillin	Semi synthetic	+	+	+	Low
Ticarcillin	Semi synthetic	+	Some	+	Low

It has been reported that most of the high yielding strains of *P. chrysogenum* are genetically unstable. Genetic unstability increases with the increase in the yield. However, it can be controlled to some extent by following suitable preservation methods. The following preservation methods are generally adopted for storing high yielding strains of *P. chrysogenum*.

1. A spore suspension is stored in a frozen state under liquid nitrogen.
2. A spore suspension can be lyophilized in an appropriate medium.

3. A spore suspension is mixed with a sterile finely divided inert material like soil or sand and desiccated.

6.1.4 Inoculum production: The microorganism which is used in a fermentation process is called as the inoculum. A high yielding strain of *P. chrysogenum* is generally employed as inoculum.

A strain of the fungus is subcultured from stock culture for inoculum development. Spores from primary source are suspended in water or in a dilute solution of a nontoxic wetting agent such as 1:10000 sodium lauryl sulfate. The spores are then added to flasks or bottles of wheat bran plus nutrient solution and these are incubated for five to seven days at 24°C so as to provide heavy sporulation. The entire process is repeated several times in order to have more sporulation.

The resulting spores are used directly to inoculate inoculum tanks or stirred fermenters. The incubation temperature is maintained at 24-27°C for 2 days with agitation and aeration in order to facilitate heavy mycelial growth, which may be added to a second or even a third stage fermentation. The resulting inoculum which is employed in a production tank is tested both by microscopic examination and by subculturing method.

Many sporulation media have been designed to obtain large number of spores. The one developed by Moyer and Coghill (1946) is most extensively used and given below (table 6.5).

Table 6.5: Composition of Moyer and Coghill (1946) sporulation medium

Component	Concentrating (glt^{-1})
Glycerol	7.5
Cane molasses	7.5
Corn steep liquor	2.5
Mg SO_4. $7H_2O$	2.5
KH_2PO_4	0.050
Peptone	0.060
NaCl	5.00
Fe-tartarate	0.005
$CuSO_4.5H_2O$	0.004
Agar	2.50
Distilled water	1.0

(i) Inoculation: Introduction of pure inoculum into the production tanks or fermenters is called as inoculation. This is done by any one of the following three methods.

1. Dry spores may be used as inoculum:

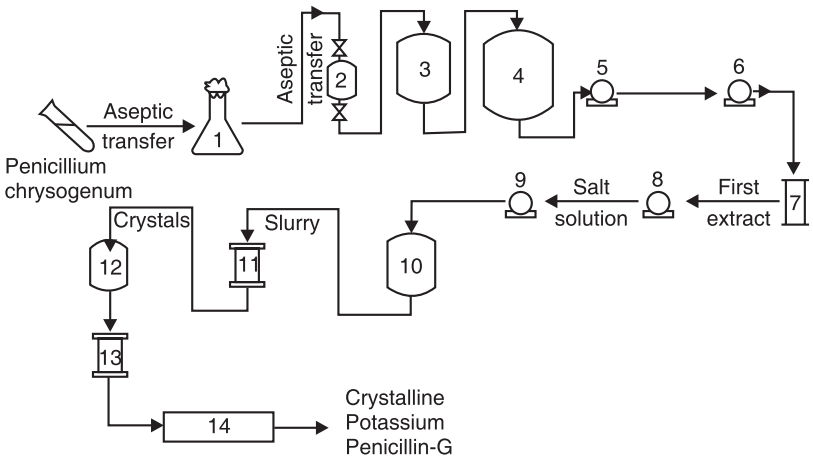
   Since the spores of *P. chrysogenum* are hydrophobic, either spores are blown deep into the medium or a wetting agent such as sodium lauryl sulphate is used.

2. Suspension of ungerminated spores:

   This suspension is made by using 1:10000 sodium lauryl sulfate solution. This suspension is fed to the fermenter by suitable techniques like spray guns or pipettes. This is followed by agitation and aeration of the fermentation medium in order to achieve equal and uniform distribution of the spores in the entire medium.

3. Feeding the fermentation tanks with pre-germinated spores or mycelial pellets which are prepared by the germination of spores. Pellets are generally fed to the fermentation medium after two or three days of spore inoculation.

Fermenters with a capacity of 40,000 to 2 lakhs liters are generally employed for the production of penicillin. Due to difficulties with the oxygen supply larger tanks are not employed. Some manufacturers use of Waldh of fermenters or air lift fermenters, but this is only possible in mutants which generate low viscosity. Depending upon the production strain, the operational temperature is maintained between 25°–27°C. A typical flow chart for penicillin production is given in Fig. 6.5.



1. Shaker flask
2. Transfer vessel
3. Seed tank
4. Fermentation tank
5. Rotary filter
6. First extractor
7. Purification column
8. Second extractor
9. Third extractor
10. Vacuum still
11. Ceramic filter
12. Crystal wash
13. Ceramic filter
14. Vacuum drier

Fig. 6.5: Flow sheet for large-scale production of Penicillin

(ii) Medium: The medium employed for penicillin production should be suitable to achieve:

1. An abundant growth of the mycelium.
2. Maximum accumulation of the antibiotic.
3. Easy and inexpensive extraction and purification of the antibiotic.

Carbon source is generally supplied in the form of lactose. Glucose, sucrose, glycerol and sorbitol can also be employed as carbon source. Nitrogen source is generally supplied in the form of ammonium sulphate or ammonium acetate or ammonium nitrate. Abundant formation of mycelium and spores takes place when a medium contains corn-steep liquor because it contains important amino acids required for mycelial growth. Potassium, phosphorus, magnesium, sulphur, zinc and copper are supplied in the form of salts. Potassium and phosphorus are supplied in the form of potassium dihydrogen phosphate, magnesium, iron and copper are supplied in the form of sulphates. All these elements may be present in corn steep liquor.

Penicillin-F and penicillin-K are the naturally produced penicillins synthesized by *P. notatum* and *P. chrysogenum,* respectively, in the absence of precursor. But, if phenylacetic acid is supplied in the medium *P. chrysogenum* produces penicillin-G instead of penicillin-K. Similarly, desired

synthetic penicillins can be obtained by adding the medium with suitable precursor.

A medium designed by Jackson (1958) which has the following composition, is generally used in fermentative production of penicillin (table 6.6).

Table 6.6: Composition of Jackson's (1958) medium

Component	Concentration (In Percentage)
Corn steep liquor	3.5
Lactose	3.5
Glucose	1.0
Calcium carbonate	1.0
Potassium dihydrogen phosphate	0.4
Edible oil	0.25

Penicillin yields with time are linear from approximately 48 to 96 hours. The final penicillin yield is in the range of 3 to 5% which largely depends upon the amount of carbohydrate consumed during fermentation process, which is approximately equal to 1500 international units per milliliter. Sylvester and Coghill (1954) have estimated that to produce 1000 gallons of fermented culture, which is capable of yielding 2.2–2.7 kg of penicillin by the submerged culture method requires approximately 227 kg of nutrients, 3400 kg of steam, 45460 lt of water, 1000 kWh of electricity and 7075 m^3 of air.

Penicillin easily get carboxylated to form penicillianic acid which is biologically inactive by the action of enzyme penicillinase. The enzyme penicillinase is widely distributed among different microorganisms. These organisms may enter into the fermenter at any stage and may convert penicillin into penicillianic acid (Fig. 6.6). Thus, in penicillin fermentation contamination is a main constraint. Hence, one has to be careful in preventing contamination. This was the one of the main problems during early times of penicillin production, when fermentation was carried out in bottles and contamination in one bottle may destroy penicillin in entire batch of bottles.

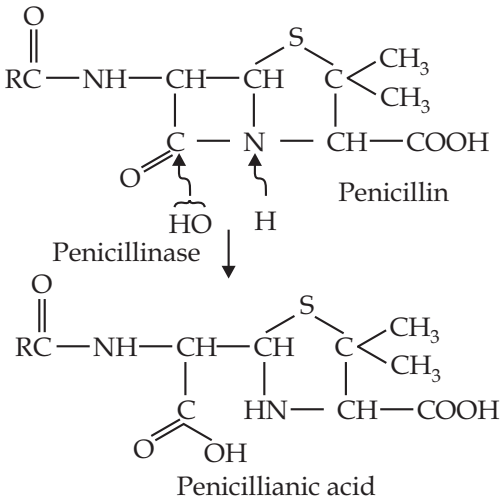


Fig. 6.6: The carboxylation of penicillin by the action of penicillinase

In the typical penicillin fermentation there is a growth of 10 hrs duration with a doubling time of 6 hrs during which the greater part of the cell mass is formed. The oxygen supply in the growing culture is critical since the increasing viscosity hinders oxygen transfer. After growth phase, the culture proceeds to actual penicillin production. The growth is sharply reduced by feeding with various culture medium components. The production phase can be extended to 120-180 hrs. Penicillin production by continuous fermentation has been attempted but it has been difficult due to instability of the production strains. A batch fill and draw system has been suggested as an alternative. In this process 20-40% of the fermentation contents is drawn off and replaced with fresh nutrient solution. This process may be repeated up to 10 without affecting yield.

(iii) Extraction and purification: After it is assessed that sufficient amount of penicillin has been produced during fermentation process, it is extracted and then purified. The entire process is carried out in three different stages. They are :

(a) Separation of mycelium

(b) Extraction of penicillin and

(c) Treatment of crude extract

(a) Separation of mycelium: Mycelium is separated from the medium by employing rotatory vacuum filter. This process should be performed carefully in order to avoid contaminating microorganisms which produce penicillinase enzyme, degrading the penicillin.

(b) Extraction of penicillin: The penicillin is excreted into the medium and less than 1% remains as mycelium bound. Extraction of penicillin is carried out by employing counter current extraction method. The pH of the liquid after separation of the mycelium is adjusted to 2.0 to 2.5 by adding phosphoric or sulphuric acid. This treatment converts penicillin into anionic form. The liquid is immediately extracted with an organic solvent such as amylacetate or butylacetate or methyl isobutyl ketone. This step has to be carried out quickly because penicillin is quite unstable at low pH values. Podbielniak counter current extractor is used for this purpose. The penicillin is then back extracted into water from the organic solvent by adding enough potassium or sodium hydroxide which also results in the elevation of pH to 7.0 to 7.5. The resulting aqueous solution is again acidified and re-extracted with organic solvent. These shifts between the water and the solvent help in the purification of the penicillin. Finally, the penicillin is obtained in the form of sodium penicillin. The spent solvent is recovered by distillation for reuse.

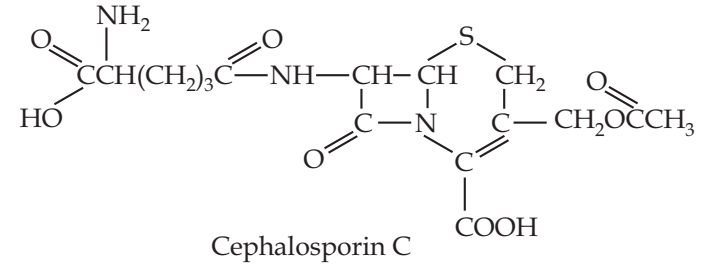
(c) Treatment of crude extract: The resulted sodium penicillin is treated with charcoal to remove pyrogens (fever causing substances). It is also, sometimes, sterilized to remove bacteria by using Seitz filter. Then, the sodium penicillin is prepared in crystalline form by crystallization. It may be packed as powder in sterile vials or prepared in the form of tablets or in the form of syrups for oral usage. The pharmaceutical grade may be used in the production of semi synthetic penicillin.

6.1.5 Uses

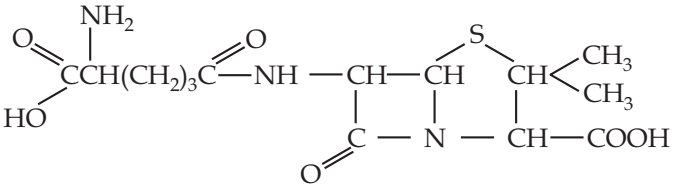
1. Most of the penicillin's are active against Gram-positive bacteria, in which they inhibit the cell wall synthesis leading to the death of bacteria.
2. Used therapeutically in the treatment of infectious diseases of humans caused by Gram (+) positive bacteria.

6.2 CEPHALOSPORINS

Cephalosporins are β-lactum antibiotics containing dihydrothiazine ring with D-α-aminoadipic acid as acyl moiety (Fig. 6.7). Cephalosporin C was discovered in culture filterates of *Cephalosporium acremonium* in 1953. These are also produced by other fungi such as *Emericellopsis* and *Paecilomyces*. In 1971 screening programme designed to discover β-lactum antibiotics, the first cephalomycins (7-methoxy cephalosporins) were produced by species of *Streptomyces* such as *S. lipmanii, S. clavuligerus* or *Nocardia lactamdurans.* These are of most value for their broad spectrum and low human toxicity. Only semi-synthetic derivatives of cephalosporins and cephamycin are useful theraupetically. These are modified to be used orally with improved β-lactum resistance and are termed as first generation cephalosporins. These are further modified so that they are active even against gram (–) ve bacteria and are termed as second generation cephalosporins. From these a third generation cephalosporins are developed which are with excellent β-lactamase stability and broader action spectrum.



Cephalosporin C



Cephalosporin N

Fig. 6.7: The cephalosporins

Thirteen therapeutically important semisynthetic cephalosporins are commercially produced by chemical splitting to form 7-aminocephalosporic acid (7-ACA) with subsequent chemical acylation as well as by modification on the C-3 site (Fig. 6.9). Cephalosporins are as economically significant as penicillins.

6.2.1 Biosynthesis: Biosynthesis of cephalosporin proceeds from α (α aminoadipyl) – L-Cystienyl –D-Valine to isopenicillin N as is the case of benzyl penicillin. In the next stage penicillin N is produced by transformation of L-α AAA side chain into the D-form via the action of a very labile racemase. After ring expansion to deacetoxy cephalosporine by expandase reaction, hydroxylation via a dioxygenase to deacetyl cephalosporin C occurs. The acetylation of cephalosporin C by an acetyl CoA dependent transferase is the end point of biosynthetic pathway in fungi (Fig. 6.8). On the other hand, in *Streptomyces* further transformation of cephalosporin C or deacetyl cephalosphorin C is converted in a two step reaction with molecular oxygen and S-adenosylmethionine to 7-methoxy-cephalospsorin or cephamycin C. In contrast to penicillin, the D-α AAA moiety cannot be changed with precursor feeding.

6.2.2 Production method: The fermentation of cephalosporin is similar to that of penicillin. Complex media with corn steep liquor, meat meal, sucrose, glucose and ammonium acetate are used. Fermentations are carried out as fed batch processes with semicontinuous addition of nutrients, at pH 6.0 – 7.0 at a temperature between 24–28°C. High aeration is necessary in main growth phase and O_2 absorption decreases sharply during production phase.

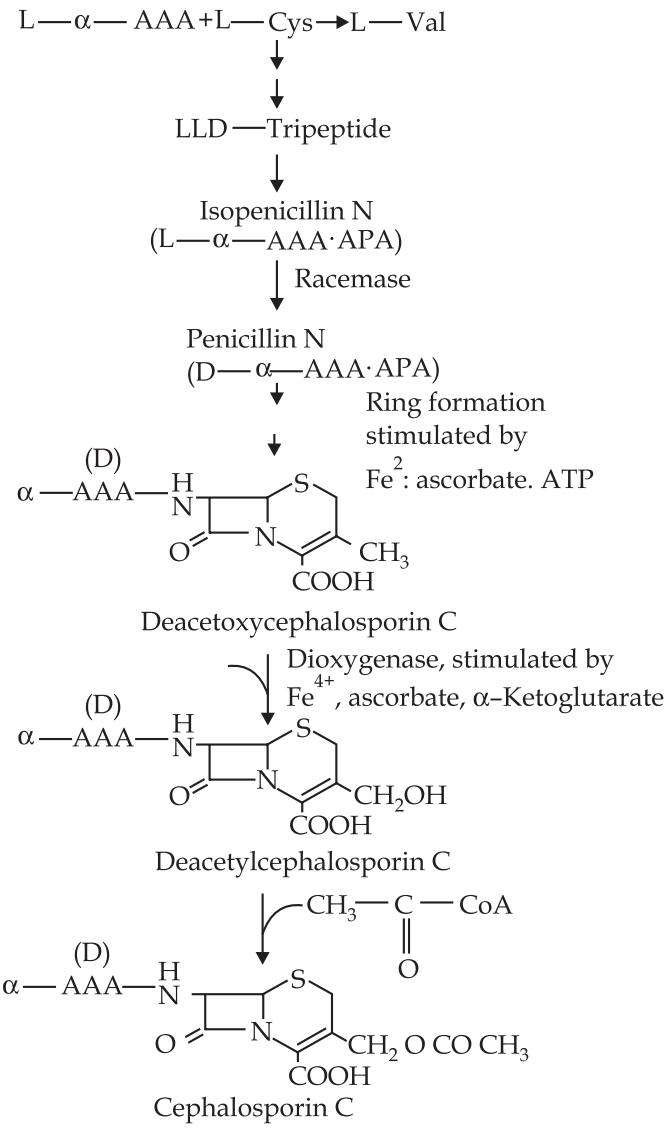
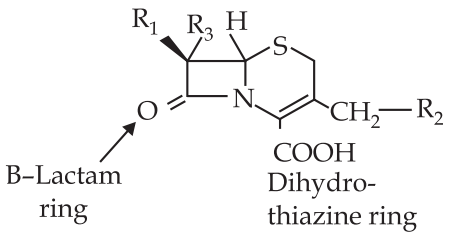


Fig. 6.8: Biosynthesis of cephalosporin by *Cephalosporium acremonium*

Chemical synthesis of cephalosporins by ring expansion of penicillin has been developed which is more economical. By using pencillin V, oraspor, an orally active cephalosporin is produced. Similarly, some of the semi-synthetic cephalosporins along with their structure and characteristics are illustrated in Fig. 6.9.



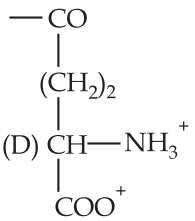
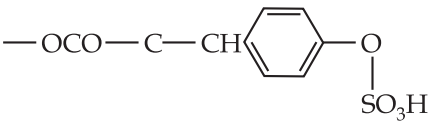
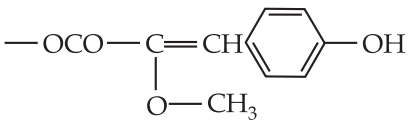
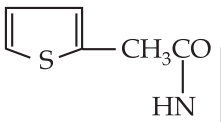
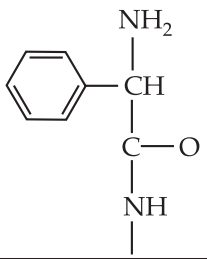
Designation	R_1	R_2	R_3
7-Aminocephalo-sporanic acid (7–ACA)	$-NH_3^+$	$-O-CO-CH_3$	—H
NATURAL CEPHALOSPORINS			
Cephalosporin C		$-O-CO-CH_3$	—H
Deacetyl–3–carbamoyl-cephalosporin C		$-O-CO-NH_2$	—H
7–Methoxy-cephalosporin C			$-OCH_3$
Cephamycin A			$-OCH_3$
Cephamycin B			$-OCH_3$
Cephamycin C		$-O-CO-NH_3$	$-OCH_3$
SEMI-SYNTHETIC CEPHALOSPORINS			
Cephalotin		$-O-CO-CH_3$	—H
Cephalexin		H	—H

Fig. 6.9: Different natural and semi-synthetic *Cephalosporins*

6.3 STREPTOMYCIN

Streptomycin, produced by *streptomyces griseus* **(Schatz et al., 1944)**, is active against Gram (–) ve bacteria and against tuberculosis bacterium, *Mycobacterium tuberculosis*. However, it proved to be useful in the treatment of infections caused by Gram (+) ve specially resistant to penicillin. It is also useful in the control of plant diseases caused by bacteria as it acts systemically in plants. One of the disadvantages of streptomycin is its neurotoxicity due to which hearing impairment and balance maintenance is lost in man due to prolonged streptomycin treatment at high dosage. Its reduction to dehydrostreptomycin results in the decreased toxicity. For this reason in recent times only dihydrostreptomycin is being produced due to ready development of resistance against streptomycin. It is used mostly in conjection with para aminosalicyclic acid or isoniazid (isonicotinic acid hydrazide) which minimizes resistance build up in sensitive microorganisms.

6.3.1 Chemical structure: Streptomycin and dehydrostreptomycin is a aminoglycoside antibiotic and basic compound which is available as hydrochloride, $C_{21}H_{39}N_7O_{12}$. 3 HCl, as a crystalline hydrochloride double salt with calcium chloride or as phosphate or sulphate and dihydrostreptomycin as the hydrochloride or sulfate. The chemical structure of streptomycin is given in Fig. 6.10. Unit of streptomycin activity is equal to one microgram of the free base. Use of precursor does not increase yields of streptomycin.

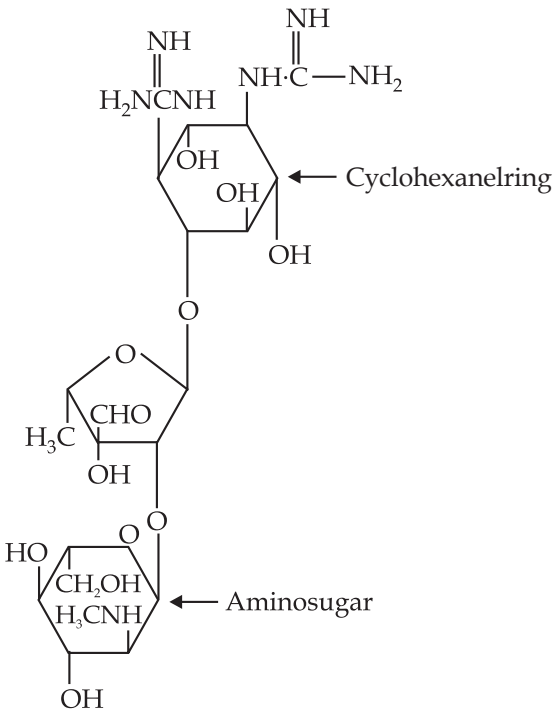


Fig. 6.10: Streptomycin

6.3.2 **Biosynthesis:** Streptomycin is directly derived from glucose. Though the enzymes involved in the synthesis of N-methyl glucosamine are not yet known, it is expected that about 28 enzymes take part in the conversion of glucose into streptomycin as précised in Fig. 6.11.

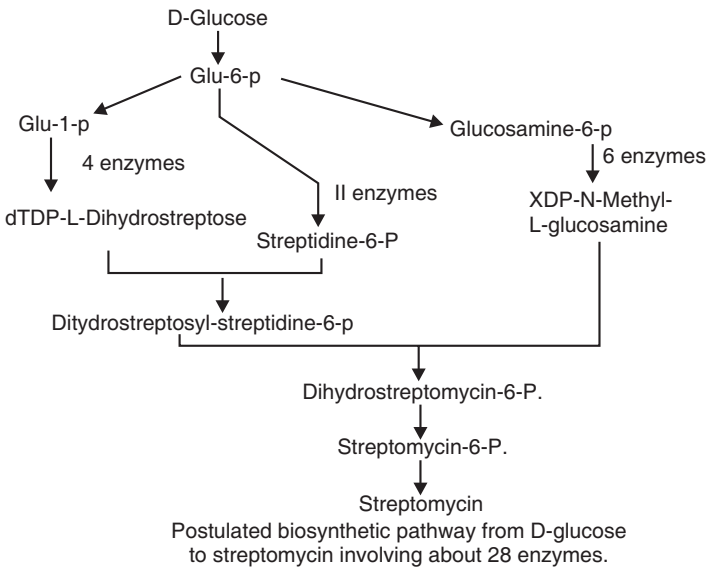
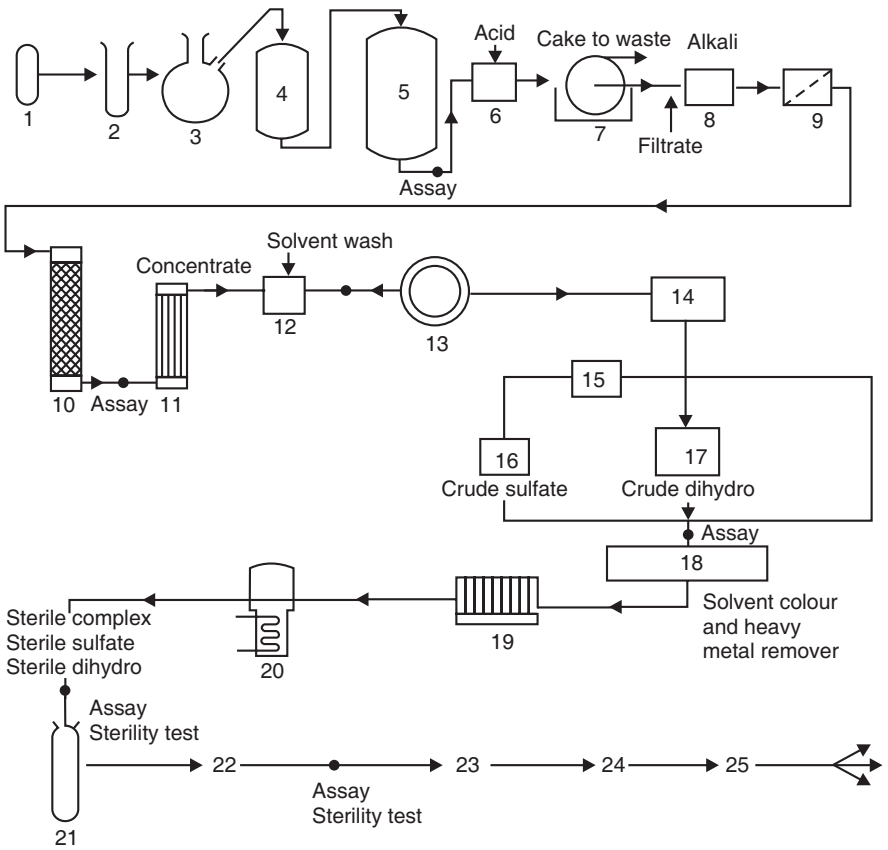


Fig. 6.11: Biosynthetic pathway of Streptomycin

6.3.3 **Fermentation:** Industrially streptomycin is produced by submerged culture method, whose flow sheet is given in **Fig. 6.12**. When **Woodruff** and **Mc Daniel** (1954) suggested medium consisting of soyabean meal (1%), glucose 1% and sodium chloride (0.5%), **Hocken hull (1963)** recommended the medium consisting of glucose (2.5%), soyabean meal (4.0%), distillers dry solubles (0.5%) and sodium chloride (0.25%) and pH 7.3–7.5 for production of streptomycin by *S. griseus.*

(i) **The inoculum production:** Spores of *S. griseus* maintained as soil stocks or lyophilized in a carrier such as sterile skimmed milk, is employed as stock culture. The spores from these stock cultures are then transferred to a sporulation medium to provide enough sporulated growth to initiate liquid culture build up of mycelial inoculum in flasks or inoculum tanks. After sufficient mycelial growth, it is fed to production fermenter.

(ii) **Preparation of the medium:** A production medium contains carbon source and nitrogen source. Glucose is one of the best carbon sources which helps in the greater yield of streptomycin, because it provides basic carbon skeleton for the streptomycin production. Apart from glucose, fructose, maltose, lactose, galactose, mannitol, xylose and starch can also be used as carbon source. Polysaccharides and oligosaccharides generally give low



1. Master culture 2. Agar slopes 3. Shaker flask 4. Seed vessel 5. Fermentor 6. Acidification
7. Filtration 8. Neutralization 9. Filter clarification 10. Ion-exchange reagent 11. Evaporator
12. Crystallization 13. Vacuum oven 14. Calcium chloride crude complex 15. Calcium chloride removal
16. Crystallization 17. Catalytic hydrogenation 18. Finishing 19. Seitz filter 20. Freeze drying
21. Vial filling machine 22. Capping 23. Labelling 24. Packing 25. Despatch

Fig. 6.12: Flow sheet of streptomycin production by submerged-method

yields. Peptones, soya extracts, meat extract, the residue from alcohol distillation, ammonium salts, nitrates and glycine may be used as nitrogen source. Magnesium, calcium, potassium, boron and molybdenum may be used as mineral source along with sulphates, phosphates and chlorides. Phenylacetic acid, L-naphthalene acetic acid may be added as growth stimulating compounds. It is better to add proline into the medium which helps in high streptomycin production. Fats, oils and fatty acids may also be used along with glucose. If necessary antioxidants such as sodium sulphate or starch or agar may also be added into the medium. There is no need of precursor in the production of streptomycin.

(iii) Fermentation: Sterilized liquid medium with all the above substances is fed to the production fermenter. Appropriate volume of inoculum (4-5%) is introduced into it. The optimum fermentation temperature is in the range of 25 to 30°C and the optimum pH range is between 7.0 and 8.0. High rate of streptomycin production, however, occurs in the pH range of 7.6 to 8.0. The process of fermentation is highly aerobic and lasts approximately for 5 to 7 days and passes through 3 phases:

(a) **The first phase:** takes about 24 hours to 48 hours. Rapid growth and formation of abundant mycelium occurs during this phase. The pH rises to 8.0 due to release of ammonia into medium, due to proteolytic activity of S. *griseus.* Glucose is utilized slowly and little production of streptomycin is witnessed.

(b) **The second phase**: lasts for 2 days. Streptomycin production takes place at a rapid rate without increase in the mycelial growth. The ammonia released in the first phase is utilized, which results in the decrease of pH to 7.6-8.0. Glucose and oxygen are required in large quantity during this phase.

(c) **Third phase:** Cells undergo lysis, releasing ammonia and increase in the pH, which falls again after a period of continuous streptomycin production. Requirement of oxygen decreases and the contents of the medium including sugar get exhausted. Finally streptomycin production ceases. An yield of 1200 micrograms per milliliter of streptomycin is obtained.

(iv) **Harvest and Recovery:** After completion of fermentation the mycelium is separated from the broth by filtration. Streptomycin is recovered by several methods. But the one which is generally employed is described below:

The fermentation broth is acidified, filtered and neutralized. It is then passed through a column containing a cation exchange resin to adsorb the streptomycin from the broth. The column is then washed with water and the antibiotic is eluted with hydrochloric acid or cyclohexanol or phosphoric acid. It is then concentrated at about 60°C under vacuum. The streptomycin is then dissolved in methanol and filtered and acetone is added to the filtrate to precipitate the antibiotic. The precipitate is again washed with acetone and vacuum dried. It is purified further by dissolving in methanol. The streptomycin in pure form is extracted as calcium chloride complex**.**

(v) **Byproduct Vitamin B_{12}:** Vitamin B_{12} is produced as a byproduct which will not affect adversely the yield of streptomycin.

6.3.4 Uses

1. Streptomycin is active against Gram(–)negative bacteria. It is used in the treatment of tuberculosis caused by *Mycobacterium tuberculi*.
2. It is also used therapeutically in the treatment of infectious diseases caused by Gram (–) negative bacteria, specially those organisms which are resistant to penicillin.
3. Prolonged treatment by streptomycin at high dosage can produce neurotoxic reactions such as hearing impairment, loss of balance maintenance in man.

6.4 TETRACYCLINES

Chlorotetracycline was discovered in 1945 produced by *Streptomyces aureofaciens.* By its dehalogenation, tetracycline was prepared which was subsequently produced by *S. viridifaciens.* Presently about 20 *streptomyces species* are known to produce mixture of different tetracyclines (table 6.7)

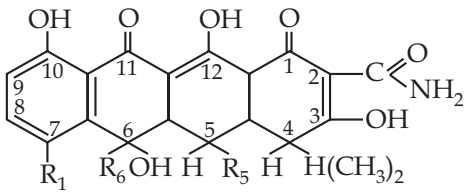


Fig. 6.13: Structure of tetracyclines

	R_1	R_6	R_5
1. Tetracycline	– H	– CH_3	– H
2. Oxytetracycline	– H	– CH_3	– OH
3. Chlortetracycline	– Cl	– CH_3	– H
4. Demeclocycline	– Cl	– H	– H
5. Doxycycline	– H	– CH_3	– OH
6. Minocycline	– $N(CH_3)_2$	– H	– H

6.4.1 **Structure:** Basic structure of the tetracycline consists of Naphthacene ring system. There are also semisynthetic derivatives in the market as well as fermentatively produced tetracyclines (Fig. 6.13).

Table 6.7: Species of *streptomyces* producing tetracyclines

Species	Tetracyclines
S.alboflavus	CTC, TC, OTC, actinomycin
S.antibioticus	OTC,TC
S.auerofaciens	CTC, TC
S.aureus	TC
S.callifornicus	TC, actinomycin
S.cellulosae	OTC, actinomycin
S.feofaciens	TC
S.flaveolus	OTC, TC, actinomycin
S.flavus	CTC, TC, OTC, actinomycin
S.lusitanus	CTC,TC
S.parvus	OTC, TC, actinomycin
S.platensis	OTC
S.rimosus	OTC, TC, rimocidin
S.Siamensis	TC, CTC
S.vendargensis	OTC, vengacide
S.viridifaciens	CTC, TC

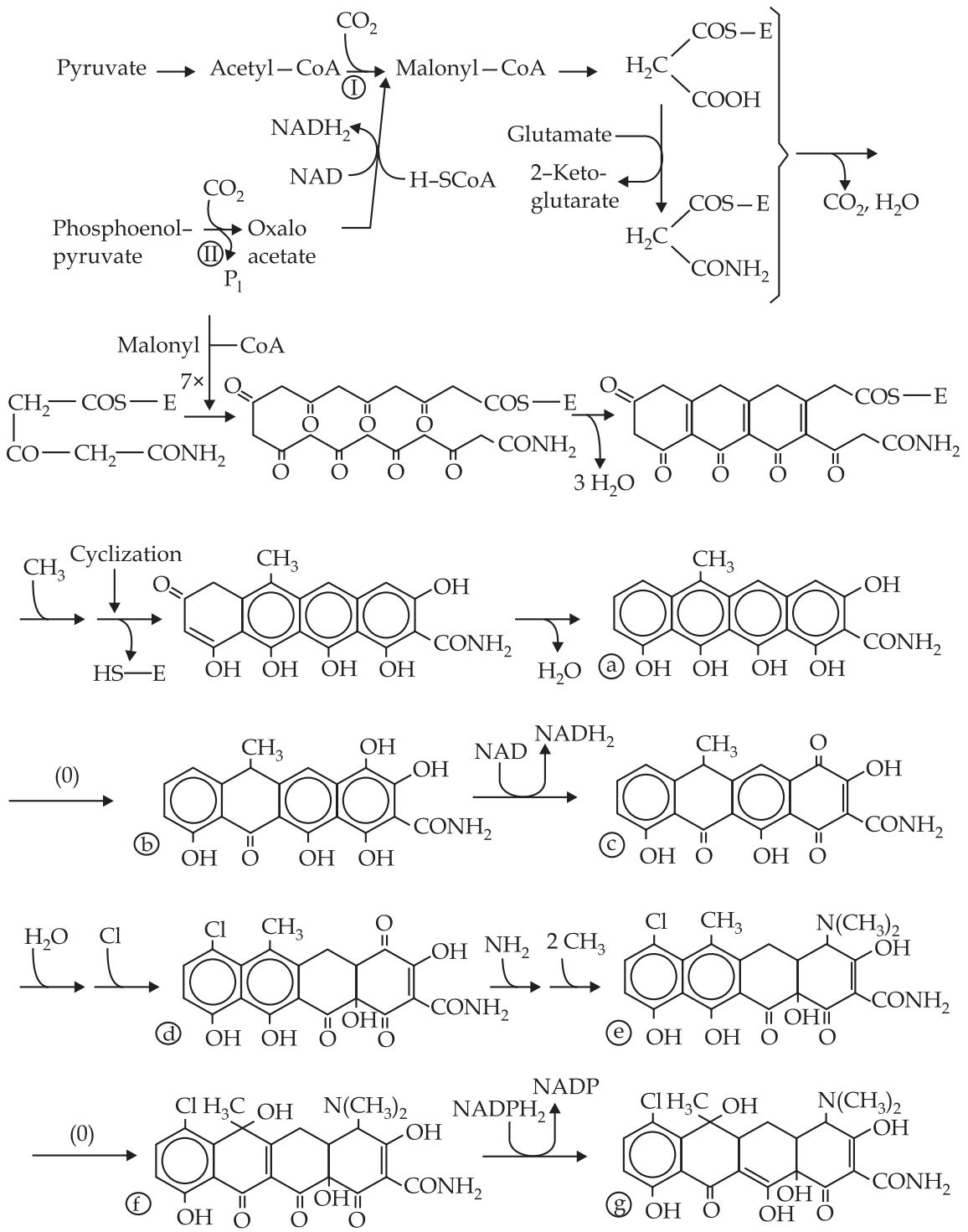


Fig. 6.14: Biosynthesis of tetracyclines

These antibiotics have broad spectrum activity and inhibit gram (+)ve and gram (–)ve bacteria as well as Rickettsias, Mycoplasmas, Leptospiras, Spirocheates and Chlamydas. There is a marked cross resistance between different tetracyclines.

Chlorotetracyclines and oxytetracyclines are commonly used in human and veterinary medicine. They are also used as nutrient supplements in poultry and swine production and also in preservation of fish, meat and poultry.

Tetracyclines inhibit protein synthesis. The site of action is the 30s ribosome where binding of aminoacyl t RNA to the ribosomal A site is inhibited.

6.4.2 Biosynthesis: Tetracyclines such as chloro and oxytetracyclines are produced by species of *Streptomyces* and tetracycline usually being only a minor component. However, *S. aureofaciens* mutants with a block in the chlorination reaction excrete tetracycline as the major product. The biosynthesis can be subdivided into three sections.

1. Glucose is converted into acetyl-Co A
2. Malonyl CoA is produced in the transformation of acetyl – CoA by means of acetyl CoA – carboxylase. After transamination to malonyl CoA, which is bound to an enzyme complex (anthracene synthase), this compound condenses with 8 mol. of malonyl CoA (Fig. 6.14). The polyketide assembly formed from this condensation has not isolated. An alternate pathway to malonyl CoA is via the production of oxaloacetate catalyzed by PEP carboxylase oxaloacetate being converted to malonylCoA by oxidative decarboxytation. Cyclization into the tricylic intermediate also takes place on the enzyme anthracene synthase.
3. The step by step transformation of the assumed tricyclic intermediate, which has not yet isolated, into variety of known tetracyclines.

The scheme of biosynthesis of chlorotetracycline, still mainly hypothetical, has been deduced from studies of mutants blocked in tetracycline biosynthesis and from co-synthesis studies with *S. aureofaceins.* Out of 72 intermediate products, only 27 substances could be characterized and the chlorination reaction is itself not yet understood.

A close correlation could be observed between carbohydrate metabolism and tetracycline production. High yielding strains have lower glycolysis rate than the low yielding strains. The addition of 0.5–3.0 μg ml^{-1} benzylthiocynate covers 50% increase in chlorotetracycline production and pentose phosphate cycle increases simultaneously. Pentose-phosphate regulates tetracycline biosynthesis. The addition of phosphate to producing culture decreases the rate of tetracycline formation by inhibition of ATC oxygenase. The rate of tetracycline synthesis is directly proportional to the activity of this enzyme, which is only detectable in the culture after phosphate has depleted completely.

6.4.3 Strain development: Over 300 genes are reported to be involved in the biosynthesis of chloro- tetracyclines. UV radiation alone or in combination with other mutagens has brought about the greatest number of high yielding strains of *S.aureofaciens* and *S.ramosus.* The increase in cholorotetracycline production with each step of mutagens treatment in the course of strain development is depicted in Fig. 6.15. Strain with increased tetracycline production have been isolated by selecting revertants of mutants blocked in antibiotic synthesis and from prototrophic revertants of auxotrophs. They may also be isolated from mutants showing increased resistance to their own product. The reported highest yield for tetracycline is 25 glt^{-1}. Hybridization alone has not yielded an improvement in tetracycline production, but in combination with mutagen treatment has

resulted an increase in its production. Protoplant fusion studies have also resulted in the increased production.

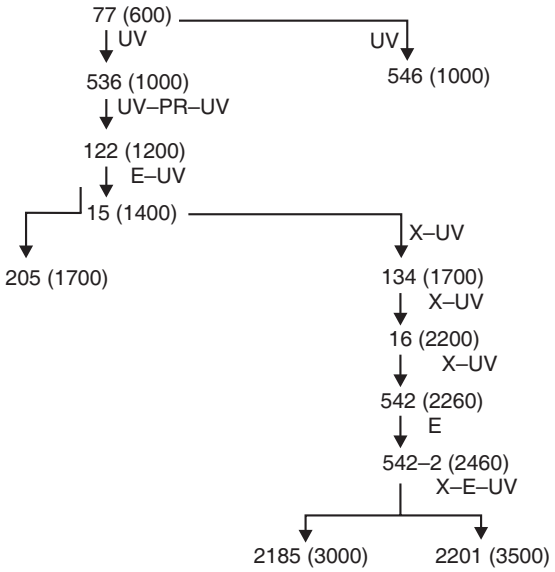


Fig. 6.15: Strain improvement programme in tetracyclines

6.4.4 Genetic regulation of tetracycline biosynthesis:

1. Three hundred genes are reported to be involved in biosynthesis of CTC.
2. First step of oxytetracycline biosynthesis is located at rib b and cyc d.
3. Genes for steps from an hydrotetracycline to oxytetracycline are located in second cluster.
4. Pro A and add A clustering of genes facilitates genetic engineering research.
5. Recombinant – DNA technology has been developed for oxytetracycline producing strain of *S.ramosus.*

6.4.5 Fermentation production:

1. By a chemical process using chlorotetracyline.
2. By fermentation in chloride-free culture medium, which can be made by pretreatment with ion exchangers.
3. By adding chlorination inhibitors to medium such as mercaptobenzothiazole, 2-thiouracil or thiourea. Bromide inhibits chlorination in some cultures but causes formation of 7-bromotetracycline in other strains.
4. With mutants which are blocked in the chlorination reactions.

Production of tetracyclines is carried out in stirred fermenters with volumes upto 1.5 lakhs liters. Typical process of production of chlorotetracycline is depicted in Fig. 6.16. If glucose is used, continuous feedings is necessary, starch can be used as an additional carbon source.

Tetracycslines are produced in phosphate limitation. This fermentation is aerobic and production increased with the increase of oxygen concentration.

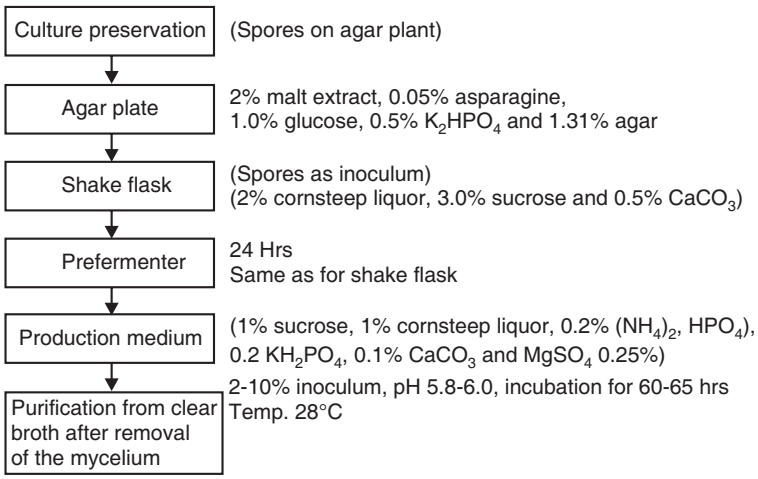


Fig. 6.16: Flow diagram of fermentative production of tetracycline

6.5 ERYTHROMYCIN

Erythromycin is one of the macrolides which are lipophilic with 14 membered macrocyclic lactone ring. Besides, erythromycin, josamycin, leucomycins, decamycin, oleandomycin, spiramycins and tylosine are other antibiotics belong to this group. Macroslides are hydrophobic and usually basic compounds.

6.5.1 Structure: Erythromycin is produced by *Streptomyces erythreus, St. griseoplanus* and species of *Micromonospora.* Erythromycin B, C and D are produced as minor products during fermentative production of erythromycin A. They consist of 12, 14, 16 and 17 numbered lactone rings with 1-3 sugar glycosidically linked with the a glycone (lactone ring) and with each other. The sugars are amino sugars and/ or 6 deoxyhexoses. The structure of erythromycin, a 14 membered macrotide, is given in Fig. 6.17.

Erythromycin is very effective against Gram (+) ve bacteria particularly *Staphylococci* and mycoplasmas, were frequently used against penicillin resistant organisms. Because of the development of the newer β-lactum antibiotics, the macrolides are no longer of great medical significance. The aglycone and sugar moieties are both necessary for antibacterial activity. Therapy with erythromycin results in a rapid development of resistance and there is also cross-resistance between the individiual antibiotics. However, because of low toxicity, efforts are being made to isolate new compounds of this class and also to modify known macrolides through biotransformation.

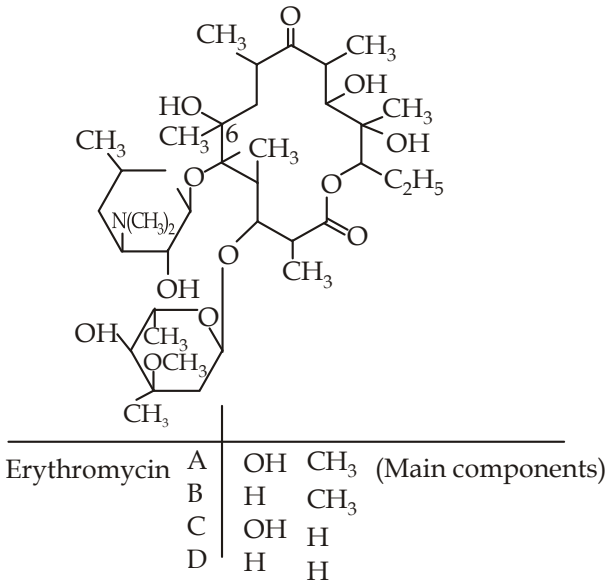


Fig. 6.17: Structure of Erythromycin

Erythromycin is a inhibitor of protein synthesis, binding to the 50s subunit of bacterial ribosomes. Erythromycin is known to inhibit the elongation factor G-dependent release of deacylated tRNA from the p-site of the ribosome.

6.5.2 Biosynthesis: The aglycone moiety of erythromycin is built either of propionate subunits or of a combination of acetate and propionate subunits. Sometimes butryate or related compounds participate in aglycon moiety formation. Details of biosythesis of erythromycin are available from labelling studies with *S. erythreus,* Erythronolide is produced from propionyl – CoA onto which 6 molecules of 2 methylmalaonyl –CoA condense. This polyketide synthesis takes place on a multi enzyme complex called lactone synthetase which is very similar to the fatty acid synthetase of *S. erythreus.* After cyclization to the aglycon, the first compound, 6 –deoxyerythronolide B can be isolated. 6-deoxyerythronolide B is further glycosylated to produce erythromycin C and erythromycin B (Fig. 6.18).

6.5.3 Fermentation: Erythromycin is produced in an aerobic submerged fermentation using *St. erythreus.* Yield in large scale industrial processes are around 20 gl^{-1}. The production medium contains glucose 50 g, soymeal 30 g, $(NH_4)_2 SO_4$ 3 g, NaCl 5 g, $CaCO_3$ 6 g, tap water 1 liter and pH 7.0. Complex culture medium ingredients may also be used in the medium. Fermentation temperature is 33°C and the incubation period ranges from 3 to 7 d. The inoculum from the culture grown on tryptone plates with suspension of at least 10^8 viable cells ml^{-1}.

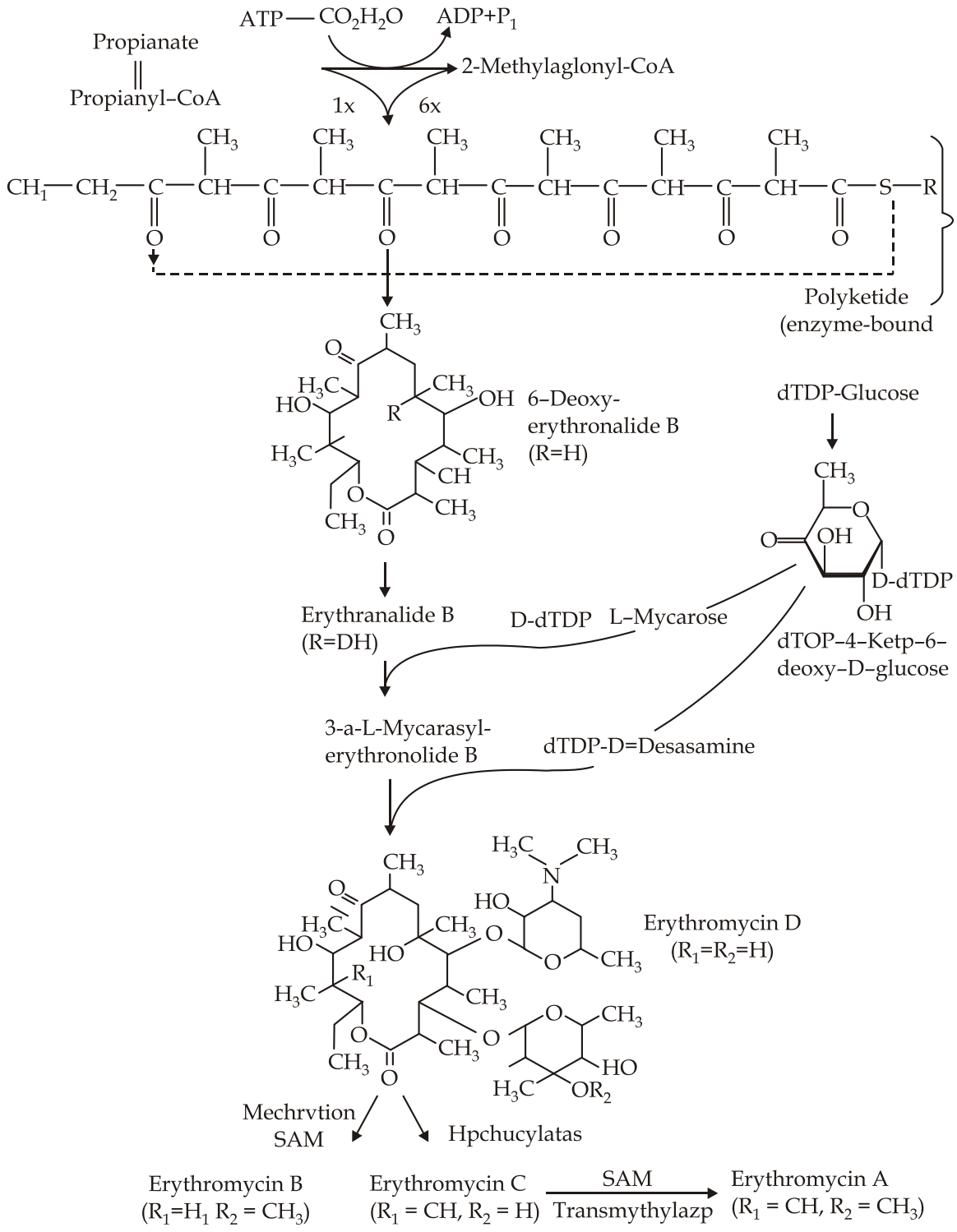


Fig. 6.18: Biosynthesis of erythromycin in *S. erythreus*. SAM: S-adenosyl-L-methionine

Erythromycin production starts when growth reaches the stationary phase, consumption capacity of inorganic phosphate, ammonium nitrogen, metal ions as well as nitrogen source increases and at high concentration of dissolved oxygen. The addition of n-propanol as precursor increases erythromycin titer. The oxygen requirement during 0-12 hrs is 0.4 v/v per min and increases to 0.9 v/v per min from 12 hrs to till harvest. Carbondioxide is added at 0.1 vol/vol/min or 11% of the inlet air which increases the erythromycin yield to 40% of the control.

Erythromycin is produced extracellularly and can be separated by filter press, centrifuge or drum filter with a filter aid. The acidic condition helps to separate mycelium from the broth. Erythromycin is extracted using methyl isobutylketone or ethyl acetate. It is then transferred to acidic water. pH is adjusted with HCl, phosphoric acid, acetic acid or citric acid. The purification and concentration is carried with ion exchange resin i.e amberlite. Further purification step of antibiotic is adsorbed on resin and is eluted by a mixture of organic solvents and water at pH 3.0-8.0.

Production of erythromycin is reported to involve three genes each of which is predicted to code for a protein of more than three lakhs daltons and six molecules each of which inturn contains acyl carrier protein, acyl transferase, keto acyl ACP-synthetase and enoyl – ACP synthetase as well as other domains needed for biosynthetic process.

6.6 CHLORAMPHENICOL

Chloromphenicol, an aromatic antibiotic Fig.(6.19), is produced by *Streptomycetes venezuelae S. Phaeochromo-genes chloromyceticus var S. omiyarnsis* and other streptomycetes. It is a broad spectrum antibiotic acting on both gram(+)ve and gram(–)ve, actinomycetes, *rickettsiae* and *Chlamydias.* Because of its toxic effect on bone marrow chloramphenicol has not been used widely. However, it is an indispensable in the treatment of persistent typhoid due to intracellular *Salmonella typhi* which does not respond to penicillin and other antibiotics. It's toxicity can be reduced if therapy is conducted carefully.

Chloramphenicol binds specifically to the 50s subunit of 70S ribosomes and blocks the peptidyl transferase reaction causing premature chain break. It is used in molecular biology as a research to study translation in prokaryotes without affecting nucleic acid synthesis.

Chloramphenicol is synthesized via aromatic pathway from chorismic acid. It has been successfully synthesized chemically since 1950 and proved to be more economical.

Fermentation: The fermentation is carried out in 30 liters fermenter containing 18 liter medium consisting of glycerol (1%), yeast extract or tryptone, sodium chloride (0.5%) and pH is adjusted to 7.5. The fermentation is carried out at 25°C for 3-4 d. The highest yield obtained was 200–300 mg l^{-1}.

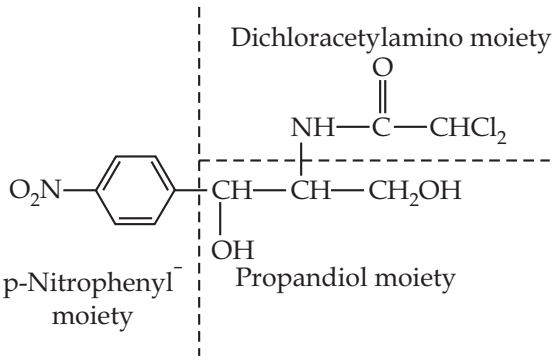


Fig. 6.19: Structure of Chloramphenicol

Chloramphenicol is extracted from the clarified broth. The filterate is extracted either with ethyl or diluted with kerosene and then washed with dilute acetic acid, sodium carbonate and water. The lipids are removed by petroleum ether. The crude product is decolorized by passing the organic solution through a column of charcoal or alumina. The purified product is recrystallized from ethylene or ether and petroleum ether mixture.

6.7 FUSIDIC ACID

Fusidic acid, a steroid antibiotic (Fig. 6.20), is produced by *Fusidium coccineum*. It is therapeutically important in the treatment of staphylococcal infections, specifically wound infections and those of skin infections.

Fusidic acid, is reported for the first time by **Godfredsen** *et al.,* (1962) to be produced by *Fusidium coccidium* by species of *Cephalosporium* and ***Mucor ramannius***.

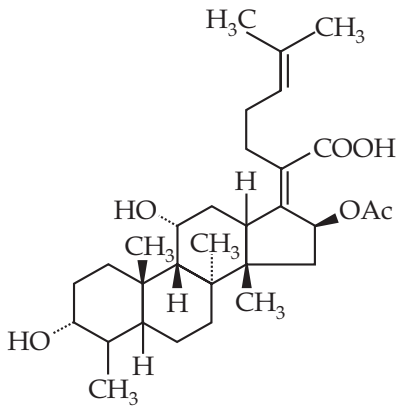


Fig. 6.20: Structure of Fusidic acid

Fusidic acid has been used extensively in the treatment of infections caused by penicillin resistant **staphylococci.**

It inhibits t-peptidyl transferase in both pro – and eukaryotes and affects EFG:GDP ribosome complex by forming stable complex. It is also a uncoupler of oxidative phosphorylation (inhibits ATP synthesis). Fusidic acid is most useful in the treatment of β-lactamase resistant staphylococci.

6.8 GRISEOFULVIN

Griseofulvin GN was reported by Oxford *et al.* (1939) from cultures of *Penicillium griseofulvum*. It is now known to be produced by several species of *Penicillium*. Commercial product of griseofulvin by high yielding strain of *Penicillium patulum*. Its chemical structure (Fig. 6.21) was elucidated by **Grove** *et al.* (1952) and is 25 trans -7 chloro -2,4,6 trimethoxy 6-methylspiro –(benzofuran -2 (3H)-1-2-cyclohexane-3,4'- dione). It inhibits net proteins, carbohydrates, lipids and RNA without affecting DNA synthesis. These levels cause regular curling of hyphae, forms of binucleate and multinucleate cells. GN caused mutation in *Microsporus gypseum*. In general GN inhibits mitosis in fungi by affecting microtubules of nuclear spindle.

Oral administration of GN controls infection of skin, hair and nails caused by *Microsporum, Trichophyton* and *Epidermophyton*. Griseofulvin is known to bind strongly to keratin of the skin. Although griseofulvin is useful in the treatment of skin infections it cannot be used in systemic mycosis.

Fermentation : *P. griseofulvum* is grown in 3 different media–(i) sporulation medium (whey powder lactose (30.0g), $NaNO_3$ (3.0g), KH_2PO_4 (1.0g), $MgSO_4.7H_2O$ (0.5g) and $FeSO_4.7H_2O$ (0.02g), (ii) germination medium (protopeptone (20g), malted cereal extract (10.0g), glucose (40.0g), starch (20.0g), $NaNO_3$ (3.0g), KH_2PO_4 (1.0g), $MgSO_4$ (0.5) and $FeSO_4$ (0.02) and (iii) seed stage medium (cornsteep liquor (3.0g) brown sugar (30g), chalk (10.0g), corn oil (10.0g) and Hodag M.F.(0.3g).

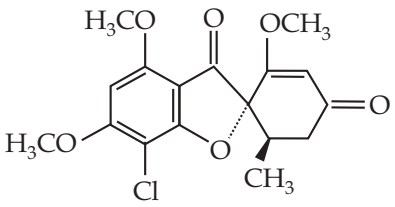


Fig. 6.21: Structure of Griseofulvin

For production of griseofulvin medium composition is cornsteep liquor (0.17g) $CaCO_3$ (0.4) KH_2PO_4 (0.4) and KCl (0.15). For good yield production of griseofulvin 12% carbon source, 0.4–0.5% phosphate and chlorine is required during fermentation. The pH is maintained at 6.8, and 3.2 incubation temperature was 25°C and it is an aerobic fermentation. Yield is 6–8g l^{-1} after 10 days after addition of methyl donors such as chlorine salts, methyl xanthate and folic acid to the medium.

Griseofulvin is extracted from the mycelium with organic solvent and butyl acetate and evaporated to dryness. The crude griseofulvin is washed with chloroform and recrystalized from aqueous methanol. Methylene chloride gives yield upto 95% and purity upto 88%.

6.9 BACITRACINS

Bacitracins, polypeptide antibiotics, are produced by a group of bacteria which are active only on closely related organisms. **Gratia** (1925) was the man who discovered bacitracin produced by *E.coli* which was effective against another strain of *E.coli*. The same was renamed as colchicine by **Gratia** and **Fedriq (1946)**. In addition to its use as a topical antibiotic, it is also used as a growth promoter in animal feeds. As a animal feed supplement, it can be used as the zinc or manganese salt, as the lignin bacitracin complex or as bacitracin methylene disalicylate. They

exhibit activity primarily against closely related species. They are attached to specific cell receptors and their genetic determents were plasmid encoded. The definition includes that the synthesis of bacitracins was lethal to producer. Bacitracins are given names according to the genus or species of producing organism eg. *Pediococcus- pedicines* and *Clostridium botulinum-bollicine*.

Table 6.8: Classification of Bacitracins from Gram-Positive Bacteria

Class	Characteristics	Producer	Example
I	Antibiotic, small peptide (19-37 amino acids), heat stable	Lactococcus loles	Nisin, lacticin 181, lactocin S, czarnocin
II	Small heat-stable non-antibiotic (<15 kDa)	Pediococcus	Pediocin PA-1, diplococcin, lactacin B, lactacin F, leuccin A, UAL 187, lactococcin A
III	Large heat		Helvetiain J, Caserem 80, lacticins A and B

Table 6.9: Bacitracins from Gram-Negative Bacteria

Class	Molecular weight (kDa)	Producer	Example
I	4–90	*Escherichia coli*	Colicins A,B,D,E1-E7, Ia,Ib, K,L,M,N,V
II	56	*Enterobacter cloaca*	Cloacin DF13
III	< 1–5	*Escherichia coli*	Microcins A15,B17,C7,D15,D140,E492,1147,15m

Different strains of *Bacillus licheniformis* produce a mixture of bacitracins such as bacitracin A.A,B,C,D,E,F,F_1,F_2,F_3 and *G. Bacitracin* is the main component (about 70%) and has the greatest biological activity. It is a dodecapeptide consisting of L and 4D –amino acids and contains both a cyclic hexapeptide and a thiazoline ring structure (Fig. 6.22).

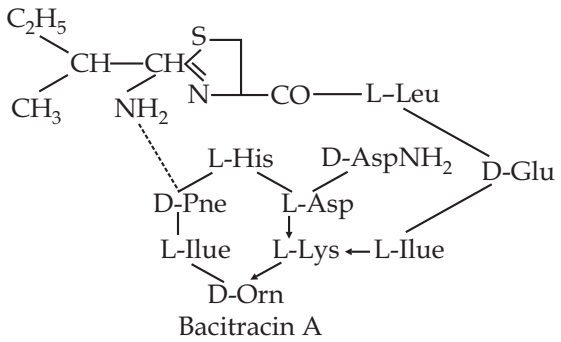


Fig. 6.22: Structure of Bacitracin

Bacitracins are also produced by both Gram (+)ve bacteria (table 6.8) and Gram (–)ve bacteria (table 6.9). Total production of bacitracins world wide was estimated to be 500 tons. Peptide formation can occur by either ribosomal or non-ribosomal mechanisms. Most peptide antibiotics

does not involve this protein-synthesizing machinery, but takes place on a multi-enzyme complex. This non-ribosomal peptide synthesis process which has also been called the thiotemplate mechanism is described here for bacitracin (Fig. 6.23).

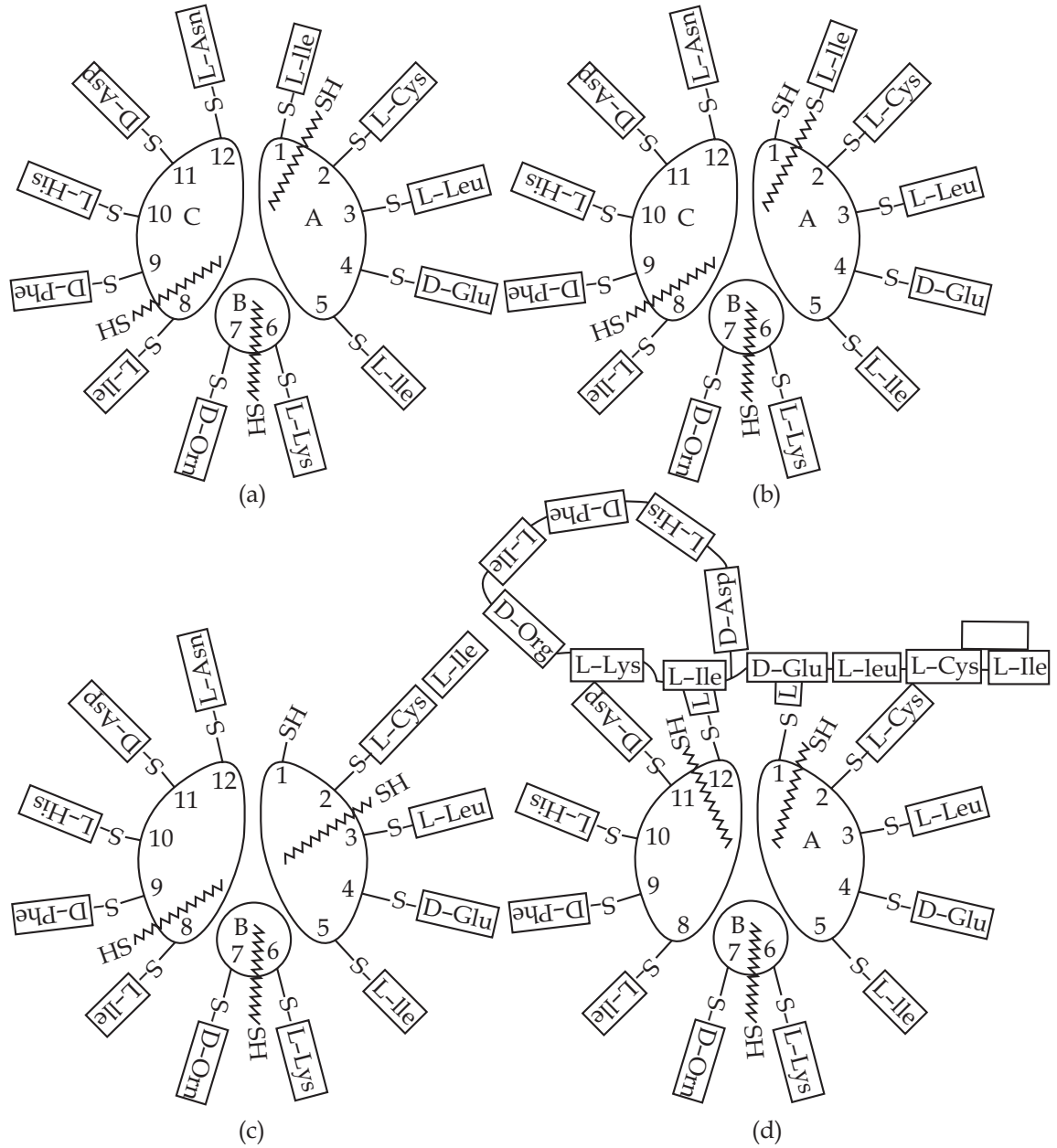


Fig. 6.23: Bacitracin biosynthesis through non-ribosomal peptide synthesis

Bacitracin synthesis consists of subunits A, B and C with molecular weights of 20,000, 2,10,000 and 3,80,000 daltons respectively. The first step is the activation of amino acids in the presence of ATP and Mg to aminoacyl adenylate and takes place at a specific activation site for

each amino acid on the enzyme complex. The activated amino acids are then bound to specific thiol groups in thioester linkages on the enzymes. Various strains of *Bacillus licheniformis* produces mixture of bacitracin A, B, C, D, E, F and G. Bacitracin A which is dodecapeptide, has greatest biological activity. It inhibits the formation of the bacterial cell wall at the level of peptidoglycan synthesis. It prevents dephosphorylation of C55 – isoprenyl pyrophosphate to C55 isoprenyl- phosphate. The amino acid sequences of the subsequent peptide is determined by the arrangement of thiol groups in thioester linkages on the enzyme. The racemization to D-enantiomers probably take place at this level. The transfer of the growing peptide chain (growth of the N-terminal toward the C-terminal end) to the next amino acid and to the next enzyme sub-unit is catalyzed by 4′ phosphopantheine, a cofactor, of each subunit. The cyclization of the peptide chain takes place in the multienzyme complex, but the formation of thiazoline ring is not yet fully understood.

6.9.1 Fermentation Production: Bacitracin is now being produced by aerobic submerged culture. Production of bacitracin is précised in (Fig. 6.24).The yields of this process were increased from about 13 mg liter^{-1} to about 9g liter^{-1} through strain development and medium optimization. A continuous fermentation process has also been developed for bacitracin production using immobilized cells.

Bacitracins are extensively used as food and feed supplement. Bacitracins intended for use as food preservatives should fulfil following conditions:

1. Proven safe for human consumption.
2. Economically acceptable cost.
3. Proven effective at relatively low concentrations.
4. No detrimental effect on organoleptic characteristics of the food.
5. Stable during storage and effective for the self life of the food.
6. No medical use.

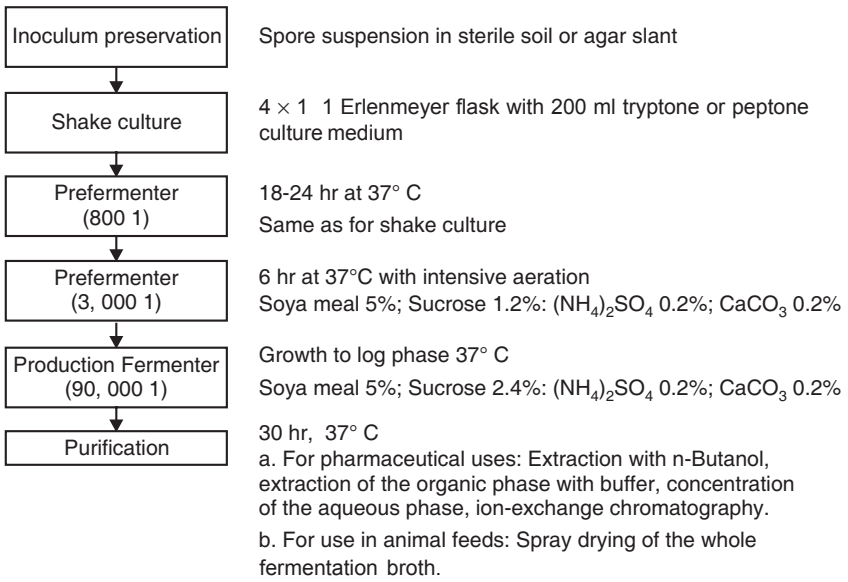


Fig. 6.24: Flow diagram for the production of bacitracin by *B.licheniformis*

6.10 NISIN

Nisin is a primary metabolite. It is a member of well characterized peptides termed lantibiotics which contain unusual dehydro residues and thioether amino acids that introduce the characteristic lanthionine rings into the peptide chain (Fig. 6.25).

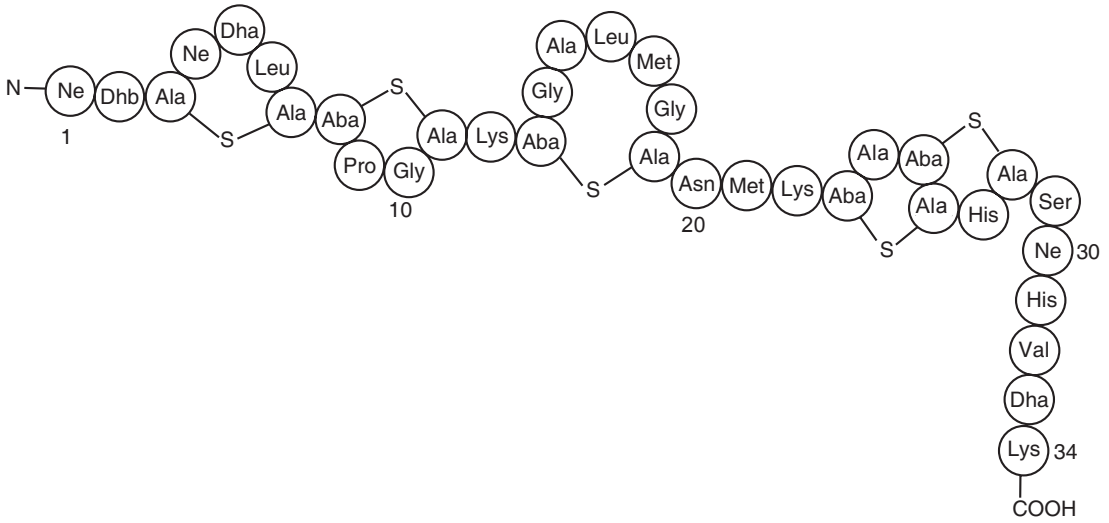


Fig. 6.25: Structure of Nisin

Nisin, a bacteriocin, is produced by *Lactococccus lactis*. It is amphilic polypeptide, pentacylic peptide of 34 amino acids with a molecular weight of 3500 daltons. It contains unusual amino acids including one lanthionine, four β- methyl lanthionine, one dehydrobutyrine (Dhb) and two dehydroalanine (Dha) residues. Dha and Dhb arise from dehydration of L-serine and L-threonine respectively. Condensation of Dha and Dhb with L-cystine generates thioether bonds to form the amino acids lanthionine and B-methyl – lanthionine, respectively. These changes result from extensive post- translational modifications to the ribosomally sysnthesized 57 amino acids, residual precursor peptide, prenisin. The active nisin molecule is finally formed and released from the cell after cleavage by protease to 34 amino acids peptide. On the post-translational modification nisins can be classifed into two types:

Type I : Lantobiotic bacteriocines undergo extensive post translation modification e.g. Nisin.

Type II: Non-Lanthionine bacteriocines undergo minimal post translational modifications e.g. lactococcine and pediocine.

The gene for nisin biosynthesis along with immunity to nisin, nisin transport, serine properties such as facility to ferment sucrose are organized as an operon that is encoded on a chromosomally located 70kb conjugative transposon. Nisin acts on membrane resulting in the leakage of low molecular weight materials i.e. ions, amino acids and ATP. Nisin molecules enter into a susceptible organism and conggregate in groups and then form pores. This also depletes the electric charge stored across the membrane and the bacterium begins metabolizing ATP to produce new protons in a futile effort to recharge the membrane eventually depleting its ATP molecules and unable to maintain its other activities and quickly dies.

It is very active against Gram (+) ve bacteria such as *Bacillus, Clostridium, Desulfomaculum, Enterococcus, Lactobacillus, Leuconostoc, Listeria, Micrococcus, Pediococcus, Staphylococcus and Sporolach.* Individually, it is inactive against G (–)ve bacteria, yeasts and filamentous fungi. It is very active against *Bacillus sporothermodurans* which is problematic in ultra high temperature treated products.

It is extensively used in controlling bacterial spoilage of both heat processed and low acid pH foods, which include dairy products, specially processed cheese, cheese spreads, egg products, dressings, along with several canned foods, nisin is particularly useful in controlling our growth of endospores of *Bacillus* and *Clostridum*. It is also used in heat sanitizers for the prevention of bovine mastitis in dairy cattle. It has potential to control oral bacteria responsible for mouth odour, plaques, acid gingivitis. It can be used as skin care protective agent in the control of skin infection of *Staphylococcus aureus* and *S.pyogenes*. It has a possible role in the control of *Helicobacter pylori* which infects the stomach mucosa and is associated with 90% of peptic ulcers.

6.10.1 Nisin production: Nisin is manufactured by controlled fermentation of *Lactobacillus lactis* in a milk based medium at pH 2.0. Above pH 3.0 nisin is absorbed by the producer cells, but it is completely destroyed at pH 3.0 or below. Consequently at the pH of the fermentation all nisin is released into the medium from which it can be extracted at the end of fermentation. Solvent extraction methods have been used and one step immunoaffinity chromatography method is highly efficient. However, in the industrial scale process, it is concentrated and separated by a simple low-cost foaming process. This involves merely bubbling nitrogen or air through a column of completed fermentation medium. As nisin is a surface active agent, it accumulates in the foam at the air aquous interface. The foam is collected, broken mechanically and the nisin is recovered. It is then spray dried before being milled into fine particle and finally standardized by the addition of sodium chloride.

6.10.2 Bacitracins are most useful in the preservation of milk and dairy products; meat and poultry products; fish and sea foods.

1. **Milk and dairy products:** (i) Reduces microbial growth in milk, (ii) inactivates mesophytic bacteria, (iii) in fermented dairy product it inhibits gas forming *Clostridium tyrobutryicum* and pathogenic bacteria such as *listeria monocytogenes, Bacillus cereus* and *staphyloeccus aureus* in cheese. It also controls over production of acid in yogurt and accelerates cheese ripening by releasing increased bacterial intracellular enzymes. It also inhibits endospore formation in processed dairy products.
2. **Meat and poultry products:** In contamination of raw meat and extends self life vacuum packaged raw meat and inhibits the growth of *listeria monocytogenes*. It inhibits the spoilage lactic acid bacteria and other bacteria including pathogenic bacteria in cooked meat products. Similarly it is useful in increasing shelf life of egg products. In fermented meats it is useful in the control of spoilage lactic acid bacteria and pathogenic bacteria like *salmonella, listeria* and *staphylococcus* and promotes the growth of starter cultures.
3. **Fish and sea products:** It is in combination of other bacitracins it is extensively used in the control of fish and other sea foods.

6.11 INTERFERONS

Interferons are antiviral, non-antibody proteins synthesized and secreted by vertebrate cells in response to virus infections and other stimuli. Cells exposed to interferons respond by acquiring resistance to subsequent virus infection. These are first described as biologically important regulating proteins called cytokines. Interferons also able to inhibit the proliferation of cancer cells because of these combined properties of inducing antiviral resistance and regulating cell growth, interferons are used as therapeutic agents for the treatment of both viral infections and cancer. Interferons are able to modulate the immune response and this has led to their clinical use in diseases underlying immunological etiologies including multiple sclerosis.

Interferons were discovered in 1957 as antiviral agents synthesized in influenza virus infected chick embryo cells. Interferons are produced by cells in response to viral infection and confer an antiviral state on cells exposed to them. In addition to avian, interferons are also present in fish, reptiles and mammals. Interferons are of two types.

1. Type I: Interferons is predominantly 1FN α and β but it also includes two other sub family members IFN ω' and IFN γ.
2. Type II: Comprises only 1FN- γ or earlier known as immune or type 2 interferon which is produced by lymphoid cells in response to mutagens and by sensitized lymphocytes when stimulated with specific antigen.

Interferons are primarily produced by leucocytes and they consist of a single polypeptide chain of 165-166 amino acid residues. Some are glycosylated with varying amount of carbohydrate moieties and their molecular mass is in the range of 16000-26000Da. The carbohydrate portion does not appear to confer any functionality on IFN α and may be removed without affecting their activity. This property allows recombinant 1 FN – α to be produced in prokaryotic systems such as *E.coli* which are not capable of the post transcriptional modifications necessary to form glycosylated polypeptides (Fig. 6.26). Recombinant 1FN-α is now manufactured by a number of companies for example Hoffmann–La Roche produce Roferon and Schering plough make intron which have combined sales worth in excess of US $ 750 millions.

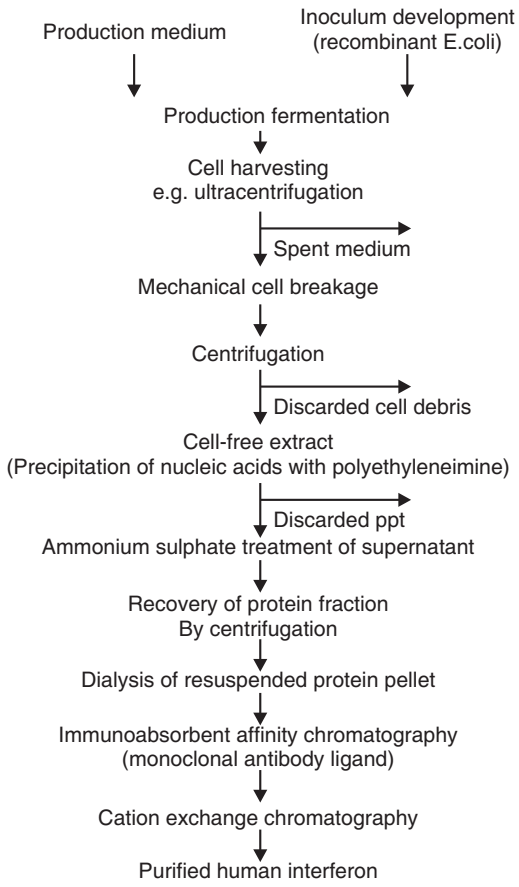


Fig. 6.26: Flow chart of production of human interferons

IFN β is naturally synthesized by mammalian fibroblast cells and can be produced recombinantly from E.coli. These products such as **Betaseon** and **Rerif** are used in the treatment of relapsing multiple sclerosis. IFN- γ sometimes referred to as immune interferon is produced naturally by activated lymphocytes. It is most important interferon involved in the immune system where it activates helper cell (TH1), natural killer cells (NK), cytotoxic T-lymphocytes (CTL) and macrophages and also exhibits antiviral and oncostatic properties. Preparations of recombinant 1FN γ from *E.coli* e.g. Genenechs Actimmune, are used in the treatment of chronic myeloid leukemia and renal cancer.

6.11.1 Applications of interferons: Though interferons hold a promise of treatment of viral diseases, it was not realized until the production and testing of recombinant interferons which are now approved for the treatment of hepatitis B and C. Interferon α is the only effective treatment for patients with chronic hepatitis C. However, it has approval of interferons (type I) for a range of malignant, viral and autoimmune diseases (table 6.10). In addition to hairy cell leukemia, both natural and recombinant interferons are used for treating chronic myelogenous leukemia, melanoma, renal cell carcinoma, lymphoma and karposis sarcoma.

Table 6.10: Approved clinical uses of Interferons

Disease category	Type
Cancer	Hairy cell leukemia
	Chronic myelogenous leukemia
	Myeloma
	Renal cell carcinoma
	Kaposis sarcoma
	Melanoma
Viral	Hepatitis B
	Hepatitis C
	Hepatitis D
	Papilomas
Immune dysfunction	Multiple sclerosis
	Chronic granulomatous disease

Interferons are either used alone or as adjunct therapy and new methods for stabilizing interferon, increasing bioavailability are impacting on the clinical utility. In addition to viral hepatitis, interferons are also used for treatment of herpetic keratitis and laryngeal and genital papillomas. Although use of interferon in multiple sclerosis was quickly terminated because of adverse outcomes, it is approved for treating chronic granulomatous diseases. Recombinant interferon β produced either as an unglycosylated form in *E.coli* or as glycosylated protein in mammalian cell cultures is used extensively in the treatment of relapsing/remitting multiple sclerosis.

6.11.2 Allergic responses: Interferons, however, cause some adverse side effects such as chills/ rigors, fever, myaglias and mild neurotropenia with initial injections, while fatigue anorexia, mild neutropenia, elevation of transaminase, weight loss and depression due to chronic administration.

REVIEW QUESTIONS

I. Essay type questions:

1. Define antibiotic. Discuss various factors contributing to their production.
2. Give mechanism of action of antibiotics.
3. Describe the process of penicillin production.
4. Give detailed account of strain improvement of *Pencillium chrysogenum*.
5. Give an account of semi synthetic penicillins.
6. Describe process of streptomycin production.
7. Give an account of tetracyclines.

II. Write short notes on:

1. Antibiotic assay
2. Spectrum of antibiotic action
3. Stock culture
4. Cephalosporins
5. Chloramphenicol
6. Griseofulvin
7. Semi-synthetic cephalosporins
8. Antibiotic assay

FURTHER READING

1. Elander, R.P.(1989). **Bioprocess technology in industrial fungi.** In Fermentation Process Development of Industrial Organisms. (ed. J.O. Neway,), 169-219.

   Marcel Dekker, New York.

2. Berbach, G.J.M., Van Der Beek, C.P. and Van Dick, P.W.M. (1984). **The penicillins properties, biosynthesis, and fermentation.** In Biotechnology of Industrial Antibiotics (ed. E. J.Vandamme), pp. 45–140. Marcel Dekker, New York.
3. Lowe, D.A. (1986). **Manufacture of penicillins**. In Beta-Lactam Antibiotis for clinical use (S.F. Queener, J.A. Webber and S.W. Queener, eds.), pp. 117–161. Marcel Dekker, New York.
4. Paradkar, A.S., Jensen, S.E. and Mosher, R.H. (1997). **Comparative genetics and molecular biology of beta-lactam biosynthesis.** In Biotechnology of Antibiotics, 2nd Edition (ed. W.R.Strohl, ed.), 241-277. Marcel Dekker, New York.
5. Queener, S. and Schwartz, R.W.(1979). **Penicillins: biosynthetic and semi-synthetic.** In Economic Microbiology, Vol. 3 (ed. A.H. Rose), 35–122. Academic Press, London.
6. Smith, A. (1985). **Cephalosporins.** In Comprehensive Biotechnology, Vol. 3 (ed. M.Moo-Young), pp. 163–185. Pergamon Press, New York.
7. Strohl, W.R. (1997). **Industrial antibiotics:** *today and the future.* In Biotechnology of Antibiotics, 2nd Edition (ed. W.R. Strohl), 1–47. Marcel Dekker, New York.
8. Vandamme, E.J. (1984). Antibiotic search and production: an overview. In Biotechnology of Industrial Antibiotics (ed. E.J.Vandamme), 3–31. Marcel Dekker, New York.

7

Vitamins

Vitamins are the most commonly needed growth factors not only by human beings but also animals. Most vitamins function as parts of coenzymes. Many microorganisms are autotrophic for vitamins and able to synthesize many of the required vitamins. Therefore, they can be exploited for the commercial production of certain vitamins such as thiamine, riboflavin, folic acid, pantothenic acid, pyridoxal, vitamin B_{12} and biotin. β-carotene, which is a provitamin A is also synthesized by certain microorganisms. Certain vitamins are produced directly by a single fermentation with the help of microorganisms, while others are produced by combined chemical and microbial methods. Although a wide variety of vitamins are produced by microbial fermentation, only some of them like Vitamin B_{12} (cyanocobalamin), Riboflavin (B_2), Vitamin C (ascorbic acid) and Vitamin A (β-carotene) are produced through fermentation process on large scale.

7.1 CYANOCOBALAMIN (VITAMIN B_{12})

Vitamin $B_{12,}$ also called as Cyanocobalamin, is synthesized in nature exclusively by microorganisms and it is required by animals for growth and metabolism. It is present in every animal tissue in a very low concentration (1 ppm in the liver cells). It remains in the form of a coenzyme (adenosyl or methyl cobalmine) in the animal tissue. Animals get the required vitamin B_{12} through feed or by absorption of it from the intestinal wall as it is produced in the intestine by the intestinal microorganisms. However, human beings obtain vitamin B_{12} only from food, since the vitamin B_{12} secreted in the intestine by microorganisms cannot be assimilated. Its daily requirement of human beings is 0.001 mg. It helps in the utilization of vegetable proteins.

7.1.1 Chemical structure: Vitamin B_{12} is not a single compound but a group of closely chemically related cobamides. It is a complex chemical compound. Its structural formula is given in Fig. 7.1.

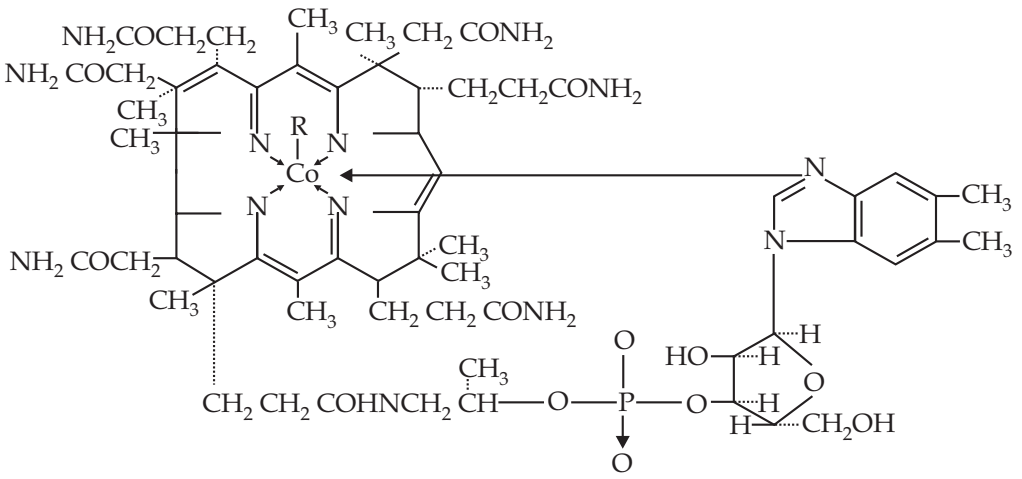


Fig. 7.1: Structure of cyanocobalamin (vitamin B_{12})

Chemically, vitamin B_{12} is called as Cyanocobalamin, which is formed by the union of cobamide and a nucleotide consisting of 5,6 dimethyl benzimiazole. The cobamide unit consists of four pyrrole molecules joined together by centrally located cobalt ion. The cobalt ion is also linked to cyanide group. Though cobamide ring resembles porphyrin ring of chlorophyll but it differs with the later in not possessing a methane bridge between A and D pyrrole rings.

7.1.2 Biosynthesis: The biosynthesis of vitamin B_{12} runs parallel with the biosynthesis of porphyrin and chlorophyll upto the formation of uroporphyrinogen – III. (Fig. 7.2).

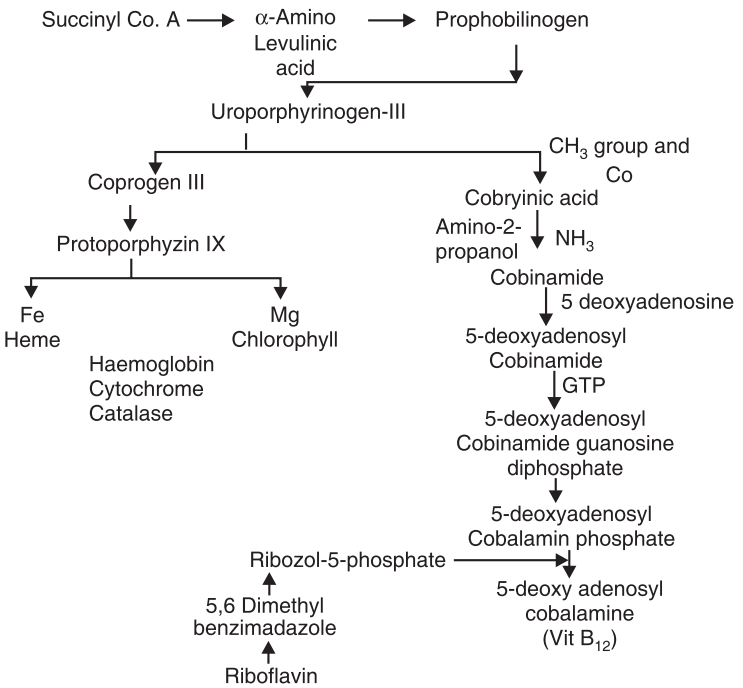


Fig. 7.2: Biosynthesis of Vitamin B_{12}

Modern genetic engineering technique is being applied to increase vitamin production. A hybrid strain called *Rhodopseudomonas protamines,* made by the protoplast fusion technique between *Protaminobacter ruber* and *Rhodopseudomonas sphaeroides*, produces 135 mg per liter vitamin B_{12} using glucose as carbon source without the addition of 5,6- dimethyl benzemidazole.

The following microorganisms may be employed in the fermentative production of vitamin B_{12} on large scale. *Streptomyces griseus, S. olivaceus, Bacillus megaterium, Propionibacterium shermanii, P. freudenreichii, Micromonospora sp. and Klebsiella pneumoniae.*

7.1.3 Fermentative production: Vitamin B_{12} is entirely produced on commercial scale by the fermentation process. The employment of a particular microorganism in the fermentative production of vitamin B_{12} differs according to the carbon source used. Production of vitamin B_{12} by *Streptomyces olivaceus* NRRL B-1125 is described where distillers solubles like soyabean meal, casein, yeast extract etc., are used as carbon source. Submerged fermentative process is generally employed. Important stages of biosynthetic pathway of B_{12} in *Propionibacterium shermanii* (Fig. 7.3) and *Pseudomonas denitrificans* (Fig. 7.4).

7.1.4 Inoculum production: In the first phase pure agar slant culture of *Streptomyces olivaceus* NRRL B-1125 is inoculated into Ehrlenmeyer conical flasks containing 100 to 250 ml of inoculum medium (Bennetts agar) whose composition is given in table **7**.1 and allowed to grow under shake culture for 72 hrs.

Table 7.1: Composition of Bennett's agar medium (inoculum medium)

Chemical	Concentration (g liter^{-1})
Yeast extract	1.0
Beef extract	1.0
N – Z – Amine – A (enzymatic hydrolysate of casein)	2.0
Glucose	10.0
Agar	15.0
Distilled water	1000 ml
pH	7.3

In the second phase, the flask culture is transferred to inoculum tanks arranged in series and containing large amount of inoculum medium whose composition is similar to the one used in the first phase.

Three such transfers are generally made in order to raise required amount of inoculum culture. Approximately, an inoculum of 5% of the volume of production medium is generally employed.

7.1.5 Preparation of the medium: A production medium should contain a source of carbon, nitrogen and cobalt. The composition of the production medium generally employed in the fermentation process is given in table 7.2.

Table 7.2: Composition of a production medium

Chemical	Concentration (percent)
Distillers solubles	4.0
Dextrose	0.5 to 1.0
$CaCO_3$	0.5
$CoCl_2 \cdot 6H_2O$	1.5 to 10 ppm
pH	7.0
Inoculum	5.0

The distiller's solubles that are used in the medium may be used either in the form of soyabean meal or casein or yeast extract. Though cobalt does not support microbial growth, it is added to the medium in order to have maximum yield of cobalamine. Sometimes cyanide is also added to the medium to facilitate the conversion of other cobalamines to vitamin $B_{12.}$ Some industries based their fermentation with the help of *Propionibacterium shermanii.*

7.1.6 Fermentation process: It is carried out in large tanks (Fig. 7.3). Approximately an inoculum of 5% of the volume of production medium is fed to the tank and fermentation process is allowed to continue for 3 to 5 days. It is important to monitor the aeration, pH and temperature for the successful completion of the fermentation. Temperature of 30 °C, oxygen 0.5 volume per minute, soyabean oil, corn oil, lard oil or silicones are added as antifoaming agent.

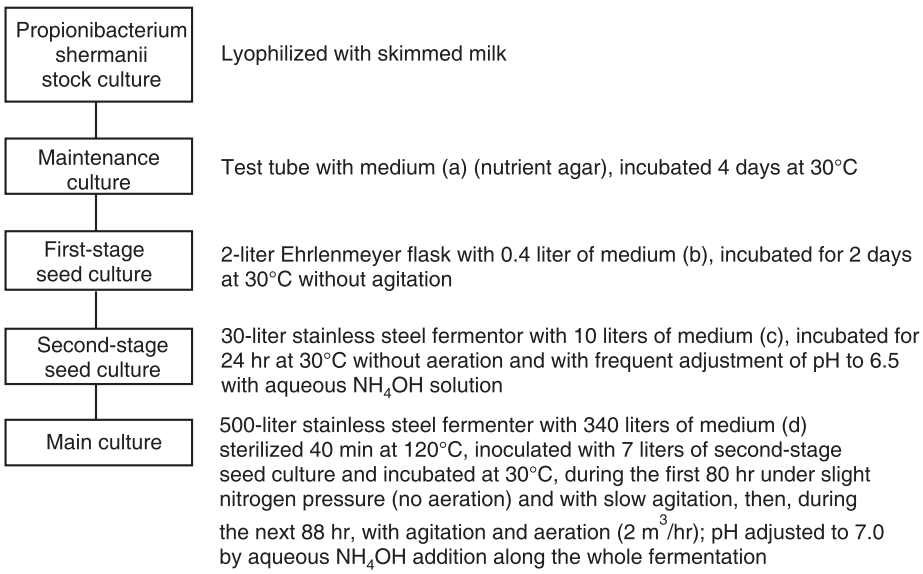


Fig. 7.3: Vitamin B_{12} production by *Propionibacterium shermanii*

7.1.7 Aeration and agitation: The aeration is achieved by sending about 0.5 volume sterile air per volume medium per minute into the production tank. The air is sterilized by passing through column containing activated charcoal. Foaming of the medium is controlled by

using any of the sterilized antifoam agents like lard oil, soyabean oil or corn oil etc., incubation temperature of 27 °C is suitable for the fermentation.

Monitoring of pH is very important in the process of fermentation. During first 24 hours, there will be rapid consumption of sugar which leads to fall in pH and shifts to alkaline side after 2 to 4 days because of lysis of the mycelium. However, it is stabilized by reducing the pH to 5 with the help of sulphuric acid and small amount of sodium sulphate. About 1 to 2 mg of cobalamine per liter of the medium is formed during the fermentation.

7.1.8 Harvest and recovery: Though considerable amount of cobalamine is in the broth at the end of the fermentation period, equal amount of it remains associated with the mycelium. It is liberated from the mycelium by heating the broth to boiling point at a pH of 5 or less. The recovery process after this step depends upon the type of product to be produced. The following recovery procedure is adapted for getting crystalline form of vitamin B_{12}.

1. The mycelium is separated from the broth by filtration.
2. The filtered broth is treated with cyanide to convert cobalamin to cyanocobalamin.
3. The adsorption of cyanocobalamin from the solution is achieved by passing it through an adsorbing agent packed in columns like activated charcoal, bentonite, fullers earth and ion exchange resins.
4. Elution of cyanocobalamin from adsorbent is done by the use of aqueous solution of organic bases like water-acetone, solution of sodium cyanide or sodium thiocyanate or hydrochloric acid.
5. Then it is extracted by counter current distribution between cresol and amyl phenol or benzyl alcohol and water or single extraction into an organic solvent like phenol.
6. Later, it is precipitated as copper or zinc cyanide- cyanocobalamin complex.
7. Thus, formed cyanocobalamin is adsorbed by passing it through a column packed with an adsorbing agent. Activated charcoal or bentonite, fuller's earth, ion exchange resins etc. are used as adsorbants.
8. Finally, it is eluted from adsorbant by the use of an aqueous solution of substances ranging from hydrochloric acid to organic bases such as sodium thiocyanate, or sodium cyanide or water acetone. Counter current distribution between cresol, amylphenol, benzyl alcohol.
9. Sometime it is also precipitated as copper or zinc cyanide – cyanocobalamin complex.

   Raw product of 80% can be used as feed additive. For medical uses, it is purified to 95 – 98%. The total yield is 75% of carbon substrate used (table 7.3).

Table 7.3: Yield of Vitamin B_{12} by different microbes

Microbe	Medium	Yield (mg liter^{-1})	Temp (°C)	Aeration	Incubation period (in hrs)
Bacillus megaterium	Molasses	0.45	30	A	18
B.coagulans	Mineral	0.5	35	A	18
Butribacterium rettgeri	Cobalt	5.0	30	A	-
Streptomyces olivaceus	-	3.3	30	A	-
Micromonospora sp p.	-	11.5	30	A	-
Klebsiella pneumoniae	-	0.2	30	A	-
Propionibacterium freuden neichii	Glucose	19.0	30	A	-
Propionibacterium shermanii	Cornsteep liquor NH_4, cobalt	30 – 40	30	A A	150
Pseudomonas denitrificans	Oxalic acid betaine cobalt mineral salt	60	30	A	-

Some industries based their fermentation of vitamin B_{12} on *Bacillus megaterium, B. coagulans* or *Pseudomonas denitrificans* (Fig. 7.4). Still many organisms use alcohols to produce economical amount of Vitamin B_{12} (table 7.4).

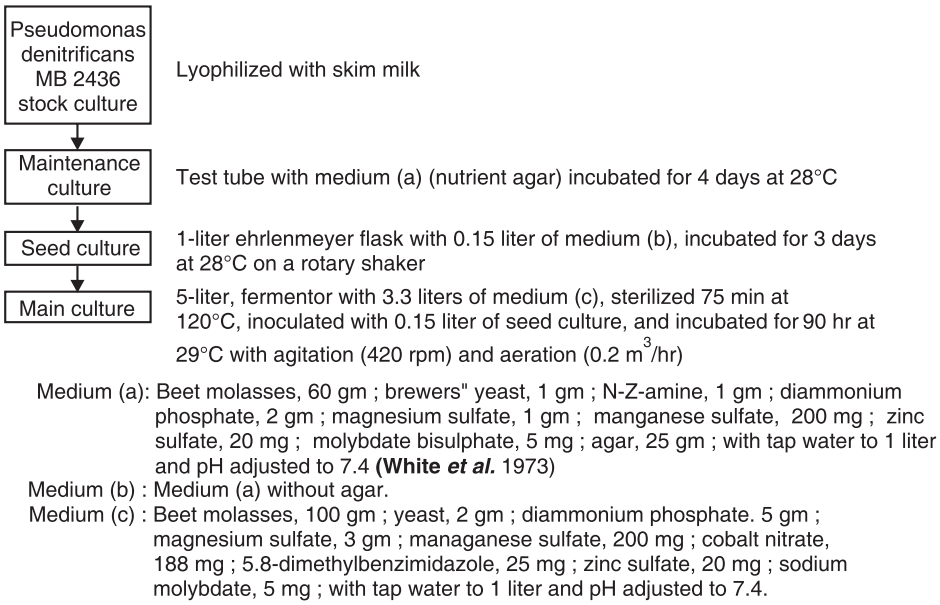


Fig. 7.4: Vitamin B_{12} Production by *Pseudomonas denitrificans*

Table 7.4: Vitamin B_{12} production from different alcohols as carbon source

Microorganism	Carbon source	Concentration (%)	Vitamin B_{12} Yield (mg l^{-1})
Bacteria from activated sludge	Methanol	0.3 – 1.2	35
Methanobacterium soehngenii	Methanol	0.8	8.0
Methanobacillus omelianskii	Methanol	0.8	8.8
Protaminobacter ruber	Methanol	1.2	2.5
Methanosarcina barkeri	Methanol	0.8	42
Fed batch culture			
Arthrobacter sp.	Isopropanol, fed Batch culture		1.1
Arthrobacter hyalinus	Isopropanol	1	0.56
Nocardia MT 2003	1,2 – Propanediol	5	0.78
Klebsiella sp. 101	Ethanol	2	0.02
	1,2 – Propanediol	2	0.06
	Methanol	2	0.16
	Methanol	0.5	1.3
Fed batch culture			
Pseudomonas sp. ATCC 14718	Methanol	1	0.05
	1,2 – Propanediol	2	0.12
	Methanol	2	0.16
	Methanol	0.7 – 0.8	3.2
Fed batch culture			
Strain XF	Methanol	2	0.23
Pseudomonas ovalis	n-Decane	1	0.08
Pseudomonas aureofaciens	n-Dodecane	1	0.04
Nocardia gardneri	n-Hexadecane	2	4.5
Mixed culture of *Corynebacterium* sp. and	n-Alkanes	10	2.3
Rhodopseudomonas sphaeroides			

7.2 VITAMIN A (β-CAROTENE)

Cartenoids are found in animal and plant tissues but originate exclusively from plants or microbes. β-carotene (pro vitamin A) is converted into vitamin A in the intestinal mucous membrane and is stored in the liver as palmitate ester. In human beings, deficiency of vitamin A (daily requirement 1.5 – 2.0 mg) causes night blindness and changes the skin and mucous membranes. It is particularly necessary for the normal biosynthesis of mucopolysaccharides.

About 100 tons of β-carotene is needed for one year mainly as food coloring agent and improving colour in meat or egg yolk in poultry industry or in the salman industry (astaxcanthine). At present, it is produced chemically as fermentative production is not economical. Halophytic alga *Dunaliella salina*, a marine algae, is employed successfully in the production of vitamin A in a competition market.

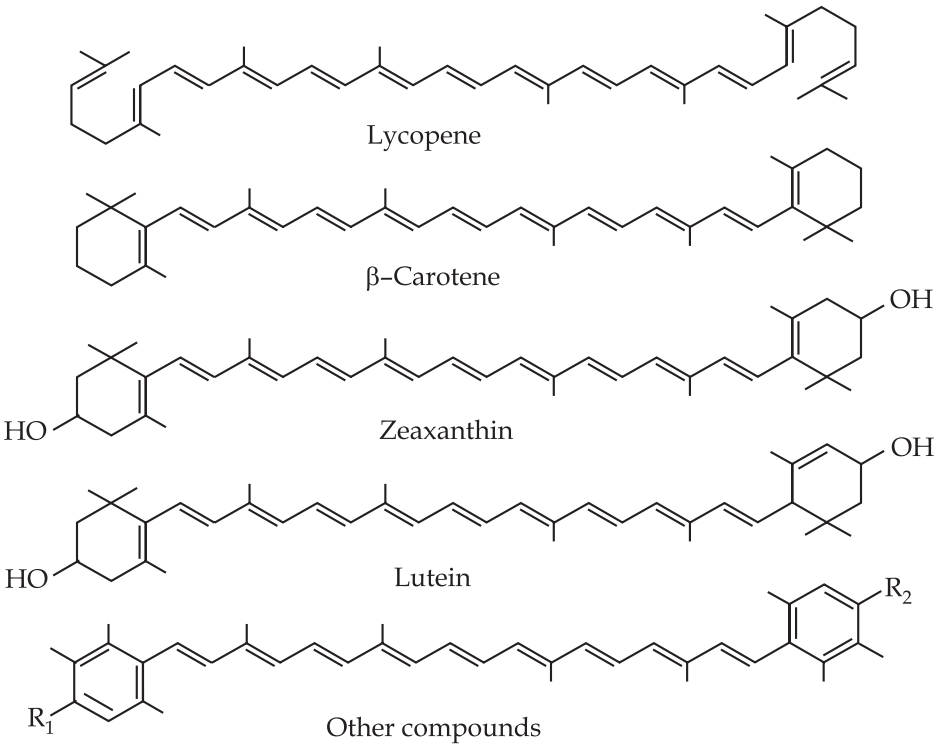


Fig. 7.5: Structure of some important carotenoids

7.2.1 Structure: Carotenoids are highly unsaturated isoprene derivatives. Naturally occurring carotenoids are tetraterpenoids consisting of 8 isoprene residues. Some of the carotenoids that can be produced by fermentation are given in Fig. 7.5.

7.2.2 Biosynthesis: Biosynthesis of β-carotene has been studied in plants and fungi is depicted in Fig 7.6. Only compounds with β-ionone structure (The ring structure tonnd at each end of the β-carotene molecule) are effective as provitamin A. Thus two molecules of Vitamin A can be formed from β-carotene, while only one molecule of vitamin A can be formed from α-and γ-carotene since each of these molecules has β-ionone ring.

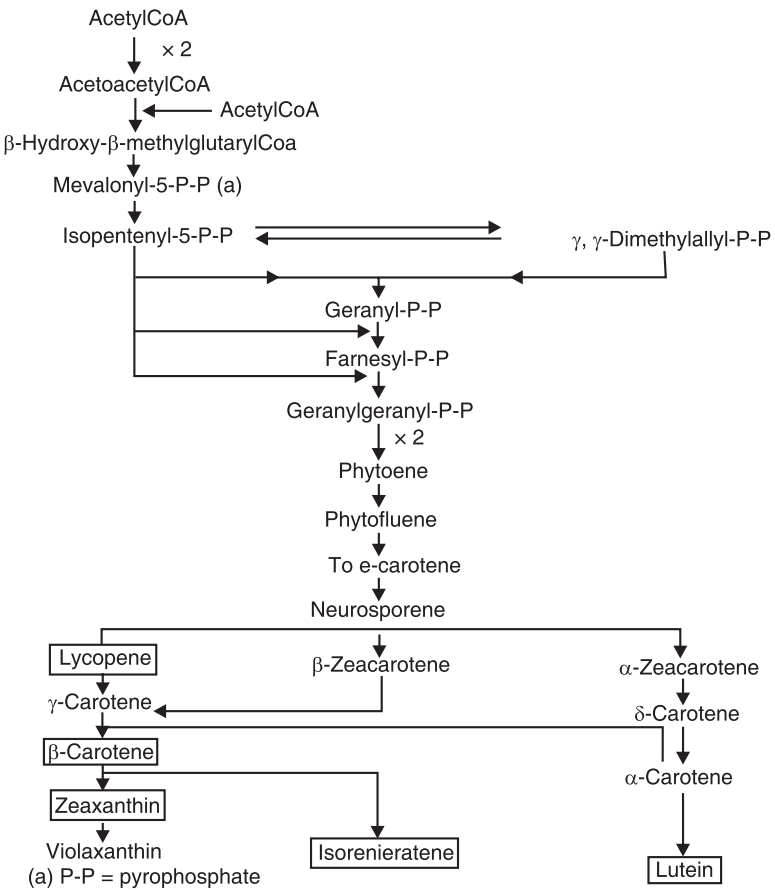


Fig. 7.6: Biosynthetic pathway of carotenoids

7.2.3 **Fermentative production:** *Blakeslea trispora* is reported to produce highest amount of vitamin A when two stains (+) and (–) are grown together. The carotene is produced during the process of zygospore formation. When cultures of both sexual forms (+) and (–) strains are mixed, a significant increase in carotene production is recorded (Fig 7.7) Production of carotene is induced by mixture of trisporic acids which act as (+) genome sex hormone. They are derived biosynthetically from β-carotene. On the other hand, *Phycomyces blakesleanus* can produce β-carotene even by one strain i.e (–) strain (**Fully et al., 1960).**

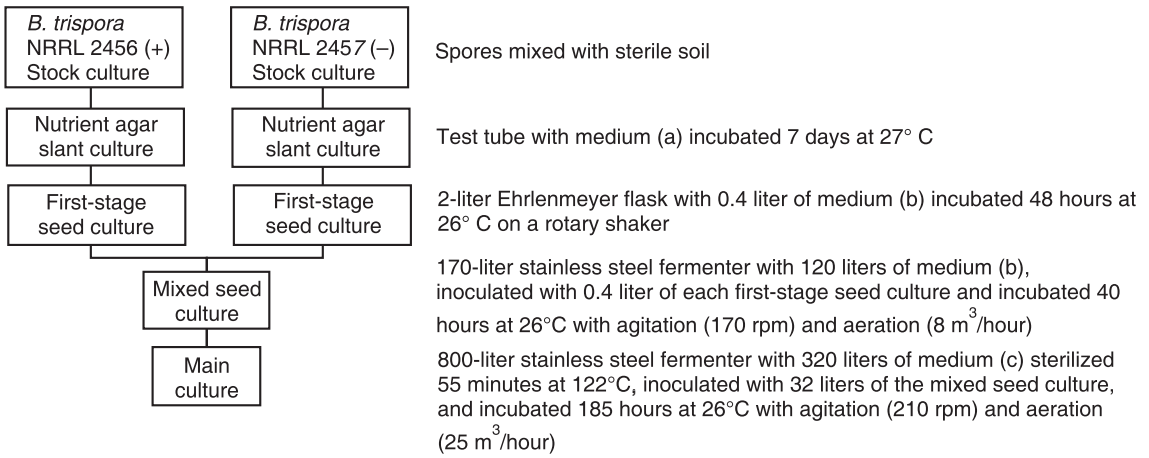


Fig. 7.7: Production of β carotenoid

The production of β-carotene carried out in a submerged fermentation using a mixed culture of (+) and (-) strains. Inoculum is produced by keeping spores in a medium with corn steep liquor 70%, corn starch 50%, KH_2PO_4 0.5% and $MnSO_4$ H_2O 0.1%, thiamine HCl 0.01% and tap water. Production of β-carotene is carried out only in 800 lit. fermenter with 320 lit. medium (Distillary solubles 70 gr, corn starch 60 gr, soyabean oil 30 gr, cotton seed oil 30 ml, antioxidant 0.35 g, $MnSO_4.H_2O$ 0.2 g. Thiamine HCl 0.5 mg; isomazid 0.6 g, kerosene 20 ml, tapwater pH 6.3. Isomiazide and kerosene are sterilized separately. After 48h 1g l^{-1} β-ion and 5 ml l^{-1} kerosene are added. Glucose feeding is continued total addition 42 g l^{-1}) until the end of fermentation. The proportions of two strains need not be equal and since it is (–) strain it should be in excess because the (–) strain in excess produces β-carotene. Another activator for β-carotene synthesis, is isoniazid particularly in combination with β-ionone.

Because of low solubility of β-carotene within the cells the addition of an antioxidant is necessary during fermentation process. To obtain pure β-carotene, the mycelium is removed dehydrated with methanol and extracted with methylene chloride (15–92%) and the crude product is further purified. Recovery and purification process of vitamin A is précised in Fig. 7.8.

Production of vitamin A is induced by trisporic acids, mixture of closely related substances. Trisporic acids are not precursors of β-carotene but rather they act as (+) gamons ionized, is another activator β-carotene particularly in combination with β-ionone. β-ionone is toxic to the producing organism but in the presence of plant oils, it promotes carotene production. β-ionone themselves are not incorporated into carotene but affects the synthesis of different enzymes involved in biosynthetic pathway. The addition of purified kerosene to medium doubles the yield by increasing the solubility of the hydrophobic substrate. The carotenoid rich mycelium can be used directly as a feed additive.

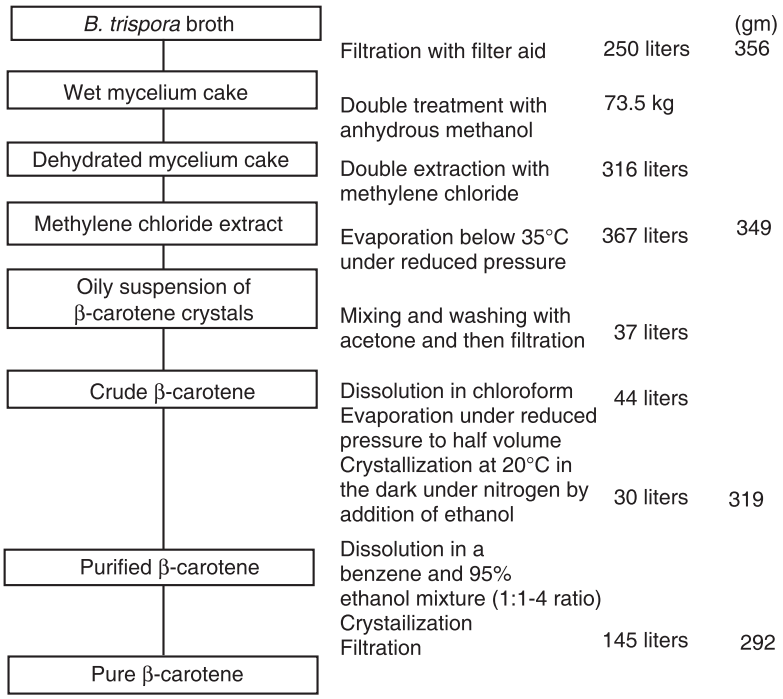


Fig. 7.8: Recovery and purification of β-carotenoids

7.3 RIBOFLAVIN

Riboflavin (lactoflavin or vitamin B_2) was first isolated from whey by **Kuhn et al**. (1933) and structure was elucidated by **Kuhn and Karrer** (1935). It is present in free state as riboflavin, in milk while in other foods (liver, kidney and eggs) it is present as a part of flavoproteins which contains the prosthetic group FMN (Flavin mononucleotide) or FAD (Flavin adenine dinucleotide).

Riboflavin deficiency in man is called as a riboflavinosis showing symptoms of dermatitis whose cure is administrating riboflavin daily at the rate of 1 mg per day. In USA it is made available to man as vitamin enriched bread (thiamine, riboflavin and nocotinic acid). Total production of riboflavin on a world –wide basis is around 2000 tons per year.

Riboflavin is produced industrially by

1. Chemical synthesis (20%)
2. Biotransformation of glucose to D-ribose by mutants of *Bacillus pumilus* and subsequent chemical conversion of ribose to riboflavin (50%)
3. Direct fermentation (30%)

7.3.1 **Structure:** Riboflavin (6,7 dimethyl -9D-1 ribityl iso alloxazine) derivative which consists of a pteridine ring condensed to a benzene ring. The side chain consists of a C_5 – polyhydroxy group a derivative of ribitol (Fig. 7.9).

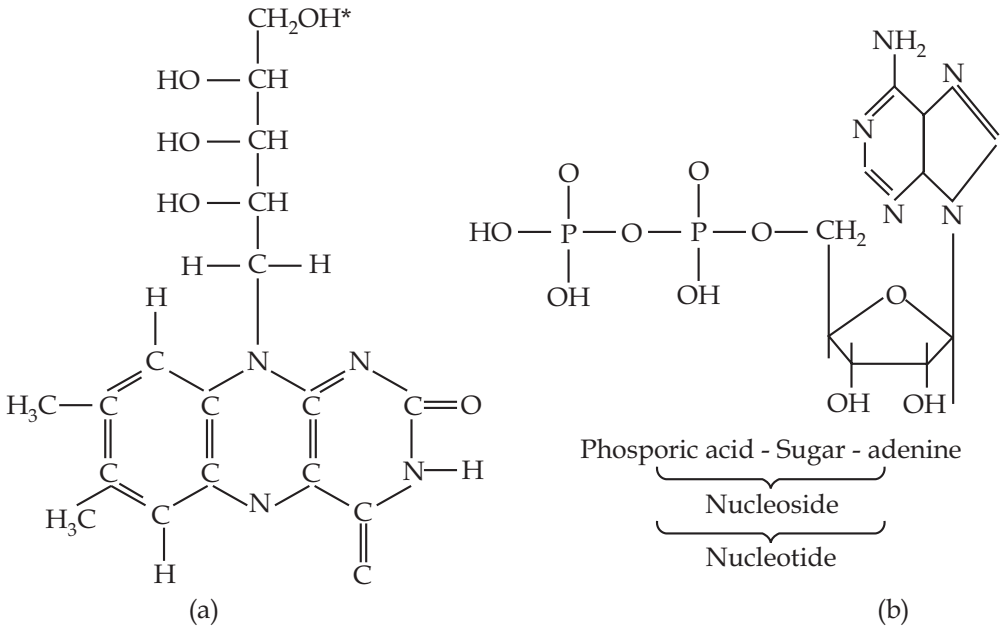
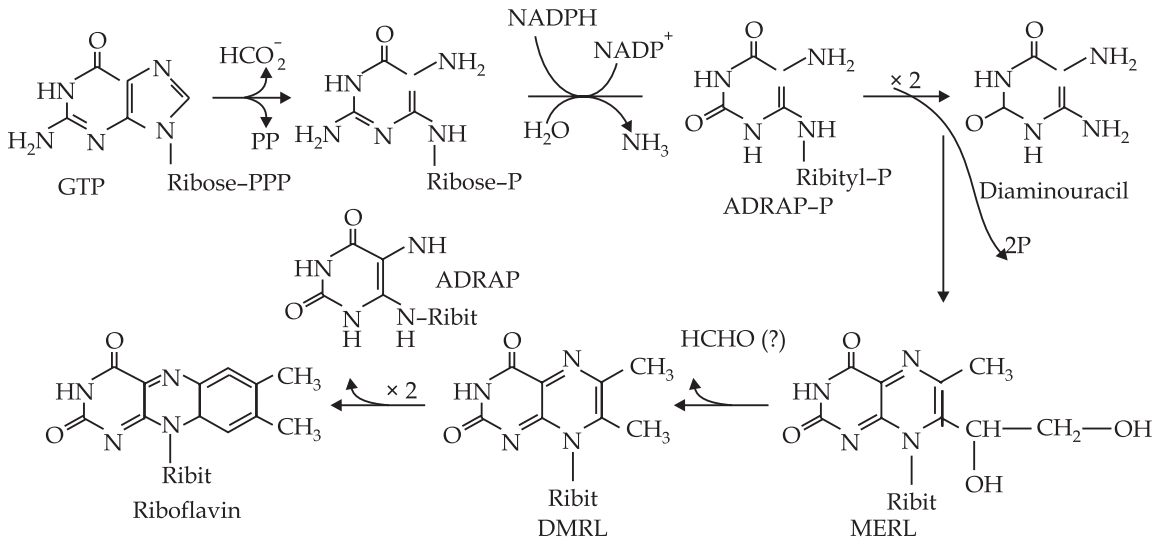


Fig. 7.9: (a) Riboflavin in free state with star (⁎) as OH group (b) Nucleotide, Replacement of star OH group by Nucleotide (b) give Flavin - Adenine-Dinucleotide (FAD) Replacement of one phosphate from FAD gives Flavin-mononucleitide (FMN)

7.3.2 Biosynthesis: Brown and **Williams** (1982) have worked on biosynthesis of riboflavin in *Ashbya gossypii* (Fig. 7.10).



Biosynthetic pathway for riboflavin GTP: Guanosine triphosphate, 2-5, Diamino -6-keto-4-(5'-phosphoribosylamino pyrimidine : ADRAP , 5' Amino -2,5-dioxy -4-(5' phophoribitylamino) pyrimidine : MERL 6-Methyl 7 (1',2' –dihydroxyethyl) -8-ribitylllumazine: DMRL, 6,7, Dimethyl 1-8-ribitylumazine : Ribit, Ribitol.

Fig. 7.10: Biosynthetic pathway of Riboflavin

Though large number of bacteria is reported to produce economical amount of riboflavin (table 7.5), only two ascomycetes, *Eremothecium ashbyii* and *Ashbya gossypii* are employed for industrial fermentations of riboflavin.

Table 7.5: Microorganisms producing riboflavin

Microorganism	Amount of Riboflavin(mgl^{-1})	Optimal iron concentration (mgl^{-1})
Clostridium acetobutylicum	97	1-3
Mycobacterium smegmatis	58	No effect
Mycocandida riboflavina	200	No effect
Candida flaveri	567	0.04-0.06
Eremothecium ashbyii	2480	No effect
Ashbya gossypii	6420	No effect

Despite high yield by two fungi, there is stiff competition between fermentation production and biotransformation and chemical synthesis. It is produced in a medium containing corn steep liquor 2.25% and glycine 0.3%, commercial peptone 3.5% soyabean oil 4.5%. The production of riboflavin is further enhanced by the addition of different peptones, glycine, distiller soluble yeast extract. Simultaneously feeding of glucose and inosital, the rate of production of riboflavin may be increased. Inoculum consists of 24 - 48 hours old actively growing cultures at 0.75-20% will be ideal. Incubation period for 5 days at 28°C and aeriation of 0.3 vvm and initial pH is adjusted to 6.0-7.5 and high aeration should be avoided as it inhibits mycelial growth and reduces riboflavin yield. Yield of 2-9 mgl can be obtained in 5 days. For foam control silicone is added at first and soyabean oil which is metabolized and is added later. Riboflavin is produced both as extracellular and intracellular. The fermentation is divisible into four phases.

Phase I: It is the initial rapid growth phase during which glucose is utilized aid pyruvic acid gets accumulated. pH turns to be acidic.

Phase II: It is a production phase. Rapid synthesis of cell bound riboflavin mainly as FAD and FMN. Pyruvic acid reduced and ammonia gets accumulated due to deaminase activity. This results into the rise in pH.

Phase III: Rapid increase in catalase activity and disappearance of cytochromes. During this phase, synthesis of cell bound riboflavin in the form of FAD or FMN occurs.

Phase IV: It is characterized by the autolysis of the cells. This causes, release of free riboflavin into the medium.

To obtain more pure form of riboflavin for drug and fine food, the broth is heated to 121°C for 1 hour at pH 4.5. The riboflavin is solubilized and the insoluble fraction is discarded. The solution is further treated with titanium chloride as a reducing agent and it will precipitate as riboflavin. It is then reoxidised in the air and dissolved at 60°C in 10% HCl. After cooling, the solution is neutralized to yield purer form of riboflavin crystals.

REVIEW QUESTIONS

I. Essay type questions:

1. Describe the fermentative production of vitamin B_{12}.
2. List out the bacteria which can produce vitamin B_{12} from alcohols.
3. Give fermentative production of vitamin B_{12} by *Propionibacterium.*
4. Describe fermentation of β-carotene.
5. Give an account of bacteria producing riboflavin.
6. Give the structural formula of riboflavin.
7. Narrate four different phases of riboflavin fermentation.
8. What are vitamins?

II. Write short notes on:

1. Fermentative production of B_{12} by pseudomonas dentrificans.
2. Give structural formulae of some important carotenoids.
3. Describe the role of trispoui acid, plants oils and kerosene in the production of vitamin-A.
4. In how many methods riboflavin can be produced industrially.
5. What is a riboflovonosis? Can it be cured?

FURTHER READING

1. Brown, C.M and Willimason, J.M (1982) *Biosynthesis of riboflavin, folic acid, thiamine and pantothenic acid* Advances in Enzymology 53: 346–353.
2. Florent, J. (1986). *Vitamins in Biotechnology,* (eds.H.J. Rehm and G. Reed) vol. 4 VCH Publishers, Deerfiled Beach, F.L. pp. 115–158.
3. Florent, J. and Ninet L. (1999) *Vitamin B_{12} in Microbial technology* (eds. H.J.Peppler and D.Perlman) vol.1 Academic Press, New York pp. 497–519.
4. Perlman, D. (1978) *Vitamins in economic microbiology* vol. 2 (eds. A.H. Rose) Primary products of metabolism. Acad. Press, London, pp. 303–326.
5. Conltate, T.P. (1996). *Food – The chemistry of its components,* 3rd edition. Royal Society of Chemistry, Cambridge.
6. Mui, Y.H and Khachatourian, G.C. (1995). *Food Biotechnology* VCH Publishers Inc., New York.
7. Goldberg, I and William, R. (1995). Biotechnology and Food Van Nostrand Reinhold, New York.
8. Nelis, H.J and De Leenheer, A.P. (1991). *A review: Microbial Sources of Carotenoid Pigments used in food and feeds* Journal of Applied Biotechnology 70: 181–191.

8

Enzymes

Microbes produce variety of enzymes. These enzymes act as catalysts for performing various biochemical reactions which occur during growth and respiration of microorganisms. In certain instances they are useful products of economic importance. Hence, they are produced in bulk quantity. The high yield of these enzymes in culture fluids and their robust properties especially their tolerance to extremes of pH and temperature led to their application in various industries particularly in starch and food processing industries, where they have replaced several unsatisfactory chemical processes and in household laundry detergents. Hence, they are produced fermentatively in bulk quantity.

Enzymes may be either adaptive or constitutive. An enzyme which is secreted in significant amount, regardless of the substratum of the enzyme is present or not in the growth medium is called as constitutive enzyme. In contrast, an enzyme that is secreted in considerable amount in response to the presence of the substratum in the growth medium is called as an adaptive enzyme.

Enzymes may be either extracellular or intracellular. An enzyme which is produced by the microbial cell and if it is released into the surrounding growth medium through cell wall is called as exocellular or extracellular enzyme.

While an enzyme which is produced within the microbial cell and is not released into the surrounding growth medium is called as endocellular or intracellular enzyme. This enzyme is released only by the breakage of the cell wall or autolysis of the cell.

Enzymes are protein molecules and hence possess physical and chemical characteristics of proteins. Some of the important characteristics are mentioned below:

1. They are denatured at high temperature.
2. They are active at specific pH and temperature optimum for enzyme activity.
3. They are substratum specific.
4. They remain active at low temperature.

Enzymes which act as biocatalysts are highly useful in industrial processes. Some of the advantages of using enzymes are prescribed as follows:

1. Have high catalytic power.
2. Are specific in their mode of action and thus do not produce unwanted byproducts.
3. Discriminate between similar parts of molecules (regiospecificity) or between optical isomers (steriospecificity).
4. Operate under mild conditions of temperature, pressure and pH.
5. Enzyme based processes have reduced energy requirements.
6. The reactions otherwise unattainable or economically unviable are feasible with enzymes (e.g. Conversion of glucose to fructose by glucose isomerase)
7. Enzymes based reactor equipment do not require expensive corrosion resistant material for their construction and thus, it reduces final manufacturing cost of the product.
8. Generally the waste (effluent) from the enzyme based processes are not toxic and its treatment is manageable i.e. use of enzyme is eco-friendly.

However, there are some constraints in exploiting these enzymes in industry, as:

1. Most enzymes are expensive materials.
2. Since recovery of enzymes from reaction mixture is difficult, it contaminates the final product.
3. Most enzymes are highly unstable i.e. they start loosing activity as soon as they are separated from their natural environment.
4. Most of the enzymes are denatured at temperature > 45°C and at high acidic or basic pH.
5. Enzymes are specific in their mode of action and thus do not produce unwanted product.

Commercial enzymes fermentatively produced with the help of fungi or bacteria along with enzymes are precised in table 8.1. Generally GRAS (generally regarded as safe) organisms are employed in fermentative production and extraction of enzymes (table 8.2). A flow chart for production of enzyme fermentatively is presented in Fig. 8.1 and application of bulk microbial enzymes used in different fields are illustrated in Fig. 8.2.

Table 8.1: Microorganisms producing different industrial enzymes

Enzyme(s)	Microorganism(s)
α-amylase	*Aspergillus oryzae, A. niger, Bacillus subtilis, B. amyloliquefaciens.*
Glucoamylase	*A. niger, Rhizopus niveus, R. delemar, Endomycopsis spp*
Proteases	*A. oryzae, A. niger, A. sydowii, Mucor pusillus, B. subtilis, Streptomyces griseus*
Lipases	*A. niger, Rhizopus spp. Candida cylindrica*
Cellulases	*A. niger, Trichoderma viride*
Pectinases	*Coniothyrium diplodiella*
Glucose oxidase and others	*Penicillium amagasakiense*
Aspartase	*Escherichia coli*
Amino acylase	*A. oryzae*
Glucose isomerase	*Streptomyces spp.*
Catalase	*A. niger*

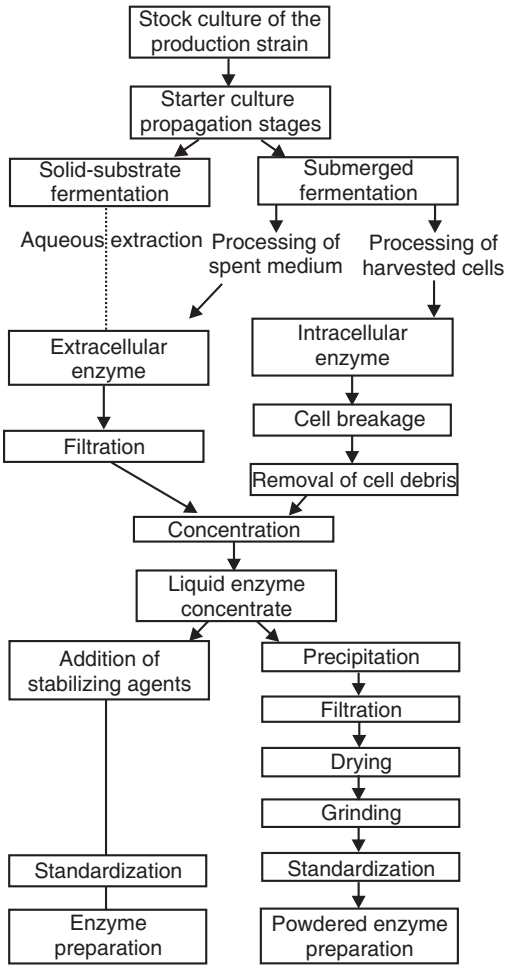


Fig. 8.1: Flow diagram of industrial production of enzymes

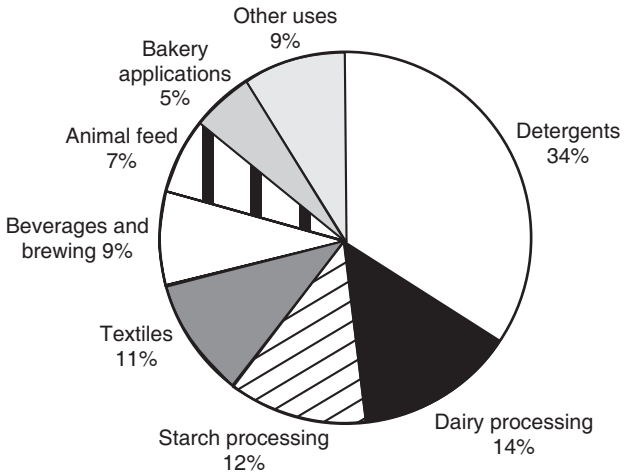


Fig. 8.2: Application of bulk microbial enzymes

As in the development of any fermentation process, enzyme production processes traditionally begin with the search for a suitable producer organism. One of the GRAS microorganisms is selected for fermentation production of desired enzyme based on its properties, pH and temperature requirements, ability to utilize inexpensive carbon and nitrogen sources, incubation period and easy downstream process. The downstream process depends on whether the enzyme is intracellular or extracellular, grown under submerged or solid state fermentation and its end use. Downstream process involves separation, purification, stabilization and preservation. The entire process of fermentation production of enzymes is illustrated in Fig. 8.1.

Table 8.2: Microorganisms classified as GRAS (generally regarded as safe)

1. Bacteria
Bacillus subtilis
Lactobacillus bulgaricus
Lactococcus lactis
Leuconostoc oenos
2. Yeasts
Candida utilis
Kluyveromyces marxianus
Saccharomyces cerevisiae
3. Filamentous fungi
Aspergillus niger
Aspergillus oryzae
Mucor javanicus (Mucor circinelloides f. circinelloides)
Penicillium roqueforti

1. Enzymes used in industry: amylases, proteases, catalases, isomerases and penicillin acylase.
2. Enzymes used in medicine: asparainase, proteases, lipases and streptokinases.
3. Enzymes used for analytical purposes: glucose-oxidase, galactose-oxidase, alcohol dehydrogenase, hexokinase, muramidase and cholesterol-oxidase.

Enzymes that are expensive to produce including many intracellular enzymes and those employed in biosensors and analytical test systems are often used in an immobilized form. Immobilization of enzymes can be accomplished by one of the processes of adsorption, entrapment ionic bonding, encapsulation and covalent bonding (Fig. 8.12) to inert inorganic or organic solid support materials such as nylon, bentomite cellulose and dextrin. Alternatively, they can be micro (–) encapsulated within semipermeable membranes, entrapped with in gels or fibers and intracellular enzymes may be immobilized within their producer cells. In industrial process, immobilization facilitates confinement and recovery and these enzyme preparations have several beneficial features such as

1. Suitability for application within continuous operations
2. Their product is enzyme free, therefore further processing to remove or inactivate the enzyme is not required.

3. Improved enzyme stability and
4. Reduced effluent disposable problem.

Industrial production of amylases, glucose isomerase, pectinases, lipases, cellulases and proteases are described in this chapter.

8.1 AMYLASES

Amylases are a complex group of enzymes that hydrolyse polysaccharides like starch and glycogen to glucose. During the hydrolysis of 1, 4-glycoside, linkages present in above polysaccharides are degraded which result first in the formation of short chain dextrins, then to maltose and glucose. Different enzymes involved in the degradation of starch to glucose are illustrated in Fig. 8.3 and precised in table 8.2a.

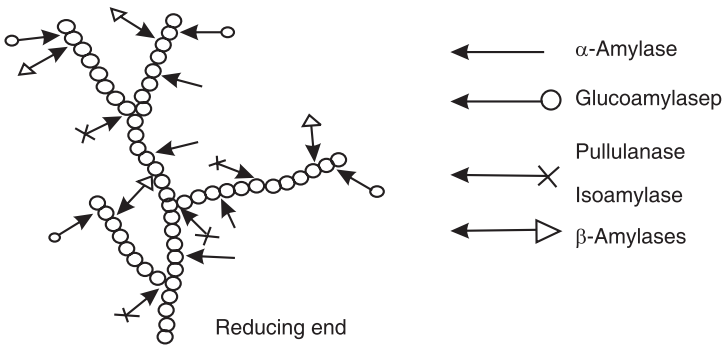


Fig. 8.3: Mechanism of starch hydrolysis by amylases

There are mainly two groups of amylases such as, α-amylose and β-amylose. α-amylases are also called as 1,4 – a – glucan - glucano hydrolase. Extracellular enzymes hydrolyze 1,4 - glycosidic bonds. These enzymes are also called as endoenzymes as they split the substratum in the interior of the molecule in a random fashion. On the other hand, β-amylases split the substrutum from one end of the molecule in a successive fashion. On every alternate 1, 4 glycosidic bond releasing maltose. These enzymes are also called exo amylases. The other enzymes involved in the degradation of starch include amyloglycosidase (*Aspergilus niger* and *Rhizopus niveus),* isoamylase and *pullulanase (Klebsiella pneumoniae* and *Bacillus acidopullulyticus)*. The molecular weight of various α-amyloses do not differ considerably (table 8.3) and require calcium as a stabilizer.

Table 8.2a: Different enzymes involved in starch degradation

Endo-amylases (dextrinizing	Act on starch randomly releasing dextrins (a) α - amylases–Act on 1, 4 glycosidic bond (b) Debranching – Act on 1, 6 glycosidic bond (i) Isoamylase – Act on dextrins (ii) Pullulanase – Act on pullulan
Exo-amylases	-Act on from one end of starch (a) α -amyloglucosidase–Act on every 1, 4-glycosidic bond releasing glucose (b) β -amylase–Act on alternate 1, 4 glysidic bond releasing maltose (c) Maltose – Act on maltose releasing glucose

Table 8.3: Molecular weight of some α-amylases from different microorganisms

Organism	Molecular weight (in × 10^3)
Aspergillus oryzae	51–52
A. niger	58–61
Bacillus acidocaldarius	68
B. amyloliquefaciens	49
B. subtilis	24–100
Thermomonospora curvata	62

α-amylases are secreted by many bacteria and fungi. They are classified according to their :

1. Starch-liquefying or saccharogenic effect
2. pH optimum
3. Temperature range
4. Stability

Saccharogenic amylases produce free sugars (β-amylose and glucoamylase), whereas starch-liquifying amyloses (**α**-amylose) break down the starch polymer but do not produce free sugars.

Bacteria which produce **α**-amylase are *Bacillus subtilis, B. cereus, B. amylo- liquefaciens, B. coagulans, B. polymyxa, B. stearothermophilus, B. caldolyticus, B. acidocaldarius, B. subtilis amylosaccharaticus, B. licheniformis,* species of *Lactobacillus, Micrococcus, Pseudomonas, Arthrobacter, Escherichia, Proteus, Thermomonospora and Serratia.* Some **α**-amylase producing fungi are species of *Aspergillus, Penicillium, Cephalosporium, Mucor, Candida, Neurospora* and *Rhizopus*.

Although all the above bacteria and fungi are capable of producing α-amylase, the most important of them from which α-amylase are produced commercially through fermentation process include *Bacillus amyloliquefaciens, B. licheniformis* and *Aspergillus oryzae*. However, *Bacillus* are used much more extensively than those of *Aspergillus*. The most important areas of application for these two enzymes are precised in table 8.4.

Table 8.4: Important applications of α-amylases

Industry	Source		Applications
	Bacillus	**Aspergillus**	
Starch industry	+	--	Liquefaction of starch for production of glucose, fructose, maltose
Milling	--	+	Modification of deficient flour
Alcohol	+	+	Liquefaction of starch before the addition of malt for saccharification
Baked goods	--	+	Increase in the proportion of fermentable carbohydrates
Brewing	+	--	Barley preparation, liquefaction of additives
	--	+	Improved fermentability of grains, modification of beer characteristics
Paper	+	--	Liquefaction of starch without sugar production for sizing of paper
Textiles	+	--	Continuous desizing at high temperature
Feed industry	+	--	Improvement of utilization of enzymatically treated barley in poultry and calf raising
Sugar	+	--	Improvement of filterability of cane sugar juice via breakdown of starch in juice
Laundry and detergent	+	--	Increase in cleansing power for laundry soiled with starch, additive in dish washer detergents

8.1.1 Fungal α–amylase: Fungal α-amylase is produced commercially by employing either *Aspergillus oryzae* or *Aspergillus niger*. Stationary culture method is used when *A. oryzae* is employed, while submerged culture method is used when *Aspergillus niger* is employed. The process of submerged culture method is only described here.

(a) Preparation of inoculum: Suitable and pure seed culture is generally selected as inoculum. In certain cases mutants capable of giving higher yield are selected as inoculum.

(b) Preparation of medium: The following medium is generally employed for submerged fermentation.

Component	Amount (g liter^{-1})
Corn starch	24
Corn steep liquor	36
Potassium chloride	0.2
Sodium monohydrogen phosphate	47
Calcium chloride	1
$MgCl_2 \cdot 6H_2O$	0.2

Amylase biosynthesis is inhibited when there is glucose in the medium. The medium is steam sterilized. The sterilized medium is passed into a production fermenter for α-amylase production.

(c) Fermentation process: A cylindrical fermenter made up of stainless steel is generally used in the fermentation process. It is equipped with an agitator, an aerating device, a cooling system and other ancillary equipment like a device for foam control, monitoring of pH, temperature and control of oxygen tension etc.

A sufficient quantity of pre-sterilized production medium is taken in the fermenter and is inoculated with spores of the selected species of the fungus. The spores are allowed to germinate and produce sufficient mycelium by controlling the fermentation conditions. Control of fermentation conditions play a vital role in the success of the process, which include pH, temperature, aeration, agitation, oxygen supply etc. The optimum pH for the fermentation is 7.0. Calcium carbonate is used as buffer to maintain pH. The fermentation process is generally operated at a temperature of 30 to 40°C. Aeration and agitation of the production medium is needed because of high viscosity of the medium due to the presence of mycelial mat.

(d) Harvest and recovery: The following steps are followed during the recovery of the enzyme after the completion of fermentation. In order to avoid denaturation of the enzyme, the fermentation broth is subjected to rapid cooling at 5°C temperature immediately and the enzyme is extracted.

1. Separation of fungal mycelium is accomplished by filtration of the refrigerated broth.
2. The suspended particles present in the broth are removed with flocculating agents like calcium phosphate.
3. The enzyme is precipitated, in order to get high degree of purity, by using acetone or alcohol or even inorganic salts like ammonium sulphate or sodium sulphate.
4. Sometimes fractional precipitation of the enzyme is done to obtain it in purest form.

8.1.2 Bacterial α–amylase: Bacterial α-amylase is produced by *Bacillus subtilis* or *B. amyloliquefaciens* or *B. licheniformis*. For industrial production of bacterial α-amylase, nowadays submerged culture method is generally employed in many countries.

(a) Preparation of inoculum: Pure culture of any of the above-mentioned species of *Bacillus* is selected as inoculum. Mutants which produce 250 times greater yields than the wild strain are preferred as inoculum.

(b) Preparation of medium: The formulation of the production medium and control of fermentation conditions play a major role in the success of enzyme fermentations. The production medium should basically contain an energy source, a carbon source, a nitrogen source and growth requirements such as essential amino acids or vitamins. For obtaining high yields of enzyme, the production medium should also contain certain inducers like lactose.

Sometimes certain compounds like glucose present in the production medium act as repressor for certain enzymes like α-amylase. In such conditions either the concentration of glucose should be kept low or it should be fed intermittently. The following production medium is generally employed in a submerged culture method (Table 8.5).

Table 8.5: Composition of medium for submerged fermentation

Substances	Amount (in %)
Lactose	4.5
Ground soyabean meal	1.85
$MgSO_4 \cdot 7H_2O$	0.04
Hodag KG –1 antifoam	0.05
Amber BYF (Autolysed brewers yeast fractions)	1.50
Distillers dried solubles	0.70
N-Zamine (enzyme casein hydrolyzate)	0.65
Water	90.40

(c) Fermentation process: The type of fermenter, that is used for fungal α-amylase is also used for bacterial α-amylase production. About 1000 to 30,000 gallons of production medium is taken in the fermenter and inoculated. The fermentation is continued upto 4-6 days. The pH of the medium is maintained at 7.0. Calcium carbonate is used as a buffer for maintaining neutral pH. The temperature is maintained at 30-40°C. The production of α-amylase starts when the bacterial density reaches 10^9-10^{10} cells ml^{-1} However, the enzyme production increases just before the growth rate of the microorganism decreases and spore formation begins.

(d) Harvest and recovery: Bacterial α-amylase is harvested and recovered by the same method that is used for the recovery of fungal α-amylase. The most active liquid enzyme preparation contains 2% amylase protein and solid preparation contains 5% amylase proteins. Flow sheet for production of α-amylase is shown in Fig. 8.4.

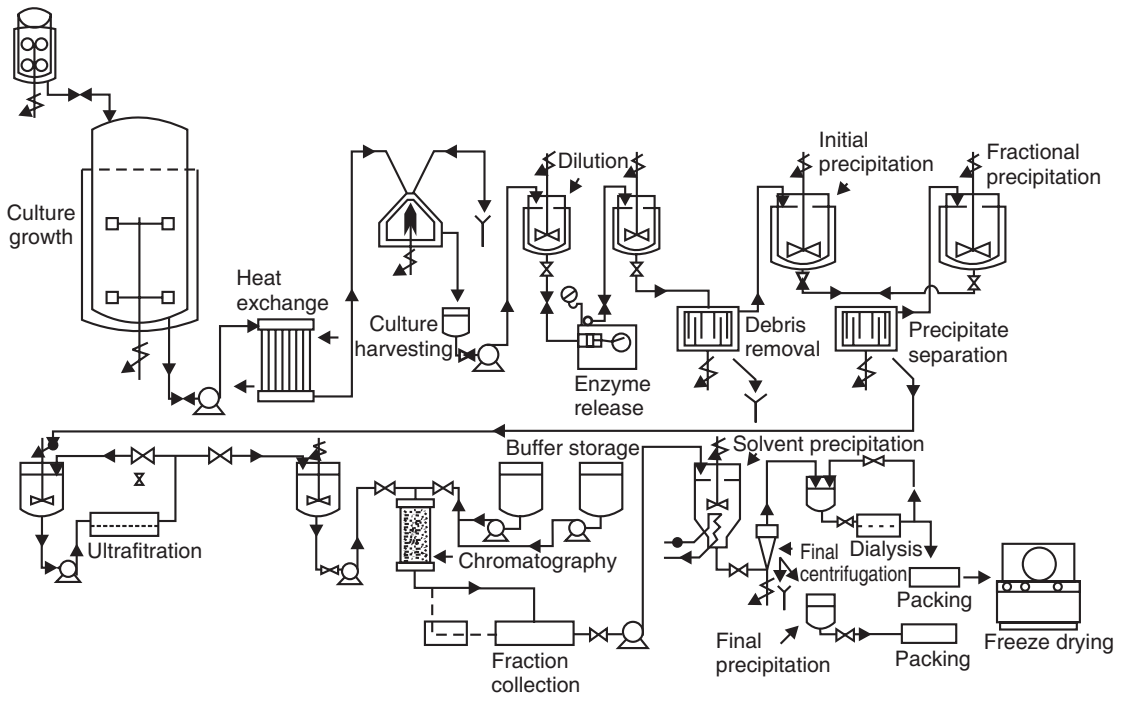


Fig. 8.4: Flow diagram of purified endocellular enzymes

8.2 PROTEASES

Proteases, second most important industrial enzymes, are produced about 500 tons per year. Proteases are primarly used in detergent, dairy, leather firms, pharmaceutical industries, the manufactured protein hydrolysates, food industry and waste processing.

Organisms: proteases are commercially produced both by fungi and bacteria based on optimum pH for their activity. They can be grouped into alkaline, neutral and acid proteases and other groups (table 8.6).

(i) Alkaline proteases: Most bacteria and some fungi excrete alkaline proteases. The most important producers of alkaline protease producers are *Bacillus licheniformis*, *B. amyloliquefaciens, B. firmus, B. megaterium* and *B. pumilus* species of *Streptomyces* such as *S. fradiae, S. griseus* and *S. rectus*. Some fungi like *Aspergillus niger*, *A. sojae*, *A. oryzae*, and *A. flavus*. Proteases of this type have many features of value for use as detergent enzymes. These enzymes are stable at high temperature, at alkaline range of pH (9.0 – 11.0), in association with chelating agents and perborates. However, their stability is low in the presence of surface active agents resulting in the limited self life. Subtilisin Carlsberg from *B. licheniformis,* subtilisin BPN and subtilisin NOVO from *B. amyloliqui-faciens* are some of the examples of this type of enzymes. These enzymes contain serine at the active site of the molecules and are not inhibited by EDTA (Ethylene diamine tetracetic acid) but are inhibited by DFP (diisopropyl fluorophosphate).

Table 8.6: Classification of proteinases (endo-types)

Main	Class	Sub type	Producing organisms
Endo-type proteinase	Serine Proteinase	Trypsin like serine	Streptomyces
		Alkaline serine	Bacillus, Streptomyces, Aspergillus
		Myxobacter a-lytic	Sorgangium
		Staphylococcal	Staphylococcus aureus
		Neutral serine	Bacillus pumilus
		Other serine	Escherichia, Saccharomyces
	Metalloproteinase	Neutral	Bacillus
		Alkaline	Pseudomonas, proteus, Serratia
	Acid proteinase	Pepsin type	Aspergillus, Mucor
		Rennet-like	Bacillus, Mucor, Endothia
	Thiol proteinase	Clostripain	Clostridium
		Streptococcal	Group A streptococci

Screening for isolation of microorganisms with better production is done by using strongly basic protein media at pH 10. As most of the wild strains are with low yield and insufficient for industrial utilization, improvement of strain is being done through genetic engineering. Protein engineering has been used to develop modified *Bacillus* subtilopeptidase with altered amino acid sequences, corresponding changes in enzymatic properties such as substrate specificity, pH optimum and stability to bleaching agents.

Protease production occurs in shaken flasks and small fermenters or in a production fermenter of 40-100 ml capacity at 30–37°C. Production of extracellular proteases is chiefly regulated by the medium composition. The fed-batch process is generally used in order to keep down the concentration of ammonia ions and amino acids which otherwise may repress protease production. High O_2 partial pressure is generally necessary for proteases titres, aeration rates are 1 V m^{-1} and incubation time of 48–72 hrs depending on the organism.

Proteases must be converted into particulate form before they are added to detergents since if dry enzyme powder is inhaled by production workers or users, allergic reaction may result. Enzyme is marketed in a microencapsulated form. To make a suitable encapsulated product, a wet paste of enzyme is melted at 50–70°C with hydrophobic substance such as polyethyl-eneglycol and then converted into tiny particles. These solidified spherical particles are not hazardous when added directly to the detergent. Immobilization in fibrous polymer is another development.

(ii) Neutral proteases: These proteases are excreted by *Bacillus subtilis*, *B. cereus*, *B. megaterium*, *Pseudomonas aeruginosa*, *Streptomyces griseus*, *Aspergillus oryzae*, *A. sojae* and *Pyricularia oryzae*.

Neutral proteases are relatively unstable.

Calcium, and sodium chloride must be added for maximal stability. The pH range of activity is narrow and unstable to increased temperature. These are also quickly inactivated by alkaline proteases which is probably the reason for their restricted industrial application. However, they are used in leather industry and in food industry for manufacture of crackers, bread and rolls.

(iii) Acid proteases: These include renin like proteases from fungi which are chiefly used in cheese production. These enzymes have optimum pH 2-4 and used in medicine, in the digestion of soya protein for soya sauce production and to breakdown wheat gluten in the bakery industry.

(iv) Renin: Milk coagulating enzyme in fourth chamber of 3-4 weeks old calves stomach which is being exploited in cheese production. Due to increased production of cheese and decline in the number of slaughtered calves intensive research has been carried out since 1960 to develop rennin products of microbial origin. A variety of fungi and bacteria have been isolated as producers.

It has also been examined for required characteristics.

Alcaligenes, Bacillus, Corynebacterium, Lactobacillus, Pseudomonas, Serratia and *Streptococcus* are some of the bacteria that have exhibited potential of rennin production. Similarly species of *Aspergillus, Candida, Coriolus*, *Endothia*, *Entomophthora, Irpex, Mucor, Penicillium, Rhizopus, Sclerotium* and *Torulopsis* are also reported to produce rennin.

Mucor pusilus var. *lindt* for solid substrate culture, while *Endothia parasitica* and *Mucor miehei* produce this enzyme in submerged culture. Rennin produced by *Endothia parastica* was first rennin marketed commercially in 1967. The enzyme is an acid protease stable at pH 4.0-5.5 with a molecular weight of 34,000 – 37,500 daltons and is produced in a medium containing 3% soyameal, 1% glucose, 1% skim milk, 0.3% $NaNO_3$, 0.05% K_2HPO_4, 0.025%, $MgSO_4.7H_2O$, pH 6.8, at 25°C incubation temperature. At the end of 48 hrs incubation period, mycelium is separated and broth which contains enzyme is concentrated and precipitated in an evaporation process.

Similarly *Mucor miehei* also produces rennin in a medium containing potato starch 4%, soyameal 3%, ground barley 10%, $CaCO_3$ 0.5%, pH 5.5, 5-6 d incubation period at an incubation temperature of 40°C. Microbial renins are stable at high temperature and remain active in the curd after precipitation and subsequently causing harmful proteolysis. Therefore, complementary cDNA from calf renin is transferred into *E. coli* making possible the first commercial production of the calf rennin by a microorganism.

Applications: Proteases are employed in variety of fields such as

1. Biological detergents—alkaline protease
2. Baking—dough modification/gluten weakening and flour improvement
3. Beer brewing—chill proofing of beer to remove protein haze
4. Leather bailing and tendering
5. Cheese manufacture—clotting of milk protein and promotion of ripening aspartic protease, calf chymopsin, aminopeptidase etc.
6. Meat tenderization and removal of meat from bones
7. Flavour control and production in food products
8. Waste treatment- treatment of silver from spent photographic film.

8.3 PECTINASES

Pectinase is an enzyme complex which contains atleast six enzymes splitting pectins at different sites of the molecule. The basic structure of pectin is α 1,4 linked galacturanic acid with upto 95% its carboxyl groups esterified with methanol. Pectinases are classified according to their attack on the point of molecule (table 8.7). The methyl ester is split by pectinesterase and the glycosidic bonds of pectin or pectic acid, and chain are split by hydrolysis with endo-polygalacturonase or exo-polygalacturonase. Another method of splitting is transelimination by means of exo-pectate lyase or endopectate lyase or exo-and endo-pectin lyase.

Commercial production of pectinases is carried out by using *Aspergillus niger* or *A. wentii* and also species of *Rhizopus* either as surface or submerged fermentation in a medium containing 2% sucrose and 2% pectin in a fed batch system. The pH is adjusted to 3.0 to 4.0 and incubation temperature 37°C and incubution period upto 60-80 hrs. The enzyme is purified by separating mycelium, by filtration or centrifugation, stabilizing agents are added. The enzyme is precipitated with organic solvents and crude protein is dried and used as enzyme.

Table 8.7: Classification of pectolytic enzymes

Type of Pectinase name of enzymes					Producing organisms
Depolymerizing enzymes	Hydrolase	Endo-hydrolase		Endopoly galacturonase	Bacteria, fungi Coniothyrium, Aspergillus
		Exo-hydrolase	Galacturono	Not cleaving	
				Cleaving	4,5-unsaturated galacturonic linkages
			Poly galacturonase		Aspergillus, Erwinia
			Digalacturono-polygalacturonase	Endo-pectic acid transeliminase	
	Transeliminase	Endotraseliminase	Endopectic acid eliminase		Erwinia, Clostridium Corynebacterium fungi
		Exotranseliminase		Endo-pectin transeliminase	
			Exotranseliminase		Erwinia, Clostridium
					Erwinia, Pseudomonas

8.3.1 **Applications:** Pectinases are used primarily for clarification of fruit juices, for maceration of vegatable and fruits, for extraction of oil. By treating with pectinase, the yield of fruit juice during pressing is considerably increased. These are also used in wine classification and coffee bean fermentation.

8.4 LIPASES

Lipases (glycerolesterhydrolases) split fats (glycerolesters) into di-or monoglycerides and fatty acids (Fig 8.5). These are usually extracellular. Most of lipases are produced adaptively in the presence of oils and fats. However, some organisms like *Penicillium roqueforti* are reported to secrete lipase constitutively and in the presence of fats its enzymes production is repressed. Species of *Aspergillus, Mucor, Rhizopus, Penicillium, Geotrichum* and yeasts (*Torulopsis* and *Candida)* are reported to be good producers. Similarly species of *Pseudomonas, Achromobacter* and

Staphylococcus are good producers of lipases. Lipases are generally bound to the cells and hence inhibit on over production, but addition of cation such as magnesium ion liberates the lipase and leads to a higher enzyme titer in the production process.

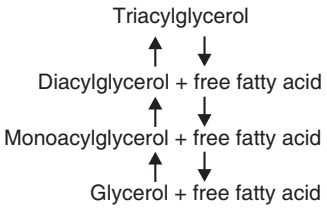


Fig. 8.5: Mechanism of action of lipase

8.4.1 Applications: Lipases are primarily marketed for therapeutic purpose as digestive enzymes to supplement pancreatic lipases. These enzymes are used in dairy industry. The cheese ripening process involves lipases. Lipases from *Candida cylindraceae* is used to hydrolyse oil in soap industry. These are useful in the synthesis of esters from acids and alcohols in non aqueous medium. Products of interesterification catalyzed by a 1, 3 – regiospecific lipase are illustrated in Fig. 8.6.

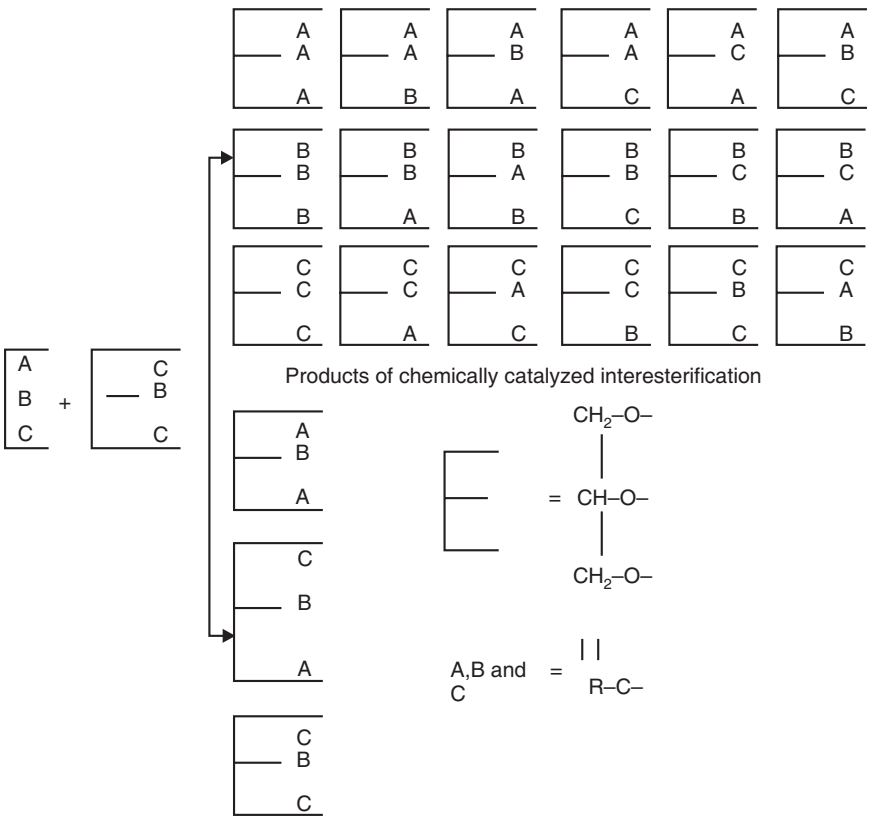


Fig. 8.6: Interesterification with chemical reaction and lipase catalysed

These are also useful in improvement of fat quality by interesterification (exchange of one fatty acid by another in ester bond) lipases are useful in removal of fat in leather processing. These enzymes are also useful in production of flavour compounds and acceleration of ripening in dairy and meat products.

8.5 CELLULASES

Cellulases are the enzymes which degrade cellulose into fermentable sugars. Although cellulose is available abundantly on the earth in the form of natural plant resources like fallen leaves, flowers, fruits, broken stems, branches and waste wood, it is mixed with substances such as lignin, hemicellulose, starch, proteins etc. (Table 8.8, Fig. 8.7). All these material are to be treated physically or chemically to breakdown cellulose into fermentable sugars. This breakdown process is called as pretreatment. The pretreatment may be done either chemically or enzymatically. In the second process cellulase enzymes are used to breakdown cellulose into simpler fermentable sugars (Table 8.9).

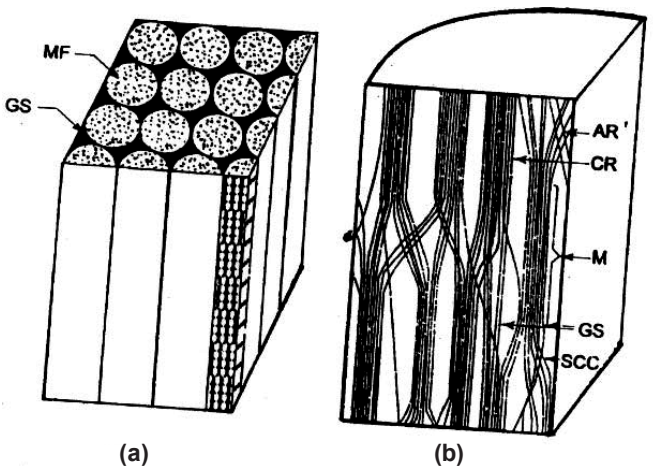


Fig. 8.7: (a) Gross structure of cellulose and microfibrils and the arrangement of cellulose molecules within a microfibril MF = microfibril, GS = ground substance (pectin, hemicellulose, or lignin), (b) AR = amorphous region of cellulose, CR = crystalline region of cellulose, M = micelle, SCC = single cellulose chain (molecule)

Table 8.8: Composition of cellulose ingredients of some plants and paper

Plant resource	Cellulose (%)	Hemicellulose (%)	Lignin (%)
Wood (Gymnosperm)	40–55	24–40	18–25
Wood (Angiosperms)	45–50	25–35	25–35
Grasses	25–40	25–50	10–30
Leaves	15–20	80–85	——-
News paper	40–55	25–40	18–30
Waste from paper manufacture	60–80	20–30	2–10

Most of these enzymes are excreted out by certain microorganisms which include both bacteria and fungi. Such enzymes are called extracellular enzymes. The bacteria include actinomycetes and *Cellulomonas* and fungi include species of *Trichoderma, Penicillium, Thermoascus, Sporotrichum* and *Humicola*. However, for commercial production of cellulases *Trichoderma viride*, *T. reesei, T. koningii, Penicillium funiculosum, P. enersonii, Aureobasidium pullulans*, *Sporotrichum* and *pulverulentum* are generally employed.

Cellulase is an enzyme complex containing at least three enzymes (table 8.9 and Fig. 8.8)

1. Exo- β-1, 4 glucanase, also called as cellobiodhydrolase or C.
2. Endo-β 1, 4-glucanase also called as carboxymethyl cellulase or endocellulase (Cx).
3. β-1, 4 glucosidase or cellobiase.

8.5.1 Fermentation production: The fungus *Trichoderma viride* is usually used for the commercial production of cellulase. Inoculum of the fungus is raised from pure or mutant culture of *Trichoderma viride* GM 9414 by a process of repeated sub-culturing.

Large fermenters of stainless steel are employed for fermentative production of cellulases. The production medium consists of the following ingredients.

KH_2PO_4	- 0.2%
$(NH_4)_2SO_4$	- 0.14%
Urea	- 0.03%
$MgSO_4 \cdot 7H_2O$	- 0.03%
$CaCl_2$	- 0.03%
$FeSO_4 \cdot 7H_2O$	- 5.0mg L^{-1}
$Mn(OH)_2$	- 1.6mg L^{-1}
$ZnSO_4$	- 1.4 mg L^{-1}
$CoCl_2$	- 2.0 mg L^{-1}

Table 8.9: Classification of cellulolytic enzymes

Name of enzymes			Producing organisms
Cellulolytic enzymes		Endo–β–1,4-Glucanase	Many Fungi
	β–1,4-Glucanase	*Trichoderma penicillium,*	
	β–1,4- Glucosidase	Exo–β–1,4-Glucanase	β–1,4-glucan cellobiohydrolase — *Fusarium*
			β–1,4-glucan glucohydrolase (=cellobiase) — *Trichoderma*, Many Fungi

The fermentation is carried out at 30°C for about 140 hours. Maximum fungal growth occurs under these conditions resulting in the maximum yield of the cellulase. However, the yield of enzyme is dependent upon the cellulose concentration (Fig. 8.6). After the completion of the fermentative process the products recovered from the fermenter and the fungal mycelium is separated by filteration. Afterwards purification of the enzyme was carried out using ion exchangers.

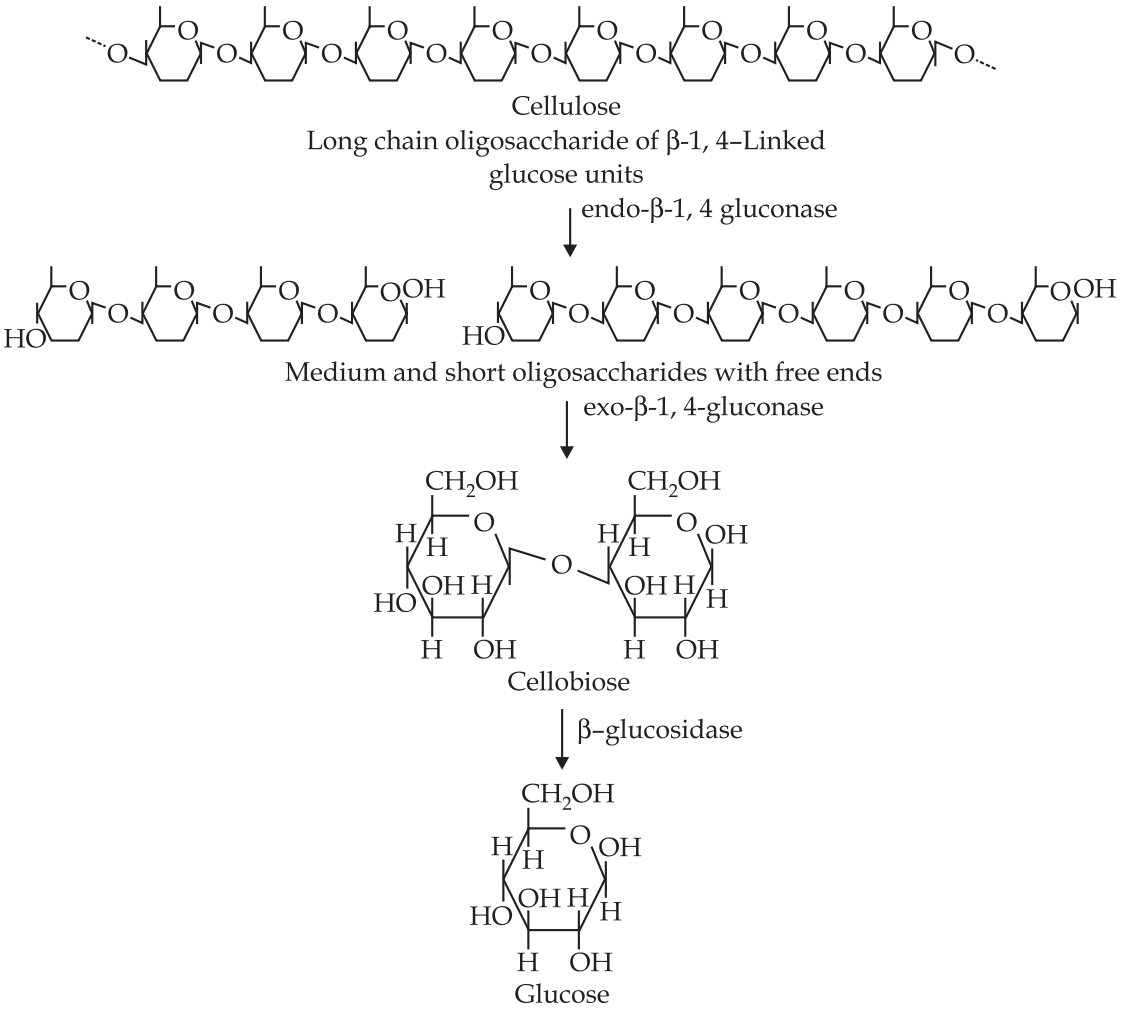


Fig. 8.8: Enzymatic degradation of cellulose

8.5.2 Applications: 1. Fruit juice and olive production and processing.

2. Wine and beer production and processing.
3. Malting-speedy modification of grains.
4. Textile processing-biopolishing of cellulose fibers.
5. Wood pulp processing.

8.6 GLUCOSE ISOMERASE

Glucose isomerase causes the isomerization of glucose to fructose. The reaction is reversible and a mixture of glucose and fructose is produced. The ratio of these products depends on the enzyme used and reaction conditions such as incubation time, pH and temperature. The enzyme has become of commercial value because price of sugar has increased as compared to that of starch. In U.S. about 4×10^6 tons of isosyrup is being produced, while in Germany 10,000–12,000 tons per year. Starch can be converted to glucose by either acid hydrolysis or enzyme hydrolysis (Fig. 8.9). The advantage of starch hydrolysis over direct sugar production (sugar beets) is that

initial materials used such as wheat, corn, cassava and to some extent potatoes are non-perishable, but sugar beets are available only 100 days per year.

Glucose has 70-75%, the sweetening strength of beet sugar (sucrose) but β-D-fructose pyranose (fructose), the sweetest monosaccharide has twice the sweetening strength of sucrose. Thus, processes for the manufacture of fructose are of considerable value. Fructose can be produced biochemically from sucrose or starch by isomerase, glucose production (Fig. 8.10 a and b). Fructose can also be produced chemically from glucose at high temperature under alkaline conditions but byproducts are formed such as psicose which cannot be metabolized in the human body. Thus chemical production of fructose is not acceptable to the food industry.

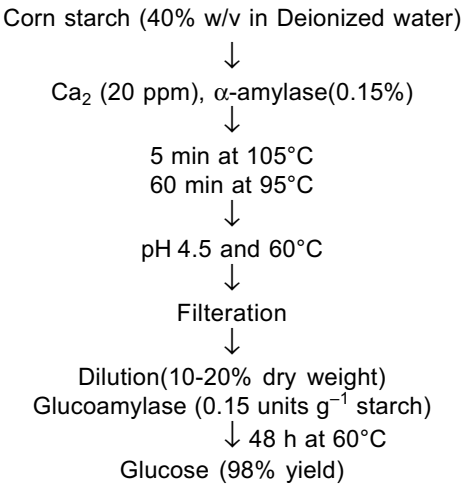
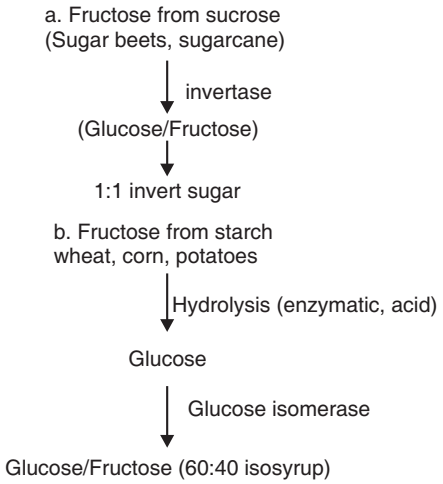


Fig. 8.9: Production of glucose from starch



(a) from sucrose (b) from starch (c) chemical isomerisation of glucose to fructose

D-glucose → intermediate → D-mannose → D-fructose

Fig. 8.10: Biochemical method of production of high fructose syrup

Fermentative production:

Glucose isomerase is used in production of high fructose syrup (HFCs) or isosyrup must fulfill the following criteria:

1. Low pH optimum to avoid side reactions.
2. High specific activity.
3. High temperature optima.

Some of the microorganisms producing glucose isomerase are listed in table 8.10. Practical conversion yields are in the range of 40-50% where fructose concentration can be increased to 55% by chromatographic enrichment. Alternatively the fructose can now be separated from glucose (+) fructose mixture and the remaining glucose fraction isomerzed to produce a final combined product containing upto 80% fructose. HFCs from corn starch is cheaper than from sucrose but have same intensity of sweetening. Consequently they have become major food ingredients particularly in North America and annual production of HFCs is now in excess of 8×10^9 kg. Some of the bacteria such as *Escherichia intermedia, E. freudii* and *Aerobacter aerogenes* can produce isomerase. Though they are able to produce glucose isomerase which is variant and requires arsenium for its activity, hence it cannot be used in food production.

Table 8.10: Glucose isomerase producers

Bacillus coagulans
Streptomyces phaeochromogenes
Streptomyces rubiginosus
Streptomyces olivochromogenes
Arthrobacter spp.
Actinoplanes missouriensis
Microbispora rosea
Micromonospora coerula
Nocardia asteroides
Nocardia dassonvillei
Brevibacterium imperiale
Flavobacterium arborescens

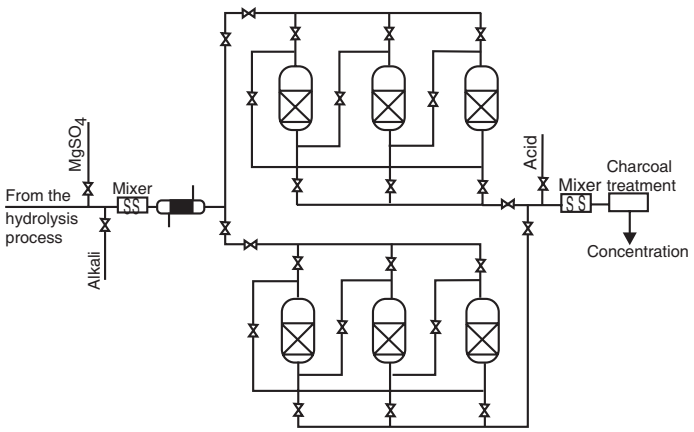


Fig. 8.11: Commercial production of fructose from glucose

The commercial process for production of fructose (Fig. 8.11) from glucose became feasible only when procedures for immobilization of the enzyme were developed (Fig. 8.12).

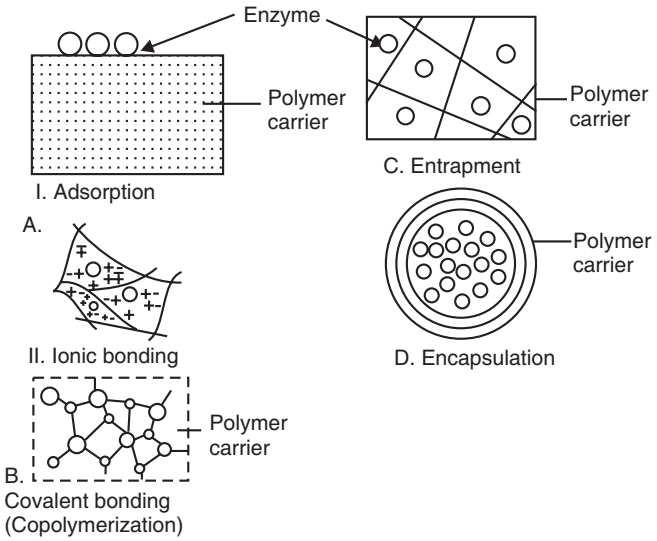


Fig. 8.12: The methods of immobilization of enzymes

Since, glucose isomerase is formed intracellularly in most strains, many commercial processes are carried out with immobilized cells or by addition of partly broken cells. Commercial production of glucose isomerase along with producing organism and industry are listed in table 8.11.

Table 8.11: Trade name, industry and organisms producing glucose isomerase

Organisms	Trade name (glucose isomerase)	Industry
Bacillus coagulans	Sweetzyme	Novo industry
Streptomyces rubiginosus	Optsweet	Miles-Kali
Actiplanes missouriensis	Ketozyme	Universal oil products
-- do --	Maxzyme	Girt Brocades
Flavobacterium arborescens	Taka-sweet	Miles Laboratories

REVIEW QUESTIONS

I. Essay type questions:

1. Describe the process of industrial enzymes production.
2. Give an account of applications of industrial enzymes.
3. Give an account of amylases and their applications.
4. Classify the microbial proteases and their methods of production.
5. Discuss the importance of microbial lipases and their role in interesterification.
6. Describe the process of high fructose syrup.
7. Give classification of pectinases and their applications.
8. What are lingocellulases and their importance in ethanol fermentation?

II. Write short notes on:

(a) Cellulases (b) Immobilization of enzymes
(c) Intracellular enzymes (d) Adaptive enzymes
(e) Downstream process of industrial enzymes.

FURTHER READING

1. Arnold, F.h. and Volkov, A.A (1999). **Directed evolution of biocatalysts**. Cury Opin. Chem. Biol.3-54–59.
2. Atkinson, B. and Mavituna, F. (eds.) (1991). **Biochemical Engineering and Biotechnology Handbook,** 2nd Edition. Stockton Press, New York.
3. Bullock, C. (1995). **Immobilized Enzymes**. Science Prog. 78: 119-134.
4. Darbyshire, J. (1981). **Large scale enzyme extraction and recovery.** In Topics in Enzyme and Fermentation Biotechnology, Vol. 5. (ed. A.Wiseman,) pp. 147–186. John Wiley, New York.
5. Demain, A.L. (1990). **Regulation and exploitation of enzyme biosynthesis.** In Microbial Enzymes and Biotechnology, 2nd Edition (eds. W.M. Fogarty and C.T. Kelly,), pp. 331–368. Elsevier, London.
6. Godfrey, T. and West, S. (eds.) (1996). **Industrial Enzymology,** 2nd Edition. Macmillan Press, London.
7. Lowe, D.A.(1992). **Fungal enzymes.** In Handbook of Applied Mycology, Vol. 4 (eds. D.K. Arora, et al.,) pp. 681–706. Marcel Dekker, New York.
8. Patel, R.N. (ed.)(1999). **Stereoselective Biocatalysis.** Marcel Dekker, New York.
9. Tanaka.A., Tosa, T. and Kobayashi, T. (eds.)(1993). **Industrial Application of Immobilized Enzymes.** Marcel Dekker, New York.
10. Zaks, A. and Dodds, D.R. (1998). **Biotransformations in the discovery and development of pharmaceuticals.** Curr. Opin. Drug Discovery Develop, 1: 290–303.

9

Beverages

Microorganisms have played a major role in the production of beverages. Production of beverages is one of the oldest fermentations throughout the recorded human history. They are manufactured world wide and their fermentation varied with the local agricultural products. Cereals are being employed for beer, fruits for wine and honey for mead. The early fermentations were spontaneous since no inoculum was employed resulting in uncertainty of these fermentations, therefore, placed them in the class of art. For that matter even today with much advanced technology, brewing is still remained as an art because of complexity of the fermentation and associated processing. The only difference is that in early times variation in flavor that could not be controlled because of lack of knowledge of fermentation and deficiencies were covered up by the addition of spices, herbs and so forth, whereas these amendments are not required today. Brewing differs from most of the other industrial fermentations because of flavor, aroma, clarity, color, foam characteristics, foam stability, percentage of alcohol and satiety are associated with the finished product. Since people differ in their likes as the relative balance of these factors in the final product of brewing process has become commercially profitable. Production of fermented beverages is a skilled and sophisticated blend of traditions and technology. Some of the common alcoholic beverages fermented in different parts of the world are précised in table 9.1.

Some alcoholic beverages are drink fresh, but more commonly they are aged to modify their flavor, whereas others are distilled to increase alcoholic strength. Generally yeasts either in single or mixed cultures are involved in some fermentations. However, species of *Zymomonas* and bacteria are also being involved in some fermentations. The fermentation products are ethanol, array of desirable organoleptic (flavor and aroma) compounds and CO_2 (provides carbonation for some products). However, in this chapter only beer and wine are discussed.

Table 9.1: Main fermented and distilled beverages

Source	Fermented beverage	Distilled beverage
Barley	Beer	Whisky (Scotland), Whiskey (Ireland)
Rye	Rye Beer	Rye Whisky
Corn	Corn Beer	Bourbon whiskey
Wheat	Wheat Beer	Wheat whiskey
Rice	Sake	Shochu, Awamori (Japan), Huangjiu Baijiu(China)
Grape	Wine	Brandy Cognac (France), Pisco (Chile)
Apple	Cider	Applejack Calvados (France)
Pear	Perry	Perry brandy
Sugarcane		Rum
Agave	Pulque	Tequila
Pomace	Pomace wine	Marc (France),Grappa (Italy), Raki (Greece)
Honey	Mead	
Potato and/or grain	Vodka	
Milk	Kefir	

9.1 BEER

The term beer is given to non-distilled alcoholic beverage made from partially germinated cereal grains referred to as malt. The various beverages in this category include beer, ale, porter, stout and malt tonics. The fermented product involving bottom yeast in the cold is called lager beer, while the beer involving top yeast and allowed to age at comparatively higher temperature are ale, stout and porter. The ale contains high level of hops and alcohol content may be as high as 8.0%. Stout and porter employ heavy wort without malt adjuncts resulting in a dark coloured, heavy bodied and high alcohol content. The main ingredients of these fermentations are hops (offering beer a characteristic flavor and aroma), water and yeast. The raw materials required for beer fermentation are: (i) Malt, (ii) Malt adjuncts, (iii) Microorganisms, (iv) Hops and (v) Water.

(i) Malt: Malt is prepared carefully from selected barley. Barley is cleaned and steeped in water for two days. Then excess water is drained and allowed to germinate upto 5 – 6 days to allow formation of a short rootlet and acrospire. This step induces the activity of α-amylase, β-amylase and protease enzymes as well as various flavor and color components. At the end of incubation period, the temperature is raised just enough to kill the seed and at the same time not inactivating desired enzymes. This becomes green malt which is dried and stored for future fermentation use. At 2–3%moisture malt can be stored for several months. The preparation of good malt is difficult task, requires careful selection of barley and close supervision of the malting process. Therefore, many leading brewing companies do not produce malt but rely on separate companies that specialized in this art. For the production of dark- coloured carmalized malt for brewing ale and stout, is prepared by raising it to a higher temperature.

Malt contributes amylases, proteases, starch, protein, additional yeast nutrients, growth

factors and flavor characteristics to medium. Nearly 65% of malt constituents contributes to beer production, while the residue is separated and used as cattle feed or discarded.

(ii) Malt adjuncts: Barley contains considerable amount of protein, which forms main source of carbon and nitrogen, which may contribute to the dark color of the beer.

It becomes unstable and filled with average taste. In order to decrease carbon and nitrogen ratio, if they are in excess starch containing malt adjuncts are added. The various sources of these malt adjuncts are grits, or meal prepared from rice, degermed corn, barley, wheat flour, prepared starches such as flaked corn, dextrose sugar and syrup. The starch of these adjuncts must first be gelatinized by boiling and saccharified by the amylases of malt. In USA upto 60% may be used, while 40% is recommended in Europe. British beer is prepared from 75% malt and 25% malt adjuncts. On the other hand, Germany the beer purity forbids the use of any adjuncts. Approximately 80 – 90% by weight of these starch adjuncts is extracted into the fermentation medium (Wort).

(iii) Microorganisms: In beer brewing both top fermenting and bottom fermenting strains of *Saccharomyces cerevisiae, S. carlbergensis* and *S. uvarum* are employed. Top fermenting yeast exhibit flotational flocculation behaviour are employed in ale and stout. The top yeasts differ from bottom yeast in their inability to ferment melibiose, on the other hand, bottom yeast perform sedimentary flocculation. Top yeasts are unable to ferment maltotriose. Yeasts are selected for their fermenting ability and their ability to flocculate at proper time at the end of fermentation. Therefore, each industry selects its own strain. However, in fermentation, yeasts are selected from previous fermentation and inoculation of yeast is known as pitching. In order to avoid contamination, the yeast cell mass is washed with tartaric acid, ammonium persulphate which inhibits the bacterial growth or phosphoric acid which makes the medium acidic (pH 2.5) and still some industries prefer to use fresh culture. 0.4536 kg of yeast is added as inoculum which yields approximately 1.39–1.84 kg at the harvest. Yeast is basically an aerobe and needs oxygen for its growth. By utilizing sugars, amino acids and different minerals present in the wort, yeast grows at faster rate which results in bubbling of fermentation medium.

(iv) Hops: Hops are the dried female flowers of *Humulus lupulus* and *H. japonicus*. Approximately 100 – 150g of hops per barrel of beer and upto 1kg per barrel of ale are added. The hops provide the beer with its aromatic, pungent character and stabilizing effect. It contains α and β resins primarily humulones (Fig. 9.1), lupulones

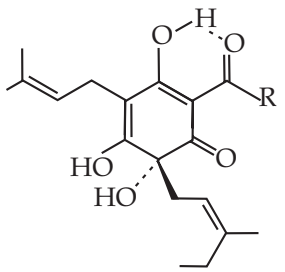


Fig. 9.1: Structure of humulones

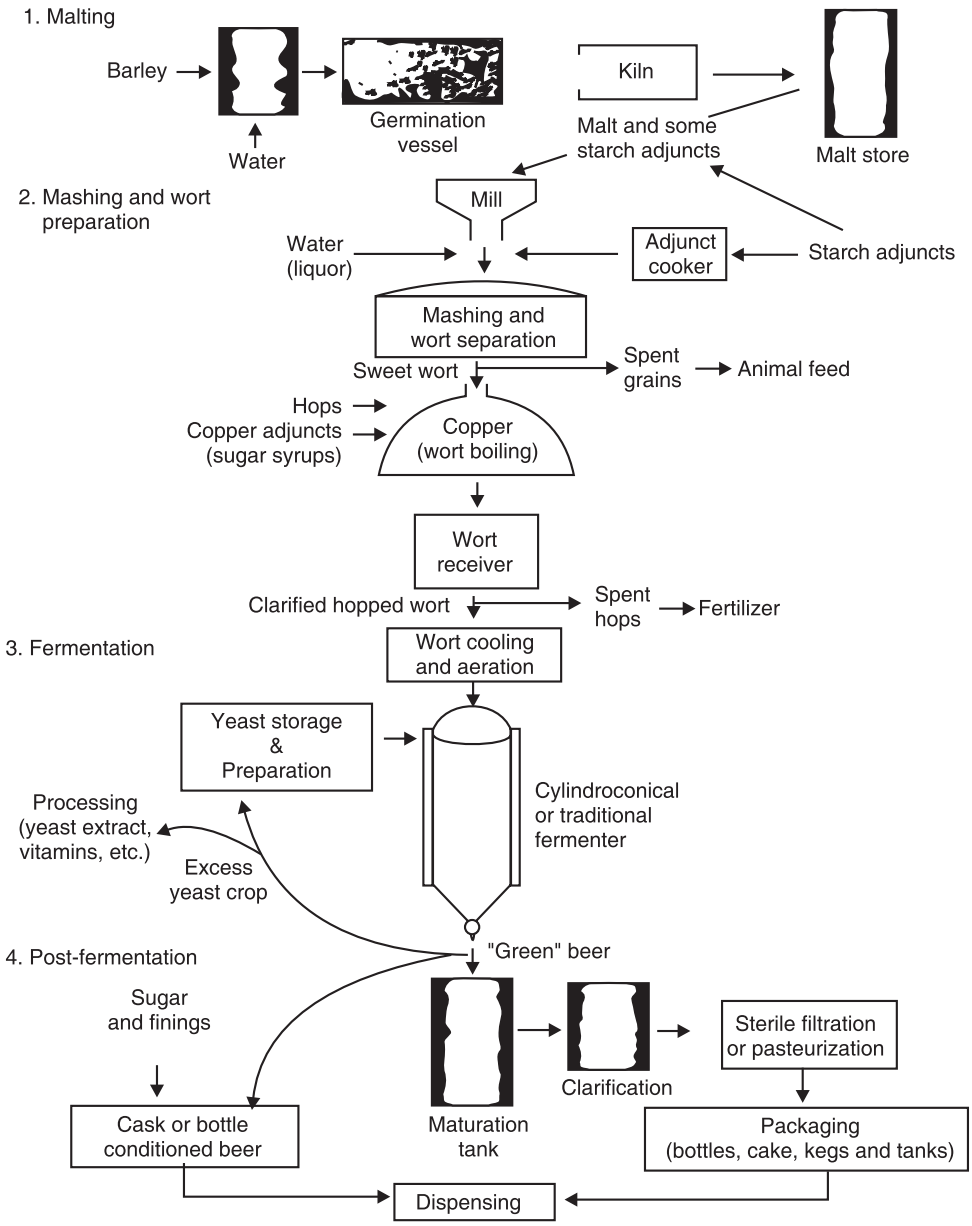


Fig. 9.2: Process of brewing of beer

and tannins which provide bitter flavour as well as preservative action against gram (+) positive bacteria. Tannins present in the bud help to coagulate protein degradation products in the medium. The pectin of buds may contribute to the foam characteristics.

(v) Water: Nearly 10–12 barrels are required to produce barrel of beer obviously only a small portion of this water ends up in the finished product. The composition of water is very important in beer production as it affects flavour and other properties of the finished product. Infact geographical location of a brewery often depends on the quality of the available water. Water containing more carbonates produces heavy flavoured and dark coloured beer, while water without carbonates and presence of sulphate results in the production of light-beer, pale-ale with a light flavour and absence of harshness. The water with pH 6.5–7.0, carbonates of calcium, magnesium less than 100 ppm, trace amounts of magnesium, calcium (preferably as sulfate) 250 - 500 ppm, calcium sulphate, sodium chloride 200–300 ppm and iron less than 1ppm is most suitable for beer production. The brewing is essentially divided into four main phases as depicted in Fig. 9.2.

1. Malting, 2. Mashing and wort preparation, 3. Yeast fermentation
4. Post-fermentation

9.1.1 Medium preparation (wort): Malt normally provides most of the potential fermentable materials and sufficient enzymes to generate a well balanced fermentation medium or wort. The wort is prepared from malt with main objective to extract sugars, amino acids and vitamins into solution. Malt and malt adjuncts are usually rolled, milled prior to mashing in such a way as to largely retain the husk intact, while reducing the remainder to coarse powder which helps in filtration aid in wort preparation. The extraction of nutrients into solution through a regulated enzyme action of malt is known as *Mashing* and the resultant clear nutrient solution which is separated from residues is known as *wort*. The resultant wort should be a well-balanced liquid medium that will supply to yeasts with all nutritional requirements for the subsequent fermentation. There are three types of Mashing systems.

(a) Infusion mashing: It is a classic method mainly for ale and stout. Solid mash ingredients (malt and malt adjuncts) are mixed with hot brewing liquor (62 – 65%) in an insulated unstirred vessel is known as mash tun (Fig. 9.3) which is 7 – 9 m in diameter and 2m deep, has slots of 1mm wide at the bottom. However, in some modern breweries these traditional vessels have been replaced by mash mixers.

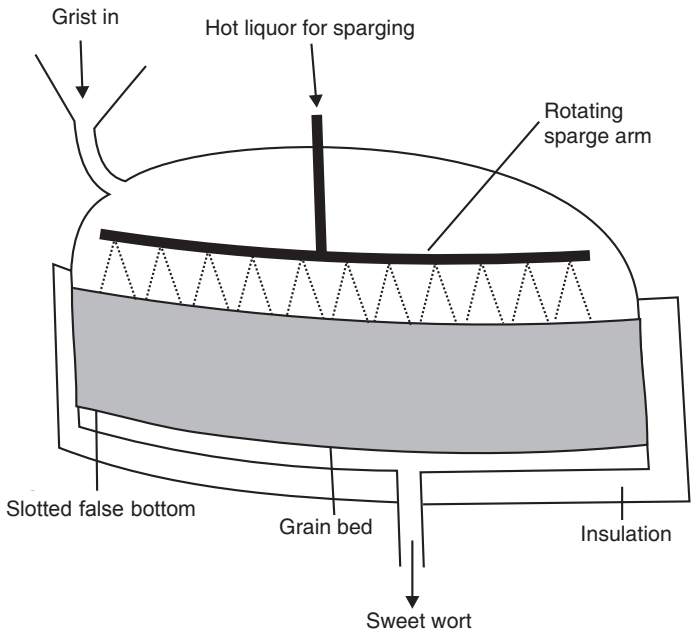


Fig. 9.3: Mash tun

The mash contains liquid and solid in the ratio of 3:1 which helps to activate as well as stabilize malt enzymes. The temperature is also suitable for starch degradation but β- glucanase and proteases are soon inactivated. Therefore, in this method a well modified malt whose cell walls and protein have already been substantially degraded during malting is used. This mashing process lasts for 1-2 hrs. which is followed by mash sparging by sparying hot water (70-75°C) to mash surface to take out residual soluble extract. The resultant liquid extract is called sweet wort. Different enzymes present in the germinating barley seeds will act on starch and protein resulting in variety of compounds which are depicted in table 9.2.

(b) Decoction mashing: This is the most suitable for less well modified malts. The initial mash temperature starts at 35-40°C with a liquor to grit ratio of upto 5% which facilitates to hydrolyse residual β-glucans and protein, is often called as protein rest. Subsequently temperature is raised to 50-55°C by removing a portion of malt which is boiled and immediately returned to mash vessel. This may be repeated once or twice with intermediate holding periods. Then temperature is gradually raised to 65°C to achieve starch degradation and finally to 75°C. Then the wort is separated with the help of lautertub or mash filter system may be used.

(c) Temperature programmed mashing: This is most modern system. In this, mash temperature is raised by heating jacket from around 40°C, to above 70°C through a number of holding stages. This method posseses greater control over mashing process. Mash separation is through lauter device or mash filter.

Table 9.2: Commercial enzymes in brewing of beer

Enzyme	Source	Stage of addition	Role
α - Amylase	*Bacillus subtilis* *Bacillus licheniformis* *Aspergillus oryzae* *Bacillus stearothermophilus* (thermostable, does not require Ca^{2+})	Mashing, adjunct cooking	Liquefaction of starch, especially maize or rice grits, has superior properties to malt enzyme
β -Amylase	Barley *(Hordeum vulgare)* Wheat *(Triticum aestivum)* Soyabean *(Glycine max)* *Bacillus cereus*	Mashing	Maltogenic, improves wort fermentability and sugar spectrum
Pullulanase	*Bacillus acidopullulyticus* *Thermus aquaticus* (thermostable, pH optimum 6.4)	Mashing, fermentation	Debranches, hydrolysis of α –1, 6 links in dextrins to produce 'diet-beers', often used with added α or β-amylase
Amyloglucosi-dase	*Aspergillus niger* *Rhziopus niveus* *Rhizopus delemar*	Mashing, fermentation, beer	Yields glucose increases wort fermentability, used to produce 'diet beers', and in enzymic priming (hydrolyses α-1, 4 and α-1, 6 linkages)
Endo-β-glucanase	*Bacillus subtilis* *Penicillium emersonii* *Aspergillus niger*	Mashing especially if using barley adjuncts	Improves extract recovery in mashing, wort separation and beer filtration rates, prevents gels and hazes
Cellulase	*Trichoderma viride* *Trichoderma reesei* *Penicillium funiculosum*	Malting, mashing	Cleaves β-1, 4 links, promotes barley modification in malting, similar role in mashing as endo- β-glucanase
Proteases (Acid) Neutral Papain Bromelain and Ficin	*Aspergillus niger* *Bacillus subtilis* Papaya *(Carica papaya L.)* Pine apple *(Ananas comosus)* Fig *(Ficus species)*	Mashing, beer	Prevents protein, polyphenol haze, aids balance of wort amino acid spectrum, also activates β-amylase
Glucose oxidase / Catalase	*Aspergillus niger*	Beer	Removal of O_2 from beer, aiding flavour stability and haze life
Acetolactate decarboxylase	*Bacillus licheniformis* *Lactobacillus species*	Fermentation of beer	Speeds maturation by eliminating diacetyl

(d) Biochemistry of mashing: The objective in mashing is to convert as much of the malt and malt adjuncts as possible to fermentable sugars. During the process, starch of malt adjuncts, glucosides which contain α-1-4 and 1,6 linkages will be acted by depolymerising, de-branching enzymes such as α-amylase, β-amylase, amylogluco- sidase, liberate fermentable sugars and some limit dextrins. The proteases of barley malt degrade wall and matrix protein of endosperm cells. The proteases which are relatively labile will act at initial starting temperatures 40-50°C, liberate peptides and amino acids. Addition of commercial proteases can aid separation and ensure sufficient levels of amino acids for yeast metabolism.

(e) Separation of wort: The husks and other grain residues in the wort as well as precipitated protein while other solids are removed from wort by passing through lautertub. Which has a fasle bottom slotted perforated to allow passage of the wort.

(f) Wort boiling: Sweet wort is boiled in a copper vessel along with dried hops or hop extracts added at the end approximately for 1.5–2.5 hrs with constant stirring or agitation. Boiling of wort is to achieve isomerization of hop α-acids to the more bitter tasting Iso-α-acids. It facilitates extraction of tannins and essential oils. Sterilization of wort is to achieve its concentration, termination of malt enzymes activity and coagulation of residual proteins, removal of volatile compounds that can impair flavour develop some flavour compounds as well as colour through formation of melanoids oxidation of phenolic compounds. The wort then is aerated and cooled while maintaining aseptic conditions. The cooled wort may or may not be filtered before used as a fermentation medium. The nutritional composition of typical wort is given in table 9.3.

Table 9.3: Nutritional status of a typical wort

Nutrient	Concentration (g lt^{-1})
1. Carbohydrate	**(g lt^{-1})**
Fermentable carbohydrates	10
Glucose + fructose	4
Sucrose	42
Maltose	15
Maltotriose	23 (not fermented by most brewing yeasts)
Higher dextrins	94
Total carbohydrates	
2. Amino acids	1.5
3. Vitamins and growth factors	**Concentration (mg lt^{-1})**
Biotin	<0.01
Inositol	60
Nicotinic acid	10
Pantothenic acid	1
Pyridoxine	0.6
Riboflavin	0.45
Thiamine	0.3
4. Minerals	**Concentration (mg lt^{-1})**
Sodium	20
Potassium	450
Calcium	40
Iron	<1
Magnesium	100
Chloride	350
Phosphate	900
5. Organic acids	**Approximately 200 mg lt^{-1}**
6. pH (wort)	5.2 - 5.4

9.1.2 Fermentation: The aerated wort is cooled to 10-11°C and placed in a closed fermentation tank of 2.0 lakhs liters. However, open tank can also be used but risk of contamination and escaping of CO_2 which is needed for carbonation is high. Further, the closed fermenter also helps in flavour development. About three quarter to one pound of yeast is added to each 141 litres of wort. The brewing yeast requires the following nutrients which are normally supplied by well balanced wort (table 9.3).

1. A carbon and energy source provided by fermentable carbohydrates.
2. A nitrogen source, comprising amino acids and peptides.
3. A source of calcium, magnesium, phosphorus, sulphur, traces of copper and zinc ions.
4. Growth factors including biotin and pantothenate.

Within 24 hrs of pitching, foam begins to appear on the surface of the medium first along the wall of the fermenter and gradually across the surface. This is followed by evolution of CO_2 which results in suspension of yeast cells in the medium. At this point, broth is transferred to a second fermenter so that dead and weak yeast cells, precipitated protein, insoluble resins are left behind. This procedure also facililtates aeration to the medium. However, if the boiled and cooled wort is filtered, this transfer may not be required. By 40-60 hrs of pitching, surface of wort is covered by foam upto 12 cm depth. During this period rapid yeast multiplication takes place and considerable heat is generated due to high metabolic activity. This may result in rise of temperature to approximately 11-13°C.

However, every care has to be taken to check further rise in temperature. In ale fermentation this temperature may be as high as 16°C. By the end of fifth day of pitching, fermentation will come to an end. This is perceptible by collapse of foam, decrease heat generation. By the end of 7–9 days yeast cells become inactivated, flocculate and is known as yeast break. However, ale fermentation is completed in five to seven days only. Yeast cells settle at the bottom. This process may be hastened with further cooling. At this time scum of the surface may be removed to improve flavour. The main factors that affect the fermentation rate and influence the beer quality are :

1. The amount of yeast used to inoculate or pitch the fermentation.
2. Yeast cell viability and yeast quality.
3. The level of dissolved oxygen in wort at pitching.
4. Wort soluble nitrogen concentration.
5. Wort fermentable carbohydrate concentration and.
6. Temperature.

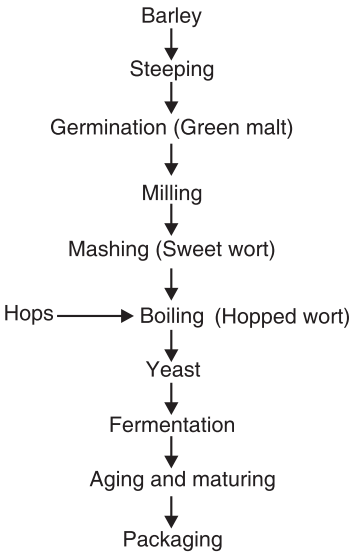


Fig. 9.4: Flow diagram of beer production

British ales are traditionally produced in relatively shallow open circular or rectangular vessels constructed of wood or stone. Now-a-days most modern brewery fermenters are closed cylindroconical vessels constructed of stainless steel with a capacity of upto 2,00,000 liters (Fig. 9.5). They normally have cooling jackets and are used to produce both ales and lagers. Advantages of these vessels include:

1. Improved fermentation rates.
2. Greater flexibility as they can be used to produce a range of beers.
3. Relatively low construction costs require little civil engineering works and occupying less land area.
4. Large fermentation capacity giving benefits of economics.
5. Lower running costs.
6. Easy CO_2 collection from the top of the fermenter and yeast removal from the conical base where yeast collects at the end of the fermentation.
7. Beer losses are reduced to a minimum.
8. Efficient temperature control.
9. Consistent product quality.
10. Easy to clean using modern cleaning-in-place (CIP) system.
11. The same vessel may also be used for beer maturation following yeast removal.

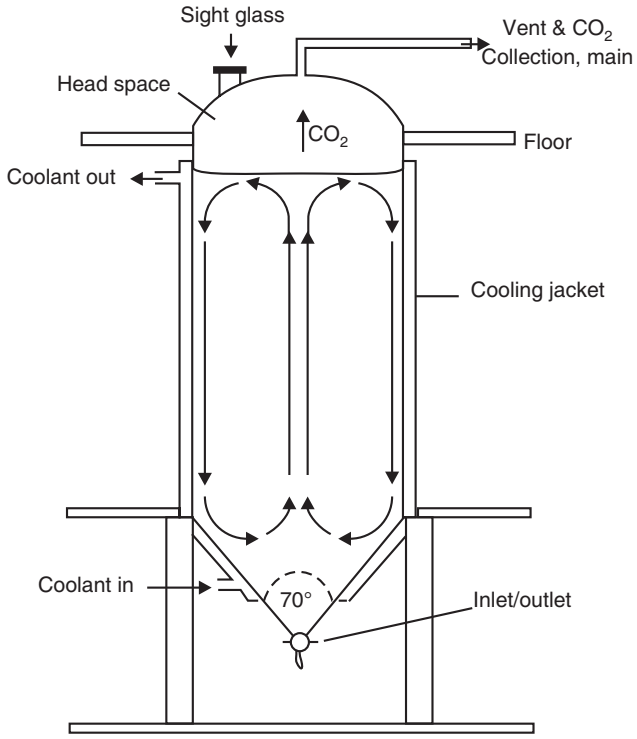


Fig. 9.5: Diagram of cylindroconical fermentation vessel showing the flow of wort during fermentation

9.1.3 **Cold storage:** The completed fermentation is transferred to storage tanks and held at 0-3°C for desired period of time. During this stage coagulated nitrogen substances, resins, insoluble phosphates and yeast cells sediment from the beer. In addition different alcohols, acids as well as esters are formed and beer matures so that beer looses its harshness (Fig. 9.6). During this process, chill proofing is practiced to prevent turbidity development. Much of their turbidity can be attributed to unstable protein in the beer and chill proofing can mean merely the removal by precipitation or adsorption of these unstable residual proteins or partially hydrolysed proteins on later exposure of the finished beer to cold. More often, proteolytic enzyme is added during this process to reduce molecular size of protein hydrolysed products. Antioxidants are also added to prevent later oxidative changes. Sulphur dioxide and ascorbic acid are commonly used to accomplish this.

9.1.4 **Carbonation:** Addition of carbon dioxide to the mature beer to make it more anaerobic and prevent oxidative changes is the common practice. In this process the evolved CO_2 from the fermentation is pumped in to finished beer sometimes-final level 0.05% and increase its stability. This is also achieved by adding active yeast cells, which produce large amount of CO_2, which can create an anaerobic condition. This process of employing active 15% yeast cells from previous fermentation is known as Krausen process. These yeast cells are allowed to slowly ferment and remove residual sugar for about 3–4 weeks so that the CO_2 evolved is vented to the final level of 0.5%. After this treatment, beer is allowed to complete the clarification.

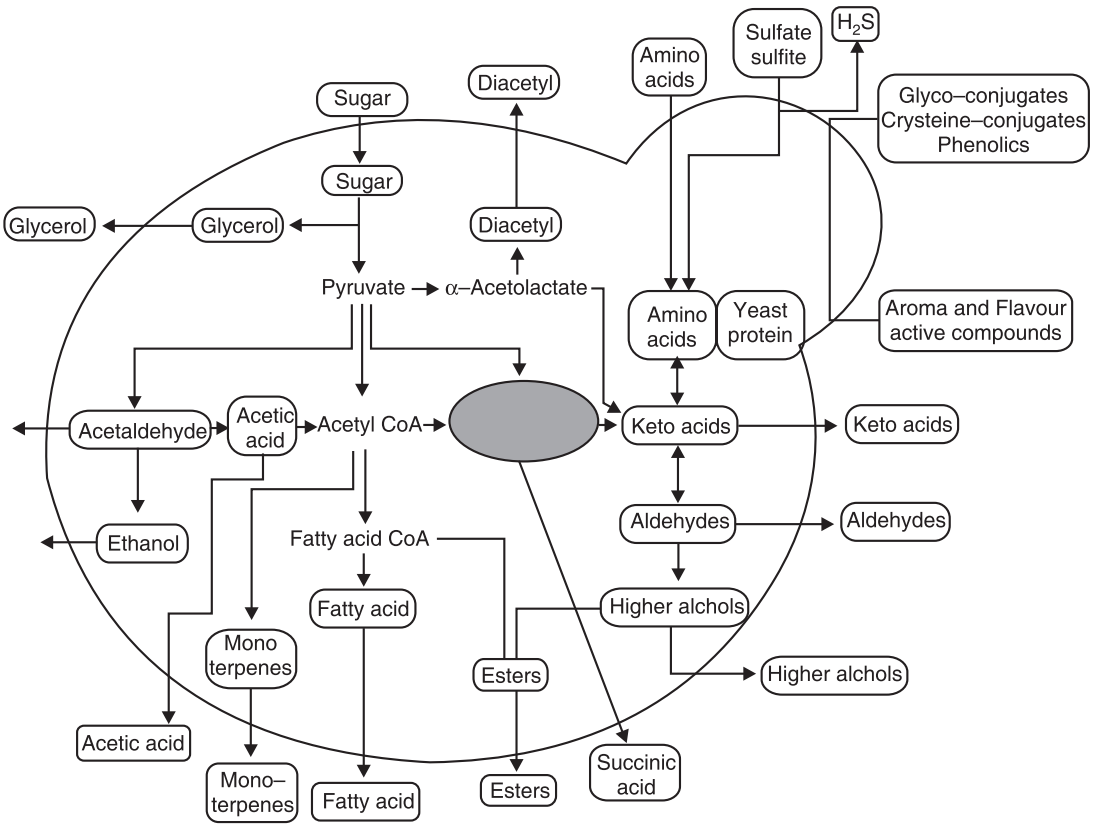


Fig. 9.6: Production of different alcohols and acids during post fermentation period

9.1.5 **Packaging:** The matured beer is passed through cooling pipes and through diatomaceous earth filter before being packed in bottles, cans, barrels or kegs. Air is excluded during packing to prevent oxidative changes in the product. Further, while bottling, beer is subjected to electronic inspection to make sure that solid impurities, turbidity or haze are not present. The bottles are pasteurized after capping the bottles or closing of cans. Beer packed in barrels or kegs is not pasteurized. Hence, it has a relatively short storage life. Pasteurization, however, affects its flavour. Some of the recent innovations help the beer sterilization without pasteurization. The entire process of brewing of beer is presented in the form of a flow chart in Figs. 9.2 and 9.4.

9.1.6 **Keeping quality:** Beer deteriorates with age specially with reference to changes in flavour and appearance. Exposure to sunlight, warm storage temperature, shaking or agitation, internal oxidation changes by residual oxygen, turbidity caused by unstable proteins, protein-tannin complexes, starch, resins being insoluble also spoil beer. Microorganisms such as yeasts, bacteria may contribute to turbidity or haze development.

9.1.7 **Contamination:** Beer fermentation basically is an anaerobic. Hence, contamination problems are minimum. The pH of wort ranges between 3.5 and 4.5, which is too acidic to allow bacterial growth. Resins of hops prevent the growth of bacteria. Production of

CO_2 and ethyl alcohol also prevents contamination. However, still contamination and spoilage of beer also occurs frequently. Some of the causes, both biotic and abiotic, for spoilage of beer are precised in table 9.4.

9.1.8 Byproducts: The residues of malt during preparation of wort, yeast cells after drying which are rich in carbohydrate, protein and vitamins can be used as cattle feed.

9.1.9 Continuous fermentation: In view of delicacy of flavor, aroma and taste, this fermentation is not suitable for continuous fermentation. However, **Royston** (1966) has designed a fermentation tank for this purpose which is yet to gain commercial acceptance.

Table 9.4: Microbial spoilage of beer

Microorganism	Stage of processing encountered	Problem
Lactobacillus spp.	Any stage	Off-flavors Haze Ropiness Acidification
Pediococcus spp.	Inoculation Fermentation Aging	Off-flavors Haze Ropiness Acidification
Acetic acid bacteria	Open fermenters Packaging	Off-flavors Haze Surface pellicles Ropiness
Hafnia protea	Wort Fermentation	Off-flavors Acidification
Other Enterobacteriaceae	Wort	Off-flavors
Bacillus spp.	Wort	Off-flavors
Zymomonas spp.	Packaging	Off-flavors
Pectinatus spp.	Packaging	Off-flavors
Negasphaera spp.	Bottled beers with pH>4.1 and <3.5% ethanol	Off-flavors
Wild yeasts	All stages	Off-flavors Haze
Moulds	Raw barley spoilage	Gushing Off-flavors

9.2 WINE

Wine can be made from any fruit juices that contain sufficient levels of fermentable sugars. However, grapes are extensively used in both temperate and tropical regions of the world including Central and Southern Europe; North and South America; South Africa; Australia and New Zealand. Total world Wine production is now approximately 10^{10}l/annum. Grape wines have become major fermentation products.

Grapes are most suitable for wine production in view of their high content of fermentable sugars (160–240 g l^{-1}), other nutrients necessary for successful yeast fermentation. Grapes have pleasant flavour, aroma components and the natural acidity (pH 2.3-3.8) which inhibit potential spoilage microorganisms. Most wines are table wines that normally contain upto 14% alcohol. However, some regions produce wines fortified with ethanol in the form of added spirit or grape brandy to give final ethanol level upto 22%. The main fortified wines are sherry, port, and

Madiera vermoths which contain different flavour components from as well as spices. The process of wine production is précised in Fig. 9.7.

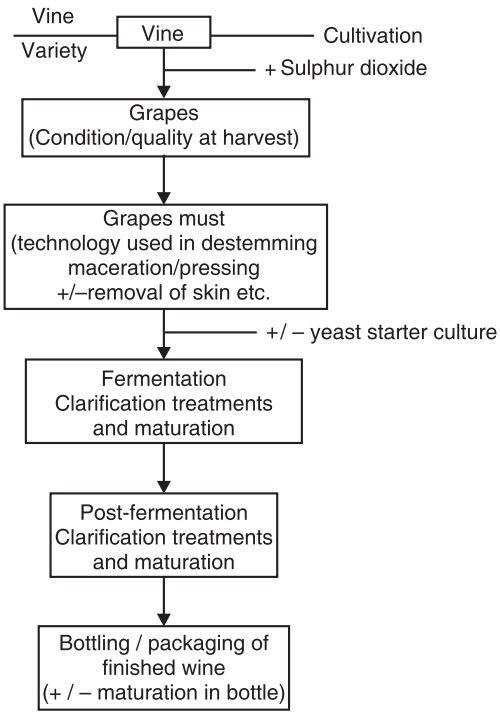


Fig. 9.7: Flow diagram of wine production

Further, alcohol content increases considerably which also discourages growth of unwanted microorganisms. The flavour components and quality factors are influenced by or derived from the wines can be classified on the basis of alcohol content, CO_2 and fermentable sugars. There is an immense range of wines which may be classified by their colour and sweetness (table 9.5). Quality of wine is influenced by:

1. Specific grape variety or mixture used (e.g. Muscat grapes contain highly characteristic flavour and aroma components, the terpene alcohols linalool and geraniol).
2. The soil, local climate and the mode of grape cultivation.
3. Grape maturity and condition at harvest.
4. The methods of grape pressing and processing.
5. The primary fermentation which generates variable quantities of ethanol, CO_2, higher alcohols, esters, aldehydes and ketones.
6. Any secondary fermentation to produce wine carbonation or for a acidity reduction via a malo-lactic fermentation, whereby malic acid is converted to lactic acid.
7. Post fermentation and maturation treatments.
8. Red wine must consists of macerated fruit skins or seeds are not usually removed, while white wine is a clarified juice until the fermentation is over.

9.2.1 Fermentation: The grape must is prepared from freshly harvested grapes. The age and health condition of grape greatly influences the quality of wine. Grapes after proper picking are then conveyed to the winery for processing. The grapes after thorough wash and degerming are subjected to mechanical stemming, crushing to form the must. For Red wine, must consists of macerated fruit, whereas for white wines it is only clarified juice that is fermented (Fig. 9.7).

For red wines, skins and seeds are not usually removed until the end of fermentation. This allows anthocyanins (red pigments that also add astringency) to be slowly extracted from the skins during fermentation. Pectinolytic enzymes may be added during processing to aid both juice and colour extraction. Carbonic maceration, an alternative to pressing for red wine production, involves holding intact grapes for several days under a blanket of CO_2. This results in the cell death and breakdown of cellular structure. Anthocyanins are slowly extracted from skins during fermentation. Pectinolytic enzymes may be added during fermentation to aid both juice and colour extraction.

White wines are produced from either red or white grapes, provided the source of colour is avoided. The early separation of skin, seeds and plant debris is essential to reduce astringency as well as browning. This can also be achieved by preventing oxidation of phenols with the help of blanket covering of CO_2 which creates anaerobic conditions. Addition of SO_2 or centrifugation of the juice remove membrane bound phenol oxidases which helps in preventing browning.

Alcoholic fermentation of wine is carried out by indigenous yeast or added starter yeast. During fermentation the yeast must grow to a sufficiently high cell density to complete the fermentation. For production of dry wines requires conversion of all sugar into alcohol. The rate of fermentation is influenced by starter yeast strain used or yeast strains that make up the natural flora. Temperature, pH, initial sugar concentration, nutritional components of the batch of grapes must, any supplements of thiamine, ammonium salts and sterols, that may be added to ensure maintenance of yeast activity until the fermentation is completed (table 9.5).

Table 9.5: Nutritional composition of medium for the growth of wine yeast

Nutrient	Concentration (mg l^{-1})
Major elements	
Magnesium	100
Nitrogen	150-200
Phosphorus	100-200
Potassium	500-1000
Minor elements	
Copper	0.1
Iron	0.01-0.2
Manganese	3-6
Sulphur	10
Zinc	5-10
Organic Compounds	
Biotin	0.001-0.003
Inositol	5-10
Nicotinic acid	0.2-0.5
Pantothenic acid	0.2-0.5
Pyridoxin	0.2-0.5
Thiamine (Vitamin B)	0.1-0.2

In ancient times open wood fermentation vessels were used which are now being replaced by stainless steel cylindroconical vessels with cooling coils or jackets. The nutritional composition of medium is given in table 9.3. Fermentation requires cooling as temperature rises by 1.3°C for every 10 g l^{-1} sugar fermented. White wines are fermented at 10 – 20°C for 2 – 4 weeks, while for red wine temperature is maintained at 24 – 29°C lasting for 3 – 5 days. Yeast fermentation stops when all available sugar has been metabolized leaving less than 1 g l^{-1} of residual sugar. However, the fermentation can be terminated to maintain sweetness of wine either by removing yeast by filtration or inactivation through cooling, heating or adding metabolic inhibitor such as CO_2, changes in composition of must and byproducts during wine fermentation are depicted in Fig. 9.8.

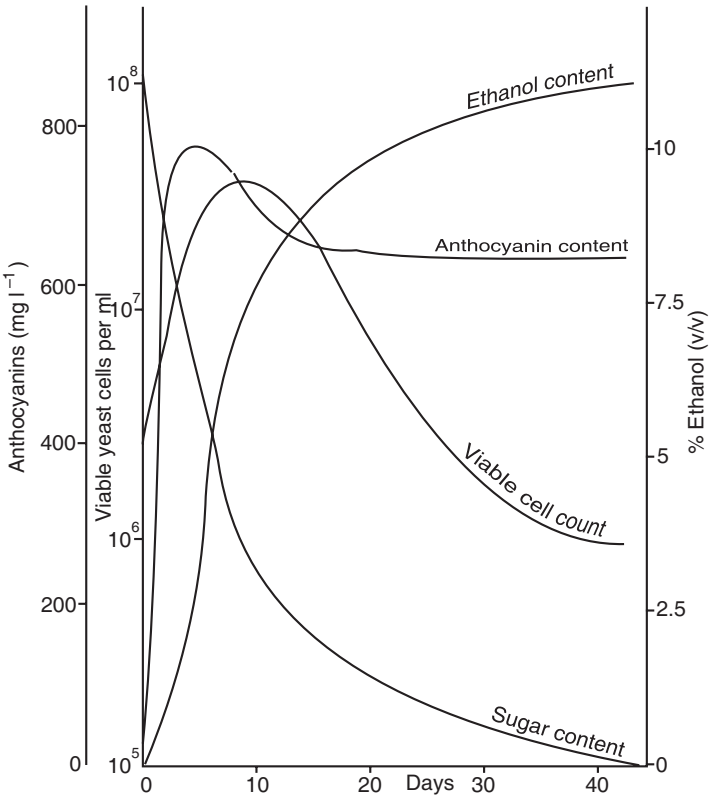


Fig. 9.8: Compositional changes during wine fermentation

9.2.2 **Secondary wine fermentation:** Following the primary alcoholic fermentation, the wine maker can encourage the further activities of yeasts or other microorganisms. Secondary fermentation include yeast alcoholic fermentation in bottles or casks to create naturally carbonated sparkling wines or the growth of aerobic surface film of yeasts to produce fine sherry or lactic acid bacteria may be encouraged to perform malo-lactic fermentation

through the action of lactic acid bacteria such as *Leuconostoc, Lactobacillus* and *Pedococcus* which are naturally present in wine. This will not only reduce acidity but also helpful to develop flavour components and the wine becomes microbiologically more stable as these organisms do not allow other microorganism to grow. Secondary fermentation reduces acidity through decarboxylation of malic acid to lactic acid, manocarboxylic acid (Fig. 9.9). *Leuconostoc oenes* may be added which promotes flavour development.

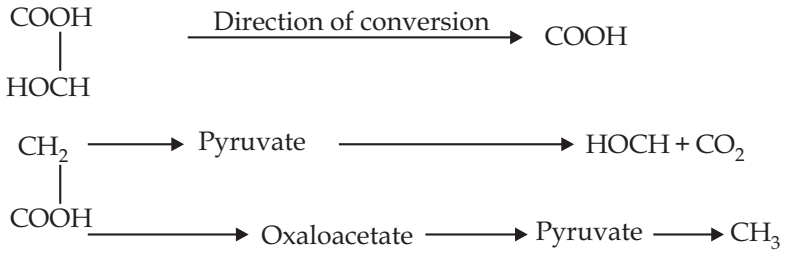


Fig. 9.9: Conversion of malolactate to lactic acid

9.2.3 Post-fermentation treatment: At the end of all fermentations, the wine is to be protected from both microbiological and non-microbiological spoilage particularly oxidation. It can be stabilized by cooling, maintenance of anaerobic conditions and preliminary clarification by settling and racking off the wine from the sediment to remove microorganisms. Alternatively, the wine may be left for 2–30 weeks so that yeasts may get sedimented and release more yeast flavour compounds. A small amount of SO_2 may be added to inhibit further microbial activity. It also protects the wine which is normally 60-200 mg l^{-1}. Wines may be allowed to mature for a period ranging from few weeks to several years in a wooden casks where it acquires additional flavour characters. Finally, the wine is filtered, pasteurized or sterile filtered and filled into bottles or other packages. The wine components continue to interact during further storage slowly modifying colour, flavour and aroma. Flow chart for production of White wine, Rose wine, Red wine, Sparkling wines are given in Fig. 9.10.

1. Traditional spontaneous wine fermentation is carried by natural microflora of vineyard, winery environment, grape skin, plant debris, soil, equipment surfaces and air develops in grape (-) must in process of succession of different microorganisms. Proliferation of yeasts is favoured by low pH, high sugar concentration and anaerobic conditions. Wild yeasts *Kloeckera* and other intolerant species, favours the growth of *S. cerevisiae.* To ensure complete fermentation and avoid unwanted flavour, aroma compounds, wine makers encourage the growth and dominance of *S. cerevisiae*. This may be achieved by addition of SO_2 at a concentration of 30-50 mg l^{-1}, as a result *S. cerevisiae* population rises to 10^3 – 10^7 cells thereby dominating the fermentation. *S. cerevisiae* is favoured at 16-20°C temperature, while below 14°C favours the growth of other yeasts like species of *Kloeckera.* Sometimes mixed population develops benefiting flavour resulting in very fine complex wines. Thus, by this fermentation wine production takes place but without definite flavour, aroma and taste.

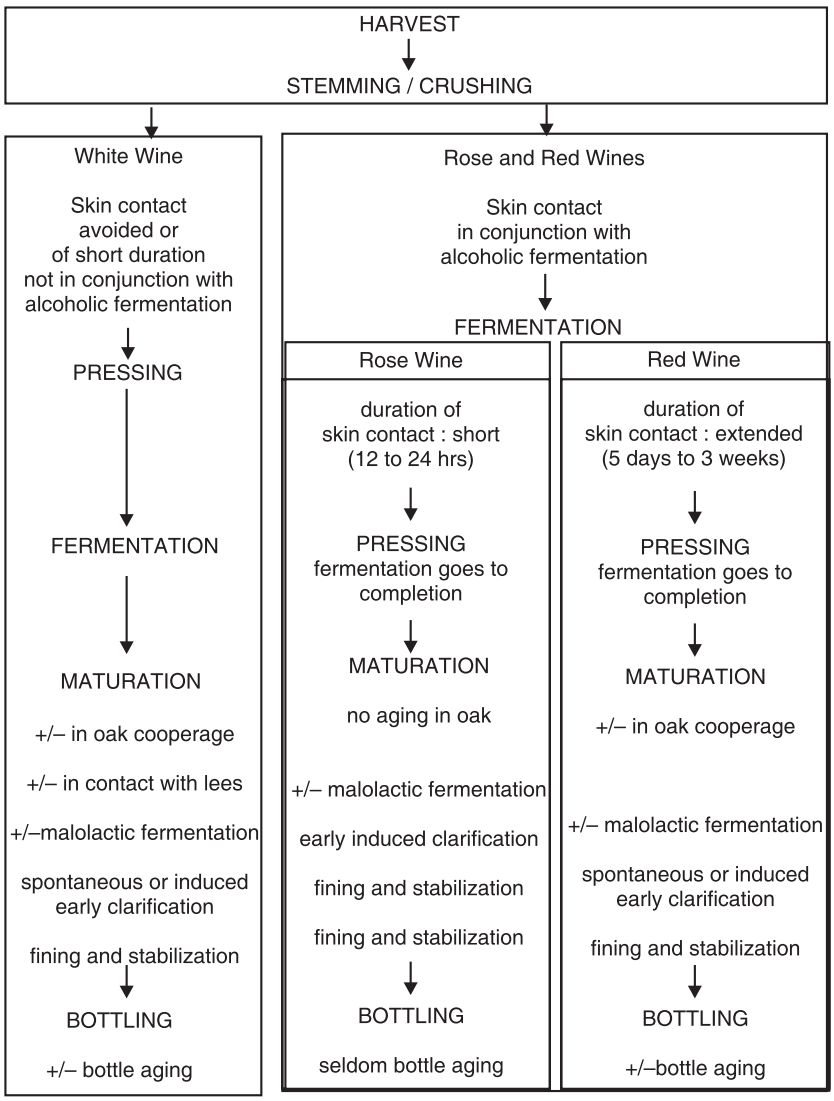


Fig. 9.10: Flow chart production of white, rose and red wines

2. Use of starter culture enables for production of desired very fine quality wines. *S. cerevisiae* termed to high quality red, white, fortified and sparkling wines. Those yeasts are homothalic diploid. Wine strains can be differentiated only by molecular techniques. Yeast is inoculated at the rate of 5×10^6 cells ml^{-1}. They grow to more than 10^7 cells ml^{-1} and dominate fermentation. An indigenous non-*Saccharomyces* yeasts are suppressed by adding 30-50 mg l^{-1} SO_2 and temperature of 16°C. Some of the advantages of using elite strain of yeasts are:

(i) It ensures completion of the fermentation and allows the fermentation rate to be controlled.

(ii) To give a shorter lag period.

(iii) Rapid fermentation

(iv) It provides less opportunity for the production of flavours by wild yeasts and bacteria.

(v) Gives more consistent flavour characteristics and quality.

One has to ensure yeasts employed for wine fermentation with following characteristics.

(i) Tolerance to alcohol

(ii) High sugar mists, SO_2, low and high temperature especially for production of sparking wines

(iii) They produce small amount of acetic acid, acetaldehyde, mercaptans, diacetyl, SO_2, and higher alcohols.

(iv) Low foam production

(v) Good flocculation properties

(vi) Suitability for drying and

(vii) Stability for long term storage of starter culture.

Construction of wine yeast strain that express additional beneficial characteristics is being persued by recombinant-DNA technology which helps to avoid use of commercial enzymes. The strain should be suitable for

(i) Production of desirable organoleptic compounds

(ii) Metabolism of a large portion of malic acid in must

(iii) Low urea excretion

(iv) Improved fermentation efficiency

(v) Suitable flocculation characteristics

(vi) Controllable ester formation

(vii) Protease production

(viii) Pectinase production to improve wine filteration rate

(ix) Production of inhibitory substances against spoilage microorganisms. E.g. resin to kill Gram (+)ve bacteria and enzymocins to kill wild yeasts.

(x) Production of glucanase and release bound terpenes to enhance aroma.

(xi) Greater dihydroxyacetone phosphate reductase and glycerol phosphatase activities to increase glycerol levels.

(xii) They must be suitable for immobilization.

9.2.4 Continuous fermentation: In view of delicacy of flavour, aroma of wines batch fermentation is most suitable. However, continuous fermentation has potential advantages over batch fermentation. But for success of this fermentation, continuous supply of grapes are to be ensured. In France, production of Red wine using over 1.5 lakhs kg of grapes are processed each day and corresponding amount of wine is produced. Average residence time of the wine in the fermenter is 3 days at 26-28°C.

9.2.5 Importance of wines: Wines are reported to have medicinal properties and responsible for improving health of man which are as follows:

1. They contain antioxidants which are useful in preventing coronary heart diseases (CHD) by following ways:
   (i) Can prevent free radical damage to the tissues
   (ii) Glutathione wheat
   (iii) Acts as a chelating agent
2. The wines increase high density lipoproteins (HDL) level in the body (also known as good cholesterol).
3. Provides nutrients and minerals (table 9.6).

Table 9.6: Mineral content (in ppm) of different fruit wines

Type of Beverages	Na	K	Mg	Ca	Fe	Cu	Zn	Mn
Apricot wine (New castle)	11	1481	18	71	2.72	0.96	1.92	0.88
Wild Apricot wine (Chudi)	43	2602	25	94	5.97	0.50	2.69	0.99
Apple wine (Golden delicious)	18	1044	16	144	3.68	0.21	0.76	0.84
Hard cidar (Golden delicious)	19	1069	17	97	3.03	0.19	0.91	0.82
Cidar (Apple juice concentrate)	61	1900	23	137	4.31	0.32	1.54	1.01
Pear wine (Sand pear)	87	1906	37	122	8.91	0.16	0.80	1.10
Plum wine (Santa Rosa)	20	1008	18	82	12.73	0.20	1.04	0.95

4. Presence of glucose tolerant factor (GTF) in wine is a cure for diabetes.
5. Presence of resveratrol prevents cancer.
6. Different wines are rich source of different minerals required by man.

REVIEW QUESTIONS

I. Essay type questions:

1. What is fermentation? Write an essay on the Industrial production of beer.
2. Give an account of raw material required for beer production.
3. Describe in detail the process of wine production.
4. Explain post fermentation treatment of wine.
5. Explain different stages of Red wine production.
6. Compare and contrast the various ways that starch is used to produce alcoholic beverages.

II. Write short notes on:

1. Malt
2. Malt adjuncts
3. Hops
4. Wort
5. Mashing
6. Cold storage
7. Carbonation
8. Secondary wine fermentation
9. Importance of wine
10. Types of wines
11. Malolactate fermentation
12. Kraussen process

FURTHER READING

1. Fugelsang, K.C. year **Wine biotechnology.** Chapman and Hall, New York.
2. Priest, F.G. and Campbell, I (eds.) (1997) **Brewing microbiology.** 2nd ed Chapman and Hall, London.
3. Stewart, G.G. and Russel, I (1985). **Modern brewing biotechnology in comprehensive biotechnology.** (eds H. M. Blanch, S. Drew and D.I.C. Warng) Pergaman Oxford Vol. 3 Chapter 17.
4. Xu, G.R. and Bao T.F. **Grandiose survey of Chinese alcoholic drinks and Beverages**: 335-382. AT http:www.wxuli.edu.cri/wine/umain.htm

10

Microbial Polysaccharides

The polysaccharides made by bacteria are quite different in function from those of higher plants. They are secreted from the cell to form a layer over the surface of the organism, which is believed to have several functions. Because of their position they are characterized as exo-polysaccharides, to distinguish them from any polysaccharide that might be found within the cell. The functions are thought to be mainly, protection, either as a general physical barrier preventing access of harmful substances, or more specifically as a way of binding and neutralizing bacteriophage. In appropriate environments they may prevent dehydration. They may also prevent phagocytosis by other microorganisms or the cells of the immune system. The capsular polysaccharides are often highly immunogenic and may have evolved their unusual diversity as a way of avoiding antibody responses.

They also have a role in adhesion and penetration of the host. The secreted polysaccharides can be involved in pathogenicity. *Pseudomonas aeruginosa,* commonly found in respiratory tract infections, produces alginate which contributes to blockage in the respiratory tract and leads to further infection, while similar blockage of phloem by pathogenic fungi in plants has been described. Some of the microbial polysaccharides commercially important are listed in table 10.1.

10.1 XANTHAN

Xanthan (or, as commonly called, 'xanthan gum') at $14/kg falls into a similar price range to other gums and exudates, has similar functional properties. It is made by the organism *Xanthomonas campestris* (which in its wild existence is responsible for cabbage blight), grown largely on glucose, itself derived from maize starch. It converts glucose with high efficiency (80 percent) to the xanthan gum. Typically for a fermentation product, the raw material costs are a small part of the total, which are mainly due to recovering the gum from the culture medium.

Glc = Glucose, Mann = Mannose, GluA = Glucuronic acid,

MannA = Mannuronic acid, Ac = Acetate

Production is around 9000 tonnes a year. It has a $\beta(1 \rightarrow 4)$ linked glucan main chain with alternating residues substituted on the 3-position with a trisaccharide chain containing two

mannose and one glucuronic acid residue. It is thus a charged polymer. Some of the mannose residues may also carry acetyl groups.

Table 10.1: Structure and microorganism producing different polysaccharides

Polysaccharide	Structure	Organism
Xanthan	β - - Glc1→ 4Glc 3 ↓ α 1 Mann6 – OAc 2 ↑ β 1 GlcA 4 ↑ β 1 Mann4,6 - OPyruvate	*Xanthomonas campestris*
Alginate	β ↓ -4D - MannA1→ 4D - MannA1: α ↓ -4I - GlulA1→ 4L - GluA	*Pseudomonas aeruginosa, Azotobacter vineland*
Curdlan	β ↓ - - Glc1→ 3Glc – –	*Alcaligenes*
Scleroglucan	β ↓ - - Glc1→ 3Glc – ↑6 β 1 Glc	*Sclerotium glucanium, S. delphinii, S.rolfsii*
Pullulan	α α ↓ ↓ → 6(Glc1 – 4Glc1 – 4Glc)1 →	*Aureobasidium pullulans*
Dextran	α ↓ – Glc1– 6Glc – several1→ 2, 1→ 3and1→ 4compounds	*Acetobacter sp., Leuconostoc mesenteroides, Leuconostoc dextranicum, Streptococcus mutans*

Glc: glucose, mann=Mannose, GlcA=Glucuronic acid, mann A=manusonic acid, GluA, Guluronic acid
Ac=Acetate

It is useful because it forms relatively rigid-rod-like structures in solution at ambient temperature, though they convert to the random configuration on heating. These rods are able to align themselves like agarose and the k- and t-carrageenans – with the unsubstituted regions of galactomannans, such as guar and its derivatives and locust bean gum, to produce fairly rigid mixed gels with applications in food manufacture.

The primary structure (as worked out by sequential degradation and methylation analysis) consists of the same backbone as cellulose:

$$\ldots \rightarrow 4)\ \beta\text{-D-Gicp}\ (1 \rightarrow \ldots$$

with the substitution at C3 of every alternate GIcp residue by the negatively charged trisaccharide

$$\beta\text{-D-Manp-}(1 \rightarrow 4)\ \beta\text{-D-GicpA-}(1 \rightarrow 2)\ \alpha\text{-D-Manp-6-O-acetyl}\ (1 \rightarrow 3)$$

with some of the terminal β-D-Manp residues substituted at positions C4 and C6 by pyruvate ($CH_3.CO.COO^-$). Thus xanthan (expect under highly acidic conditions) is a highly charged polyanion. Figure 10.1 shows the Haworth projection of a (pyruvylated) alternate repeat section of xanthan. The biosynthetic pathway of xanthan is depicted in Fig. 10.2.

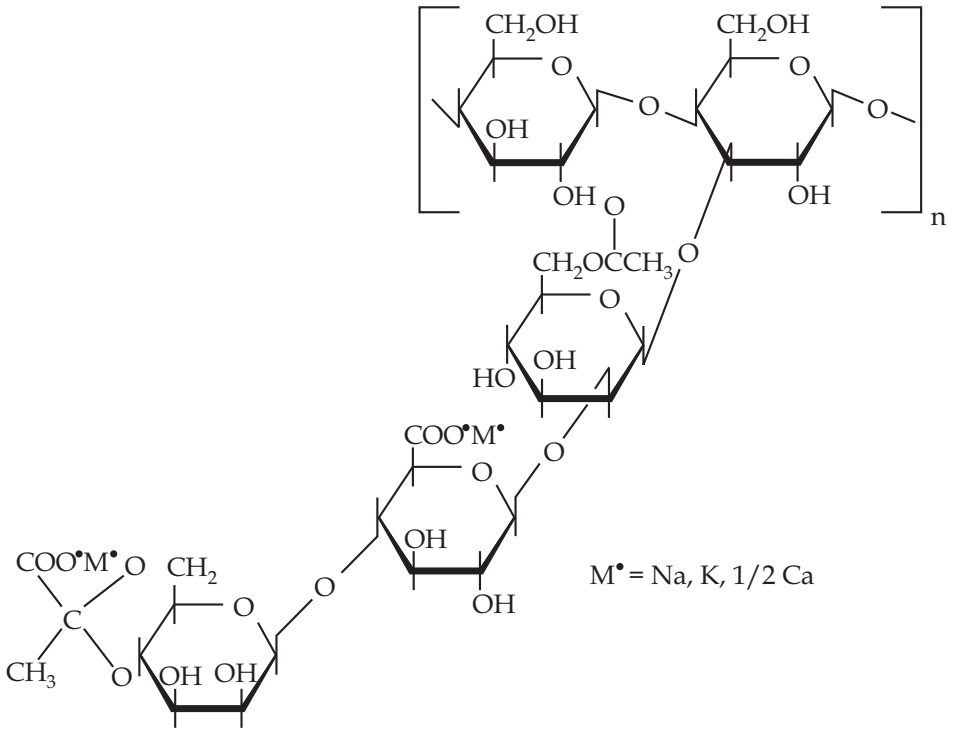


Fig. 10.1: Structure of Xanthan

(i) Extraction and production: The primary laboratory and commercial fermentation medium – as has been known for over fifty years for *X. campestris* growth and xanthan production is a phosphate-buffered (pH ≃ 7) broth containing D-glucose (30 g l^{-1}) (or sucrose, starch, hydrolysed starch), NH_4CI, $MgSO_4$, trace salts and 5 g l^{-1}caesin (or soybean) hydrolysate and the fermentation process takes place aerobically at a

temperature of ≃ 28°C. Xanthan production is further stimulated by the presence of pyruvic, succinic or other organic acids. The xanthan produced in this way is very similar to the xanthan produced naturally by the microbes living on a cabbage. In the commercial process the oxygen uptake from the broth is controlled to a rate of 1 mmol l^{-1} min^{-1}. Treated in this way the bacterium becomes an extremely efficient enzyme mini-factory converting >70 per cent of the substrate (D-glucose or related substrates) to polymeric xanthan. The bacterium having done its work is then removed in a rather undignified way by centrifugation and the xanthan precipitated with methanol or 2-propanol at 50 per cent weight concentration. The xanthan slurry is then dried and milled for use. The flow chart for production of xanthan is depicted in Fig. 10.2. The original commercial producer of xanthan was Kelco Ltd. (now MSD-Kelco) and together with other suppliers the annual world-wide production is now over 10000 tonnes.

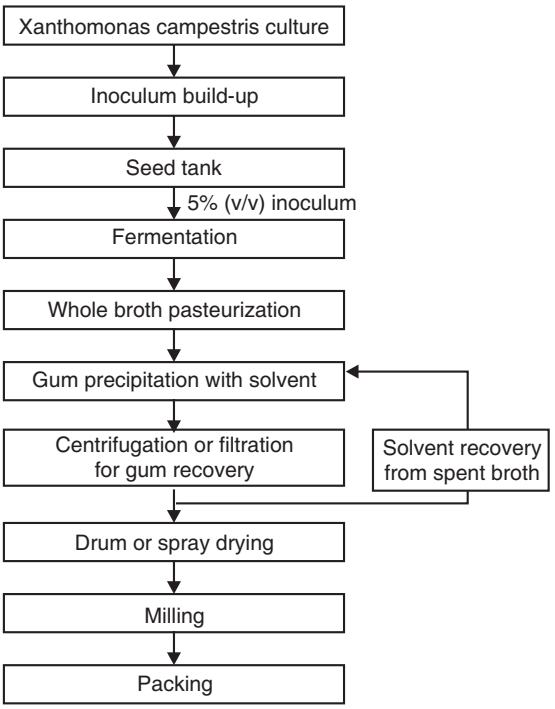


Fig. 10.2: Flow sheet of production and extraction of Xanthan

The biosynthetic process by the bacterium worked out by Sutherland (1989) and later confirmed by others, follows the same basic pattern proposed for other microbial polysaccharides. (Fig. 10.3).

1. Substrate uptake
2. Substrate metabolism
3. Polymerization
4. Modification and extrusion

It involves lipid carriers Fig. 10.3 although there is still uncertainty over their precise role and how the whole process is controlled.

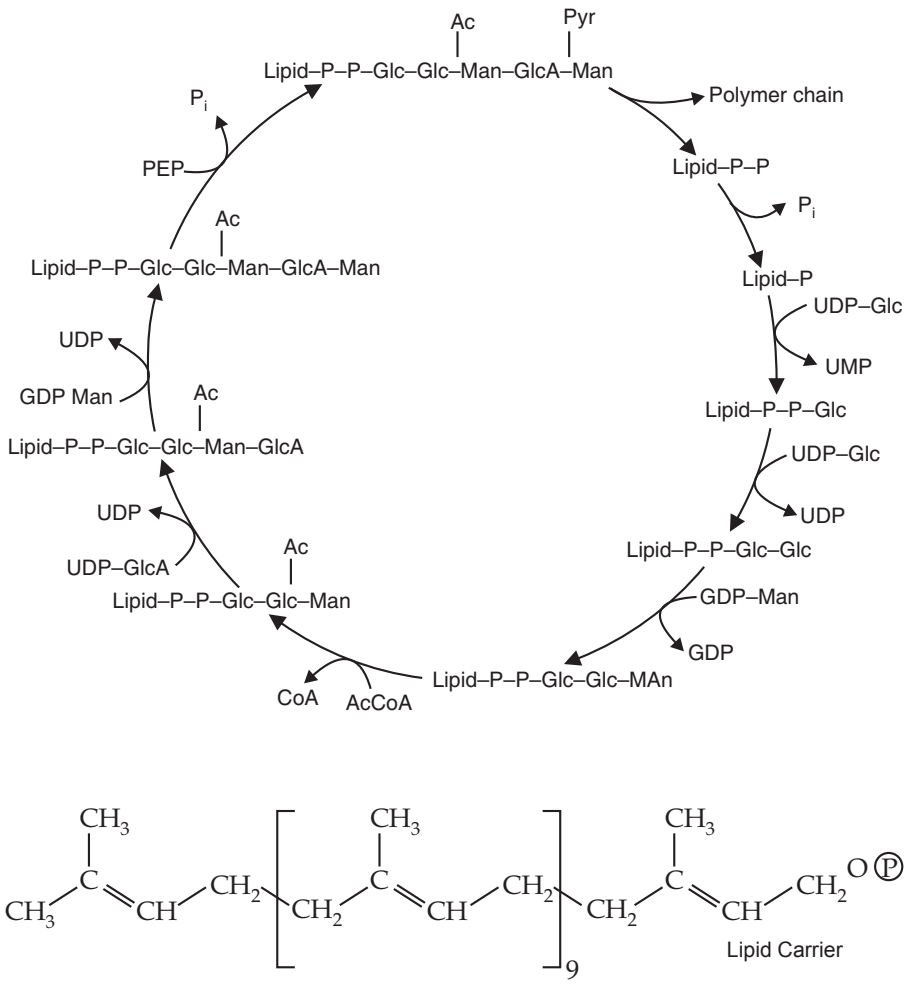


Fig. 10.3: Biosynthesis of Xanthan

(ii) Properties: Table 10.2 summarizes the fundamental properties of a popular commercial xanthan. Xanthan is one of the largest of the aqueous soluble polysaccharides. Solutions are correspondingly extremely viscous. The intrinsic viscosity is one of the highest known for a polysaccharide and the dilution solution concentration limit, is very low. The very high viscosity at low concentrations makes it ideal as a thickening and suspending agent. By itself xanthan, however, only forms transient weak gels since the junction zones are weaker than in those used for networking in carrageenan and agarose.

Table 10.2: Fundamental properties of a commercial xanthan

Property	
Polymer type	Polyanionic; complex repeat
Conformation type	Zone A/B (Extra-rigid/rigid rod)
Persistence length, L_p	1500 Å
Solubility	Cold or hot aqueous systems
Molecular weight	(5.9 ±1.1) x10^6 Da
Intrinsic viscosity [η]	(7500 ± 2700) ml g^{-1}
Sedimentation coefficient, $S^o_{20,w}$	(13.0 ± 0.3) S
Wales-van Holde ratio, k_s/[η]	0.3 ± 0.1
Mark-Houwink a coefficient	1.2
Gelation properties	Transient weak gels
Key sites for biomodification	C3 trisaccharide side chains; helical backbone-synegistic interactions with galactomannans

(iii) Modification: The key sites for modification are the first and terminal mannose residues on the trisaccharide side chains and the helical backbone by forming non-covalent interactions with galactomannans. These interactions appear to be also affected by the substitutions in the side chains. As to the trisaccharide side chains themselves, there are two approaches for alteration or control: one is changing the physiological conditions of fermentation; the other is the use of different pathovars or strains of *campestris*. The pathovar p. *phaseoli* and *oryzae* yield virtually acetyl-free or pyruvate-free xanthan respectively. The other approach (see e.g. Vanderslice et al., 1989) has been to look at the genetics of the enzymes, controlling the biosynthetic pathway, attempt to produce and isolate in sufficient quantity.

Genetic mutants deficient or defective in one or more of these enzymes to give 'polytetrame' (i.e. lacking in the terminal mannosyl or pyruvalated mannose group) and 'polytrimer' (lacking in addition the adjacent glucuronic acid residue). Xanthan reacts with carrageenan and agarose synergistically and give stronger gels. Deacetylating the xanthan side chains seems to enhance these synergistic interactions.

(iv) Uses of xanthan: Xanthan which has food grade approved by the USA Food and Drug Administration, makes it not only attractive as a food product but also for use in packaging material in contact with food, pharmaceutical and biomedical applications that involve ingestion. Its uses chiefly derive from its solubility in hot and cold water, its very high thickening, suspending potential which in turn derives from the very high viscosity of its suspensions. Despite the high viscosity, xanthan suspensions exhibit high shear thinning which means they also flow easily.

Applications

(a) Food industry: Besides high viscosity, thickening and suspending ability, xanthan suspensions have high acid stability. This makes them highly popular in sauces, syrup, toppings and salad dressing. In drinks, the addition of xanthan together with carboxymethylcellulose adds 'body' to the liquid and assists with uniform distribution of fruit pulp etc. It is also to add body to dairy product. The high freeze – thaw stability of xanthan suspensions makes them particularly attractive for the frozen food industry. The high suspending and stability properties are also taken advantage of the animal feed industry for transporting liquid feeds with added vitamins and other supplements

that would otherwise sediment out with transport or storage time. Specific uses of xanthan gum in food industry is given in Table 10.3.

Table 10.3: Applications of Xanthan in food industry

Function	Application
Adhesive	Icing and glazes
Binding agent	Pet foods
Coating	Confectionery
Emulsifying agent	Salad dressings
Encapsulation	Powdered flavours
Film formation	Protective coating, sausage casings
Foam stabilizer	Beer
Stabilizer	Ice cream, salad dressings
Swelling agent	Processed meat products
Syneresis inhibitor	Cheeses, frozen foods
Thickening agent	Jams, sauces, syrups and pie fillings
Synergistic xanthan-galactomannans mixtures	
Gel formation and stabilization	Ice cream Cheese and cream cheese Dessert gels Milk shakes Pudding and pie fillings
Viscosity control	Ice cream Instant soups Chocolate shake Milk shakes

(b) Pharmaceutical and cosmetic uses: Microencapsulation Xanthan has recently been added to the list of hydrophilic matrix carriers, along with chitosan, cellulose ethers, modified starches and scleroglucan. Tablets containing 5 per cent xanthan gum were shown under low shear conditions to give successful controlled release of acetaminophen into stomach fluid, and tablets containing 20 per cent xanthan successfully carried a high loading of the drug theophyline.

The high suspension stability is made use of in pharmaceutical cream formulations and in barium sulphate preparations. This high cream stability takes advantage of cosmetic industry including toothpaste technology, where the toothphaste will hold its ingredients (high viscosity) and then easily brush onto, off the teeth (high shear thinning). Uniform pigment dispersal, along with other ingredients and long-term stability, make xanthan a good base for shampoos.

(c) Oil industry: Xanthans are used in a number of aspects of well-bore technology such as (i) a proppant into fissures for rock fracturation, (ii) the suspension and removal of debris from the bore area even in the presence of harsh environment (sea water included), pipelines, (iii) enhancing oil recovery using the technique of polymer 'flooding'.

(d) Wallpaper adhesives: The high stability of suspensions makes xanthan an ideal base for suspending adhesive agents for wallpapers.

(e) Textile industry: Like alginates the suspension stabilizing property of xanthan makes them ideal for producing sharp prints from dyes with a minimum risk of running, for application to textiles and carpets.

10.2 PULLULAN

Pullulan is a glucan made up from maltotrise units linked by 1,6 bonds (→6)-α-D-Glc-(1→4)-α-D-Glc-(1→4)-αD-Glc-(1→). It is obtained from *Aureobasidium pullulans* and is hydrolyzed by pullulanase to yield maltotiose. It is not attacked by digestive enzymes of the human gut, and is used to form films. Production is now substantial, and has found particular application in formulating snack foods in Japan based on cod roe, powdered cheese and as a packaging film for ham.

It is water soluble, with molecular weights in the range of 5000 Da to 900000 Da, with straight unbranched chains, behaves as a random coil according to a combination of sedimentation coefficients and intrinsic viscosity measurements. Similarly, light scattering and sedimentation equilibrium measurements show that pullulans above about 50000 Da molecular weight behave to a close approximation like a classical random coil (Kawahara et al., 1984). It has been proposed as a 'standard polysaccharide' in the sense that it is so near to random coil behaviour and readily obtainable in very reproducible form that it could be used for comparative tests with other polysaccharides. The idea is presumably that deviation from pullulan-type behaviour must imply a more complex structure. It might also be used to standardize instruments. The solution in water are liable to bacterial attack with rapid degradation of the molecular weight and preservatives such as azide may be needed.

Pullulan gel is heat stable and remain unaffected by pH and compatable with most electrolytes. Barium and titanium increases viscosity dramatically. Its viscosity is not changed in presence of NaCl and 100°C. It is highly biodegradable and soluble in water. Properties of pullulan can be changed by esterification, etherfication or cross linking. It is first commercially produced in 1976 by Hayastihara chemical co. It is produced by *A. pullulans* in 15% acid hydrolysed starch, 0.2% K_2HPO_4, 0.2% NaCl, 0.2% Peptone, 0.04% $FeSO_4$, pH 6.5, incubation tempearture 30°C aerobic, incubation period 100-125 hrs. It is diluted with water and precipitated with isopropane.

Applications: It is extensively used in food, pharmaceutical and industrial sector. It is considered to be safe and non-toxic product. It is a good substitute for low caloric additive to wheat flour or starch in bakery products and dietary foods. It is used as a thickener in beverage, creams, frosting, fillings, sauces, dried vegetables and meats. Package with this polysaccharide helps to retain freshness and prevent oxidation over long periods. It is used in tablet coatings, contact lenses, pharma expanders. Pullulan coatings are for lithographic printing and plate protection. It acts as a binder for the preparation of tobacco sheets. Acts as a boundary sand binder.

10.3 DEXTRAN

Bacterial dextrans are produced in substantial quantities by *Leuconostoc mesenteroides* and are familiar to laboratory workers as the basis for cross-linked dextran beads used in gel filtration columns. They are mostly α(1 → 6)D-glucopyranosyl polymers with molecular weights up to $\simeq 1 \times 10^6$ Da more or less branched through (1 → 2), (1 → 3) or (1 → 4) links (Fig. 10.4).

→1) – α-D-Glc-(1→

→2) – α-D-Glc-(1→

→3) – α-D-Glc-(1→

→4) – α-D-Glc-(1→

Fig. 10.4: Structure of Dextran the predominant linkage is α-(1-6)

In most cases the length of the side chains is short, and branched residue vary between 5 and 33 per cent. The major commercial dextran is about 95 per cent (1 → 6) linked, 5 per cent (1 → 3) linked, and is made by selected strains of *Leuconostoc*. After ethanol precipitation from the culture medium acid hydrolysis is used to reduce the overall molecular weight, though fungal dextranases can also be used. Sequential degradation of dextran established that 40% of side chain is one unit long, 45% are two units long, 15% contain more than two units and overall 5% degree of branching molecular weight of native dextran ranges from 2×10^6 to 5×10^6 daltons. The product with an average molecular weight of about 60000 Da is used in medicine as a blood extender, while fractions of defined molecular weights (e.g. the Pharmacia 'T'-series, where 'T500 Dextran' would stand for dextran of weight average molecular weight 500 kDa) are familiar in laboratories and to some extent, like the pullulan 'P-series, serve as polysaccharide standards in molecular weight calibrations. They are also used as part of incompatible phase separation systems, usually with polyethylene glycol. 'Blue dextran' is a well-known marker for the void volume for gel filtration studies.

Dextran is produced in a medium containing 2% sucrose which induces dextran sucrose, vitamin cofactors, amino acid supplements as yeast extract, corn steep liquour, acid hydrolysed casein, malt extract, peptone, tryptone broth, phosphate in 0.1 – 0.5% and pH 6.7 – 7.2, incubation temperature is 25°C. Dextrans have been found very wide applications in laboratory work because they are particularly free from positive interactions with proteins. The interaction can be almost entirely characterized as coexclusion. This has been found application, as already noted, in gel filtration media, but cross-linked dextran gels show other effects. For example, they swell and shrink in a way related to the osmotic pressure of the solvent system and can be used to make miniature osmometers. Dextran 70 is used for treatment of shock reading risks thrombiasis, post-operative pulmonary emboli. Dextran 40 is used in reduction of blood viscosity and erythrocyte agglutination.

10.4 CYCLODEXTRINS

Cyclodextrins have attracted more interest than any other bacterial polysaccharide because of their unique structures. The interest has been more scientific than industrial, since xanthans are much more important in commercial terms, but there always seems to be a potential application of the cyclodextrins on the horizon which would lead to a major expansion in demand for them. Formerly called Schardinger dextrins after their discoverer's still called cyclomyloses name cyclomaltoses here and there, they are cyclic molecules formed by glucopyranose linked by α-D-(1 → 4) linked bonds. They contain six, seven, eight or nine glucopyranose residue. It is impossible to form cyclic structures with less than six residues, while those with more than nine

have so far been found to have at most a nine-membered ring with other glucose residues attached as branches through (1 → 6) links. The structures (for a seven-membered ring, $cGIc_7$ or in common notation $cGIu_7$) are illustrated in Fig. 10.5.

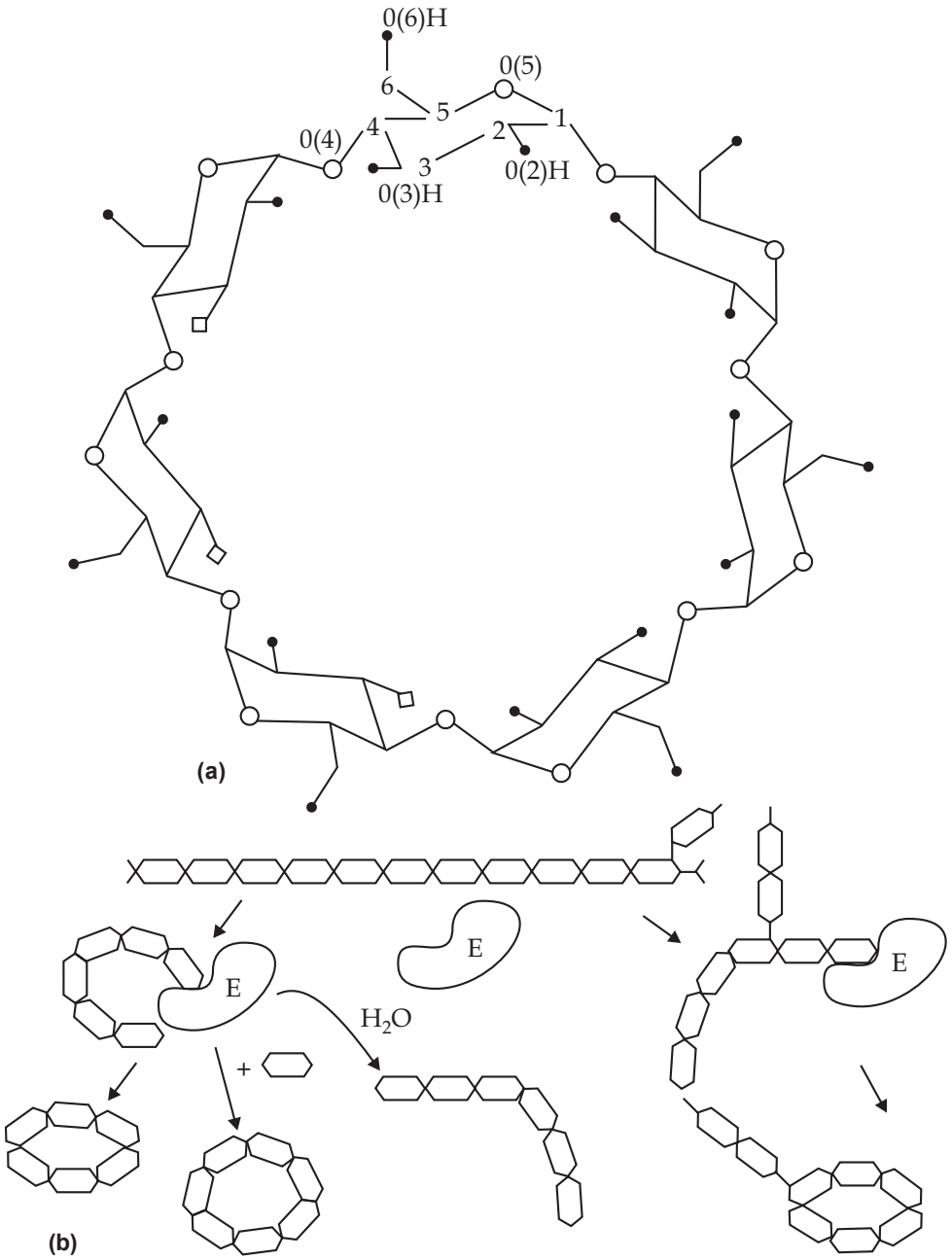


Fig. 10.5: (a) Diagramatic structure of Cyclodextrin with seven residue ring
(b) Mode of action of Cyclodextrin synthases

The nine-membered ring is uncommon and found only as a minor by-product of enzymatic synthesis of the other. The six, seven and eight-membered rings are sometimes called as α-, β-, and γ-cyclodextrins, abbreviated as $cGIu_6$, $cGIu_7$, $cGIu_8$.

They are water soluble, easily crystallized and their structures have been determined. The great interest in them is because the cavity is of a size to include many small molecules of interest and is also fairly hydrophobic in character. The outside of the toroidal cone is full of hydrophilic hydroxyl groups. As might be expected, most of the potential applications involve inserting small molecules into the cavity. The dimensions are roughly those of a cylinder with length of 7 Å, diameter of 4.5 Å for the six-membered ring and 8.5 Å for the eight-residue ring. Thus they can accommodate an aromatic ring like benzene, as well as alkyl chains.

Applications: 1 cyclodextrins are competitive insulators of α-and β-amylases, potato phosphoxylase, pullulanase are used as low calorie.

10.5 GELLAN

Gellan is an extracellular polysaccharide produced by *Pseudomonas elodea*. Gellan consist of a linear tetrasaccharide sequence containing two D-glucopyranosyl units, one D-glucano pyranosyl unit and one Rhamnopyranosyl unit plus the acyl groups, acetyl, L-glyceryl units attached to 0-6,0-2 respectively of one of the D-glucopyranosyl units (Fig. 10.6).

$$\rightarrow 3)\text{-}\beta\text{-D Glc-}(1\rightarrow 4)\text{-}\beta\text{-D-GlcA-}(1\rightarrow 4)\text{-}\beta\text{-D Glc-}(1\rightarrow 4)\text{-}\alpha\text{-L-Rhac-}(1\rightarrow$$

Fig. 10.6: Structure of gellan

This glucose carries 0-acetyl and glyceryl residues in the native polymer. The molecular weight of gellan is estimated to be 1 or 2 x 10^6 daltons. The polysaccharide is characterized as a parallel, half staggered double helix in which each polymer chain is in a left-handed, 3-fold helical confirmation and in which two such duplexes are packed antiparallel to each other in the unit cell. It is sensitive to calcium levels but has rhelogical properties similar to those of xanthan which has a similar charge density. It was clearly intended to be a xanthan competitor though before permission for food use was obtained it was promoted as an agar substitute particularly for use in growth media. It is produced in submerged aerobic fermentation. The medium consists of D-glucose, NH_4NO_3, soy protein hydrolysate, KH_2PO_4 and trace element solution. The viscosity of medium reaches 5000-6000 centipolses. Gellan is recovered from broth by isopropanol precipitation which is dried, milled and packed.

(i) Properties: Gellan by deacylation produces firm, brittle gels. It shows thermo reversible viscosity changes. Cooling with ion free water can be heated and cooled without gelation. It has a good thermal stability and can be autoclaved without loss of viscosity. It requires either monovalent or divalent cations for gel formation. Texture and strength of gel can be changed with the anion and its concentration.

(ii) Applications: It can serve as a substitute for agar in preparation of solid media for microorganisms. It is used in plant tissue culture experiments. Its requirement is $1/4^{th}$ to that of agar. It is used in antigen-antibody diffusion technique and as a matrix for enzyme and cell immobilization.

10.6 WELAN

Welan commercially known as Biozan. It is similar to gellan having single side chain unit. It has high thermostability (at 135°C/hr). It is produced by *Acaligenes sp*. It is produced under submerged aerobic fermentation. The medium composed of glucose, soyprotein, hydrolysate, phosphate buffer, NH_4NO_3 and trace elements is used for production of welan. Extracellular polysaccharide is recovered by precipitation with isopropyl alcohol.

Applications: It is used in drilling fluid additive. It is used in oil recovery.

10.7 CURDLAN

Curdlan, α β (1-3) linked D-glucan discovered in 1996, is a thermogelable polysaccharide (1→3) -β-D Glc-(1→. This glucose carries residues in the native polymer. It is produced by *Alcaligenes faecalis v. myxogenes* in a medium containing 4% glucose, 0.1% citrate, succinate or fumerate, 0.55% (NH_4) HPO_4 and mineral salts. It is insoluble in acidic and neutral water but soluble in alkaline water. It is soluble in formic acid, dimethyl sulfoxide, saturated solution of urea or thiourea or 2% potassium. At 55°C it becomes clear solution. Curdlan gels are stable over wide pH range and stable to freeze thaw. It forms a firm resistance high set gels that melts at 140 + 160°C. Its annual production exceeds 200-300 tons. It is nutritionally inert and non-toxic. Mutants of *Agrobacterium* which produces curdlan in pure form and in high yields, is being used in Japan for commercial production.

Applications: It helps to improve texture of different foods such as tofu, yokan, boiled fish paste, noodles, sausage, jellies and jams. It masks odors, prevents scording aroma and adhesion. Curdlan is a good binder in animal feed, carrier for immobilized enzymes and substitute soil for rice plants.

10.8 POLYHYDROXYBUTYRATE

Polyhydroxy alkanoates (PHAs) have considerable potential as biogradable alternatives to petroleum derived plastic PHAs are linear homochiral thermoplastic polysters produced as intra cellular energy reserves by numerous microorganisms. The most widely encountered PHAs are poly β hydroxybutyrate (PHB) and polylactic acid (poly hydroxy propionate) formed from the monomers of hydroxybutytic acid and lactic acid respectively.

Bacteria differ in their food reserve with respect to genotype of the organism and nutrient limitation if the carbon to nitrogen ratio is high and if nitrogen or oxygen is limiting. Many bacteria accumulate glycogen/aliphatic polyster (poly 3–hydroxyalkanoates) in amounts ranging from 30-80% of their cellular dry weight. Polyhydorxy alkalnoates produced from naturally occurring organic substrates have the general structure (Fig. 10.7).

General formula is to be given:

Where α = 0–8 or higher. The asterisk indicates an asymmetric centre. α-cmoigne (1926) for the first time recorded poly-(RR) (hydroxybutryate) (PHB) in *Bacilus megaterium***.** It was only after many years it came to know this thermoplastic material which melts when heated to certain temperature but harden again as they cool. This cycle can be repeated many times. This in contrast to familiar thermoplastic material such as polyethylene, polypropylene, polystyrene, polycarbonate and neston which are non-biodegradable are readily biodegradable. Some of the differences between two materials are listed table 10.4.

Table 10.4: Comparison of plastics and polyhydroxy butryate properties

Property	Polypropylene	Poly (3 hyrdoxybutryate)
Mol. Weight	2.27×10^5	$1\text{-}8 \times 10^5$
Melting point	171 – 186	171 – 182
Crystallinity (%)	65 – 70	65 – 80
Density (g.cm-3)	0.905 – 0.94	1.23 – 1.25
Flexural modulus (Gpa)	1.7	3.5 – 4.0
Tensile strength (Mpa)	39	40
Extension at break (%)	400	6-8
Resistance to UV radiation	Poor	Good
Biodegradability	Very poor	Good

The difference in density of two materials makes polypropylene to float on water while poly (3-hydroxybutryate) sink to the bottom which facilitates to degradation. Further biodegradable polyhydroxy butyrate has many advantages in the manufacture of packing containers, bottles, wrapping films, bags and the like. They have important medical uses such as sensing as surgical pans, staples wound dressings, bone replacements, plates and biodegradable carriers for the long-term release of medicine derivates of polylactic acid which have been used in medicine as template for tissue growth and in plastics for replacement fornts. It is also used in food storage materials and shampoo bottles.

Polyhydroxy alkanoates (PHAs) have considerable potential as biodegradable alternatives to petroleum derived non-biodegrade plastics such as poly propylene, ethyleneglycol. These are linear homocheral thermoplastic polysters produced as intra cellular energy reserves by numerous microorganims. These biopolymers accumulate as distinct 0.2–0.7 μm diameter granular inclusion bodies in response to nutrient limitation especially in *pseudomonads* (Fig. 10.7).

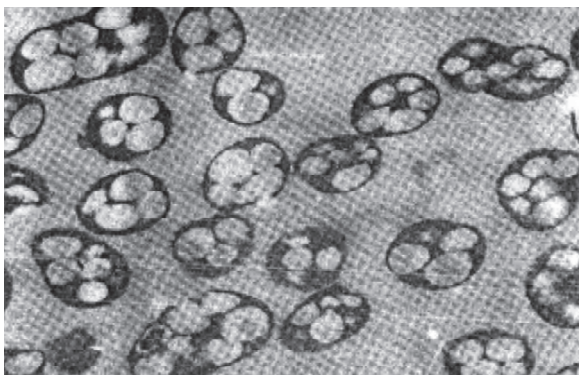


Fig. 10.7: PHA granules in cells of *Ralstonia eutropha*

The most widely encountered PHAs are polyhydroxy butyrate (PHB) and polylactic acid (polyhydroxy propionate) formed from the manomers hydroxybutyric acid and lactic acid respectively. The manomenic units are fined by enter hinkeen.

The biosynthetic pathway of PHB Arabidopsis thaliana is depicted in Fig. 10.8.

Acetyl CoA→pathways leading to lipids and flavanoids.
↓3-ketothialase
Aceto acetyl CoA→Mevalonate→pathways leading to phytohormones, carotenoids, sterols
↓
Acetoacetyl CoA and quinines
↓
3-Hydroxybutryl CoA
↓
Polyhydroxy butyrate synthase
↓
Poly (3-hydroxy butyrate)

Fig. 10.8: Biosynthesis of poly 3-hydroxybutyrate in transgenic *Arabidopsis thaliana* expressing *Acaligenes eutrophus* genes

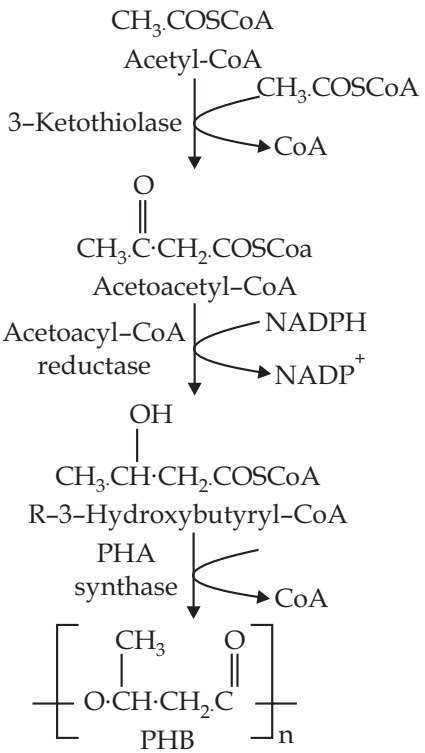


Fig. 10.9: Biosynthesis of PHB from glucose in *Ralstonia eutropha*

First is the biosynthesis of the hydroxyacyl CoA manomers followed by their head to tail polymerization to form the polymer chain which exceeds 10,000 units in length. The most fully characterized pathway is that of PHB biosynthesis in *Ralstonia eutropha (= Acaligenes eutropus)* Fig. 10.9. This involves three enzymes thiase catalyses a claisen condensation of two molecules of acetyl CoA to form acetoacetyl CoA which is reduced to the chiral intermediate R-3-hydroxybutyl CoA by a reductase. Polymerization is then performed by a PHA synthetase. PHAs are synthesized at stationary phase where nutrient exhaustion is a common feature and the granules get accumulated.

In view of its high cost, efforts made to cheaper means of production are being sought. PHAs are now synthesized by recombinant microorganisms i.e. *E. coli* which contain the genes encoding the enzymes necessary for PHA biosynthesis. Such microbial recombinants may become an economically attractive source of PHAs. Alternatively transformation of a hygen plant in these genes provide an even cheaper means of PHA production in the longer term likely transgenic hosts which include *Arabideopsis thaliana* and *Brasssica napus* or *Zea mays.*

REVIEW QUESTIONS

I. Essay type questions:

1. List out some of the microbial polysaccharides along with their applications.
2. General mechanism of microbial polysaccharide synthesis.
3. Describe industrial production of xanthan. Add a note on its applications.
4. Describe the process of accumulations of 3-polyhydroxy butyrate.

II. Write short notes on:

1. Pullulan 2. Curdlan 3. Dextrans 4. Gellan 5. Lipid Carrier
6. Welan 7. Cyclodextrins

FURTHER READING

1. Carlson, S.E (1995). The role of PUFA in infant nutrition. Inform 6, 940–946.
2. Doi, Y. (1990). Microbial Polysters. VCH, Weinheim.
3. Gill, I. & Valivety, R. (1997). Polyunsaturated fatty acids, part 1: Occurrence, biological activities and applications. Trends Biotechnol, 15, 401–409.
4. Madison, L.L. and Huisman, G.W. (1999). Metabolic engineering of poly(3-hydroxyalkanoates): From DNA to plastic. Microbial, Mol.Biol, Rev. 63. 21–53.
5. Ratledge, C. (1997). Microbial lipids. In Biotechnology, Vol. 7 (Rehm, H-J and Reed, G. eds.), pp. 135–197. VCH, Weinheim.
6. Stryer, L. (1995). Fatty acid metabolism. In Biochemistry, 4th Edition, pp. 603–628. W.H. Freeman & Co, New York.
7. Sutherland, I.W. (1990), Biotechnology of Microbial Exopolysaccharides. Cambridge University Press, Cambridge.
8. Sutherland, I.W. (1998), Novel and established applications of microbial polysaccharides. Trends Biotechnol 16, 41–46.
9. Tombs, M. and Hardings, S.E. (1998). An Introduction to Polysaccharide Biotechnology. Taylor & Francis, London.

11

Hybridoma Technology

Hybridoma technology is one of the most important techniques of antigen and antibody reactions resulting in the production of monoclonal antibodies which have wide range of applications in research as well as medical science, especially for diagnostic analysis and treating certain cancer diseases.

In this process β-lymphocytes of spleen are made to fuse with myelomal (tumor) cells to form hybrid cells that are capable of growing in a culture and retain the ability to produce antibodies. This technique of β-cell-myeloma cells fusion is known as the Hybridoma technology. As β-lymphocytes play a very important role in the hybridoma technology, their isolation and subsequent growth in a culture medium is a critical stage in this technique.

11.1 β-LYMPHOCYTES

Leukocytes that are able to recognize and react with a specific antigen are called as lymphocytes. Lymphocytes are one of the most prevalent mammalian cell types. There are two categories of lymphocytes, β-lymphocytes and T-lymphocytes, which are involved in antigen-specific immune responses and are derived from stem cells present in the bone marrow. The differentiation of stem cells into mature lymphocytes is determined by the organ in which precursor lymphocytes become established. As β-lymphocytes mature in a special organ called the Bursa of fibricius fetal liver and adult human bone marrow, as they are isolated for the first time from bursa of fibricius they are called β-lymphocytes. So also T-lymphocytes are designated with that name because they mature in an organ called as thymus (Fig. 11.1).

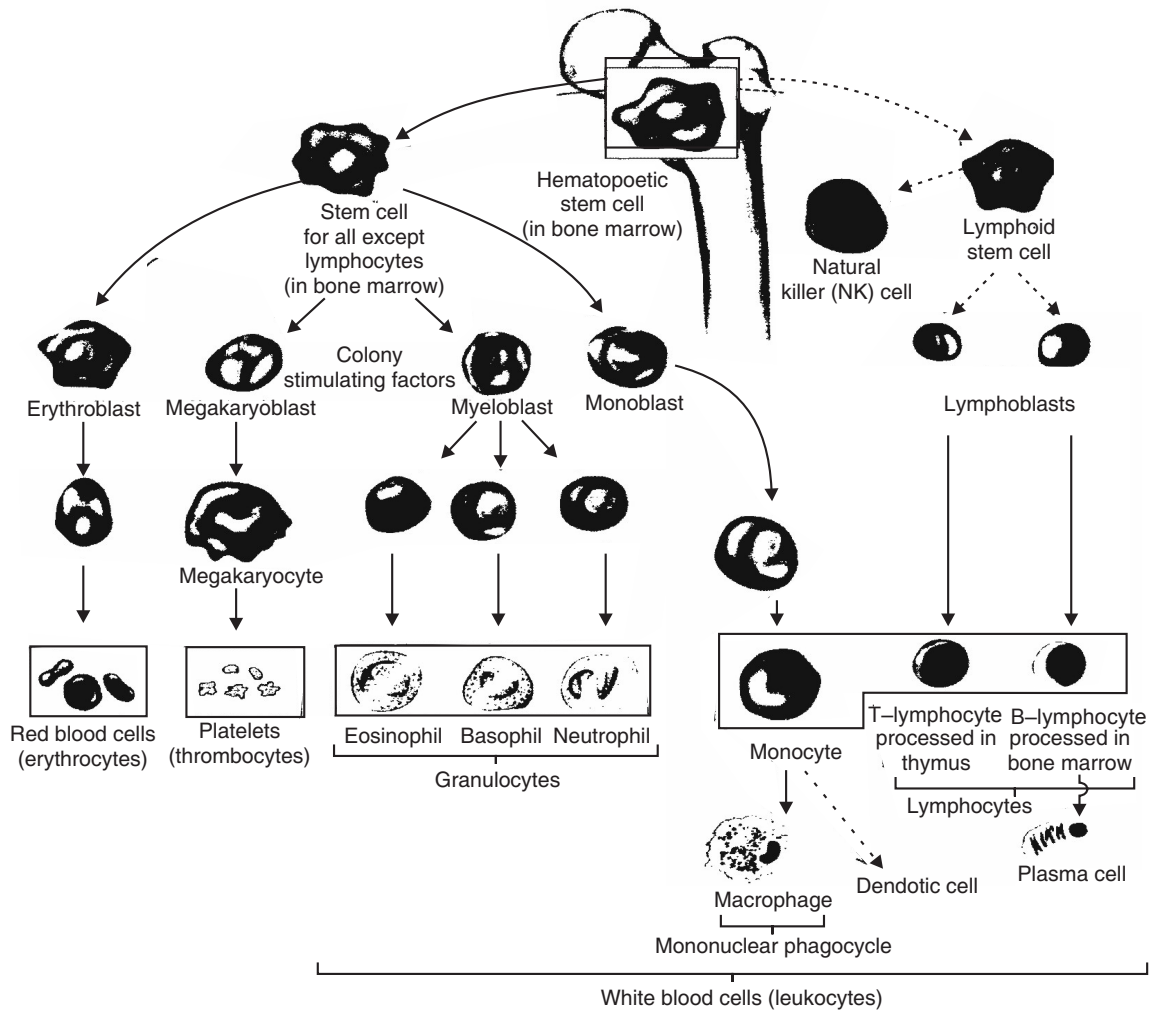


Fig. 11.1: The different types of human blood cells, Pluripotent stem cells in the bone marrow divide to form two lineages

As β-lymphocytes play an important role in hybridoma technology, some of their important characters are given here. They are responsible for antigen interaction, antibody production and immune memory. Antibody molecules are present on their surface. This surface antibody recognize antigen, β-lymphocytes are concentrated in the cortex of lymph nodes (Fig. 11.2) where they can contact antigens. After antigen exposure, β-cells divide into memory cells or plasma cells. The memory cells are long lived and may remain in the cortical area for several years. On restimulation with antigen memory cells quickly proliferate to produce many more memory cells and plasma cells. The differentiated antibody producing plasma cells live for several days. They are found in the medulla where antibodies can directly be drawn into an efferent lymph vessel (Fig. 11.3).

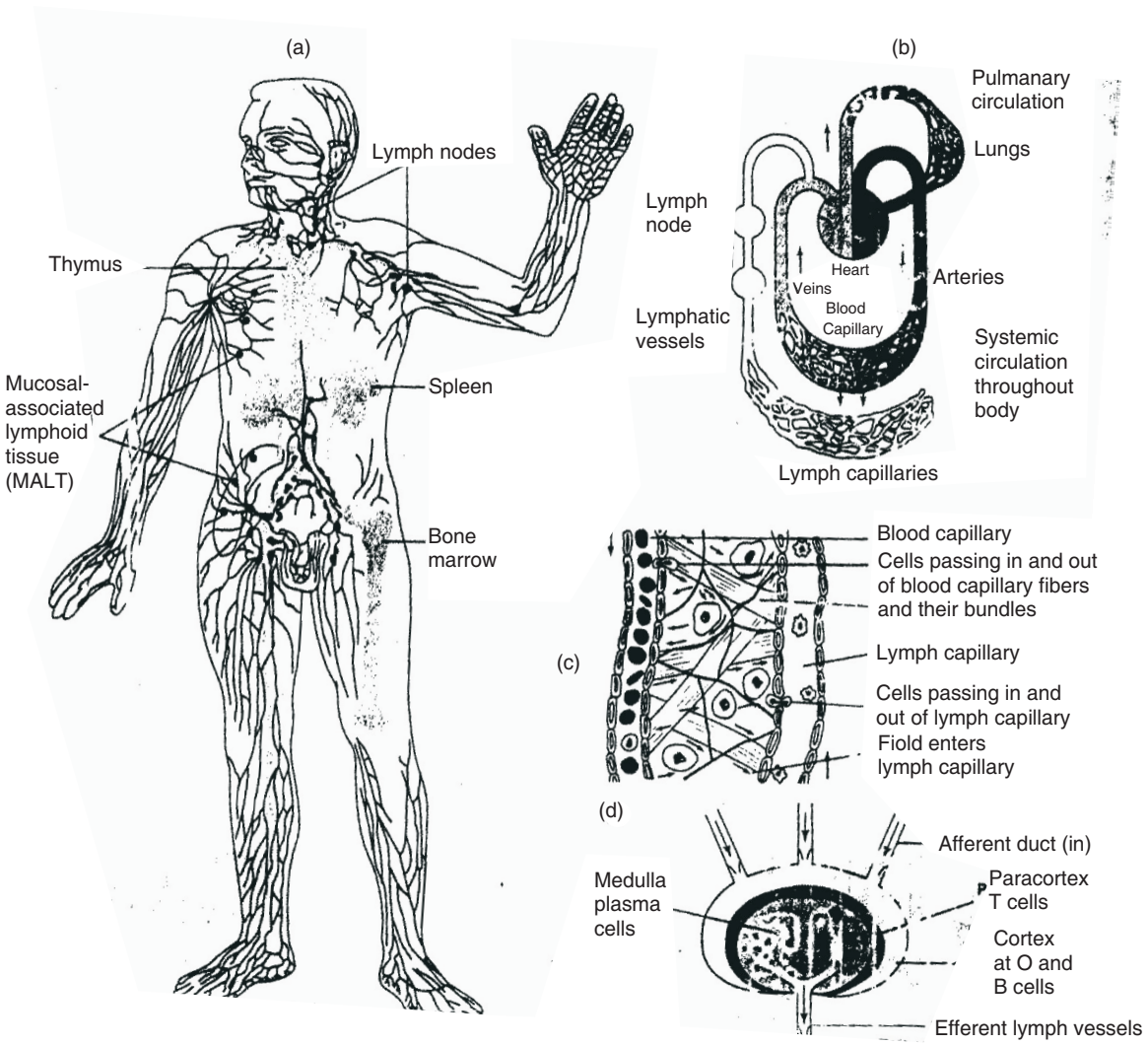


Fig. 11.2: The blood and lymph systems. (a) Overall view of the major lymph systems showing the locations and major organs. (b) Diagrammatic relationship between the lymph (c) Connection between the blood and lymph systems is shown microscopically. Both blood and lymph capillaries are closed vessels, but cells and fluids can pass from one vessel to another by a process known as e: ravasation (d) A lymph node. The diagram depicts major anatomical areas and the immune cells

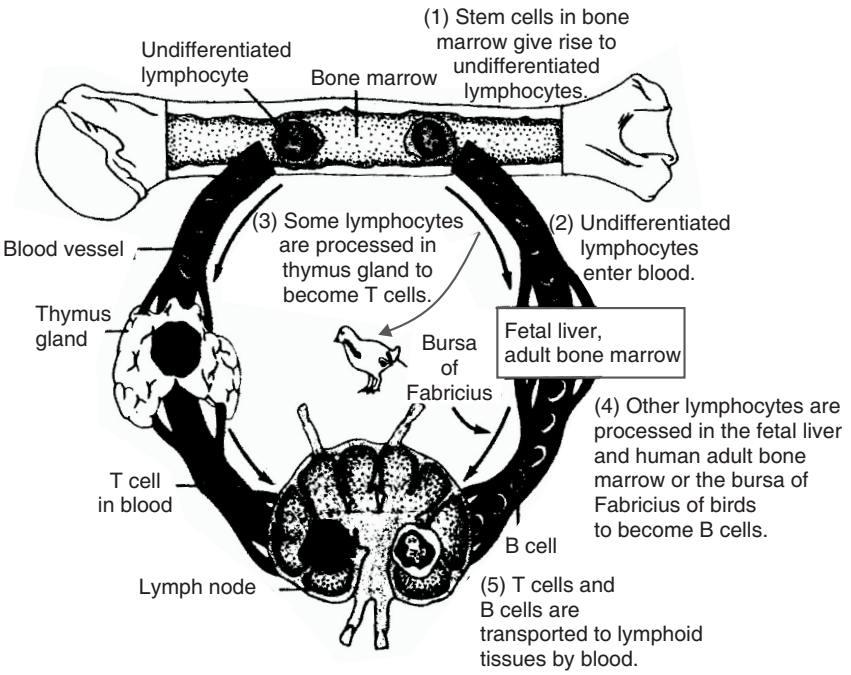


Fig. 11.3: Schematic representation of development of β-lymphocytes

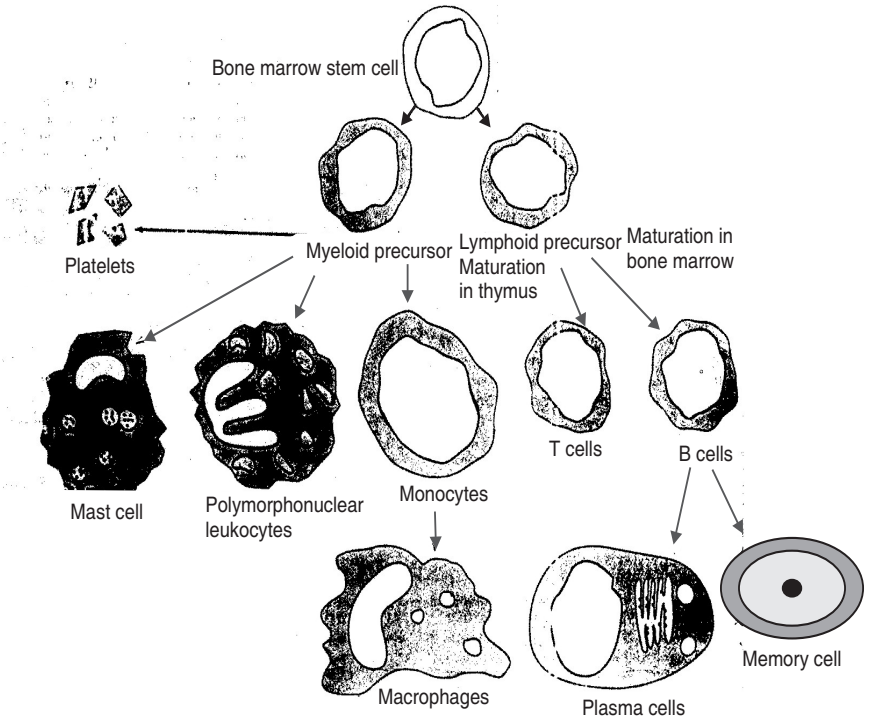


Fig. 11.4: Origin of major cells involved in the immuno response. Two major precursor lines are generating phagocytic cells (myeloid precursors) and the other generating lymphocytic cells (lymphoid precursors) participate directly in immune responses

11.2 MONOCLONAL ANTIBODIES

Antibodies are proteins synthesized in blood against specific antigen (pathogen/parasite) just to combat and give immunity in blood. Such antibodies are heterogenous and contain a mixture of antibodies. They are called polyclonal antibodies (Fig. 11.4). They are not specific in their action. If a specific lymphocyte after isolation and cultured *in vitro* becomes capable of producing a single type of antibodies which bear specificity against specific antigen, it is known as monoclonal antibody and is used in the diagnosis of disease. Antibodies of only a single specificity and derived from a single cell clone are called as monoclonal antibodies. Monoclonal antibodies possess the following characters.

1. It contains only a single antibody capable of recognizing only a single determinant.
2. It produces only a single class of antibodies.
3. It can make a specific antibody using an impure antigen.
4. It is highly reproducible.

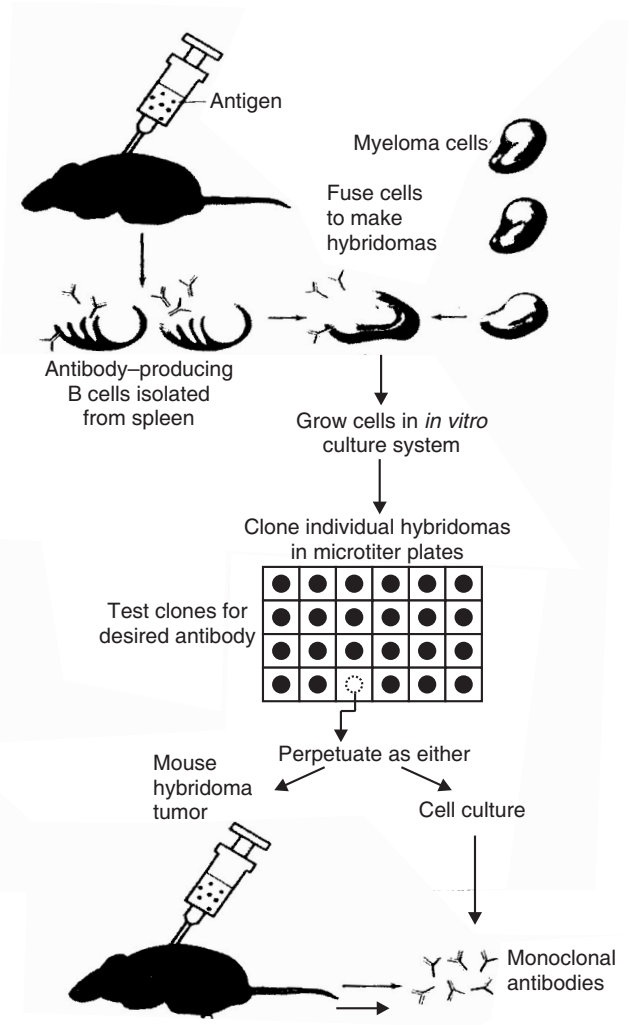


Fig. 11.5: The hybridoma technique and production of monoclonal antibodies

11.2.1 Production of monoclonal antibodies: First a mouse is immunized with the antigen of interest and a period of weeks time is allowed for specific B cell clones to proliferate and begin to produce antibodies by normal process. Thus large number of β-lymphocytes are produced and stored in the spleen. Then spleen tissue rich in β-lymphocytes is removed from the mouse and are fused with myeloma (tumor) cells, the mixture is allowed to grow in the HAT medium containing the compounds hypoxanthine, thymidine and aminopetrin. It has been observed that only a small fraction of total cell population (β-lymphocytes and myeloma cells) present in the mixture can fuse and form fused hybrids. The hybrid cells alone are able to grow and proliferate in HAT medium because they received the genetic information, due to fusion, to utilize the compounds hypoxanthine and thymidine present in the medium.

Whereas the unfused cells cannot grow due to toxic effect of aminopetrin present in the HAT medium and die off within a week or two. The hybrid cells are then identified, isolated, placed in the wells of microtiter plates, allowed to grow and produce antibodies of interest (Fig. 11.5).

***In vitro* propagation:** Production of hybridoma cells outside the mouse in tissue culture is called ***in vitro* propagation.** The method of its production is outlined below. From a single fusion several distinct clones are isolated, each making a monoclonal antibody to a different determinant of the antigen. Once the clones of interest are identified, they are injected into mice and be perpetuated as a mouse myeloma tumor. As a mouse tumor line hybridomas secrete large amount of monoclonal antibodies in mice over 10 mg of pure monoclonal antibody can be obtained per milliliter of mouse peritoneal fluid. Specific hybridomas of interest can also be preserved by allowing them to freeze. The hybridoma when required are carefully defreeze and injected back into mice or grown in tissue culture when more of a specific monoclonal antibody is needed.

11.2.2 Applications: Monoclonal antibodies are highly useful reagents for research and medical science.

(a) Applications in Research:

(i) Monoclonal antibodies are directed against specific cell markers. For example the CD9 and CD8 lymphocyte markers, can be used to identify and separate mixtures of T-cells from one another.

(ii) Monoclonal antibodies are employed to study active sites of enzyme activity when treated with monoclonal antibodies to specific determinants on the enzyme.

(iii) They are useful in the immunological typing of bacteria and in the identification of cells containing foreign surface antigens, for example, a virus infected cell.

(iv) They are also used in genetic engineering for identifying and measuring levels of gene products which are not detectable by other methods.

(b) Applications in Medicine:

They are preferentially used in clinical diagnostics and medical therapeutics.

(i) They are employed in the detection and treatment of human malignencies or cancer diseases. Malignant cells contain a variety of surface antigens, which are not present on the surface of normal cells. Major are tumor specific cell surface antigens. Monoclonal antibodies prepared against cancer related antigens are allowed to bind covalently to specific cancer cells diffusing toxins. Such monoclonal antibodies bounded with toxins are able to destroy and prevent the

proliferation of cancer cells. The specificity of the monoclonal antibody treatment can greatly improve cancer chemotherapy, avoid damage to normal cells by the toxic chemical and radiation treatments used in conventional cancer chemotherapy.

(ii) They are used in organ transplantation where they improve tissue-matching process, which is critical for the success of transplantation.

(iii) Monoclonal antibodies can also be used for deriving more accurate results in blood typing in the determination of rheumatoid factor, in the determination of damaged heart muscles in an heart attack.

(iv) They are also employed in certain conventional tests like pregnancy test.

REVIEW QUESTIONS

I. Essay type questions:

1. Describe the process of monoclonal antibodies production.
2. Give an account of importance of monoclonal antibodies.

II. Write short notes on:

1. Lymphocytes
2. Antigens
3. Antibodies
4. B-Cells
5. T-Cells

FURTHER READING

1. Birch, J.R. and E.S. Lennox (1995) Monoclonal antibodies, principles and applications, John Wiley, New York.
2. Capra, J.D. (ed) (1997) Antibody Engineering, Vol. 65, chemical immunology Karger, Basel.
3. Goldsby, R.A., T.J. Kndt and BA Osborne 2000. Kuby immunology 4th edition W.H. Freeman and Company, New York.
4. Harris, W.J. and J.R. (eds) (1997) Antibody Therapeutics CRC Press, New York.
5. King, D.J. (1998). Applications and Engineering of Monoclonal Antibodies Tayler and Francis, London.

12

Bioleaching

Leaching process was first observed in pumps and pipelines installed in mine pits containing acid water. This process was later on employed for recovering metals from ores containing low quantity of the metal. Presently certain metals from sulfide ores and other ores are extracted by employing only leaching method.

Extraction of metals from low-grade ores by employing microorganism is called as bio-leaching. Large quantities of low-grade ores are produced during the separation of higher-grade ores and are generally discarded in waste heaps. Metals from such ores cannot economically be processed with chemical methods. There are large quantities of such low-grade ores especially copper ores, which can be processed profitably by bio-leaching.

Copper and Uranium are presently produced commercially by employing bioleaching process. However, problems may creep in when the large scale bioleaching process of a waste dump is improperly managed because seepage of leach fluids containing low pH and metals into natural water supplies and ground water causing metal pollution.

12.1 MECHANISM OF BIO–LEACHING

The process of bioleaching is accomplished by two ways.

(i) Direct bioleaching (ii) Indirect bioleaching

(i) Direct bioleaching: *Thiobacillus ferrooxidans* is oftenly used in microbial leaching. It is an autotrophic, aerobic, gram (–) negative rod shaped bacterium. It synthesizes its carbon substances by CO_2 fixation. It derives the required energy for CO_2 fixation either from the oxidation of Fe^{2+} to Fe^{3+}or from the oxidation of elemental sulphur or reduced sulphur compounds to sulfates.

1. $4Fe + 6H_2SO_4 + 3O_2 \rightarrow 2Fe_2 (SO_4)_3 + 6H_2O$ (1)
2. $2S + 3O_2 + 2H_2O \rightarrow 2H_2SO_4$ (2)
3. $2FeS_2 + 7O_2 + 2H_2O \rightarrow 2FeSO_4 + 2H_2SO_4$ (3)

Thiobacillus thiooxidans oxidizes insoluble sulphur to sulphuric acid, which takes place in the periplasmic space. It is possible to dissolve iron through direct bacterial leaching as shown in the above reactions.

(ii) Indirect bioleaching: This leaching process takes place without direct involvement of microorganisms but they indirectly support the leaching by producing agents responsible for oxidation of minerals. It can be explained by the process of oxidation of pyrite. Pyrite is a common rock mineral that is found in association with many ores. The pyrite is initially oxidized to elemental sulphur equation 4, which is subjected to further oxidation by *Thiobacillus ferrooxidans* due to which sulphuric acid is formed which is shown in equation 2.

$$FeS_2 + Fe_2(SO_4)_3 \rightarrow 3FeSO_4 + 2S \quad (4)$$

Thiobacillus thiooxidans and *Thiobacillus ferrooxidans* are generally seen associated with leaching dumps. In pilot plant reactors of 50 liter capacity, leaching can be performed continuously in a cascade series with recycling of cells and leachates.

In the laboratory better yields of bioleaching products can be obtained under optimal conditions, like control of temperature, O_2 and CO_2 adjustments, maintenance of pH between 2 and 3, and eh around – 300mv with very finely ground ores in a tower (percolator). However, conditions and yield cannot be achieved in a commercial scale because it is expensive.

12.2 BIOLEACHING ORGANISMS

Thiobacillus thiooxidans and *Thiobacillus ferrooxidans* are generally used in bioleaching methods. However, a number of other microorganisms such as *Thiobacillus concretivorus*, *Pseudomonas florescens, P.putida, Achromobacter, Bacillus licheniformis, B. cereus, B luteus, B. polymyxa, B. megaterium* and several thermophilic bacteria like *Thiobacillus thermophilica, Thermothrix thioparus, Thiobacillus* THl and *Sulfolobus acidocaldarius*. Because of more rapid growth rate the thermophilic bacteria may significantly accelerate the bioleaching process.

12.3 COMMERCIAL PROCESSES

Methods described below are generally employed in large scale bioleaching processes.

(a) Slope leaching: In this method finely powdered ore, approximately 10,000 tons are made into large piles along the slopes of a mountain, and water containing *Thiobacillus* is continuously sprinkled. Metals are extracted from the water that collects at the bottom of the mountain. The water is recycled again after metal extraction and regeneration of the bacteria in an oxidation pool (Fig. 12.1a).

(b) Heap leaching: The ore is arranged in a big heap, which is treated with water as in slope leaching. The recovery of metals and other processes are conducted just like in slope leaching (Fig. 12.1b).

(c) In-situ leaching: This process is carried with an ore which remains in its original location in the earth. The permeability of ore is increased by sub-surface blasting. Several passages, as shown in Fig. 12.1c are drilled through the ore. A well like pit is also made out at the bottom of the ore. Now acidic water containing *Thiobacillus* is pumped through drilled passages of the ore. The acidic water percolates through the ore and collects in the pit at the bottom of the ore. The water is pumped out from the pit and the

minerals are extracted. The water after extraction of minerals, is reused after regeneration of bacteria.

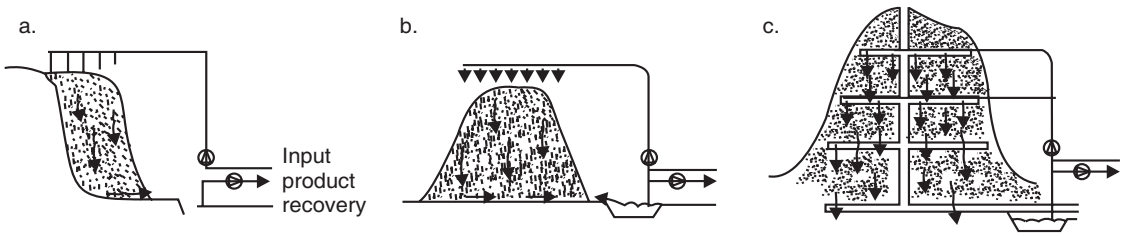


Fig. 12.1: Processes of bioleaching (a) slope bioleaching (b) heap bioleaching (c) in-situ bioleaching

12.4 COPPER LEACHING

To have an idea of bioleaching process copper leaching by bacteria is described as an example. Covallite, chalcolite and chalcopyrite are generally used as copper ores for bioleaching processes. Apart from containing copper, the ores also contain other elements like iron, zinc and sulphur. For example Chalcopyrite contains 26% copper, 25.9% iron, 20.5% zinc and 33% sulphur.

Mechanism of Leaching: During the oxidation of Chalcopyrite the following reaction occurs.

$$2CuFeS_2 + 8\ ½\ O_2 + H_2SO_4 \rightarrow 2\ CuSO_4 + Fe_2\ (SO_4)_3 + H_2O$$

Similarly covallite is oxidized to copper sulphate

$$CuS + 2O_2 \rightarrow CuSO_4$$

Generally heap leaching process is employed in copper leaching process but sometimes a combination of heap leaching and *in situ* leaching processes are used. The solution (Sulphate/Fe^3 solution) is sprinkled over the heap which percolates through the ore and collect at the bottom pit. The solution collecting in the bottom pit will include copper metal, which is removed by precipitation. The remaining water with Fe^{3+} is used again in the leaching process after adjusting the pH to 2.0 with the help of H_2SO_4. An outline of microbial leaching of copper is illustrated in Fig. 12.2.

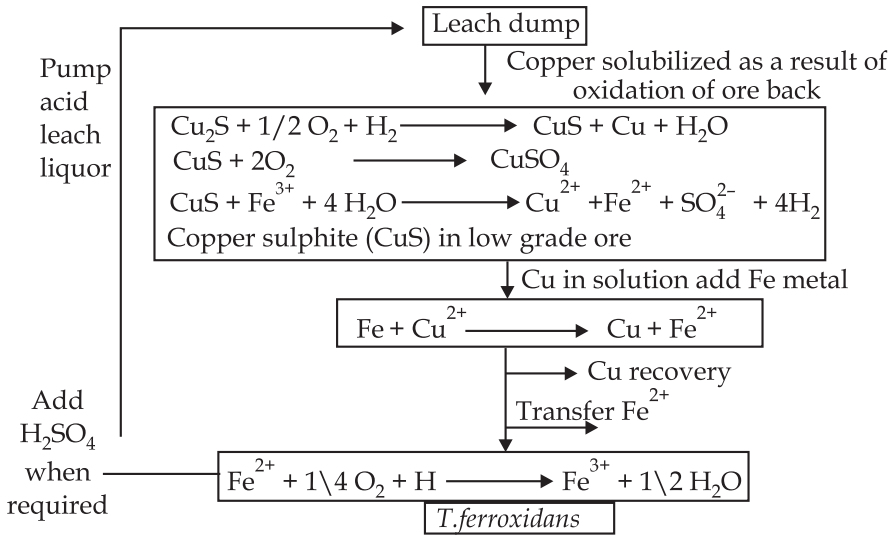


Fig. 12.2: Microbial leaching of copper

Bioleaching of copper has been used in the United States, Australia, Canada, Mexico, South Africa, Portugal, Spain and Japan. About 5% of the world copper production is obtained through bioleaching.

12.5 URANIUM LEACHING

In the uranium leaching process, insoluble tetravalent uranium is oxidized with a hot H_2SO_4/Fe^{3+} solution to soluble hexavalent uranium sulfate.

$$UO_2 + Fe_2(SO_4)_3 \rightarrow UO_2SO_4 + 2FeSO_4$$

Uranium bioleaching process is more significant economically. In situ microbial leaching is greater acceptance since it eliminates the expense of moving vast amounts of material. For instance thousand tons of uranium ore must be handled in other than bioleaching processes, to obtain one ton of uranium. This is an indirect leaching process since the microbial attack is not on uranium ore directly but on the Iron oxidant. Ferric sulphate and sulfuric acid can be produced by *T.ferrooxidans* from the pyrite within the uranium ore.

$$2FeS_2 + H_2O + 7.5\ O_2 \rightarrow Fe_2\ (SO_4)_3 + H_2SO_4$$

For the initial production of the Fe^{3+} leach solution the pyrite reaction is used. For carrying this reaction pilot plants with surface reactors are used which are similar to tuckling filters used in sewage operations.

For getting optimum uranium leaching the incoming air should passes a pH of 1.5-3.5, temperature of 35°C and CO_2 0.2%. However, certain thermophilic strains require a temperature optimum of 45-50°C.

The dissolved uranium is extracted from the leach liquor, in commercial processes with organic solvents like tributyl phosphate and the uranium is subsequently precipitated from the organic phase. The organic solvents which remain in the water system after extraction of uranium may be toxic and hence cause problems when the microbiological system is reused.

Microbial leaching of refractory precious metal ores to enhance recovery of gold and silver is one of the most promising applications. Gold is obtained through bioleaching of arsenopyrite/pyrite ore and its cyanidation process. Silver is more readily solubilized than gold during microbial leaching of iron sulphide.

Similarly silica is leached from ores like magnesite, bauxite, dolomite and basalt by *Bacillus licheniformis.* The silica is accumulated by *B.licheniformis* by process of adsorption which is readily separated. This technology of obtaining silica from magnesite is being adapted by *Salam works* of Burn, Standard Co. Ltd, Tamil Nadu in collaboration with the department of Biotechnology, Govt. of India. Ore leaching by microbes has potential for use in the extraction of other metals such as zinc, cobalt and nickel. New reactor systems are likely to be developed to increase the efficiency of bioleaching in terms of cost and kinetics. These innovations are expected to extend the scope of bleaching applications.

REVIEW QUESTIONS

I. Essay type questions:

1. What is bioleaching? Describe the general process of bioleaching.
2. Explain the process of copper bioleaching.
3. Write in detail bleaching of copper by giving methods and mechanisms of bioleaching.

II. Write short notes on:

1. Types of bioleaching
2. In-situ bioleaching
3. Uranium bioleaching
4. Microorganisms associated with bioleaching
5. Slope bioleaching

FURTHER READING

1. Brierley, C.L. (1982) Microbiological mining. Sci Amer. 247: 44–53.
2. Gentina, J.C. and F. Aceveda (1985) Microbial ore leaching in developing countries Tr. Biotechnol 3: 86–89.
3. Torma, A.E. (1987) Impact of Biotechnology on metal extractions. Mineral processing and extractive metallurgy, Rev. 2: 289–330.
4. Brierley, J.A. (1990) Biotechnology for the extractive metal industries, JOM 42: 28–30.

13

Biosensors

Biosensor is an analytical device consisting of a biocatalyst (enzyme, cell or tissue) and a transducer which can convert a biological or biochemical signal or response into quantifiable electrical signal (Fig. 13.1). The biocatalyst component of most of biosensors is immobilized into a membrane or within a gel such that the biocatalyst is held in intimate contact with the transducers and may be reused. The biological recognition system in biosensors is microbial cell or an enzyme, antibody, hormone, nucleic acid that are immobilized in microchip devices for quantitative estimation of a substance. The principle of simple glucose sensor using glucose oxidase is illustrated in Fig. 13.2. The biosensor does not directly detect the target analyte (glucose) instead measures the change in the concentration of a co-reactant (oxygen) or a co-product (hydrogen peroxide) of the reaction catalyzed by immobilized biological sensing material (GOD).

Combination

Biological sensing element (biological reaction) Enzyme (enzyme reaction) Antibody (immuno reaction) Microorganism (biodegradation) DNA (hybridization) etc.	**Transducer (example)** Electrode (oxygen electrode) Optics (photon counter) Plezo electric elements etc.	**Biosensor-enzyme sensor** Immuno sensor Microbial Sensor DNA sensor etc.

Fig. 13.1: Biological elements are combined with transducers when biosensors are fabricated. Transducers convert a change in the concentration of the compound into an electronic signal

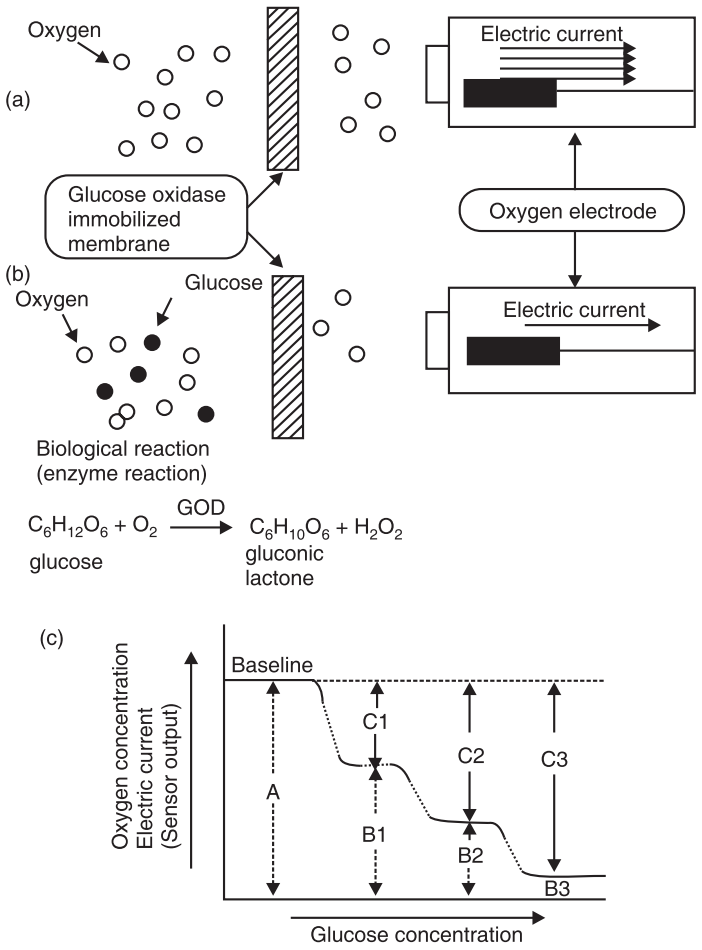


Fig. 13.2: Principle of glucose sensor, glucose oxidase [GOD] is a biosensor

GOD is combined with an oxygen electrode as a transducer

(a) The base line current of sensor

(b) Oxygen concentration is lower than I a and electric current decreases

(c) As glucose concentration increases sensor response will also increase

Biosensors can be categorized based on degree of intimacy between biocatalyst and transducers into:

1. **First generation instruments:** The two components (biocatalyst and transducer) may be easily separated and both may remain functional in the absence of the other.
2. **Second generation instruments:** The two components interact in a more intimate fashion and removal of one of the components affects the usual functioning of the other.

   Biosensor configurations are categorized into two types. Batch type (Fig. 13.2a) and flow type (Fig. 13.2b). In flow type sensors a column is shifted with biomaterial – immobilized leads can be incorporated separately from a transducer. These are very useful for continuous monitoring of target analytes. Typical immobilization carners are glass,

alginate, membrances and resin beads. Flow injection analysis (FIA) is a technique sometimes used in flow type biosensors. Biosensors using FIA yield very rapid and accurate measurements and very useful in different fields like environmental monitoring and food quality control. Disposable biosensors such as amperometic glucose sensors for medical diagnosis are example of a batch-type sensor.

3. **Third generation instruments:** The biochemistry and electro-chemistry are even more closely linked and where the electrochemistry occurs at a semi-conductor, the term biochip, may be applied to describe such instruments.

The biosensor may be of different kinds like electrochemical, amperometric, thermister bioaffinity, whole cells (microbial), optoelectronic etc. The sensor which may be carbon electrode, an ion-sensitive electrode, oxygen electrode, photocell or thermister, analyses the biological signals and converts them into electrical signals in the readable from which is recorded on a meter. The substance to be analysed passes through the membrane and interacts with immobilized material to yield the biological signals (electrons, heat, ions, gases etc.). These biological signals are then converted into electrical signals by the sensors.

The natural biosensors are the sense organs primarily the chemical sensors of smell and taste. Technical biosensors have been under intense development since the middle of 1960 with the prospects of commercial potential offered by biotechnology. These are the combination of biologically active material displaying characteristics especially with chemical or electronic sensor to convert the response into electrical signals. Some of the common biosensors, used extensively are listed in table 13.1.

Biosensors are, infact biocatalysts. They are used as an immobilized biological molecule (usually an enzyme or antibody) or a whole microbial cell to detect or sense a particular substance. The biosensor does this by reacting specifically with the substance to be detected so as to give a product which is used to generate an electrical signal by means of a device called transducer. The response of biosensor is measured in terms of substrate used or product formed. They are of different types like carbon electrode, glucose electrode ion sensitive electrode, photocell, oxygen electrode, adenosine electrode. For example a glucose electrode is constructed by immobilizing a layer of glucose in polyacrylamide gel around a platinum oxygen electrode. When a solution of glucose is brought into contact with electrode, glucose and oxygen diffuses into an enzyme layer and are converted into gluconolactone and hydrogen peroxide lowering oxygen concentration around the electrode. Oxygen concentration read by electrode is proportional to glucose concentration in the sample (Fig. 13.3).

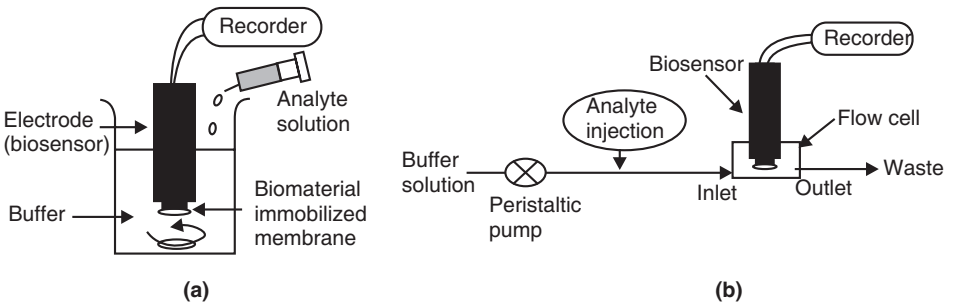


Fig. 13.3: Typical biosensor measurements (a) Batch type measurement (b) Flow type measurement

Table 13.1: Some common biosensors

Analyte	Biocatalyst	Transducer	Immobilization	Stability (in days)	Response (in mins)
Alcohol	Alcholooxidation Streptococcus	O_2	Glutaraldehyde	>14	1–2
Arginine	Faecium	NH_3	Physically entrapped	20	20
Cholesterol	Nocardia erythropsis	O_2	Physically entrapped	28	35–70
D-glucose	Glucose oxidase	O_2	Chemical	21	1
Glutamate	Glutamate decarboxylase	CO_2	Glutaral dehyde	7	10
NAD+	NADase & *E.coli*	NH_3	Dialysis membrane	7	5–10
Nitrate	(*Azatobacter*) *vinelandii*	NH_3	Physically entrapped	14	7–8
Pencillium	Penicillinase	H^+	Polyacrylanide	14	15–30 sec
Urea	Urease	NH_4^+	Polyacrylamide	>13	20–30 sec

Biosensor technology couples our knowledge of biology with advances in microelectronics. A biosensor is composed of a biological component such as a cell or antibody linked to a tiny transducer. Biosensors are detecting devices that rely on the specificity of cells and molecules to identify and measure substances at extremely low concentrations. When the substance of interest collides with biological components, the transducer produces a digital electronic signal proportional to the concentration of the substance. Biosensors are used for following purposes.

Applications:

1. Measure the nutritional value fresheners, flavor and safety of food.
2. Measure vital blood components.
3. Locate and measure environmental pollutants.
4. Detect glucose, acetic acid, glutamic acid, ethanol and biochemical oxygen demand (BOD) in a sample.
5. Measurement of cephalosporins, nicotinic acid and several B-Vitamins.
6. Measuring food quality based on aflatoxin contaminatin. An automated unit based on a new column based immunoaffinity recharged at 0.1 to 50 ppb aflatoxins in 1.0 ml sample in less than 2 minutes.

It is some times useful in monitoring the specific chemical both accurately and rapidly.

In Medical field such as detection of hepatitis antigens in blood during infection by virus can be detected. Abnormal amount of urea in blood or urine in kidney, such diseases can be measured. Hormone, gonadotropin produced during pregnancy can be tested and measured. High concentrations of creatine produced after heart attack can be analysed.

Biosensors are used to know about the nature of industrial products of acids, alcholos, phenols, pollutants etc. The industrial workers can know the presence and concentration of hazardous chemicals in the environment surrounding them. Biosensors can detect various chemical warfare agents, nerve gas etc.

In environmental analysis such as measurement of BOD, ammonia, nitrate, sulfite. The BOD sensor, which uses an omnivorous yeast, Trichosporon cutaneum is a well known example of a microbial biosensor applied to environmental monitoring. The microbial BOD sensor measures the oxygen uptake by the respiratory system of the microorganism which changes as a result of the biodegradation of organic compounds in the sample.

An intensive research is going in Central Electro-chemical Research Institute (CECRI) Karaikudi, Tata Institute, Fundamental Research and Technology (TIFRT) New Delhi, Council of Scientific and Industrial Research (CSIR) Chandigarh and Central Electrical Engineering Research Institute (CEERI), Pilani.

Military research have been taking more interest as a means of detecting nerve gases and other chemical warfare toxins.

In recent times microorganisms are being used as molecular recognization elements in bisensors. The principle of a microbial sensor is based either on the change of respiration or the amount of produced metabolites as the result of assimilation of substrates by microorganisms. Further, more use of auxotrophic mutants can selectively determine many kinds of substances. For example vitamin B_{12} sensor was constructed by using immobilized *E.coli* 215 which is auxotroylic to B_{12}. The linear relationship was obtained in the range between 5 × 109 and 25 × 109 g/ml within 25 days. The decrease in the response was only 8% approximately. Recently thermophilic bacteria which minimizes contamination and stable for long-term are being exploited for this purpose. Some of the biosensors based on microorganisms are listed in table 13.2. Microbial biosensors have many advantages such as:

1. Implantation in human body and are suitable to *in vivo* measurement.
2. Can be integrated on one chip and are useful for measuring various substances in small amounts of sample solution simultaneously.
3. Since semiconductor fabrication technology is applied to microbiosensors, it is possible to develop disposable transducers for biosensors through mass production.

Biosensor technology can be used:

1. To determine quality of fresh food such as meat and fish.
2. In online measurement of alcohol concentration in culture broth as required in brewing and fermentation industries.

Table 13.2: Microbial biosensors containing oxygen electrode

Microorganism	Sensing for	Response time (min)	Range
Brevibacterium lactofermentum	Assimilate sugars	1–10	Linear above 1 m mole dm^{-3}
Bacillus subtilis	Mutagen screening	90–100	1-6 μ cm^{-3}
Methylomonas flagellata	Methane	1	Upto 6.6 m mol dm^{-3}
Trichosporon brassicae	Acetate	6–10	Linear upto 22.5 mg dm^{-3}
T. brassicae	Ethanol	10	Below 22.5 mg dm^{-3}

3. It is possible to detect toxins based on bioluminescence. Certain bacteria can be genetically engineered to grow in dark. These bacteria are used as living biosensor for detecting a variety of toxic agents.

4. Biosensor containing immobilized, *Trichosporon cutaneum,* is used to test the extent of water contamination by measuring the B.O.D.

Biosensors have replaced conventional methods, which are often complicated, time consuming expensive and require pretreatment or clean-up of real samples prior to analysis. Biosensors have following advantages over other analytical methods:

1. Rapid and convenient detection
2. Direct measurement of real sample
3. Very specific detection

However, these biosensors are instable and lack reproducibility, a major concern.

Rapid advances are being made in all areas of biosensor technology. These include major improvements in the stability and durability of these units, which are being made more portable and sensitive. Microorganisms and metabolites such as glucose can be measured, thus meeting critical needs in modern medicine.

Biosensors can also be used in the following fields:

1. Clinical diagnosis and biomedical monitoring.
2. Agricultural, horticultural and veterinary analysis.
3. Detection of pollution and microbial contamination of water
4. Fermentation analysis and control.
5. Monitoring of industrial gases and liquids.
6. Measurement of toxic gas in mining industries.
7. Direct biological measurement of flavors, essences and hormones.

REVIEW QUESTIONS

I. Essay type questions:

1. Define biosensor. Describe the methods of building a biosensor.
2. Give an account of applications of biosensor.

II. Write short notes on:

(a) Transducer
(b) Biocatalysts
(c) Types of biosensor measurements

FURTHER READING

1. Bureic D.G. (1993) Biosensors. Technomic Lancaster, Pennsylvania, USA.
2. Cass, A.E.G (1990) Biosensor IRL/Oxford University Press, New York.
3. Karube, I, Y. Nomura and Y. Arikawa (1995) Biosensors for environmental control. Trends Anal Chem. 14:295-299.
4. Suzuki, S. (1990) Biosensor 5th edition Kohdan-Sha, Tokyo.

14

Biosurfactants

Biosurfactants can be defined as amphipathic molecules with both hydrophilic and hydrophobic (generally hydro carbons) moieties that partition preferentially at the interface between fluid phases with different degrees of polarity and hydrogen bonding such as oil/water or air/water interfaces. This allows the monomers to form micelles or to aggregate into micellar tubes, micellar bilayers and vesicles (Fig. 14.1). Surfactants are also characterized by an ability to enhance the formation of emulsions of immiscible liquids such as hydrocarbons and water (Fig. 14.2).

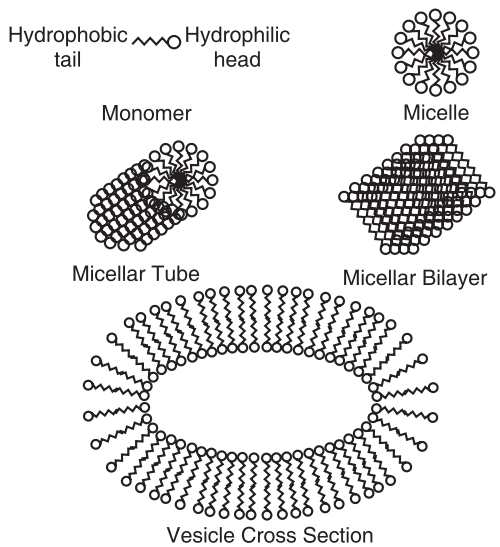


Fig. 14.1: A Surfactant having both hydrophobic and hydrophilic moieties present within the same molecule. This allows the monomers to form micelles or to aggregate into micellar tubes, micellar bilayers and vesicles

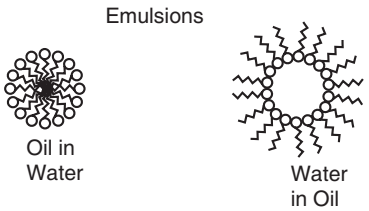


Fig. 14.2: Surfactants are characterized by an ability to enhance the formation of emulsions of immiscible liquids such as oil and water

These properties render surfactants capable of reducing surface and interfacial tension and forming microelusion where hydrocarbons can solubilize in hydrocarbons. These characteristics confer excellent dertergency, emulsifying, foaming and dispensing traits. They readily form foam. Biosurfactants can be categorized on the energy charge associated with hydrophilic group in the non ionic, anionic, cationic or amphoteric. These have a great utility as wetting agents and for their effectiveness as emulsifier. Biosurfactants at present have 9.4 billion dollars sales only in USA and is likely to increase at a rate of 3-5% annually. At present surfactants are in use chemicals derived from petroleum. But in recent times microbial surfactants are receiving increased demand due to the following features. (1) Their diversity and environment friendly nature, (2) Possibility of their production through fermentation, (3) Their potential applications in the environmental protection, (4) Crude oil recovery, (5) They are useful in health care and food processing.

Biosurfactants are superior to chemical surfactants due to the following features:

(1) They have lower toxicity, (2) They are easily biodegradable,

(3) Better environmental compatibility, (4) Higher foaming ability,

(5) Higher selectivity and specific activity at extreme temperature, pH and salinity.

(6) They can be synthesized from renewable feed stocks.

Methods of microorganisms screening for their biosurfactant production potential are:

(a) Aximetric drop shape analysis.

(b) Colorimetric estimation (anionic surfactants react with cationic indicator to form coloured complex).

(c) A rayeid drop collapsing test.

(d) Direct TLC method.

(e) Estimation of the emulsification index value.

14.1 CLASSIFICATION OF BIOSURFACTANTS

According to their Polarity, chemical composition and microbial origin biosurfactants can be classified into: (1) Glycolipids, (2) Lipopeptides or lipoprotein, (3) Phospholipids and fatty acids (mycolic acids), (4) Polysaccharide–lipid complexes, (5) Complete cell surface, (6) Polymeric surfactants, (7) The particulate type.

(i) Biosurfactants can also be classified according to their source of production (Fig. 14.3).

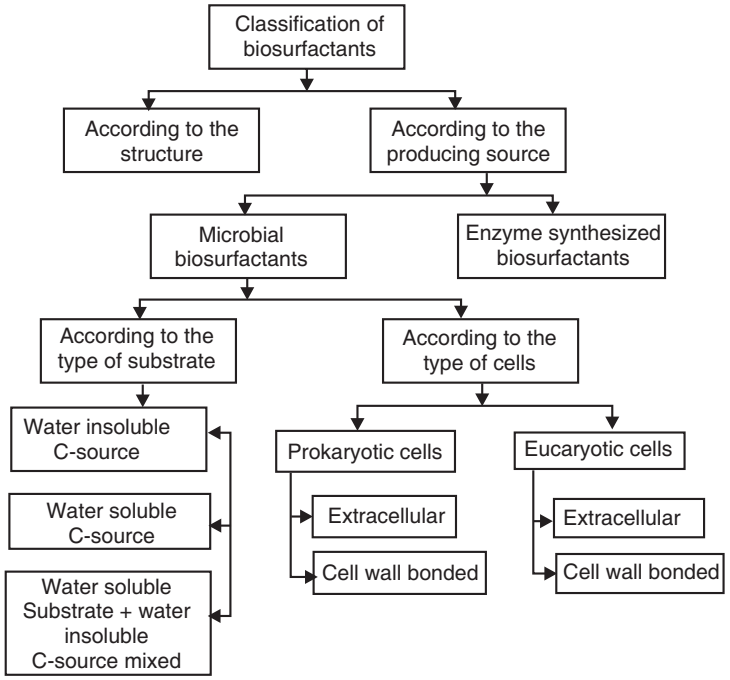


Fig. 14.3: Classification of biosurfactants

(ii) Biosurfactants can be classified based on chemical composition and microbial origin as:

(1) Hydrophilic moiety containing amino acids or polypeptides anions or cations (mono, di or polysaccharides),

(2) Hydrophobic moiety containing unsaturated, saturated fatty acids

(i) Lipopeptides (ii) gramicidins (*B.brevis*) (iii) fatty acid, phospholipids, and (iv) polymeric.

(iii) Biosurfactants are classified on the basis of producing source, as :

(a) Microbial biosurfactants, and (b) Enzymatic synthesized surfactants

(iv) Classification by type of substrate used

1. Those producing biosurfactants with alkanes as carbon sources (*Corynebacterium* sp and *Arthrobacterium* sp)
2. Those producing biosurfactants with water soluble substrates as carbon sources (*Bacillus* sp)
3. Those producing biosurfactants with alkanes and water soluble substrates as carbon sources (*Pseudomonas* sp)

14.2 ENZYME SYNTHESIZED BIOSURFACTANTS

Many isolated enzymes that catalyze hydrolysis, alcoholysis, condensation, acylation or esterification reactions have been used for the production of various surfactants including

monoglycerides, phospholipids, glycolipids and amino acids based surfactants from relatively inexpensive raw materials such as fats and plant oils. Enzymatic methods of synthesis of surfactants have several advantages over conventional chemical synthesis. The enzymatic methods have low energy requirement, minimal thermal degradation, high biodegradability and radioactive immobilized microbial lipases such as lipozyme from *Rhizomucer miehei* or lipase sp 435 from *Candida antarctica* are the most frequently used enzymes.

Using this technology Glycolipids such as fructose oleate, monocapryloyl-1 D-fructofuronase, B-1-1fructopyrasanose-oleate, a mixture of C-1 and C-6 monopalmintoyl fructose, 6-octano glucose and various primary monoesters of sugar alcohos such as sorbitol, mannitol and lytitol have been produced. Aliphatic alcohol glycosides, phenol glycosides, vitamin glycosides, glycerol glycolipids and glycolipid biosurfactant, fructose monooleates, monoole aleoyl glycerol and uncommon hydroxyl compounds such as 2-hydroxy ethyl-trimethyl silane which is attached to C1 glucose are among the recently synthesized enzymatic surfactants. Major characteristic features of microbial and enzyme synthesized surfactants are précised in Table 14.1.

Table 14.1: Comparison of microbial biosurfactants and enzyme-synthesized surfactants

Feature	Microbial bio-surfactants	Enzyme-synthesized surfactants
Advantages	Bio-degradability	Biodegradability
	Diversity	Ease of structural modification
	Low production costs	Low recovery costs
	In sity applications	Ease of purification
Disadvantages	High recovery costs	High enzyme costs
	High waste volume	Low solubility of substrates
Key points for further developments	Strain improvement Whole cell immobilization	Enzyme immobilization Enhanced enzyme stability and activity
	Improved fermentation technology	Multiple-phase systems
	Metabolic engineering	Supercritical fluid technology

However, disadvantages of enzyme technology together with non-aqueous phase kinetics will immensely contribute towards the elucidation of these issues. Similarly main disadvantage of high recovery costs and large liquid waste volumes are likely to be eliminated as a result of future research in strain development, metabolic engineering and fermentation technology.

14.3 MICROBIAL BIOSURFACTANTS

Some of the important classes of biosurfactants are listed along with producing microorganism in table 14.2 and described below:

Table 14.2: Major biosurfactant classes and microorganisms involved

Surfactant class	Microorganism
1. Glycolipids Rhamnolipids	*Pseudomonas aeruginosa* *Rhodococcus erithropolis*
Trehalose lipids	*Arthobacter sp.*
Sophorolipids	*Candida bombicola, C.apicola*
Mannosylerythritol lipids	*C.antartica*
2. Lipo-peptides Surfactin/iturin/fengycin	*Bacillus subtilis*
Viscosin	*P.fluroscens*
Lichenysin	*B.licheniformis*
Serrawettin	*Serratia marcescens*
Phospholipids	*Acinetobacter sp.* *Corynebacterium lepus*
3. Surface-active antibiotics	
Gramicidin	*Brevibacterium brevis*
Polymixin	*B.polymyxa*
Antibiotic TA	*Myxococcus xanthus*
4. Fatty acids/neutral lipids	
Corynomicolic acids	*Corynebacterium insidibasseosum*
5. Polymeric surfactants	
Emulsan	*Acinetobacter calcoaceticus*
Alasan	*A.radioresistens*
Liposan	*C.lipolytica*
Lipomanan	*C.tropicalis*
6. Particulate biosurfactants Vesicles	 *A.calcoaceticus*
Whole microbial cells	*Cyanobacteria*

1. **Glycolipids:** These are carbohydrates in combination with long chain aliphatic acids or hydroxyaliphatic acids. The linkage is by means of either ether or an ester group.

   Rhamnolipids are biosurfactants in which one or two molecules of rhannonose linked to one or two molecules of β-hydroxy decanic acid (Fig. 14.4). These are produced by *Pseudomonas aeruginosa*.

Trehalolipids are other group of biosurfactants. Species of *Mycobacterium, Nocarchia* and *Corynebacterium* are reported to produce these biosurfactants which include disaccharide trehalose linked at C6 and C6 associated with mycolic acid (Fig. 14.5). Mycolic acids are long chain and branched β-hydroxy fatty acids. *Rhodococcus erythropolis* and *Arthrobacter sp* are reported to elaborate with lowered surface and interfacial tension in culture.

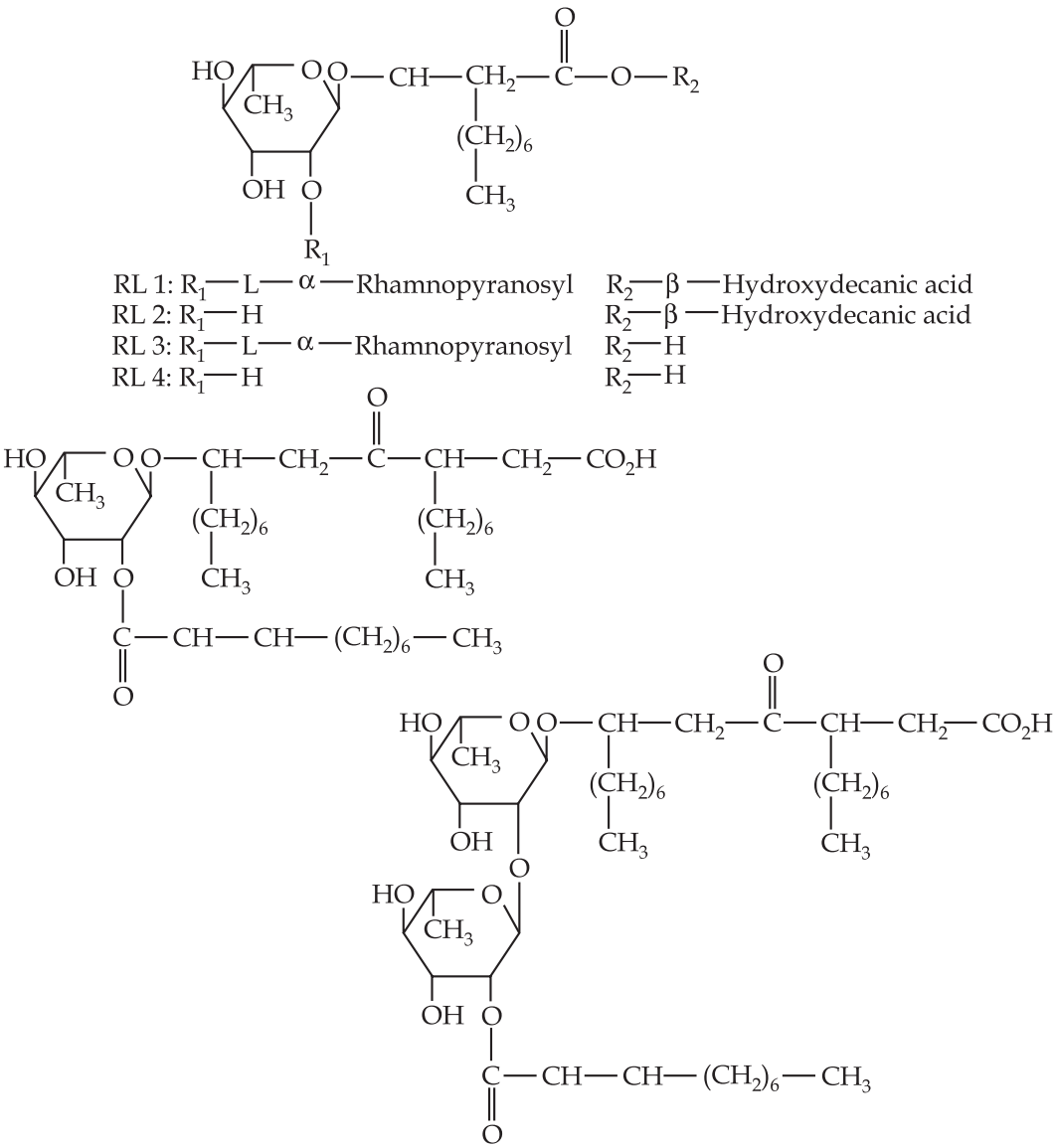


Fig.14.4: Rhamnolipids from *Pseudomonas sp*

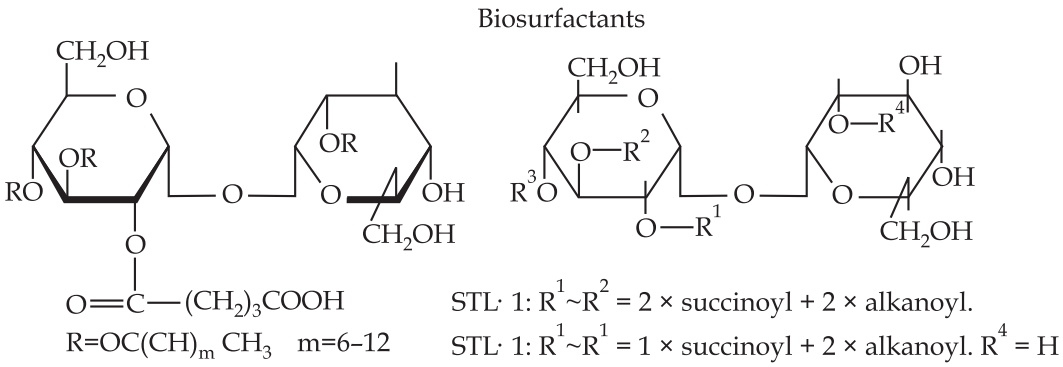


Fig. 14.5: Trehalolipids from *Rhodo erythropolis* (right) and Arthrobacter sp (left)

Sophorolipids are other category of glycolipids which are mainly elaborated by yeasts such as *Torulopsis bombicola*, *T.petrophilum* and *T.apicola* in which sophore is linked to a long chain hydroxy fatty acid by glycosodic linkage consisting of atleast 6-8 different hydrophobic sophorolipids (Fig. 14.6). Glucose lipids contain 3-hydroxy fatty acids Rubiwettin RG1 (Fig. 14.7) and one produced by the marine bacterial strain MM1 are two important glucose lipids. Rubiwettin RG1 lowers from strain tension of saline to 26 mN/m.

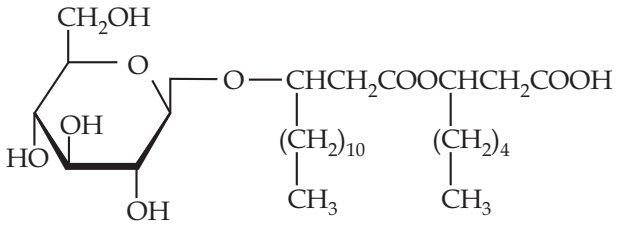


Fig. 14.6: Glucose lipids of *serratia rabidae* and *Torulopsis bombicola*

2. **Lipopeptides and lipoproteins:** In these biosurfactants lipids are attached to a polypeptide chain. Surfactin which is produced by *Bacillus subtilis* ATCC21332 is a seven amino acid ring structure coupled to a fatty acid chain via lactone linkage (Fig. 14.7). Its surface tension is low (27.9 mN/m), at concentration as low as 0.005%. Lichenysin is another Lipopeptide biosurfactant elaborated by *Bacillus lichenifermin*. This biosurfactant lowers surface tension of water to 27 mN/m and the interfacial tension between water and n-hexadecane to 0.36 mN/m.

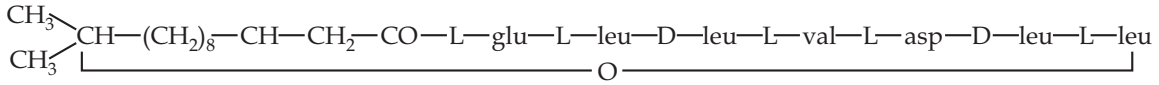


Fig. 14.7: Surfactin from *Bacillus subtilis*

3. **Fatty acids, phospholipids and neutral lipids:** Several yeasts and bacteria such as *Acinetobacter sp* produces phosphatidyl ethanolamine rich vesicles which form optically clear micro emulsions of alkanes in water.
4. **Particulate biosurfactants:** Extracellular membrane vesicles partition hydrocarbons to form a microemulsion which plays an important role in alkane uptake by microbial

cells. Vesicles of *Acinetobacter sp* with a diameter of 20-50 nm and a buoyant density of 1.158 cubic g/cm and composed of protein phospholipids and lipopolysaccharides.

5. **Surface active antibiotics:** Gramicidins, polymixins and antibiotic TA comes under this category and they are elaborated by *Brevibacterium brevi, B. polymxya* and *Myxococcus xáanthus* respectively. Gramicidin S is cyclosymmetric decapeptide and has antibacterial activity which is attributed to its high surface activity. Polymixin D is a decapeptide in which amino acids 3 through 10 form a cyclic octapeptide A branched chain fatty acid is connected to the terminal 2,4-diaminobutyric acid (DAB). Polymixins are able to solubilize certain membrane enzymes. Antibiotic TA inhibits peptidoglycan synthesis by interfering with polymerization of the lipid disaccharide pentapeptide. It has interesting chemotherapeutic applications.
6. **Polymeric microbial surfactants:** Most of these polymeric biosurfactants are heterosaccharide containing proteins. The best studied polymeric biosurfactants are emulsan, liposan, alasan and lipomanans produced by *Acinetobacter calcoacetices*. Emulstin is an effective emulsifying agent for hydrocarbon in water even at a concentration as low as 0.001 to 0.01 percent. Liposan produced by *Candida lipolytica* is an extracellular water soluble emulsifier composed of 83% carbohydrate and 17% protein.

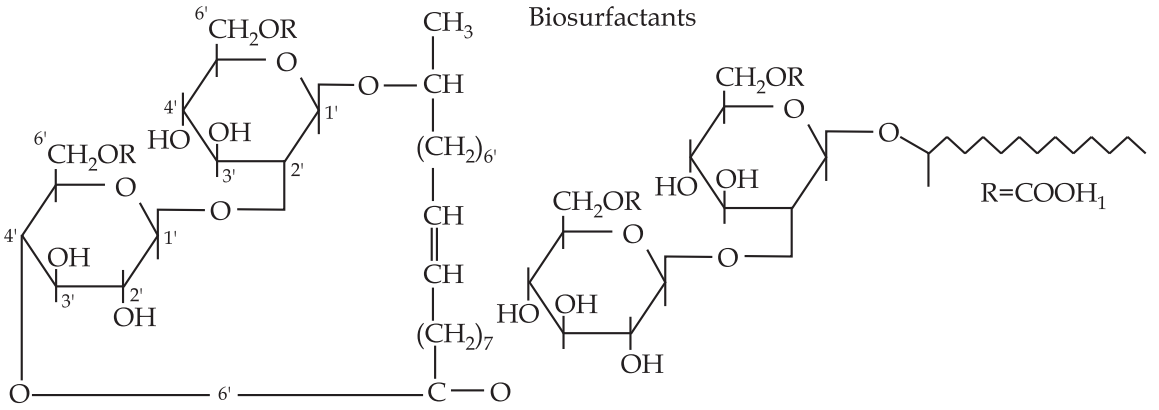


Fig. 14.8: Classical-type (left) and novel-type (right) sophorose lipid from *candida bombicola*

Table 14.3: Various biosurfactants produced by microorganism

Organism	Types of surfactant
Acinetobacter caloceticus RAG-1	Lipoteteropolysaccharide (Emulsan)
Alcanivorax borkumensis	Glycolipid
Arthrobacter paraffineus	Sucrose and fructose glycolipids
Bacillus licheniformis	Lipoprotein, lipopeptice (lichenysin A)
Bacillus pumilis A1, Bacillus subtilis	Surfactin
Candida antartica WSH 112	Glycolipids mannosylerythriol lipids

contd...

Organism	Types of surfactant
Candida bombicola Candida apicola, Candida lipolytica Y-917, Candida gropengiesseri	Glycolipid (sophorose lipids) Lipsan
Candida tropicalis	Polysaccharide-fatty acid complex, Mannan-fatty acid
Clostridium pasteurization	Neutral lipids
Cornebacterium diphteri.e	Acyl glucoses
Corynebacterium hydrocarboclastus	Polysaccharide-protein complex, protein-lipid-carboxy, sucrose ester
Corynebacterium insidiosum	Phospholipids
Corynebacterium lepus	Corynomycolic acids, fatty acids
Corynebacterium salvonicum SFC	Neutral lipid
Nocardia corynebacteroides	Glycolipid (Pentasaccharide lipids)
Nocardia erythropolis	Neutral lipids, fatty acid+neutral lipids
Pseudomonas aeroginosa	Glycolipid (rhamnose lipid)
Pseudomonas cepacia	Glycolipids
Pseudomonas fluoroscens	Rhamnose lipid, vicosin
Pseudomonas rubescens, Thiobacillus thiooxidans	Ornithin
Rhodococcus aurantiacus	Glycolipid
Rhodococcus erythropolis	Trehalose dimycolates, trehalose dicorynomycolate, PE amines
Serratia marcescens	Serrawettin
Serratia rubidae	Rubiwettins
Streptococcus thermophilus B	Glycolipid
Streptomyces tandae	Peptides (Streptofactin)
Torulopsis petrophilum	Glycolipid and/or protein

Biosurfactant production: Though the biosurfactants are produced by hydrocarbon degrading bacteria, they are readily produced on water soluble compounds such as glucose, sucrose, glycerol or ethanol. Biosurfactant production are either growth associated, growth limiting condition or resting or immobilized cells or they are produced with addition of precursors. Generally their production is influenced by the carbon source, nitrogen, phosphate source and other environmental factors. List of microorganisms producing different types of biosurfactant are listed in table 14.3. Different methods of biosurfactants production are precised in Table 14.4.

Culture conditions such as pH, temperature, dissolved oxygen and ionic strength also influence biosurfactant production. Temperature is especially important in the case of *Arthrobacter paraineus* ATTCC 1958, *Rhizomucor erythropolis* and *Pseudomonas* sp Bacillus subtilis CO_2 is reported to produce three fold more lipo peptide under anaerobic condition when compared to aerobic condition. Carbon, nitrogen and phosphate sources, metal ions and other additives used in media formulation may play a critical role in the production and yield of biosurfactant.

Table 14.4: Methods for production of biosurfactants by microorganisms

1.	Cell growth-associated production of biosurfactants.
	(i) Induction of production by lipophilic substrates.
	(ii) Increase in production by optimization of medium composition.
	(iii) Increase in production y optimization of environmental factors such as pH, temperature, aeration and agitation speed.
	(iv) Increase in production by addition of reagents such as alkanes, kerosene and EDTA which cause a detachment of cell wall-bounded biosurfactants into the medium.
	(v) Increase in production by addition of reagents such as penicillin, ethambutol and EDTA which cause a change in cell wall permeability.
2.	Biosurfactants production by growing cells under growth-limiting conditions.
	(i) Production under N limitation.
	(ii) Production under limitation of multivalent cations.
	(iii) Increase in production under growth-limiting conditions by a change of environmental conditions such as pH or temperature.
3.	Biosurfactant production by resting cells.
	(i) Production by resting-free cells.
	(ii) Production by resting-immobilized cells.
	(iii) Production by resting-immobilized cells with simultaneous product removal.
4.	Biosurfactant production by growing, resting-free and resting-immobilized cells in the presence of precursors.

Carbon substrate is reported to act in an order to reduce production costs, use of cheaper substrates such as rice hull hydrolysate, starch waste liquors, domestic wastes, potato processing wastes and they has been recommended. Similarly use of olive oil mill effluents and peat pressate have also been recommended.

Fermentative production: Since biosurfactants are diverse group of compounds produced by a variety of microbial species, it is difficult to develop general guidelines for process development. Although, most biosurfactants are released into the culture medium throughout the exponential phase, some can also be produced by resting cells or by immobilized biocatalysts.

The production of biosurfactants can be carried out using batch or continuous fermentation. The production of biosurfactants that can be produced by resting cells such as glycolipids by *Torulopsis bombicola* can be carried out by immobilized biocatalyst. Its production can be increased by online removal using an adsorption column or by foam fractionation, application of air lift fermenter and aqueous two phase fermentation.

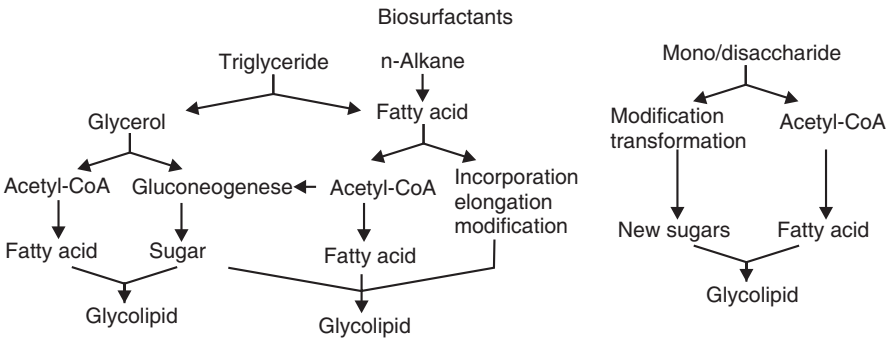


Fig.14.9: Production of glycolipids from a single or in combination of different carbon sources

Critical role of carbon source in biosurfactant production: These carbon sources are used either singly or in combination and produce glycolipids as shown in Fig. 14.9 and Table 14.5.

Solid state fermentation using soya bean, curd residue using a recombinant B *subtilis* has been reported to yield four times higher than that in submerged fermentation. Multi-organism strategy for biosurfactant production such as microbial single oil is obtained from *Lipomyces* or *Chlorella*. The single cell oil obtained from microorganism is used as a substratum for production of glycolipid by *Candida bombicola.* Similarly sophorose lipids are produced by two stage fed batch process using *C bombicola* and *Cryptococcus curvatus.* Some of the cheaper raw materials employed in biosurfactant production are listed in Table 14.6.

Table 14.5: Various microbial biosurfactants, their C sources and surface properties

Biosurfactant	Microorganism	Carbon source	Surface tension (Nm/m)	Interfacial tension (Nm/m)
Glycolipids	*Rhodococcusa urntiacus*	n-Alkanes	26	0.35
Glycolipids	*Rhodococcus sp.strain H13A*	Hexadecane	na	0.02
Glycolipids	*Torulopsis apicola*	Alkane/carbohydtrate	30	< 0.9
Pentasaccharide lipid	*Nocardia*	n-Alkane	26	<1
Rhamnolipid	*Pseudomonos aeruginosa*	Glucose	29	0.25
Rubiwettins	*Serratia rubidoea*	Glycerol	25	na
Sophorose lipids	*Candida bombicola*	Glucose/oleic acid	33	1.8
Trehalose mono-and di-corynomycolates	*Nocardia corynebacteroides Rhodococcus erythropolis*	n-Alkanes	32–36	14–17
Lipopeptides	*Bacillus licheniformis JF2*	Glucose	27	0.016
Lipopeptides	*Bacillus licherniformis 86*	Glucose	27	0.36
Viscosin	*Pseudomonas fluoresccens*	Glycerol	26.5	na
Serrawettin	*Serratia marescens*	Glycerol	28.8–33.9	na
Surfactin	*Bacillus subtilis*	Glucose	27-32	1
Fatty acids	*Corynebacterium lepus*	Kerosene/alkanes	< 30	2
Fatty acid+ neutral lipids	*Nocardia erythropolis*	Hexadecane	32	<3
Protein-carbohydrate complex	*Pseudomonas fluorescens 378*	n-Alkanes	27	na
Phosphatidylethanol amines	*Rhodococcus erythropolis*	n-Alkanes	30	< 1
na	*Corynebacterium insidiosum*	Hexadecane	28.5	0.55

Table 14.6: Use of inexpensive raw materials for the production of biosurfactants by various microbial strains

Low cost or waste raw material	Biosurfactant type	Maximum		
		Producer microbial strain	Yield (g/l)	Reference
Rapeseed oil	Rhamnolipids	Pseudomonas sp DSM 2874	45	41
Babassu oil	Sophorolipids	Candida lipolytica IA 1055	11.72	42
Turkish corn oil	Sophorolipids	Candida bombicola ATCC 22214	400	43
Sunflower and soya bean oil	Rhamnolipid	Pseudomonas aeruginosa DS10-129 4.31		44
Sunflower oil	Lipopeptide	Serratia marcescens	2.98 44	45
Soya bean oil	Mannosylerythritol lipid	Candida sp. SY16	95	
Oil refinery waste	Glycolipids	Candida antartica, Candida apicola	10.5 53	
Curd whey and distillery waste	Rhamnolipid	Pseudomonas aeruginosa strain BS2	0.92 48	
Potato process effluents	Lipopeptide	Bacillus subtilis 2.7		51
Cassava flour waste water	Lipopeptide	B.subtilis ATCC 21332, B.subtilis LB5a	2.2 35	

The separation and concentration of biosurfactants can be achieved by a few steps such as precipitation, organic extraction and absorption chromatography. Ultrafiltration can also be used to recover biosurfactant from fermentation broth. General methods of down stream process for isolation and purification of biosurfactants are precised in table 14.7.

Table 14.7: Physico-chemical property-based biosurfactant recovery methods and their relative advantages

Downstream recovery procedure	Biosurfactant property responsible for separation	Instrument/apparatus/set-up required	Advantages
Acid precipitation	Biosurfactants become insoluble at low pH values	No set-up required	Low cost, efficient in crude biosurfactant recovery
Organic solvent extraction	Biosurfactants are soluble in organic solvents due to the presence of hydrophobic end	No set-up required	Efficient in crude biosurfactant recovery and partial purification, reusable nature
Ammonium sulphate precipitation	Salting-out of the polymeric or protein-rich biosurfactants	No set-up required	Effective in isolation of certain type of polymeric biosurfactants
Centrifugation	Insoluble biosurfactants get precipitated because of centrifugal force	Centrifuge required	Reusable, effective in crude biosurfactant recovery

contd...

Downstream recovery procedure	Biosurfactant property responsible for separation	Instrument/apparatus/set-up required	Advantages
Foam fractionation	Biosurfactants, due to surface activity, form and partition into foam	Specially designed bioreactors that facilitate foam recovery during fermentation	Reusable in continuous recovery procedures, high purity of product
Membrane ultrafiltration	Biosurfactants form micelles above their critical micelle concentration, which are trapped by polymeric membranes	Ultrafiltration units with porous polymer membrane	Fast, one-step recovery, high level of purity
Adsorption on polystyrene resins	Biosurfactants are adsorbed on polymer resins and subsequently desorbed with organic solvents	Polystyrene resin packed in glass columns	Fast, one-step recovery, high level of purity, reusability
Adsorption on wood-activated carbon	Biosurfactants are absorbed on activated carbon and can be desorbed using organic solvent	No setup required. Can be added to culture broth. Can also be packed in glass columns	Highly pure biosurfactants, cheaper, reusability, recovery from continuous culture
Ion-exchange chromatography	Charged biosurfactants are attached to ion-exchange resins and can be eluted with proper buffer	Ion exchange resins packed in columns	High purity, reusability, fast recovery
Solvent extraction (using methyl tertiary-butyl ether)	Biosurfactants dissolve in organic solvents owing to the hydrophobic ends in the molecule	No set-up required	Less toxic than conventional solvents, reusability, cheap

Application of Biosurfactants: Biosurfactants are used in variety fields and have wide range of applications (Table 14.8). In view of ecofriendly nature they have received great interest in the recent past in their production. The market is extremely competitive and there is a need to develop cost effective process of production of high yield and activity. Microorganisms of extremophilic nature and genetically engineered bacteria are need of the hour. As we further understand both the applications, limitations of biosurfactants and their specific properties such as emulsification and demulsification foaming; water binding capacity spreading and wetting properties, viscosity and consistency will likely find application in an increasingly broad range of industrial, household processes and products. A further understanding of the genetics and biochemistry of biosurfactant synthesis will make possible more effective in situ and ex situ production, application and assessment. As new biosurfactants continue to be discovered and existing ones become better characterized their utility is likely to expand dramatically through a multitude of specialized applications optimized conditions and modified processes.

Table 14.8: Applications of biosurfactants

Area	Use	Effect
Metals	Concentration of ores, cutting and forming, casting, rust and scale removal, plating	Wetting, foaming, emulsifying, lubrication, corrosion, inhibition in rolling oils, cutting oils, mold release additives in pickling and electric cleaning, electrolytic plating
Paper	Pulp treatment, paper machine, calendar	Deresinification, washing, defoaming, color leveling and dispersing, wetting and levering, coating and coloring
Paint and protective coating	Pigment preparation, latex paints, waxes and polishes	Dispersion, wetting of pigment during grinding, emulsification, stabilize latex, retard sedimentation, pigment separation, stabilize emulsions and antistat
Petroleum products production	Drilling fluids, work over of producing wells, second recovery, refined products	Emulsify oil, disperse solids, modify rheological properties of drilling fluids for oil and gas wells; emulsify and disperse sludge; sediment in cleanout of wells, demulsifying crude petroleum, inhibit corrosion of equipment in flooding operations, preferential wetting detergent sludge dispersant, corrosive inhibitor in fuel oils crank case oils and turbine oils
Textiles	Preparation of fibers, dyeing and printing, finishing of textiles	Detergent and emulsifier in raw wool scoaring, dispersant in viscous rayon spin bath, lubricant and antistat in spinning of hydrophobic filaments, wetting penetration, solubilization emulsification, dye leveling, detergency, finishing formulations, softening, antistatic additive to finishes
Building construction	Paving, concrete, ceramic	Improve bond of asphalt to gravel and sand, promote air entrainment
Agriculture	Phosphate fertilizers, spray application	Prevent caking during storage, wetting, dispersing, suspending of powdered pesticides and emulsification of pesticides solutions
Elastomers and plastics	Emulsion polymerization, foam polymers, latex adhesive, plastic articles, plastic coating and laminating	Wetting, solubilizaion, emulsification of monomers, introduction of air, control of cell size, improve bond strength, antistatic agents
Food and beverages	Food processing plants, fruits and vegetables, bakery and ice cream, crystallization of sugar, cooking, fat and oils	For cleaning and sanitizing, improve removal of pesticides and wax coating, solubilization flower oils, control consistency, regard stalling, improve washing, reduce processing time, prevent spattering due to superheat and water
Industrial cleaning	Janitorial supplies, descaling, soft goods	Detergents and sanitizers, wetting agents and corrosion inhibitors in acid cleaning of boiler tubes and heat exchangers

contd...

Area	Use	Effect
Cosmetics and pharmaceuticals	Insect repellent, antacid, bath products, acne pads, antidandruff products, contact lense solution, odorant, nail care, body massage accessories, lipstick, lipmaker, eye shadow, mascara, soap, toothpaste and polish, denture cleaner, adhesives, antiperspirant, lubricated condoms, baby products, food care, mousse, antiseptics, shampoo, conditioner, shave and depilatory products, moisturizer, health and beauty products	Emulsifier, foaming agent, solubilizer, wetting agent, cleanser, antimicrobial agent, mediator of enzyme action
Pollution controls	Soil bioremediation, oil storage tank cleanup, removal of oil spills	Emulsifiers, deemulsifiers

REVIEW QUESTIONS

I. Essay type questions:

1. Define biosurfactant. Discuss the merits over chemical surfactants.
2. Give classification of biosurfactants.
3. Give general account of biosurfactants.
4. Describe the methods of production of biosurfactants.
5. Give an account of applications of biosurfactants.
6. Discuss the economics of biosurfactant production.
7. Discuss the role of carbon source in biosurfactant.

II. Write short notes on:

(a) Emulsion (b) Surfactin (c) Polymeric surfactants
(d) Particulate biosurfactants (e) Rhamnolipids (f) Gramicidin

FURTHER READING

1. Aboul-Kassim, T.A and Simoneit, B.R.T., 1993. Detergents: A review of the nature, chemistry, and behavior in the aquatic environment. I. Chemical composition and analytical techniques. Crit. Rev. Environ. Sci. Technol., 23: 325.
2. Atlas, R.M. and Bartha, R., 1992. Hydrocarbon Biodegradation and Oil Spill Bioremedidation. In: K.C. Marshall (Ed)., Advances in Microbial Ecology, Vol. 12, Plenum Press, New York, pp. 287-338.
3. Bertrand, J.C., Bonin, P. and Goutx, M., 1994. The potential application of biosurfactants in combating hydrocarbon pollution in marine environments. Res. Microbiol., 145: 53–56.
4. Fiechter, A., 1992. Biosurfactants: moving towards industrial application. Trends Biotechnol., 10: 208–217.
5. Gutnick, D.L., 1987. The emulsan polymer; perspectives on a microbial capsule as an industrial product. Biopolymers, 26: 223–240.

6. Inoue, S. and Ito, S., 1982. Sophrolipids from Torulopsis bonbicola a microbial surfactants in alkane fermentations. Biotech. Lett., 4: 3–8.

7. Ishigami, Y., 1993. Biosurfactants face increasing interest. Inform, 4: 1156–1164.

8. Kosaric, N., Gray, N.C.C and Cairns, W.L., 1987. Introduction: Biotechnology and the Surfactant Industry. In: Kosaric, N., Cairns, W.L and Gray, CC (Eds.), Biosurfactants and Biotechnology. Marcel Dekker, New York. pp. 1–19.

9. MacDonald, C.R., Cooper, D.G. and Zajic, J.E., 1981. Surface-active lipids by Corynebacterium lepus. Appl. Environ. Microbiol., 37: 4–10.

10. Rosen, MJ., 1989. In Surfactants and Interfacial Phenomena. Wiley, New York.

11. West, C.C. and Hartwell, J., 1992. Biosurfactants and subsurface remediation. Environ. Sci. Technol., 26: 2324–2330.

15

Cell and Tissue Culture

15.1 PLANT CELL AND TISSUE CULTURE

Plant tissue culture is not a separate branch of plant science like taxonomy, cytology, plant physiology etc., rather it is an experimental method of growing large number of isolated cells or tissues under sterile and controlled conditions. The cells or tissues are obtained from any part of the plant like stem, root, leaf anther etc. which are encouraged to produce more cells in culture and to express their totipotency (i.e. their genetic ability to produce more plants). Cells or tissues are grown in different types of glassware containing a medium with mineral nutrients, vitamins and phytohormones. Therefore, to carry out experiments using tissue culture techniques, a well equipped laboratory is first required.

In recent year there has been a phenomenal increase in number of research laboratories using tissue culture techniques to investigate many fundamental and applied aspects of plants. However, the use of these techniques is not confined to research alone. Tissue culture techniques are now being exploited for commercial purposes. Even many horticultural companies are setting up small tissue culture units to multiply plants which are difficult to propagate by conventional methods.

Tissue culture laboratory

The general laboratory for tissue culture techniques should be provided with the following: a washing area with a large sink, running hot and cold water, brushes of various sizes, detergent and a bucket of distilled water for a final rinse of the washed glass materials. A number of plastic buckets are required for soaking the glassware to be washed. Another separate bucket with lid is also required for disposing off the used or infected media before cleaning. Only this bucket should has to be kept outside the room or cleaning area.

15.1.1 List of apparatus required for tissue culture work:

1. Flasks (100, 250, 500 ml, 1 litre, 5 litre).
2. Volumetric flasks (500 ml, 1 litre, 2 litre, 3 litre).
3. Measuring cylinders (25, 50, 100, 500 ml, 1 litre).
4. Graduated pipettes (1, 2, 5, 10 ml).
5. Culture vials (culture tubes, screw – cap bottles of various size, Petri dishes, nipple flasks etc.) with suitable closure.

6. Plastic or steel buckets to soak glassware for washing.
7. Hot-air oven to dry washed labware.
8. Oven for dry-heat sterilization of glassware.
9. Wire-mesh baskets, to autoclave media in small vials and for drying labware.
10. Water distillation unit or demineralization unit, to obtain high quality water.
11. Plastic carboys (10, 20 litres) to store high quality water.
12. Balance, one to weigh small quantities and the other to weigh comparatively larger quantities.
13. Hot plate-cum-magnetic stirrer, to dissolve chemicals.
14. Vacuum pump, to facilitate filter sterilization.
15. Plastic bottles of different sizes, to store and deep-freeze solutions.
16. Refrigerator, to store chemicals, stock solutions of media, plant materials etc.
17. Deep freeze, to store stock solutions of media for longer periods, certain enzymes, coconut milk etc.
18. Steamer, to dissolve agar and melt media.
19. pH meter, to adjust pH of media and solutions.
20. Autoclave or domestic pressure cookers, for steam sterilization of media and apparatus.
21. Heat-regulated hot plate, domestic pressure cooker for steam sterilization.
22. Membrane filters and their holders to filter sterilize solutions.
23. Hypodermic syringes, for filter sterilization of solution.
24. Trolley with suitable trays, to transport cultures, media and apparatus.
25. Laminar air-flow cabinet, for aseptic manipulations.
26. Spirit lamp or Bunsen burner.
27. Automizer, to spray spirit in the inoculation chamber.
28. Screw-cap bottles, to sterilize plant material.
29. Instruments stand, to keep sterilized instruments during aseptic manipulations.
30. Large forceps with blunt ends, for inoculation and subcultures.
31. Forceps with finetips, to peel leaves.
32. Fine needles, for dissection.
33. Scalpels, for shredding of the tissues.
34. Spatula, to subculture friable tissues.
35. Cork-borer, for excising tissue cylinders of standardized size.
36. Binocular microscope, for dissecting out microscopic explants.
37. Air-conditioners, to maintain temperature of the culture room.
38. Shaker, to grow suspension cultures.
39. Stainless steel or Teflon sieves of various pore size to separate cell clumps of various dimensions.
40. Low speed bench centrifuge, to sediment cells for determining cell-packed volume and cleaning of protoplasts.

41. Haemocytometer, for cell counting.
42. Cavity slides, for hanging – drop cultures.
43. Ordinary microscope slides and cover-glass, to make microscopic preparations of cells and tissues.
44. Compound microscope, to observe cells and tissues.

15.1.2 History: The history of **in vitro** culture of plant tissues goes back to Schwann and Schleiden (1836) who put forward the theory of 'totipotency', which states that cells are autonomic and capable of regenerating to give a complete plant. This theory was the foundation of plant cell and tissue cultures. The first attempts by * Hyberlandt, the father of plant tissue culture, in 1902 on plant tissue culture techniques failed. However, between 1907 and 1909, Harrison, Burrows and Carrel succeeded in animal and human tissues in vitro. Although earlier workers have achieved **in vitro** culture of orchid seeds (seedlings), culturing of roots done successfully by Nobecourt, Gautheret and White independently in 1939. Plant tissue culture lagged behind, animal and human tissue culture because of the late discovery of plant hormones. The first regulator to be discovered, the auxin (IAA) created great opportunities for the **in vitro** culture of plant tissues. The discovery of the plant regulator kinetin (a cytokinin) in 1955 was a further stimulus in this direction.

A few important milestones in understanding the **in vitro** propagation of plants are :

1838 : Schleiden and Schwann proposed theory of Totipotency.

1892 : Sacchs observed that plants synthesize organ forming substances which are polarly distributed.

1902 : Haberlandt made first attempt to grow plant tissues **in vitro.**

1904 : Hanning made an attempt to culture cruciferous embryo.

1909 : Kuster demonstrated fusion of two plant protoplasts but the product failed to survive.

1922 : Knudson could make orchid seed to germinate **in vitro** without mycorrhizae.

1922 : Robbins could culture root tips.

1925 and 1929 : Laibach demonstrated interspecific crosses through embryo culture.

1934 : White cultured tomato root tips.

1934 : Gautheret demonstrated **in vitro** culture of the cambium tissue of a few trees and shrubs but failed to sustain since auxin had not yet been discovered.

1936 : La Rue cultured embryos of gymnosperms.

1939 : Gautheret, Nobecourt and White independently succeeded in maintaining growth of callus culture.

1940 : Gautheret studied adventitious shoot formation from cambial tissues through tissue culture.

1941 : Braun could grow Crown-gall tissues.

1948 : Skoog and Tsui demonstrated that formation of adventitious shoots and roots of tobacco is determined by the ratio of auxin/adenine.

1952 : Morel and Martin could get virus – free Dahlias by meristem culture.

1954 : Muri et al. were the first to get a plant from single cell.

1959 : Gautheret published first hand book on plant tissue culture.

1960 : Kanta for the first time demonstrated fertilization in papaver in test tube.

1964 : Giha and Maheshwari developed haploid plants from pollen grain of Datura.

1970 : Power et al. were successful in protoplast fusion technique.

1971 : Takebe et al. regenerated plants from protoplast.

In plant tissue cultures several techniques are being adopted. They are preparation of culture medium (nutrient medium), sterilization, aseptic manipulation and maintenance of culture. By these techniques one can get an organ culture, cellus culture, anther and pollen culture, protoplast culture, embryo culture etc. The details of the procedure is described below:

15.1.3 Preparation of medium: Plant cells or organs or tissues are cultured **in vitro** on a suitable nutrient medium which is known as culture medium (table 15.1).

A culture medium is composed of inorganic salts, vitamins, amino acids, growth substances and a carbohydrate supply. Inorganic salts are supplied in two groups as macro salts and micro salts. The salts needed in higher amounts are called macro-salts and they include nitrogen, phosphorous, sulphur, magnesium, calcium and potassium. The other essential inorganic salts needed in trace amounts are called micronutrients. Nitrogen is mostly provided in two forms as nitrates and as ammonium compounds. When nitrate is used along, the pH of the medium drifts towards alkalinity, but adding ammonium compounds together with nitrate, checks the drift of pH. In most media, iron is added in the form of sodium ethylene-amine tetra-acetate (Fe-EDTA) facilitating the gradual release of iron into the culture medium. Vitamins used in culture media are mesoinositol, nicotinic acid, pyridoxine, thiamine etc. Carbohydrate is supplied usually as sucrose. The most commonly used amino acid is glycine. In addition, auxins and cytokinins or their synthetic counterparts are added either singly or in combination to initiate and maintain cell division. The concentration and ratio of hormones may vary from plant to plant. The auxins that are commonly used in tissue culture medium are IAA (indole-3-Acetic acid), 2,4-D (2,4-dichlorophenoxy acetic acid), BTOA (2-benz-thiozyl acetic acid), NAA (x-naphthalene acetic acid) and IBA (3-indole butyric acid). The cytokinins are kinetin (6-furfuryl-amino purine), 6-BAP (6, Benzyl amino purine) Zeatin and 2 IPA (2, isopentenyl adenine). Gibberellic acid is rarely used in the medium.

The inorganic or organic chemicals used in the preparation of media should be analytical grades i.e., AR or GR (Guaranteed reagent).

Some of the plant tissues are grown in the presence of complex additives such as coconut milk, casein hydrolysate, yeast extract, ripe tomato extract, orange juice etc. Yeast extract is a good source of organic nitrogen and vitamins. Casein hydrolysate is obtained from milk by removing the cream and acidifying the skimmed milk which causes casein to precipitate without any decomposition. It contains all the common amino acids. Potato extract, ripe tomato extract, orange juice and water melon juice contribute a number of essential nutrients and vitamins. Diphenyl urea, a growth factor found in coconut milk, exhibits cytokinin – like responses. So, as a source of cytokinin, 10-15% V/V coconut milk is added to the medium.

It is not possible to weigh and mix all the constituents just before the preparation of medium. So it is convenient to prepare the concentrate stock-solutions of macrosalts, microsalts, vitamins, amino acids, hormones etc., all stock solutions should be stored in a refrigerator for a limited period.

15.1.4 Sterilization of nutrient media: Before placing on a medium, the seeds, parts of plants, organs and tissues etc., must be sterilized i.e., make free from all microorganisms.

Sterilization can be carried out as follows :

1. Physical destruction of microorganisms by dry hot air, steam or irradiation (UV light or gamma irradiation).
2. Chemical destruction of microorganisms by using microbicides (ethylene oxide, alcohol, hypochlorite etc.) or antibiotics.
3. Physical removal of microorganisms through bacterial filtration or washing.

Generally the medium is sterilized with the help of autoclave or large pressure cooker, less often by filtration and seldom by irradiation. The sterilized nutrient medium, glassware etc., should be stored in a sterile cupboard or metal box which has previously been disinfected.

(i) Autoclave : Usually nutrient media are sterilized with the help of an autoclave. In this, media are exposed to pressurized steam resulting in the elevation of boiling point of water up to 121°C which is sufficient to destroy all microorganisms. An autoclave has a temperature range of 115-135°C. Good sterilization relies on: time, pressure, temperature and volume of the object to be sterilized.

(a) Advantages of an autoclave are: Speed, simplicity, the additional destruction of viruses and no adsorption (this occurs with filter sterilization).

(b) Disadvantages of an autoclave are: Change in pH, components can separate out and chemical reactions can occur resulting in a loss of activity of media constituents.

Some of the guidelines to use an autoclave are:

1. Test tubes and flasks containing 20-50 ml nutrient media must be kept for 20 minutes at 121°C.
2. Flasks containing 50-500 ml nutrient media must be kept for 25 minutes at 121°C.
3. Flasks containing 500-5000 ml nutrient media must be kept for 35 minutes at 121°C.
4. Empty test tubes, flasks and filter paper must be kept for 30 minutes at 130°C.

It must be realized that the heat penetration is very important in an autoclave and large volumes must in principle be sterilized for longer periods. Material that can be dry sterilized (such as test tubes, empty flasks and petridishes, paper, instruments etc.) require 2-3 hours exposure to dry sterilization at 160°C.

The nutrient media and empty object such as glass, paper etc., should be sterilized separately. The same holds true for large and small flasks.

During and after autoclaving the following points should be taken into account:

1. The pH of the media is lowered by 0.3-0.5 units.
2. Autoclaving at too high temperature can burnt sugars, which may then be toxic.
3. Autoclaving for too long can precipitate salts and at the same time depolymerize the agar.
4. Care should be taken to use the correct duration of pressure and temperature (effective temperature).
5. It must be realized that volatile substances can be destroyed by the use of an autoclave (e.g. ether, ethylene).

6. If nutrient media slopes are needed (e.g. For embryo culture and culture of orchid seeds) the test tubes must be placed on a slope to set after autoclaving (at a temperature of 45-50°C).
7. It is recommended that de-ionized water is used in the autoclave as tap water usually contains too much calcium which gets precipitated and autoclave gets corroded.

(i) Sterilization by Irradiation: Sterilization of nutrient media through irradiation (via gamma rays) is hardly ever-used in the growth of tissue cultures because it is extremely expensive when compared with the usual method of autoclaving. Although gamma ray sterilization is as effective as autoclaving, tissue culture growth is significantly less on media sterilized by this method. When sterilizing plastic containers, boxes, tubes etc., use of an autoclave is not possible gamma ray sterilization is used.

(ii) The Laminar Air-Flow Cabinet: (Fig. 15.1)

The preparation and cutting of explants, the cutting of calluses etc., is done on a sterilized glass plate or between sterile filter paper, it is necessary to carry out these tasks in a laminar air-flow cabinet (also called an inoculation cabinet). Research laboratories and commercial tissue culture laboratories always use laminar air flow cabinets to limit the possibilities of contamination.

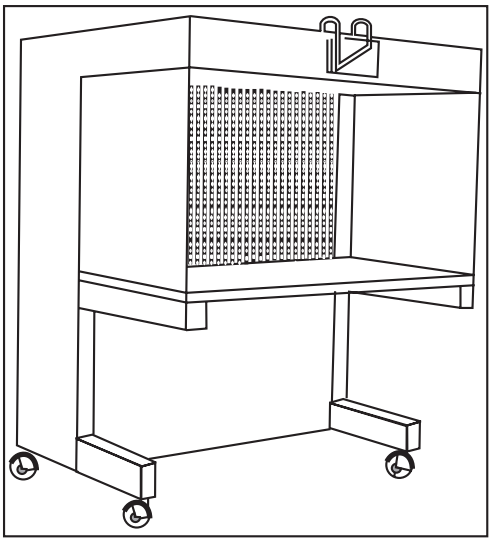


Fig. 15.1: A laminar air flow

A laminar air-flow cabinet is one in which the air is sucked from outside, first being filtered through very fine filters before reaching the table top of the inoculation cabinet. This filtering system ensures that the air flow over the table is (this flow being laminar giving the cabinet it's name) completely sterile. Since there is a continuous air flow through the inoculation cabinet, it is practically impossible that anything can pass from outside room into cabinet itself. When not in use the air flow cabinet can be closed from the outside air with a plastic cover. The air flow can be regulated and fluorescent tubes, for illumination are fixed in the roof of the cabinet. A gas supply is needed on the table top for use in flaming. However, flaming may be substituted by a spirit lamp or so called dry sterilization process.

In modern laboratories the laminar air-flow cabinet is built in a special (clean) isolation room, which is kept sterile and dust free by the use of filters. The room is vacuumed so that non-sterile air from outside cannot enter.

The filters of the laminar air-flow cabinet should be regularly vacuumed and replaced annually. The floor of the room containing cabinet should be decontaminated every day and only clean indoor shoes and clean laboratory coats should be used. Visitors from outside are particularly a source of infection.

(iv) Sterilization by filtration: All the particles, microorganisms and viruses which are bigger than the pore diameter of the filter used in a medium are removed. The greatest advantage of this method of sterilization is that thermolabile substances (broken down during autoclaving) can be sterilized without any harm. Disadvantages can be adsorption of substance into the filter. Sometimes virus particles pass through the filter, this procedure is time consuming and not as simple as autoclaving.

15.1.5 Types of cultures (Fig. 15.2)

Plants consist of different organs, each being composed of different tissues, which inturn are made up of individual cells. If the cell walls of these cells are enzymatically digested, protoplasts are produced. For growing the above plant materials different techniques have to be employed.

1. **Culture of intact plants:** A seed may be sown **in vitro** from which a seedling and finally a plant develops.
2. **Embryo culture:** An isolated embryo is grown after removal of the seed coat on a nutrient medium.
3. **Organ culture:** An isolated organ is grown **in vitro**. They are called meristem culture, shoot tip culture, root culture, anther culture etc., after a part (tissue mass, organ) which has been isolated from a plant is referred to as an explant and the culture of this is an explant culture.
4. **Callus culture:** (Fig. 15.2) If a mass of tissue is isolated, allowed to grow **in vitro** it gives rise to callus tissue and the process is known as callus culture.
5. **Single-cell culture:** The growing of individual cell which is obtained from a tissue or callus with the aid of enzymes or mechanically is called as single-cell culture.
6. **Protoplast culture:** The culture of protoplasts obtained from cells by enzymatic digestion of the cell wall.

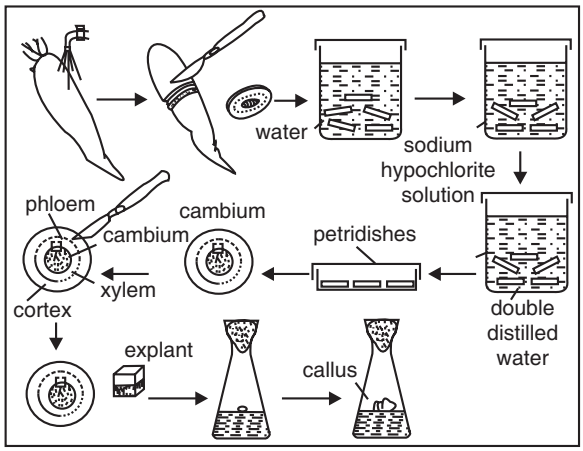


Fig. 15.2

15.2 ORGAN CULTURE

Culturing of plant organs on nutrient medium is called as organ culture. It is of different types:

Vegetative organs (growing from vegetative parts)	Root culture leaf culture Shoot culture
Reproductive parts (growing from Reproductive organs)	Anther – pollen culture Ovary culture Ovule culture Embryo culture

15.2.1 Root Culture: The culturing of root tips of aseptically germinated seeds in a nutrient medium is known as root culture (Fig. 15.3).

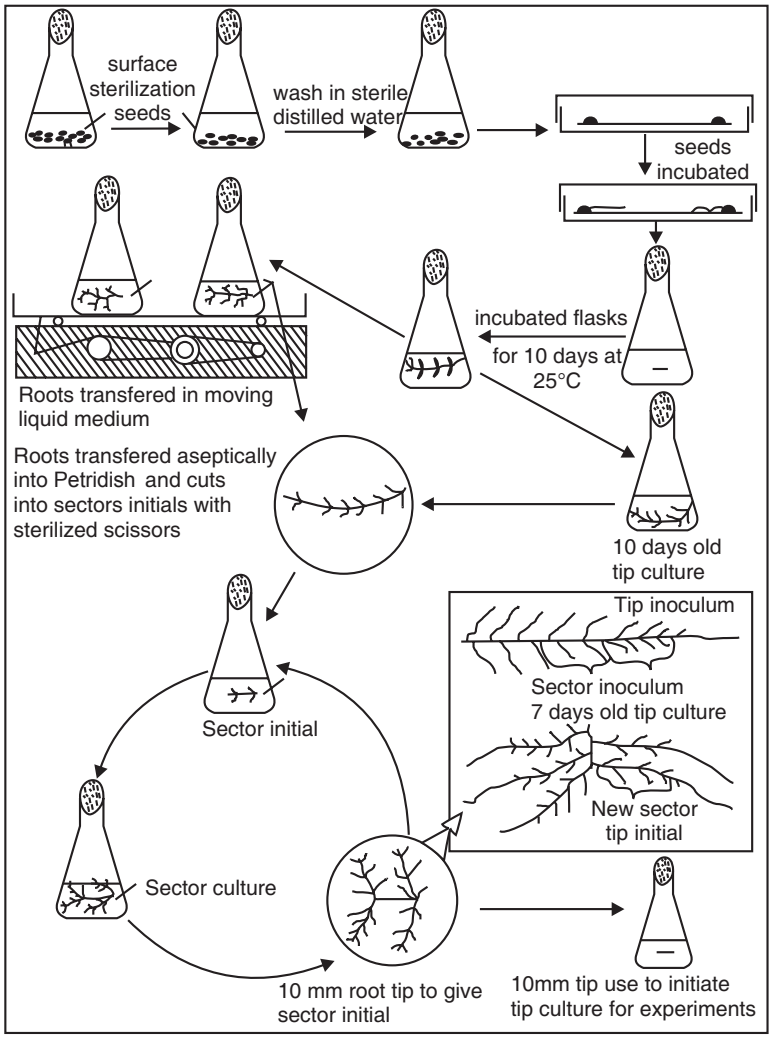


Fig. 15.3: Procedure for the maintenance of isolated roots in continuous culture

The root tip culture was developed by Robbins of USA and Kotte of Germany in 1922. Kotte cultured the root tips of maize and pea on a variety of nutrient media but he could not succeed in the root culture experiments. In 1934, White was successful in culturing the root tips of tomato plant. He prepared a medium in which the yeast extract was replaced by pyridoxine, thiamine and nicotinic acid of B-vitamin complex. This medium is commonly known as White's medium. Later, this medium was proved as one of the best basic media for **in vitro** root culture experiments.

Importance of root culture:

1. Root tip cultures increase our knowledge of carbohydrate metabolism, role of mineral ions, vitamins etc., in the growth of roots.
2. These cultures proved the dependence of roots on shoots for the growth hormones.
3. Root culture of Leguminous plants help us to study the process of nodule formation by nitrogen fixing bacterium i.e., **Rhizobium** sps.
4. Root cultures also help us in locating the biosynthesis of alkaloids by the roots and also to get increased production of such compounds ex: Hyoscyamine is an alkaloid commercially produced from the root cultures of **Datura stramonium**.

15.2.2 Leaf culture: This culture is obtained from excised young leaf primordia or immature leaves on a chemically defined medium. T.A Steeves and I.M. Sussex (1966) successfully cultured the leaf primodia of Ferns Ex: **Osmunda**. They have choosen the very young leaves or leaf primordial for the leaf culture. Mature leaves are not useful for the **in vitro** culture experiments (Fig. 15.4).

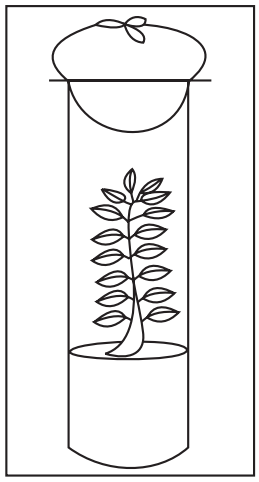


Fig. 15.4

Importance of leaf culture:

1. Leaf culture is useful to study the effects of various nutrients, growth regulators and environmental factors on leaf growth.
2. In Ferns, leaf culture helps us in studying the various developmental stages of sporangia.

15.2.3 Shoot tip culture: The culturing of terminal portion of shoot tip with meristem is called shoot tip or meristem culture. Loo (1945) reported the culture of shoot tips **Asparagus.** Ball (1946) devised a method to identify the exact part of the shoot systems that gives rise to a whole plant. Morel and Martin (1952) showed that virus free plants can be obtained by this method. The excised stem tip or meristem can be cultured aseptically on nutrient medium. The meristem grows directly into a simple leafy shoot or multiple shoots (Fig. 15.5).

Importance of shoot tip culture:

1. Generally the meristematic tissue is free from virus due to their fast mitotic activity. Therefore shoot tip culture of a plant is ideal for producing virus free stock.
2. Asexual or vegetative propagation of whole plants using shoot tip culture is known as micropropagation.
3. Many plants produce seeds that are highly heterozygous in nature. Such seeds are not suitable for storing genetic resources. Genetic resources of such plants can be stored by shoot tip culture.

4. Propagation of haploid plants. Haploid plants derived from anther/pollen culture always remain sterile until and unless they are made homozygous diploids. Meristem or shoot tip culture of haploid plants can be used for the propagation of such haploid plants.

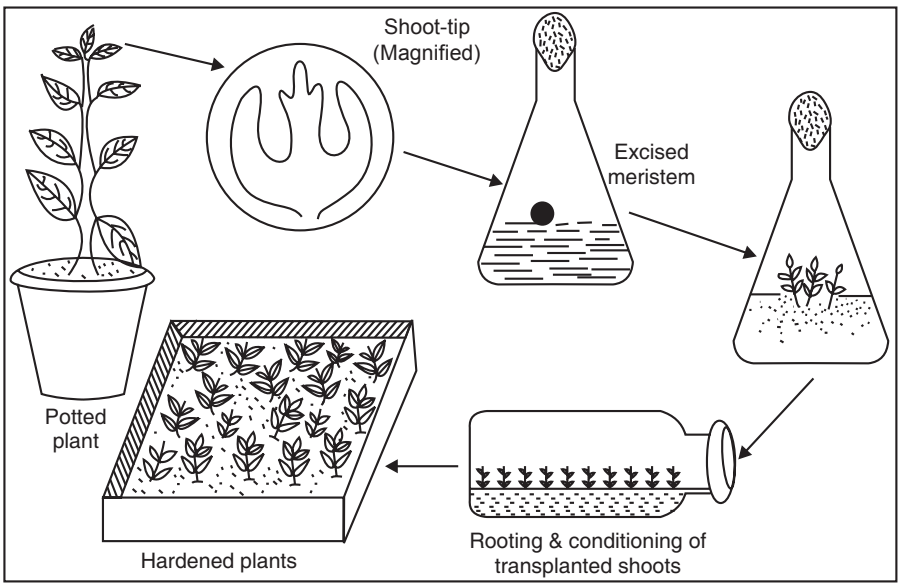


Fig. 15.5: Flow diagram illustrating the technique of shoot tip

15.2.4 Anther culture: The anther culture was first developed by Shimakura in 1934 while studying the physiology of meiosis. The first haploid callus was obtained from the pollen grains of **Ginkgo biloba** by Tulecke in 1953. Subsequently haploid tissues were initiated from a large number of gymnosperms and angiosperms. In 1964, Guha and Maheshwari from the University of Delhi reported the formation of direct embryos from the cultured anthers of **Datura innoxia (**Fig. 15.6). The origin of these embryos was traced to pollen grains and the plants formed were haploids. Later, with the technique developed by Nitsch, it became possible to culture isolated microspores and generate haploids from them. It also became possible to double the chromosome number of microspores using chemicals like colchicine and generate homozygous diploids from cultures. The technique of haploid cultures resulted in the selection of improved varieties of paddy, tobacco, wheat, tomato and mustard etc. Some excellent contributions in anther culture and pollen embryogenesis came from Sunderland and his co-workers. They developed float anther culture technique where the anthers with Uni or binucleate pollen were floated on a nutrient medium and after a period of time the anthers burst and released the pollen into the nutrient medium. These pollen grains later developed into embryoids.

One of the most significant developments in plant biology during last two decades has been the discovery of **in vitro** androgenesis i.e., the formation of sporophyte from the male gametophyte. e.g. Solanaceae and Gramineae members.

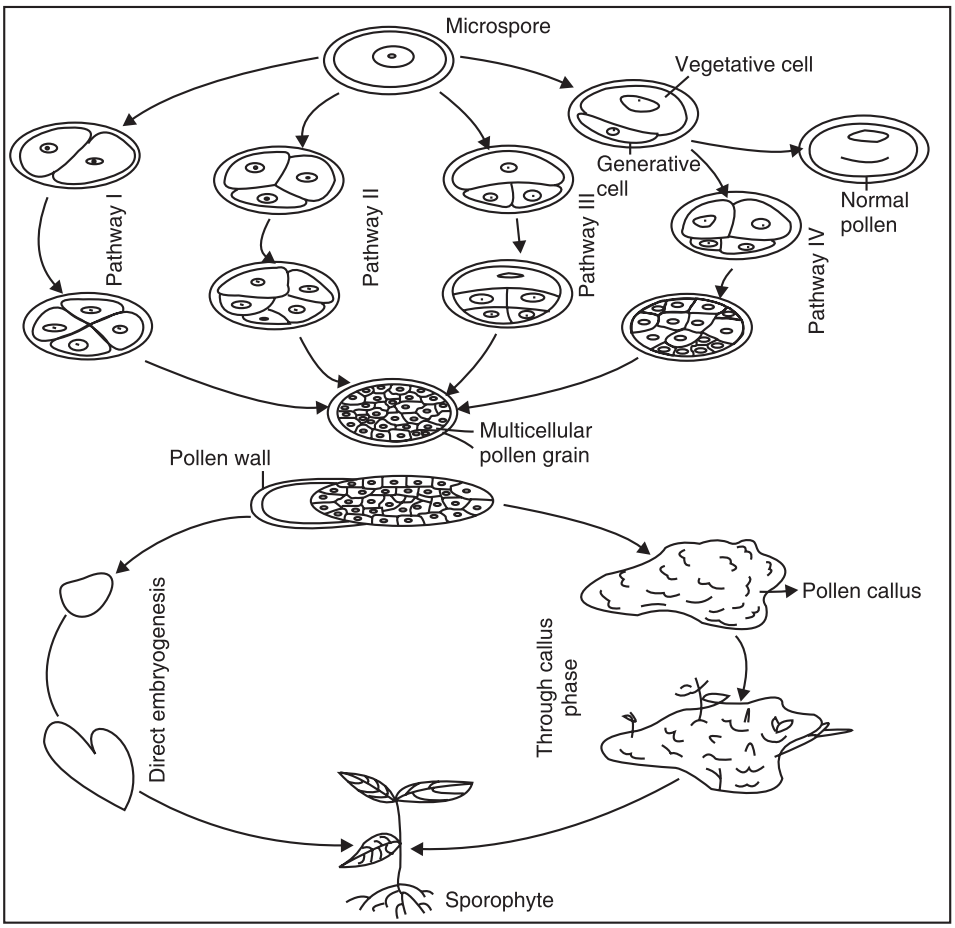


Fig. 15.6: Diagram showing the origin of sporophytes

The haploid callus, embryo and plantlet may originate in any one of the following ways:

1. From the continued divisions of the vegetative cell of the pollen. The generative cell here plays no role and soon degenerates.
2. From the division and redivision of the generative cell, not the vegetative cell.
3. From the division products of both the vegetative and generative cells.

Importance of Anther and Pollen culture:

1. Haploid plants derived from anther or pollen culture and useful in cytogenetic studies.
2. By comparing the heterozygous diploid with haploid homozygous diploid population recessive phenotypic characters can be identified very easily.
3. Critical genetic analysis of haploid population derived from individual microspores of pollen tetrad is very useful for the study of genetic recombination in higher plants.
4. The series of cell divisions and mode of differentiation (embryogenesis or organogenesis), starting from single cell (microspore) and ending in whole organism can be studied under microscope.

5. Double haploid, that are homozygous and fertile are readily obtained, ending the selection of desirable gene combination.
6. Culture of isolated pollen provides a novel experimental system for the study of factor controlling pollen embryogenesis of higher plants.
7. Study of meiotic behaviour of haploids provides valuable clues to measure chromosome duplication within a species and for understanding a phylogenetic relationship among species. It also provides information for the interpretation of chromosome homology.
8. Genetic analysis could be performed on haploid population to establish inheritance patterns.
9. The use of haploids in the production of monosomics, nullisomics and other aneuploids.

15.2.5 Ovary culture: Tissue culture raised from the ovaries of pollinated or unpollinated flowers is called ovary culture.

The technique of ovary culture was first developed by Nitsch (1951). He cultured the detached ovaries of **Lycoperisicon esculentus**, **Cucumis anguria, Nicotiana tobacum** and **Phaseolus vulgaris** on synthetic medium. The ovaries of **Cucumis** and **Lycopersicon** excised from pollinated flowers developed into matured fruits with viable seeds. Maheshwari (1958) succeeded in rearing the ovaries of **Iberis amara** excised from flowers one day after pollination. Nitsch (1951) obtained seedless fruits of tomato from unpollinated ovaries cultured on a medium with synthetic auxins like 2, 4-D (2,4-dichlorophenoxy acetic acid) or NOA (2, Naphthoxy acetic acid) or 2,4,5-trichloro phenoxy acetic acid).

Sachar and Guha (1962) observed that the achenes of **Ranunculus** produced in the fields remain dormant for about an year, but the achenes formed in ovary culture germinated without any dormant period.

Technique:

1. Collect the pollinated or unpollinated flowers from a healthy plant.
2. Wash them thoroughly with tap water, dip into 5% Teepol solution for 10 minutes and again wash to remove the trace of teepol.
3. Transfer the flowers to laminar air flow cabinet, surface sterilize the flowers by immersing in 5% sodium hypochloride solutions for 5–7 minutes. Wash them with sterile distilled water.
4. Transfer the flowers to a sterile petridish using a flamed forcep and a surgical scalpel, dissect out the calyx, petals, anther filaments etc., leaving only the pistil. During isolation of pistils, care should be taken to ensure that the ovaries are not injured in any way. Damaged pistil should be discarded as they often form callus tissue from the damaged parts.
5. Place the ovaries on nutrient agar medium.
6. Incubate cultures at 25°C for 16 hrs in day light. The light is provided by fluorescent lamp.

Importance of ovary culture

1. To study the early development of embryo, fruit, different aspects of fruit physiology including respiration, maturation and disease.

2. To study the effect of phytohormones on parthenocarpic fruit development from the culture of unpollinated pistil.
3. To study the role of floral organs in the fruit development. It has been found that pollinated ovary produce the normal fruits **in vitro**, if the sepals are not removed before culture. If sepals are removed before culture, addition of sucrose (5%) is necessary to obtain satisfactory growth of ovaries in culture. In barley, lemma and palea are very important. In onion, the growth of ovaries without perianth is markedly inhibitory.
4. Ovary culture has been successfully employed to obtain hybrids of diploid **Brassica chinensis** and autotetraploid of **B. pekinensis** which are normally cross incompatible.
5. Ovary culture has also been successful in inducing polyembryony. Polyembryony may develop in culture from the various parts of the ovary. These polyembryos give rise to many shoots instead of a single plantlet.
6. The culture of ovaries of apomicts may, therefore, help in understanding the nature of stimulus provided by pollination.

15.2.6 Ovule culture: Ovule culture is an elegant experimental system by which ovules are aseptically separated from the ovary and grown on nutrient medium.

In 1932, White for the first time cultured ovules of **Antirrhinum majus**. However, the technique of ovule culture was perfected by Maheshwari in 1958 by culturing of **Papaver somniferum.** Maheshwari and Lal (1961) could raise ovule culture of **Papaver somniferum** from ovules excised 6 days after pollination. Seeds formed in these cultures germinated immediately.

Technique:

1. Collect the open flower where the anthers are dehisced and pollination has taken place. To ensure the fertilization, collect the flower after 48 hours of anther dehiscense.
2. Separate sepals, petals, androecium etc., from the ovary containing either fertilized or unfertilized ovules.
3. Soak the ovary in 6% NaOCl solution.
4. Rinse the ovaries 3-4 times with sterile distilled water.
5. Ovules are gently prodded aseptically with the help of spoon shaped spatula by breaking the funicules at its junction with placental tissue.
6. The spatula with ovules is gently lowered into the sterile solid or liquid medium as the culture vial is slanted about 45°.
7. Incubate the ovule culture in either dark or light at 25°C.

Importance of ovule culture

Isolated ovule culture as early as the zygote of two to four celled proembryo stage is of considerable importance. It has the following important applications.

1. **Test tube pollination and fertilization:** By this technique, it may be possible to germinate pollen in the same culture as the excised ovule and to induce **in vitro** fertilization. Excised unfertilized ovules of **Argemone, mexicana, Papaver somniferum, Nicotiana tabacum** have been cultured along with their respective pollen grains. All the stages of development starting from the germination of pollen to double fertilization have been observed and the mature seeds containing viable embryos have been obtained by this method. Using the same method, it has been possible to fertilize the ovules of

Melandrium album with the pollen grains from other species of caryophyllaceae and subsequently even with pollen of **Datura stramonium.**

2. **Application of ovule culture in hybridization:** In many interspecific and intergeneric crosses, the F1-hybrid embryos frequently become abortive in the developing seeds or the F1 seeds are not capable to support the development of embryos. Ovule culture has been successfully employed to obtain hybrid seedlings in **Abelmoschus** and **Gossypium** etc.
3. **Production of haploid callus through ovule culture**: Uchimiya et al. (1971) cultured unfertilized ovules of *Solanum melangena* and obtained vigorous callus formation on a medium supplemented with IAA or kinetin. Although the origin of the callus tissue was not known, a cytological observation revealed it to be haploid in nature. So, it is an important attempt to obtain a haploid cell line or plant from an alternative source rather than anther or pollen culture.
4. **Ovule culture and angiospermic parasites :** It is generally believed that in obligate root parasites such as **Striga** and **Orobanche**, the formation of seedlings is dependent on some stimulus from the host root. Studies on ovule culture of **Orobanche** have demonstrated that the formation of shoots **in vitro** can be induced in the absence of any stimulus from the host.
5. **Ovule culture in orchid plants:** In nature the seeds of orchids germinate only in association with a proper mycorrhizal fungus. Thus many seeds fail to give rise to plants. Further, the seeds of many orchids take long time to mature. To overcome such problems several attempts have been made to culture the fertilized ovule of orchids **in vitro**.
6. **Induction of polyembryony:** In horticultural practices, the artificial induction of polyembryony holds a great potential. The nucellus of monoembryonic ovule of citrus can be induced to form embryos in culture.
7. **Virus irradiation through ovule culture:** In citrus, the ovule culture has proved decisively advantageous to make them virus free.

15.2.7 Embryo culture: Embryo culture is the sterile isolation and growth of an immature or mature embryo **in vitro** with the goal of obtaining a viable plant. Hanning (1904) for the first time obtained viable plants **in vitro** from excised embryos of **Cochleria** and **Raphanus.** Overbeek et al. (1941) discovered that coconut milk stimulated the growth of Datura embryos. This facilitated embryo culture **in vitro**. Randolf (1945) isolated **Iris** embryos to shorten the life cycle. In 1945 Cox and his co-workers grew embryos of different cultivars of cabbage to speed up seed germination.

In principle there are two types of embryo culture:

1. **Culture of immature embryos originating from unripe seeds:** This type of embryo culture is mainly used to avoid embryo abortion (early death of the embryo), with the purpose to produce a viable plant. This type of culture is difficult, not only because expertise in dissection is necessary, but also a complex nutrient medium is needed. The chances of success of this type of culture depend strongly on the developmental stage of embryo when it was isolated.
2. **Culture of mature embryos derived from ripe seeds:** This type of culture is relatively easy and is used to eliminate the inhibition of seed germination. The use of a simple nutrient medium with agar, sugar and minerals for this type of culture is sufficient.

If the development of immature embryos **in vitro** and **in vivo** is compared starting from the globular stage then the embryos **in vitro** regularly have (Monnier, 1980)

1. A more bulky growth and are pear shaped.
2. A longer globular stage.
3. Initially only one cotyledon (in dicotyledonous plants) while two develop **in vivo** simultaneously.
4. The possibility of showing polycotyledonous development (more than two cotyledons).

In vitro grown embryos usually exhibit early germination (Jensen 1976) since there is a loss of inhibitors when the seed coat is removed, another reason might be that the (negative) osmotic potential **in vivo** has a much higher value than **in vitro.**

Techniques: With embryo culture there are normally no problems with disinfection. Single mature seeds are externally disinfected, then embryo is removed after the seed coat is cut open. The dissection of the embryos produces more problems. It is possible to dissect large embryos without the use of a microscope, but this must be used with a cold light source, for small embryos. Inoculation needles and empty holders with a piece of razor blade mounted at the end are used during the dissection procedures. Care should be taken when cutting the seed as it is easy to see where the small embryos are located and consequently there is a less chance of damaging them, although care should be taken not to damage the suspensor.

Isolation of cherry embryos is as follows: (Theilur 1971) First of all the stones are removed from the fruits, then the stones are placed into a water bath to see which embryos are good (those stones that sink in water). These are then disinfected. The sterile cherry stones are then cracked and the embryo can be seen. The embryos are then removed with the help of pointed forceps and inoculated on to a solid medium.

Application of Embryo Culture:

1. Due to the inhibition of seed germination in a few species it is absolutely impossible to obtain germination invivo. In these cases embryo culture is essential. **Musa balbisiana** (Cox et al. 1960) **Pinus armandii × P.Koraiensis** (Bulard and Degivry, 1965) and **Colocasia esculenta** (Raghavan, 1977).
2. Seeds of obligate parasites will not be able to germinate without host, but it can be achieved with embryo culture. Johri and Bajaj (1962) successfully cultured the young embryo of **Cuscuta reflexa.**
3. Shortening of the breeding cycle or life cycle. The life cycle of **Iris** was reduced from 2-3 years to less than one year by using embryo culture techniques.
4. **Production of Haploids:** With the cross ***Hordeum Vulgare* × *H.bulbosum*,** fertilization occurs but there after the chromosomes of *H. bulbosum* are eliminated. The result is that the haploid embryo of **H.Vulgare** remains which is viable only by embryo culture.
5. Prevention of embryo abortion with early ripening of stone-fruits like cherry, peach, apricot and plum. The transport of water and nutrients to the yet immature embryo is sometimes cut off too soon, resulting in the abortion of the embryo. To overcome this problem embryo culture is the only solution.
6. The prevention of embryo abortion as a result of incompatibility with interspecific crosses, intergeneric crosses and with crosses between diploids and tetraploids the endosperm often develops poorly or not at all. The results in embryo abortion **in vivo,** which can only be avoided by embryo culture.

7. **Vegetative propagation:** The embryos of gramineae and conifereae are often used as a starting material for vegetative propagation.

15.2.8 Applications of plant cell and tissue culture: The importance and application of plant cell and tissue culture in plant science are vast and varied. During last few years of research in plant cell, tissue and organ culture have seen the emergence of technology from technique. Plant tissue culture has become an important tool in biotechnology. The establishment of micropropagation for rapid propagation, the use of shoot-tip culture to produce nuclear stock free from pathogens especially viruses and the application of a variety of procedures including anther and pollen culture to speed up the process of producing better varieties, protoplast culture for hybrid plant production and genetic manipulation have all contributed to the acceptance of plant tissue culture as a valuable tool for plant improvement. Further, a considerable commercial interest has been developed in the exploitation of this new technology. The principle applications of plant tissue culture can be discussed under the following subheadings.

(i) Micropropagation: The regeneration of whole plant through tissue culture is popularly called micropropagation. This is a technique where a callus mass has been initiated from a single explant taken from any living part of a donor plant and within a very short time and space, a large number of plantlets can be produced from such callus tissue. Again, it is possible to make a large number of callus pieces from the origin stock culture during subculturing. Then it is possible to produce hundreds of plantlets that develop on each of these callus pieces. Therefore, the most obvious advantage of micropropagation is the numerical one. Large numbers of plant species or varieties can be propagated all the year around. The plant breeder or grower is no longer restricted by season in the production of large numbers of plants.

(ii) Clonal propagation: In vitro clonal propagation is a type of micropropagation. The cultured plants raised from tissue cuture are derived asexually and also multiply within culture vessel by asexual means. Asexual reproduction gives rise to plants which are genetically identical to the parent plant. Multiplication of genetically identical copies of a cultivar by asexual reproduction is called clonal propagation and a plant population derived from a single donor plant in tissue culture constitutes a clone. So, the variability that can arise from sexual reproduction and seed formation in a seed plant is omitted. More specifically a single plant with desirable characters can be selected from a breeding programme and propagated so that further trials and selections can be carried out as quickly as possible. Plants with long seed dormancy can be raised faster by **in vitro** clonal propagation than **in vivo** seed propagation. **In vitro** clonal propagation is the only commercially viable method of micropropagation in orchid plants. Clonal multiplication of cultivar is very important in horticulture and silviculture.

(iii) Production of genetically variable plants: There is a major tendency of the callus tissue towards the numerical variation of chromosomes in the cells that occur after a number of serial subcultures. These variations may be due to: (1) The cells of various ploidy and genetic constitution of the initial explant may be induced to divide. (2) Cultural conditions may contribute to irregularities. The chromosomal instability in the cultured cells play an important role in polyploidization of cells and genetically variable plants can be raised from such polyploidized cells by subsequent micropropagation. Thus, tissue culture is, providing to be rich and novel source of variability with a great potential in crop improvement without resorting to mutation or hybridization. Variants

selected through tissue culture are calliclones (from callus culture), protoclones (from protoplast culture) but Larkin and Scowcroft proposed a general term Somaclones for plant variants achieved from tissue cultures, irrespective of their origin. Such variant plants may show some useful characters such as resistance to a particular disease, herbicide resistance, stress tolerance etc. Such changes are valuable for crops which are normally propagated by vegetative methods. Plant breeders can exploit such variants for their breeding programme.

(iv) Plant pathology and plant tissue culture: There have been many valuable contributions of plant tissue culture to the problems of plant pathology.

One outstanding success is the virus eradication by apical meristem culture and second success of tissue culture in plant pathology is the result of its application to their problems of plant tumors especially crown gall.

Virus eradication: In virus infected plants, the distribution of viruses in plant body is uneven. It is well known that the apical meristems are generally either free or carry a very low concentration of viruses. The apical meristems culture is the only way to obtain a clone of virus free plant which can be multiplied vegetatively under control conditions that would protect them from the chance of reinfection.

Virus eradication by apical meristem culture has enormous horticultural and agricultural value. In the horticultural world, the production of plants for the cut flower industry must be maintained virus free and are used for the propagation and breeding.

In the agricultural world, the production or yield of a crop can fall due to viral infection. Tissue culture techniques could be of value in restoring the original characters of the variety by removing the infection and so bringing it back into the commercial market. These virus tested stocks could provide ideal material for the national and international distribution of plants either for further propagation or use as breeding material.

(v) Study of crown gall by plant tissue culture: Smith and Townsend (1907) discovered that crown gall or plant tumor formation can be induced by a bacterium, **Agrobacterium tumefaciens.** Braun (1941) showed that in sunflower **Agrobacterium** could induce tumors not only at the inoculated point but at considerable distance, where secondary tumors free of bacteria are formed. Cells of these medium without adding auxin and cytokinin, where as normal tissue requires auxins and in some cases cytokinins. Crown gall tissues deprived of bacteria give rise to tumors by grafting. But the basic mechanism of tumor transformation has not been clarified. It is hoped that plant tissue culture technique can throw some light on the basic mechanism of tumor transformation.

(vi) Plant breeding, plant improvement and plant tissue culture: The conventional breeding methods are most widely used for the crop improvement. But these methods have to be supplemented with plant tissue culture techniques to increase their efficiency or to be able to achieve the objective which is not possible through the conventional methods. Embryo culture is useful for shortening the breeding cycle, overcoming seed dormancy and incompatibility. It is also useful in the propagation of orchids.

In meristem culture, shoot apical meristem along with some surrounding tissue grown **in vitro** is used for clonal propagation recovery of virus free plants, potentially useful in germplasm exchange and long-term storage of germplasm through freeze preservation. Anther and Pollen culture has a potential application in plant breeding and plant

improvement programme for the production of haploid as well as homozygous diploid plant.

All year round rapid clonal propagation using plant tissue culture techniques has highlighted the possibilities for new plant improvement techniques. Another most important approach is the mutation of tissue culture cells to produce a mutant line from which plants can be raised. Production of mutant line is highly desirable for plant breeding.

(vii) Production of useful biochemicals: Man depends on plants for many compounds other than food such as medicines, pigments, vitamins, hormones, flavoring agents latex and tannins. It is possible to culture cells from a plant that naturally produces a certain biochemical and cause the culture to produce that chemical **in vitro** conditions. We do not yet understand the regulation mechanisms that control the production of most biochemical substances and so that we can manipulate them.

Number of cell cultures have been found to produce specialized biochemicals which are found in the intact parent including alkaloids such as nicotine, atropine, ephedrine, caffeine, codeine, their precursors and derivatives. Production of cordiac glylosides, other steroids, benzquinones, latex, phenolics, anthocyanins, organic acids, anti-tumor agents, anti-microbials and various flavours, odours have also been reported.

(viii) Application of plant tissue culture in forestry: Forest is an important renewable natural resource for man because it provides several forest products like fuel, timber, paper, fodder etc. But with the spread of industrial civilization and the rapid growth of population, unfortunately forests tend to disappear. This should be stopped immediately and the conservation of forests should be done by adopting some scientific and technical methods. The most common traditional measure is the rapid propagation and scientific plantation of forest trees. But there are some problems to propagate forest trees by the conventional methods. To overcome such problems, plant tissue culture technique has been employed and exploited for rapid multiplication of forest trees within very short time and to produce new varieties. By this technique, millions of potential forest trees could be maintained in a few test tubes. For ex: micropropagation of forest trees like **Sequoia**, sempervirens, **Tectona grandis, Eucalyptus citriodora** etc.

Micropropagation means the shoot multiplication cycle is very short i.e., within 2-6 weeks each cycle results growth in the number of shoots. By apical meristem culture in **Manihot esculenta** commonly known as cassava, we can eliminate virus and raise the number of virus free plants. By embryo culture, a hybrid between **Pinus lambertiana** x **P.armandi** and **P.lambertiana** x **P.koraiensis** can be grown successfully. It is a disease resistant pinus plant. Endosperm culture in *Populus tremuloids* (plywood yielding plant) resulted in producing a better pulp wood quality plant compared to diploid plant obtained from seeds.

(ix) Importance of plant tissue culture in biotechnology: Biotechnologists are trying to modify the genetics of the cultured cells by one of the three ways such as:

(i) Mutagenesis and selection of cell lines in cell suspension culture.

(ii) Transplantation of foreign genetic material in protoplasts by means of genetic engineering.

(iii) Somatic hybridization by the fusion of protoplasts brings together in a single plant, genes of different species that are too unrelated to allow mixing, thereby allowing creation of hybrid or cybrid plants for the improvement of crop species.

Another goal of biotechnology using plant tissue culture technique is to produce self fertilizing plant that would provide their own usable nitrogen. Generally plants rely on a few types of bacteria and cynaobacteria to fix atmospheric nitrogen into a biologically usable form. Genetically engineered plants that contain the bacterial genes for nitrogen fixation could be grown well even in nitrogen deficient soils. Genetic engineers are also developing plants that produce their own pesticides. The value of plant protein to human diet is being improved by creating corn or beans that manufacture a complete protein, one with all amino acids essential to the human diet.

In another effort the luciferase gene from the firefly **(Photinus pyralis)** was used for light production in transected plant cells and transgenic plants. A complementary DNA (cDNA) clone of the firefly luciferase gene under the control of a plant virus promoter (cauliflower mosaic virus 35 S-RNA promoter) was introduced into plant protoplast cells (Daucus carota) and into plants **(Nicotiana tabacum)** by the use of **(Agrobacterium tumefacience**) tumor induced plasmid. In the above experiment stable expression of the firefly luciferous gene in plant cell and transgenic plants has been achieved. The transgenic plant incubated in luciferin also emits light like firefly. The successful introduction of animal DNA into plant genome has opened a new avenue in the field of biotechnology.

15.3 ANIMAL CELL AND TISSUE CULTURE

The most important fields of today, research which have potential economic value and prospects for commercialization are cell and tissue culture based production of vaccines, monoclonal antibodies, pharmaceutical drugs, cancer research etc. Now-a-days several valuable products of medical use have been produced through animal cell culture or genetically engineered cells.

In 1907, Ross Harrison made first attempt to culture animal cells and cultivated embryonic nerve cells of frog by using hanging drop method. Thereafter this method was extended and a wide range of mammalian cells were cultured in vitro. The significance of animal cell culture was increased when viruses were used to produce vaccines on animal cell cultures in late 1940's. Similarly during 1950's polio vaccine production on cultured cells further increased the significance of animal cell culture technology.

The term tissue culture refers to the culture of whole organs, tissue fragments as well as dispersed cells on a suitable medium. Cell cultures are obtained either by enzymatic or mechanical dispersal of tissues into individual cells or by spontaneous migration of cells from an explant.

Freshly isolated cell cultures are called primary cultures. They are usually heterogeneous and slow growing, but are more representative of the tissue of their origin both in cell type and properties. Once a primary culture is subcultured, it gives rise to cell lines which may either die after several subcultures (finite cell lines) or may continue to grow indefinitely (continuous cell lines). Normal tissues gives rise to finite cell lines, while tumours give rise to continuous cell lines.

In 1913, Carrel developed a complicated methodology for maintaining cultures free from contamination, especially by bacteria. Subsequently, suitable culture media were developed and the techniques of cell culture were refined.

15.3.1 Requirement for animal cell and tissue culture: Culture desirable requirements are:

1. Air conditioning, 2. Hot room with temperature recorder, 3. Microroom for carrying out microscopic work, 4. Dark room, 5. Service room, 6. Sterilization room for sterilization of glassware and culture media, 7. Preparation room for media preparation etc.

15.3.2 Substrates for cell growth: There are many types of vertebrate cells that require support for their growth *in vitro* otherwise they will not grow properly. Such cells are called anchorage dependent cells. A large number of substrates which may be adhesive (e.g. plastic, glass, palladium, metallic surfaces etc.) or non adhesive (e.g. Agar, Agarose, etc.) types may be used.

(i) Plastics: Plastic wares are non autoclavable, are supplied sterile and are for a single use, polystyrene is the most commonly used, but polyethylene, polycarbonate, Perspex, PVC, Teflon, cellophane and cellulose are all suitable substrates.

(ii) Glass as a substrate: Glass is an important substrate used in laboratory in several forms such as test tubes, slides, coverslips, pipettes, flasks, rods, bottles, Petri dish several apparatus etc. These are sterilized by using chemicals, radiations, dry heat and moist heat.

(iii) Metals: Both stainless steel and titanium are suitable substrates, since they are relatively chemically inert and negatively charged. The proper grade of stainless steel must be chosen to avoid release of toxic ions and it should be acid washed to remove surface impurities etc.

Substrate Treatment: To increase the efficiency of cultured cells through increased cell attachment, the surface of the substrates is treated with purified fibronectin or collagen. The chemical is poured on to the surface of dish, excess amount drained, chemical is dried and sterilized by using UV light. In addition, the surface of substrates is also treated with monolayer of special cells which is called feeder layer because of cells. The special cells are used as feeder layer, chemical foetal, intestine, mouse embryo fibroblast, etc.

15.3.3 Culture media: Culture of animal cells and tissue is rather more difficult than that of microorganisms and plants because the latter synthesize certain chemical constituents (unlike microbes). There are two types of media used for culture of animal cell and tissue. The natural media and artificial media.

(A) Natural media: Natural media are the natural sources of nutrient sufficient for growth and proliferation of animal cells and tissue. These are of three types:

1. Plasma clots 2. Biological fluids 3. Tissue extracts

(i) Plasma clots: The most commonly used clots are plasma clots which have been in use for a long time. Plasma is now commercially available either in liquid or lyophilized state. It may also be prepared in the laboratory usually from the blood of male fowl.

(ii) Biological fluids: The various biological fluids used as culture medium (e.g.: aminiotic fluid, pleural fluid, aqueous humour from eye, insect haemolymph etc). Serum is most widely used. Serum may be obtained from adult human blood, placental cord blood, horse blood or calf blood. Of these foetal calf serum is commonly used.

(iii) Tissue extracts: Chick embryo extract is most commonly used but bovine embryo extract is also used. Others are spleen, liver, bone marrow, leucocytes etc. Tissue extracts can often be substituted by a mixture of amino acids and certain other organic compounds. For cell cultures, artifical media with or without serum are used.

(B) Artificial media: Synthetic media are prepared artificially by adding several nutrients (organic and inorganic), vitamins, salts, O_2 and CO_2 gas phases, serum proteins carbohydrates, cofactors etc. However, different types of synthetic media may be prepared for a variety of cells and tissues to be cultured. It can be prepared for different functions. Synthetic media are of two types: 1. Serum containing media and 2. Serum free media. Examples of some of the media are essential medium (MEM) (Eagle, 1955), 199(1950), CMRL 1066 (parker et al., 1957), RPM1 1640 and F12 (Han, 1965) (Table 15.1).

Table 15.1: Composition of different media (with serum)

	Eagles Mem	Dulbeccols Modification	Aams F12
1. Amino acids			
L.amparagine	126	84	211
L.cystine	24	48	--
L.glutamine	292	584	146
Glycine	--	30	75
L.Histidine	42	42	21
L.Isoleucine	52	42	21
L.leucine	52	105	13.1
L.lysine-HCL	73.1	146	36.5
L.Methoionine	15	30	4.48
L.phenylalaine	33	66	4.96
L.proline	--	--	34.5
L.serive	--	42	10.5
L.Threonine	48	95	11.9
L.Tryptophan	10	16	2.04
L.Typosine	36	72	5.4
L.Valine	47	94	11.7
2. Vitamins			
Biotin	--	--	0.0073
D-ca-patothenate	1	4	0.48
Choline chloride	1	4	14
Folic acid	1	4	1.3
Iositol	2	7.2	18
Nicotinamide	1	4	0.04

contd...

	Eagles Mem	Dulbeccols Modification	Aams F12
Pyridoxal, HCl	1	4	0.062
Riboflavin	0.1	0.4	0.038
Thiamin. HCl	--	--	1.36
Vit B12	--	--	1.36
Pyridoxin.HCl	--	--	0.062
3. Inorganic salts			
$CaCl_2$ (anhydrous)	200	200	44
$CaCl_2.2H_2O$	--	--	--
$Fe(NO_3)_3.9H_2O$	--	0.1	--
KCl	--	400	--
$MgCl_2.6\ H_2O$	--	--	--
$MgSO_4.7\ H_2O$	--	200	100
NaCl	--	6800	60
$NaHCO_3$	--	2200	3700
$Na_2H_2PO_4.H_2O$	--	140	125
$Na_2HPO_4.7H_2O$	--	--	--
$CuSO_4.5H_2O$	--	--	0.00249
$FeSO_4.7H_2O$	--	--	0.834
$ZnSO_4.7H_2O$	--	--	0.863
4. Other Chemicals			
D.Glucose	1000	4500	1802
Lipoic acid	--	--	0.21
Phenol red	10	15	12
Sodium pyruvate	--	110	110
Hypoxanthine	--	--	4.1
Linoleic acid	--	0.084	--
Putrescine.2HCl	--	--	0.161
Thymidine	--	--	0.73
CO_2(gas phase)	5%	10%	5%

When synthetic media are devoid of serum in culture medium, it is called serum free media. Examples are given in table. (Table 15.1) Serum itself has several disadvantages as:

1. It deteriorates within a year and differs with batches.
2. A number of batches are required if more than one cell types are used, which makes difficult for maintaining and co-culturing of cells.
3. Undesirable growth stimulation and inhibition may occur.

4. It is not chemically defined and therefore, it is of variable composition.
5. It may be source of contamination by viruses, mycoplasma, prions etc.
6. It is most expensive ingredients of the culture media.

However, it has some advantages such as:

1. Serum contains a complete set of essential growth factors, hormones, attachment and spreading factors, binding and transport proteins.
2. It binds and neutralizes toxins.
3. It contains protease inhibitors.
4. It increases buffering capacity.
5. It provides trace elements and other nutrients.

15.3.4 Sterilization: The thoroughly washed glassware, equipments (Table 15.2) and carefully prepared culture media are sterilized by steam or millipore filter paper (Table 15.3).

Table 15.2: Sterilization of glassware and equipments

Sterilization	Items
Autoclaving	Apparatus containing glass and silicon tubing, disposable tips or micropipettes dispenser tubing for pippet, glass bottles with screw caps, magnetic stirrer bars, screw caps, silicon tubing stopper, rubber, silicon, Millipore filters etc.
Dry heat	Glassware glass cover slips, shifted instruments, partewr pipette test tubes etc.
15 lb/in (=121°C)	For 20 minutes; ** 160°C/1hr

Table 15.3: Sterilization of liquids by different method

	Storage	Solution
Autoclaving	Room temperature	Agar, Antibiotics EDTA, Glucose, Glycerol, HETPES, lactal bumin, $NaHCO_3$, NaOH solution, phenol red salt solution (with glucose) tryptose, water
Autoclave	4°C	Methocel
Filter	Room temp.	Glucose solution HCl, $NaHCO_3$ Solution
Filter	20°C	Collagenase NaOH solution, Antibodies glutanine, sodium pyruvate solution, transferin, trypsin, vitamins
Filter Stacked Filters	4°C	Bovine serum albumin
Filter Stacked Filters	– 20°C	Serum
Steam (30 min at 100°C)	4°C	Carboxyl methyl cellulose

15.3.5 Isolation of animal material explants: Following an appropriate protocol developed to avoid contamination and retain cell survival. In general embryos and other contents of uterus are free from contamination, while in adult tissues bacterial contamination may be common. In simple terms, the site of opening is sterilized with 70% alcohol, tissues are removed and placed in balanced salt solutions EBSS or HBSS, supplemented with antibiotics.

(A) Disaggregation of explants: Cell culture are started from disaggregated explants occasionally, cells do migrate out of the culture whole tissues: such cells are separated by trypsin or other enzymatic treatment to establish cell cultures. There are three ways of disaggregation: 1. Mechanical, 2. Enzymatic, and 3. EDTA treatment.

(i) Mechanical disaggregation: Tissue is chopped into small pieces approximately 1 mm in size. They are then cultured into a suitable vessel. The pieces attach to the substrate either due to their own adhesiveness, the dish may be scratched to facilitate attachment or clotted plasma may be used. Cells grow out from the tissue pieces which are typsinized and subcultured. The explants themselves may be retained in the same dish or transferred to a new vessel to obtain further cell growth.

(ii) Enzymatic disaggregation: Trypsin and collagenase are the most commonly used enzymes, but other enzymes like elastase, mucase, papain etc. have also been used. Dissociated cells are collected, washed free of the enzyme and cultured. The trypsin and collaginase solutions are generally prepared in a balanced salt solution.

15.3.6 Subculture: When mechanically or enzymatically disaggregated tissue pieces or cells are cultured, they attach to the substrate and a monolayer of cells develop from them. These cultures are called primary cultures. Primary cultures do not contain many types of cells present in the tissue because they are unable to attach to the substrate and survive. The dead cells are easily eliminated when the medium is poured and subcultured.

After a period of time, the available subculture becomes occupied by the monolayer cells. Then they inoculated or seeded into a fresh vessel from the primary cultures. The cultures obtained by this method are called cell lines. The process of inoculating cultured cells into fresh culture vessels is termed as subculturing.

15.3.7 Establishment of cell cultures: There are many types of animal cells that can grow *in vitro* such as tumour cells, pigmented melanoma cells, steroid producing adrenal cells, growth hormone prolactin secreting cells, teratoma cells capable of differentiating in artificial conditions, pigmented or cortilage cells etc. On the basis of purpose of experiment a continuous (immortal) cell line can be developed from cultured tissues (Fig. 15.7). It is not necessary that the cells which are plated into cultured medium will start immediate growth. On the basis of growth responses, culture cells are divided into 3 types.

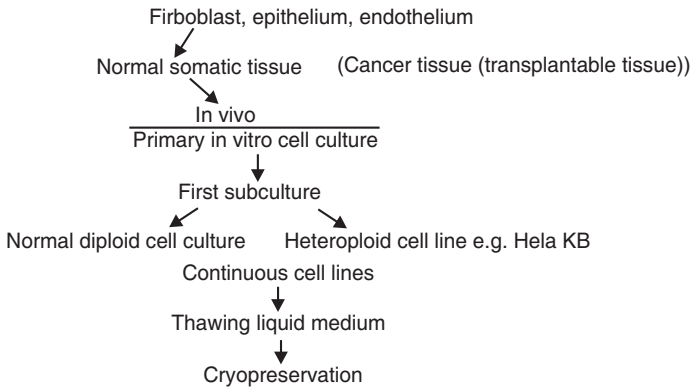


Fig. 15.7: Cell culture from somatic tissue (Foetus/Adult tissue)

1. Precursor or (system cells), 2. Undifferentiated but committed cells, and 3. The mature differentiated cells: The precursor cells have the ability to proliferate but they do not differentiate until the proper conditions for induction are applied in the medium. This is done to facilitate some or all cells to mature and differentiate. These are totitpotent cells which have the ability to differentiate into different types of cells. The entire blood and immune systems are produced by the totipotent cells. Cell cultures at some stages consist of above three types of cells.

(i) Evolution of cell lines: The mixture of cells in single suspension may be used as a primary culture or starter culture. The primary culture is sub-cultured in special growth nutrients at optimal growth conditions. Consequently, some cells attach to the surface and proliferate to yield single cell line, inspite of being damaged in suspension. So while subculturing the supension should be diluted with fresh medium at certain ratio and transferred into a flask.

The primary culture becomes a cell line only after its first subculture. The subculture is needed when the nutrients present in medium for cell growth diminish. These may be subcultured several times on the fresh medium and propagated accordingly. By doing so a continuous growth of cells is maintained. This results in multiple copies of a single type of cell with a negligible amount of non-proliferating cells. During the course of repeated subculture and selection the cell line gets evolved and properly established consisting of rapidly proliferating cells. Thus unaltered form of cell line is called continuous cell line which propagates in logarithmic way.

A cell line consists of several similar or dissimilar lineages. A defined cell lineage having specific properties is called cell strain. On the basis of life span of culture, the cell line or cell strains in a gas or finite.

(ii) Large scale culture of cell lines: Cell cultures are obtaining useful products like biochemicals (interferons, hormones, enzymes, antibodies etc.). For these objectives large scale cell cultures are essential. Fermentaters of upto 10000 lit are used for this purpose. The scalingup of cell cultures may be done as: 1. monolayer cultures, 2. suspension cultures, and 3. immobilized cell systems.

(iii) Monolayer culture: Monolayer cultures are essential for anchorage dependent cells. Scaling up of such cultures is based on increasing the available surface area or by using plates, spirals, ceramics and microcarriers.

15.3.8 Suspension culture: In scalingup both chemical and physical factors have to be optimized for good results. The medium must be suitably stirred to keep the cells in suspension and to make the culture homogeneous. The objective of stirring is to achieve good mixing without causing damage to cells. It is important to supply sufficient O_2 without damaging the cells. The reactors used for large scale suspension cultures are of 3 main types: 1. stirred bioreactors, 2. continuous flow reactors, and 3. air lift fermentators.

15.3.9 Immobilized cell cultures: Cultures based on immobilized cells offer several advantages: 1. Higher cell densities, 2. Stability and longivity of cultures, 3. Is applicable to both suspension and monolayer cultures, 4. Many systems protect the cells from shear forces due to medium flow, 5. Less dependence of cells at higher densitities on external supply of growth factors which saves culture cost.

15.3.10 Organ culture: Different organs of animals can be cultured on artificial medium. For organ culture care should be taken to handle in such a way that tissues should not be damaged. Two types of culture media are used for organ culture:

1. Natural media.
2. Synthetic media.

(i) Natural media: Natural sources of nutrient sufficient for growth and proliferation of animal cells or tissues. They are of three types: 1. Plasma clots, 2. Biological fluids—most commonly used are human, placental, foetal calf serum, and 3. Tissue extracts: extract from some tissues such as embryo, liver, spleen, tumours, bone marrow etc. are used.

(ii) Synthetic media: Prepared artificially by adding several nutrients, vitamins, salts, O_2 gas phases, serum proteins, carbohydrates, cofactors etc. However, different types of synthetic media may be prepared for variety of cells or tissues to be cultured. It is more easy to culture embryonic organs than the adult animals. Embryonic organs can be cultured by applying any one of the following four methods:

(a) Organ culture on plasma clots: A plasma clot is prepared by mixing 5 drops of embryo extract with 15 drops of plasma in a watch glass placed on a cotton wool pad. The cotton wool pad is put in petri dish. Time to time cotton is moistened so that excessive evaporation should not occur. Later, a small piece of organ tissue is placed on the top of plasma clot present in the watch glass.

(b) Organ culture on agar: Solidified culture medium with agar is also used for organ culture. The nutrient agar medium may or may not contain serum. When agar is used in medium, no extra mechanical support is required. Agar does not allow to liquefy the support. The tumours obtained from adults fail to survive on agar media, whereas embryonic organs grow well. The media consists of ingredients: agar, chick embryo extracts and horse serum in the ratio of 7:3:3.

(c) Organ culture in liquid media: The liquid medium consists of all the ingredients except agar. When liquid media are used for organ culture, generally perforated metal gauze or cellulose acetate or raft of glens paper is used.

(d) Whole embryo culture: For embryo culture, a suitable medium prepared is poured in to watch glasses, which are then placed on moist absorbent cotton wool pad in petri dishes. In case of chick embryo, the eggs are incubated at 38°C for 40-42 hours so that a dozen of embryos could be produced. The egg shell sterilized with 70% ethanol is broken into pieces and transferred into 50 ml of BSS. The vitelline membrane covering the blastoderm is removed and kept in petri dish containing BSS. Vitelline membrane is removed with the help of forceps. Developmental stage of blastoderm could be found when the embryo is observed under microscope. Blastoderm is placed on the medium in watch glass placed on sterile absorbant cotton wool pad in petri dishes. Excess of BSS is removed from medium and embryo culture of chick is incubated at 37.5°C for further development.

REVIEW QUESTIONS

I. Essay type questions:

1. Plant cells are cultured in the laboratory on a suitable medium. Describe the process of medium preparation.

2. Define sterilization. Describe the process of sterilization with the help of Autoclave.
3. Describe the process of Root culture and add a note on its importance.
4. Write an essay on Anther culture with its importance in tissue culture experiments.
5. Describe the technique of Ovary culture and write its importance.
6. Write about the Ovule culture and add a note on its importance.
7. What do you mean by Embryo culture? Write the applications of it.
8. Write an essay on the importance and applications of plant cell and tissue culture in plant science.
9. Describe different methods of animal cell and tissue cultures.
10. Give an account of animal organ culture.
11. Write an essay on the biotechnological applications of plant cell and tissue culture in agriculture.
12. Give a detailed account of production of secondary metabolites for cell culture.
13. What do you know about transgenic plants? Write in detail on selectable markers and their use in production of transgenic plants.
14. Write an essay on somatic embryogenesis.
15. How will you isolate protoplasts from plant cells? Give the applications of protoplast culture?
16. What is an explant? How will you induce callus from it?

II. Write short notes on:

1. Tissue culture laboratory.
2. Laminar air flow.
3. Write the different types of cultures.
4. Write about leaf culture.
5. Define shoot tip culture and add note on its importance.
6. Write the advantages of propagating the plants with the help of tissue culture.
7. Substrates for animal cell culture.
8. Natural media for animal cell culture.
9. Encapsulated seeds.
10. Production of virus free plants.
11. Edible vaccine.
12. Micropropagation.
13. Totipotency.
14. In vitro androgenesis.
15. Somaclonal variants.
16. Mention the basal ingredients of a minimal nutrient medium.
17. Why Agar-agar is used in the media?
18. Name the instrument generally used to sterilize the nutrient media.
19. Name the natural products which are added to the medium in certain tissue culture.

20. What should be the favorable pH of the nutrient media?
21. Name the indifferentiated mass of tissue produced during tissue culture.
22. Name the phytohormone used in the development of root from the callus.
23. Name the chemical used to obtain diploid homozygous plants from a haploid one.
24. Name the process by which organs are produced by culturing the plant parts without the formation of callus and embryoids.
25. Why cuticle is not formed in the plants grown by tissue culture?
26. Name the process where diploids (homozygous) are obtained from haploids without the application of colchicines.
27. Define clone.
28. What are calliclones?
29. Mention the disadvantages of an autoclave.
30. Who is regarded as father of plant tissue culture?
31. Name the apparatus used to count the cells.
32. Biological fluids in animal cell culture.
33. Animal embryo culture.

FURTHER READING

1. Biotol sense (1992) in vitro cultivation of plant cell: Butterworth – Heinemann, Oxford.
2. Biotol series (1992) Biotechnological Prentice Hall, New York.
3. Butler, M. (ed.) (1991). Mammalian Cell Biotechnology. A Practical Approach. Oxford University Press, New York.
4. Ignacimuthu, S. (1998) Plant Biotechnology Oxford and IBH Publishing Co. Pvt. Ltd., New Delhi.
5. Johri, B.M. (1982) Experimental embryology of vascular plants springer-verlag, Berlis Spier, R.E.(ed.) (2000). The Encyclopedia of Cell Technology. John Wiley, New York.
6. Lindsey, K. and Mak Jones 1990 Plant Biotechnology in agriculture, Prentice Hall, New Jersey.
7. Pierik, RLM (1987) In vitro culture of Higher plants. Martinus Nighoff Publishers Drodrecht.
8. Thorpe, T.A. (1983) Plant tissue culture methods and applications in Agricultural Academic Press inc. New York.

16

Single Cell Protein

Proteins which are compounds of carbon, hydrogen, oxygen, nitrogen and may also contain sulphur, have an important role in life processes. They occupy the second position in our diet after carbohydrates. There are certain proteins which we directly obtain from food. They perform different functions in our body, majority of the proteins are synthesized from the free-amino acids. Still some proteins are formed from regrouping of amino acids resulted from the digestion of proteins in the cells of a body for development of cells and tissues such as skin, muscle, blood and bones. Some of the important functions performed by protein are precised in Table 16.1.

Table 16.1: Important functions of proteins

Type of protein	Functions
Enzymes	Catalyse or help in biochemical reactions
Transport proteins	Carry different substances in the blood to different tissues
Contractile proteins	Help contraction of muscles and other cells
Hormones	Regulate many body functions
Structural proteins	Form structural elements of cell and tissues
Protective proteins	Help to fight infections

Thus protein is an important component of our diet. Deficiency of this component causes a disease called 'Kwashorkar' (A rejected child). The symptoms of the diseases include skin cracks and becomes scaly. Abdomen swells with thin legs and hands, hair become reddish and life becomes miserable. The cure is simply to provide sufficient amount of protein in the form of grams, peanuts and animal protein.

The conventional sources of protein such as groundnuts, beans, pulses and cereals (rice, wheat and maize etc.) from plants and meat, fish, eggs and cheese from the animals, are produced on a very large scale to feed increasing population. However, there is a growing demand for more and more protein for human nutrition in view of increasing human population.

As a result, protein deficiency diseases are prevalent in many parts of the world specially in developing countries. Hence, search for non-conventional source of protein began. It is Wilson (1966) of Massachusetts institute of technology, Boston, USA considered that single cell organisms like bacteria, yeasts and algae can serve as protein.

It can supplement human food, animal feed and coined the term single cell protein (SCP). Roth (1982) defined single cell protein as cellular constituents of selected microorganisms which are rich in protein and is referred as single cell protein. But international union of pure and applied chemistry has taken a serious objection as microbial cells contain other than protein also. Further, it suggested that it can be best called as single cell biomass (SCB) instead of SCP. However, SCP is termed differently in different parts of the world such as: Medical Academy of France prefers to call it as proteins debiosynthese. In Japan it is called as petroprotein and in Italy it is called as Bioprotein.

Microbes are considered to be live-stock of future era, because the microorganisms are good converters of organic matter. Further, microbes grow fast, utilize simple, cheap food and produce a quality protein. Comparative efficiency of microbial protein production over animal protein production is given in Table 16.2.

Table 16.2: Efficiency of microbial protein production over animal protein production (1000 gm feed)

Source	Protein production (dry wt. in g)	Doubling time (in weeks)
Chicken	49	3
Pigs	42	8
Bullock	14	25
Fungi	136	0.04

16.1 WHY MICROORGANISMS AS A SOURCE OF PROTEIN?

Microorganisms are more suited for developing the non-conventional source of protein as food and feed because of the following reasons:

1. Conventional sources are not adequate to meet the requirement of growing population.
2. Microorganisms have a very fast growth rate. A protein doubling time is as short as two hours has been recorded in *Neurospora crassa.*
3. They can easily be grown on a variety of substrates including wastes of agriculture, urban and industrial.
4. They can be genetically modified easily for growth on particular substratum under particular cultural conditions.
5. Their protein content are quite high, varying from 35 to 80% on dry weight basis.
6. They can be grown in slurry or solid state fermentation.
7. Their nutritional values are as good as those of other conventional foods rich in proteins.
8. The processes occupy little land area.
9. Production is independent of seasonal and climatic variation.
10. Consistent product quality.

16.2 MICROORGANISMS

When microorganism has to be used as human food and animal feed, it would have to meet certain minimum requirements in order to become popular and also compete with conventional protein source. Some of which are:

1. It should have satisfactory biological value (BV) as demonstrated in animal feeding studies and in human clinical studies.
2. It should have satisfactory nitrogen, protein content, amino acid profile, lipid content, minerals and vitamins.
3. SCP product must be free from toxic substances in the cells themselves or introduced into the cells through residues in the substratum such as polyaromatic hydrocarbons and from contamination by pathogenic microorganisms or their toxins.
4. It should have a satisfactory flavour, aroma, colour and texture.
5. Methionine supplementation is required to obtain satisfactory feeding performance in non-ruminants.
6. It should have high protein efficiency ratio (PER).
7. Toxic residues should not accumulate in the tissues of animals to be subsequently consumed by man.
8. It should not show adverse effect upon reproductive performance in animal.
9. Should have acceptable physical characterstics such as formulation, blending and storage stability.

Microorganisms selected for SCP should also possess following physiological and other characteristic features.

A. Physiological characteristics:

1. It should be able to grow fast.
2. It should be able to grow on simple media with no expensive growth requirements.
3. It should have high affinity and yield coefficient on a carbon substratum.
4. It should have the ability to utilize complex substrates when required.
5. It should have the ability to grow at high cell densities.
6. It should have resistance to substrate or product toxicity.
7. It should grow stably in a continuous culture.
8. It should have high optimum temperature.
9. It should be resistant to common contaminents.
10. It should be feasible to genetical modification.
11. It should be able to utilize ammonia as a source of nitrogen.
12. It should be pH tolerant.

B. Other characteristics

1. Protein, fat and carbohydrates of high quality.
2. Nucleic acid content should be low.
3. High nutrient content.

4. High digestibility.
5. It should not have toxicity.
6. Good taste and ease recovery.
7. It should be amenable to further processing (drying without changes in color, taste and smell etc.).

Some of the microorganisms employed in SCP production are precised in Table 16.3. Advantages and disadvantages of employing different groups of microorganisms are given below:

Table 16.3: Single cell protein producing microorganisms

Major group	Genus
A. Bacteria	
i. Phototrophs	*Rhodosprillum, Rhodopseudomonas*
ii. Gram (–ve), aerobes	*Alcaligenes, Pseudomonas*
iii. Gram (+ve), facultative anaerobes	*Aeromonas,Chromobacterium, Flavobacterium, Klebsiella, Hydrogenomonas, Methanomonas*
iv. Coryneform bacteria	*Arthrobacterium, Corynebacterium, Microbacterium*
B. Actinomycetes	*Actinomyces, Mycobacterium, Nocardia, Streptomyces*
C. Yeasts	*Rhodotorula, Saccharomyces, Candida, Rhodosporidium, Debaromyces, Endomyces, Saccharomycopsis, Torulopsis* and *Sporobolomyces*
D. Filamentous fungi	*Cunninghamella, Mortierella, Mucor, Chaetomium Aspergillus, Aureobasidium, Botrytis, Trichoderma* and *Fusarium*
E. Algae	
i. Green algae	*Chlorella, Scenedesumus, Euglena* and *Microactinum*
ii. Blue green algae	*Spirulina*

(i) Algae: Autotrophic nature of algae makes these organisms ideal and economical. *Spirulina meseima, chlorella, Microoclanum, Euglena* and *Senedesmus* are some of the organisms being exploited. General method of processing algal SCP is illustrated in Fig. 16.1**.** It Illustrates methods of upstream and downstream process of algal protein production and flow chart of *Spirulina* protein production is given in Fig 16.2. However, these organisms have advantages with certain limitations.

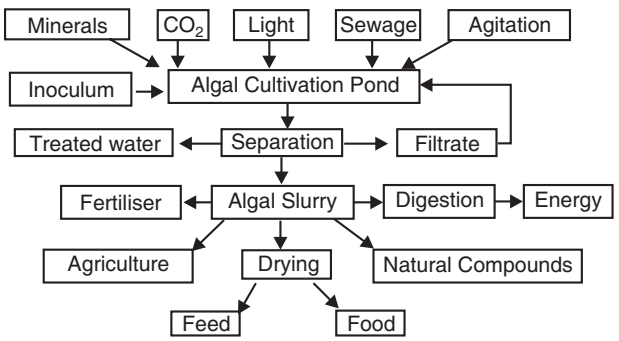


Fig. 16.1: Flow diagram of the cultivation and processing of microalgae

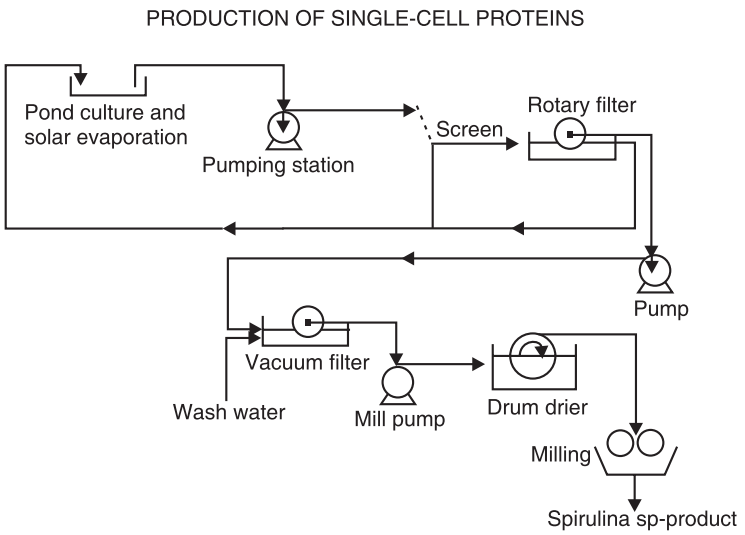


Fig. 16.2: Flow diagram of Spirulina sp. SCP Production

Advantages:

1. Growth with simple inorganic substances without the need for energy and carbon source.
2. Some algae utilize atmospheric nitrogen.
3. SCP production can be carried out in large open pools, natural lakes or ponds and do not require aseptic conditions.
4. Collection or separation procedures are easier.
5. Lysine content of algae is high.

Disadvantages:

The limitations of these group of organisms include:

1. Algal growth rate is slower than that of fungal growth.
2. Biological values of algal proteins are less than that of bacteria and yeasts.
3. Algal proteins are deficient in S–amino acids.
4. Poor mammalian digestibility of algal protein except ruminant.
5. Relatively high concentration of unsaturated fatty acids, which cause adverse effect on animals.

(ii) Fungi: Fungi in view of their ubiquitous nature, rapid growth on variety of waste substrates, are receiving greater attention in the exploitation as a source of SCP. At present mycoproteins of human food are being met from mushrooms, *penicillium notatum, Fusarium* and *Trichoderma harzianum*. Merits and demerits of fungi as SCP organisms are:

Advantages:

1. Grow on wide range of substrates as they produce wide range of enzymes.
2. Can be recovered by single filtration which reduces the processing costs.
3. Because of filamentous nature of the product, it can be used as textured food.

Disadvantages:

1. Growth rate of fungi are slower than those of yeasts and bacteria.
2. Protein content is lower.
3. Fungal proteins are deficient in sulphur containing amino acids.
4. The digestibility of fungal protein is problematic.
5. Fungi are unknown entity with regard to human nutrition and toxicology.

(iii) Yeasts: Infact the concept of SCP is developed from yeasts. They are not only ubiquitous and most familiar to man as a components of beverage or bakery products. Three types of food yeasts used extensively are brewer yeasts (*Saccharomyces sp*), bakers yeast (*S. cervisiae*) and dried yeast (C*andida utilis* and *Torula thermophila*). Flow scheme for production of bakers yeast is given in Fig. 16.3. Dried yeast is a byproduct of beverage fermentation is being used *Candida utilis* was employed as food during World War I and II by Germans as meat substitute for army, prisoners of war and civilians. They have several good qualities to choose as SCP. They are:

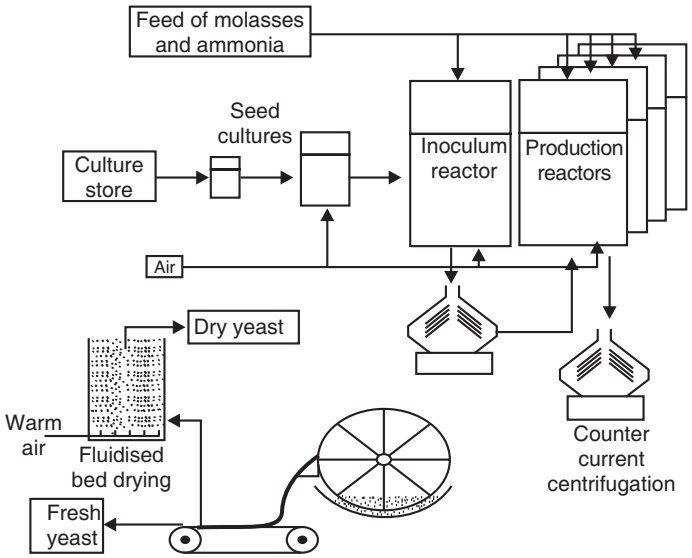


Fig. 16.3: Flow diagram for bakers and dry yeast SCP production

Advantages:

1. Grown on any type of carbohydrate.
2. Sufficient quantities of essential amino acids (Lysine, tryptophan and threonine).
3. Good source of vitamin B, D and E.
4. Low nucleic acid content.
5. Larger size makes SCP separation easy.
6. They are with lower toxicity potential.
7. Greater acceptance by consumers.

Disadvantages:

These organisms are reported to contain low concentration of sulphur containing amino acids.

(iv) Bacteria: Bacteria which are microscopic, fast growing and versatile in nutrient requirement are rich in proteins. Hence, many bacteria such as species of *Bacillus* are being exploited for SCP. Several advantages of bacterial SCP are:

Advantages:

1. Efficient in conversion of substrate carbon to cell mass, 2. Higher protein conversion efficiency, 3. Lower oxygen requirement, 4. The proteins of bacteria have fairly good amino acid profile. Similarly they are rich in sulphur containing amino acids.

Disadvantages:

They are beset with some limitations such as

1. Only limited number bacterial species are exploited for food purpose, 2. Small size of the bacteria makes the separation of protein difficult, 3. In view of their growth under limited oxygen condition, they emit unpleasant odour which may make it unacceptable, and 4. Contamination with pathogenic species.

(v) Mixed Cultures: Mixed cultures proved to be more efficient in SCP the than monoculture in view of following the reasons:

1. Better utilization of a complex carbon source as carbon source.
2. Removal of organic carbon excreted by one component microorganisms.
   (a) Removal of toxic compounds effecting one component microorganisms.
   (b) Increase of the total biomass yield from the primary carbon source.
   (c) Elimination of foam.
   (d) Increase the growth rate of one component microorganism.
   (e) Enable water recycling.
   (f) Resistance to perturbation in culture environment.
   (g) Resistance to contamination by bacteria, yeast, fungi and phages.
   (h) Lower maintenance energy requirement.
3. Attainment of desired protein composition.

However, in choosing a substrate, consideration must be given to:

1. The cost of the substrate.
2. Biomass yield generation.
3. Oxygen requirement during fermentation.
4. Heat produced and level of fermenter cooling requires.
5. Down stream processing costs.

Some of the industrial processes of SCP production are described below:

16.3 RAW MATERIALS

A variety of substances (in physical and chemical characteristics) are used as raw materials for the production of SCP. Some of these include wastes from all the three major sources like urban,

agricultural and industrial. This will serve two purposes i.e. production of value added protein and disposal of waste and pollutant materials. Some of the raw materials being exploited for production of **SCP** are given in Table 16.4.

Table 16.4: List of raw materials for SCP production

A. Agricultural wastes	Straw (Rye, wheat, barley, rice, oat and grass)
	Stover (Sorghum, Soyabean, Corn)
	Sugarcane bagasse
	Cotton gin
	Husks from grains
	Vegetable and fruit pellings
B. Industrial wastes	Paper Mill sludge
	Lignocellulose based industrial by products and wastes
C. Municipal waste	Methane, ethane, propane and butane
D. Hydrocarbons	Liquid hydrocarbons (Gas oil, paraffins,
	Petroleum, Kerosine) Alcohols (Methanol, ethanol)

The selection of the microorganisms is mainly dependent upon the available raw materials. Agricultural wastes are rich in lignocellulosic substances. Hence, filamentous fungi, capable of producing lignases and cellulases are employed for SCP production. However, when the raw material is in a complex state, it is subjected to either acid or enzymatic hydrolysis before being used as substrate for SCP production. Whenever, suitable inorganic substances are available at cheaper rate and sunlight is available in plenty, algae and photosynthetic bacteria appear to be more promising SCP microorganisms than others due to many reasons such as:

1. They utilize solar energy,
2. Make use of cheap inorganic substances,
3. SCP from this source is superior to others such as it is free from toxic substances,
4. They purify the environment.

16.4 SCP FROM HYDROCARBONS

When hydrocarbons are used as raw materials we should employ certain bacteria, yeasts and filmentous fungi which are capable of utilizing them. Again, depending upon the type of hydrocarbon, one should use particular species or strain. SCP from hydrocarbons is more desirable in view of the following reasons:

1. Inexpensive substrates.
2. Agents for corrosion and biodeterioration can be removed from environment.
3. Useful in the recovery of oil deposits.

Some of the microorganisms capable of utilizing hydrocarbons are given in Table 16.5. Generalized formula of utilization of hydrocarbons by microorganisms can be represented as:

$$2nCH_2 + 2nO_2 + 0.19\ n\ NH_4 \rightarrow n(CH_{1.7}\ O_{0.5}\ N_{0.19}) + n\ CO_2 + 1.5\ n\ H_2O + n\ (48000)\ cal.$$

Table 16.5: List of hydrocarbons and microorganisms utilizing them

Hydrocarbon	Microorganism
Methane	*Methanomonas methanica, M. methanoxidans Methylococcus capsulatus, Pseudomonas methanica, Graphium spp.*
Propane and Butane	*Arthrobacter simplex, Nocardia paraffinica*
LPG gas	*Candida rigida*
Liquid hydrocarbons	*Achromobacter delvaricata, Acinetobacter cerificans Mycobacterium phlei, Pseudomonas aeruginosa*
Alkanes	*Candida guilliermondii, C.intermedia, C.tropica*
Methanol	*Hyphobacterium sp. Methanmonas methanica, Streptomyces sp., Candida boindinii, Torulopsis glabrata*
Ethanol	*Candida enthanothermophilum*

However, in choosing a substrate, consideration must be given to

1. The cost of substrate
2. Biomass yield generates
3. Oxygen requirement during fermentation
4. Heat produced and level of fermentor cooling requires
5. Down stream processing costs

16.5 MIXED CULTURES

Mixed cultures proved to be more efficient in SCP production than monoculture in view of following reasons.

1. Better utilization of a complex carbon source or of carbon source.
2. Removal of organic carbon excreted by one component microortanisms.
   a. Removal of toxic compounds effecting one component microorganisms.
   b. Increase of the total biomass yield from the primary carbon source.
   c. Elimination of foam.
   d. Increase the growth rate of one component microorganism.
   e. Enable water recycling.
   f. Resistance to perturbation in culture environment.
   g. Resistance to contamination by bacteria, yeast, fungi and phages.
   h. Lower maintenance energy requirement.
3. Attainment of desired protein composition.

16.6 STEPS IN SCP PRODUCTION

The SCP is produced in the following steps:

1. Selection of SCP microorganism
2. Substrate
3. Process

4. Product recovery
5. Purification
6. Drying
7. Supplementation of nutrients
8. Packing
9. Marketing

For producing quality SCP one has to take care in every step. The first and foremost important is the selection of microorganisms. The selected microorganism should be fast growing, wide substrate utilizing capacity, should be non pathogenic and do no t produce any type of toxin, should produce protein not only of high quality but also good quantity and contain limited quantity of nucleic acid etc. The next step would be selection of sub-stratum. The sub stratum is not only cheaper, but readily utilizable. The organisms employed do not contribute for liberation of toxic substances. Then the growing condition of organism should be ambient and do not required any additional requirements. The incubation period should be short and downstream process should be simple and in apposition to yield pure protein with minimum number of steps. The purification and drying of protein should be a simple and should not affect the quality of protein. If protein is deficient in some respects that can be improved by supplementation of required substances. Thus the post harvest stage of SCP should do simple and amenable for processing. This will help to get a quality SCP and will be useful as feed additive for other purposes.

16.6.1 Bel process: Dairy industry generates over 100 million tonnes of whey each year. This by product of cheese manufacture has pollution load with a chemical oxygen demand (COD) of 60 g of O_2 per liter. Whey contains approximately 45 g/L lactose and 10 g/L protein. Hence, it is particularly suitable for SCP utilizing lactose utilizing yeast, *Kluyvermyces lactis* or *K. marxianus*. Bel industries in France has developed Bel process with the aim of reducing the pollution load of dairy industry waste, while simultaneously producing most acceptable SCP (Fig 16.4).

16.6.2 Pekilo process: This process began operating in 1975 and was the first commercially continuously operating process for the production of a filamentous fungal proteins. This process was developed in Finland for utilization of spent sulphite waste liquor derived after wood processing that contains monosaccharides and acetic acid. Supplement of other carbon sources, usually molasses and whey added prior to inoculation with *paecilomyces varotii* (Fig. 16.5). This continuous process is operated aseptically and produces over 10,000 tonnes of SCP a year from bio 360 m^3 fermenters. The dried pekilo protein containing up to 59% crude protein is used in the preparation of compounded animal feed.

16.6.3 Bioprotein process: Bioprotein process developed in 1990 by Norferm uses methane rich natural gas, as a sole carbon and energy source for the growth of *Methylococcus capsulates* (Fig. 16.6). A mixture of heterobacteria is also present which helps to stabilize the process. This manufacturing plant is at Tedbergoden in Norway. This highly aerobic continuous fermentation is performed in a loop fermenter with a medium containing ammonia, minerals and methane obtained from North Sea gas fields. The biomass is continuously harvested by centrifugation and ultrafilteration prior to inactivation and spray drying. The final product contains 70% protein and currently marketed as *pronin*. It is approved for use as fish and animal feed but may be used as human foods in the future.

16.6.4 Pruteen process: Attempts to develop methanol based processes were made in Europe, the former society union, Japan and the USA. They involved bacterial species (*Hyphomicrobium, Methylococcus, Methylophilus* and *Methylotrophus*), yeasts (*Candida boidinii, Pichia angusta* and *P.pastorn*) and even filamentous fungi (*Gliocladium deliquescans, paecilomyces variotii* and *Trichoderma linganus*). The most adventurous process was developed by ICI in UK in 1980. This process used the methylotrophic bacterium, *M.methylotrophus,* to produce a feed protein for chicken and pigs called pruteen. It is a world largest continuous aerobic bioprocess system involving 3000 m^3 pressure cycle airlift fermenter with inner loop and working fluid volume of 1.5×10^6 L. capable of producing upto 50,000 tonnes of pruteen protein (Fig. 16.7).

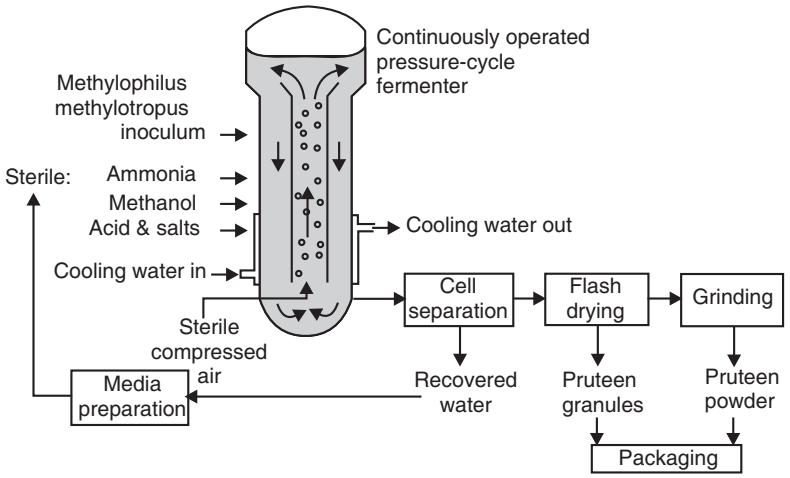


Fig. 16.4: The SCP production through pruteen process

16.6.5 Quorn production: The RHM company developed process for production of mycoprotein utilizing *Fusarium vevenatum* in 1964, known as Quorn, a food grade SCP. Following a joint venture with ICI, UK, 1000 tonnes of quorn are now produced each in the 40 m^3 airlift fermenter that was formerly used as pruteen pilot fermenter. This fermenter is operated continuously at 30°C and pH 6.0. Food grade wheat starch is used to control the pH and acts as nitrogen source. Oxygen is supplied as sterile compressed air and if it goes below, byproducts are formed resulting in unacceptable flavour and aroma. During fermentation biomass doubles for every 4-5 hrs.

The filamentous structure of harvested organism is a critically important factor related to eating quality. Mycoprotein is different from other noval bacterial and yeast protein because of its microfilamentous structure which can be partially digested to resemble the microfibrils of meat. This enables it to be processed and flavored to form meat substitute foods that have a meat texture.

The fungal biomass generated contains 10% RNA which is too high for human consumption. RNA levels can be reduced by a thermal shock at 40°C for 30 min. which kills the organism and activates organisms to secrete RNA ases. This results in the breakdown of RNA into nucleotides that diffuse out of cells into the medium. Thus RNA concentration is reduced to an acceptable level of less than 2% (W/W). Following RNA reduction the mycelium is continuously harvested by vacuum filtration. The filter cake formed is a mat of inter fungal hyphae which can be frozen as sheets into various shapes, granulated or powdered.

16.6.6 Symba process: The symba process was developed in Sweden to produce SCP for animal feed from potato processing wastes to make it more attractive and economical. The process was developed with two microorganisms that grow in symbiotic association (Fig. 16.8). The yeast (*Saccharomycosis fibuligera*) which produces copious amount of amylases necessary for starch degradation, while *Candida utilis* utilizes resultant sugars. The process is operated in two stages. In the first stage *S. fibuligera* is grown in a small reactor on the sterilized waste supplemented with a nitrogen source and phosphate. At this point starch is hydrolysed. The resulting broth is then pumped into second larger fermenter of 300 m^3 capacity where both organisms are present. However, *C.utilis* dominates and constitutes 90% of the final product. The symba process operates continuously and the pollution load of the waste is reduced by 90%. Resultant protein rich biomass (45% protein) is concentrated by centrifugation and finally spray or drum dried (Fig. 16.5).

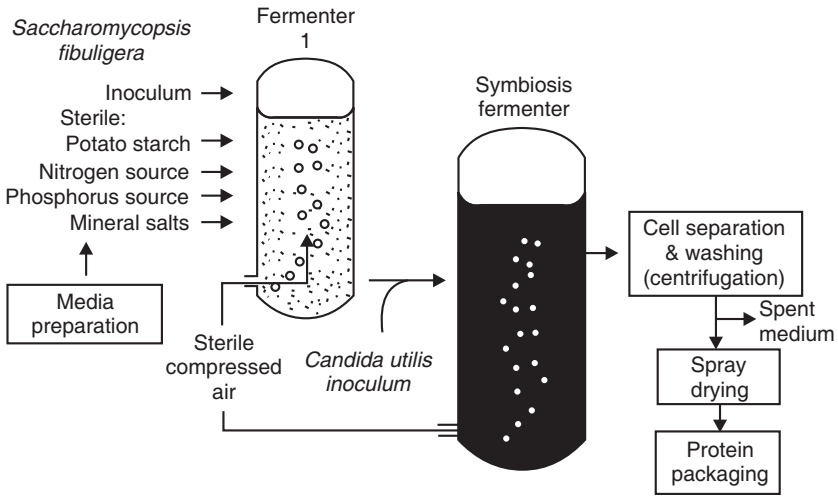


Fig. 16.5: The symba process

16.6.7 The waterloo process: Production of SCP from waste biomass is based on the cellulolytic fungus *chaetomium cellulolyticum*. It utilizes agricultural wastes, crop residues and forestry residues (wood, saw dust, paper pulp and mill sludges) for SCP production. It was developed by university of Waterloo, Ontario, Canada.

The process operation involves the following steps:

1. Thermal and/or chemical pretreatment of cellulosic materials.

2. Aerobic fermentation of pretreated materials with nutrient supplements with *C.cellulolyticum*. The SCP produced is recovered by the usual process. This fermentation process has several advantages such as:

(a) The raw material is cheap.

(b) The process employes low cost technology operation, efficient mass and energy exchanges.

(c) By products like methane can be obtained.

(d) SCP is satisfactory as animal feed protein rations in terms of safety, digestibility and nutritive value.

16.6.8 High–rate algal ponds (HRAP): This was developed at Technion–sherman Environmental Engineering Research Centre, Haifa, Israel. In this process algal SCP is produced to supplement animal feed in conjection with the treatment of sewage in HRAP. This system involves symbiotic interaction between algae and bacteria. The decomposition of sewage by bacteria releases CO_2 which is utilized by algae for photosynthesis. Algae grow by utilizing solar radiation, CO_2 and releases O_2 which is utilized by bacteria for sewage decomposition. The biomass thus produced will serve as SCP. This process has several advantages such as:

1. Waste water treatment with less cost and producing useful biomass.
2. It is non-aseptic process which does not require sterilization or additional carbon source.
3. Less oxygen is required thus reducing the aeration costs.
4. Sludge treatment is not required.

However, this process has a limited utility where solar radiation is not abundantly available. Further, for algal pond large land area is required.

16.7 NUTRITIONAL STATUS OF SCP

Microbial biomass which is harvested as a part of SCP is reported to have both inorganic and organic nutrients which are needed for healthy growth of man or animals. It is generally fiberous, low in fats and carbohydrates and contains variety of vitamins and minerals which is desirable from health point of view. Chemical analysis of pruteen SCP from *Pseudomonas* are précised in (table 16.6). It contained most of B-complex vitamins, vitamin A, inosital and choline etc in more than required quantity. Mineral analysis also revealed the presence of macroelements (P,Mg,K,Na,Ca, and Fe)(table16.7) and microelements(mn, Zn, Cu, Se, Co and Cl) in required quantity(table 16.8), at the same time some of the heavy metals (As,Hg, Sb,Pb, Cd and fluorine are below the toxic limits(Table 16.9) Variety of fats both qualitatively and quantitatively are also present in a desirable quantity (Table 16.10). The protein value is on par with the protein rich food such as soybean, meat etc. (table16.11) Quality of protein produced is also good in term of essential amino acid index (98, Fourth position) amino acids profile (89, fourth) and finally nutrition index (28 fourth) is also superior.

Table 16.6: Vitamin (mg/kg) composition of ICIS Protein from *Pseudomonas*

Ca-d pantothenate	1.58
Folic acid	8.74
Nicotinic acid	58.0
Choline	10.0
Biotin	3.12
Inositol	50
A	0.00
B_1	5.75
B_2	74.30
B_6	0.93
B_{12}	23.00
E	28.00

Table 16.7: Ash analysis of pruteen ICI'S (% of dry wt.)

P_2O_5	6.58
MgO	0.53
K_2O	0.30
Na_2O	1.53
CaO	0.04
Fe_2O_3	0.12
Total Ash	9.00
Protein	35-58

Table16.8: Trace elements(in ppm) in Pruteen ICI'S

Mn	12-36
Zn	8-30
Cu	3-7
Se	< 0.1
Co	<10
Cl	<0.0.05

Table 16.9: Heavy metals(in ppm)in pruteen ICI'S

AS	0.01
Hg	0.01
Sb	0.1
Pb	0.05
Cd	0.1
F	10

Table 16.10: Lipid composition of pruteen ICI'S (% of total phospholipids)

Phosphatidylethanolamine PEA	76.2
Phosphatidylglycerol PG	12.8
Phosphatidyl serine	8.8
Di phosphatidyl glycerol	2.2
Phosphatidyl choline	0
PEA/PG	6.0

Table 16.11: Nutritive value of SCP organisms and other traditional foods

EAA indexes	Amino acids scores	Nutrition indexes
100 pork, chicken	100 pork	59 chicken
Beef	98 chiken, beef	43 Beef
99 Milk	91 Milk	35 Pork
98 SCP(High)	89 SCP(High)	28SCP(High)
91 Potato	63 Cabbage	28 Spinach
Kidney beans	59 potatoes	26 Spinach
88 Corn	53 Peanuts	25 Milk
86 cucumbers	-	-
79 peanuts	50 corn	21 kidneybeans
76 Spinach	46 kidney beans	20 peanuts
72SCP(low)	42 Cucumbers	17 Cabbage
Cabbage	32 SCP (low)	14Cucumbers
68Turnips	31 Carrots	10 Turnips
53 Carrots	28 Spinach	9 Potatoes
44Tomatotes	18 Tomatotes	8 Tomatoes

16.7.1 Toxicological Status of SCP: Major drawback with SCP is comparatively more amount of nucleic acid. Though ruminants (herbivores) posses the metabolism of nucleic acid which are oxidized to uric acid which in turn oxidized to allantoic acid (which is water soluble) and get excreted in urine and never reach the toxic level, On the other hand, carnivorous animals and man have no mechanism of metabolizing uric acid to allantoic acid. As a result uricaemea /gout disease is caused besides having interference to native nucleic acids. Risk of presence of mycotoxins if the SCP organism happens to be a mycotoxin producing fungus. Some of the SCP bacteria are known to produce endotoxins which may cause health hazards. If organism selected for SCP produces fatty acids of uneven numbered carbon chain or relatively high content of palmitic acid are likely to have ill effects. Presence of unused carbons may turn out to be carcinogenic. If the SCP produced contains D-amino acids it will not be of much useful as these amino acids will not be absorbed by human beings.

16.7.2 Nucleic acid status: The main hurdle in the utilization of SCP as food and feed is its comparatively high nucleic acid content. However, an attempt is being made to remove excess nucleic acids by different physico-chemical methods. (Table 16.12). If these methods are successful without affecting nutritive value, SCP can be good substitute to natural proteins. At present the SCP is extensively used as feed supplement to livestock.

Nutritive quality of SCP can be improved by the application of genetic manipulation techniques including R-DNA technology. Significant improvement in several aspects of SCP production can be also be achieved.

(1) Hydrolysis of sugar polymers and utilization of the resultant sugar monomers
(2) Utilization of all constituents of a multiple carbon sources by one organism.
(3) Saving on energy required for SCP production
(4) Increasing the growth rate of a desired organism
(5) Isolation of mutants which are able to grow at elevated temperature
(6) Changing the morphological properties of the cells to improve recovery process.
(7) Alteration in the amino acid composition of the SCP product.

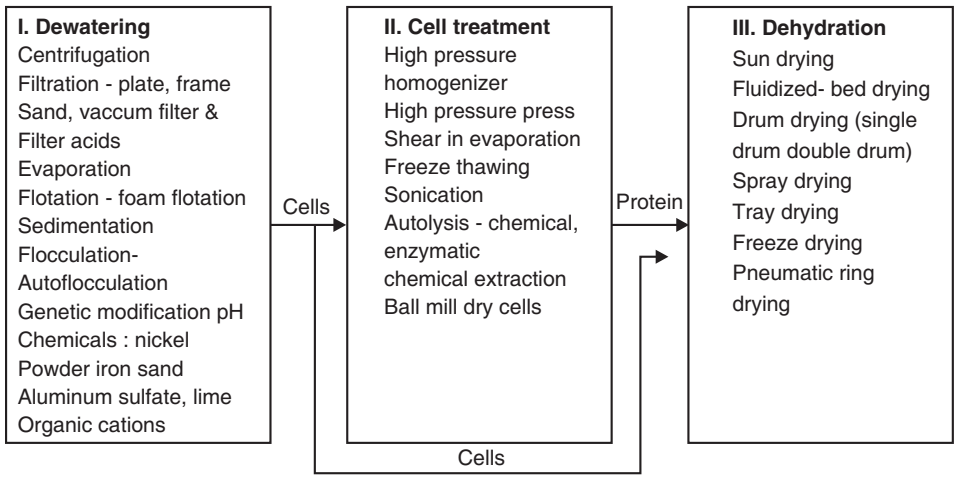
Table 16.12: Methods for RNA Reduction in the SCP product

Method	Protocol	Organism	Initial and final % RNA (of dry wt)
Base-catalyzed	0.12 N NaOH for 30 min at 50 C	*P. variotii*	9% to 3%
Chemical extraction Cell disruption	5% Nacl at 120C Disintegration	Bakers Yeast Different Microorganism	Complete removal Final % RNA =23
Physical properties Enzymatic treatment	glass beads Malt sprout nuclease	Yeast	Final % RNA<3
(b) Chemical	Succinic anhydride pH 4.2-4.4	S.carlbergensis	6.8% to 1.8%
(c) Chromatography	Cellex E-Ecteola Cellulose	*C.utilis* *S.cereuisiae* *S.mobilis*	Less than 3% of initial content
Exogenous RNAase	Bovine pancreatic Ribo nuclease 55-56C	*C.utilis* *C.intermedia*	Final %RNA =1.5
Endogenous RNAase Heat shock	90C,pH2, 20min 45C,pH5.8-6.9,2h 68C for few sec 50 C 1 h,55C 1h 50-60C +NaCl	*R.glutinis* *C.utilis* *C.utilis* *S. cervisiae*	6.5% to 1.2% 6.8%to 2.0% 7.8% to 1.2% 7.8% to 1.4%

16.8 DOWNSTREAM PROCESS

At the end of fermentation, the micro-organism is harvested by suitable method depending on the microorganism employed. After harvest care should be taken to remove excess water by employing one of the methods as detailed in table 16.13 The dehydrated biomass is subjected to cell breakage by one of the methods listed in table 16.13 At the end, separation and purification of protein should be taken to make sure it is free from water by one of the suitable methods listed in table16.13 The procedure by this method will make the protein of good quality and whatever nutrient is deficient can be supplemented before final packing. For instance BP yeast Taphrina G protein is improved by supplementing with methionine.

Table 16.13: Recovery and drying methods for SCP manufactures



16.9 CONCLUSION

It is evident from the above SCP has greatest potential to meet increasing shortage of protein due to explosive population growth. However, much is to be known in order to harvest maximum benefits. Some of the important areas to be given attention of the potential source.

(1) Construction and operation costs have to be minimized

(2) Use of cheaper raw materials such as hydrocarbons and problems associated in this potential source of carbon are to be tackled.

(3) Photosynthetic bacteria and algae which are autotrophic and require minimum investment are to be tapped for this purpose.

(4) There is a need to develop agro based SCP processes.

(5) Extensive and coordinated research in respect of microorganisms, substrate and process, recovery, purification etc.

(6) Education of consumers is of prime importance.

REVIEW QUESTIONS

I. Essay type questions:

1. Trace evolution concept of single cell protein.
2. Why SCP is needed? Mention its merits and demerits.
3. Give an account of suitability in different microorganisms for SCP production.
4. What criteria are required to be a source of SCP?
5. Describe downstream process for SCP.
6. Give an account nutritive quality of SCP.
7. Describe briefly different fermentations of SCP.

II. Write short notes on:

(a) Substrates for SCP production
(b) Toxicological aspects of SCP
(c) Limitations of SCP
(d) Mixed cultures in SCP production
(e) Methods of containing nucleic acids
(f) Nucleic acid metabolism

FURTHER READING

1. dejong-Gubbels, P., Vanrolleghem, P., Heijnen, S., van Deijken, J.P. and Pronk, J.P. (1995). Regulation of carbon metabolism in chemostat cultures of *Saccharomyces cerevisiae* grown on mixture of glucose and ethanol, Yeast 11:407-418.
2. Pham, H., Larsson, G. and Enfors, S.O. (1998). Growth and energy metabolism in aerobic fed-batch cultures of Saccharomyces cerevisiae: simulation and model verification, Biotechnol. Bioeng. 60: 474-482.
3. Rose, A.H. and Harrison, J.S. (eds) (1989). The Yeasts Vol. 3, 2nd Edition, Academic Press, London.
4. Rose, A.H. and Vijayalakshmi, G. (1989). Baker's yeasts. In The Yeasts, Vol. 5, 2nd Edition (A.H. Rose and J.S. Harrison, eds.). Academic Press, London, pp. 357-397.
5. Sonnleitner, B. and Kappeli, O. (1986). Growth of *Saccharomyces cerevisiae* is controlled by its limited respiration capacity: Formulation and verification of a hypothesis. Biotechnol. Bioeng. 28:927-937.

17

Biotransformation

Biotransformation is the process by which an organism or its enzyme bring out chemical changes on compounds that are not part of their metabolism and they result in the formation of novel or useful products that are often difficult or impossible to obtain by conventional chemical means. The total chemical transformation of one steroid to another not only requires many stages but an expensive process although provide only low yield. The biotransformation is used for the preparation of products of defined chemical structure that are related to the substrate or starting material for the reaction by only a small number of chemical changes and in many cases the changes are brought about by the action of only a single enzyme. Biotransformation reactions reported in the chemical literature of nineteenth century was developed as part of the synthetic routes for the production of L-ascrorbic acid (vitamin C) and ephedrine. Oxidation of alcohol to acetic acid by bacterium *Acetobacer xylinum* (Brown, 1886); oxidation of glucose to gluconic acid by *Acetobacter acetii* (Bontronk, 1880), sorbitol to sorbose by *Acetobacter* (Bentroank, 1896) but were not fully utilized. Biotransformation is realised beyond doubt in early part of 20th century when the conversion of D-sorbitol to L-sorbose by *Acetobacter suboxidans* and benzaldehyde to phenyl (lactyl carbinol) by yeast. Mamoli and Vercellone (1937) were the first to demonstrate the oxidation of nuclear hydroxyl group of steroid and reduction of nuclear double bond of steroid by yeast. Welsch and Hongshem (1948) have not only confirmed but also enlarged the above result by employing a *streptomces* sp. Kramli and Horvath (1949) could oxidize cholesterol to hydroxyl cholesterol by *Pencillium roseum* and *Azatobacter* sp. Hench and his associates (1949) demonstrated the curative effect of cortisone on rheumatid arthritis was possible only by introducing O_2 at its 11th carbon atom with the help of *Rhizopus arrhizus* which chemically was very difficult. Subsequently several people could accomplish this task by using different fungi (Peterson and Murray, 1952; Fried et al., 1952), actinomycetes (Thoma and Fried 1957) and bacteria (Noble and Kita, 1955). Presently variety of biocatalysts are in use for carrying biotransformation reaction. They are listed in table 17.1.

Table 17.1: Different biocatalysts used in biotransformation reactions

Biocatalyst	Number
Microorganism	161
Isolated enzymes	189
Plant cells	3
Catalytic antibiotics	2

The distinguishing feature of biotransformation is its use for the preparation of products of defined chemical structure that are related to the substratum. These biotransformations are being effected either by isolated enzymes or microorganisms (whole cell), plant cells and catalytic antibiotics. The general goals of biotransformation, are specific modification of the substrate structure via selective transformation reactions. Partial degradation of substrates into desirable metabolites by means of controlled microbial reactions or reaction pathways extension of the substrate structure by the use of biosynthetic reactions to artificial structures. Parallel with the growth of steroid hydroxylation during 1950 and 1960, was the discovery and development of other whole cell catalyzed biotransformation reactions notably the side chain degradation of steroids, Baeyer – villiger oxidation of ketones to esters, redox reactions resulting in the interconversion of alcohol and carbonyl groups, dehydrogenation reactions, hydrolytic reactions of esters and amides particularly those applicable to the production of β-lactum antibiotics which are illustrated in Fig. 17.1.

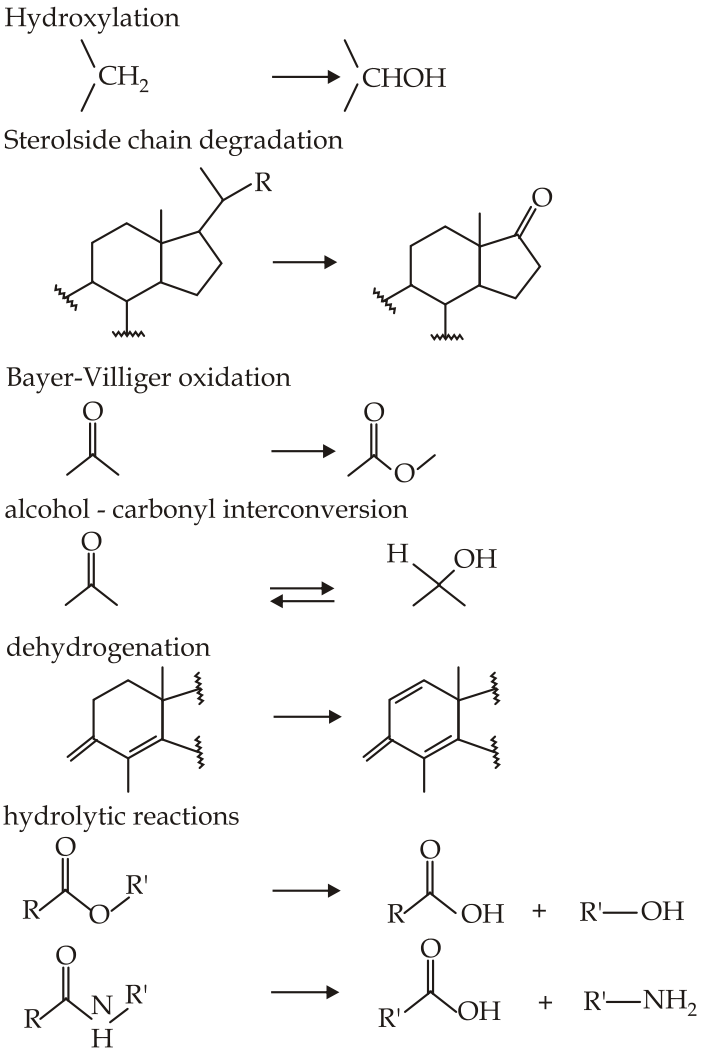


Fig. 17.1: Common whole cell-catalyzed biotransformations.

Seres (1992) and Kelley (1998) have given detailed account of different types of biotransformations achieved with the help of variety of microorganisms and their enzymes. The interest in this area is growing because of the following reason:

1. The ease with which the biological approach can be exploited with every day organic chemical laboratory equipment.
2. The commercial availability of enzymes, bakers yeast and a variety of microorganisms.
3. The ease with which one can straight away get into complex chiral products for assessing their biological, industrial and medicinal utility.
4. Generally enzymes (biocatalysts) are more efficient catalyst than chemical catalysts. Very low concentration (10^{-3} to 10^{-4} mol % of catalysts) of biocatalysts is required for an enzymatic reaction as compared to the chemical catalyst (0.01 - 1 mol %).
5. The availability of guides to procedures for selection of the most suitable microorganism for desired conversion.
6. Microbial transformations take place under mild conditions, often at a room temperature and close to neutral pH thus minimizing the problems of isomerisation, racemisation, epimerization and rearrangement that often plague the traditional chemical method.
7. Enzyme systems are highly efficient, very selective in terms of the type of reactions catalyzed and with respect to structure and stereochemistry of the substrate. Use of enzymes minimizes undesired side reactions such as decomposition, isomerization, racemization and rearrangement which can be a problem in chemical reactions.
8. Contrary to the stereospecific enzymatic conversions which are easy, the stereospecific chemical conversions of even a simple organic compound needs a complex organometallic reagent which is expensive, difficult to synthesize in simple laboratory and quite not possible for industrial exploitation. Enzymes can distinguish between enantiomers of a racemic substrate. Therefore, only one enantiomer is attacked leading to the formation of one selective product.
9. Biotransformations exhibit reaction specificity, regiospecificity, stereospecificity and can be carried out under mild reaction conditions. The substrate molecule is usually attacked at a specific site, even if several groups of equivalent or similar reactivity centres are present.
10. Most biocatalysts show catalytic activity in both aquous and organic solvents, hence suitable over conventional chemical reactions.
11. Enzymes, mediators of biotransformations, are environment friendly. They can degrade completely in the environment, unlike chemical catalysts which often require special processes for their disposal. e.g. disposal of toxic heavy metals.

Disadvantages with the use of biotransformation processes (Leuenberger, 1990).

1. The necessary expenditures for the development of a biotransformation process including the product isolation are usually high.
2. In most cases the reaction time is rather long.
3. The substrate/product concentrations are low and the stability of the biocatalysts is limited.

With the advances in fermentation technology and the advent of cheaper enzymes, the majority of biotransformation reactions using isolated enzymes, are being adopted from the post 1970 era. The range of reactions currently known to be catalyzed by isolated enzymes is vast, apparently limited only by the ease of isolation, the stability and the cost of these biocatalysts. Influence of genetic engineering, recombinant DNA techniques in this field hold the potential to facilitate production of expensive enzymes and perhaps to partially or completely remove many of the current limitations on the application of isolated enzymes for biotransformation. For instance chemical transformation of bile steroid to cortisone required 37 steps, whereas the same is accomplished under simple conditions with biotransformation with in a short time under simple conditions. Since 1970 there is steady progress in number of biotransformations during 1982-1992 (Fig. 17.2).

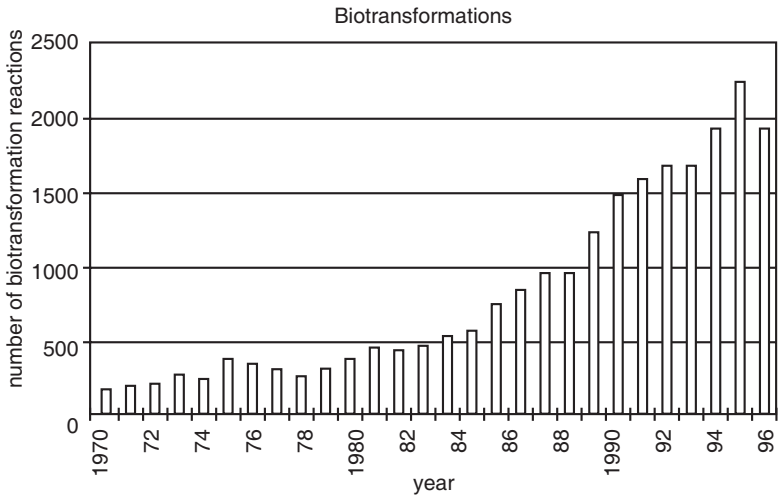


Fig. 17.2: Biotransformation reactions, 1970-1997

The biotransformations can be accomplished by employing number of biocatalysts. Probably this is the reason why there was rapid growth in the field of biotransformation since 1970. Further, interest in this area has been recognised all over the globe (Fig. 17.3).

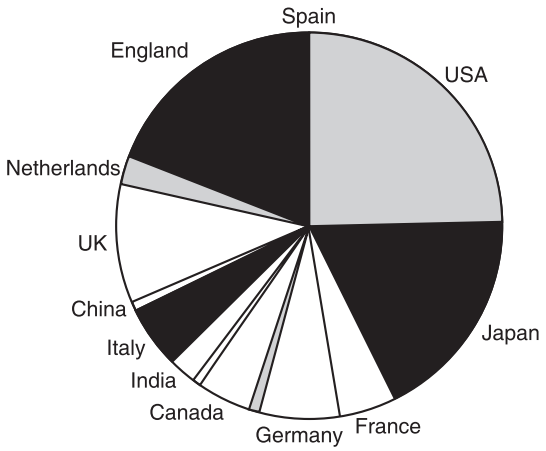


Fig. 17.3: Source of whole cell biotransformations, post-1980

17.1 MICROORGANISMS

Some of the important microorganisms employed in biotransformation are precised in table 17.2. The importance of bakers yeast is attributable to its ease of availability and use with commercial dried preparation often being used without any other processing. Bakers yeast, *pseudomonas putida* and *Aspergillus niger* are for 50% of the usage of whole cell biocatalyst. Similarly *Beaveria bassiana* (*B. sulfurastum*) and *Sporotrichum sulfuricans* are probably more versatile of the known biocatalyst in terms of the number of different chemical reactions that they can perform. These can perform biotransformation of more than 300 substrates involving wide range of reactions such as hydroxylation of saturated, unsaturated carbon compounds, aromatic carbon keto alcohol redox reactions, Baeyer villager oxidations, conjugate reactions, epoxide, hydrolysis, the hydrolysis of esters, amides and other functional groups.

Table 17.2: Most frequently used microbial catalysts

Biocatalyst	Percentage of usage
Bakers yeast	30
Pseudomonas putida	11
A. niger	8
Beaveria bassiana	6
B. Sulfurescens	3
Sporotrichum sulfurescens	5
Cunninghamella elegans	5
C.echinulata	3.5
Rhizopus migricans	5
R.Stoloniner	3
R. arrhizus and *R. oryzae*	4
Curvularia lunata	4
Mortierella isabellina	3

17.2 ISOLATED ENZYMES

Biotransformations catalysed by isolated enzymes is dominated by the use of various lipases for the hydrolysis or formation of esters. Hydrolytic enzymes constitute the most frequently used biocatalysts which account for about 30% biotransformations. Some of the isolated enzymes mediate biotransformations are precised in table 17.3.

Table 17.3: Most oftenly used isolated enzymes for biotransformation

Enzyme	Total usage (in %)
Lipase (all types)	21.09
Chymotrypsin	3.0
Acylases	2.7
Transferases	2.5
Subtilsin	2.1
Aldolases	2.1
Alcohol dehydrolases	1.4
Hydrolases	1.1
Glycosidases	1.0

17.3 OTHER BIOCATALYSTS

Use of plant cells for biotransformation is not common. However, cells of *Datura carnata, Nicotiana tabacum* and *Catharanthus roseus* are being used for relatively specialized applications. Catalystic antibodies raised in response to a small molecule transition state analogue such as antigen has received considerable attention in the past decade. However, its application for preparative biotransformations is still in an early stages of development. Some of the reactions, which can be catalysed by microorganisms or their enzymes, cover nearly all types of chemical reactions as detailed below:

17.4 TYPES OF REACTIONS

Some of the reactions which can be catalysed by microorganisms or their enzymes covers nearly all types of reactions and are precised in table 17.4 and table 17.5.

(i) **Oxidation:** Hydroxylation, epoxidation, dehydrogenation of C-C bonds, oxidation of alcohol and aldehydes, oxidative degradation of alkyl, carboxyalkyl or ketoalkyl chains, oxidative removal of substituents, oxidative deamination, oxidation of heterofunctions and oxidative ring fission.

(ii) **Reductions:** Reduction of organic acids, aldehydes, ketones and hydrogenation of C-C bonds, reduction of heterofunctions, dehydroxylations and reductive elimination of substituents.

(iii) **Hydrolysis:** Hydrolysis of esters, amines, amides, lactones, ethers, lactams etc.

(iv) **Condensation:** Dehydration, O- and N-acylation, glycosidation, esterification, lactomization and amination.

(v) **Isomerization:** Migration of double bonds or oxygen functions, racemization, rearrangements, formation of C-C bonds or hetero-atom bonds.

(vi) **Mixed reactions:** Hydroxylation with reduction; Hydroxylation with oxidation; hydroxylation with side chain degradation; rupture of C-C linkages with oxidation of side chain.

Table 17.4: Types of biotransformation reactions

Reaction	Substratum	Product
Reduction	C=C	CH_2=CH_2
	C=C–CO	CH_2–CH_2-CO
	CO	CHOH
	CHO	CH_2OH
Oxidation	CH_3	CH_2OH or COOH
	CH_2OH	CHO
	CHOH	CO
	=S	S=O
Hydroxylation	CH	C-OH
	CH_2	CH-OH
	NH_2	NHOH
Epoxidation	CH=CH	Epoxide

contd...

Reaction	Substratum	Product
Glycosylation	OH COOH- N	O-glucose COO-glucose N-arabinose
Esterification	OH COOH NH	O-palmate COO-malate N-acetate
Methylation and demethylation	 OH N N-CH_3	 O-CH_3 N-CH_3 N-CH_3
Isomerisation	trans Dextro β-OH	cis laevo-rotation α-OH
Mixed reactions	Hydroxylation with reduction. Hydroxylation with oxidation. Hydroxylation with side chain degradation Rupture of C-C linkages with oxidation of side chain	

17.5 METHODS OF BIOTRANSFORMATIONS

Transformation of organic compounds may be accomplished by use of microorganism, isolated enzyme, immobilization techniques and solvent selection. The submerged fermentation is carried out in a stainless steel tank with minimal nutritional quantities to allow maximum transformation and use of easy extraction and purification of transformation product.

The microorganisms are grown in a suitable medium for 12-72 hrs depending on bacterium and fungus at optimum temperature, pH, aeration and agitation. The fermentation is carried out in two phases.

1. Growth phase
2. Product formation phase

At the end of suitable incubation period (growth phase), measured quantity of organic compound to be transformed is added to the growing culture. The enzyme produced by the microorganism act upon the organic compound and does the desired function (product formation phase). At the end of suitable incubation period, the microbial biomass is separated from the fermentation broth. The broth is subjected to separation of both added substratum and product formed by the transformation. For analysis, if the product samples are obtained at regular intervals upto end of incubation period (1-5 days), which are analysed by using TLC, paper chromatography, gas chromatography or HPLC technique. The extraction of product is done by appropriate organic solvents such as methylenechloride, chloroform, ethylacetate and methyl isobutylketone. Product obtained from cell and substratum should be extracted separately. Different factors like pH, temperature, addition of steroid and mineral content are reported to influence biotransformations.

Use of isolated enzymes for biotransformations is limited by the availability of an enzyme for the desired transformation and subjective optimization of parameters such as temperature, pH

and substrate protecting groups (if appropriate). These are most conveniently performed at milligram to gram scale. However, some are amenable to scale up, to process tens or even hundreds of grams of substrate.

Table 17.5: Most frequently reported whole cell biotransformations

Type of Reaction	Percentage
Hydroxylation at saturated carbon	23.1
Carboxyl reduction	14.8
Alkene oxidation	3.7
Alkene reduction	3.7
Hydroxylation at carbon 11	3.4
Arene dioxygenation	3.3
Ester hydrolysis	3.2
Baeyer – villiger oxidation	2.3
Sulfide oxidation	2.0
Heteroatom dealkylation	1.6

Procedures for using isolated enzyme are dictated by the nature of the enzymes, the biotransformation reaction to be catalysed and the requirements (if any) for cofactors or cofactor recycling. For preparative uses, an immobilized form of the enzyme is often desirable to facilitate catalyst recovery and product isolation.

Whereas use of an organic solvent or co-solvent for the reaction may be indicated when dealing with substrates or products of low water solubility and is necessary in the application of hydrolytic enzymes for ester or amide formation.

Immobilization of an isolated enzyme or whole cell leads to increased ease of biocatalyst recover, thus facilitates reuse and often results in greater stability leading to an increase in the durability of the catalyst. In some instances immobilization can result in a catalyst of lower activity.

Conversion time is related to the type of reaction, the substrate conversion and the microorganisms involved. Oxidation and dehydration reactions using bacteria are often completed in a few hours, while yeasts and fungi require several days. Hydrolytic reactions with most of the microorganisms can be accomplished in a few hours.

Biotransformations in a large scale are carried out under sterile conditions in aerated and stirred fermentor. The conversion process is being monitored chromatographically or spectorscopically. The process is terminated when a maximal titer is reached. Sterility is required because contamination can suppress the desired reaction, induce the formation of faulty conversion products or cause total substrate break down.

If enzyme induction by the added substrate is not necessary, resting cells may be used. This has the considerable advantage that growth inhibition by the substrate is eliminated. High cell densities, which promote increased productivity, may be used and at the same time risk of contamination is reduced. Since the transformation reaction occurs predominantly in the buffer solution, the recovery of the product is relatively easy. A number of transformation processes can be carried out continuously and cells can be used over and over again. Immobilized bacterial cells are being used commercially in the production of L-asparatic acid, L-alanine and malic acid. Microorganisms are very convenient enzyme source which can easily be manipulated to

produce increased amount of enzyme either through environmental factors or genetic manipulations.

Range of Biotransformation

Wide range of reactions are being catalysed through biotransformation. Almost 50 of such biotransformations reported since 1950 are concerned with only two reactions – hydroxylation at saturated carbon and the reduction of carboxyl groups.

Stereochemical features

Biotransformations are used for a variety of chemical purposes many of which rely on the regioselectivity (Fig. 17.4), diastereo selectivity or enantio selectivity inherent in enzyme catalysed reactions for the formation of a product with defined stereochemistry.

Progesterone 11-α hydroxy progesterone.

Majority of biotransformations involve some aspect of stereochemical selectivity. This may be expressed as mesoselectivity, prochiral selectivity enantio selectivity in either substrate utilization or product formation, diastereo selectivity in either substrate utilization or product formation or as a combination of several previously mentioned features.

Green chemistry: The conversion of acrylonitrite to acrylamide by *Rhodococus modrochrous*, the hydroxylation of nicotinic acid and related compounds at C_6 by *pseudomonas,* other bacteria and the conversion of indole to indigo by *E.coli* expressing naphthalene dioxygenase gene are important examples of valuable biotransfomations with regiochemical but no explicit stereochemical features. Such biotransformations represent green alternatives to conventional chemical procedures involving extreme conditions of temperature and pressure or harsh reagents, as such they exemplify the current emphasis placed on the development of environmentally responsible industrial process.

Unique reagents: Biocatalysts are frequently used to perform reactions for which no analogous chemical method is available. Biotransformations are also used for production of intermediates in the environmental biodegradation or mammalian metabolism of other xenobiotic compounds. Some of the fungi carry out biotransformation of polycylic aromatic compounds resulting in the production of oxidized intermediates present in the biodegradation pathway.

Large scale applications: Large scale application of biotransformations for enzymatic processes are summarized in Table 17.6. These are restricted to the use of inexpensive enzymes with high activity for desired conversion, almost invariably hydrolytic enzymes with no factor requirements. Similarly these are applied to whole cell catalyzed transformations (Table 17.7).

Table 17.6: Large scale biotransformations by isolated enzymes

Enzyme	Conversion	Approximate scale (tons p.a. of product)
Aminoacylases	Resolution of amino acids	Ca. 1000
Penicillin acylases	Hydrolysis of penicillin-G or penicillin-V	5000
Thermolysin	Synthesis of aspartame	2000
Trypsin	Conversion of porcine to human insulin	<1
Various lipases	Preparations of chiral esters (e.g., glycidyl butyrate and isopropyl myristate)	Ca. 2000

Table 17.7: Large-scale biotransformations by microbial cells

Microorganism	Process	Approximate scale (tons p.a. of product)
Acetobacter suboxydans	Sorbital oxidation	50,000
Achromobacter, Pseudomonas or *Alcaligenes*	Hydroxylation of nicotinic acid	1000
Agrobacterium radiobacter or *Bacillus brevis*	Hydantoin hydrolysis in amino acid production	1,000
Arthrobacter	Steroid dehydrogenation	50
Aspergillus sclerotiorum	Hydroxylation in oxamniquine production	<1
Curvularia lunata or *Rhizopus stolonifer*	Steroid hydroxylation	50
Escherichia coli	Conversion of fumaric to L-aspartic acid	400
Mycobacteria or *Nocardia*	Steroid side chain degradation	50
Pseudomonas putida	Dehalogenation in (S) –2-chloropropionate production	2,000
Rhizopus arrhizus	Carbonyl reduction in steroid synthesis	1,000
Rhodococcus	Nitrile hydrolysis in acrylamide production	8,000
Various yeasts	1. L-Phenylacetyl carbinol production	300
	2. Conversion of fumaric to L-malic acid	50
	3. Hydration of crotonobetaine to L-carnitine	150

Applications: Although a vast array of biotransformations have been described, only a few of these processes have found industrial applications either because of insufficient yields or market is too limited. However, in future wide range of applications is expected to arise as most cost effective process.

Transformation of steroids: Steroids are a group of organic compounds which have four membered ring, are produced by testis, ovaries, adrenal cortex and placenta. The steroids which differ in the nature of functional group or side chain, exhibit different biological properties. Naturally occurring steroids have hormonal properties. Adrenal cortex hormones androgens, estrogens and progesterone's are some such steroids. These steroids are widely used medically as anti-inflammatory agents, anesthetics, antifertility agents and in the treatment of sterility. Cortisone is useful because of anti-inflammatory action in arthritis and skin diseases. By introducing bond in ring A of cortisone or cortisol is converted to prednisolone or prednisone with increased anti-inflammatory action.

Microorganisms help in biotransformation of steroids as follows:

1. *Rhizopus arrhizus* and *R. nigricans* hydroxylates progesterone to produce 11-α hydroxy progesterone.
2. *Cunninghamella blakesleeana* hydroxylates cortexotene at 11 carbon atom.
3. *Corynebacterium simplex* dehydroxylates cortisone to produce prednisone. It can also bring about dehydrogenation of hydrocortisone or cortisol to produce prednisolone.
4. *Nocardia restrictus* biotransforms cholestene-19-hydroxy-3-one into estrone.

5. Androstenodione is converted into testosterone by yeast.
6. Peterson and his associates have demonstrated that compounds like progesterone and Reichstein compound S may be biotransformed to testolactone through action of *A.flavus* and *P. adametzi.*

Some of the biotransformations resulted from progesterone involving microorganisms are illustrated in Fig. 17.4.

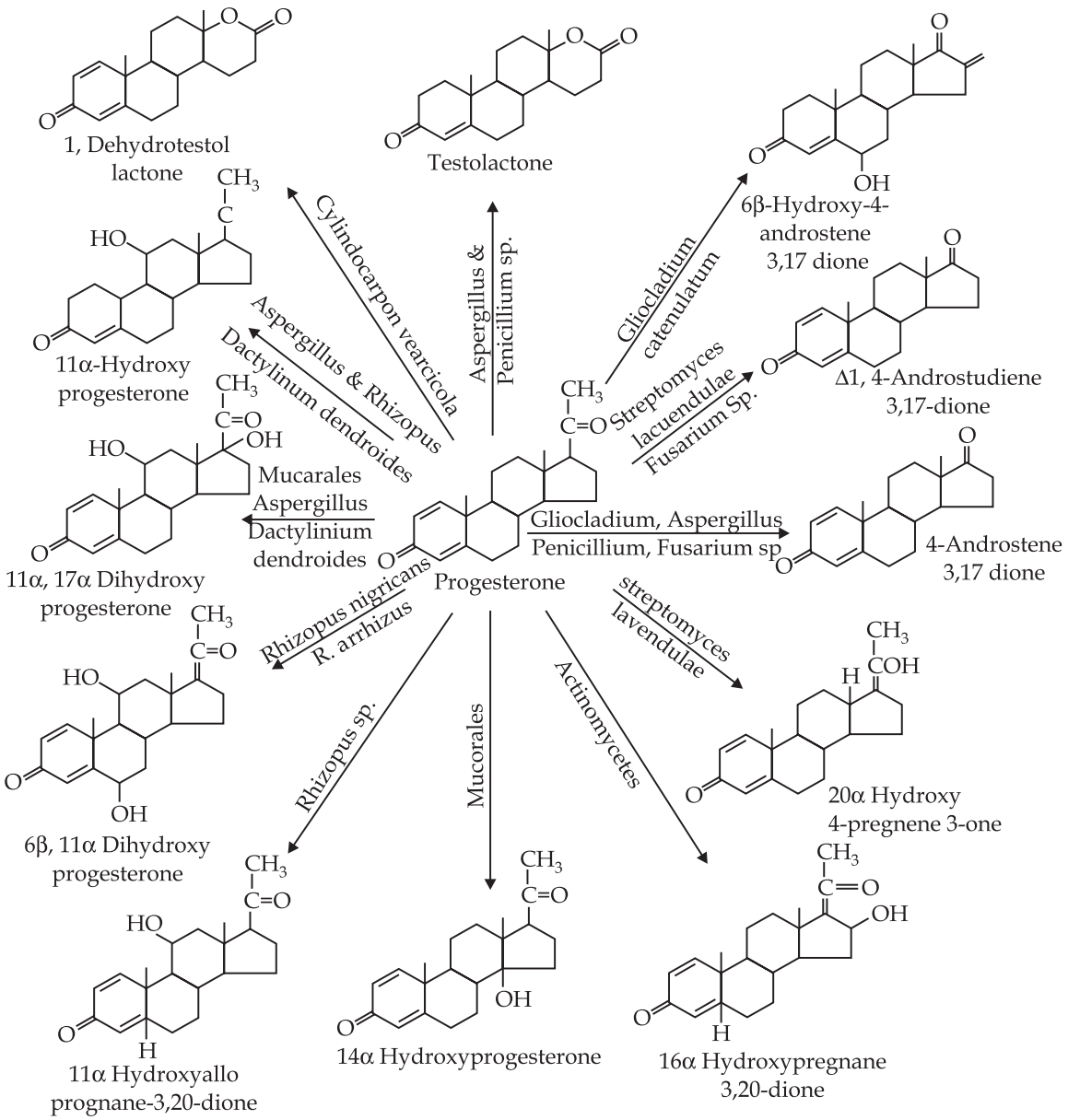


Fig. 17.4: The transformation of progesterone by different fungi

Microorganisms are capable of carrying out a wide variety of other steroid transformation reactions besides hydroxylations (table 17.8).

In recent times thousands of modified steroids produced by a combination of chemical and microbial reaction steps have been tested for their therapeutic effectiveness. Production of cortisone and its 1-dehydro-derivatives from diosgenin via Reistein's S (11-deoxycortisol) is illustrated in Fig. 17.5.

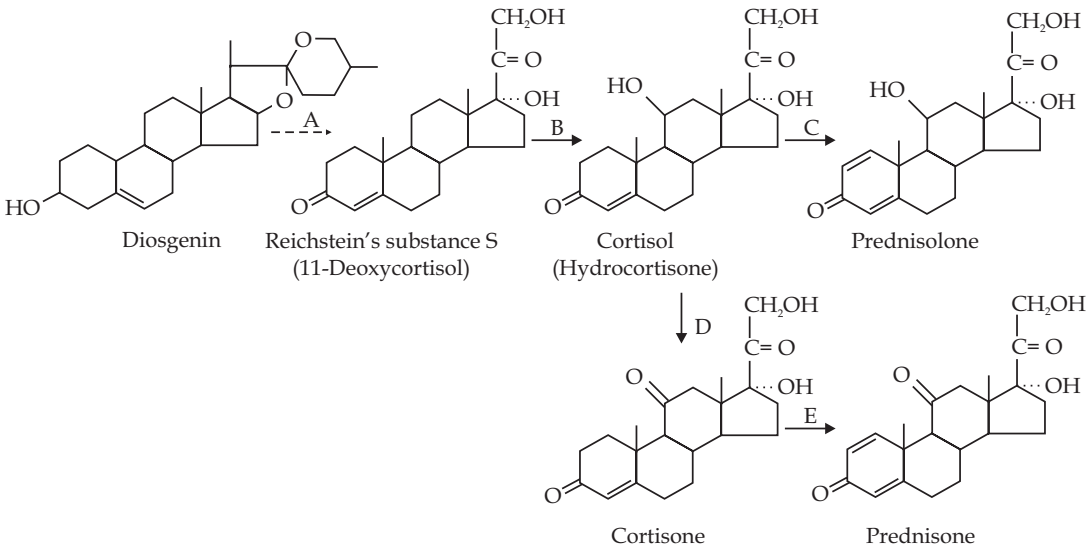


Fig. 17.5: Production of cortisol, cortisone and the 1-dehydro compounds from diosgenin via Reichstein's substance S (A, several steps; B, 11β-hydroxylation with *curvularlia lunata*; C, 1-dehydration with *Corynebacterium simplex*; D, chemical oxidation; E, 1-dehydration with *Corynebacterium simplex*)

Products of microbial transformation of steroids proved to be of great economic importance (table 17.8).

Table 17.8: Examples of some commercial steroid processes

Reaction	Substrate Product	Microorganism	Manufacturer
11α -Hydroxylation	Progestorone → 11 α -Hydroxyprog-esterone	*Rhizopus nigricans*	Upjohn company
11β -Hydroxylation	Components S→ Cortisol	*Curvularia lunata*	Pfizer Inc., Gist-Brocades
16 α-Hydroxylation	9 α Fluorocortisol → 9 α Fluoro -16 α -hydroxycortisol	*Streptomyces* roseochromogens	E.R.Squibb and Sons, Lederle Laboratories
1-Dehydrogenation	Cortisol→ Prednisolone Diendiol[a] → Triendiol[b]	*Arthrobacter simplex* *Septomyxa affinis*	Schering Corp. Upjohn Company
1-Dehydrogenation Side chain splittting	Progesterone →1-Dehydrotestol olactone	*Cylindrocarpon radicicol*	E.R.Squibb and Sons
Side chain splitting	β -Sitosterol→ Androstadiendione, and/or 9 α -hydroxyndrostendione (see Fig. 15.5 for structure)	*Mycobacterim sp., M.fortuitum* mutants	G.D.Searle and Co. Upjohn Company

Biotransformations involving different types of chemical reactions in steroids mediated by different microorganisms are given below.

HYDROGENATION:

Δ^4 : Actinomycete

Δ^{16} : *Rhizopus nigricans*

Ketone : *Streptomyces lavendulae, Bacillus Putrifaciens, Epicoccum oryzae*

Aldehyde : *R. nigricans*

DEHYDROGENATION:

Didymella lycopersici, Alternaria sp., Calonectria decora, Fusarium solani and *Corynebacterium simplex.*

EPOXIDATION:

Mucor giseocyanas, M. parasiticus, Cunninghamella blakesleana and *Helicostylum pyriiforme are able to transfer epoxile group.*

CLEAVAGE:

Gliocladium catenulatum, Pycnodothis sp, Streptomyces lavendulae, Fusarium solani and *Cephalothecium subverticellatum.*

REVIEW QUESTIONS

I. Essay type questions:

1. Define biotransformation. List out possible types of biotransformations.
2. Give an account of different agents causing biotransformations.
3. Describe an account of biotransformaton of progesterone.
4. List out advantages and disadvantages of biotransformations.
5. Describe methods of biotransformation.

II. Write short notes on:

(i) Microorganism as biotransformation agent.
(ii) Biotransformation by cell free enzymes.
(iii) Factors affecting biotransformation.
(iv) Green chemistry.

FURTHER READING

1. Blanch, H.W. and D Clark, D.S. (eds.) (1991). *Applied Biocatalysis,* Vol. 1. Marcel Dekker Inc., New York.
2. Cabral, J.M.S., Best, D., Boross, L. and Tramper .J. (eds.) (1994). Applied Biocatalysis. Harwood Academic Press, Switzerland.
3. Kelly, D.R. (vol.ed) (1998). Biotechnology Series. (H.J. Rehm and G. Reed, eds.), Vol. 8a, Biotransformations I, 2nd Edition. Wiley-VCH Verlag GmbH, Weinheim.
4. Lilly, M.D. (1992). The design and operation of biotransformation processes. In Recent Advances in Biotechnology (F.Vardar-Sukan and S.S.Sukan,eds.) Kluwer Academic,Amsterdam, pp. 47-68.

5. Straathof, A.J.J. and Adlercreutz, P. (eds.) (2000). Applied Biocatalysis, 2nd Edition. Harwood Academic Press, Switzerland.
6. Tanaka, A., Tosa, T. and Kobayashi, T. (eds.) (1993). Industrial Applications of Immobilized Biocatalysis.Marcel Dekker,Inc.,New York.
7. Tramper, J., Vermue, M., Beeftink, H.H. and van Stocksar,U. (eds.) (1992). Biocatalysis in non-conventional media, Elsevier, Amsterdam.
8. Wells, J.A. and Estell, D.A. (1988). Substilisin:an enzyme designed to be engineered, TIBS 13,291-297.
9. Wingard, L.B., Katchalski-Katzir, E.and Goldstein, L. (eds.) (1976). Immobilized Enzyme Principles. Academic Press, New York.
10. Zakas, A. and D.R, Dodds (1998). Biotransformations in the discovery and development of pharmaceutical *Curr.opin. Drug Discovery Develop* 1: 290-303.

18

Biopesticides

18.1 MICROBIAL INSECTICIDES

In natural ecosystems the population of any species is controlled by natural enemies maintaining the ecological balance. However, in man made environment, intensive cultivation of agricultural crops, birds and organisms thriving on these monocultures increase steeply. This uncontrolled growth of thriving species on crops referred as pests and become limiting factor and inturn benefits derived by man.

Pest is defined as any organism that causes economic loss in terms of human welfare resources which may be plants raised for food, feed and fiber. It is estimated that annual loss to agricultural crops due to pests is around seven thousand crore rupees in India. Of which 33%, 26%, 26% and 15% loss is reported to be due to weeds, diseases and insects, rodents and birds respectively.

The traditional way to control these pests is the application of chemical pesticides. In India about 90,000 tonnes of chemical pesticides are used annually. The major portion of pesticides used as insecticides (70%), fungicides (12-15%), weedicides (4-5%) and miscellaneous chemicals (10-14%). Majority of the pests are suppressed to significant extent by application of chemical pesticides. However, use of such huge quantitities of pesticides is not only harmful but also lead to various hazardous effects in ecosystem as:

(i) Pest varieties develop resistance in due course and need higher dose of chemical pesticides to control them.

(ii) Chemical pesticides are non-biodegradable or degraded very slowly in nature.

(iii) Pesticides may affect non-target organism.

(iv) Bio-amplification of chemical pesticide residues in animal and human occur resulting in health hazards.

(v) Cost of production of some of the chemical pesticides is very high.

Excess use of synthetic pesticides in agriculture to control pest and diseases leads to destruction of not only soil organisms (responsible for soil fertility) but also causes environmental pollution.

Rachel Carson exposed the dangers of excess of chemical methods of plant disease control. In view of the above facts, it is desirable to choose a natural enemy of the pest species, to control it. The natural enemy may be a predator, parasite or pathogen so that it kills and reduces natural population. This method of control is known as Biological control. According to Freeman (1981) when one biotic agent acts upon another in such a manner to limit its population, then a state of biological control operates. To be more precise the use of one specific organism to control another specific organism is called biological control.

The natural enemy of the pest species may be selected from any group of organisms like predators, parasites or pathogens (fungi, bacteria, viruses and rickettsiae). The biochemical's produced by the pest itself or other organisms can also be used to suppress the pest population. A careful selection of biocontrol organism is necessary to maximize damage to target species without affecting the other organisms.

In order to minimize the hazardous effects of these pesticides an endeavor has been made for the past few years to develop biological control of pests and plant diseases. A range of microorganisms have been found to be ideal in controlling pests especially insect and nematode pests. Any microbial preparation that can be used as biopesticide should possess the following requirements:

1. Large scale production must be possible.
2. It should maintain its viability.
3. It must not be pathogenic or toxic to other plants and animals.
4. It should be less expensive than chemical pesticides.
5. It should not develop resistance in the target pest.

Protozoa, bacteria, viruses and fungi are being used in pest control as well as plant protection. Some of these microorganisms are naturally antagonistic to a particular pest and are called as natural pesticides others are produced commercially in large scale fermentation or in insect hosts.

This practice is prevalent since ancient times such as cats are reared for the control of rats. Chinese people fostered predaceous ants and marketed them to control certain citrus diseases in the early 900 A.D. In recent times cottony cushion scale is being successfully controlled by vedalia bettle. The use of microscopic organisms for controlling the pests is of recent. Bassi (1834) isolated *Beauveria bassiana* from dead silkworm which is one of the frequent used pathogens employed today in the control of insect pests. A real boon to this, however, is the isolation of *Bacillus thuringienesis* from diseased caterpillar of flour moth by Malts (1903). Since then large number of pathogens have been isolated and a few of them have been successfully used to control various pests. Some of the agents that are used in the control of plant pests include

1. Predators, 2. Protozoa, 3. Viruses, 4. Fungi and 5. Bacteria.

18.1.1 Predators

Insect predators such as *Scymmus* and *menochilus* feed on other pest insects like mealy bugs, coccids, scales and mites on citrus plants, grapevine. The most effective *Chrysopa* insect predator feeds on aphids, mealy bugs and young caterpillars. The insects belonging to lipidoptera, hemiptera and diptera are attacked by sporozoa of protozoan parasites. *Farinocystus trifolii* has been successfully used to control red flour bettle in India. *Neoaplectana carpocapsae* is reported to attack about 200 species of insects. Entomogenous nematodes are cultured on solid substrate medium *in vitro* and dispered on crops having insect pests. The nematode parasitizes pest insect and kills (Table 18.1).

Table 18.1: Examples of predation and parasitism

Mode of Antagonism	Plant pathogens	Antagonists (hosts)	Post-infection event
Mycoparasitism	*Botrytis alli*	*Gliocladium roseum*	Penetration of hypae and Antibiosis
	Cocchliobolus sativus	*Myrothecium verrucaria* and *Epicoccum purpurascens*	penetration
	Rhizoctonia solani and *Fomes annosus*	*Trichoderma viride*	Coiling, cytoplasm Coagulation
	Sclerotium rolsfsii	*T.harzianum*	Coiling, penetration and lysis
Nematophagy	*Heterodera Rostochiensis*	Phialospora *heteroderae*	Penetration of cysts and egg killing
Mycophagy	*Cocchliobolus Sativus*	Soil amoebae	Perforation in conidia
	Gaeumannomyces graminis var. tritici	Soil amoebae	Penetration and lysis of hyphae

18.1.2 Protozoa

The protozoa*, Nosema louistae, N. algeru, N. pyrausta, Vairimorpha necatrix* and *Matteria trogodermae* have been tested but have not yet found practical application.

18.1.3 Viral insecticides

Viruses which kill insects are called as viral insecticides. They are pathogenic to insects. About 650 insect viruses have been described which belong to family Baculoviridae (Baculovirus). Baculorvirus is rod shaped and dDNA, containing protein coat as double layered envelope. Further, these are not pathogenic to vertebrates and plants. These viruses can be grouped into three types, on the basis of morphological characters (Fig. 18.1 and Table 18.2).

Prototype member: Autographa californica nuclear polyhedrosis virus (AC NPV)

1. Infect insects of the orders – Lepidoptera (butterflies, moths), Hymenoptera (saw flies), coleoptera (bettle).
2. Rod shaped, Genome ds DNA.
3. In nature, they are found occluded within proteinaceous crystals known as polyhedra on plant foliage, plant debris and soil.
4. Upon ingestion by the larvae, the viruses replicate in gut cells and causes lysis.
5. Host and insect specific.
6. A good expression vector.

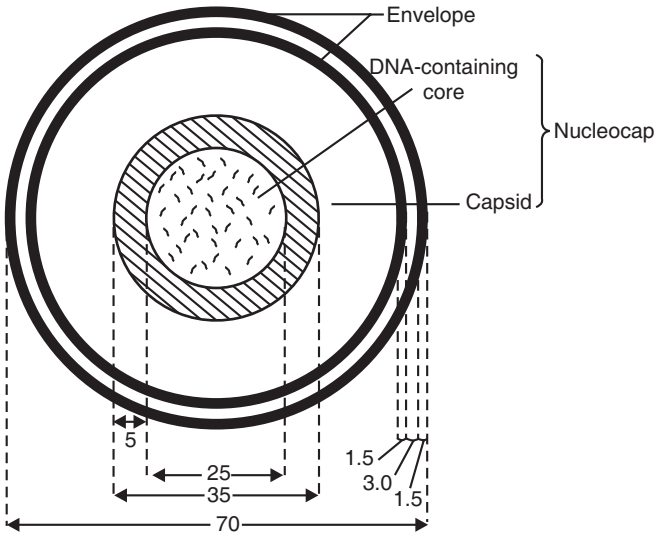


Fig. 18.1: Structure of baculovirus

Table 18.2: Characteristics of baculoviruses as bioinsecticides

(i)	**Nuclear Polyhedrosis Viruses (NPV):** The genome of these viruses code for a protein called polyhedron which forms aggregate or occlusion bodies around virions. These bodies are polyhedral, hold many virions and infects only larvae.
(ii)	**Granulosis viruses (GV):** They develop either in nucleus or cytoplasm of host trachea or epidermal cells. The occlusion body is generally ellipsoid and infects only larvae.
(iii)	**Oryctes:** These develop in the cytoplasm of epithelial cells of host gut. These viruses do not form occlusion bodies, infect both larvae and adults.

These viruses have restricted host range to arthropod insects which cause important pests of different crops (table 18.3). Similarly these viruses have been successfully employed in the control of pests of some important crops (table 18.4).

Table 18.3: List of baculoviruses in the control of insects of different crops

Virus	Target host	Crop
Nuclear polyhedrosis viruses		
Anticarsia gemmatalis NPV (soybean lopper)	*A.gemmatalis*	Soyabeans
Autographa californica NPV	*Orgyia pseudotsuga*	Forests
(alfalfa semilooper)	*Trichoplusia ni*	Cabbage
Gilpinia hercyniae NPV	*G.hercyniae*	Forests
(Spruce sawfly)		
Heliothis spp. NPV (cotton bolloworm)	*Heliothis spp.*	Cotton, cabbage and other vegetables
Lymantria dispar NPV	*L.dispar*	Forests
(gypsy moth)		

contd...

Virus	Target host	Crop
Mamestra brassicae NPV (cabbage moth)	*Mamestra, Heliothis and Diparopsis spp*	Cotton, cabbage, and other vegetables
Neodiprion sertifier NPV (pine sawfly)	*N.sertifer*	Forests
Neodiprion lecontei NPV (redheaded pine sawfly)	*N.lecontei*	Forests
Orgyia pseudotsugata NPV (Doughlas fir tussock moth)	*O.pseudotsuga*	Forests
Spodoptera littoralis NPV (cotton leafworm)	*S.littoralis*	Cotton
Trichoplusia ni NPV (Cabbage looper)	*T.ni*	Cabbage
Granulosis viruses		
Cydia pomonella GV (codling moth)	*C.pomonella*	Orchards
Plodia interpunctella GV (Indian meal moth)	*P.interpunctella*	Stored grain
Nonoccluded baculoviruses		
Oryctes rhinoceros virus (coconut rhinoceros bettle)	*O.rhinoceros*	Coconut and oil palm

Table 18.4: Some insect pests controlled by baculoviruses

Pest	Common name	Crop
Neodiprion sertifer	European pine sawfly	Pine
Heliothis sp	Cotton ballworm	Cotton, sorghum
Orgyia pseudotosuagata	Tussock moth	Douglas fir
Cydia pomonella	Codling moth	Walnuts, apples
Oryctes rbinoceros	Rhinoceros beetle	Coconutus

Table 18.5: Some genes that have been expressed in baculoviruses to increase insecticidal activity

Gene	Effect on host insect of introduced gene
Discretic hormone	Reduced hemolymph volume
Juvenile hormone esterase	Feeding ceasation
B.thuringiensis toxin	Feeding ceasation
Scorpion toxin	Paralysis
Wasp toxin	Premature melanization, low weight gain

Table 18.6: Entomocidal activity of some genetically engineered nuclear polyhedrosis viruses

Type of gene inserted	Transformed Virus	Mode of action	Insecticidal effects
Androctonus australis Nector neurotoxin	Ac NPV	Paralysis	Significant decreases ST 50
Pyemotes tritici TxP-1 neurotoxin	Ac NPV	Paralysis	Paralysis within 2 days
Buthus epeus Insect toxin I	Ac NPV	Paralysis	None
Heliothis virescens	Ac NPV	Hydrolyzes JH To initiate Moulting	Reduction in weight, No decrease in LT_{50} / LD_{50}
Mand sexta Diuretic hormone	Bm NPV	controls water balance	20 percent decrease in LT_{50}

The NPVs are being genetically modified to increase their efficiency (Table 18.5). Some of these viruses have been genetically transformed to change its mode of action and increased insecticidal activity (Fig. 18.6). Some of the strains of these NPV are registered for their application in the protection of crop (Table 18.7). Further, these viruses are not pathogenic to vertebrates and plants. Sequence of events in infection of insects are illustrated in Fig. 18.2.

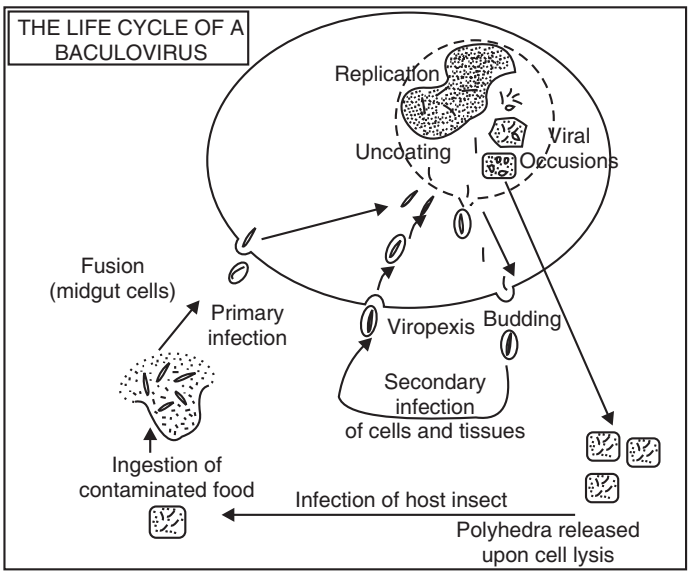


Fig. 18.2: Sequence of contamination virus

Table 18.7: Registered baculoviruses insecticides

Virus	Target Insect
Dendrolimus NPV	Dipteran sibericus
Limantria NPV	Lepidoptera disper
Heliothes NPV	Heliothis sp
Spodotera NPV	Many insects

The viruses which are budded viron enclosed by cytoplasmic membrane or occluded or viron enclosed by polyhedron nuclear envelope enter the digestive track and get dissolved by alkaline content of gut of the larval host, virus particles are released. The coat of DNA is also gets dissolved at nuclear pores disclosed double stranded circular and super coiled DNA enters the nucleus. It starts replicating within 6 hrs of entering after about 10 hrs, virus particles are released from the cell by budding into hemolymph. These extracellular viruses transmit the disease to other tissues of the host and kills the host. The occlusion bodies protect the viron from the long term infection loss. The disease spreads with in the insect population. Capsid of most of the insect viruses not only insoluble in water but also serve to protect the virus particle, make them effective even under normal storage conditions for many years and give them remarkable potential as insecticides.

Out of three categories of viruses described above, research has been done on nuclear polyhedrosis viruses. Following their ingestion they develop within the nuclei of host cells (Fig. 18.3). They cause damage to the skin, fat body and hemolymph of infected larvae. Infected tissue becomes packed with the characteristic polyhedral bodies. Subsequently millions of these polyhedral bodies are Liberated when the insect dies and its skin ruptures, in the form of liquefying body content, which finally leak out from the captured strain. This fluid content is highly infective in nature and easily disseminated among the healthy insect population. The dissemination takes place by wind or rain.

The application of viral preparation to control insect pests is one of the most rapidly expanding fields. All the above groups are safer to man, invertebrates, other higher animals and useful to suppress the insect pests. Following the ingestion of NPV, the viral particle replicates in the nuclei of the host cells resulting in the death of insect larvae. The rupture of the larval skin liberates millions of NPVs in the environment spreading disease in healthy insect larvae. About three ecotypes of insect specific NPVs have been identified and are being tested to develop potential viral insecticides. In Canada NPVs were sprayed over 50 acres to control European saw fly and after 24 days 94% larvae were infected and died. In India no commercial preparation of NPVs are available. However, field trails of NPVs of cotton bollworm, *Heliothis,* which also attacks chick pea, millets, groundnut and several other crops proved to be successful to suppress insect pests. Viron H is registered viral insecticide available in USA to control cotton bollworm and the closely related insects (Table 18.7). It is also used against tobacco budworm, cabbage worm and the closely related insects. Recently efficiency of Baculoviruses have been improved by transformation and could overcome resistance (Table 18.6). Some of the patented insect pathogens are listed in Table 18.8.

Table 18.8: Some patented insect pathogens

Organism	Target pest	Organization
Bacteria:		
Bacillus cereus	Lepidopterna larva	M.G Gandman [RU]
B. thuringiensis	Lepidopterna larva	29 Organizations
B. insectus	Lepidopterna larva	A.B Gukasyan [RU]
B. papillaiae	Japanese bettle	US Agr.Res.Corp
B. sphaericus	Mosquito larvae	Sumitomo [J]
B. moritai	Dipterna larvae	S.Morita [UK]

contd...

Organism	Target pest	Organization
Fungi:		
Beauveria bassiana	Fleas	Restov pest
B.bassiana	Colorado beetle	Forschzentrum Graz [DT]
Coelomomyce siliensis	Culex larvae	Kazakhstan zool Inst. [RU]
Viruses:		
Dendrolimus NPV	*D.sibiricus*	Katukara Ind.Co.[J]
Lymantria NPV	*L.dispar*	Katukara Ind.Co.[J]
Hehiothis NPV	*Heliothis sp*	Sandoz/IMC [US]
Spodoptera NPV	*S.frugiperda*	MGK [US]
Nematodes:		
Heterorhabditis Bacatriophora	*Numerous* targets	CSIRO [DT]
Neoaplectana carpocapsae	Numerous targets	CSIRO [J]

In recent times cross protection method has attracted the considerable attention. A plant infected by one strain of virus cannot be infected by another one. The plants to be protected are artificially inoculated with a wild virus that does not cause visible symptoms. Such plants become resistant to challenging virus that causes a severe disease.

A survey conducted in 1991 based on seventy countries revealed that biological pest control methods achieved commendable success. Eighty three pests are completely eradicated, while a substantial reduction in 120 pest severity and 110 pests have been partially controlled.

Limitations: Most of the viral insecticides have a narrow range of action and are capable of killing larvae at a definite phase. Therefore, the timing of application is an important factor. For example, NPV of *Heliothis* is able to kill young larvae more readily than old larvae. It has been reported that NPV kills first instar larvae within a short span of time ranging from one day to two days after application, while four to eight days are required for killing third instar larvae.

18.1.4 Fungal insecticides

More than 700 entomogenous fungi have the potentials of being used as mycoinsecticides. These are relatively non-specific in host range. Out of these, the infectious strains of *Beauveria, Nomeurea, Metarrihizium, Verticillium, Hirsutella, Acremonium, Fusarium, Paecilomyces and Aschersonia* are being produced on commercial scale. Many of these fungi can easily be cultured in Petri plates in semi solid media such as rice bran etc. *Metarrhizium* has been found to destroy completely the sugarcane plant hopper (*Pyrilla perspsuila)* in North India. Production technology of *Metarrhizium anisopliae* has been developed which showed complete destruction of coconut rhinoceros bettle pest in Tamil Nadu. Some of the fungi which have been successfully employed in the control of pests are detailed in Table 18.9.

Table 18.9: List of fungal insecticides

Fungus	Production technique	Production method
Aschersonia sp	Submerged	Pilot plant
Beauveria bassiana	Submerged	Small scale
Conidiobolus boscurns	Submerged	Pilot plant
Calcinomyces clavosporus	Submerged	Small scale
Entomophthora grylli	In insects	Small scale
Erynia neoaphidia	Submerged / insects	Small scale
Hirsultella thompsonii	Semi-solid	Pilot plant
Leginidium giganteum	Submerged	Small scale
Metarhizium anisopliae	Semi-solid	---
Nomurea rileyi	Solid	Small scale
Verticillium lecanii	Submerged	---
Zoophthora radicans	Submerged	Laboratory studies

(i) Mass Production of Fungal Insecticides: Generally semi-solid fermentation is employed for the production of fungal biomass on large scale. Fungal insecticides of *Acrostalagmus apludum, Metarhizum anisopliae* and *Beauveria bassiana* can be produced by semi-solid fermentation. The procedure employed for the production of fungal insecticides is similar to the one employed for *Bacillus thuringiensis*. Three different media are also used in this procedure. The composition of inoculum medium and production medium are similar to the one used for *Bacillus thuringiensis*. But the following medium is employed for the production of seed tank culture.

Glucose	15.0 g
Soyabean meal	10.0 g
Corn meal	10.0 g
Calcium carbonate	2.00
Sodium chloride	5.00
Distilled water	1000 ml

Though the production medium is generally similar to the production medium of *Bacillus thuringiensis* except that, half of the bran is replaced with corn meal. It is necessary to the fermenting bran medium once or twice before sporulation takes place in order to facilitate maximum mycelium binding to the bran. After the mycelium is well developed the culture is dried as soon as possible to obtain maximum sporulation. The dry conidia-bran mixture is stable but cannot be stored for longer time. General methods of production of mycoinsecticide is illustrated in Fig. 18.5. The fungal insecticides can be used in the field either in the form of liquid or dust preparations.

(ii) Applications: *Beauveria bassiana* and *Metarhizium anisopliae* are used for control of the colaroido beetle and leaf hopper in sugarcane plantation respectively. *Verticillium lecanii* and *Entamophthora* sp. are used for the control of aphids in green houses and field environment. Individual entomagenous fungus is tested for its pathogenicity against target insect as depicted in Figs. 18.3 and 18.4 and their production, commercialization and application is illustrated in Figs. 18.4 and 18.5. Some of the mycoinsecticide employed in the control of plant pests are listed in Table 18.10.

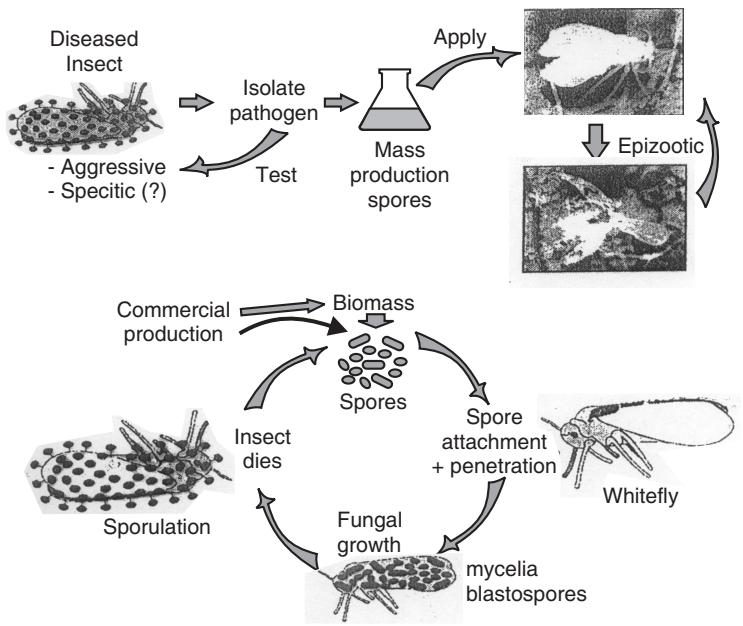


Fig. 18.3: Inoculum production and testing of mycoinsecticides

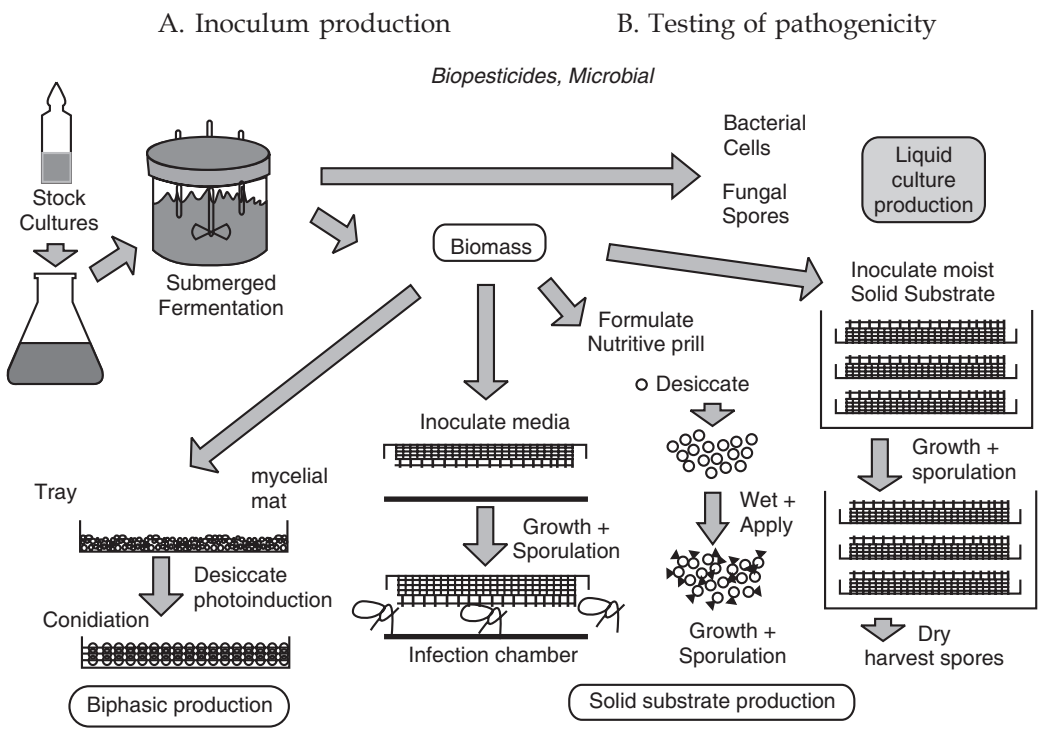


Fig. 18.4: Outline of the production methods used for the production of living microbial biopesticides. Liquid culture fermentation, solid substrate fermentation, and biphasic production are the most widely used commercial production methods

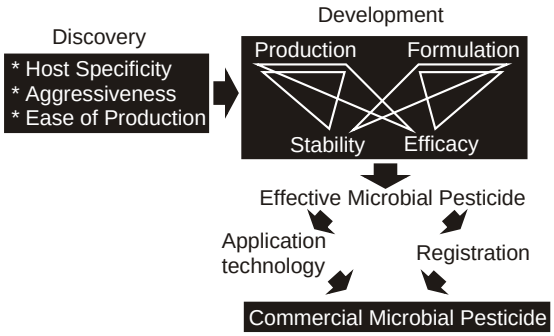


Fig. 18.5: The developmental sequence for the commercialization of microbial biopesticides

Table 18.10: Some mycoinsecticides in the control of insect pests

Fungus	Insect
Beauvaria bassiana	Codling moth
Nomaria spp	Black rice bugs
Metarrhizium anisopliae	Sugarcane hopper
Verticillium lacani	Broom soft scale insect
Hirsutella thompsoni	Citrus rust mite
Aschersonia elirodis	Citrus white fly
Coelomyces spp.	Mosquito larvae
Entomophthora sp	Scale insects
Paecilomyces anasinus	Potato tuber borer

18.1.5 Bacterial insecticides

It has been reported that several bacteria act as pathogens of insects and can be used as bacterial insecticides. They include both endospore forming and non-endospore forming bacteria. The former include species of *Bacillus* and *Clostridium* and the later include *Pseudomonas, Enterobacter, Proteus, Serratia* and *Xenorthobolus.* Of these bacteria, *Bacillus thuringiensis* has been most extensively studied and used as insecticide for controlling about 150 insect pests. In addition, it is also used to control certain insects which are carriers of human diseases, for example *Anopheles* sp for malaria and *Simulium damnosum* for river blindness.

B. thuringiensis which produces delta endotoxin causing paralysis of gut when ingested and subsequent death of larvae. This bacterium also produces exotoxin capable of killing insect. The toxin crystal dissolves under conduction in mid-gut and attacks epithelial lining of gut wall and paralyses larval body within 60 min. of toxin intake. Thuricide formulation containing *B. thuringiensis* spores in the form of emulsion or dust are being manufactured by atleast 12 manufactures in five different countries. *B. thuringiensis V. israelensis* produces a crystalline toxin that kills mosquito and black fly larvae. *B. sphaericus* possess the toxin in the cell wall and proved to be a promising pathogen against mosquito larvae. *B. papillae* and *B. lentimrbus* cause milky disease in insects and these bacteria have been used to control some insects. In practice, the adult beetles are injected with spores of these bacteria, then released into the field to spread the disease. About 62 species of insects are known to pick up the disease from infected bettles.

Bacillus thuringiensis, now generally called as Bt-bacterium was first discovered by a Japanese scientist Ishiwata in 1902 from diseased silk worm larvae and he named it as *Bacillus Satto*. A German microbiologist Berliner in 1915 isolated a similar bacterium from the diseased caterpillars of the flour moth and named it as *Bacillus thuringiensis*. The importance of this organism was realized after a gap of 50 years and commercial production of Bt biopesticide was started in 1960 in USA. Now hundred of Bt strains are available which are employed for the commercial production of biopesticide. The characteristic features of *B. thuringiensis* are precised in Table 18.11.

The *Bacillus thuringiensis* is reported to kill wide range of insects which include, moths, beetle, mosquitoes flies, aphids, insects, ants, termites, midges, butterflies, cockroaches, snails and protozoan's. Some pathogenic fungi are also killed by this bacterium, which include *Pythium ultimum* and *Fusarium oxysporum*, f. sp *lycopersii.*

Bacillus thuringiensis is able to kill a wide range of hosts by producing several toxins. Of which four toxins are reported to have insecticidal properties. They are

1. d-exotoxin,
2. d-exotoxin,
3. d-endotoxin and
4. louse factor.

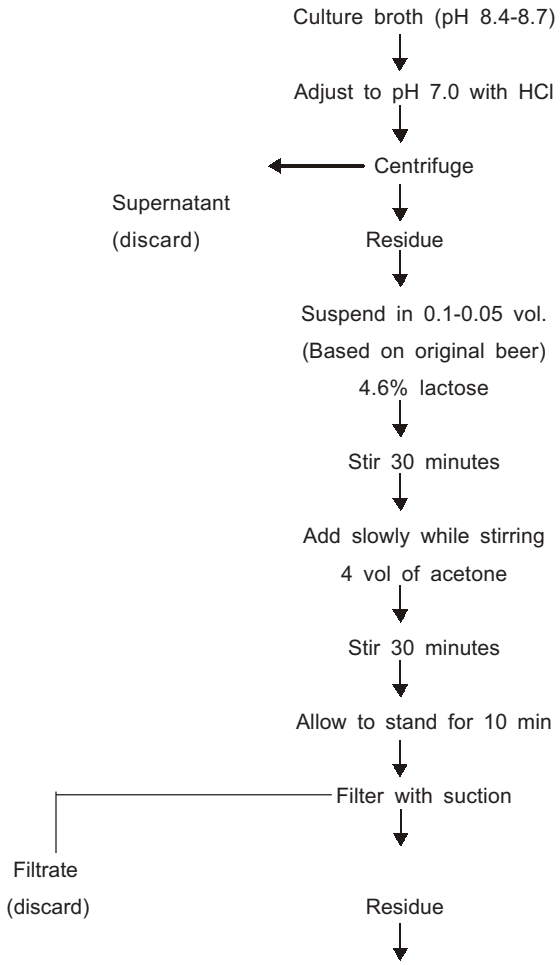
Table 18.11: Characteristic features of *Bacillus thuringiensis*

1. First report	:	*B.thuringiensis* sub sp. Kursataki 1902
2. First commercial preparation	:	1957, Thuricide (Sandoz label)
Toxin	:	Parasporal Glycoprotein crystal
3. Toxin types		(a) Cy-I-(Lepidoptera specific)
		(b) Cry-II-(Lepidoptera and Diptera Specific)
		(c) Cry-III (coleopteran specific)
		(d) Cry-IV (Diptera specific)
4. Prototoxin	:	135 K Da
5. Active toxins	:	60-70 K Da
6. Mode of action	:	Toxins bind to specific receptors and lysis the epithelial cells. Gut paralysis and starvation. Death within 24-48 hours.
7. Limitations of Bt.biopesticide	:	
		(a) Low stability of protein in the field
		(b) High production cost
		(c) Limited insecticidal spectrum

The -exotoxin is identified as leutherase-c, which is water soluble, heat labile and toxic to insects. -exotoxin is identified as thuringienin which is water soluble and heat stable toxin. The -endotoxin is a crystal, produced during sporulation. All these toxins are produced in greater amounts during spore formation. The extraction and characterization of individual toxins is detailed in Figs. 18.6 and 18.7 and Table 18.12.

Table 18.12: Some properties of the insecticidal toxins from various strains of *B.thuringiensis*

B.thuringiensis strain serotype or subspecies	Toxin class class	Prototoxin size (K Da)	Target insects	Serotype
bertiner	Cry I	130-140	Lepidoptera	1
kursataki KTO,HD-1	Cry I	130-140	Lepidoptera	3
entomocidus 60.1	Cry I	130-140	Lepidoptera	6
aizawai 7.29	Cry I	130-140	Lepidoptera	7
aizawai IC 1	Cry I	135	Lepidoptera, Dipthera	7
kursataki KTO,HD-1	Cry II	71	Lepidoptera, Dipthera	3
tenebrionis(san diego)	Cry III	66-73	Coleoptera	8
morrisoni PG 14	Cry IV	125-145	Diptera	8
israelensis	Cry IV	68	Diptera	14



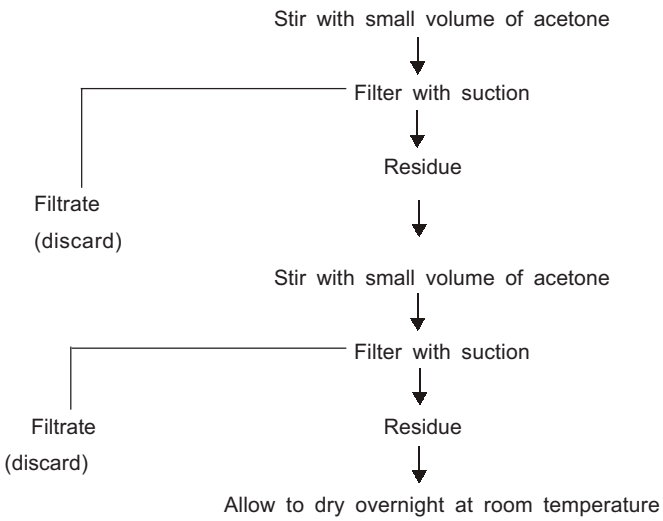


Fig. 18.6: Extraction and purification of B.thuringiensis toxins

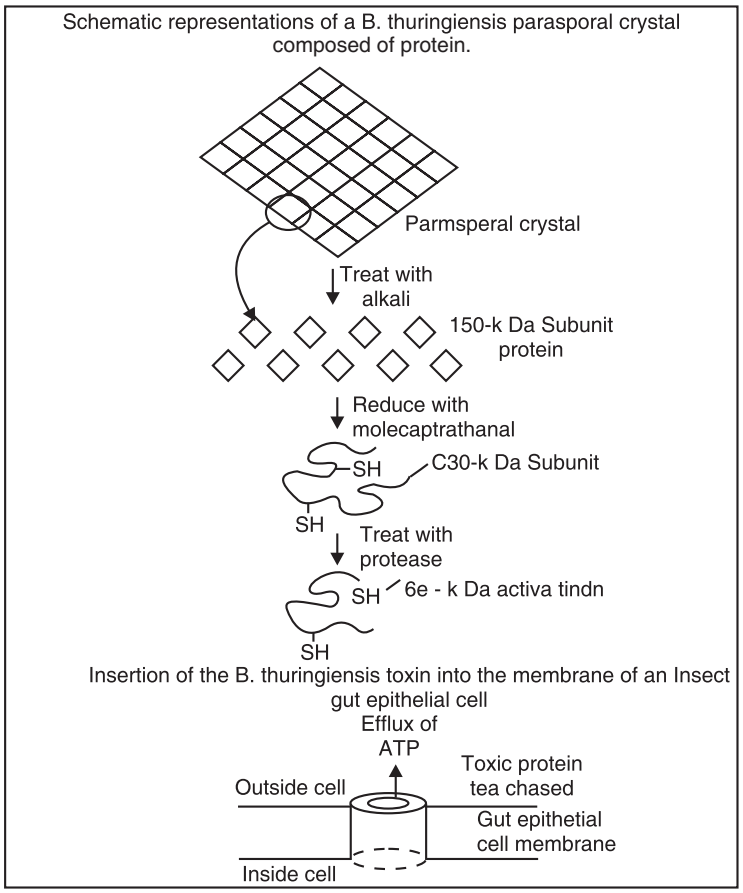


Fig. 18.7: Structure of thuricide

(1) Mechanism of action of toxins: Presently 12 groups of *Bacillus thuringiensis* are recognized. All of them produce these toxins. They are in the form of protein crystals and exhibit similar mechanism of action. They dissolve under alkaline reducing conditions which are present in the mid gut of most of the insects. On ingestion these crystals dissolve in the mid gut fluid of an insect and are also hydrolysed by the proteolytic enzymes released into the mid gut leading to the release of the toxins Fig. 18.8. The released toxins attack the cementing substances present in the mid gut wall which leads to loosening of epithelial cells of the gut wall (Fig. 18.9),

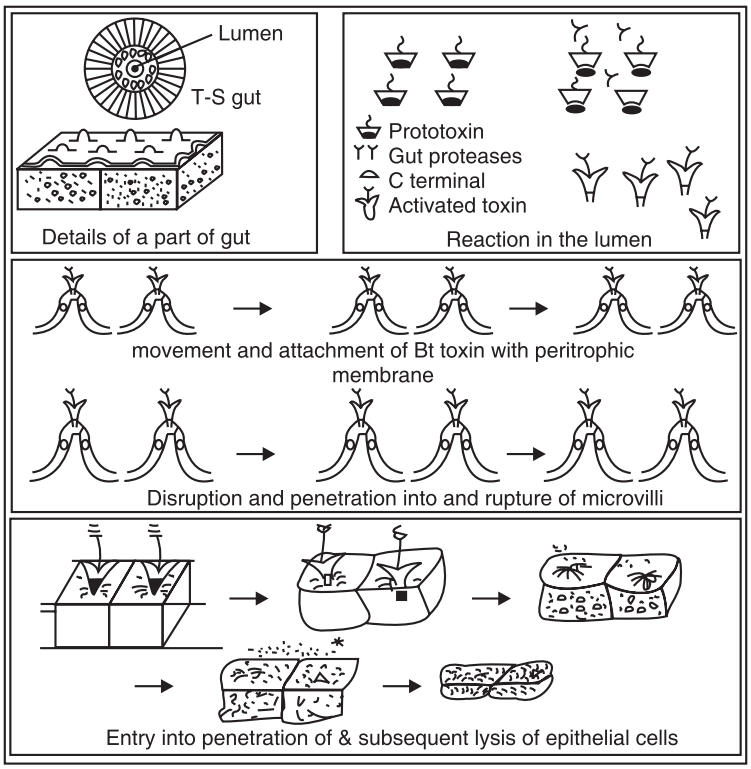


Fig. 18.8: Mechanism of action of bacillus thuringiensis toxin

due to which liquid diffusion occurs from gut wall to the blood making which is highly alkaline. This leads to gut paralysis followed by general body paralysis and death of the insect often occurs in as little as 60 minutes. In this way mixture of spores and crystals are pathogenic for a wide range of lepidopterous larvae and for a few sawflies. However, such mixture is non-toxic to other life forms like man, animal and plants. The insecticidal toxins from several strains of *B. thuringiensis* have been isolated and characterized based on the insecticidal activity of the toxin can be grouped into four major classes Cry I, Cry II, Cry III and Cry IV which are generally toxic to, both lepidaptera and diptera, coleoptera and deptera respectively. Further, based on immunological differences *B. thuringiensis* has been divided into 30 serotypes. A separate serotype is due to a particular set of antigenic/differences (epitopes) on the surface of one *B. thuringiensis* strain in contrast to another *B. thuringiensis* are also being transformed with improved efficiency (Table 18.13).

Table 18.13: Toxicity of naturally occurring and transformed subspecies of *B.thuringiensis*

Source of toxin		Toxicity		
Host DNA	**Introduced DNA**	**Pieris**	**Aedes**	**Phaedon**
B. aizawai	None	++	+	_
B. israelensis	None	_	++	_
B. israelensis	aizawai	++	++	_
B. israelensis	tenebrionsis	+	++	++
B. kurstaki	None	++	+	_
B. Kurstaki	tenebrionis	++	+	++
B. Tenebrionsis	aizawai	++	+	++

(2) Mass production of *B.thuringiensis:* The following two methods are generally employed for the mass production of *Bacillus thuringiensis.*

1. Submerged fermentation
2. Semisolid fermentation

(a) Submerged Fermentation: It consists of the following stages.

(i) Inoculum Preparation: Two different media are employed for the preparation of inoculum of *Bacillus thuringiensis*. The composition of these media is indicated below.

(a) Composition of the first medium

Beef extract	3 g
Peptone	5 g
Agar Agar	15 g
Distilled water	1000 ml

The pH of the medium is adjusted by adding few drops of 10 N sodium hydroxide.

Table 18.14: Some registered uses of *Bacillus thuringienesis*

Pest species	Crop variety
(i) Fruit crops:	
Fruit-tree leaf roller (*Archips argyrospillus*)	Oranges
Winter moth (*operophtera brumata*)	Grapes
Orange dog (*Papitio cresphosites*)	Oranges
Grape leaf folder (*Desmia funeralis*)	Grapes
(ii) Vegetable and field crops:	
Tomato horn worm (*Manduca quinque morculata*)	Tomato
Tobacco horn worm (*Manduca exta*)	Tobacco
Tobacco bud worm (*Helianthis virescens*)	Tobacco
Diamond back moth (*Plutella maculipennis*)	
European corn-borer (*Ostrinia nubulalis*)	Cabbage
Imported cabbage worm (*Pieris rapae*)	Sweet corn
Cabbage looper (*Trichoplusia ni*)	Cabbage and cauliflower
	Beans, cabbage and cauliflower, cotton, cucumber, lettuce melons, potatoes, tobacco
Alfa lalfa cater pillar (*Colias eurytheme*)	Alfa-alfa
Boll worm (*Helothis zea*)	Cotton
Artichoke plume moth (*Platyptilia carcuidactyla*)	Artichokes

(b) Composition of the second medium.

Corn-steep solids	0.85%
Beet molasses	1.00%
$CaCO_3$	1.00%
Distilled water	1000 ml

First medium is taken in a first shake flask and the second medium in the second shake flask. Inoculum of *Bacillus thuringiensis* is prepared by passing cells through two media. Each flask is incubated for 24 hours. The inoculum is passed into seed tank which is incubated until vigorous vegetative growth occurs. Then the cells are passed into a fermenter containing the production medium whose composition is given below.

Cotton seed flour (Pharma media)	1.40%
Corn steep solids	1.70%
Beet molasses	1.86%
$CaCO_3$	0.10%
Distilled water	1000 ml

The production medium is designed in such a way that fermentable carbohydrates and available nitrogen get exhausted at the time of commencement of sporulation.

(ii) Recovery: After sufficient biomass of the bacterium is produced in the production medium, it is recovered as described below.

The production medium is added with HCl to lower pH to 7.0. It is centrifuged and the resultant supernatant is discarded. The residue is suspended in 1.0 to 1.05 volume of 4-6% lactose solution and the contents are stirred for 30 minutes. Then 4 volumes of acetone is added while stirring slowly for 30 minutes. The contents are allowed to stand for 10 minutes and then filtered with suction. The residue is again stirred with small amount of acetone and filtered with suction. The filtrate is discarded. The above procedure is repeated again. The residue thus obtained is allowed to dry over night at room temperature. A suspension containing high amount of spore-crystal complex is formed. They can be used as sprays. Liquid formulation of the complex used to deteriorate fast and require storage under refrigeration. But presently these preparations have been stabilized, thus can be stored at room temperature (Fig. 18.9).

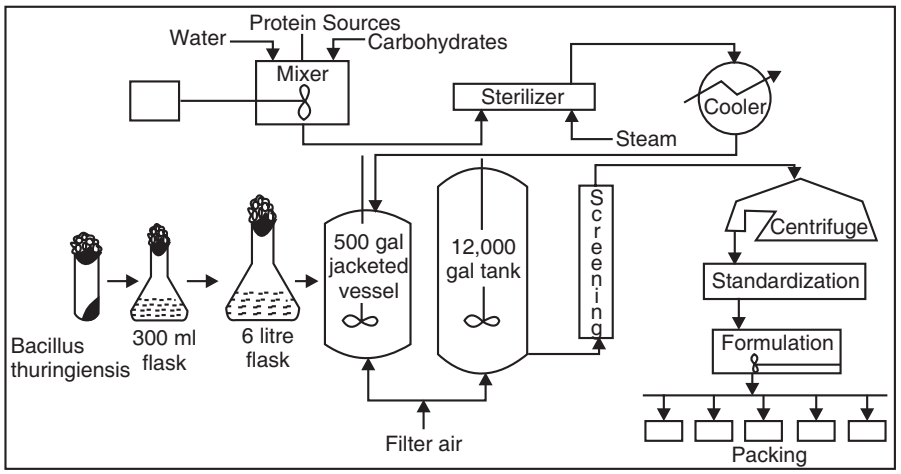


Fig. 18.9: Mass fermentative production of bacterial thuricide

(b) Semi-solid Fermentation: Three different media are used in this process for large-scale production of *Bacillus thuringiensis,* one for propagating the culture in flask. Second for propagating the culture in seed tank and the third for biomass production in the fermenter. The composition of the three media is as follows.

Table 18.15: Bts on the Market

Organism	Target pest	Product	Company
Bt.kurstaki	Caterpillars	Dipel	Abott labs
		Thuricide /Javelin	Sandoz
		Biobit /Foray	Nova Nordisk
		Bactospeine/Biocot	Nova Nordisk
		Cutlass/condor	Ecogen
		MVP	Mycogen
		Agree/Able	Ciba
		Delta-Bt	Deltabiological
		Raptor	American Cyanamid
Bt.israelensis	Mosquitoes	Acorbe	American Cyanamid
	Blackflies	Teknar	Sandoz
		Vectobac	Abbot labs
		Skeetal	Novo Nordisk
	Fungus gnats	Bactimos	Novo Nordisk
		Gnatro	Abbot labs
Bt.tenebrionsis	Colorado potato	Novodor	Novo Nordisk
	Bettle	Trident	Sandoz
		Foil	Ecogen
		M-trak	Mycogen
Bt.aizawai	Wax Moth	Certan	Sandoz
	Diamond back moth	Florbac	Novo Nordisk
		Xentari	Abbot labs

(i) Flask medium

Dextrose	15.0 g
Yeast autolysate	5.0 g
Corn steep liquor	5.0 g
Potassium orthophosphate	4.0 g
Distilled water	1000 ml

pH is adjusted to 7.0-7.2 with sodium hydroxide

(ii) Seed tank medium

Corn-steep liquor	4.5 g
Yeast autolysate	6.0 g
Dextrose	10.0 g
Calcium chloride	0.10 g
Potassium orthophosphate	3.5 g
Sodium hydroxide	0.43 g
Distilled water	1000 ml

(iii) Production medium

Dextrose	036 g
Wheat bran	545 g
Soyabean meal	062 g
Expanded peslite	380 g
Lime	3.6 g
Calcium chloride	0.29 g
Sodium chloride	0.9 g
Water	160 ml

About 0.1 percent by volume of the flask culture is added to the seed tank and is incubated at 30°C for 16 hours. About 400 ml of seed tank culture per kilogram of the production medium is employed.

About 1 kg of the production medium after partial sterilization with flowing steam for 60 minutes is inoculated with 400 ml of seed tank culture giving a final mixture with moisture content of 60% by weight and pH of 6.9. Fermentation is generally carried out by placing 250 kg of inoculated bran in bins with perforated bottoms, through which 0.4-0.6 volume air/volume with 95-100% relative humidity and a temperature of 30-34°C is passed into the inoculated production medium for first 3 hours and later on, the air flow is increased to 1.0-1.2 volume/volume. The bins are incubated for 36 hours. The pH rises to 7.5 and the moisture content decreases to about 53% during incubation. After incubation at 50-55°C dry air is passed through the bran for another 36 hours. The bran medium is then harvested and ground through 80 mesh screens. The moisture content of the final product is adjusted to 4% and the pH at 7.0. Finally bran cake is obtained by drying and grinding. However, it is also produced in the form of bran dust. The final product contains spores, crystals and exotoxins of *Bacillus thuringiensis*.

3. Application:

1. Commercial production of *Bacillus thuringiensis* are presently registered and used in the USA for more than 20 agricultural crops in addition to trees, ornamental shrubs and forests. They are used to control at least 23 insect pests (Table 18.14)

Table 18.16: Commercial Biopesticides developed from *Bacillus thuringiensis*

Bacterium	Target insect	Trade name
B.thuringiensis	Caterpillars	Dipel, Thuricide, Biobitagri, Delto BT
B.thuringiensis sub sp israelensis	Mosquitoes Block flies	Technar, Acorbe Skeetal, Vectobac
Bt.lentemorbus	Coloradopotato Beetle	Novodar Trident Foil
Bt.aizawai	Wax moth Diamond Diamond Black moth	Certan Florbac Xentari

2. A microbial insecticide called Depel was developed from *Bacillus thuringiensis*, strain HD-1 and tested for its effectiveness against various pests.
3. Promising results have been secured from field trials of Dipel combined with small amount of chemical insecticides.
4. Dipel is caliberated for field use at 16,000 ucn per liter. A minimum of viable spore count of 2.5×10^{10} per liter is required for effective control.

5. Formulations from *Bacillus thuringiensis* are also used to control *Aedes, Anopheles* mosquitoes and black fly which cause yellow fever, malaria and river blindness respectively.
6. Presently more than 50 Bt genes have been isolated, cloned and characterized. Making use of microorganisms, genetic engineered transgenic microbes and transgenic plants are developed. Many crop plants are colonized by harmless bacteria. Such bacteria are identified and genetically transformed by vectors carrying Bt genes. These bacterial formulations are then sprayed on the crop plants, which provide a protection cover to the crop. Some of the Bts available in the market are listed in Table 18.5. Certain crop plants are also being protected against insects by genetic manipulation that is transgenic plants with Bt genes. For example, boll worm resistant cotton, stem borer resistant rice, corn borer resistant maize, potato beetle and tuber moth resistant potato, tomato resistant to pinkworm, Bt transgenic cabbage and cauliflower plants.

Bacillus papilliae and *Bacillus lentimorbus* are the other species of *Bacillus* which are pathogenic to insects, especially they are able to kill larvae of the insect, *Papillia japonica,* which causes milky disease in certain plants. Both of them have been developed for microbial control of the pest. But *Bacillus papilliae* has been studied more extensively since it has proved to be most effective in the field.

It has been reported that the microbicidal effect on the pest is more in the spore formation phase of the bacterium. After ingestion of the viable spores of the bacterium by larvae, germination of spores takes place in the gut. The resultant cells penetrate the gut wall and enter the body cavity where they multiply. Extensive growth of the bacterium occurs in the haemolymph of infected larvae.

The mechanism of pathogenicity of these bacteria is not clearly known. Presence of a refractile body called parasporal body is reported from the sporangium of *Bacillus papilliae.* But there is no evidence of participation of this parasporal body in causing pathogenicity. However, it has been reported that the major pathogenicity may be due to removal of nutrients and essential growth factors from the blood by the actively growing bacterial cells. Toxins are also involved, since cell free filtrates of bacterial cultures exhibit toxicity, if injected in small amounts into larvae.

The spore containing material is applied to soil in 2-gr amounts at the intervals of a few feet. Such a treatment programme resulted in the spread of the infection throughout the Japanese beetle population in the treated area, which leads to the reduction of larval population control of the pest and the disease. These bacteria have also been employed in controlling pests other than Japanese beetle. After a detailed study of these bacteria, it has been emphasized that *B. papilliae* and *B. lentimorbus* are only commercially important pest control agents in this large group of microorganisms (Table 18.16).

18.1.6 Miscellaneous methods

Radiation sterilized adults of mass reared population of screw-wormfly were released in the environment. The copulation with sterile adults proved to be futile and resulted in the reduction of fly population. Million sterile adults of screw worm and 1.3 million pink bollworm were released daily by USDA to bring these pests under control. Successful attempts have also been made to reduce the moth population using sex attractants (pheromones), growth inhibitors (keromones) and light traps.

18.1.7 Biotechnology of microbial insecticides

Upto date large number of fungi, bacteria, viruses, protozoa and rickettsiae are being exploited for controlling insect pests. However, only a few could be mass cultured, formulated and marketted (Table 18.17).

Table 18.17: Status of biopesticides developed

Insect pathogen	Described	Mass production	Available	Production technology		
				Whole organism	Fermentation technology	Tissue culture
Fungi	530	30	2	7	27	0
Bacteria	100	14	4	4	10	0
Viruses	721	33	3	33	0	9
Protozoa	1000	15	0	11	3	2

The application of biopesticides to control pest species show some benefits as well as drawbacks over the use of traditional chemical pesticides which are listed below:

(i) Advantages:

1. They can be applied in various ways some of them can be directly introduced and colonized in the insect pathogen which leads to permanent control. Some of them can be applied as liquid or dust preparations.
2. The production is relatively easy and inexpensive.
3. No hazardous effect on non-target organisms.
4. High degree of host specificity kill only the specific host.
5. They can be used synergistically in combination with chemical pesticides.
6. Absence and non-accumulation of toxic residues in the environment.
7. Less possibility of resistance to microbial insecticides by the insect pests.
8. Efficient microbicidal action at low dosage in certain cases.
9. Biopesticides persists in the field for some time and infect insects when it reappears again.

(ii) Disadvantages:

1. The difficulty of production of some microbial insecticides in large quantities inexpensively.
2. The difficulty in maintaining the microbial insecticide in viable and virulent state till it makes contact with the target insect.
3. Timing of the application in regard to disease incubation period should be correct, otherwise the application becomes waste.
4. Ineffective action of the formulation, if the disease is caused by more than one pests, because of specificity of the host.
5. Requirement of specific climatic conditions by some of the insecticides in order to invade and infect the host insect.

18.1.8 Status of bioinsecticides in India

With establishment of Commonwealth institute of biological control (CIBC) laboratory at Bangalore in 1959, research on Biological control has started in India and union directorate of plant protection (UDPP) extended biocontrol activities at national and state levels. Biocontrol research laboratory, of pest control (India) Ltd., Bangalore is actively engaged in mass culturing the native predators, parasites in order to supply them to farmers for the control of grape mealy bugs, control of leaf caterpillars and sugarcane borer. Tamilnadu co-operative sugarcane federation, Chengleput has set up main biocontrol laboratory for supply of egg parasite, *Trichogamma* for the control of sugarcane borer. Tamilnadu Agriculture University, Coimbatore has also set up a pilot plant facility for the mass production of *Trichogramma* predators, *Chrysopa*, viruses NPVs and Gvs. Union department of biotechnology has established seven similar centers in different parts of the country.

18.1.9 Future

At present biopesticides control is used only 1% of total insecticides and needs attention of several disciplines of biology and strengthen this field. The efforts to clone *Bacillus thuringiensis* toxic genes in *E. coli* and *B. subtilis* to make their expression in these bacteria for large scale production of toxin have already begun. It is certain that biological control of pests will play vital role in the management of various pests. There is an urgent need to find out additional target insects, selection and strain improvement of microbial pesticide, safety of microorganisms, mass production and their effectiveness, impact of microbial pesticides on environment, formulation and application technology and improvement of application technology.

18.2 BIOLOGICAL CONTROL OF PLANT DISEASES

Biological control may be defined as any condition or practice whereby survival or activity of a pathogenesis is reduced through the agency of any other living organism except man himself, with the result that there is a reduction in the incidence of the disease caused by the pathogen. Garrett (1965) defined biological control of plant disease as the reduction of inoculum density or disease producing activities of a pathogen or parasite in its active or dormant state, by one or more organism accomplished naturally or through manipulation of the environment, host or antagonist or by mass introduction of one or more antagonists.

W. Roberts (1874) was the first to demonstrate the antagonism between *Penicillium glaucum* and bacteria which paved the way for biological control of plant pathogens. Demonstration of Hartley (1921), Sanford (1926), Millarrd and Taylor (1927), Grossback and Broadfact (1931) and Rishbeth (1970) clearly established the way for biocontrol of plant diseases.

Through successful competition, production of enzymes (glucanases proteases and chitinases) to lyse pathogen cell walls, antibiotic and inhibitory metabolite production indirect effect resulting in modification of the environment adverse to pathogen. Biological control of plant pathogen occurs as, both resident and introduced antagonists are able to diminish plant diseases (Mathre et al., 1999), but its general utility was skeptically regarded (Kubicek 1998). The ecological balance of soil microorganisms can be manipulated by modifying the organic matter content, pH, temperature (Hasson *et al*., 1997), by competition of microorganisms for limited nutrients (Chaube and Singh, 1991), parasiting the pathogen by microorganisms (Mukherji and Sen, 1992), production of antibiotics by antagonists (Guetsky *et al*., 2001), suppresive soil to soil-borne pathogens (Lorito *et al*., 1996), introduction of microorganism of suppressive soils to conducive soils (Sawant *et al*., 1995), inoculation of plant with mild strain of virus can protect it

against subsequent inoculation with a more virulent strain (cross protection). Biocontrol agent may induce silent defense genes that make plant system resistant against plant pathogens (Thrane *et al.*,1997), application of more than one biocontrol agent (Harman and Bioorman, 1998), genes from biocontrol agents can be a good source for improving plant resistance to pathogens (Benftez et al., 1998), transformation of biocontrol agents with extragenes could increase the antimicrobial activity of the strain (Gupta *et al*., 2001).

Though the term biological control was used for the first time in relation to plant pathogens by C.F. Von Tubeuf in 1914, it is in 1921 when C. Hartley introduced microorganisms into soil to control root disease. Subsequently several plant pathologists including Sanford (1926); Millard and Taylor (1927); Sanford and Broadfood (1931) contd. Researches from 1950 to 1970 clearly established possibility to control plant diseases through microorganisms specially fungi like species of *Trichoderma* and *Gliocladium*. The mechanism of biological control operates in variety of ways such as both resident and introduced antagonist are able to diminish plant diseases. Ecological balance of soil microorganisms can be manipulated by modifying the organic content, temperature, pH of soil. Microorganisms compete for limited available nutrients, microorganisms may parasitize other microorganisms (Chaube and Singh, 1991), antagonists produce antibiotics that may play a part in biocontrol of pathogen (Elad 1994). Some soils are naturally suppressive to some soil-borne plant pathogens or become so following prolonged occurrence of disease caused by pathogen, inoculation of plant with a mild strain of virus can protect it against subsequent inoculation with a more virulent strain and/or biocontrol agents may induce silent defense genes that make plant system resistant against plant pathogen induced resistance (Thrane et al., 1997). Total microflora of some suppressive soils can be transferred to conducive soils making them suppressive (Kleifeld and Chet, 1992), application of more than one biocontrol agents (mixed formulation) could be reliable means of reducing variability and increasing the reliability of biological control (Harman and Bloorfman, 1998), genes from biocontrol agents can be a good source for improving plant resistance to pathogens. Chaoube *et al*., (2003) discussed elegantly various aspects of biological control of plant pathogens.

A good and stable biocontrol is possible when a potential organism is employed. The biocontrol organism should possess following characteristics:

1. Utilize available nutrients, increase in number and colonize the region in rapid fashion.
2. The ability to produce an antibiotic is a desirable character.
3. It should be able to elicit resistance response at infection site.
4. Prolonged survivability is a quality.
5. Antagonist must be compatable with chemical used.
6. Biological agents should not be adversely affected by the storage environment specially with reduced O_2, elevated CO_2 and nitrogen levels.
7. Antagonist should not be health hazardous.

The mechanism of protection of plants by biocontrol is either by direct action against pathogen or indirectly by reducing host susceptibility (Fig. 18.10). Plant diseases are controlled by antagonists involving variety of mechanisms which varies both with the pathogen and host as precised in table 18.18.

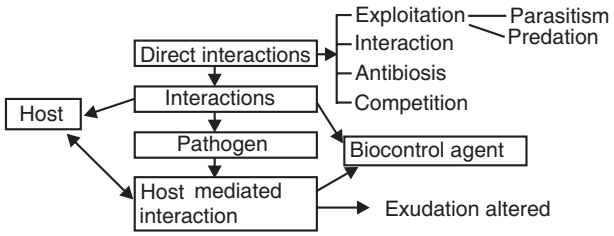


Fig. 18.10: Modes of action of biocontrol agents

Table 18.18: Mechanism of action of fungal biocontrol agents against plant pathogens

Antagonists	Pathogens	Mechanism of action	References
Aspergillus flavus *A.niger, A.mudulans*	*Melodogyne incognita, M.phaseolina,Xantho-monas oryzae pv. oryzne, Sclerotium rolfsii, Glomerella cingulata, R.solani, F.oxysporum*	Non-volatile compounds of culture filtrate, Antibiosis, Mycoparasitism, induced systemic resistance and production of siderophores	Varshney and Chaube (2000)
Acremonium obclavatum	*Puccinia arachidis*	Hyperparasitism	Goldman et al., (1994)
Chaetomium globosum	*Alternaria alternate, F.moniliformae*	Antibiotics	Singh and Misra (2001)
Epicoccum purpurescens	*M.phseolina*	Antibiotics	Siddiqui and Hussain (1995)
Gliocladium roseum, Gliocladium sp.	*Sclerotinia sclerotiorum, Sarocladium oryzae*	Mycoparasitism and antibiosis	Sen (2000)
Igaspora margarita	*Heterodera cajani, F.udum*	Antibiosis	Shrestha (1992)
Glomus claroidium, G. mosseae, G.etunicatum, G.fasciculatum, G.vessiforme	*M.phaseolina,F.oxysp-orum f.sp. lycopersici, S.rolfsii, Meloidogyne incognita, R.solani, R.bataticola, Sclerotium oryzae*	Changes in population of microorganisms in rhizosphere, Alteration in root exudation, Antibiosis, Thickning of cell wall by lignification and production of polysaccharides	Rajappan (1997)
Paeculomyces lilacinus	*Rotylenchulus reniformis M. inocognita. R. solani*	Antibiosos	Kaur and Mukhopadhyay (1992) Nakkeeran and devi (1997)

contd...

Antagonists	Pathogens	Mechanism of action	References
Penicillium citrinum, P.oxalicum, P.simplicissimum, P.funiculasum,P.conyl-ophilum,P.islandicum, P.pinophilum,P. purpurescens	*Colletotrichum gloeosporioides, Xanthomonas compestris, M.phaseolina,Sarocla-dium oryzae, Alternaria brassicicola,F.oxysp-orum f.sp.ciceri, Mycosphaerella arachidis, R.solani, Ralstonia solanacearum, ustilago scitaminea*	Parasitism and antibiosis	Nakkeeran and devi (1997)
Trichoderma aureoviride,T.haemat-um,T.harzianum,T.virid e, T.koningii, T.lignorum,T.longibrac hidatum,T.virens	*R.solani, Meloidogyne javanica, M.phaseolina, Sclerotium rolfsii, F.oxy sporum. f.sp. zingiberi,F.oxy.f.sp.vasi nfectum,Cercospora monicola,F.solani,Ustil ago segetum tritici, Pythium aphanidermatum, Claviceps fusiformis, Phytophthora cinnamoni, P.cactorum,P.nicotian-eae, P.fragariae, Sclerotinia americana, Aspergillus flavus, Colletotrichum gloeosporioides, Neovossia indica, Drechslera sorokiniana, Botrytisalli, Sarocladium oryzae,F.moniliforme*	Parasitization on mycelium and sclerotia, Direct parasitization of second juvenile larva and eggs of nematode, Induction of systemic resistance, Competition	Schirmbock et al., (1994)

A. Natural Biological Control:

The natural reduction in pathogens population and disease incidence occur due to

(i) Suppressive soils and (ii) Monoculture

(i) Suppressive soils: Suppressive soils are unfit for the development of certain diseases. In these soils the saprophytic growth, pathogenic activity and survival of pathogen is reduced, if pathogen establishes it cause little or no damage. When continuous cropping with susceptible crop results decrease in disease severity. This is also called induced natural suppression. The chlamydospore germination and germ tube growth are highly suppressed in suppressive soils than in conducive soils (Albavette *et al.,* 1980). There are now several evidences, that by the use of suppressive factors it may be possible to

establish microbially based resistance in certain soils conductive for diseases. The mechanism of suppressiveness has not been fully resolved. However, the explanation given for the suppressive phenomenon include antibiosis, competition, parasitism and predators (Hornby, 1983). Albonnette *et al.* (1979) have concluded that the suppressiveness is linked with microbial activity and thus it involves biological control. Take all (Weller and Cooke, 1985), fusarial wilt (Scher and Baker, 1980), damping off (Hancock, 1977) and potato scab (Menzies, 1959) have been effectively controlled by this principle. Baker (1950) demonstrated that small quantities of wilt suppressive soil when added to steamed soil or natural soil provides suppressiveness to *Fusarium* wilt of carnation.

(ii) Monoculture: Significance of monoculture for biological control of soil-borne plant pathogens has been extensively discussed by Cooke and Baker (1983). King (1923) reported for the first time the decline of cotton root-rot caused by *P. amnivorum* due to monoculture. All disease of wheat was also controlled by suppressive factor which was fastered by wheat monoculture. These changes are:

(i) Development of specific antagonism,

(ii) Change in pathogens population and

(iii) Change in the microflora, which affect the pathogen. Soil factors help in building up of suppressiveness in soil due to monoculture.

B. Induced Biological Control

1. **Annulment of fungistasis:** Soil-borne plant pathogens survive in the soil for long time mainly because of soil fungistasis which inhibits germination of spores, sclerotia and chlamydospores of pathogens. Germinating spores are highly susceptible to lysis due to other microbes. Hence, to induce lysis of pathogens propagules the fungistasis should be first annulled. Stimulation of fungal propagules to germinate by organic materials without supporting formation of new resistant propagules may provide the basis for biological control. Sclerotia of the onion wilt, root pathogen (*Sclerotum cepivorum*), do not germinate in soil because of fungistasis. Volatiles such as organic sulfides are exuded by the roots of onion and they stimulate germination of sclerotia of *S. cepivorum*. The germlings die either from nutrient exhaust or of lysis by microorganisms. Similarly volatile compounds liberated during alflfa hay decomposition increase microbial activity in soil, stimulate germination and lysis of sclerotia of *S. rolfsii*. Sanford (1926) reported the control of apple scab by green manuring and heavy application of organic matter controlled *Phymatotrichum omnivorum* of cotton are some of the examples of annulment of fungistasis. It has several advantages such as it may have multiple pathogen suppression, lasting effect and involve less cost.
2. **Elevation of fungistasis:** Instead of annulling fungistasis it can also be elevated and control soil-borne diseases. Soil can be made increasingly fungistatic to chlamydospores of *Thielaviopsis basicola* by amending it with alfalfa hay and allowing decomposition. Similarly rye and corn residues enhance fungistasis to chlamydospores of *Fusarium solani* f. sp *phaseoli.* These treatments elevate the level of toxicity of soil to such an extent that the pathogens propagules become inactivated even in the presence of root excudates.
3. **Inhibitory Volatiles:** Crucifers when added to the soil, many volatiles such as methanethiol, dimethyl sulfide and dimethyl disulfide are released. These compounds

control pea root-rot caused by *Aphanomyces euteiches* by inhibiting growth, zoospore mortality and germination of fungal propagules.

4. **Antagonist Organisms:** Introduction of antagonist into the soil to reduce the activity of pathogen has been most frequently used practice because of increase in population of antagonist. Seed treatment (Broadbent *et al.,* 1977) and pelleting of seeds (Merriman et al., 1973) with antagonist proved to be successful in the control of root rot of rye caused by *R. solani. Bacillus subtilis, Trichoderma* spp., *Pencillium* spp. and *chaetomium globosum* were considered highly effective for protection of corn, peas and soya bean then control of crown gall caused by *Agrobacterium tumifaciens* by *A. radiabacter.* Organic amendments with high C/N ratio increases antagonists, *Streptomyces* spp., and control root-rot of bean caused by *Rhizoctonia solani.*
5. **Cross protection:** When avirulent strain of the pathogen is inoculated into the soil, the disease caused by virulent strains is reduced. *Verticillium* wilt of cotton is controlled by incorporating avirulent strain of the pathogen in the soil. The phenomenon involves the prior colonization of infection count by the antagonists. This is also called as induced resistance. Similarly *Verticillium dahliae* causing wilt of mint can be controlled by use of *V.nigricans* (Melank and Horner, 1975). However, drawbacks of this method are :
   (i) Avirulent strain of pathogens may have negative effect on plant health and vigour.
   (ii) The crop protecting strains may have pathogenic effect on other plants grown in the same field.
6. **Mycorrhizal fungi:** Ectomycorrhizal fungi like *Boletus variegata* produce volatiles which inhibit the growth of *Phymatotrichum cinnamomi* and *H. annosus* (Markers, 1975). Similarly Davis et al. (1979) have reported that *Verticillium* wilt of cotton was less severe in VA mycorrhiae colonized roots. Commercial soil inoculum of *Glomus deserticola* is now distributed in California for biological control. However, careful study is needed before considering these fungi for biological control.
7. **Amoebae:** The soil amoebae feed on fungal spores and hyphae. The perforation and lysis of the spores have been described in *Thielaviopsis basicola.* Baltruochat and Shanbeck (1975) have recorded decrease in the population of *T. basicola* and *Cochliobolus sativum* by addition of cysts or active vanpyrellid amoebae into soil. Chakrabarty and Warcup (1983) have also reported reduction of take-all by mycophagous amoebae.
8. **Mycoparasites:** Ayens and Solams (1981) have listed mycoparasites which have been successfully employed in the control of soil-borne diseases. *Trichodema* spp. have been exploited for this purpose as they readily parasitizes many soil-borne fungi. Four species of *Trichoderma* viz., *T. viride, T. haziaanum*, *T. koeningii* and *T. hamatus* are reported to be mycoparasites. *T. hamatum* is reported to be a hyperparasite of *R. solani* and protected the seeds of pea and radish. Trutmonn has employed *Gliocladium roseum, Myrothecium verrucaria* in the biological control of *Sclerotinia scleriotarum. Sporidesmium sclerotiorum* has been reported to parasites the sclerotia of *Sclerotinia minor* and *Sclerotium cepivorum* (Adams and Ayers, 1980). Biological control of *Sclerotinia lettuce* crop in the field by *S. sclerotivorum* and *R. solani* on sugar beet by *Corticum* sp. is due to the pathogenic activity on respective plant pathogens. The mycoparasites are generally isolated from the soil by introducing sclerotia of fungi or other fungal structures.

   The fungi parasiting plant pathogens are called hyperparasites. These fungi are being exploited by applying spore suspension to diseased plant (Table 18.19). Mycoparasitic species of *Trichoderma* (*T.harzianum* and *T.Viride*) have been successfully developed as

biocontrol agents. *T.viride* grow on wide variety of substrates and on selerotia of other fungi. Conidia of *Trichoderma* sp do not survive well in soil but Chlamydospores survive for long time. *Trichoderma* cannot compete with other saprophytes and it is only a secondary invader. Green manure, compost supports *Trichoderma* well. In soil fumigated with carbondisulfide or formaldehyde or methyl bromide, *Trichoderma* multiplees well. These chemicals are non-inhibitory to *Trichoderma* but inhibit other saprophytes. Treatment of soil with steam enhances *Trichoderma* in soil. Similarly soil solarization and sub-lethal heating of soil encourages the *Trichoderma.* Thiram, Captan, PCNB and chloronel are inhibitory to *Trichoderma* but suppress other saprophytes. However, benonyl, Carbondazin, thiophanate methyl and thiabendazole are inhibitory to *Trichoderma. Trichoderma* does not survive in the rhizosphere and anaerobic soils. Characteristic features of *Trichoderma* are:

Table 18.19: Mycoparasties of plant pathogens

Pathogen	Mycoparasite (Hyperparasite)
1. *Pythium aphanidermatum*	*Lactisaria aralis*
2. *Plasmopara viticola*	*Trichothecium plasmoparae*
3. *Sphaerotheca fuliginea*	*Acremonium alternatum*
4. *Erysiphe graminis v. sesam)*	*Ampelomyces quisqualis (syn. Cicinnobolus)*
5. *Hemileia vastarix*	*Verticillium hemileiae*
6. *Puccinia arachidis*	*Darluca fulvum Eu*
Verticillium	
	Tuberculina costaricana
	Penicillium islandicum
	Acremonium persicinum
7. *Puccinia recondite*	*Aphanocladium album*
8. *Uromyces apperndiculatus*	*Verticillium lecanii*
9. *Rhizoctonia solani*	*Trichoderma haematum*
	Lactisaria aralis
10. *Sclerotium rolfsii*	*Trichoderma harzianum*

(i) Methods of application: Wheat bran plus peat, barley grain and composted wood bark are good substrates for production of *Trichoderma.* Alginate pellet with bran are useful for introduction of *Trichodera* into soil. Diatomaceous earth granules impregnated with molasses also serve as carrier of *Trichoderma* into the soil. The fungus survives well in spermosphere. Hence, seed treatment with the fungus controls many seed-rot and dampling off diseases.

(ii) Mechanism of action: *Trichoderma* produces antibiotic (trichodermin, gliotoxin and viridins) in culture but not in soil. *Trichoderma* invades hyphae of *Pythium* and *Rhzoctonia* produce produce haustoria. It produces β (1-3)-glucanase and chitinase then degrades the glucans in the cell wall of *Pythium* sp. while chitin and glucans in the cell walls of *Rhizoctonia.*

(iii) Control of soil-borne diseases by *Trichoderma*: Treatment of seeds of peas, tomato and tobacco with *Trichoderma*, controls the damping off by *Pythium* and *R. solani*. It controls the root rot of ground nut (*Sclerotium rolfsii*).

(iv) Limitations:

1. *Trichoderma* does not survive in the rhizosphere soil as well as in unamended soil. The fungal inoculum should be in young mycelial stage and old cultures with spores will be inactive.
2. *Trichoderma* does not survive in the soil.
3. *Trichoderma* controls soil-borne diseases only in early stages of crop growth.

Other fungi: *Gliocladium virens* acts similar to *Trichoderma* as a biocontrol agent against *Pythium, Sclerotium* and *Rhizoctonia* diseases. *Sporidesmium sclerotiorum, Sclerotinia minor* and *Coniothyrum minutans* control *Sclerotima sclerotium* population in soil.

9. **Bacteria:** Many bacteria produce siderophores. Siderophores are low molecular weight ferric ion transport agents and supply iron to the bacterial cell. Siderophores are relatively complex with iron of very high affinity and made it unavailable to other microorganisms including pathogens. Inspite of wide variation in structure of siderophore with the producing organism, it forms six coordinate octahedral complex with ferric iron and reported to play an important role with iron nutrition. Siderophore chelate Fe(III) and microbial membrane receptor proteins specifically recognize and take up the Fe complex. This results in making Fe unavailable to rhizophere microorganisms including plant pathogens which produce less siderophores or different siderohhores with lower binding coefficiencies. The result is less pathogen infection and control the disease. *Pseudomonas florescens* and *P. putida* produce siderophores control black-rot of potato caused by *Erwinia cartovara.* The siderophores of *Pseudomonas fluorescens* are called *Pseudobacterin* has been found to control all disease of wheat, barley caused by *Gaumanomyces* var *tritici* and flax wilt disease caused by *Fusarium oxysporum* f. sp. *lini.* Addition of *Pseudobacterin* to the soil also controls wilt diseases. Pseudobacterin converts pathogen conducive soils into pathogen suppressive soils. It suggests that suppressiveness of the soil is mainly due to siderophores.

10. **Other approaches**

(a) Solarization:

(i) Solar heating of soil by covering it with polythene sheet results in the change of microbial activity. This process leads to increased microbial activity of the soil against pathogen,

(ii) Decrease the survability of the pathogen, and

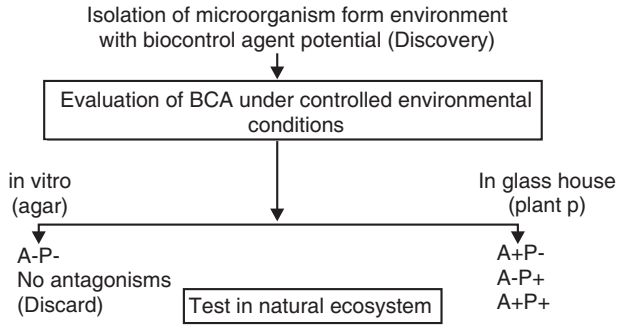
(iii) Increased susceptibility of pathogen to soil microorganisms. Solarization has been effectively used in the control of *Verlicillium dahlae* (Puttman *et al.,* 1980), *Sclerotium rolfsii* and *R. solani* (Katan, 1980).

(b) Flooding: Flooding of soil has been found to eliminate or reduce the population of *F. oxysporum* f. sp. *cubens* (Stover, 1962), *V. dahliae* (Bufferfield *et al.,* 1998) and *Sclerotinia sclerotiorum* (Naiki and Cooke, 1983).

(c) Crop rotation: Baker and Cork (1979) reported that crop rotation is one of the most oldest approaches. Crop rotation for 1 to 3 years is adequate for the control of *Gaumonomyces graminis V. tritici* and *Cercosporella herpotrichoides* (Baker, 1981). However, Cook and Baker (1983) feel that this approach is being replaced by tillage, crop sanitation and changing planting dates (Baker, 1981).

11. **Development of mixed formulations of Biocontrol agent:** Most of the studies on biological control of plant diseases deal with single biocontrol agent as against a single pathogen. Due to variety of situations and host specificity even at subspecies level, most of the times inconsistant performance of biocontrol agent is reported. Thus single biocontrol agent is not likely to be active in all soil environments or against all pathogens that attack the host plant. This situation may be achieved either by (i) selecting a strain of biocontrol agent with wide host range, (ii) modify the genetics of the biocontrol agent to add mechanisms of disease suppression that are operable against more than one pathogen, (iii) alter the environment to favour the biocontrol agent and disfavour the competitive microflora or develop a strain mixtures with superior bicontrol agent activity, (iv) Several strategies for developing mixtures of biocontrol agents could be developed including mixtures of organisms with different plant colonization patterns, mixtures of antagonists with different mechanisms of disease suppression, mixtures of taxonomically different organisms.

18.2.1 Commercialization of biocontrol agent: Commercialization of biocontrol agent is a multistep process involving wide range of activities as depicted in Fig. 18.11.



(+) 1. Mechanisms
2. Strain improvement
3. Commercialization
   1. Mass production
   2. Formulations
   3. Delivery
   4. Compatibility
   5. Registration and release

(+) – Antagonism/Disease suppression

(–) – No evidence of antagonism/Disease suppression

Fig. 18.11: Components and stages in developing biocontrol activity

Screenings for pesticidal organisms involves selection of healthy plant from areas otherwise infested with targeted pathogen disease and isolate organisms from leaf surface and rhizosphere. Isolates are then screened for activity against the pathogen in the laboratory assay, green house condition and optimization of conditions. It is then evaluated under field conditions. Potential biocontrol agents are maintained and supplied to target agencies. In India biocontrol agents are maintained at project directorate, biological control, Bangalore and G.B. Pant University of Agriculture and Technology, Pantnagar.

18.2.2 Mass production: One of the greatest obstacles to biological control by introduced antagonists has been scarcity of methods for mass culturing and delivering the biocontrol agents. The unique problem of biocontrol agent is that represents a living system which must be able to stand the process of formulation and should remain sufficiently viable for a period until it reaches farmer. Despite the limited scope, scientists are engaged in developing effective experimental systems for growth and delivery of antagonist. Methods of Mass production of widely used antagonist, *Trichoderma* are precised in table 18.20. The unique problem in developing biopesticide, living system, must sustain the process of formulation and remain sufficiently viable till it reaches farmers field.

Table 18.20: Mass production of fungal biocontrol agents

BCA	Base material	Form of formulation	Reference
Trichoderma harizianum	Black gram shell, shelled maize cob, peat, gypsum, coffee fruit skin, biogas slurry, coffee husk, molasses-yeast, mushroom grown waste, poultry manure, sugarcane, straw spent tea waste, wheat bran etc.	Powder pellets	Becker and Schwinn (1993) mihuta-Grimm and Powe (1986) Chaube and Singh (1991)
T.viride	Barley grain, groundnut shell, farm yard manure	Pellets	Mukhopadhyay (1996)
T.virens	Coffee chery husk, soil, talc, wheat bran, saw dust	Pellets	Kerr (1980)
T.longibrachiatum	Talc, Wheat bran + saw dust	Powder	Roy (1977)
A.niger AN27	Citrus pomace, Talc+CMC	Pallet, powder	Selvakumar and Srivastava (2000)
A.terreus	Maize meal + sand	Powder	Elad et al. (1999)

Development of safe, easy to handle, cost effective formulation of biocontrol agent is of paramount importance. Blending of active ingredients such as fungal spores with inert carriers such as diluents and surfactants in order to alter physical characteristics to a more desirable form. Formulation must have a minimum shelf life of 2 years at room temperature. At present, kaolin and bentonite are being used as carrier materials of *T.harzianum* (papavizas et al., 1984).

Biocontrol agents which have an easier and quicker passage for registration are with indigenous microorganisms that are never recorded as being plant, animal or human pathogens and specific to a defined group of target pathogens. The information required for registration of any biocontrol agents are systematic name and common name, natural occurrence and morphological description, details of manufacturing process (active and inert ingredients of formulation), mammalian toxicity, environmental toxicity and residual analysis. Some of the commercial formulations of fungal control agents are listed in Table 18.21.

Table 18.21: Commercial formulations of fungal biocontrol agents

Bioagent	Product(s)	Use	References
T.viride	Biocon, Bioguard, Defence-SF, Ecofit, Funginil, Trichoguard, Tricho-X	As seed/soil/foilage application against soil-borne plant pathogens	Siddiqui and Mahmood (1995)
T.harzianum	Plant biocontrol agent – 1, F-STOP, Trichodex, Rootshield, Biodac	As seed/soil/foilage application against soil-borne plant pathogens	Ramray et al. (1989)
T.virens	Soilguard	As seed/soil/foilage application against soil-borne plant pathogens	Ramray et al. (1989)
T.viride+T.harzianum	Bioderma	As seed/soil/foilage application against soil-borne plant pathogens	Siddiqui and Mahmood (1995)
T.harzianum+P.fluoresc -ens	Plant Biocontrol Agent – 3	As seed/soil/foilage application against soil-borne plant pathogens	Siddiqui and Mahmood (1995)
Aspergillus niger	Kalisena SD, Kalisena SL	For control of *Pythium ultimum* in sugarbeet	Selvakumarate et al. (2000)
Pythium oliandrum	Polygandron	For control of *Botrytis* spp. and *Penicillium* spp. on citrus and pome fruits	Baker and Cork (1974)
Candida oleophila	Aspire	For control of *Botrytis* spp. and *Penicillium* spp. on citrus and pome fruits	Ramraj et al., (1989)

18.2.3 Improvement of biocontrol agents

Once the suitable antagonist is obtained, it may be further improved for desirable characteristics like better antagonisitic activity, wider host range, tolerance to pesticides, survivability in the environment atmosphere competence, tolerance to adverse environmental conditions, vigorous growth and its long life in the laboratory. The most important and challenging biological control is the methodology of preparing and delivering the antagonist into the soil. Impregration of clay granules or diatomaceous earth for introducing inoculum of the antagonist is one such attempt. Commercial inoculum of *P. gigantea* spores are suspended in strong sugar solution and is now distributed in sacheti of biological control. For application, the contents of sachets are mixed with 5 liters water. Strains of *Trichoderna* spp. mixed in bettle powder and given a trade name B1, NAB control in France. A. *radiobacter* strain K 84 is also being distributed by several companies in a finely ground peat preparation for control of crown gall pathogen.

Genetic manipulation of biocontrol agents can play a great role in improving their potential. To improve biocontrol agents for desirable characteristics like better antagonistic ability, wide host range, tolerance to pesticides, survival ability in the environment, rhizosphere competence, tolerance to adverse environmental conditions, vigorous growth and self life. Different strategies adapted for this purpose are described below.

(i) Mutation: Ahmad and Baker (1988) and Carsolio et al. (1999) improved *Trichoderma harzianum* for rhizosphere competence and *T.viride* to hypersensitivity respectively. Mukerjee et al. (1997) and Starz et al. (1988) have developed vinclazolin and benomyt tolerant mutants of *Trichoderma viride* respectively through mutagenesis. Selvakumar et al., (2000) have developed carboxin tolerant mutants of *Trichoderma viride* by exposing the tolerant biotypes of *T.viride* developed by UV irradiation and ethyl methane sulfonate. Mukherjee and Mukhopadhya (1993) developed stable mutants of *T.viride*

which differed from the wild type strain in phenotype growth rate, sporulation and antagonistic potential by exposing to r-rays.

(ii) Protoplast fusion: Protoplast fusion is one of the best techniques for improvement of strain. *T.viride* strains are well known producers of chitinases, glucanases, cellulases and different other mycolytic enzyme combination of these desirable traits from different species or strains may give rise to superior strain. Protoplast fusion technique has advantage as the possibility of obtaining recombinants is more and also allows testing of large number of recombinants in a short time (Singh et al., 2001).

(iii) Genetic Engineering: Transgenic microorganisms in which specific strains are constructed to express genes from another microorganism whose products are inhibitory to plant pathogens are likely to have high utility as biocontrol agents. Lo et al. (1998) have developed transformant of *T.harzianum* strain 1295-22 by integrating β-glucouridinase (GUS) and hygromycin (hygB), phospotransfererase genes that exhibited increased biocontrol activity against *R.solani* as compared with wild type.

12. Prospects: Though the biocontrol way of plant protection is well established but it has yet to become an integral part of plant protection and needs further attention in the following areas:

1. Exploration of biodiversity and their conservation is an utmost important as, more potential organisms are likely to be discovered.
2. Genetic improvement of bioagents through molecular techniques.
3. Improvement in mass production technology, shelf life and delivery system.
4. More understanding of relationship among plant, pathogen, fertilizers and biocontrol agent is of basic need. How and why they work and interact with each other must be understood to evolve and formulate strategies.
5. There is need to select not only strains of biocontrol agent but also good growth and plant defense inducers.
6. Integration of biocontrol with other management practices under field condition is need of the hour.
7. Bioprimning of seeds with biocontrol agents, bioferitilizers and micronutrients must be explored and exploited.

REVIEW QUESTIONS

I. Essay type questions:

1. What are biopesticides? Discuss merits and demerits of their applications.
2. Describe methods of fermentation production of biopesticides.
3. Describe methods and application of viral insecticides.
4. Describe methods of *Bacillus thuringiensis* production. Add a note on mechanism of action of thuricide toxins.
5. Describe different mycoinsecticides, problems in their methods of production and formulation.
6. Give an account of limitations of biopesticides.
7. Give briefly developments of biopesticide technology.
8. Give an account of entamogenous fungi and their utility in insect pests management.

9. Describe different biological methods of plant disease control.
10. Discuss limitations of biocontrol of plant diseases.
11. Give an account of botanicals in the control of insect pests.
12. Define microbial insecticide. Describe different types of microbial insecticide.
13. Define biological control why this method is superior to chemical control.
14. Give an account of *Trichoderma* as biocontrol agent.
15. Trace the history of biopesticides.
16. Define biological control. Discuss different strategies adapted in plant disease control.
17. Discuss integrated disease management.
18. Describe process of commercialization of biocontrol agent.

II. Write short notes on:

(1) Biopesticide
(2) Predators
(3) Radiation as biocontrol method
(4) Application of biopesticides
(5) Formulation of biopesticides
(6) Protozoans as biopesticides
(7) Nematodes in the control of plant diseases
(8) Fungistasis
(9) Antagonist
(10) Hyperparasite
(11) Entomogenous fungus
(12) Characteristics of a good biological control agent
(13) Limitations of biopesticides
(14) Solarization
(15) Crop rotation
(16) Siderophere
(17) Suppressive soil
(18) Parasoral body

FURTHER READING

1. Chaube, H.S., Mishra, D.S., Varshney S. and Singh, U.S. (2003). Biocontrol of plant pathogens by fungal antagonists: Historical background present status and future prospects Ann. Review of plant pathology 2:1-42
2. Chaube, H.S. and Singh U.S. (1991) Plant disease management principles and practices CRC press Boco Raton, USA.
3. Goldman, G.H. Hayas, C. and Harman, G.E. (1994), Molecular and Cellular biology of biocontrol by Trichoderma harzianum Trends Biotechnol 12:478-482
4. Granados, B.R. and Fedrici (1986) The biology of baculoviruses Vol. 1 Biological properties and molecular biology vol. II practical application for insect control CRC press.
5. Greer, F., Ignoffo, C.M. and Anderson, R.F. (1990) The first viral pesticide chemtech 347:606-611.
6. Hansen, T.A. (1988). The history of pest control. From traditional agriculture through pesticide era. In escape from the pesticide treadmill; Alternatives to pesticides in developing countries institute for consumer policy research pp. 9-29.
7. Harman, G.E. (2000). Myths and dogma of biocontrol: Changes in the perceptions derived from research on Trichoderma harzianum T-22 Plant Dis. 84: 377-393.

8. Kurstak, E. (ed) (1982) Microbial and viral pesticides Marcel Decker.
9. M.C. Roy, C.W. (1990) Entemogenous fungi as microbial pesticides in new directions in biological control (eds. RR Baker and PE Dunn), New York pp. 139-159.
10. Metcalf, R.L. (1980) Changing role of insecticides in crop protection Ann. Review of Entomol 25:219-256.
11. Milner, R.J. 2004. From lab to field. The critical factors in the development and application of mycoinsecticides in proc. International Workshop on entenogenous fungi.
12. Wood, RKS and Way, M.J. (edi) (1988). Biological control of pests, pathogens and weeds: Developments and prospects. The Royal Society.

19

Vaccines

A vaccine is a collection of immunological determinants which are presented to the immune system as killed or live antigens which provoke protective immune responses. The development of vaccines against viral and bacterial diseases is one of the greatest achievements of human endeavours. Vaccination is the most cost effective and efficient way to control viral diseases. The use of conventional viral vaccines has saved humans and animals from dreaded disease epidermics.

Infectious diseases which account for about 30-50% deaths due to lack of effective chemotherapeutic agents and that do exists are often too costly. Thus vaccines have become important tools for fighting infectious diseases in many parts of the world. On the other hand, in developed countries these infectious diseases account only 4-8% of all deaths. This low incidence of infectious diseases is largely due to the wide spread use of vaccination. The well known example is eradication of smallpox with the help of smallpox vaccine. Similarly use of several other vaccines like diphtheria, poliomyelitis, measles and rubella resulted dramatic decrease of diseases incidence and mortality. Further vaccination is much inexpensive than treating people who are already sick with modern antibiotics and other chemotherapeutic agents. Some of the commonly used vaccines are listed in Table 19.1. Vaccines continue to play an important role in veterinary medicine. However, sometimes these traditional vaccines become serious problem. Recombinant DNA techniques and synthetic organic chemistry have had major impact in overcoming, some of these problems and developing vaccines of a new type. Apart from this, some modern molecular approaches (Table 19.2) are applied. These methods enabled us to develop vaccines against diseases for which traditional vaccines do not exist.

Table 19.1: Some of the commonly used vaccines

1.	**Live viral vaccines** (a) Poliomyelitis (b) Mumps (c) Measles (d) Rubella (e) Yellow fever
2.	**Inactivated viral vaccine** Influenza
3.	**Viral subunit vaccine** Hepatitis B
4.	**Killed bacterial vaccines** (a) Pertussis (whooping cough) (b) Typhoid fever
5.	**Inactivated bacterial toxins (toxoids)** (a) Diptheria (b) Tetanus

Table 19.2: Molecular approaches to the design of human viral vaccines

Genetically engineered attenuation
Genome rassortants
Host range variants
Replication defective viruses
Recombinant viral vectors
Recombinant proteins and peptides
Virus-like particles
DNA vaccines

Limitations of Traditional Vaccines:

Live and attenuated vaccines of either bacteria or viruses are sometimes ineffective or not entirely safe. The other possibility that the organism or the toxin in the vaccine may not be completely killed or inactivated. Sometimes the vaccines revert to the virulence strain. Live attenuated vaccines may cause clinical disease if not attenuated sufficiently or revert back to virulence. The over attenuation of the virus may render it less immunogenic. The inactivated vaccines are often unable to generate protective levels of immune response due to loss of important antigenic determinants during inactivation and poor antigen load. Further, we lack vaccines for many important diseases. Another risk run by the workers who cultivate dangerous pathogens in large amounts to manufacture the virus vaccines. Over 200 years ago, Edward Jenner (1796) could demonstrate virulent human disease such as small pox could be checked with infections with mild cowpox by inoculating James Phipps, an 8-year old boy with exudate from a cowpox pustule. Communicable diseases such as tuberculosis, small pox, cholera, typhus, bubonic plague and poliomyelitis have in the part been a scourge for humankind. With advent of vaccination, antibiotics and effective public health measures these epidemic diseases have for the most part been brought under control (Fig. 19.1). However, occasionally protective measures become ineffective and devastating new outbreaks occur. Further, for many diseases of man and animals there are no vaccines (Table 19.4).

Table 19.3: Types of vaccines available for different diseases

1. Live attenuated bacterial vaccines	
(*i*) *Bacillus anthracis (spore vaccine)*	Anthrax
(*ii*) *Brucella abortus S19*	Brucellosis
(*iii*) *Salmonella typhi*	Typhoid
(*iv*) *Shigella sonnei*	Shigellosis
2. Inactivated bacterial vaccines	
(a) Capsular antigens	
(i) *Escherichia coli K88, kgg and 98 7p*	Diarrhoea and food poisioning
(*ii*) *Neisseria meningitites A&C*	Meningitis
(*iii*) *Dasteurella multocida*	Pasteurellosis
(b) Cell wall antigens	
(*i*) *Bordetella pertussis*	Whooping cough
(*ii*) *Salmonella paratyphi*	Paratyphoid
(*iii*) *Salmonella typhi*	Typhoid fever
(*iv*) *Vibrio cholera*	Cholera
3. Taxoid antigens	
(*i*) *Clostridium tetani*	Tetanus
(*ii*) *Corynebacterium diphtheria*	Diphtheria
4. Recombinant vaccines	
(a) Bacterial	
(*i*) *Chlamydia species*	Chlamydia
(*ii*) *Clostridium tetani*	Tetanus
(*iii*) *Corynebacterium diphtheria*	Diphtheria
(b) Protozoal	
Plasmodium vivax	Malaria
(c) Viral	Hepatitis B Herpes simplex Influenza Measles Poliomyelitis, Rabies
5. DNA Vaccines	Malaria

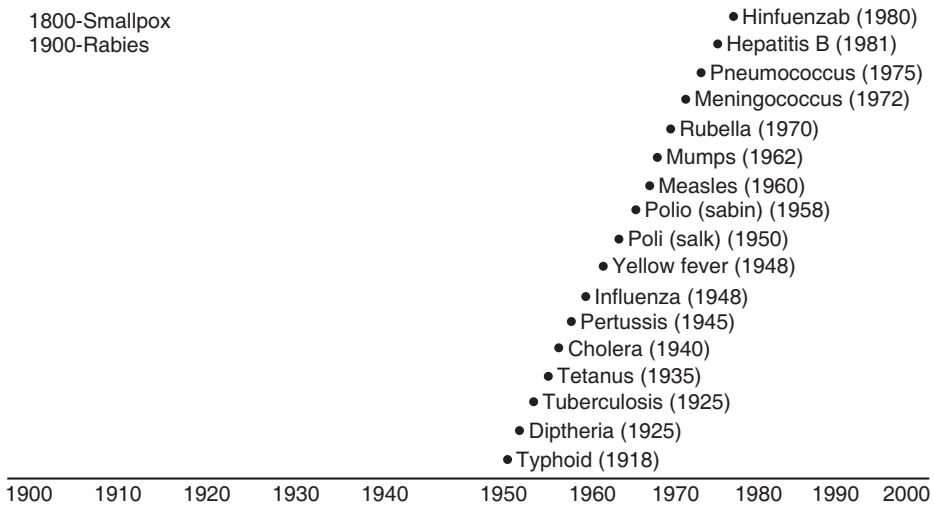


Fig. 19.1: Chronology of development of vaccines for different diseases

Table 19.4: Some diseases for which effective vaccines are not yet available

Disease	Pathogen	Infections (Millions/Years)	Deaths (thousands year)
AIDS	Virus	0.75	670
Diarrheal disease	Usually bacteria	3000-5000	5000-1000
Malaria	Protozoa	800	1200
Schistosomiasis	Trematodes	200	500-1000
Chagas disease	Protozoon	12	60
Leishmaniasis	Protozoon	12	5
African trypanosomiasis	Protozoon	1	5
Leprosy	Bacterium	12 (Estimated total number of patients)	Low

Today over 2 billion people suffer from diseases which could be curtailed by vaccination (table 19.3).

Table 19.5: Live viral vaccines for prevention of human disease

Current vaccine	Vaccine under development
Measles	Influenza A and B
Mumps	Respiratory syncytial virus
Rubella	Parainfluenza virus 1-3
Varicella	Herpes simplex virusus 1 and 2
Polioviruses 1-3	Cytomegalovirus
Rotavirus	
Yellow fever	
Vaccinia	

In addition an acquired immune deficiency syndrome (AIDS) for which vaccine might be useful. Live viral vaccines are developed to some of human diseases (Table 19.5). To control some other viral diseases, non-infectious viral vaccines are developed (Table 19.6).

Table 19.6: Noninfectious vaccines for prevention of human viral diseases

Current vaccine	Vaccines under development
Polioviruses 1-3	Humanpapilloma virus
Influenza A and B	Human immuno deficiency virus
Hepatitis A and B	Herpes simplex virus 1 and 2
Japanies encephalitis virus	Respiratory syncytial virus
Tick-born encephalitis	Hepatitis C

Notwithstanding the considerable success that has been achieved in creating effective vaccines against diseases such as German measles, diphtheria, whooping cough, tetanus, smallpox and poliomyelitis, there are number of limitations to the current mode of vaccine production such as:

1. All infectious agents cannot be grown in culture and so no vaccines have been developed for a number of diseases.

2. Production of animal and human vaccines requires animal cell culture which is expensive.
3. Both yield, rate of production of animal and human vaccines in culture are often quite low, thereby making vaccine production costly.
4. Extensive safety precautions are necessary to ensure that laboratory and production personnel are not exposed to a pathogenic agent.
5. Some diseases such as AIDS cannot be prevented through traditional vaccines.
6. Most current vaccines have a limited shelf life and often require refrigeration to maintain potency. This requirement creates storage problem in countries with large unelectrified rural areas.

The development of recombinant DNA technology has provided the possibility of creation of new generation of vaccines that overcome the drawbacks of traditional vaccines. The availability of gene cloning has enabled research to contemplate various noval strategies for vaccine development.

(1) Virulence genes could be deleted from an infectious agents that retains the availability to stimulate an immunological response.

(2) Live nonpathogenic carrier systems that carry discrete antigenic determinants of unrelated pathogenic agent can be created. In this form the carrier system facilitates the induction of storing immunological response directed against the pathogenic agent.

(3) For infectious agents that cannot be maintained in culture, the genes for the proteins that have critical antigenic determinants can be isolated, cloned and expressed in an alternative host system such as *E.coli* or mammalian cell line. These cloned gene proteins can be formulated into a subunit vaccine.

(4) There are some infectious agents that do not damage host cells directly. Instead the disease condition results when the host immune system attacks its own (infected) cells. For these diseases it may be possible to create targeted cell specific killing system. Although not a true vaccine, this type of system attacks only infected cells, thereby removing the source of the adverse immunological response. In these cases, the gene for a fusion protein is constructed. First, one part of this fusion protein binds to an infected cell, while the other part kills the infected cell.

Though these vaccines developed on recombinant DNA technique, that are being utilized for immunization of animal diseases but gradually they are improved to cure human diseases.

Recombinant technology can be used in various ways to create reliable vaccines. Immunologically active, non-infectious agents can be produced by deleting the genes that cause virulence. With this deletion a live vaccine would never be able to revert to the infectious form. It is possible to clone gene(s) that encodes the major antigenic determinant(s) from a pathogenic organism that can be harvested, purified and used as a vaccine. With this strategy, complete genes produce subunit vaccines and cloned domains of major antigenic determinants produce peptide vaccines. Peptide vaccines may also be produced by chemical peptide synthesis.

Modern molecular methodologies have provided hope for designing better vaccines which will be devoid of the problems associated with existing conventional vaccines. The advent of recombinant DNA (r DNA) technology created a revolutionary impact on the science of vaccinology. Some of the noval approaches currently in the process of development of vaccines are:

1. Synthetic vaccines, 2. Recombinant subunit vaccine, 3. Genetically altered live vaccines, 4. Vectored vaccines, 5. DNA vaccines and, 6. Plant and plant viruses based vaccines.

19.1 SYNTHETIC VACCINES

Two decades long research have revealed that small synthetic peptide having sequences of immunogenic epitopes on a protein will react with antibody to the intact native antigen and in some cases neutralize the biological activity. Anderer and Schlumberger (1965) demonstrated that chemically synthesized peptides corresponding to protein fragment would also produce the virus neutralizing antibody. This observation led to an extensive research on synthetic antigens. Sale et al. (1992) demonstrated that a synthetic fragment of 20 amino acids corresponding to 89-108 of the coat protein of MS_2 (a bacteriophage) elicited antibodies which reacted with the intact virus particle. These initial observations provided a ground for the subsequent exploration of peptide immunogen. The identification of genes encoding immunogenic proteins and availability of nucleic acid sequences allowed derivation of amino acid sequence of the immunogenic peptides. There are several advantages of synthetic peptide vaccines: (i) they possess indefinite shelf life, (ii) would have precise composition, (iii) no possibility for the adventitious presence of live virus, (iv) costly handling facilities are not necessary. A number of different diseases have been the target for synthetic vaccines e.g. FMDV, influenza and Hepatitis B.

Though significant protective immune responses have been achieved in experimental conditions, the synthetic vaccines are far from the commercialization mainly due to poor immunogenicity, short-term immunity and high cost of vaccination. Now the focus is on improving the immunogenicity by incorporating B and T lymphocyte epitopes and better adjuvants in the synthetic peptides.

19.2 RECOMBINANT SUBUNIT VACCINES

The understanding that isolated proteins of viruses can provoke protective immune response, led to application of genetic engineering in cloning, expression of critical viral genes in prokaryotic and eukaryotic expression vectors. Thus, the immunogenic peptides of the viral origins could be expressed in bacteria and yeasts. The concept of a subunit vaccine contain only those immunogenic components that are necessary to elicit a protective response and excluding unnecessary that has several alternative features. Subunit vaccines lack infectivity and hence no complication of arising out of vaccination in immunocompromised individuals. Subunit vaccination eliminates allogenic immunosuppressive and other undesirable reactogenicity. These vaccines also exhibit a great deal of stability.

The first successful recombinant vaccine was produced against hepatitis B. This has paved the way for the development of other recombinant peptide vaccines. The other recombinant vaccine which is in advanced stage of development is against parvovirus B19. Though the recombinant subunit vaccines are quite safe, there are still considerable problems like poor immunogenicity, high cost of production, short term immunity etc. Until these problems are tackled effectively, it is unlikely that the recombinant subunit vaccines will hit the market in near future.

19.3 GENETICALLY ALTERED LIVE VACCINES

Conventionally attenuated or naturally found avirulent strains of viruses have been used to develop live vaccines. Now with the development of rDNA technology, it is possible to attenuate a virus by deleting or altering the virulent genes. This kind of gene manipulation reduces its virulence or makes it unable to replicate completely and thus make a live vaccine with precisely defined modifications. A live simian immunodeficiency virus (SIV) vaccine, attenuated by deleting net gene exhibited the most impressive protection against SIV in macaques (Roy, 1996). Investigations are in progress to generate live attenuated vaccines for the use in human including improved polio (Mettenlieter, 1995) and dengue vaccines (Strube et al., 1995). Recently Pirkin et al. (1997) demonstrated the feasibility of using genetically modified live vaccine against influenza-A. Genetically altered live vaccines could be used veterinary vaccines but human factors are to be considered. The clinical trials of such products in human have not been carried out.

19.4 VECTORED VACCINES

Vectored vaccines approach allows expression of heterogenous genes into an avirulent viral/ bacterial organism which are harmless to vaccinated animals and induce cellular and humoral immune response against the foreign product. This approach also enables expression of more than one foreign genes encoding immunogenic proteins, thus paving the way for development of multivalent recombinant vaccines against human and animal viral infection. In the past one decade, a wide range of viruses and bacteria have been used for expression of relevant foreign genes. The most extensively used vector has been vaccinia virus.

Vectored vaccines against viral diseases are generally based on infectious, semi infected viral, bacterial vectors into which the genes of interest have been cloned. Vaccinia virus has been used as a vector for development of vaccine against rabies and rinderpest. A number of relevant genes of other viruses have been expressed in a variety of vectors and their vaccine potential has been studied extensively.

19.5 DNA VACCINES

A relatively recent approach in vaccine development is based on the observation that when plasmid containing suitable promoters and the genes encoding immunogenic proteins are injected into tissues, the gene may be expressed and generate humoral and cellular immune responses (An and Whitton, 1997). The same response can be generated by inoculating relevant nucleic acid into the skin or mucosa. This approach has generated a great deal of commercial interest due to several advantages including duration of immunity generated and stability of the vaccine. DNA vaccine technology provides exciting new approach for prevention and control of variety of viral diseases.

Delivery of DNA vaccines require expression of immunogenic proteins in tissues accessible immuno system such as muscle or skin or mucous membranes. For effective transfection and expression of the gene within these tissues, it is imperative that the introduced DNA is supercoiled and it includes strong tissue specific promoter and the transcribed mRNA has poly. A tail to ensure its stability in the host cell. Like plasmid DNA, naked mRNA encoding immunogenic protein can also be directly inoculated into host to induce immune response against the protein product.

Though the nucleic acid vaccine approach appears very attractive, there are a number of constrains including the possibility of integration of the nucleic acid, persistence and immunotolerance, which can be addressed.

19.6 PLANT AND PLANT VIRUSES BASED VACCINES

The results of recent studies have suggested that genetically engineered plants and plant viruses could be used as vaccines. In recent years, several attempts have been made to produce various antigens and antibodies in plants. Antigens and antibodies expressed in plants can be administered orally as any edible part of the plant, or by parenteral route after isolation and purification from the plant tissue. The edible part of the plant to be used as a vaccine is fed raw to experimental animals or humans to prevent possible denaturation during cooking and avoid cumbersome purification protocols. Thus, plants like tomato, banana and cucumbers are generally, the plants of choice. Virus based vectors can also be used to express the gene transiently to develop the products in a short period.

Plant system has the capability of producing any vaccine in large amounts and in a less expensive manner. However, the purification of the product may require the use of existing or even more cumbersome procedures. Attention, therefore, has been paid to mainly those antigens that stimulate mucosal immune system to produce secretory IgA (S-IgA) at mucosal surface, such as gut and respiratory epithelia. Thus, an antigen produced in the edible part of plant can serve as a vaccine against several infectious agents which invade epithelial membranes. The first report of the production of edible vaccine (a surface protein from *Streptococcus)* in tobacco appeared in 1990 in the form of patent application. Subsequently, a number of attempts were made to express various antigens in plants.

One of the utilities of producing antigens in plants in large amounts is in treatment of autoimmune diseases like diabetes mellitus which involve production of antibodies against glutamic acid dehydrogenase (GAD) and insulin, leading to destruction of insulin producing pancreatic cells. The antigens targeted for autoimmune response can be fed to the animals to induce immune tolerance.

19.7 VACCINES AGAINST BACTERIA

Due to readily availability of wide range of antibiotics against bacterial diseases, not much efforts are put on the development of vaccines against these diseases. Need of such vaccines being felt due to following reasons:

(1) Not all bacterial diseases are readily treated with antibiotics

(2) The use of antibiotics has resulted in purification of bacterial strains that are resistant to several antibiotics.

(3) Refrigeration facilities for the storage of such antibiotics are not available in many tropical countries.

(4) It is often difficult to ensure that individuals receiving antibiotic therapy have to undergo the full course of treatment.

Thus different strategies have to be adapted depending upon the type of bacterial disease. For instance Rocky mountain spotted fever caused by *Rickettsiae rickeltsie* can not be cultured. In this case a cloned 155-Kda protein that is a major surface antigen of *R.rickettsie* was used as a

subunit vaccine. The incidence of bacterial diseases of man has resulted total eradication or reduced to near total (Table 19.7). In view of these facts, several other vaccines for the control of other bacterial diseases have also been permitted (Table 19.8).

Table 19.7: Effect of vaccination on the incidence of bacterial diseases

Disease	Baseline (Prevaccine) annual cases	1998 Provisional Cases	% Decrease
Diphtheria	175.885	1	100
Pertussis	147,271	6279	95.7
Tetanus	1,314	34	97.4
Haemophilus Influenzae type b	2000	54	99.7

Table 19.8: Some bacterial vaccines licensed for use

Vaccine	Reference for ACTP recommendation
Diphtheria, tetanus, pertussis (DTP)	mmwR 40 (RR-10), 1991
Acellular pertussis (DTaP)	mmwR 46 (RR-7), 1997
Haemophilus influenza type b	mmwR 40 (RR-1), 1991
Pnemococcal	mmwR 46 (RR-8), 1997
Meningococcal	mmwR 46 (RR-5), 1997
Typhoid	mmwR 43 (RR-14), 1994
Cholera	mmwR 37 (40), 1988
Anthrax	mmwR 40 (RR-12), 1991
Plague	mmwR 45 (RR-14), 1996
Tuberculosis (BCG)	mmwR 45 (RR-4),1996
Lyme	mmwR 48 (RR-7), 1999

19.8 FUTURE

The developments in the past ten years have clearly pointed towards immense possibility of developing more efficacious, safer, thermostable and economically affordable viral vaccines using molecular approaches. Already some efficacious recombinant human and animal viral vaccines have been commercialized and some are at different stages of development. Recombinant plants carrying viral immunogens could easily immunize humans and animals against the targeted viral diseases.

Despite the great strides in vacciniology, a number of problems related to stability, safety, efficacy, environmental concern and economic feasibility remain unresolved. Once technical problems are solved regulatory and public issues must be tackled.

REVIEW QUESTIONS

I. Essay type questions:

1. Define vaccination. Bring out the developments in vaccination.
2. List out different types of vaccines and their limitations.

3. Discuss the role of vaccines in the disease management.

II. Write short notes on:

1. Vectored vaccines
2. Edible vaccines
3. DNA vaccines
4. Recombinant subunit vaccine

FURTHER READING

1. Arnon, R. and R. Levi (1996). In Novel strategies in design and production of vaccines (Kohen, S. & A. Shafferman eds), Plenum Press, New York.
2. Kaper, J.B., J.G. Morris Jr. and M.M. Levine (1995) cholera clin. Microbial Rev. 8: 48-86
3. A. K. Sharma, A. Mohanty, Y. Singh and A. K. Tyagi (1999). Transgenic plants for the production of edible vaccines and antibodies for immunotherapy. Curr.Sci, 77(4) : 524.
4. Tang,D.C., M.Devit and S.A. Johnston (1992) Genetic immunization is a simple method from eliciting an immune response Nature 356: 152-154.
5. Titus, R.G., G. Gueiros-Filho, L.A.R.De Freitas and S.M. Beverley (1995). Development of a safe live Leishmania vaccine line by gene replacement proc Natl. Acad. Sci USA 93: 10267-10271.

20

Biofertilizers

Man began to cultivate land in an organized way for foodgrains around 8000 B.C. Very soon he realised that the same land cannot support the growth of the plants endlessly and this led him to think about the ways and means of improving the fertility of the soil. In the early days of agriculture, man replenished the soil fertility with organic manure with crop and cattle residues. At the same time inadvertently he identified and practiced the composting process. This process continued almost till the end of the last century. Man made fertilizers as N, P and K containing fertilizers in the present century for increasing the output of agricultural productivity and to meet the needs of increasing human population has been further accentuated, by the limited availability of additional fertile land. The world's population, food production and fertilizer consumption have increased considerably during last few decades. To guarantee enough food for all either the population growth has to be checked or more fertilizer has to be used. The demand for chemically fixed nitrogen is increased and the nitrogen gap is likely to widen. The increase in world fertilizer requirement by 2050 A.D. would be approximately 2 times the rate of current consumption.

The new millennium brings with it renewed hope for a better livelihood for all the organisms of this earth. Hence, the themes often discussed at international fora on human food, security and safety to the provision of a productive and healthy environment to human kind and its future generations. Current scientific strategies to maintain and improve yields in support of high input agriculture place great emphasis on 'fail face' techniques for each component of the production sequence with little consideration of the integration of these components in a holistic systems approach. Green revolution in India during 1960's no doubt brought about self sufficiency in food, however, excessive use of inorganic fertilizers, plant protection chemicals for maximizing crops yields as well as change in traditional cultivation practices resulted in deterioration of chemical, physical and biological health of cultivated land. Major drawbacks to use skewed chemical fertilizer are:

1. Enhanced energy load for tonne of grain from 4.5 in 1983-84 to 147 Kcal. in 2005.
2. The widening green house effect.
3. A steep fall in nutritional quality of grain.
4. Steep rise in vulnerability of crops to pests and diseases.
5. Soil microbial activity and nutritional dynamic disturbances.

6. Underground water nitrate pollution among others resulted in the elimination of desirable type of microorganisms from soil, allowed enrichment of undesirable pathogens and pests which forced the farmers to use innumerable types of plant protection chemicals. Due to high yielding varieties we are still self sufficient in rice and wheat but it would not remain for long. High input agriculture is broadly recognized as an environment degrading, unprofitable, considerable crop damage by insects and pathogens will make the situation alarming if the dynamics of useful microorganisms are allowed to deteriorate further, indirectly affect crop productivity. Use of such useful microorganisms as bioinoculants may help in economizing crop production without affecting adversely the soil. In an Indian context inappropriate use of biofertilizers at farms in peroration to the demand of growing population have gained specific consideration.

The farmers are struggling to meet minimum requirements as cost of input is rising while price of output declining. This situation forcing us to think of alternative biological methods of using biofertilizers to chemical fertilizers, fungicides and pesticides. Microbes gained importance for sustainable agriculture systems. The bioinoculants are pre-adapted to fit into long-term sustainable agricultural systems. Biofertilizers as name indicates to the preparation containing useful microorganisms intended to use as bioinoculants. Some of the soil microorganisms which play crucial role in plant health and growth are listed in table 20.1. Still some microorganisms help the plant in growth and development either by close association or by forming specialized structures such as root and leaf nodules, mycorrhizae, coralloid roots etc. (Table 20.2). These microorganisms have greater potential for biocontrol, plant growth promotion and biodeterioration of xenobiotics. Therefore, it is proposed to discuss the challenges that must be met if we have to be successful in using microorganisms as bioinoculants for sustainable agriculture.

Table 20.1: Major types of beneficial interactions/associations between plants and soil microorganisms

Nature of Interactions/associations	Examples of higher plants involved	Examples of micro-organisms involved
Rhizosphere, Rhizoplane and Phyllosphere microflora	All plants with roots and leaves	Bacteria,fungi and *Actinomycetes*
Ectomycorrhizae	Forest trees–*Pinus*	Mostly basidiomyceto fungi Armillaria Boletus,Lactarius,
Endomycorrhizae (VAM fungal association)	Certain orchids, cereals grasses & legumes	Rhizoctonia,Endogor and *Glomus*
Root nodules of nodulating legumes	Soyabean, gram, Lucerne etc	*Rhizobium* spp. and *Bradyrhizobium sp.*
Root nodules of plants other than legumes	*Alnus*, *Myrica, Casuarina*	*Actinomycetous endophytes*(*Frankia*)
Leaf nodules	*Psychotria*, *Pavetta*	Klebsiella
Algal associations with higher plants	*Cycas, Zamia, Heterozamia Gunnera scabra, Azolla*	*Anabaena*, *Nostoc*
Associative symbiosis	*Grasses, Soghum and Millets*	*Azospirillum*

Table 20.2: Major microbiological processes in soil by freeliving microorganism which indirectly influence plant growth

Nature of microbial processes	Examples of microorganisms involved
Aerobic decomposition of organic matter (cellulose, lignin,chitin etc.)	*Trichoderma, Fomes, Armillaria, Achromobacter Nocardia, Streptomyces*
Anaerobic decomposition of organic matter	*Clostridium, methane bacteria (Methanobacter and Methanococcus)*
Non-symbiotic nitrogen fixation	*Anabaena, Azotobacter, Beijierinckia Bacteria, fungi and actinomycetes*
Nitrogen immobilization	*Pseudomonas, Bacillus, Serratia*
Nitrogen mineralization	*Nitrosomonas, Nitrobacter*
Nitrification	*Pseudomonas, Achromobacter*
Denitrification	*Pseudomonas, Bacillus, Aspergillus*
Phosphate solubilization	*Thiobacillus, Beggiatoa, Desulfovibrio*
Sulphur transformations	*Gallionella, Ferribacterium, Leptothrix*
Iron transformations	*Aerobacter, Corynebacterium, Flavobacterium*
Manganese transformations	*Cladosporium*
Copper transformations	*Desulfovibrio, Clostridium, Escherichia*

Nitrogen fertilizers will continue to serve for increasing grain production until a foreseeable future. In the area of chemical fixation no major breakthrough is yet visible to minimize the energy requirements of conventional Haber-Bosch process for the production of NH_3. Further, it has been experimentally proved that chemical fertilizers and pesticides are causing harm to the soil, resulting in definite productivity loss. Therefore, efforts should be oriented towards augumenting biological nitrogen fixation mediated by microorganisms.

20.1 WHAT ARE BIOFERTILIZERS?

The term biofertilizers denote all nutrient inputs for plant growth which are of biological origin. Biofertilizers or more accurately microbial inoculants can be defined as preparations containing live or latent cells of efficient strains of nitrogen fixing, phosphorus solubilizing or cellulolytic microorganisms used for application to seed, soil or composting areas with the objective of increasing numbers of such microorganisms and accelerate certain microbial processes to augument the extent of availability of nutrients in a form which can easily be assimilated by plants. Such microbiological processes may be as complex as that of nitrogenase mediated reactions in nitrogen fixing microorganisms which reduce elemental nitrogen into ammonia or as simple as the organic acid secretions by phosphate dissolving bacteria. An ideal fertile soil is not only characterized by optimum physical properties and chemical constituents conducive for plant growth but also by microbiological processes which are maintained in an equilibrium. Maintenance of this equilibrium in soil is important in order to increase crop productivity. Infact, wild ecosystems have sustained through centuaries by means of such natural interconversions of essential elements. In view of these naturally proved beneficial roles, recent price escalations in fertilizer price, potential hazards, due to use of fertilizers in long run, the use of biofertilizers have gained more significance than ever before and becomes imperative. Although the concept of using bioinoculants to increase crop yields goes back to 100 years, relatively very few reliable bioinoculant products are currently available in the market. This is mainly due to lack of

confidence among end users about the performance of the products being marketed. Because of simplicity and low cost of production of the bioinoculants, markets inferior products thereby giving bad name to the whole inoculant industry. Some of the reasons for the failure of optimal performance and wider use of bioinoculants are:

1. Soil may not be conducive for the propagation of the bioinoculants.
2. Inoculum may not be able to compete with indigenous microflora.
3. Lack of reproducible performance of the bioinoculants.
4. Poor quality of bioinoculants.
5. Treating bioinoculants in a chemical rather than a biological paradigm.
6. Lack of coordination among researches and bioinoculant manufacturers.
7. Lack of proper knowledge about bioinoculants in the farming community.

Obviously there is no simple or single solution to the complex, ecological, socio-economic and technological problems faced by those who are engaged in promoting sustainable advances in agricultural biotechnology.

In view of above facts even govt. is forced to redraft the policy based on the experience of the past, reasonable expectations of the future and accordingly the use of biofertilizer by our farmers being promoted. The ministry of Dept. of agriculture and cooperation has set up a National Project on development and use of biofertilizer with the National centre at Ghaziabad and six other regional centres in different parts of the country. Apart from production, this project, (1) provides financial assistance to bioinoculant production units, (2) organizes a number of training and field demonstrations on bioinoculants with the publication of technical literature, and (3) acts as quality control agency. Thus serious efforts are made to promote a message that only quality of product is the key to success. Fortunately, several companies mainly private manufacturers have entered into the market with a strong quality base.

20.2 POTENTIAL ORGANISMS FOR BIOFERTILIZERS

Quite a good number of microorganisms both symbiotic and non-symbiotic are being identified and exploited as potential source of biofertilizers. Attempts are being made to large scale production and mass inoculation of these organisms to the soil. Some of these microorganisms are listed below.

I. Nitrogen fixing organisms

A. Symbiotic organism

(i) Associative organisms – *Azospirilum*

(ii) *Azolla*

(iii) *Rhizobium*

B. Non-symbiotic nitrogen fixers

1. Heterotrophs

(i) *Azotobacter*

2. Phototrophic

(i) Blue green algae

(ii) Phototrophic bacteria

II. Phosphate solubilizing or phosphate augumenting organisms

(i) *Bacillus* species

(ii) Mycorrhizae

20.2.1 Rhizobium

The role of legumes and part played by species of *Rhizobium* in enriching the fertility of soil was scientifically demonstrated only in the latter half of the 19th centuary. With subsequent understanding of legume – *Rhizobium* interactions *Rhizobium* as legume inoculant was first developed and marketed in USA during early part of this century. During subsequent three decades, work on the methodology of mass cultivation of rhizobia, preparation of carrier based inoculants and application of rhizobia to soil or seeds of legume has been carried out in Australia which have been adopted in Asian, Latin American, African countries with modifications to suit their local needs and conditions.

(i) Cultivation and mass production of *Rhizobium* inoculant : Rhizobia are maintained on yeast extract mannitol (YEM) agar medium (Yeast extract-1 g; Mannitol 10 g.; K_2HPO_4 3.5 g, $MgSO_4$. 7 H_2O – 0.2 g, NaCl-0.1 g; agar 15 g, D. water -1 litre pH -6.5-7.0) either by subculturing at frequent intervals or by lyophilization to be reconstituted into agar-based cultures whenever necessary. The primary culture is designated as the Mother Culture. For large scale cultivation YEM liquid medium is employed. The selection of suitable strains of *Rhizobium* species dependent upon many criteria, a single strain or more than one strains for a particular cultivar or a group of cultivars or crops. The selected strain is grown for 3-4 days on YEM agar slants depending on the fast or slow growing nature of the strain. The culture is tested for purity by well known tests and transferred to large flasks containing sterile solid or liquid medium for 4-9 days. This is called 'starter culture' which is transferred to a seed tank fermentor and incubated for 4-9 days. By about the same time, a large quantity of liquid broth is formulated in the fermentor, pH adjusted to 6.5 to 7.0 and sterilized. After cooling to 30°C, inoculum from the seed tank fermentor is transferred aseptically to the production fermentor at the rate of 1 percent by volume. The factors influencing the output of cells are aeration, volume, initial inoculum level, bacterial strain, temperature and incubation time. The main objective is to attain high populations in the minimum time.

In USA large automatic fermentors are used, whereas in Australia a container such as drum of 10 to 100 litre capacity is used. In India, most manufacturers use indigenously made shakers which can hold many conical flasks to produce inoculants on a small scale. The resulting broth from fermentors is checked for purity according to methods outlined by Vincent. The general method of Rhizobial inoculum production is outlined in Fig. 20.1.

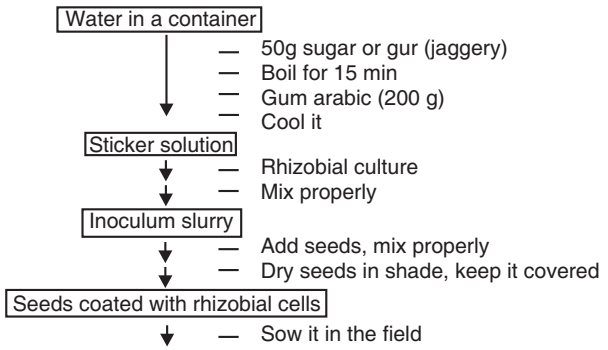


Fig. 20.1: Seed dressing with rhizobial cells

(ii) The Product: The product is free flowing carrier based preparation containing live cells of specific rhizobia. In the USA diluted broth having rhizobial cell population in excess of 10^9 cells/ml is blended with the peat carrier so as to bring its final moisture level to 35-40 percent on wet basis.

Table 20.3: Response to *Rhizobium* in farmers' fields (range of percent increase in grain yield)

Crop	Location	1978	1979
Chickpea	Hissar	7-22	5-11
	Durgapura	11-33	9-31
Pigeonpen	Hissar	20	10-16
Moong	Ludhiana	11-18	12-16
Urid	Durgapura	20	13
Cowpea	Durgapura	30	26-30

Azolla: Can also be grown with rice seedlings after transplantation, then the ferns are buried with hands in the soil and the process is repeated as often as necessary so that the Azolla mat may not cause choking of rice plants causing oxygen starvation.

The final product should have atleast 300 million rhizobia per g of peat. In the process of blending, the broth is sprayed to powdered peat and left for curing. The final product is again milled and packed in polythene sheets where the number of rhizobia multiply. Finely powdered farmyard manure and charcoal powder are good alternatives to peat. The expiry date of the product is six months in India. Indian Standards Institution (ISI) is entrusted with the task of quality control. The rhizobial inoculum is applied to seeds by adding gum arabic (40%) or Carboxymethyl cellulose (20%) is added to inoculum slurry before mixing seeds (Fig. 20.2).

(iii) Benefits of Inoculation: Trials in farmer's fields in India have proved a great success and serve to depict the importance of legume inoculation in the improvement of grain legume yields (Table 20.3). There have been many advances in the method of inoculating seeds with rhizobia. Adhesive and seed pelleting agents such as gum arabic, lime etc. have been advocated to improve the survival of bacteria on seed.

20.2.2 Azolla

Azolla is a free floating aquatic fern plant with branched stem, bilobed leaves and true roots which penetrate water. The dorsal, fleshy chlorophyll containing lobe has an algal symbiont, *Anabaena azollae* within a cavity which fixes nitrogen. It naturally grows in tropical fresh water ponds and rice fields. The value of *Azolla* as a biofertilizer in rice cultivation was first demonstrated in North Vietnam, where *A.pinnate* is grown in 400,000 ha. In recent years, it has become a common input in rice cultivation in Thailand, Indonesia, China and Phillippines.

(i) Mass Cultivation: The Chinese grow small nurseries where *Azolla* initially grown for four weeks. At the time of rice cultivation, *Azolla* is seeded in the flooded rice fields at the rate of 7.5 t/ha. After 5-10 days, the water in the field is drained and the *Azolla* is ploughed. This process of flooding the field, draining water and ploughing is repeated before transplanting the rice seedlings.

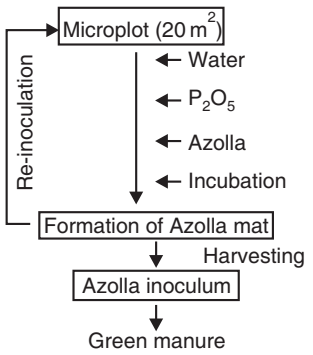


Fig. 20.2: Production of azolla inoculum

Table 20.4: Dual cropping of *Azolla pinnata*

Treatment	Azolla growth Fresh wt. after 25 days of inoculation (t/ha)	Tillers / hill after 53 DAP		Ricecrop response Effective tillers/m^2	Grain yield (t/ha)
T_1-	Control	-	5.32	224	2.1
T_2-	*A.pinnata* (India) unincorporated	11.0	5.10	238	2.2
T_3-	*A.pinnata* (India) incorporated after cover	8.3	5.43	224	2.5
T_4-	*A.pinnata* (Bangkok) unin corporated after cover	16.0	5.27	238	2.5
T_5-	*A.pinnata* (Bangkok) in- corporated after cover	12.9	6.30	280	2.9

20.2.3 Azospirillum

Azospirillum lipoferum (earlier known by *Spirillum lipoferum*) as a nitrogen fixer was known since 1963. It was Deobereiner and his associates in Brazil who in 1975 highlighted and attributed the nitrogen fixation potential of some tropical sorghum, wheat and rye to the activity of *A. lipoferum* in their roots. Subsequently, the bacterium has been isolated from many tropical countries, found near roots and aerial parts of plants. Dobereiner coined the name 'Associate symbiosis' to denote the occurrence of nitrogen fixing *Azospirillum near the roots* in plants. Three more species viz., *A.amazonense, A.barsilense* and *A.serapedica* are added to this genus.

Azospirillum is a gram negative and contains poly-B-hydroxy butarate granules. It shows polymorphism and spirillar movements. For fixation of molecular nitrogen, it needs microaerophilic conditions. The ability of the organism to fix nitrogen has been demonstrated by the acetylene reduction test and uptake of 15_{N_2} gas. This organism is also known to produce growth substances such as IAA, kinetins and gibberellins.

(i) Mass cultivation of Azospirillum and inoculant preparation: For large-scale cultivation of *Azospirillum*, bottles or flasks with ammonium chloride containing liquid media are being used. The composition of the medium is as follows (g/l). A solution

(a) K_2HPO_4 – 6.0, KH_2PO_4 – 4.0, Dist. Water – 500 ml.

(b) MgSO4 – 0.2, NaCl – 0.1, $CaCl_2$ – 0.02, NH_4Cl – 1.0, Malic acid – 5.0, NaOH – 3.0, Yeast extract – 0.05, $Na_2Mo\ O_4$ – 0.0002, $MnSO_4$ – 0.001, H_3BO_3 – 0.0014, $Cu(NO_3)_2$ – 0.0004, $ZnSO_4$ – 0.0021, $FeCl_3$ – 0.002, Dist. Water – 500 ml, Bromothymol blue - 2 ml

The phosphate buffer portion of the medium was made in half of the total volume require part (a) and (b) were sterilized separately, mixed while hot and poured into plates. These containers are incubated at 35°C on a rotary shaker and cells are harvested after three days. The broth is incorporated into a carrier material consisting of powdered, sterilized farmyard manure

and soil in the ratio of 1:1 then packed in polythene packets. This carrier is the most suitable among different, tested carriers.

(ii) Benefits of inoculation: Several field experiments have been carried out in India by inoculating seeds of different crop plants with carrier-based culture of strains of *Azospirillum brasilense*. The response of sorghum and pearl millet to *A.brasilense* inoculant under several agroclimatic conditions of India is presented in Table 20.5. Such responses to *Azospirillum* inoculant have also been confirmed in Israel.

Table 20.5: Summary of the *Azospirillum* inoculation trials conducted on large plots

Crop	No. of Experiments	Increase in yield Percentage		No. of experiments showing	
		Range	Average	Significant Increases	Increase above 5%
Corn (forage and sheet corn and grain corn)	13	0-36	16	7	11
Wheat (forage and grain)	10	3-19	9	5	8
Sorghum (forage and grain)	5	12-20	17	5	5
Setaria italica	2	38-47	43	2	2
Panicum milliaceum	1				

20.2.4 Azotobacter

It is a heterotrophic free-living nitrogen fixing bacterium belonging to the family *Azotobacteriaceae.* Several species of *Azotobacter* are recognized such as *A.chroococcum, A.agilis, A.uinelandii* and *A.beijerinckii*. Among different species of *Azotobacter, A.chroococcum* and *A.Vinelandii* are the most intensively investigated species, that have been commercially exploited. Bacterial preparations containing *Azotobacter* cells under the name *'Azotobacterin'* are being produced, used in USSR, East European Countries such as Czechoslovakia, Rumania, Poland, GDR, Bulgaria and Hungary where bacterization of seeds with Azotobacterin has proved beneficial in increasing crop yield.

(i) Mass cultivation of *Azotobacter* and the preparation of inoculants: In India large scale cultivation of *Azotobacter chroococcum* has been carried out either by in large flasks on rotary shakers or in batch fermentors. The medium used for *Azotobacter chroococcum* is as follows (g/l) Sucrose – 20, K_2HPO_4 – 01, $MgSO_4.7H_2O$ – 0.5, NaCl – 0.5, $FeSO_4$ – 0.1, $CaCO_3$ – 2.

At the end of the desired incubation period, the broth is poured into a powdered carrier and mixed uniformly to bring the level of moisture in the carrier to 40 percent. After curing for 2-5 days, the carrier based inoculant is packed in polythene bags for transport. Seed inoculation is done in the same way as have done for *Rhizobium*. However, in transplanted crops such as rice, cauliflower, cabbage etc. the roots of seedlings are dipped for 10 to 30 minutes in the watery slurry of the inoculant before transplanting. For other crops, manufacturer recommends multiple inoculations during the early stage of plant growth by pouring the slurry of the inoculant near the root zone. Mixing the cultures in farmyard manure and broadcasting near the root zone is also recommended.

(ii) Benefits of inoculation: Recent work on the response of field crops such as maize and cotton to inoculation with new strains of *A.chroococcum* has shown the feasibility of using *Azotobacter* inoculants to minimize the use of nitrogen fertilizer. Besides the ability

to fix atmospheric nitrogen, *Azotobacter* is also known to synthesize biologically active substances such as B-vitamins, indole acetic acids and gibberellins in pure cultures. The organism possesses fungistatic properties even on certain pathogenic ones such as *Alternaria* and *Fusarium*. These attributes of *Azotobacter* explain the observed beneficial effects of the bacteria in improving seed germination, plant growth, plant stands and vegetative growth (Table 20.6).

Table 20.6: The effect of *Azotobacter chrococcum* inoculation in India on the yield of three crops

Crop	Location of field Trials in India	Without *Azotobacter*	With *Azotobacter*	C.D. % at 5%	increase due to *Azotobacter*
Sorghum (g/ha)	Pali	12.8	14.0	1.22	9.3
	Dharwar	23.6	32.6	10.56	38.1
Maize (g/ha)	IARI	7.8	13.4	4.8	71.7
	Dharwar	32.0	43.7	9.90	36.5
Cotton (g/ha)	Surat	1254	1339	241	6.7

20.3.5 Blue-green algae

Blue-green algae constitute an important group of microorganisms capable of fixing atmospheric nitrogen. They comprise unicellular, colonial and filamentous forms. Most of the nitrogen fixing BGA belong to the orders Nostocales and stigonematalses under the genera *Anabaena, Anabaenopsis, Aulosira, Chlorogloea, Cylinderospermum, Nostoc, Calothrix, Scytonema, Tolypothrix* and *Fischerella*. In general, nitrogen fixation is associated with forms possessing heterocysts, although there are reports of N_2 fixation by unicellular and filamentous non-heterocystous strains. The number of heterocysts could be taken as a rough parameter to indicate the nitrogen fixing capacity of BGA. The importance of BGA (Cynobacteria) as biofertilizers was recognized as early in 1939, when De reported the restoration of soil fertility in rice fields by cyanobacteria. Thenceforth agronomic potential of cyanobacteria has been further proved from the studies at various laboratories. Restoration of soil fertility in tropical paddy fields has been attributed mainly to nitrogen fixing, heterocystous forms.

(i) Mass Cultivation of BGA: In India, a simple farm oriented open air method for bulk production of BGA from a starter culture consisting of a soil based mixture of *Aulosira, Tolypothrix, Nostoc, Anabaena* and *Plectonema* has been developed. Shallow trays of galvanized iron sheet (6′ × 3′ × 9′) or brick mortar or its pits lined with polythene sheets are constructed into which 10 kg soil plus 200 g superphosphate, pH of mixture is adjusted to 7.0 by the addition of lime. When the soil settles down, saw dust is sprinkled along with the starter culture of algae. On hot sunny days, the algal growth in the open air tanks is quick and within a week a thick scum of algae is formed. The water is allowed to drain and the dried algal flakes from the surface are scraped and stored in bags. The production of dried algae on farmer's fields can be a continuous process.

Dried algae are broadcasted at the rate of 10 kg/h over the standing water in rice fields, one week after the transplantation of rice seedlings. The cost of algal material has been calculated at Rs. 30 for 10 kg sufficient for one hectare of land. Mass production of carrier based immobilized cyanobacterial inoculants are depicted in Fig. 20.3.

(ii) Benefits of inoculation: Field trials conducted in different parts of India have shown significant increase in grain yields of rice due to the inoculation of rice fields with BGA.

The mean increase in yield of grain due to algal inoculation works out to be 300 kg for Rs. 30 investment. It is now well known that any form of nitrogen is detrimental to the process of nitrogen fixation by microorganism. The increased yields of rice due to algal inoculation, even under heavy doses of nitrogenous fertilizers could be attributed to the combined effect of biologically fixed nitrogen and the growth substances secreted by BGA (Table 20.7).

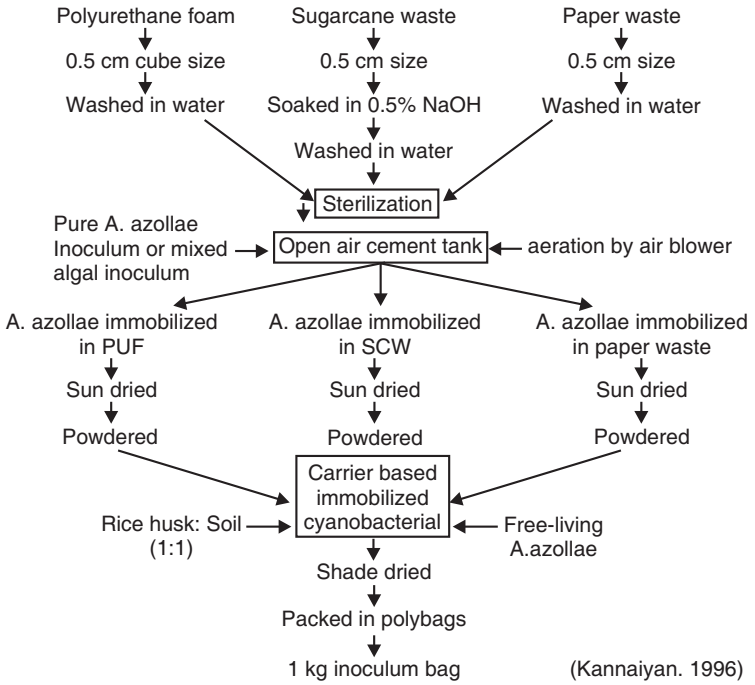


Fig. 20.3: Mass production of carrier based immobilized cyanobacterial inoculants

Table 20.7: Blue-green algal trials in farmer's fields

Details				Mean yield (kg/ha)		
No.		P (Kg/ha)	K (kg/ha)	Vellore (11 trials)	Trichy (8 trials)	Cuddalore (10 trials)
1.	50	25	25	4235	4557	3789
2.	50	25	25+Algae	4455	4785	3884
3.	75	37.5	37.5	4400	4485	3953
4.	75	37.5	37.5+Algae	4565	5117	4156

20.2.6 Photosynthetic bacteria

Though some photosynthetic bacteria like *Rhodopseudomonas* are known to fix the atmospheric nitrogen in a photoautotrophic manner, these are not yet commercially exploited as biofertilizers.

20.2.7 Phosphate solubilizing or phosphate augmenting organisms

The deficiency of phosphorus may occur in crop plants growing in soils containing adequate phosphates. This may be partly due to the fact that plants are able to absorb phosphorus either by plant roots or by soil microorganism through secretion of organic acids. Therefore, phosphate dissolving soil microorganisms play some part in correcting phosphorus deficiency of crop plants. They may also release soluble inorganic phosphate (H_2PO_4) into soil through decomposition of phosphate rich organic compounds. In this regard phosphate solubilizers such as fungi and bacteria and mycorrhizae have been investigated as promising sources of biofertilizers.

(i) Phosphate solubilizers: Many fungi *(Aspergillus, Penicillium*) and bacteria *(Bacillus megatherium, Pseudomonas striata*) are potential solubilizers of bound phosphates as revealed by experiments in pure culture. Though fungi seem to be better agents in the dissolution of phosphates, bacteria have been used in the commercial preparation of phosphate dissolving cultures to improve the growth of plants.

(ii) Mass cultivation of phosphate solubilizers and the preparation of inoculants: In India large scale cultivation of *Bacillus megaterium* or *pseudomonas striata* is being carried out either by growing cultures in large flasks on rotary shakers or in batch fermentors. The carriers that are used for *Rhizobium* and *Azotobacter* are also suitable for phosphate – solubilizing bacteria. The medium used for large scale production is as follows. (g/l). Glucose 10, $Ca_3(PO_4)_2$ 5, $(NH_4)_2\ SO_4$ 0.5, KCl 0.2, $MgSO_4$. $7H_2O$ 0.1, $MnSO_4$ trace, yeast extract 0.5. Preparation and application of inoculum are essentially similar to *Azotobacter* inoculant.

(iii) Benefits of inoculation: It has been observed that vegetables respond better than cereal crop to the application of phosphate dissolving microorganism. A commercial preparation under the name 'Phosphobacterin' containing bacterial cells of *Bacillus magatherium* was widely used in USSR. Field trials conducted at IARI, New Delhi with wheat, bersem, maize, arhar and rice have shown significant increase in the yield in 10 out of 37 experiments (Table 20.8).

Table 20.8: Effect of phosphate-solubilizing microorganisms on various crops in some field

Name of the crop	Organism	No. of Field experiments	No. of experiments showing significant increase	% yield increase over control
Berseem (*Trifolium alexandrinun*)	Phosphobacterin (*Bacillus megatherium*) *var. phosphaticum*	6	4	10-20
Maize (*Zea mays*)		3	2	0-14
Wheat (*Triticum aestivum*)		6	1	0-37

contd...

Name of the crop	Organism	No. of Field experiments	No. of experiments showing significant increase	% yield increase over control
Rice (*Oryza sativa*)		12	2	12-31
Gram (*Cicer arietinum*)		1	0	-
Arhar (*Cajanus cajan*)		1	0	-
Urid (*Phaseolus aureus*)		3	1	16–19
Soybean (*Glycine max*)		1	0	-
Groundnut (*Arachis hypogea*)		1	0	-
Potato (*Solanum tuberosum*)		1	0	-
Wheat (*Triticum aestivum*)	*Pseudomonas striata*	1	1	10
Rice (*Oryza sativa*)	*Bacillus polymyxa*	1	1	9.5
Potato (*Solanum tuberosa*)	*Pseudomonas striata*	1	1	25.0

20.2.8 Mycorrhizae

The symbiotic association between fungi and root systems of higher plants come under the general name, mycorrhiza which literally means 'fungus roots'. These fungus roots were first discovered by Frank (1855) in pine, but subsequent work has pointed out that such a symbiotic association with fungi exists under natural conditions in root systems of many other economically important crops and play a very important role in plant nutrition. The correlation between mycorrhizal formation and mineral deficiency in the soil led to the conclusion that mycorrhizae must help in the absorption of nutrient of plants from the soil. The increase in nutrient absorption has been attributed to several reasons.

There are two kinds of mycorrhizae the ectomycorrhizae and endomycorrhizae. The ectomycorrhizae is not as common as the endomycorrhiza even though the former has been studied more thoroughly. In the ectomycorrhizae, the fungus completely encloses each feeder rottlets in a sheath or mantle of hyphae. The hyphae penetrate only between the cells of the root cortex (Harting net). Ectomycorrhizal associations are common in most forest trees primarily in the families Pinaceae, Betulaceae and Fagaceae. In endomycorrhizae, the fungus lives within the cells of the root (intracellular) and establishes direct connections between the cells of the roots and the surrounding soil. Endomycorrhiza produced by nonseptate fungi is more commonly known as vesicular arbuscular mycorrhiza (VAM) or arbucular fungi (AM). Throughout the world, its widespread occurrence is found in almost all family's plants.

The beneficial effects of VAM mycorrhizae on plant growth are numerous. It is now well established that mycorrhizae can improve the P nutrition of host particularly in low fertility due to exploration of the soil by the external hyphae beyond the root hairs and phosphorus depletion zone. Mycorrhizal fungi also stimulate plant, uptake of zinc, copper, sulphur, potassium and calcium. They play an important role in water economy of plants and bestow the drought tolerance to the plants. VAM infections alleviate heavy metal toxicity, increase the tolerance of the crops to high acidity and temperature. It also decreases the growth depression and root rots caused by fungal pathogens. Mycorrhizal inoculations stimulate rooting and growth, thereby transplant survival of cuttings and seedlings, which is essential for the successful reclamation and forestation programmes.

Due to these beneficial attributes the mycorrhizae are widely used as biofertilizers for improving the growth of agricultural and horticultural plants. They have been exploited to save the costly phosphatic fertilizers. They are also employed in wasteland reclamation and afforestation. However, the most important constraint in their wide application is the commericialization of their inoculum because of their inability to grow and develop in the absence of their host. At present, VAM fungi are cultured in bulk only on the roots of the living plants grown either in soil (pot culture) or in nutrient solutions. These systems are crude and expensive, require a lot of money and space. Nevertheless, attempts are being made to refine these systems. An outline for VAM inoculum production is given in Fig. 20.4. produced at different centre including Terri, New Delhi and Forest Research Institute, Dehradun. Inocula of these can be procured as needed and used in crop production, horticulture and forestry.

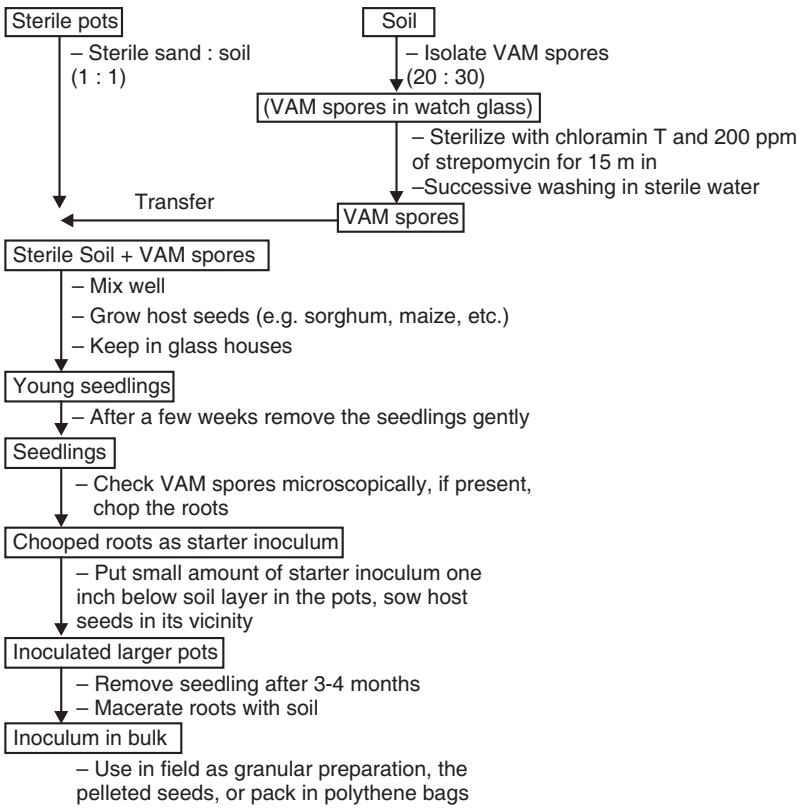


Fig. 20.4: Inoculum production of VAM fungi

Finally it is concluded that bioinoculants have a tremendous potential in India as a cheap source of plant nutrients provided the quality of product is assured. Another important thing is to discover and use as many bioinoculants as possible rather to evaluate the subject by quibbling or sophistsy. The subject is so challenging that one should approach it hopefully and with optimism, not with skeptism and doubts. As Samuel Johns observed long ago. 'Nothing will ever be attempted if all possible objections must first be overcome'. Each successful application, no matter how limited increases the level of familiarity.

REVIEW QUESTIONS

I. Essay type questions:

1. What are biofertilizers? Discuss the advantages of these fertilizers.
2. Describe different biofertilizers employed as nitrogenous fertilizer. Add a note on their limitations.
3. Give an account of phosphate solubilizing bioinoculants.
4. Discuss the role of mycorrhizae in growth and development of crop plants.
5. Give general account of formulation and application of biofertilizers.
6. Discuss the constraints in the utilization of these fertilizers.
7. What is bacterization? Give an account of bacteria used as bioinoculants.
8. Give an account of mass cultivation of Rhizobium and its use as biofertilizer.

II. Write short notes on:

(a) Phototrophic bacteria
(b) Frankia
(c) Cyanobacteria
(d) Arbuscular mycorrhizae
(e) Azolla
(f) Azospirillus
(g) Azotobacter
(h) Biofertilizers for paddy cultivation
(i) Phosphobacterin

FURTHER READING

1. Kannaiyan, S. (2002) Biotechnology of biofertilizers, Narosa Publ. House, New Delhi.
2. Subba Rao, N. S. (1993) Biofertilizers in Agriculture and Foresty 3rd edition, Oxford and IBH Publ. Co., New Delhi.
3. Venkataraman, G.S. (1972) Algal biofertilizers and rice cultivation Today and Tomorrow Printer, New Delhi P 75.
4. Rangswami, G. and G. Oblisami (1962). Studies on some legume root nodule bacteria J. Indian soil Sci. Sco. 10: 175-186.

21

Mushroom Cultivation

The term mushroom means in general a fungus but commonly it is the fruiting body of some fungi which produce and disseminate spores. Like all other fungi, they lack Chlorophyll and thus cannot produce their own food. They grow saprophytically or sometimes symbiotically upon other dead and living plants respectively to obtain organic matter as food. Mushrooms are variable in size and shape. Many have cap and stalk but some varieties are devoid of stalk. Some varieties even produce fruit bodies below the ground. There are large number of species growing wild in nature, while many are edible, some are highly poisonous. The collection of colourful and variety of shapes of mushrooms is in practice since time immemorial. *Auricularia, Lentinus edodus, Agaricus bisporus* and *volvariella volvacea* were collecetd in China and France and elsewhere in the world. In India they have started to cultivate *mushrooms* from 1943. The world production of different edible mushrooms are summarized in Table 21.1.

21.1 IMPORTANCE OF MUSHROOM CULTIVATION

Mushrooms are popular for their delicacy and flavour. They are excellent sources of vitamins, proteins and minerals. They are good source of Vitamin 'B', folic acid, the blood building vitamin, useful in anaemic condition.

They also contain pantothenic acid, vitamin B-12, ascorbic acid and the precursor of vitamin A and D. They are also good source of phosphorus, potassium, iron, copper, contain all essential amino acids particularly L-lysine and L-tryptophan. Mushrooms contain least quantity of carbohydrates as well as fats, hence very valuable diet for those suffering from diabetes and heart problems. They also contain compounds capable of preventing heart attack, diabeties, cancer, infections due to bacteria, fungi, viruses and protozoa. Edible mushrooms have been recommended by the FAO as food, contributing to the protein nutrition of the developing countries depends largely on cereals. Mushrooms are useful in treating many human ailments.

With increasing population, food demand problems in developing and under developed countries, mushrooms can play an important role to enrich human diet particularly in India where a large section of the population are vegetarians. It is an ideal means of recycling agrowastes which are available abundantly. The spent compost from mushroom farms is a good

organic manure and a better substrate for bio-gas production. It is a labour intensive indoor activity which, help the landless, small and marginal farmers to raise their income, diverse economic activity, can create gainful employment, especially for the unemployed/underemployed youth, weaker sections of the society and women. Our country has resources, potential for large scale production of mushrooms both for domestic consumption and export. These are cultivated all over the world. Though mushroom cultivation is known from time immemorial its cultivation in India started in 1943 when *Volvariella voilvacea* was cultivated at Coimbatore, *Agaricus bisporus* and *Pleurorus sojar caju* were cultivated in 1966 and 1970 respectively. In spite of it, mushroom cultivation in India is only 0.09% of world production. However, Indian Govt. has recognized the importance of mushrooms under changed conditions and established a national research centre at Solan, Himachal Pradesh where an intensive research have been carried out on different aspects of mushrooms, their cultivation to motivate people of India for the use of mushroom as a part of their diet.

Table 21.1: World production of cultivated edible mushroom

Sl. No.	Common Name	Botanical name	Substrate	Production (in 10^3 tonnes)	Contribution (in%)	Purpose
1	Button mushroom	Agaricus bisporus A. bitorquist	Straw	1424	37.8	Food
2.	Oyster mushroom	Pleurotus sojar-caju	Straw	909	24.2	Food
3.	Straw mushroom	Volvareilla volvacea	Straw	207	5.5	Food
4.	Shiitake mushroom	*Lentinus edodes*	Shii and Oak tree	393	10.4	Food and medicine
5.	Nameka	*Pholior nameka*	Saw dust	53	1.4	Food
6.	Inkap	*Coprinus comatus*	Straw	-	-	Food
7.	Wood ear or Jews ear	*Auricularia polytricha* and other species	Wood	400	10.6	Food
8.	Winter mushroom	*Flammulina velutipes* (Fig. 21.3d)	Wood	143	3.8	Food
9.	Silver ear	Tremella fusiformis	Wood	105	2.8	Medicine
10.	Bracket	*Ganoderma Luciderma*	Wood	-	-	Medicine

21.2 CLASSIFICATION OF EDIBLE MUSHROOMS

Edible mushrooms are classified according to taxonomic position as well as their natural habitat. The taxonomy of mushrooms is a fascinating field, both morphological characters of fruit body, spore production and spore color proved to be useful. The spore print which can be obtained by placing cut fruit body on white paper are covered with a beljar for over night. The next day one

can get spore print which reveals spore colour and arrangement of gills. These help in the identification of genus (Fig. 21.1). The shape and attachment of basidiocarp to the stipe in the identification of mushroom is also useful (Fig. 21.2).

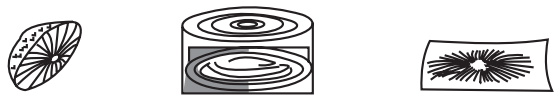


Fig. 21.1: Spore print

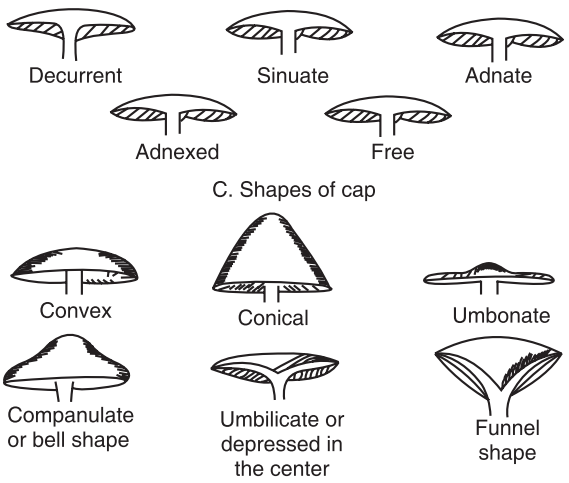


Fig. 21.2: Cap Shape attachment of basidiocarp to the stipe

Mushrooms are classified on the basis of natural habitat as precised here:

(A) Humicolous or folicolous (humus inhabiting)

(i) Saprophytic :	*Lepista nuda,Volvariella* spp,
	Marasmium spp
	Polyporus,Tuberaster
(ii) Symbiotic :	*Boletus, Lacterius,*
	Tricholoma, Tuber
	and *Morchella*

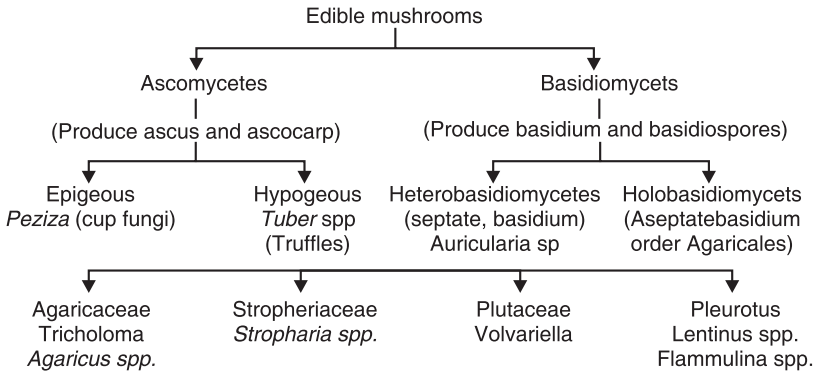
(B) Lignicolous (Wood inhabiting)

(i) Saprophytic :	*Agrocybe, Pleurotus*
	Auricularia spp
	Lentinus edodes
(ii) Parasitic:	*Armillaria mellea*
	Cyttaria

(C) Coprophilous (Dung inhabiting)

Agaricus spp
Coprinus spp

Similarly mushrooms are classified on the basis of taxonomic position as detailed below:



21.3 GENERAL STEPS IN MUSHROOM CULTIVATION

(i) Pure culture: Mushroom cultivation is carried out in the following manner. The selection of quality mushroom is an important aspect . It must have good quality in respect of growth conditions, taste, aroma, fruiting characters, disease and pests resistant along with long preservation quality. The mushroom cultivation is carried out partly in laboratory under aseptic conditions and other in mushroom house.

(ii) Spawn production: Good spawn production from monosporeculture is one of the methods of production. It should be fresh, fast growing, free from insects, moulds and mites.

(iii) Preparation of substrate/beds: Preparation of bed is another event. Substrate preparation varies with the mushroom. Either short term period or long term period of substrate preparation can be adapted. Finally it should be decomposed and support the growth of mushroom as well.

(iv) Spawning and spawn running: Healthy spawn should be spread on the bed in any one of the methods of spawning and allowed to grow. Sufficient moisture has to be maintained.

(v) Cropping: Once budding appears, one should be ready to harvest. The bud grows to full fledge mushroom and just bloomed should be harvested. Care should be taken not to break mushroom mycelium.

(vi) Canning: Mushrooms as far as possible are to be consumed freshly. If needed, they may be stored for a few days after canning as detailed in latter part.

The details of each stage is depicted in Fig. 21.3.

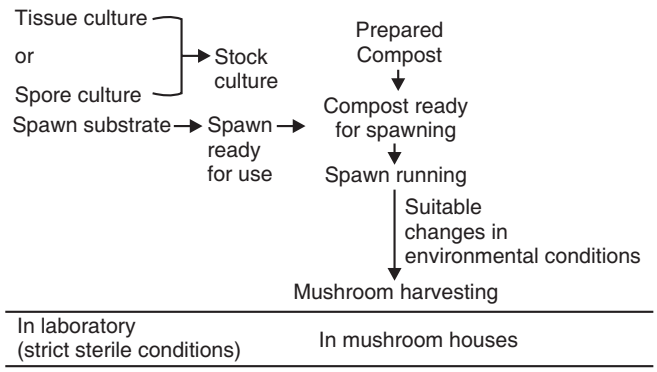
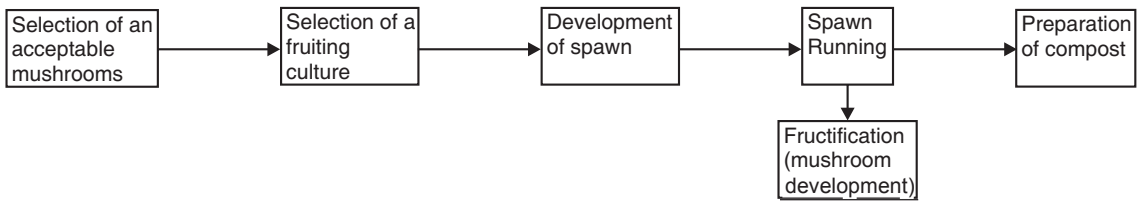


Fig. 21.3: Major steps in mushroom cultivation



- Location
- Climate
- Raw materials
- Acceptability
- Tissue culture
- Spore culture
- (a) without mating for homothallic species
- (b) mating with compatible isolates for heterothallic species
  - Mixed culture
  - Preservation
- Substrate Vigorous Growth
- Free contamination
- Avoid use of senescent and degenerate spawn
- Good survival of storage
- Establishment of mycelium
- Environmental requirements
  (a) temperature
  (b) light
  (c) aeration (O_2, CO_2)
  (d) pH
  (e) moisture
- Casing
- Watering and care
- Concept of composing
- Microbial activity
- Softening of substrate for case of colonization
- Physical characteristics
- Chemical components
- Aeration
- Water content

Fig. 21.4: Major steps in mushroom cultivation

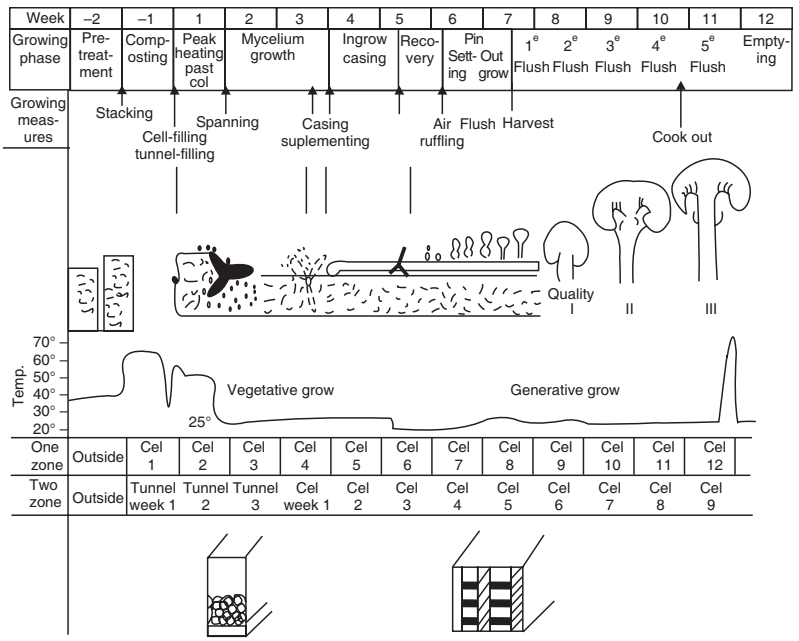


Fig. 21.5: Mushroom growing stages

21.4 MUSHROOM CULTIVATION IN INDIA

There are three major types of mushrooms cultivation in India. They are: (1) European button mushroom (2) Oyster mushroom and (3) Paddy straw mushroom. European mushroom requires more specific temperature, costly equipments and more investment. The two other types of mushrooms can be very easily cultivated in India. The mushroom cultivation proved to be profitable both at industrial scale as well as part time for unemployed youth and household wives. The economics of mushrooms cultivation are given in table 21.2.

Table 21.2: Economics of mushroom cultivation

(a) Non Recurring Expenditure In ₹		**(b) Recurring Expenditure IN ₹**	
(1) Drums with heating coil	2000	(1) Paddy straw 16 tonnes	24000
(2) Chaff cutters	3000	(2) Pesticides and fungicides	1200
(3) Wooden table	1000	(3) Spawn bottles	16000
(4) Wire mesh frame	500	(4) Polythene bags	10000
(5) Bamboo racks	2000	(5) Labourer (Wages)	27000
(6) Sprayers and buckets	2000	(6) Electricity and water	3500
(7) Drums for soaking straw	2000	(7) Miscellaneous	2000
		(8) Rent for building	5000
	12500		88700
Interest 15% on 12500	1875		
Depreciation 20%	2500		
	4375	Say:	89000
Anticipated yield at @ 30% of 16 tonnes	:	4800 kg	
Sale ₹ 25/kg		120000	
Total expenditure 89000 + 4315	=	93375 (94000)	
Net income : ₹ 120000 – 94000	=	26000	

21.4.1 Oyster mushroom cultivation

There are several edible varieties of *Pleurotus* namely *P.ostreatus, P.florida, P.sojor-caju,* etc. which are well known for their delicacy and flavour. These species are being grown on commercial scale in various countries. However, in India *P.sajor-caju* and *P.florid*a are most popular for commercial cultivation. Cultivation of mushrooms are being carried out in mushroom houses. Part of the cultivation requires aseptic conditions, while the other does not require perfect sterile rooms (Fig. 21.3).

(i) Substrate preparation: Oyster mushroom can be grown on various substrates viz., wheat/paddy straw, maize stalks, maize cobs, cotton waste, wooden logs, saw dust and vegetable plant residues. Since paddy straw is easily available throughout the year in most parts of the country, it is widely used. One should always prefer to use fresh and well dried paddy straw from which compost is prepared. Both bacteria and fungi of mesophilic and thermophilic are involved in compost preparation. (Table 21.3) Major

steps in mushroom cultivation are precised in Fig. 21.3 and mushroom growing stages are given in Fig. 21.4.

Table 21.3: Dominant Thermophilic and thermotolerant microorganisms involved in composting process

	Species dominant at	
	Start of establishment stage	End of maturation stage
Bacteria	*Bacillus subtilis Flavobacterium sp.*	*B.staerothermophilus Pseudomonas sp.*
Actinomycetes	*Streptomyces thermovulgaris*	*S.thermorulgaris S.rectus*
Fungi	*Mucor pusillus Humicola griseus Aspergillus Humicola languinosa*	*Streptomyces sp. Torula thermophila*

(ii) Soaking: Paddy straw chopped into 3-5 cm pieces is soaked in fresh water for 8-24 hours. Old, broken, rotten straw and stagnant water should never be used. Wet substrate is spread on wire mesh to drain off excess moisture.

(iii) Pasteurization: Water is boiled in a wide mouth container such as a tub or drum. The wet substrate is filled in Gunny bags or basket and closed. The filled bag/basket is dipped in hot water (80-85°C) for about 30-60 minutes and to avoid floating it is pressed with the help of wooden piece. After pasteurization, excess of hot water should be drained off in a container so that it can be used for other sets. Care should be taken to maintain boiling water temperature 80-85°C, for all sets to achieve pasteurization. Pasteurized substrate is kept inside the chamber where bag filling and spawning has to be done. Once it cools down to room temperature it is filled in bags. The moisture content should be 70%.

(iv) Spawning: When pasteurized substrate has cooled down to room temperatue, it is ready for filling and spawning. At this stage the moisture content should be about 70%. Polythene bags (35 × 50 cm, 150 gauge) or polypropylene bags (35 × 50 cm 100 gauge) may be used for its cultivation. Two percent spawn on the basis of wet straw i.e. one 500 ml bottle spawn (200-250 gm) for 10^{-12} kg wet straw is used. Spawning can be done by one of the methods as follows.

(a) Surface spawning: Bags are opened and 2% spawn is broadcasted on top and a little ruffling is done to mix it in top layer (2-5 cm thick) and quickly closed.

(b) Layer spawning: Substrate is filled and gently pressed at a depth of 8-10 cm and spawn is broadcasted above it. Similarly 2nd and 3rd layers are put simultaneously spawned then bags are closed. This is more suitable when pasteurized straw is filled in bags.

(c) Through spawning: Pasteurized straw is mixed with 2% spawn and filled in bags. It is gently pressed and closed for spawn running. This will not be convenient for sterilized polypropylene bags.

Spawned bags should be stacked in racks in neat and clean place in a closed position. Temperature of 25 ± 5°C and relative humidity 70-85% should be maintained by spraying water twice a day on walls and floor. At this stage, fresh air requirement is minimum. It takes 20-25 days when bags will be fully covered with white mycelium.

(v) Cropping picking and packing: Once bags are fully covered with mycelium, they are transferred to cropping room, and polythene/polypropylene covers are removed. The

open blocks are kept in racks about 20cm apart. Rack should be 60 cm wide and a gap of 50-60 cm between two shelves. There should be 60-75 cm gap between two rows of racks for working. It grows in a temperature range of 20°-33°C (Optimum 25+2°C). Relative humidity 80-85% is maintained by spraying water twice a day but watering on blocks should be avoided for first 2-3 days. After 2-3 days, light mist spray of water is given on blocks. This is the time when small pin heads appear. Once pin heads are of 2-3 cm size, a little heavier watering is done on blocks and further watering is stopped to allow them to grow. This helps to avoid bacterial rotting. However, the optimum relative humidity must be maintained inside the cropping room. Once mushrooms are 6-8 cms in size, they are plucked, mushrooms should not be allowed to produce spores as it results in poor quality of mushrooms (Fig. 21.6). After harvesting the first flush about 0.5–1.0 cm outer layer is scraped. This helps to clean remanants of the 1st crop and to initiate 2nd flush. Second flush will appear in about 10 days of 1st flush, similarly, 3rd and 4th flushes will appear in 8-10 days interval. However, almost 80% of crop will be over in 1st and 2nd flushes. Hence, many growers take only 2 flushes. Care must be taken that the beds do not become too wet which may otherwise cause rotting.

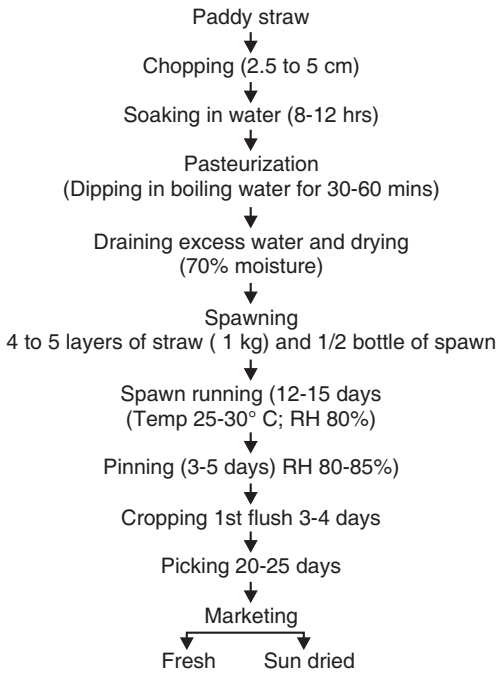


Fig. 21.6: Flow chart of cultivation of oyster mushroom

Mushrooms should be harvested before watering and their lower portion is cleaned with dry cloth. Mushrooms are packed in perforated (5-6 small holes) polythene or polypropylene bags. It should be sent to the market while fresh. This variety can also be sun-dried by keeping fresh mushrooms in sun for 2 days. Even polythene sheet may be spread to about 30-40 cm above the mushrooms for quicker dehydration in sun. It can be mechanically dried at 40-45°C. The dried product can be packed in polythene bags for marketing. Dried mushrooms should be soaked in water for 10 minutes before use. Details of oyster mushrooms are precised in flow chart (Fig. 21.6).

21.4.2 White Button Mushroom *(Agaricus bisporus)* Cultivation

(i) Spawn Production: The seed material of mushroom fungi is called spawn. It is in fact, the actively growing vegetative mycelium in a suitable sterile organic matter. The spawn is generally produced in milk or saline bottles. The entire process of spawn preparation should be aseptic from the beginning to the end. The rice straw cuttings, cotton wastes, cotton seed hulls, rice hulls, sorghum grains, rye grains etc. are generally used either singly or in combination for spawn making. The method of spawn production from jawar grains is explained further.

Jawar grain is boiled in an equal volume of clean soft water till the entire water is absorbed by the grain. Later calcium carbonate at the rate of 20 g/kg of grain is added

to the cooked material and the mixture is filled in clean dry empty saline bottles upto ¾ full. The mouth of the bottle is plugged with non-absorbant cotton. The cotton plug and a part of the bottle neck is covered with a clean 10 cm square paper bit and tie it with twine or rubber band. Sterilize the bottles in a pressure cooker or autoclave at 15 lb pressure for 1 ½ hours. After cooling, a small quantity of the fungal culture should be inoculated to the bottle in an inoculation chamber. On incubation at 25°C for about 2 weeks the fungus covers the grain in the bottle. This is called spawn.

Button mushroom is the most popular in all the countries. U.S.A. is the maximum button mushroom producing country. *Agaricus bisporus* is cultivated in many countries. Recently *A.bitorquis* is also introduced. It requires slightly higher temperature for cropping. White button mushroom cultivation includes the following steps (Fig. 21.7).

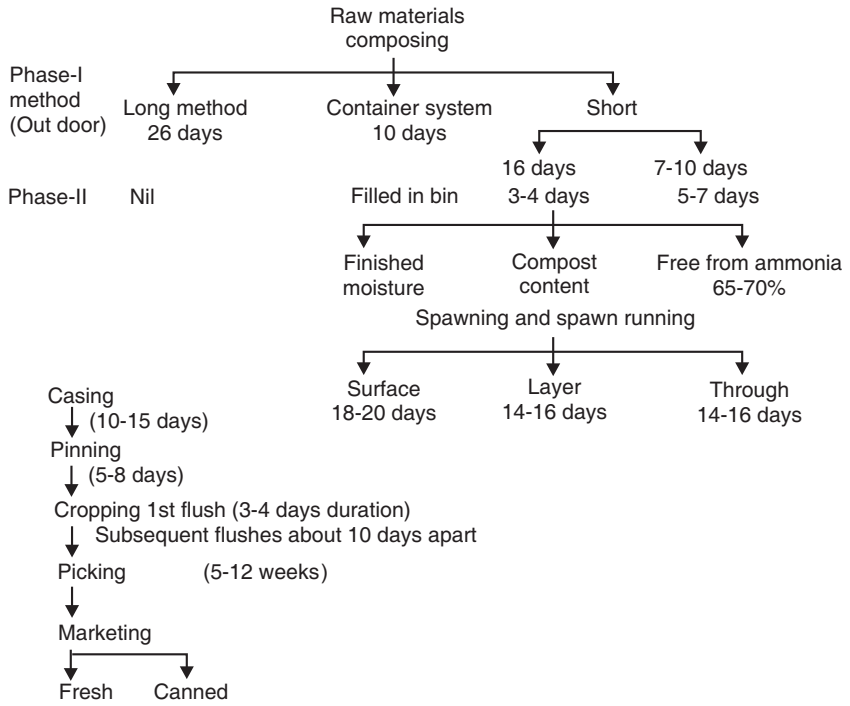


Fig. 21.7: Flow chart of white button mushroom cultivation

(ii) Spawn production:

In ancient times composts made of horse dung and cow dung were used. But now-a-days synthetic compost made out of grain crops straw is widely used. In south India locally available substrates like paddy straw, ragi straw, maize straw, horse dung and chicken manure are used. Paddy straw or mixture of paddy straw and maize straw form a good compost. Two compost formulae and the method of compost preparation are given below (long method).

Formula I		Formula II	
Paddy straw	1000 kg	Paddy straw	500 kg
Ammonium sulphate	30 kg	Maize stalk	500 kg
Super phosphate	10 kg	Ammonium sulphate	30 kg
Urea	13 kg	Super phosphate	10 kg
Rice bran	150 kg	Urea	13 kg
Cotton seed	20 kg	Rice bran	150 kg
Chalk powder	33 kg	Molasses	40 kg
Gypsum	40 kg	Cotton seed	20 kg
		Chalk powder	33 kg

Paddy straw watered for 2 days to have moisture content 75-77% is mixed with fertilizers and stacked in a heap of 1.65 – 1.8 × 1.65 – 1.8 m or required length and turned on 6th, 10th, 13th, 16th, 19th, 22nd, 25th and 26th day. On 10th day chalk powder and on 13th day gypsum are added. On 26th day the substrate compost will be ready. It is filled into trays, boxes or shelves to make the so called beds and transferred into the room of pasteurization. Pasteurization is a partial sterilization process operated at low temperature. Live steam generated from a water boiler is introduced to compost for 2 hrs so as to have air temperature 60-62°C. Maintain this temperature for 2 hrs and then introduce a gentle steam of fresh air to lower the temperature for the next 6-8 hrs. The objective of pasteurization of compost is to keep away the insects then pests in the substrate then spores of contaminating microorganisms then bring the temperature of compost uniformity to 50-55°C which promotes decomposition of the substrates by thermophilic microorganisms (Table 21.4). Through this final adjustment a more selective medium favouring the growth of the mushroom is accomplished. Carelessness at this stage may lead to a crop failure.

(iii) Spawning: The spawn is spread over the surface and covered with a thin layers of compost. The growth of active mycelium in the beds is called spawn running. Spawn running requires 15-20 days at 90 to 96 percent relative humidity R.H. and 25 ± 2°C temperature.

(iv) Casing: After spawn running, the beds are covered by casing material (thin layer of pasteurized material) continued incubation for another 8-10 days, i.e., till pin heads start appearing. At this stage the temperature is lowered to 16 ± 18°C and fresh air is introduced. Watering the beds is to be carried out as and when required.

(v) Cropping : The crop starts producing mushrooms in 3rd week after casing and continue for 10-12 weeks. Mushrooms are picked by griping the cap and twisting when they are in button stage.

Mushrooms or fruit bodies appear in rhythemic cycles which are called flushes or breaks. Generally, a large and more number of mushrooms are produced in first four flushes. After the fourth flush smaller and less number of mushrooms are produced. Hence, beds are usually removed after the fourth flush. The spent out bed material can be used as a manure.

21.4.3 Paddy straw mushroom cultivation: The paddy straw mushroom *Volvariella volvacaea* prefers to grow on paddy straw, hence it is known as paddy straw mushroom. Though its cultivation started in China during 18th century, its cultivation in India is only in 1960 at Coimbatore, Tamilnadu. Three species of this genus, *V.esculenta,V.volvacea* and *V.diplasia* are being cultivated in India. At the beginning it used to be grown on twisted 10 kg paddy straw soaked in water and placed on wooden platform (75 × 75 × 30 cm) in a haphazard manner. The paddy bed is spawned with 3rd twist. The whole bed is

covered with the polythene sheet to maintain moisture. Recently polybag method was developed by Bahl (1982). The method has following steps:

1. Chop the paddy straw and soak in water for 24 hrs.
2. Cut waste paper into small pieces and soak for the same period.
3. Decant water after 24 hrs.
4. Mixed thoroughly the chopped straw, paper pulp and spawn.
5. Fill the mixture in polythene bags.
6. Puncture the bags with needle to facilitate the exchange of air and drain the water.
7. Tie the month of polyethylene bag. Keep them on wooden rock at 35-40°C
8. Maintain the humidity of the contents to about 80-90°RH by sprinkling water
9. At the end of 10-15 days small pin heads of mushrooms appear which grow into matured mushroom within 2-3 days.
10. Harvest mushroom while they are tender before starting spore production.

21.5 PESTS AND DISEASES OF MUSHROOMS

21.5.1 Pests: Different insects and nematodes grow on mushroom bed and will be responsible for yield loss. Even rats eat away fruit bodies along with grains and other things causing yield loss.

(A) INSECTS

(i) Spring tails - *Lepidocystus cyaneus*
(ii) Sciarid flies - *Lycoriella solani*
(iii) Phorid flies - *Megasella halterata*
(iv) Coccids - *Heteropeza phagmae*
Mycophila brunnesi
(v) Mites - *Rhizoglyphus phylloxerae*

(B) NEMATODES

(i) *Dactylenthus myceliophagus* affect mycelial growth

(C) RATS

(i) Damage the beds and eat grain spawn

(D) CONTROL MEASURES OF PESTS

(1) Maintain cleanliness
(2) Prevent compost and straw from coming in contact with soil
(3) Treat the tools with 2% formalin
(4) Spray dichlorovas (Nuvan) at the rate of 0.6 ml/litre
(5) Spray Dicofeel (0.01%) to control mites
(6) Baiting and killing rats

21.5.2 Diseases: Different parasites grow on growing mushrooms in beds, cause characteristic diseases and responsible for considerable loss to fruit bodies harvest.

(A) Fungal Diseases

(i) Dry bubble - *Verticillium malthousia*

(ii) Wet bubble - *Mycogone penniciosa*

(B) Bacterial Diseases

(i) Bacterial pit - *Pseudomonas sp*

(ii) Bacterial brown blotch - *Pseudomonas tolassii*

(C) Viral Diseases

(i) Elongated bend stipes - seven types of viral particals

(ii) Disintegration of mycelium

21.5.3 Weed Moulds and Competitors

The following moulds grow on growing mushrooms and will be responsible for yield loss.

(i) FALSE TRUFFLES - *Deihliomyces microsporus*

(ii) WHITE PLASTER MOULD - *Scopulariopsis fimicola*

(iii) INK CAP - *Coprinus sp*

21.6 CANNING OF MUSHROOMS

Mushrooms have a good taste when cooked fresh. However, canning is required when consumers are located in far off places. Following steps are involved in canning process:

1. **Pre-cleaning:** Mushrooms are cleaned to remove foreign particles, soil etc.
2. **Washing:** Mushrooms are washed in water
3. **Blanching:** Mushrooms are blanched in hot water having 0.2% citric acid for 3-5 minutes. This process results in loss of 30% weight.
4. **Cooling:** Blanched mushrooms are cooled through continuous counter flow cooling system.
5. **Grading:** Mushrooms are graded according to size.
6. **Slicing:** Mushrooms are sliced for the required size.
7. **Filling:** The cans are filled and weighed.
8. **Brining and exhausting:** Hot brine solution (salt 2%+sugar 2%+citric acid 0.3%) is added and temperature is raised to 80°C in the centre of can to exhaust.
9. **Can sealing:** Cans are sealed with lid.
10. **Retorting:** The sealed cans are sterilized at 15 PSI for 15-20 minutes.
11. **Labelling and packing:** The cans are labeled and packed in cartons. Different sizes of cans are used as per requirements.

21.7 NUTRITIONAL AND MEDICINAL ASPECTS OF MUSHROOMS

Mushrooms, rich in nutrients, are being used as nutraceuticals. They are considered to provide strength to warriors in battles as believed by Greeks, while Chinese feel that they are health food and treat them as elixir of life. Romans consider them as God given food, while pharha, European tribes, considers them as food of delicacy. Mexican Indians eat them during festive

occasion as hallucinogens. They are food of fibrous nature, low in fat, rich in proteins, vitamins, this food preferred for diabetes and heart patients. Similarly mushrooms are rich in all essential vitamins and contains full compliment of mineral composition (Table 21.4). The nutritive value is superior to egg, meat and pulses. In nutritional index (NA) it stands 4th, while in essential amino acid index (EAI) and protein efficiency ratio (PER) it stands third. Thus these mushrooms proved to be quality food. They are also reported to have medicinal value and used in the treatment of human ailments (Table 21.5).

The following recipes such as mushroom puree, mushroom paneer, mushroom pulaw mushroom omlette and mushroom soup which are for delicacy and tasty are made. They are also rich in different nutrients.

Table 21.4: Nutritive value of some common mushrooms

Nutrient	*Agaricus bisporus*	*Pleurotus ostreatus*	*Lentinus edodes*	*Volvariella volvacea*
1. Protein	3.9	2.8	2.8	5.0
2. Fat	0.2	0.6	0.4	0.7
3. Crude Fibre	1.1	1.1	1.2	1.4
4. Vitamins				
Thiamine	1.1	4.8	7.8	1.2
Riboflavin	5.0	4.7	4.9	3.3
Niacin	44.7	108.7	54.9	91.9
Ascorbic acid	81.9	-		
5. Minerals				
Calcium	23.0	98.0	33.0	71.0
Phosphorus	1429.0	476.0	1248.0	677.0
Sodium	-	61.0	837.0	374.0
Potassium	4762.0	3756.0	3793.0	3455.0
Iron	0.2	8.5	15.2	17.1

Note: Each kg of mushroom is equal to 0.5 kg meat, 2 ½ egg, 1.5 kg potato, 0.5 kg soyabean and 1.0 kg pulses.

Table 21.5: Medicinal properties of mushrooms

Activity	Species used
Antibacterial	*Pleurotus sp.*
	Agaricus bisporus
Antifungal	*Lentinus edodes*
Antitumer	*Lentinus edodes*
	Flammulina velutipes
	Pleurotus ostreatus
	P.spodoleucus
	Agaricus bisporus
	Auricularia auriculata
Antiviral	*Volvariella spp*
	Piptoporus setulina
	Lentinus edodes

contd...

Activity	Species used
Hypolipidemic	*Lentinus edodes*
General health complaint	*Polyporus officinalis (Trade name Agerick)*
Bleeding	*Polyporus officinalis (Trade name Agerick)*
High sweating in TB	*Polyporus officinalis (Trade name Agerick)*
Inflamed eyes	*Auricularia auriculata*
Bleeding (Operation)	*Lycoperdon giganteum*
Anesthesia	*Clavatia gigantia*
Swollen glands, Rheumatid Arthritis	*Amanita muscaria*

21.8 FUTURE OF MUSHROOM CULTIVATION

Increasing awareness of nutritive and medicinal value of mushroom, a boost in cultivation of mushrooms resulted in mass scale. Cultivation of mushrooms helps to convert agrowastes into human food. Their cultivation provides labour employment as they are fast growing and are responsible for production of quality food. Mushrooms represent untapped source of nutraceuticals and valuable palatable food. Substrate/compost preparation with special reference to fermentation, collection of strains of mushrooms from different geographical regions and their evaluation use in breeding work. Breeding for high yielding strains of species will be of immense value of *Agarics*, *Pleurotus,* other promising mushroom for both cold and hot climate is of an urgent need. Improvements in prolongation of shelf life and canning and processing will also boost the prospects of mushroom cultivation. However, lack of awareness, shelf life maintenance of pure culture, unpredictable yield, shorter shelf life are some of the limitations in mushroom industry. Indifference of academicians, Govt. and institutions adds to the constraints of mushroom cultivation. Developing sporeless or low spore shedding ***Pleurtus*** and other mushrooms with desired traits will also help the mushroom industry. The methods of cultivation and other associated problems in cultivation of untapped mushrooms such as *Amamita, Agrocybe, Armillaria, Boletus, Cantherellas, Lactarius, Lepiota, Marasmius, Morchella, Peziza, Hydnum, Psalliota, Rhodopaxillus, Russula* and *Termitomyces* should be taken up for cultivation as well as nutritive quality determination. Protoplast fusion technique in developing quality mushroom is also need of the hour.

REVIEW QUESTIONS

I. Essay type questions:

1. Give general account of mushrooms and their economic importance.
2. Give detail account of cultivation of button mushrooms.
3. Describe the method of cultivation of Oyster mushroom.
4. Give an account of nutritional value and medicinal importance of mushrooms.
5. List out major steps in the cultivation of mushrooms.

6. Discuss the taxonomy of mushrooms.
7. Give an account of insect pests and diseases of mushrooms.
8. What are edible mushrooms? What role could they play in facing the challenges of world food shortage. Discuss in the light of their nutritional status.
9. Discuss the status of mushroom cultivation in India.
10. Write an essay on formulation and preparation of composts in India studied by you.
11. Describe in brief cultivation of straw mushroom.

II. Write critical notes on:

(a) Spawn
(b) Spawn running of mushroom
(c) Grain spawn
(d) Recipes of mushroom
(e) Casing
(f) Blanching
(g) Canning of mushrooms
(h) Nutritive value of mushrooms

FURTHER READING

1. Chang, S.T. and W.A. Hayes (1978) The biology and cultivation of edible mushrooms, Acadamic Press, New York.
2. Chang, S.T. and T.H. Quimio (1982) Tropica (mushrooms Chinese university Press, Hong Kong.
3. Flegg, P.B., D.M. Spencer and D.A. Wood (1985). The biology and the technology of the cultivated mushrooms, John Wiley and Sons Chickester.
4. Anke, T. (1996). Fungal biotechnology, Chapman and Hall, Weinheim.
5. Stamets, P. (1993) Growing gourmene and medicinal mushroom Ten Spreed Press, Berkeley.
6. Zadrazil, F. and K. Grabble (1983) edible mushroom in Biotechnology (H.J. Rehm and G. Reed) Verlag Chemie, Weinheim.

22

Intellectual Property Rights and Patents

A patent is a form of protection issued by the Government to an inventor for a publicly disclosed and details of product or process by a way of granting a limited period legally enforceable right to exclude others from exploiting it commercially or otherwise. This is done to promote innovations of human importance and also justify the risks of development and assures the inventor a reasonable profits on his invention/innovations. The inventor makes his/her inventions public by disclosing how it works and the manufactures may utilize the knowledge for commercial production. This concept was in existence since 1332, when revocable patent for 12 years was granted for a wind mill discovered by Bartolomeo verdice of venice. This patent act was enacted in 1474 in Venice for providing exclusive rights for 10 years to an inventor of new art or machines. Subsequently Adlus obtained a patent for his italic type in 1501 and Galilo got a patent for a machine meant for raising water from a deep well. Patents have played a critical role in the economic policies of many countries for over 650 years.

For many scientists patent laws are confusing and would like to avoid if possible. However, to exploit the economic value of invention, patent protection may determine the success or failure of the commercial utilization of invention. An understanding of the fundamentals and functioning of the patent system specially recent amendments in the law concerning biological inventions is of paramount importance.

In England patent was in the form of monopolies granted by Queen Elizabeth as personal favour. In 1623 statue of monopolies was enacted. In USA the patent concept came into existence in 1565 and it was enacted in 1790 and comprehensive patent laws came into existence. In India patent system was introduced in 1856 which underwent a series of enactments and exist till 1970 which also underwent amendments in 1999 and 2002. Patent laws of these three countries influenced subsequent development of patent protection in other countries like Egypt, Switzerland, Italy, Belgium, Luxamberg and France. In China these laws were promulgamated in 1984. From this time onward patent protection continues to promote the technical innovations and industrial development in different countries of the world.

Intellectual property (IP) is any creative work or invention, non-tangeable possession that can be protected by an intellectual property right (IPR). The legal protection for IP prevents others

from exploiting it without the owners permission for a definite period. Examples of IP includes, patents, designs, trade mark, copy right, data bases and geographical indications. A patent is a document granted by patent office which describes an invention and claims exclusive rights in the subject matter of invention. By granting the patent, the state makes both the details of the invention and the legal rights associated with it to public record. The effect of the patent is such that the patentee is authorized to produce or market, offer for sale or use the subject matter of the invention commercially. Patents are offered as a form of intellectual property or they may be bought or sold (i.e. assigned), licensed and also inherited. The grant of license governs the right to use the inventions between the owner of the patent (Licensor) and a third party (licensee).

In order to get patent protection, the invention must be filed with the patent office giving details of invention and claim what is to be protected. The invention must be described in sufficient details so that any person skilled in that art can put into practice and it is known as specification.

The person who gets patent rights in the form of transfer or purchase is called assignee. If more than one person is involved in patent rights, others are called co-inventors. Each joint owner owns an undivided interest in the whole property unless otherwise agreed upon. The scope of the rights granted by a license can vary greatly since the parties to a contract such as license are in principle, free to determine its terms. It may range from mere permission to use invention as a right to exclude others from utilizing it. Such non-exclusive and exclusive licenses, may inturn be further limited by various criteria such as geographical area, time constraints, limitations as to specific use of the invention or restriction to a particular subject matter of the patent.

Patent system is desirable in the public interest, on one hand the inventor has the opportunity to reap a just reward for his invention through exploitation of it and on the other hand, that technological progress, competition and the invention efforts of others be spurred upward through the disclosure of the invention to the public. The rights under a patent do not constitute a monopoly in the strict legal sense of the term. The patentee does not take anything from the society which it previously enjoyed. Since among the requirements for obtaining the patent is that invention to be new non-obvious. By granting patents the state does not intend to restrict the composition. The disclosure of the invention by the applicant for a patent faster competition in that other inventors are made aware of the inventors contributions and are encouraged to make other possible patentable contribution. Indeed the inventor is rather a public benefactor. It is because he adds to the wealth and comforts of the society and promotes the progress of utilization that he is awarded the exclusive right to his invention for a limited period. In order to encourage improvements in inventions and also to encourage their disclosure in preference to their use in secret, any person who has to make an invention may apply for a patent that gives him the right to exclude others for commercial exploitation of the invention.

22.1 REQUIREMENTS FOR PATENTABILITY

There are four universally accepted requirements to be met for the grant of a patent to any invention. 1. There invention must be novel and useful. 2. The invention must contain inventive step and it must contain full description of invention so that a knowledgeable person can implement. 3. Patents shall be granted for any inventions which are susceptible of industrial applications which are new and involves an inventive step.

The patent law of most countries requires that the inventions show some utility in order to be patentable. Any new and useful process, machine use or composition of matter, therefore, can be patented. Thus a claim that the invention is useful in having biological properties is not sufficient disclosure of utility. On the other hand, a live vaccine against measles possess utility. In recent times issue of biotechnological patents has become a common practice. Biotechnological patents include products, composition, process; use or methods of use, techniques that use living organisms in making products to improve patents or developing microorganisms for specific use. Louis Pasteur was the first to get a patent in 1880 for fermentation of acetic acid. Therapeutic patents came into existence in 1895 by a license may vary greatly since parties to a contract such as a license are in principle free to determine its terms. It may range from mere permission to use the invention to the right to exclude others from utilizing it. Such non-exclusive and exclusive licenses may inturn be further limited by various criteria e.g. Geographical area, time constraints, limitations as to specific uses of the patent. Further, it is to be noted that licensing agreements may present anti-trust problems which go beyond the scope of this chapter. Nevertheless, the reader should be aware of the potential questions of law which such licensing agreements may pose. A good introduction to the patent law of the USA is provided by P.D.R. Osenberg in patent law fundamentals. The more details of patents can be obtained from a book "Common patents for Common market" published in chemical and Engineer News, It gives the information on:

(i) What is considered as a prior Art?
(ii) What cannot be patented?
(iii) Language of patent written in.
(iv) Types of examination.
(v) Opposition.
(vi) Life of patent.
(vii) Maintenance fee.
(viii) Working requirements.

However, the following can not be patented as they are not regarded as inventions.

(i) Discoveries, scientific theories and mathematical methods. For instance the discovery of DNA by Crick and Watson, though useful was not patentable.
(ii) Aesthetic creations
(iii) Schares rules and methods or performing mental acts, playing games or flourishing business and programmers for computers.
(iv) Presentation of information.
(v) Methods for treatment of the human or animal body by surgery or therapy and diagnostic methods practiced on them.
(vi) Discovery of microorganism is not patentable but method of its culture or product can be patented.

Similarly patents shall not be granted to inventions, publication or exploitation by public order or mortality. Plant or animal varieties or essentially biological process for the publication of plants or animals. However, this does not apply to microbiological process or products.

22.2 TYPES OF PATENTS

The patents can be categorized into:

(i) Product patents:

(a) Substance: cloned genes, recombinant protein, monoclonal antibodies, plasmids, promoters, vectors, **C-DNA** sequences and monovalent vaccines.

(b) Composition of matter: multivalent vaccines, biofertilizers, bioinsecticides, pharmaceutical mixtures, microorganisms and transgenic organisms.

(c) Devices: pulsed field get electrophoretic apparatus, DNA sequencing apparatus, microprojectile gene transfer machine.

(ii) Process Patents:

(a) Process of preparation: Isolation of double stranded DNA, vector insert construction, PCR applications and purification of recombinant protein.

(b) Method of working: Nucleic acid hybridization assays, diagnostic procedures, detection systems using PCR and mutant assays.

(c) Use: Applying biofertilizers and bioinsecticides, fermentation by genetically modified organisms (GMO) and therauptic animal treatment systems.

There are three types of patents based on:

(a) Utility patents - to any one who invents or discovers any new and useful process, machine article of manufacture or composition of matter or any useful improvements there of.

(b) Design patents - may be granted to any one who invents a new, original and ornamental design for any article manufacture.

(c) Plant patents - may be granted to any one who invents or discovers and sexually reproduces any distinct and new variety of plants.

22.3 COMPOSITION OF PATENT

For obtaining a patent following procedure has to be adapated. Patent application consists of three parts:

(a) The grant – It is a signed agreement between inventor and patent sanctioning authority. It is not published and filed in sanctioning authority office.

(b) Specifications – It is published along with data and available to public at nominal charges. It gives a narrative description of the subject matter.

(c) Claims – It gives scope to protect the invention. It should give how this process differs from the existing one. At the same time others do not practice without prior permission. It is very critical for the validity of a patent. The claims should be precise and accurate so that it can withstand for rigorous testing.

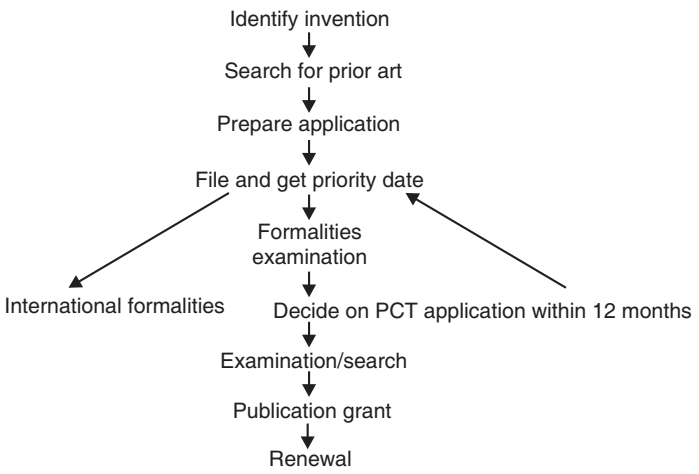


Fig. 22.1: Procedural flow diagram for obtaining patent

22.4 PROCEDURE FOR OBTAINING A PATENT

The procedure for obtaining a patent is precisely presented in Fig. 22.1. Following documents are to be furnished with the patent application.

(a) Application form (in triplicate)
(b) Fee form
(c) Application data sheet
(d) Complete specifications which includes
  (i) Title of the invention
  (ii) Abstract of the disclosure
  (iii) Field of invention
  (iv) Background of invention
  (v) Object of invention
  (vi) Summary of invention
  (vii) Brief description of the accompanying drawing
  (viii) Detailed description of the invention
  (ix) Claim or claims, sequence listing (if any).
(e) Drawing in triplicate.
(f) Declaration

In microbiological invention it may be necessary to describe the invention with biological and physical data. In identifying a noval antibiotic the following parameters should be given.

(a) Microbiological parameters such as antibacterial spectrum, co-existence spectrum, special effects of experimental conditions, pH, presence of serum, concentration, kinds of ions, inoculum size and the frequency of resistance occurrence.

(b) Biological (pharmacokinetic) parameters such as ED50 and LD 50.

(c) Mechanism of action.

(d) Solubility behaviour in various solvent systems. Effect of pH and absorption of ion exchange resins.

(e) Behaviour during chromatography, paper chromatography, TLC with various solvent systems.

(f) Physico-chemical properties such as melting, fusion, decomposition point optical rotation, elemental analysis, principal functional groups, isoelectric point, IR, UV and visible spectra, NMR, Mass Spectra and stability (pH, light, specific enzymes).

A patent application is processed through a patent office. There are two major categories of patent processing. In the registration or non-examining system the patent office limit their examination to formal matters only proper filing of patent application and payment of government fees automatically result in the granting of a patent. The merit of the invention is not examined and hence if need arises, the patent validity must be determined by the court. Some of the countries like France, Switzerland, Italy, Belgium and Luxenberg have adapted this system.

In examination system the application is examined critically to determine the novelty of invention. Further, the invention must shown to be useful. An expert committee will examine and verify the claims. On recommendation of the expert committee, the patent will be granted. One or two witnesses who understand the invention. Persons associated with invention as supporting or helping can not be granted patents. The persons who contribute mentally can share the patent right.

A published patent states at the top of its first page the type of ownership that has to be associated with the patent. There are three categories:

1. In this category of patents, the commercial rights are retained by the inventor or the concern to which the patent has been assigned or sold.
2. The patents are assigned to the federal government. This assignment allow some control over the license to use these patents.
3. The patents dedicated to public. These patents do not provide monopoly for any individual or any industrial concern but make the invention available for all. The absence of exclusivity for these patents may not provide adequate incentives to investor in the development or marketing of products.

A valid patent is granted only to the first inventor regardless of filing dates for patent application. Most of the times it is not clear who really the first inventor. Therefore, patent laws provide for interference proceedings to determine the priority of invention. These interference proceedings may be initiated by the patent officer or copending applicant within one year after the patent is issued. The inventor is that individual who first conceived the idea of invention regardless of whether the idea at that time been reduced to practice. It is of utmost important for the inventor to establish the actual date on which he conceived the idea of the invention. To establish the date, the inventor should immediately get the inventive idea put down in writing, sign and the paper on which it is written should have witness signed, date of the document who can verify the date and content of the description of invention. This recording of the invention is often done in a research book and it should include what is considered to be the results of the invention. In addition to recording the invention, every page in the research book on which further experimental work has been recorded should also be signed and dated by the inventor.

22.5 SUBJECT MATTER AND CHARACTERISTICS OF PATENT ON MICROBIAL PROCESS OR PRODUCTS

The patent application should disclose the full information regarding invention/process as illustrated by lysine production. The lysine production patent was obtained by Lyster E.Casida and Jr Baldwin for their application No. 551-987 dated, approved and assigned to ChasPfizer & Co., inc, New York, a corporate of Delware.

(i) **Process of preparation of L-lysine:**
   (a) Fermentation broth containing glycerol, cornsteep liquor under submerged and aerobic culture of auxotrophic mutant of *E.coli.*
   (b) Treating diaminopimelic acid (DAP) with enzyme produced by wild strain of *E. coli* or *Aerobacter aerogenes* and pH near neutrality.

(ii) Process of preparation of DAPA, while maintaining a neutral pH.

(iii) Preparation of L-lysine from DAPA by enzyme produced by *A. aerogenes* or wild strain of *E. coli.*

(iv) Conversion of DAPA with *E.coli* with improved pH.

(v) Improvement of medium by addion of cornsteep liquor.

(vi) Improvement of production of lysine by *A. aerogenes* or *E. coli* by adding chelating agent.

(vii) The added chelating agent is citric acid.

(viii) The chelating agent is EDTA.

(ix) As in claim 2 with *E. coli* ATCC.

(x) As in claim 3 with *A. aerogenes* ATCC.

Claims in the application are:

(i) Production of DAPA

(ii) Alternate method of production of DAPA

(iii) Conversion of DAPA into lysine

(iv) Alternate method of conversion of DAPA to lysine.

22.6 PATENTS INVOLVING MICROORGANISMS

Majority of today's innovation are made in the field of biotechnology. These inventions broadly include any technique that uses living organisms to make or modify products, to improve plants or animals or to develop microorganism for specific use. However, patents on biotechnological systems are not new. The first patent dealing with microorganisms was issued to Louis Pasteur for a claim to a biologically pure culture of a microorganism employed in acetic acid fermentation. Therapeutic patents were issued from 1895. The patent application for any new invention involving microorganism be supported by a deposit of that culture in a recognized international depository authority (IDA) which must fulfil following obligations:

(i) It should issue official receipt for the culture deposited.

(ii) It should test the viability and issue certificate.

(iii) It should keep the deposit of culture secret.

(iv) Maintain the culture atleast at 30% viability. Check the viability frequently or as depositor demands.

(v) Be impartial to any depositor.

The culture collection centres, on the other hand, should fulfil following conditions in order to get designation as a IDA.

(i) It should be located on territory of contracting member state.
(ii) It should be in continuous existence.
(iii) It should have impartial policy concerning all deposits.
(iv) It should possess appropriate staff and facilities.
(v) It should accept specified types of microorganisms or cell lines.
(vi) Determination of viability of deposit culture and issuance of viability statement.
(vii) Storage of deposited strains for at least 30 years.
(viii) It should adapt sufficient safety measures.
(ix) It should maintain the secrecy of deposits.
(x) It should furnish the samples in appropriate timely manner.

ATCC (USA), NRRL (USA), AGL (Australia), CBSC (Netherlands), CNCM (France), CAB, IMI (UK), CCAP (UK), DSM (FRG), ECACC (UK), FRI (Japan), IBFM (USSR), *in vitro* International (INC), (USA), MIMNG (Hungary), NB IMCC (Bulgaria), VNIAA (Soviet) are some of the culture collection centers recognized as IDA by world international patent office (WIPO). This will act as reference point from which patent and trade mark office committee examines claims in a patent application and get convinced. Examiner could determine the adequacy of total information provided and resolve any disputes as to the nature, identity and operability that was claimed in the specification.

The depositor should fulfil following requirements so as to enable him to deposit a culture in an IDA.

(i) Indication that the deposit is being made under treaty.
(ii) He should enter an agreement not to withdraw the deposited culture for 30 years.
(iii) He should provide name and address of the deposit.
(iv) He should designate deposited strain.
(v) Provide details necessary for growth.
(vi) He should submit the culture in a required form and quantity.
(vii) He should make one time payment of fee.

Further the depositor should provide the following information to IDA:

(i) A signed statement of cultures being submitted, the same is not submitted earlier.
(ii) A number and indication of the date on which the depositor secured a certificate.
(iii Reasons for making new deposit.
(iv) A copy of the receipt.
(v) A copy of the most recent scientific description of the culture.

For maintaining a patent deposit culture in American type culture collection (ATCC) 6 samples of bacteria, fungi and other microorganism, 25 ampules of cell cultures including hybridomas, plasmids and viruses; 25 ampules of embryo samples and 2500 seeds. They should be deposited in a frozen or freeze dried form.

The Patent Act 1970 of India made many changes that have significant bearing on the patent law. Patentability requirements as per the Indian Patent Act 1970 are:

1. It should be new (Novelty).
2. It should involve an inventions step (non obviousness of the invention).
3. It should be capable of industrial application.

In the field of biotechnology a number of patents have been granted world wide in the area of r-DNA technology, microorganisms, vaccines, polymerase chain reaction (PCR), DNA/RNA and protein sequences, DNA sequencing methods, hybridoma technology, stem cell technology, plant tissue culture, genetically modified plants and animals, cloning of animals such as Dolly and many more with Indian institutes and companies attempting to pioneer the world wide IP race.

Some of the Indian companies active in the area of IPR include Ranboxy, Biocon, Dr. Reddy's labs, Shantha Biotech, Bharat Biotech, Panacea, Dabur, Ayur and Himalaya. The traditional knowledge resources of India have also seen the light of patentability since a long time now with different institutes of CSR such as Indian Institute of Chemical Technology (IICT) Hyderabad, Regional Research Laboratory (RRL), Johrat, Assam National Botanical Research Institute (NBRI) Lucknow, Central Institute of medicinal and aromatic plants (IC-MAP) Lucknow contributing towards it. This shows that the concept of patent protection has boosted technological innovations through out the world.

On the other hand, following are not inventions and cannot be patented:

1. An invention which is frivolous.
2. An invention the primary or intended use or commercial exploitation is contrary to public order or morality.
3. The mere discovery of scientific principle or the formulation of theory.
4. The mere discovery of any new property or new use of known substance
5. Mere admixture resulting a substance.
6. Mere arrangement or rearrangement or duplication of known device.
7. A method of agriculture or horticulture.
8. A mathematical or business method or a computer program per se.
9. Topography of integrated circuits.
10. A presentation of information.
11. Plants and animals in whole and thereof other than microorganism.
12. A literacy, dramatic, musical or artistic work including cinematographic works and television production.
13. Inventions relating to atomic energy.

As per the Indian Patent Act 1970 the term of patent shall be 20 years from the date of filing of application for the patent. Therefore, during these 20 years the patent holder can commercially exploit the invention in the market since he/she holds monopoly or can authorize any other person to do so, or license such usage.

22.7 COST OF PATENT

Cost of patent differs from country to country. In USA the fees was $ 30 at the time of application and at the time of patent issue. But it has been enhanced to $65 as application fee and extra

charges $ 10.00 for claims and additional charges $10 per page of specifications. Additional fees for drawings, figures and fee for Attorney General.

22.8 PATENT IN DIFFERENT COUNTRIES

An inventor should apply for grant of patent in different countries. Patent system in different countries varies significantly. In many countries an inventor is considered as the individual who first files a patent application, not necessarily the one who first conceives the invention and reduces it to practice. At present pharmaceuticals and the microbiological processes cannot be patented in Italy. On the other hand, Germany and Netherlands provide patent grants for processes of making chemicals and pharmaceuticals but the products can not be patented. Most countries, other than USA, require payment of periodic maintenance fee to keep a patent inforce. Many countries require that a patent actually be held with in the country granting the patent is to stay in force. In few countries the patent may be recovered if not used. If Govt. feels in the interest of Govt., it may request the licensing authority to issue license to other who are willing to use the invention. On January 1, 1980 at Paris agreed that once the application is filed the applicant enjoys for the purpose of priority for 12 months running from the priority date. He has to take a decision whether to file in a convention country within that one year period.

Violation of patent Rights: If any person violates inventors protection rights acts, he is liable to infringement. The patent owner gets triple damages and court fee.

22.9 PROSPECTS

It is envisaged that granting of patent may be changed to patent grant the first who file rather than actual inventor. Preliminary application prior to finalization of discovery or process which will be published within 18-24 hrs. However, the patent will be confirmed after verification. In USA the validity of patent may be raised from present 17 to 20 years.

REVIEW QUESTIONS

I. Essay type questions:

1. What is patent right? Trace the evolution of patent law.
2. Give an account of procedure for obtaining a patent.
3. Discuss merits and demerits in obtaining a patent.
4. What is intellectual property right? Discuss in detail the different forms of its protection.
5. What are important criteria for determining the patentability of an invention?
6. What is the necessity of depositing of microorganism in a depository (IDA) recognized under Budapest treaty? Does India have such a depository?
7. Differentiate between IPR and patent.
8. Write an essay on patenting biological materials.

II. Write short notes on:

(a) Impringement of patents
(b) Trade secrets and Trade marks
(c) Copy rights
(d) Patent claims

(e) International status of patents
(f) Patent specifications
(g) Plant variety protection
(h) Patent systems
(i) Validity of patent
(j) Plant variety protection
(k) Patenting trends in the area of recombinant DNA technology
(l) Patenting live forms

FURTHER READING

1. Irwing Marcus (1979). Fermentation processes and products problems in patenting in microbial technology, vol. 2 (ed.) Academic Press, pp. 497-529
2. John Candlish ( ), Patenting Inventions in microbiology in microbial biotechnology: Principles and applications (eds) pp. 749 – 766.
3. Jennifer Gorden Intellectual property in Downstream processes (ed.) pp. 309–319
4. Balasubramanian, D., Rao, A.N., and Thomas, J. (1997) Intellectual Property rights in biology, Indian National Research Centre, New Delhi.
5. Cain, B (2003). Legal aspects of Gene Technology, Thomson Sweet and Maxwell, London.
6. Crepsi, S. (1995) Biotechnology Patenting. The wicked animal must defend itself, EPIR 9: 431-441.
7. Gopala Krishnan, N.S. (1995). Biotechnology and Intellectual property rights protection. Academy law review (vol. 19, P 1-30), school of legal studies, Cochin university of science and technology.
8. Gopal Krishnan, N.S. (2000). Patenting of microorganisms under TRIPS-concerns of India, in Biotechnology and Development Review, pp. 55-63.
9. Grub, P.W. (1999). Patents for chemicals, pharmaceuticals and biotechnology, Fundamentals of colobal law, practice strategies Oxford University Press, New York, pp. 224-242, 245-260.

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